START

ENN581 Robot Motion, Control & Planning: revision booklet

Written exam · 2 hours · any calculator · printed notes only. Print at 100 % (actual size), A4 landscape, double-sided (flip on the short edge).

How to use it. Find the question type on the **premises page** (what you are given, what you must find) or in the **lookup index** ("if the question says…"), then go to the week. Each week follows the slides' order and ends with *How this is examined: worked solutions*. The edge tab on the right shows where you are: START, RECIPES, BASICS (maths builders, tables), ROBOTS (standard robots), W1–W12, SAMPLE (the W13 sample questions solved). **Colour key.** Boxes: **blue** formula (with a SYMBOLS table), **green** steps, **purple** worked example, **grey** derivation, **teal** alternative approaches side by side, **red dashed** trap, **amber** slide check (a slide error and its correction), **slate** premise (given → find), **black** exam-style question with full solution. Tags: [p. 12] = slide page of that week's lecture; [added] = not on the slides; [checked] = number verified by script. Figures marked [W2 lec p. 28] are cut from the slides; [drawn for this booklet] figures were drawn from the same maths.

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Read the question, find its row, write the *given* and the *find* at the top of your answer (it earns method marks), then use the core relation. Symbols: $\mathbf{q} = (q_1, \ldots, q_n)$ joint variables ($q_i = \theta_i$ for a revolute joint in rad, $q_i = d_i$ for a prismatic joint in m); {0} or {S} base/station frame; {E} or {T} end-effector/tool frame; ${}^{A}\mathbf{T}_{B}$ pose of {B} expressed in {A}; subscript d = desired.

Problem	Given	Find	Core relation	Go to
Mechanisms and kinematics (W1–W6)				
Mobility (DoF) of a mechanism	a drawing: links N (count the ground), joints J, freedom f_i of each joint; planar ($D = 3$) or spatial ($D = 6$)	mobility M = number of independent inputs needed	$M = D(N - J - 1) + \sum_{i=1}^{J} f_i$; extended: $+R_C - R_M$	W1, W4
Forward kinematics (FK)	joint variables $\mathbf{q}$; geometry (link lengths, DH table or a drawing)	tool pose ${}^{0}\mathbf{T}_{E} = \begin{bmatrix} {}^{0}\mathbf{R}_{E} & {}^{0}\mathbf{p}_{E} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}$ (position and orientation)	${}^{0}\mathbf{T}_{E} = {}^{0}\mathbf{T}_{1}(q_1)\,{}^{1}\mathbf{T}_{2}(q_2)\cdots{}^{n-1}\mathbf{T}_{n}(q_n)\,{}^{n}\mathbf{T}_{E}$ (ETS or DH)	W2
Transform between frames	frames drawn (or described) relative to each other	${}^{A}\mathbf{T}_{B}$, a point in another frame ${}^{A}\mathbf{P} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{P}$, or the inverse ${}^{B}\mathbf{T}_{A}$	${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C}$; ${}^{B}\mathbf{T}_{A} = \begin{bmatrix} \mathbf{R}^{\top} & -\mathbf{R}^{\top}\mathbf{t} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}$	W2
Inverse kinematics (IK)	desired tool pose ${}^{0}\mathbf{T}_{E}$, or only position (x, y, z) / planar (x, y, ϕ)	*all* joint vectors $\mathbf{q}$ that reach it (0, 1, 2, … solutions; say which branch)	solve $\mathrm{FK}(\mathbf{q}) = {}^{0}\mathbf{T}_{E}$: geometric (cosine rule, atan2) or algebraic	W3
Workspace	link lengths, joint types (and limits)	reachable workspace (every position the tool can reach) / dexterous (reach with every orientation); sketch it	distance range from the shoulder $\lvert L_1 - L_2\rvert \ldots L_1 + L_2$; sweep about the base axis	W3
Parallel (closed-chain) IK	desired output pose of the platform / end link (e.g. four-bar output angle)	actuated (and passive) joint angles, every assembly mode	loop-closure: the vector sum round each loop = $\mathbf{0}$	W4
Parallel FK	actuated joint values	platform pose (usually several solutions)	same loop-closure equations, solved the other way	W4
Trajectory	start, goal (and via) values θ_0, θ_f (θ_v), duration t_f, boundary velocities (accelerations), or limits $\dot{\theta}_{\max}, \ddot{\theta}$	$\theta(t), \dot{\theta}(t), \ddot{\theta}(t)$: cubic coefficients $a_0..a_3$, or LSPB blend time t_b	cubic: 4 boundary conditions → 4 coefficients; LSPB: $s = ut + \frac{1}{2}at^2$ per segment	W5
Velocity kinematics	$\mathbf{q}$, joint rates $\dot{\mathbf{q}}$	tool velocity $\boldsymbol{\nu} = (\boldsymbol{\upsilon}, \boldsymbol{\omega})$ (linear, angular)	$\boldsymbol{\nu} = \mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}}$	W6
Inverse velocity (resolved rate)	desired tool velocity $\boldsymbol{\nu}$ (e.g. trace a line/circle), $\mathbf{q}$	joint rates $\dot{\mathbf{q}}$	$\dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\,\boldsymbol{\nu}$ (square, non-singular $\mathbf{J}$)	W6
Singularities	$\mathbf{J}(\mathbf{q})$	poses where the arm loses a direction of motion	$\det \mathbf{J}(\mathbf{q}) = 0$	W6
Statics	wrench at the tool $\mathbf{F}_E$ (force N, moment N m), $\mathbf{q}$	joint torques/forces $\boldsymbol{\tau}$ that hold it	$\boldsymbol{\tau} = \mathbf{J}^{\top}(\mathbf{q})\,\mathbf{F}_E$	W6
Dynamics and control (W7–W9)				
Equations of motion (EoM)	masses m_i, centre-of-mass positions, inertias I_i, geometry, gravity g	$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q})$ in full	Lagrange: $\mathcal{L} = K - P$, $Q_i = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i}$; or Newton–Euler	W7
Inverse dynamics	motion $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$	actuator torques/forces $\mathbf{Q}$	substitute into the EoM	W7, W9
Forward dynamics	torques $\mathbf{Q}$, state $\mathbf{q}, \dot{\mathbf{q}}$	accelerations $\ddot{\mathbf{q}}$ (then integrate)	$\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g})$	W7
Single-joint control (servo)	plant $m\ddot{x} + b\dot{x} + kx = f$, specification (critically damped, stiffness, disturbance)	gains k_p, k_v (k_i), closed-loop response, steady-state error	$f = \alpha f' + \beta$, $f' = \ddot{x}_d + k_v\dot{E} + k_pE$; critical: $k_v = 2\sqrt{k_p}$	W8
Model-based (computed-torque) control	robot model $\mathbf{M}, \mathbf{C}, \mathbf{g}$; desired trajectory $\mathbf{q}_d(t)$	control law $\mathbf{Q}$; error dynamics	$\mathbf{Q} = \boldsymbol{\alpha}\dot{\mathbf{Q}} + \boldsymbol{\beta}$, $\boldsymbol{\alpha} = \mathbf{M}(\mathbf{q})$, $\boldsymbol{\beta} = \mathbf{C} + \mathbf{g}$, $\dot{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$	W9
Cartesian / force control	desired Cartesian path $\mathbf{x}_d(t)$ or contact force $\mathbf{f}_d$	joint torques; which directions are position- vs force-controlled	$\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$; error $\mathbf{E} = \mathbf{x}_d - \mathbf{x}$ or $\mathbf{f}_d - \mathbf{f}_e$	W9
Wheeled robots (W10–W11)				
Wheeled FK	wheel speeds $\dot{\varphi}_i$ (and steering angles β_i), wheel layout	chassis velocity $\zeta = (u, v, \dot{\theta})$ in {C}; world velocity $(\dot{x}, \dot{y}, \dot{\theta})$ in {O}; ICR	invert the stacked wheel equations; $\nu = \mathbf{J}(\theta)\,\zeta$	W10
Wheeled IK	desired chassis (or world) velocity, e.g. drive a circle or a square	wheel speeds $\dot{\varphi}_i$ (and steering)	each wheel's rolling equation, one row per wheel	W10
Wheeled-robot dynamics	mass m, inertia ${}^{G}I$, wheel forces, COM offset	chassis accelerations; the forces/torques needed	Newton–Euler in {C}: $\mathbf{F} = m\,\mathbf{a}_G$, $N_z = {}^{G}I\ddot{\theta}$	W11
Planning (W12)				
Graph search	graph/grid, edge costs, start, goal (heuristic h)	order of expansion, path, its cost; is it optimal?	BFS (queue), DFS (stack), Dijkstra (cost g), A* ($f = g + h$, h admissible)	W12
Sampling-based planning	C-space, obstacles, start, goal	a collision-free path (not optimal)	PRM (roadmap, then search); RRT (grow a tree towards samples)	W12

If the question says… go to

"transformation matrix", "indicate frames", ETS .... p. 5, 33
"DH table", "link parameters" ..... p. 5, 33
"sketch the workspace" ..... p. 5, 45
"derive the inverse kinematics" ..... p. 5, 45
"how many solutions" ..... p. 5
"mobility", "degrees of freedom", Grübler ..... p. 5, 26, 56
"parallel", "four-bar", "loop closure" ..... p. 5, 56
"cubic", "via point", "spline", "blend" ..... p. 5, 67
"Jacobian", "singularity", "joint torques for a force" p. 5, 78
"equations of motion", Lagrange ..... p. 5, 91
"inertia tensor", "Newton–Euler" ..... p. 91
"critically damped", "gains", "steady-state error" .. p. 5, 102
"computed torque", "partitioned", "force control" . p. 5, 111
"wheel speeds", "omni", "differential drive", ICR . p. 5, 122
"holonomic" ..... p. 122
"dynamics of a mobile robot" ..... p. 134
"A*", "Dijkstra", "PRM", "RRT", "C-space" ..... p. 5, 142
"what is given / what to find" ..... p. 3
"trace a circle / line / square" (joint rates) ..... p. 78
"drive a circle / rectangle" (wheel speeds) ..... p. 122
"block diagram", "model-based", "servo" ..... p. 102, 111
"BFS", "DFS", open/closed list ..... p. 142
RR, RP, RRP, SCARA, Puma, Panda: tables, ETS ..... p. 15
skew matrix, cross product, $s = ut + \frac{1}{2}at^2$**,** atan2 ..... p. 6
the W13 sample questions, solved ..... p. 153
trig identities, rotations, inertia of shapes ..... p. 6

Answer layout (method marks)

1. **Draw and define**: frames, axes, positive angle directions, assumptions (no joint limits; point masses; rest to rest; $\gamma = 0$).
2. **State the general tool** before using it: ETS, cosine rule, Euler–Lagrange, G–K, boundary conditions.
3. **Substitute and simplify**, one step per line, symbols first, numbers last, with units.
4. **Box the result**; list *every* branch (IK) or mode (four-bar).
5. **Sanity check** at $\theta = 0$, $t = 0$, or a symmetric case. Write "check:".

Time (2 hours)

Read all questions first (5 min). Split time by marks. Leave 10 min to check signs and units. If stuck: write the set-up and the equation you would solve, then move on.

Ten checks before you hand in

1. Calculator in **radians**? Angles labelled ° or rad?
2. atan2, not $\tan^{-1}(y/x)$ (quadrant).
3. Both IK branches (and "no solution if $|\cos| > 1$").
4. Order of matrix products (right = about the current frame).
5. Transposes: $\mathbf{R}^{-1} = \mathbf{R}^\top$; inverse $\mathbf{T}^{-1} = [\mathbf{R}^\top \;\; -\mathbf{R}^\top\mathbf{t}]$.
6. $\mathcal{L} = K - P$; gravity sign; product rule in $\frac{d}{dt}$.
7. $k_v^2 = 4k_p$ for critical damping (unit mass).
8. Ground counted as a link; $D = 3$ planar, 6 spatial.
9. Wheel row: $-\dot\theta\, ^C y$ in the x part, $+\dot\theta\, ^C x$ in the y part.
10. Units: mm vs m; rad/s vs rpm; N m vs N.

Notation

${}^A\mathbf{T}_B$	pose of $\{B\}$ in $\{A\}$; ${}^A\mathbf{P} = {}^A\mathbf{T}_B\, {}^B\mathbf{P}$
${}^A\mathbf{R}_B$	columns = B's axes written in A
$R_x, R_y, R_z,$ t_x, t_y, t_z	elementary transforms (ETS)
$a_{i-1}, \alpha_{i-1}, d_i, \theta_i$	proximal (Craig) DH
c_i, s_i, c_{ij}	$\cos\theta_i$, $\sin\theta_i$, $\cos(\theta_i+\theta_j)$
$\mathbf{J}$	Jacobian: $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$
$\mathbf{M}, \mathbf{C}, \mathbf{g}$	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$
$E = q_d - q$	servo error (desired minus actual)
M, D, N, J, F_i	mobility; 3/6; links incl. ground; joints; joint DoF
R_C, R_M	over-constraints (add); local mobility (subtract)
$u, v, \dot\theta$	chassis velocity in $\{C\}$; $\dot\varphi_i$ wheel spin
β_i, γ_i	wheel drive angle from x_C; roller angle
g, h, f	A*: cost so far, heuristic, $f = g + h$

Elementary rotations

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Use: R_y's minus sign is bottom-left. A translation $t_x(a)$ is $\mathbf{I}$ with a in row 1, column 4.

Sample-paper map (W13 revision): solved in full on p. 153

Q1 FK of RRR (ETS)	W2	p. 5
Q2 workspace sketch	W3	p. 5
Q3 IK derivation	W3	p. 5
Q4 equations of motion	W7	p. 5
Q5 four-bar IK	W4	p. 5
Q6–Q7 mobility	W1, W4	p. 5
Q8–Q9 cubic splines	W5	p. 5
Q10 omni robot	W10	p. 5

STEPS 1 · Forward kinematics by ETS (W13 Q1)

1. Draw {S}; mark each joint axis and its *drawn* positive sense.
2. Walk base → tool. Translations: t_x, t_y, t_z. Joint: $R_{axis}(\pm\theta_i)$; minus if the drawn arrow opposes the right-hand rule about your axis.
3. End with the constant rotation that makes your last frame match {T}.
4. Group per joint and write each ${}^{i-1}\mathbf{T}_i$ (= "indicate intermediate frames").
5. Multiply; give at least the position column. Check: all $\theta = 0$ reproduces the drawing.

STEPS 2 · Proximal DH table

1. z_i along joint axis i. Origin of {i} at the common normal to axis $i{+}1$ (or the intersection).
2. x_i along that normal towards axis $i{+}1$; $y_i = z_i \times x_i$.
3. Row i: a_{i-1} ($z_{i-1} \to z_i$ along x_{i-1}), α_{i-1} (about x_{i-1}), d_i ($x_{i-1} \to x_i$ along z_i), θ_i (about z_i).
4. ${}^{i-1}\mathbf{T}_i = R_x(\alpha_{i-1})t_x(a_{i-1})R_z(\theta_i)t_z(d_i)$; multiply rows.
5. Add a constant tool transform if {N} ≠ {tool}. Write "proximal (Craig)".

STEPS 3 · Sketch the workspace (W13 Q2)

1. Find the joints that only *spin the plane* (base rotation) and the planar sub-arm.
2. Planar 2-link sub-arm: distance from its first joint ranges $|L_a - L_b| \ldots L_a + L_b$ (cosine rule) → annulus.
3. Spin the annulus about the base axis. If the axis passes through the annulus's centre → spherical shell; otherwise → a torus-like solid. Joint limits cut wedges; a prismatic joint extrudes.
4. Draw side and top views; label radii, centre height. Say "no joint limits, collisions ignored".

STEPS 4 · Derive the inverse kinematics (W13 Q3)

1. Write FK first. Spherical wrist? Wrist centre $\mathbf{p}_W = \mathbf{p}_E - d_6\,\mathbf{R}\hat{z}$.
2. Top view: $\theta_1 = \operatorname{atan2}(y, x)$, second branch $\theta_1 + \pi$ (reach back, $r < 0$).
3. Side view: subtract base height; planar 2R in (r, z'): $\cos\theta_3 = \dfrac{r^2 + z'^2 - L_a^2 - L_b^2}{2L_aL_b}$ (relative angle; interior angle is $\pi - \theta_3$).
4. $\theta_3 = \operatorname{atan2}(\pm\sqrt{1-c_3^2}, c_3)$; $\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_b s_3,\ L_a + L_b c_3)$.
5. Wrist: ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_6$ → Euler angles (two: wrist flip).
6. Count: shoulder × elbow (× flip). No solution if $|c_3| > 1$. Check one branch by FK.

STEPS 5 · Mobility (W13 Q6, Q7)

1. $D = 3$ planar, 6 spatial (S, U joints or "still 3D" → 6).
2. Number bodies; ground = 1; a prismatic pair = 2 bodies → N.
3. List joints and F_i (R, P = 1; U, C = 2; S = 3) → J, $\sum F_i$.
4. $M = D(N - J - 1) + \sum F_i$. Implausible? Extended: $+R_C - R_M$.
5. R_M: idle spins (rod with S at both ends → 1 each). $R_C = \sum F_i - \sum u_k$ (u_k: distinct platform DoF per loop, union).

Delta: $6(11 - 15 - 1) + 39 = 9$, then $9 + 0 - 6 = 3$. *4RRR:* $-6 + 7 - 0 = 1$.

STEPS 6 · Parallel robot: loop closure, IK, FK (W13 Q5)

1. Break the loop (at a point K, or both ends of a link) into serial chains.
2. Write the break point's pose from each chain; equate (3 equations at K; 1 length equation if broken at both ends).
3. **IK** (pose of K, ϕ given): $A = K - a(\cos\phi, \sin\phi)$, $\theta_1 = \operatorname{atan2}(A_y, A_x)$, $\varphi_2 = \phi - \theta_1$; $B = K + b(\cos\phi, \sin\phi)$, $\varphi_1 = \operatorname{atan2}(B_y, B_x - L_0)$, $\varphi_3 = \phi + \pi - \varphi_1$.
4. **FK** (θ_1 given): intersect circles (L_2 about A, L_3 about D): two assembly modes, or none.

STEPS 7 · Trajectories (W13 Q8, Q9)

1. Rest-to-rest cubic: $a_0 = \theta_0$, $a_1 = 0$, $a_2 = \dfrac{3\Delta}{t_f^2}$, $a_3 = -\dfrac{2\Delta}{t_f^3}$; peak speed $1.5\Delta/t_f$ at $t_f/2$.
2. Given end speeds: $a_2 = \frac{3\Delta}{t_f^2} - \frac{2\dot\theta_0}{t_f} - \frac{\dot\theta_f}{t_f}$, $a_3 = -\frac{2\Delta}{t_f^3} + \frac{\dot\theta_0 + \dot\theta_f}{t_f^2}$.
3. Two segments, continuous acceleration: list 8 conditions (pos at both ends ×2, rest at start and end, $\dot\theta$ and $\ddot\theta$ match at the via) and solve.
4. Blends (LSPB): need $\ddot\theta \geq 4\Delta/t_f^2$; $t_b = \dfrac{t_f}{2} - \dfrac{\sqrt{\ddot\theta^2 t_f^2 - 4\ddot\theta\Delta}}{2\ddot\theta}$; linear speed $\ddot\theta t_b$.
5. Check: positions and speeds at both ends.

STEPS 8 · Jacobian, singularity, statics

1. Position FK $\mathbf{x}(\mathbf{q}) \to \mathbf{J} = \partial\mathbf{x}/\partial\mathbf{q}$ (column j = how the tool moves for joint j). Geometric: revolute column $\hat{z}_j \times (\mathbf{p}_E - \mathbf{p}_j)$; prismatic $\hat{z}_j$.
2. Singular where $\det\mathbf{J} = 0$ (square) or rank drops. 2R: $\det = L_1L_2\sin q_2$.
3. $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}}$; near-singular → huge joint rates.
4. Statics: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ (same frame for J and F).

STEPS 9 · Equations of motion by Lagrange (W13 Q4)

1. Coordinates $\mathbf{q}$; gravity direction drawn.
2. COM positions from FK; differentiate → velocities.
3. $K = \sum \frac{1}{2}m_i v_{C_i}^2 + \frac{1}{2}\omega_i^\top I_i \omega_i$; $P = \sum m_i g h_i$; $\mathcal{L} = K - P$.
4. Per joint: $\partial\mathcal{L}/\partial\dot{q}_i \to \frac{d}{dt}$ (product rule!) → minus $\partial\mathcal{L}/\partial q_i$.
5. Collect M ($\ddot{q}$ terms), C (velocity products), g. Check M symmetric, g has no $\dot{q}$, C = 0 at rest.

2R point masses: $M_{11} = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2)$, $M_{12} = m_2(L_1L_2c_2 + L_2^2)$, $M_{22} = m_2L_2^2$.

STEPS 10 · Control design (W8, W9)

1. Plant $m\ddot{x} + b\dot{x} + kx = f$ (or $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} = \mathbf{Q}$).
2. Partition: $f = \alpha f' + \beta$, $\alpha = m$ (**M**), β = everything else.
3. Servo: $f' = \ddot{x}_d + k_v\dot{E} + k_pE$, $E = x_d - x \to \ddot{E} + k_v\dot{E} + k_pE = 0$.
4. Critical: $k_v = 2\sqrt{k_p}$ ($\omega_n = \sqrt{k_p}$, $\zeta = k_v/2\sqrt{k_p}$).
5. Constant disturbance: $E_{ss} = f_{dist}/(k_pm)$; add $k_i\int E$ to remove it (stable if $k_vk_p > k_i$).
6. Model error: hats on α, β; the error equation gets a right-hand side.

STEPS 11 · Wheeled robot kinematics (W13 Q10)

1. For each wheel: $({}^Cx, {}^Cy)$, β (drive direction from x_C), γ, r.
2. Row: $r\dot{\varphi}_i = [\cos\beta \ \ \sin\beta]\,[u - \dot\theta\,{}^Cy,\ v + \dot\theta\,{}^Cx]^\top$ ($+\tan\gamma$ term for omni).
3. Standard wheels: add no-slide $[-\sin\beta \ \ \cos\beta]\,(\cdot) = 0$.
4. Stack → invert. Inertial: $\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$. ICR in {C}: $(-v/\dot\theta,\ u/\dot\theta)$.

STEPS 12 · Planning by hand (W12)

1. Table: node, g, h, $f = g + h$, parent; open and closed sets.
2. Expand the lowest f (Dijkstra: lowest g); relax neighbours; record improvements.
3. Stop when the goal is *expanded*, not when discovered. Trace parents.
4. A* optimal if h admissible ($h \leq$ true cost); consistent: $h(n) \leq c(n, m) + h(m)$.
5. PRM: sample, connect, search (multi-query). RRT: grow towards random samples (single query). Both probabilistically complete, not optimal.

How to use this part

Every box below is a *builder*: a formula you will need inside a week's recipe. Each has the formula in full, a SYMBOLS table, a "Use" line (which week needs it, and for what) and a tiny numeric example. Everything here is standard first-year maths and physics, so it is marked [added](not from the lecture slides) unless a slide page is given.

Builder	Where the unit uses it
1 Vectors	W6 Jacobian columns, $\boldsymbol{\omega}\times\mathbf{r}$; W7 torque, W10 ICR
2 Matrices	W3 IK, W6 $\mathbf{J}^{-1}$, $\mathbf{J}^{+}$, W7 $\mathbf{M}>0$, W10 wheel matrix
3 Rotations	W2 $\mathbf{R}$, ETS, DH; W3 wrist angles; W10 $\mathbf{R}(\theta)$
4 Homogeneous $\mathbf{T}$	W2 FK, W3 IK, W4 frames, W7 Newton–Euler
5 Trigonometry	W2–W4 every arm; $a\cos\theta + b\sin\theta = c$ in IK
6 Algebra	W3, W4 quadratic roots, number of IK solutions
7 Calculus	W6, W7 (Lagrange), W5 (cubic derivatives)
8 Particle kinematics	W5 trajectories, W10 wheel speeds, W11
9 Dynamics basics	W7 K, P, inertia; W8 gear and motor
10 Second-order systems	W8, W9 PD gains, W11 error dynamics
11 Units and signs	every numeric answer
12 Graphs	W12 search
13 Notation table	whole booklet

1. Vectors

PREMISE vectors **Given:** two or three vectors $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$, all written in the same frame
Find: lengths, angles, a perpendicular direction, a rotational velocity or a volume

Dot product, norm, unit vector, angle, projection [added]

$$\mathbf{a}\cdot\mathbf{b} = \mathbf{a}^{\top}\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\varphi, \qquad |\mathbf{a}| = \sqrt{\mathbf{a}\cdot\mathbf{a}}$$

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}, \qquad \varphi = \arccos\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}, \qquad \mathrm{proj}_{\mathbf{b}}\,\mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}\,\mathbf{b}$$

SYMBOLS

$\mathbf{a},\mathbf{b}$	vectors in one frame (3×1 columns)	any
a_i, b_i	components along x, y, z	any
$\lvert\mathbf{a}\rvert$	length (norm) of $\mathbf{a}$	same as $\mathbf{a}$
$\hat{\mathbf{a}}$	unit vector along $\mathbf{a}$ (length 1)	–
φ	angle between the vectors, $0\le\varphi\le\pi$	rad
$\mathrm{proj}_{\mathbf{b}}\,\mathbf{a}$	component of $\mathbf{a}$ along $\mathbf{b}$ (a vector); the scalar part is $\mathbf{a}\cdot\hat{\mathbf{b}}$	same as $\mathbf{a}$

Use: lengths and angles in arm triangles (W3), speed = $|\mathbf{v}|$ (W10); $\mathbf{a}\cdot\mathbf{b}=0$ means perpendicular; columns of a rotation matrix are unit and perpendicular (W2).

Example. $\mathbf{a}=(1,2,2)$, $\mathbf{b}=(2,0,1)$: $\mathbf{a}\cdot\mathbf{b}=2+0+2=4$, $|\mathbf{a}|=3$, $|\mathbf{b}|=\sqrt{5}=2.236$, $\varphi=\arccos(4/6.708)=53.4°$, $\mathrm{proj}_{\mathbf{b}}\,\mathbf{a}=\frac{4}{5}(2,0,1)=(1.6,0,0.8)$. [checked]

Cross product: determinant expansion (mind the $\hat{\mathbf{j}}$ sign) [added]

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix}\hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix} = \hat{\mathbf{i}}\begin{vmatrix}a_2 & a_3\\ b_2 & b_3\end{vmatrix} - \hat{\mathbf{j}}\begin{vmatrix}a_1 & a_3\\ b_1 & b_3\end{vmatrix} + \hat{\mathbf{k}}\begin{vmatrix}a_1 & a_2\\ b_1 & b_2\end{vmatrix}$$

$$= \begin{bmatrix} a_2b_3 - a_3b_2 \\ -(a_1b_3 - a_3b_1) = a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1\end{bmatrix}, \qquad |\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\varphi, \qquad \mathbf{a}\times\mathbf{b} = -\,\mathbf{b}\times\mathbf{a}$$

SYMBOLS

$\hat{\mathbf{i}},\hat{\mathbf{j}},\hat{\mathbf{k}}$	unit vectors along x, y, z	–
$\mathbf{a}\times\mathbf{b}$	vector perpendicular to both, direction by the right-hand rule (fingers $\mathbf{a}\to\mathbf{b}$, thumb along the result)	unit of $a\cdot b$
$\lvert\mathbf{a}\times\mathbf{b}\rvert$	area of the parallelogram on $\mathbf{a},\mathbf{b}$; zero if parallel	–

Use: W6: linear velocity of a point $\mathbf{v}=\boldsymbol{\omega}\times\mathbf{r}$ and Jacobian column $\hat{\mathbf{z}}_i\times(\mathbf{p}_e-\mathbf{p}_i)$; W7: torque $\boldsymbol{\tau}=\mathbf{r}\times\mathbf{F}$; $\mathbf{a}\times\mathbf{a}=\mathbf{0}$.

Example. $\mathbf{a}=(1,2,2)$, $\mathbf{b}=(2,0,1)$: $\hat{\mathbf{i}}$: $2\cdot1-2\cdot0=2$; $\hat{\mathbf{j}}$: $-(1\cdot1-2\cdot2)=+3$; $\hat{\mathbf{k}}$: $1\cdot0-2\cdot2=-4$. So $\mathbf{a}\times\mathbf{b}=(2,3,-4)$. Checks: $\cdot\mathbf{a}=2+6-8=0$, $\cdot\mathbf{b}=4+0-4=0$, $|\cdot|=\sqrt{29}=5.385=3\cdot2.236\cdot\sin53.4°$. [checked]

Cross product as a matrix product: the skew matrix $[\mathbf{a}]_\times$ and vex [added]

$$\mathbf{a}\times\mathbf{b} = [\mathbf{a}]_\times\mathbf{b} = \begin{bmatrix}0 & -a_3 & a_2\\ a_3 & 0 & -a_1\\ -a_2 & a_1 & 0\end{bmatrix}\begin{bmatrix}b_1\\ b_2\\ b_3\end{bmatrix}, \qquad \mathbf{a} = \vee_\times([\mathbf{a}]_\times) = \begin{bmatrix}S_{32}\\ S_{13}\\ S_{21}\end{bmatrix}$$

$\mathbf{S}=[\mathbf{a}]_\times$: $\mathbf{S}^{\top}=-\mathbf{S}$ (skew) $\qquad$ $\mathbf{S}\,\mathbf{a}=\mathbf{0}$

$[\mathbf{a}]_\times\mathbf{b}=-[\mathbf{b}]_\times\mathbf{a}$ $\qquad$ $\det\mathbf{S}=0$ (rank 2)

$[\mathbf{a}]_\times^2=\mathbf{a}\mathbf{a}^{\top}-|\mathbf{a}|^2\mathbf{I}$ $\qquad$ $\mathbf{R}[\mathbf{a}]_\times\mathbf{R}^{\top}=[\mathbf{R}\mathbf{a}]_\times$

SYMBOLS

$\mathbf{S}_{ij}$	row i, column j entry of $\mathbf{S}$	–
$\vee_\times$	"vex": reads the vector back from a skew matrix (zero diagonal)	–
$\mathbf{I}$	3×3 identity	–

Use: W6 [p. 10]: $\dot{\mathbf{R}}=[\boldsymbol{\omega}]_\times\mathbf{R}$ needs a matrix, not a cross product; a vector crossed into itself vanishes ($\mathbf{S}\mathbf{a}=\mathbf{0}$).

Example. $\mathbf{a}=(1,2,2)$: $[\mathbf{a}]_\times=\begin{bmatrix}0&-2&2\\2&0&-1\\-2&1&0\end{bmatrix}$; times $\mathbf{b}=(2,0,1)^{\top}$: $(0-0+2,\ 4+0-1,\ -4+0+0)=(2,3,-4)$, the same as above. [checked]

Scalar triple product (volume) [added]

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \det\begin{bmatrix}\mathbf{a}^{\top}\\ \mathbf{b}^{\top}\\ \mathbf{c}^{\top}\end{bmatrix} = \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \mathbf{c}\cdot(\mathbf{a}\times\mathbf{b}), \qquad \mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$$

SYMBOLS

$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$	signed volume of the box on $\mathbf{a},\mathbf{b},\mathbf{c}$ (zero if coplanar)	m^3 if lengths

Use: test whether three vectors are coplanar (a singularity-style check); the second identity expands a double cross product (velocity of a point on a link of a rotating link, W6).

Example. $\mathbf{a}=(1,2,2)$, $\mathbf{b}=(2,0,1)$, $\mathbf{c}=(0,1,0)$: $\mathbf{b}\times\mathbf{c}=(-1,0,2)$, so $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=-1+0+4=3$. [checked]

2. Matrices

PREMISE matrices **Given:** a square or rectangular matrix, e.g. a Jacobian, an inertia matrix, a wheel matrix

Find: det, inverse, eigenvalues, rank, a definiteness test, or a generalised inverse

2×2: determinant, inverse, eigenvalues, eigenvectors [added]

$$\mathbf{A}=\begin{bmatrix}a & b\\ c & d\end{bmatrix},\quad \det\mathbf{A}=ad-bc,\quad \mathbf{A}^{-1}=\frac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\lambda^2-(a+d)\lambda+(ad-bc)=0$$

$$\lambda_{1,2}=\frac{(a+d)\pm\sqrt{(a+d)^2-4(ad-bc)}}{2},\qquad \mathbf{v}=\begin{bmatrix}b\\ \lambda-a\end{bmatrix}\ (b\neq0)$$

SYMBOLS

λ	eigenvalue: scalar with $\mathbf{Av}=\lambda\mathbf{v}$	as entries
$\mathbf{v}$	eigenvector (a direction that $\mathbf{A}$ only stretches)	–
$a+d,\ ad-bc$	trace and determinant: $\lambda_1+\lambda_2=a+d,\ \lambda_1\lambda_2=ad-bc$	–

Use: $\det=0$*: no inverse (singular Jacobian, W6). Eigenvalues of* $\mathbf{JJ}^\top$ *give the squared semi-axes of the velocity ellipse (W6); of* $\mathbf{M}$ *the sign test (W7).*

Example. $\mathbf{A}=\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$: $\det=5$, $\mathbf{A}^{-1}=\frac{1}{5}\begin{bmatrix}3 & -1\\ -1 & 2\end{bmatrix}$; $\lambda^2-5\lambda+5=0$, $\lambda=(5\pm\sqrt{5})/2=3.618,\ 1.382$; eigenvectors $(1,\lambda-2)=(1,1.618)$ and $(1,-0.618)$, perpendicular since $\mathbf{A}$ is symmetric. [checked]

3×3: determinant and inverse by the adjugate [added]

$$\det\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}=a(ei-fh)-b(di-fg)+c(dh-eg)$$

$$\mathbf{A}^{-1}=\frac{1}{\det\mathbf{A}}\begin{bmatrix}ei-fh & ch-bi & bf-ce\\ fg-di & ai-cg & cd-af\\ dh-eg & bg-ah & ae-bd\end{bmatrix}\quad\text{(the matrix is }\operatorname{adj}\mathbf{A}\text{, the transposed cofactors)}$$

SYMBOLS

$a\ldots i$	the nine entries, read row by row	–
$\operatorname{adj}\mathbf{A}$	adjugate; $\mathbf{A}\operatorname{adj}\mathbf{A}=\det\mathbf{A}\,\mathbf{I}$	–

Use: 3D Jacobians and inertia tensors (W6, W7). The minus sign on the middle term of the determinant is the same cofactor sign as the $\hat{\mathbf{j}}$ *term of the cross product.*

Example. $\mathbf{B}=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 3\\ 2 & 0 & 1\end{bmatrix}$: $\det=1(1-0)-2(0-6)+0=13$; $\operatorname{adj}\mathbf{B}=\begin{bmatrix}1 & -2 & 6\\ 6 & 1 & -3\\ -2 & 4 & 1\end{bmatrix}$, so $\mathbf{B}^{-1}=\frac{1}{13}\operatorname{adj}\mathbf{B}$. Check: row 1 of $\mathbf{B}$ times column 1 of the adjugate $=1+12+0=13$. [checked]

Transpose and inverse rules, rank, positive definiteness [added]

$$(\mathbf{AB})^\top=\mathbf{B}^\top\mathbf{A}^\top,\quad (\mathbf{AB})^{-1}=\mathbf{B}^{-1}\mathbf{A}^{-1},\quad (\mathbf{A}^\top)^{-1}=(\mathbf{A}^{-1})^\top\ \text{(order reverses)}$$

SYMBOLS

rank	number of independent rows (= columns); full rank for square $\mathbf{A}$ means $\det\mathbf{A}\neq0$	–
$\mathbf{M}\succ0$	positive definite: $\mathbf{x}^\top\mathbf{Mx}>0$ for all $\mathbf{x}\neq\mathbf{0}$ ($\mathbf{M}$ symmetric)	–

Test for $\mathbf{M}=\mathbf{M}^\top\succ0$: all eigenvalues >0; or all leading minors >0 (Sylvester: $M_{11}>0$, $\det\begin{bmatrix}M_{11} & M_{12}\\ M_{12} & M_{22}\end{bmatrix}>0$, …).

Use: W7: $\mathbf{M}(\mathbf{q})$ *is symmetric positive definite, so* $\mathbf{M}^{-1}$ *exists for forward dynamics. W6: rank of* $\mathbf{J}$ *drops at a singularity.*

Examples. $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$: minors $2>0$, $5>0$, so positive definite (eigenvalues $3.618, 1.382>0$). $\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}$: $\det=-3<0$, eigenvalues $3,-1$: indefinite. $\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 6\\ 1 & 0 & 1\end{bmatrix}$ has rank 2 (row 2 = 2× row 1). [checked]

Pseudo-inverse (right inverse) for a redundant arm [added]

$$\mathbf{J}^+=\mathbf{J}^\top\left(\mathbf{JJ}^\top\right)^{-1}\quad(\mathbf{J}:\ m\times n,\ \operatorname{rank}m<n),\qquad \dot{\mathbf{q}}=\mathbf{J}^+\boldsymbol{\nu}\ \Rightarrow\ \mathbf{J}\dot{\mathbf{q}}=\boldsymbol{\nu},\ \ |\dot{\mathbf{q}}|\ \text{smallest}$$

SYMBOLS

$\mathbf{J}$	Jacobian, m task rows by n joint columns	m, 1
$\mathbf{J}^+$	right pseudo-inverse, $n\times m$; $\mathbf{JJ}^+=\mathbf{I}_m$	–
$\boldsymbol{\nu}$	wanted task velocity ($m\times1$)	m/s, rad/s

Use: W6: more joints than task rows (e.g. planar 3R tracking a point); W10: wheel speeds from chassis velocity when there are more wheels than $(u,v,\dot\theta)$*. Tall* $\mathbf{J}$ *(*$m>n$*, rank* n*):* $\mathbf{J}^+=(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$ *(least squares).*

Example. $\mathbf{J}=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}$: $\mathbf{JJ}^\top=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$, inverse $\frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$, so $\mathbf{J}^+=\frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\\ 1 & 1\end{bmatrix}$. For $\boldsymbol{\nu}=(1,2)$: $\dot{\mathbf{q}}=(0,1,1)$; check $\mathbf{J}\dot{\mathbf{q}}=(0+1,\ 1+1)=(1,2)$, $|\dot{\mathbf{q}}|=\sqrt{2}$. [checked]

3. Rotations

PREMISE rotation **Given:** an axis and an angle, or a sequence of turns
Find: the 3×3 rotation matrix $\mathbf{R}$, its use on a vector, and the angles back from $\mathbf{R}$

Basic rotations about x, y, z (right-hand rule) [p. W2 10–12]

$$\mathbf{R}_x(\theta)=\begin{bmatrix}1&0&0\\0&c\theta&-s\theta\\0&s\theta&c\theta\end{bmatrix},\quad \mathbf{R}_y(\theta)=\begin{bmatrix}c\theta&0&s\theta\\0&1&0\\-s\theta&0&c\theta\end{bmatrix},\quad \mathbf{R}_z(\theta)=\begin{bmatrix}c\theta&-s\theta&0\\s\theta&c\theta&0\\0&0&1\end{bmatrix}$$

$$\mathbf{R}^{\top}\mathbf{R}=\mathbf{I},\quad \mathbf{R}^{-1}=\mathbf{R}^{\top},\quad \det\mathbf{R}=+1,\quad {}^{A}\mathbf{P}={}^{A}\mathbf{R}_{B}\,{}^{B}\mathbf{P}$$

SYMBOLS

θ	turn angle, positive anticlockwise looking down the axis towards the origin	rad
$c\theta, s\theta$	$\cos\theta$, $\sin\theta$	–
${}^{A}\mathbf{R}_{B}$	orientation of $\{B\}$ in $\{A\}$; its columns are $\hat{x}_B, \hat{y}_B, \hat{z}_B$ written in $\{A\}$	–
${}^{B}\mathbf{P}$, ${}^{A}\mathbf{P}$	the same point as a column in $\{B\}$ and in $\{A\}$	m

Use: W2 every FK; W3 wrist; W10 $\mathbf{R}(\theta)$ *between chassis* $\{C\}$ *and world* $\{O\}$. *Same-axis angles add:* $\mathbf{R}_z(a)\mathbf{R}_z(b)=\mathbf{R}_z(a+b)$. *Different axes do not commute. The minus sign sits in the upper right of* $\mathbf{R}_x, \mathbf{R}_z$ *but the lower left of* $\mathbf{R}_y$.

Example (2D, top-left block of $\mathbf{R}_z$). $\mathbf{R}_z(30^\circ)\,(1,0)^{\top}=(\cos30^\circ,\sin30^\circ)=(0.866,0.5)$. [checked]

ALTERNATIVE APPROACHES

A. Current (moving) axes: post-multiply

First turn $\mathbf{R}_1$, then turn about the *new* axes by $\mathbf{R}_2$:

$$\mathbf{R}=\mathbf{R}_1\,\mathbf{R}_2$$

Read left to right: the order you write is the order you do. This is how ETS and DH chains are written.

B. Fixed (world) axes: pre-multiply

First turn $\mathbf{R}_1$, then turn about the *original* axes by $\mathbf{R}_2$:

$$\mathbf{R}=\mathbf{R}_2\,\mathbf{R}_1$$

Read right to left: the last matrix written is the last turn, about the fixed frame.

Same two turns, two orders (90° about z and 90° about x) [added]

$$\mathbf{R}_z(90^\circ)\mathbf{R}_x(90^\circ)=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix},\qquad \mathbf{R}_x(90^\circ)\mathbf{R}_z(90^\circ)=\begin{bmatrix}0&-1&0\\0&0&-1\\1&0&0\end{bmatrix}$$

The two answers differ, so the order is part of the answer. Check the first by its columns: the new x-axis is $(0,1,0)$ (the old x-axis after the z-turn), the new y-axis is $(0,0,1)$, the new z-axis is $(1,0,0)$. [checked]

Rotation about an arbitrary unit axis (Rodrigues) [added]

$$\mathbf{R}(\hat{\mathbf{n}},\theta)=\mathbf{I}+\sin\theta\,[\hat{\mathbf{n}}]_{\times}+(1-\cos\theta)\,[\hat{\mathbf{n}}]_{\times}^{2}$$

SYMBOLS

$\hat{\mathbf{n}}$	unit vector along the rotation axis	–
$[\hat{\mathbf{n}}]_{\times}$	skew matrix of $\hat{\mathbf{n}}$ (builder 1)	–

Use: angular velocity $\boldsymbol{\omega}=\dot{\theta}\hat{\mathbf{n}}$ *for a turn about a fixed axis (W6). For* $\hat{\mathbf{n}}=\hat{\mathbf{z}}$ *it reproduces* $\mathbf{R}_z(\theta)$ *exactly;* $\mathbf{R}\hat{\mathbf{n}}=\hat{\mathbf{n}}$ *(the axis does not move).* [checked]

Angles from a matrix: ZYZ and ZYX (roll–pitch–yaw) [p. W3 38] [added]

$$\mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_z(\gamma)=\begin{bmatrix}c_\alpha c_\beta c_\gamma-s_\alpha s_\gamma & -c_\alpha c_\beta s_\gamma-s_\alpha c_\gamma & c_\alpha s_\beta\\ s_\alpha c_\beta c_\gamma+c_\alpha s_\gamma & -s_\alpha c_\beta s_\gamma+c_\alpha c_\gamma & s_\alpha s_\beta\\ -s_\beta c_\gamma & s_\beta s_\gamma & c_\beta\end{bmatrix}=[r_{ij}]$$

$$\beta=\operatorname{atan2}\!\left(\sqrt{r_{13}^2+r_{23}^2},\,r_{33}\right),\quad \alpha=\operatorname{atan2}(r_{23},r_{13}),\quad \gamma=\operatorname{atan2}(r_{32},-r_{31})\qquad (s_\beta\neq0)$$

Second solution: $-\beta,\ \alpha+\pi,\ \gamma+\pi$ (then $\alpha=\operatorname{atan2}(-r_{23},-r_{13})$, $\gamma=\operatorname{atan2}(-r_{32},r_{31})$). If $s_\beta=0$ only $\alpha\pm\gamma$ is fixed.

$$\mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi)=\begin{bmatrix}c_\psi c_\theta & c_\psi s_\theta s_\phi-s_\psi c_\phi & c_\psi s_\theta c_\phi+s_\psi s_\phi\\ s_\psi c_\theta & s_\psi s_\theta s_\phi+c_\psi c_\phi & s_\psi s_\theta c_\phi-c_\psi s_\phi\\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi\end{bmatrix}$$

$$\theta=\operatorname{atan2}\!\left(-r_{31},\sqrt{r_{11}^2+r_{21}^2}\right),\quad \psi=\operatorname{atan2}(r_{21},r_{11}),\quad \phi=\operatorname{atan2}(r_{32},r_{33})\qquad (c_\theta\neq0)$$

Second solution: $\pi-\theta,\ \psi+\pi,\ \phi+\pi$. Singular at $\theta=\pm90^\circ$.

SYMBOLS

α,β,γ	ZYZ Euler angles (first z, then new y, then new z)	rad
ψ,θ,ϕ	yaw (about z), pitch (about y), roll (about x)	rad
r_{ij}	entry of $\mathbf{R}$ in row i, column j	–

Use: W3 p.38: the three wrist joints of a spherical wrist are a ZYZ-type compound rotation, so the algebra is this. Always use atan2 *and report both solutions.* [checked]

Example. $\mathbf{R}=\mathbf{R}_z(30^\circ)\mathbf{R}_y(40^\circ)\mathbf{R}_z(50^\circ)$ gives back $(\alpha,\beta,\gamma)=(30^\circ,40^\circ,50^\circ)$; the second solution $(-150^\circ,-40^\circ,-130^\circ)$ gives the same $\mathbf{R}$. For ZYX, $(\psi,\theta,\phi)=(20^\circ,35^\circ,-25^\circ)$ round-trips.

4. Homogeneous transforms

PREMISE pose **Given:** rotation ${}^{A}\mathbf{R}_{B}$ and the position of the origin of $\{B\}$ in $\{A\}$, ${}^{A}\mathbf{P}_{B}$
Find: one 4×4 matrix that maps points of $\{B\}$ into $\{A\}$, its inverse, and chains of them

Pose matrix, inverse, composition [p. W2 12–14]

$$ {}^{A}\mathbf{T}_{B} = \begin{bmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\mathbf{P}_{B} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}, \qquad \begin{bmatrix} {}^{A}\mathbf{P} \\ 1 \end{bmatrix} = {}^{A}\mathbf{T}_{B} \begin{bmatrix} {}^{B}\mathbf{P} \\ 1 \end{bmatrix}, \qquad {}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C} $$

$$ \left({}^{A}\mathbf{T}_{B}\right)^{-1} = {}^{B}\mathbf{T}_{A} = \begin{bmatrix} {}^{A}\mathbf{R}_{B}^{\top} & -{}^{A}\mathbf{R}_{B}^{\top}\,{}^{A}\mathbf{P}_{B} \\ \mathbf{0}^{\top} & 1 \end{bmatrix} $$

SYMBOLS

${}^{A}\mathbf{T}_{B}$	pose of frame $\{B\}$ in $\{A\}$ (4×4)	–
${}^{A}\mathbf{R}_{B}$	3×3 orientation block	–
${}^{A}\mathbf{P}_{B}$	position of the origin of $\{B\}$, written in $\{A\}$	m
$\mathbf{0}^{\top}$	row of three zeros	–

Use: W2–W4 every FK chain: the inner frame letter cancels: ${}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{C}$. To move a direction (not a point) use only $\mathbf{R}$ (fourth entry 0).

Example. ${}^{A}\mathbf{T}_{B}$ with $\mathbf{R} = \mathbf{R}_{z}(90°)$, $\mathbf{P} = (2,1,0)$: the point ${}^{B}\mathbf{P} = (1,0,0)$ maps to $\mathbf{R}(1,0,0)+\mathbf{P} = (0,1,0)+(2,1,0) = (2,2,0)$. The inverse has $\mathbf{R}^{\top} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and translation $-\mathbf{R}^{\top}\mathbf{P} = -(1,-2,0) = (-1,2,0)$; check: ${}^{B}\mathbf{T}_{A}(2,2,0) = (2,-2,0)+(-1,2,0) = (1,0,0)$. [checked]

The elementary transforms as full 4×4 matrices [p. W2 15–16]

$$ t_{x}(a) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad t_{y}(b) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad t_{z}(c) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ \mathbf{R}_{x}(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta & -s\theta & 0 \\ 0 & s\theta & c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{R}_{y}(\theta) = \begin{bmatrix} c\theta & 0 & s\theta & 0 \\ 0 & 1 & 0 & 0 \\ -s\theta & 0 & c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{R}_{z}(\theta) = \begin{bmatrix} c\theta & -s\theta & 0 & 0 \\ s\theta & c\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

a, b, c	slide distances along the current x, y, z axis	m
θ	turn about the current x, y, z axis	rad

Use: an ETS is a product of these, read left to right; e.g. planar RR: ${}^{0}\mathbf{T}_{E} = \mathbf{R}_{z}(q_1)\,t_{x}(L_1)\,\mathbf{R}_{z}(q_2)\,t_{x}(L_2)$. A prismatic joint is a $t_{z}(d)$ whose d is the variable.

Order matters. $t(2,1,0)\,\mathbf{R}_{z}(90°)$ (move, then turn about the new origin) has position $(2,1,0)$; $\mathbf{R}_{z}(90°)\,t(2,1,0)$ (turn, then move along the *turned* axes) has position $(-1,2,0)$. [checked]

WORKED chain of two planar frames, intermediate frame shown

${}^{A}\mathbf{T}_{B}$: $\{B\}$ at $(1,0)$ in $\{A\}$, turned $30°$. ${}^{B}\mathbf{T}_{C}$: $\{C\}$ at $(0.5,0)$ in $\{B\}$, turned a further $60°$. Then ${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C}$ has rotation $\mathbf{R}_{z}(90°)$ and position $(1,0)+\mathbf{R}_{z}(30°)(0.5,0) = (1+0.433,\ 0.25) = (1.433,\ 0.25)$:

$$ {}^{A}\mathbf{T}_{C} = \begin{bmatrix} 0 & -1 & 0 & 1.433 \\ 1 & 0 & 0 & 0.25 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$ [checked]

5. Trigonometry

Identities used in every arm

$$ \sin(a\pm b) = \sin a\cos b \pm \cos a\sin b \qquad \cos(a\pm b) = \cos a\cos b \mp \sin a\sin b $$

$$ \tan(a\pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a\tan b} \qquad \cos^2 a + \sin^2 a = 1 $$

$$ \sin 2a = 2\sin a\cos a \qquad \cos 2a = \cos^2 a - \sin^2 a = 2\cos^2 a - 1 = 1 - 2\sin^2 a $$

$$ \tan 2a = \frac{2\tan a}{1-\tan^2 a} \qquad \cos^2 a = \tfrac{1}{2}(1+\cos 2a),\ \sin^2 a = \tfrac{1}{2}(1-\cos 2a) $$

$$ \cos a\cos b = \tfrac{1}{2}[\cos(a-b)+\cos(a+b)] \qquad \sin a\sin b = \tfrac{1}{2}[\cos(a-b)-\cos(a+b)] $$

$$ \sin a\cos b = \tfrac{1}{2}[\sin(a+b)+\sin(a-b)] \qquad \sin(-a) = -\sin a,\ \cos(-a) = \cos a $$

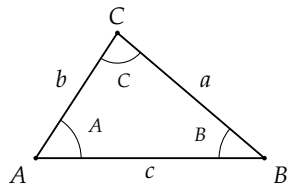
$$ \sin(\pi-a) = \sin a,\ \cos(\pi-a) = -\cos a \qquad \sin(\tfrac{\pi}{2}-a) = \cos a,\ \cos(\tfrac{\pi}{2}-a) = \sin a $$

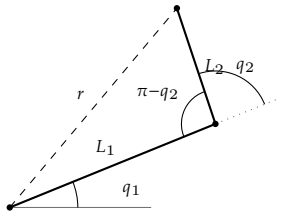
SYMBOLS

a, b	any two angles	rad
c_1, s_{12}	short forms $\cos q_1$, $\sin(q_1+q_2)$ (as in the weeks)	–

Use: simplifying arm algebra (W2–W4): $c_1c_{12}+s_1s_{12} = c_2$ and $s_1c_{12}-c_1s_{12} = -s_2$; expand $c_{12} = c_1c_2-s_1s_2$, $s_{12} = s_1c_2+c_1s_2$ before you solve. [checked]

Triangle rules: cosine rule, sine rule (the arm triangle)





[drawn for this booklet] left: triangle with side a opposite A, etc.; right: a 2-link arm closes a triangle L_1, L_2, r with interior angle $\pi - q_2$ at the elbow

$$ c^2 = a^2+b^2-2ab\cos C, \qquad \cos C = \frac{a^2+b^2-c^2}{2ab}, \qquad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \qquad A+B+C = \pi $$

SYMBOLS

a, b, c	side lengths, a opposite angle A, etc.	m
A, B, C	interior angles	rad
r	distance from base to tip, $r^2 = x^2+y^2$	m

Use: planar RR / RRR IK (W3): the interior elbow angle is $\pi - q_2$, so $r^2 = L_1^2+L_2^2-2L_1L_2\cos(\pi-q_2) = L_1^2+L_2^2+2L_1L_2\cos q_2$. The sign flip is the classic trap. Four-bar and delta triangles (W4) use the same two rules. [checked]

Example. $A = 40°$, $B = 60°$, $c = 10$: $C = 80°$, $a = c\sin A/\sin C = 6.527$, $b = 8.794$. Sides $5, 6, 7$: $\cos C = (25+36-49)/60 = 0.2$, $C = 78.46°$.

atan2 and why $\tan^{-1}(y/x)$ fails

$$\theta = \operatorname{atan2}(y,x) \in (-\pi, \pi] \quad \text{(the angle of the point } (x,y) \text{ from the } +x \text{ axis)}$$

quadrant	x	y	atan2(y,x) in terms of arctan(y/x)
I	>0	≥ 0	$\arctan(y/x)$
II	<0	≥ 0	$\arctan(y/x)+\pi$
III	<0	<0	$\arctan(y/x)-\pi$
IV	>0	<0	$\arctan(y/x)$
axis	0	$\gtrless 0$	$\pm\pi/2$ (y/x undefined)

SYMBOLS

y, x	the two coordinates; y **is the first argument**	m
θ	angle in radians; add 2π if you want $[0, 2\pi)$	rad

Use: every position angle in IK (W3): $q_1 = \operatorname{atan2}(y,x)$, $\phi = \operatorname{atan2}(b,a)$. $\arctan(y/x)$ cannot tell $(1,1)$ from $(-1,-1)$ because the ratio is the same.

Example. $(x,y) = (-1,1)$: $\operatorname{atan2}(1,-1) = 135°$ but $\arctan(1/-1) = -45°$ (wrong by $180°$). $(-1,-1)$: atan2 $= -135°$, $\arctan(1) = +45°$. [checked]

Solving $a\cos\theta + b\sin\theta = c$: two methods, both solutions

$$r = \sqrt{a^2+b^2}, \quad \varphi = \operatorname{atan2}(b,a), \quad a\cos\theta + b\sin\theta = r\cos(\theta-\varphi) = c$$

$$\Rightarrow \theta = \varphi \pm \arccos\frac{c}{r} \qquad (|c| \leq r)$$

Tangent half-angle: $t = \tan\dfrac{\theta}{2}$, $\cos\theta = \dfrac{1-t^2}{1+t^2}$, $\sin\theta = \dfrac{2t}{1+t^2}$, giving

$$(c+a)\,t^2 - 2b\,t + (c-a) = 0, \qquad \theta = 2\arctan t \ \text{ (two roots } t)$$

SYMBOLS

a, b, c	known numbers (often lengths or products of lengths and coordinates)	as the problem
r, φ	amplitude and phase of $a\cos\theta + b\sin\theta$	as a; rad
t	$\tan(\theta/2)$	–

Use: IK where the unknown angle appears as $\cos\theta$ and $\sin\theta$ together (W3 algebraic IK, W4 four-bar and delta): $|c| > r$ means out of reach (no solution), $|c| = r$ one (boundary), $|c| < r$ two. The t-method misses $\theta = \pi$ (when $c + a = 0$): test it separately.

Example. $3\cos\theta + 4\sin\theta = 2.5$: $r = 5$, $\varphi = 53.13°$, $\arccos 0.5 = 60°$, so $\theta = 113.13°$ or $-6.87°$. Half-angle: $5.5t^2 - 8t - 0.5 = 0$, $t = 1.5146$ or -0.0600, $\theta = 2\arctan t = 113.13°$ or $-6.87°$. Same answers. [checked]

Small angles, degrees and radians

$$\sin\theta \approx \theta, \quad \tan\theta \approx \theta, \quad \cos\theta \approx 1 - \frac{\theta^2}{2} \ (\approx 1)$$

$$\pi \text{ rad} = 180°, \quad 1° = 0.017453 \text{ rad}, \quad 1 \text{ rad} = 57.296°$$

SYMBOLS

θ	angle in **radians** (the approximations only hold in rad)	rad

Use: linearising a pendulum or a wheel steering angle ($\tan\beta \approx \beta$, W10). Errors at 0.1 rad (5.7°): sin 0.17%, tan 0.33%, cos ≈ 1 0.5%. At 20°: sin 2.1%, tan 4.1%. Set the calculator to the unit you mean (DEG/RAD) before every trig key. [checked]

6. Algebra

Quadratic formula and the discriminant (how many IK solutions)

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{\Delta}}{2a}, \qquad \Delta = b^2 - 4ac, \qquad x_1 + x_2 = -\frac{b}{a}, \quad x_1 x_2 = \frac{c}{a}$$

SYMBOLS

a, b, c	coefficients ($a \neq 0$)	as the problem
Δ	discriminant	–

$\Delta > 0$	two real roots: two solutions (elbow up and elbow down, two assembly modes)
$\Delta = 0$	one double root: the boundary of reach, a singular pose
$\Delta < 0$	complex roots: no real solution, target unreachable

Use: W3/W4 IK: $\cos q_2 = \dfrac{x^2+y^2-L_1^2-L_2^2}{2L_1L_2}$ and $\sin q_2 = \pm\sqrt{1-\cos^2 q_2}$; the radicand $1-\cos^2 q_2$ is the discriminant: <0 unreachable, $=0$ arm stretched or folded (one solution), >0 two solutions. W8: $ms^2 + bs + k$ has $\Delta = b^2 - 4mk$ (over/critical/under-damped).

Example. $x^2 - 5x + 6 = 0$: $\Delta = 1$, $x = (5 \pm 1)/2 = 3, 2$. RR, $L_1 = L_2 = 1$, target $(1,1)$: $\cos q_2 = (2-2)/2 = 0$, $q_2 = \pm 90°$ (two solutions); target $(1.2, 1.2)$: $\cos q_2 = (2.88-2)/2 = 0.44$ fine; $\cos q_2 = 1.2$ would give $1 - 1.44 = -0.44 < 0$: out of reach.

Two linear equations (Cramer's rule), elimination, completing the square

$$\begin{aligned} \alpha x + \beta y &= e \\ \gamma x + \delta y &= f \end{aligned} \Rightarrow x = \frac{e\delta - \beta f}{\alpha\delta - \beta\gamma}, \quad y = \frac{\alpha f - e\gamma}{\alpha\delta - \beta\gamma} \ \text{(denominator} \neq 0)$$

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$$

Square and add (removes an angle): if $x = L_1c_1 + L_2c_{12}$ and $y = L_1s_1 + L_2s_{12}$ then $x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2(c_1c_{12} + s_1s_{12}) = L_1^2 + L_2^2 + 2L_1L_2c_2$, with no q_1.

SYMBOLS

$\alpha, \beta, \gamma, \delta$	coefficients of the unknowns x, y	as the problem
e, f	right-hand sides	as the problem

Use: eliminating one unknown to reach a single-angle equation (IK, W3); two velocity or force equations in two unknowns (W6, W10); locating a minimum or a circle centre by completing the square.

Example. $2x + y = 5$, $x - y = 1$: denominator $2(-1) - 1 \cdot 1 = -3$; $x = (5(-1) - 1 \cdot 1)/(-3) = 2$, $y = (2 \cdot 1 - 5 \cdot 1)/(-3) = 1$. Square: $x^2 - 6x + 13 = (x-3)^2 + 4$, minimum 4 at $x = 3$. [checked]

BASICS

7. Calculus

Chain rule and product rule with joint angles (time derivatives)

$$\frac{d}{dt}f\big(q_1(t),q_2(t)\big)=\sum_i\frac{\partial f}{\partial q_i}\dot q_i,\qquad \frac{d}{dt}(uv)=\dot u v+u\dot v$$

$$\frac{d}{dt}\cos q_1=-\sin q_1\,\dot q_1 \qquad \frac{d}{dt}\sin q_1=\cos q_1\,\dot q_1$$
$$\frac{d}{dt}c_{12}=-s_{12}\,(\dot q_1+\dot q_2) \qquad \frac{d}{dt}s_{12}=c_{12}\,(\dot q_1+\dot q_2)$$
$$\frac{d}{dt}\dot q^{\,2}=2\dot q\,\ddot q \qquad \frac{d}{dt}(\cos q\,\dot q)=-\sin q\,\dot q^{\,2}+\cos q\,\ddot q$$

SYMBOLS

$q_i, \dot q_i, \ddot q_i$	joint variable, its rate and acceleration	rad or m; per s; per s^2
c_{12}, s_{12}	$\cos(q_1+q_2)$, $\sin(q_1+q_2)$; the inner angle is q_1+q_2 so its rate is $\dot q_1+\dot q_2$	–
$\partial f/\partial q_i$	partial derivative: differentiate in q_i, hold the others fixed	–

Use: W6: the tip velocity is the time derivative of FK; the row of partials is the Jacobian row. W7: velocities of link centres before $K=\frac12 mv^2$. Do not forget the inner rate (e.g. $\dot q_1+\dot q_2$ in c_{12}).

Example (planar RR tip velocity). $x=L_1c_1+L_2c_{12}$, $y=L_1s_1+L_2s_{12}$:

$$\dot x=-L_1s_1\dot q_1-L_2s_{12}(\dot q_1+\dot q_2),\qquad \dot y=L_1c_1\dot q_1+L_2c_{12}(\dot q_1+\dot q_2)$$

$L_1=L_2=1$, $\mathbf{q}=(0,90^\circ)$, $\dot{\mathbf{q}}=(1,0)$: $\dot x=-0-1\cdot1=-1$, $\dot y=1+0=1$. The tip is at $(1,1)$ and turns about the origin at 1 rad/s: $\boldsymbol{\omega}\times\mathbf{r}=(-1,1)$. [checked]

Partial derivatives in Lagrange's equation [p. W7 15–17]

$$\mathcal{L}=K-P,\qquad Q_i=\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot q_i}\right)-\frac{\partial\mathcal{L}}{\partial q_i}$$

Treat q_i and $\dot q_i$ as *independent symbols* while taking ∂; only afterwards differentiate $\partial\mathcal{L}/\partial\dot q_i$ with respect to time (builder above). If $K=\frac12\dot{\mathbf{q}}^{\top}\mathbf{M}(\mathbf{q})\dot{\mathbf{q}}$ then $\partial K/\partial\dot{\mathbf{q}}=\mathbf{M}\dot{\mathbf{q}}$.

SYMBOLS

K, P, $\mathcal{L}$	kinetic, potential energy, Lagrangian	J
Q_i	generalised force (torque N m for a revolute joint, force N for a prismatic one)	N m or N

Use: W7 every equation-of-motion question.

Example (simple pendulum, q from the downward vertical). $K=\frac12 mL^2\dot q^2$, $P=-mgL\cos q$, $\mathcal{L}=\frac12 mL^2\dot q^2+mgL\cos q$ (L = pendulum length). $\partial\mathcal{L}/\partial\dot q=mL^2\dot q$, so d/dt of it is $mL^2\ddot q$; $\partial\mathcal{L}/\partial q=-mgL\sin q$. Therefore $Q=mL^2\ddot q+mgL\sin q$. At rest at $q=90^\circ$: $Q=mgL$ (holding torque). [checked]

Time derivative of a rotation matrix [p. W6 18–19]

$$\dot{\mathbf{R}}=[\boldsymbol{\omega}]_\times\mathbf{R},\qquad [\boldsymbol{\omega}]_\times=\dot{\mathbf{R}}\mathbf{R}^{\top},\qquad \frac{d}{d\theta}\mathbf{R}_z(\theta)=\begin{bmatrix}-s\theta & -c\theta & 0\\ c\theta & -s\theta & 0\\ 0 & 0 & 0\end{bmatrix}$$

SYMBOLS

$\mathbf{R}(t)$	rotation matrix as a function of time	–
$\boldsymbol{\omega}$	angular velocity, in the frame $\mathbf{R}$ is measured from	rad/s

Use: angular velocity of the tool from its orientation (W6). Example in the W6 part: $\theta=30^\circ$, $\dot\theta=2$ rad/s gives $\dot{\mathbf{R}}_z=\begin{bmatrix}-1 & -1.732 & 0\\ 1.732 & -1 & 0\\ 0 & 0 & 0\end{bmatrix}$ *and* $\boldsymbol{\omega}=(0,0,2)$. [checked]

8. Kinematics of a particle and of a rotating body

Constant acceleration (SUVAT)

$$v=u+at,\qquad s=ut+\tfrac12 at^2,\qquad v^2=u^2+2as,\qquad s=\tfrac12(u+v)\,t$$

$$s=\int_0^t v\,dt\ \ \text{(area under } v(t)\text{)}$$

SYMBOLS

u, v	initial and final velocity	m/s
a	acceleration (constant)	m/s^2
s	displacement in time t	m
t	elapsed time	s

Use: trapezoidal and blended trajectories (W5), a driving robot's stopping distance (W10–W11). The same four equations hold for rotation with $s\to\theta$, $v\to\omega$, $a\to\alpha$. Pick the equation that omits the unknown you do not need.

Example. Start from rest, $a=2$ m/s^2 for 5 s: $v=10$ m/s, $s=25$ m, and $v^2=100=2\cdot2\cdot25$. Trapezoid profile (accelerate 2 s at 1 m/s^2 to 2 m/s, cruise 3 s, brake 2 s): distance $=\frac12\cdot2\cdot2+3\cdot2+\frac12\cdot2\cdot2=10$ m (area under the v–t plot). Given $u=2$, $v=8$, $s=15$: $t=2s/(u+v)=3$ s, $a=(8-2)/3=2$ m/s^2. [checked]

Circular motion, a point on a rotating body, relative velocity

$$v=\omega r,\quad a_c=\frac{v^2}{r}=\omega^2 r\ \text{(toward the centre)},\quad a_t=\alpha r\ \text{(along the path)},\quad |\mathbf{a}|=\sqrt{a_c^2+a_t^2}$$

$$\mathbf{v}_P=\mathbf{v}_O+\boldsymbol{\omega}\times\mathbf{r}_{OP},\qquad \mathbf{v}_{B/A}=\mathbf{v}_B-\mathbf{v}_A$$

SYMBOLS

r	radius from the axis to the point	m
ω, α	angular velocity and acceleration of the body	rad/s; rad/s^2
$\mathbf{v}_O$	velocity of the reference point O of the body	m/s
$\mathbf{r}_{OP}$	vector from O to P	m
$\mathbf{v}_{B/A}$	velocity of B as seen from A	m/s

Use: W10: a wheel of radius r turning at $\dot\varphi$ rolling without slipping has centre speed $v=r\dot\varphi$; its top point moves at $2v$ and its contact point at 0 (the no-slip condition). A differential-drive wheel at distance a from the centre has speed $u\pm a\dot\theta$ (W10). W6: $\mathbf{v}=\boldsymbol{\omega}\times\mathbf{r}$.

Example. $r=0.5$ m, $\omega=4$ rad/s, $\alpha=2$ rad/s^2: $v=2$ m/s, $a_c=8$ m/s^2, $a_t=1$ m/s^2, $|\mathbf{a}|=\sqrt{65}=8.06$. $\boldsymbol{\omega}=(0,0,3)$, $\mathbf{r}=(2,0,0)$: $\boldsymbol{\omega}\times\mathbf{r}=(0,6,0)$ m/s. Body with $\mathbf{v}_O=(1,0,0)$, $\boldsymbol{\omega}=(0,0,2)$, $\mathbf{r}_{OP}=(0.5,0,0)$: $\mathbf{v}_P=(1,0,0)+(0,1,0)=(1,1,0)$. Wheel $r=0.1$ m at 5 rad/s: centre 0.5 m/s, top 1.0 m/s, contact 0. [checked]

9. Dynamics basics

Newton, torque, momentum, energy, work, power

$$\mathbf{F} = m\mathbf{a}, \quad \boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \quad \tau = I\alpha \text{ (fixed axis)}, \quad \mathbf{L} = \mathbf{I}\boldsymbol{\omega}, \quad \boldsymbol{\tau} = \dot{\mathbf{L}}$$

$$K_{\text{trans}} = \tfrac{1}{2}mv^2, \quad K_{\text{rot}} = \tfrac{1}{2}\boldsymbol{\omega}^{\top}\mathbf{I}\boldsymbol{\omega} \; (= \tfrac{1}{2}I\omega^2), \quad P = mgh$$

$$W = \int \mathbf{F} \cdot d\mathbf{x} = \int \tau \, d\theta, \quad \text{power} = \tau\omega = Fv$$

SYMBOLS

m	mass	kg
$\mathbf{F}$, $\mathbf{a}$	force, linear acceleration of the centre of mass	N; m/s²
$\mathbf{r}$	vector from the axis or reference point to where $\mathbf{F}$ acts	m
$\boldsymbol{\tau}$	torque (moment)	N m
I, $\mathbf{I}$	moment of inertia about the axis; inertia tensor	kg m²
$\boldsymbol{\omega}$, α	angular velocity, angular acceleration	rad/s; rad/s²
g, h	9.81 m/s²; height of the centre of mass above the reference	m/s²; m
W	work	J

Use: W7: K and P of every link, Newton–Euler (force and moment on each link); W8: motor power and torque. Energy check: a frictionless swing keeps $K + P$ constant.

Example. $\mathbf{r} = (0.5, 0, 0)$ m, $\mathbf{F} = (0, 10, 0)$ N: $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = (0, 0, 5)$ N m. At 20 rad/s that torque delivers $5 \times 20 = 100$ W; a 10 N force at 2 m/s delivers 20 W. $m = 2$ kg at 3 m/s: $K = 9$ J; lifted 0.3 m: $P = 2 \cdot 9.81 \cdot 0.3 = 5.886$ J. [checked]

Moments of inertia of standard bodies (about the centroid) and the scalar parallel-axis theorem

Body (mass m)	Moment of inertia
Point mass at distance r from the axis	mr^2
Slender rod, length L, axis ⊥ rod	centre $\frac{1}{12}mL^2$; end $\frac{1}{3}mL^2$
Solid cylinder or disc, radius r, own axis	$\frac{1}{2}mr^2$
Solid cylinder (radius r, length h), diameter axis	$\frac{1}{12}m(3r^2 + h^2)$; thin disc: $\frac{1}{4}mr^2$
Thin ring or hollow tube, own axis	mr^2
Solid sphere	$\frac{2}{5}mr^2$ (any axis); thin shell $\frac{2}{3}mr^2$
Box $a \times b \times c$, axis along c	$\frac{1}{12}m(a^2 + b^2)$ (corner: $\frac{1}{3}m(a^2 + b^2)$)

$$I = I_C + md^2 \qquad \text{(axes parallel; } d = \text{distance between them)}$$

SYMBOLS

I_C	moment of inertia about the parallel axis through the centre of mass	kg m²
d	perpendicular distance between the two axes	m

Use: W7 link inertias; a rod pivoted at its end is $\frac{1}{3}mL^2$. Never add inertias about different points. Gearbox of ratio N (motor:load): load sees $N^2 I_m$ and torque $N\tau_m$.

Example. $m = 2$ kg, $L = 0.4$ m: centre $\frac{1}{12} \cdot 2 \cdot 0.16 = 0.02667$; end $= 0.02667 + 2 \cdot 0.2^2 = 0.10667 = \frac{1}{3} \cdot 2 \cdot 0.16$ kg m². Motor $I_m = 0.001$, $N = 10$: load sees 0.1 kg m². [checked]

Inertia tensor in full, and the tensor parallel-axis theorem [p. W7 34–39]

$$ {}^{A}\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}, \quad \begin{aligned} I_{xx} &= \iiint (y^2 + z^2)\rho \, dv, & I_{xy} &= -\iiint xy \, \rho \, dv \\ I_{yy} &= \iiint (x^2 + z^2)\rho \, dv, & I_{yz} &= -\iiint yz \, \rho \, dv \\ I_{zz} &= \iiint (x^2 + y^2)\rho \, dv, & I_{xz} &= -\iiint xz \, \rho \, dv \end{aligned}$$

$$ {}^{A}\mathbf{I} = {}^{C}\mathbf{I} + m\left(\mathbf{p}^{\top}\mathbf{p}\,\mathbf{1}_3 - \mathbf{p}\mathbf{p}^{\top}\right) = {}^{C}\mathbf{I} + m\begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix}, \qquad {}^{B}\mathbf{I} = \mathbf{R}\,{}^{A}\mathbf{I}\,\mathbf{R}^{\top}$$

SYMBOLS

${}^{A}\mathbf{I}$	tensor about the origin of $\{A\}$, written in $\{A\}$ (symmetric)	kg m²
I_{xx}, I_{yy}, I_{zz}	moments about the axes (always ≥ 0)	kg m²
I_{xy}, I_{yz}, I_{xz}	products of inertia (note the minus sign in the slide definition)	kg m²
ρ, dv	density, volume element	kg/m³; m³
$\mathbf{p} = (x, y, z)$	vector from the centroid C to the new origin A, in $\{A\}$ axes	m
$\mathbf{1}_3$	3×3 identity	–

Use: W7: kinetic energy $\frac{1}{2}\boldsymbol{\omega}^{\top}\mathbf{I}\boldsymbol{\omega}$ and Euler's equation. Entries at the centroid vanish for a body symmetric about the axes.

Example (box $W \times L \times H = 0.2 \times 0.3 \times 0.1$ m, $m = 2$ kg, frame at a corner, $\mathbf{p} = (0.1, 0.15, 0.05)$). Centroid tensor $\mathrm{diag}(0.01667, 0.00833, 0.02167)$; adding $m(\mathbf{p}^{\top}\mathbf{p}\,\mathbf{1} - \mathbf{p}\mathbf{p}^{\top})$:

$$ {}^{A}\mathbf{I} = \begin{bmatrix} 0.06667 & -0.03 & -0.01 \\ -0.03 & 0.03333 & -0.015 \\ -0.01 & -0.015 & 0.08667 \end{bmatrix} \text{ kg m}^2$$

equal to the W7 slide formula $I_{xx} = m(L^2 + H^2)/3$, $I_{xy} = -mWL/4$, …. A rod along x pivoted at its end: $\mathrm{diag}(0, \frac{1}{3}mL^2, \frac{1}{3}mL^2)$. [checked]

BASICS

10. Second-order systems and stability

Mass–spring–damper: characteristic equation, ω_n, ζ, the three cases [p. W8 16–26]

$$m\ddot{x} + b\dot{x} + kx = 0, \qquad ms^2 + bs + k = 0$$

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{b}{2\sqrt{mk}}, \quad s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

case	roots	free response, $x(0) = x_0$, $\dot{x}(0) = v_0$
over, $\zeta > 1$	real $s_1 \neq s_2$	$x = A_1 e^{s_1 t} + A_2 e^{s_2 t}$; $A_1 + A_2 = x_0$, $s_1 A_1 + s_2 A_2 = v_0$
critical, $\zeta = 1$	real double $s = -\omega_n$	$x = (A_1 + A_2 t)\, e^{-\omega_n t}$; $A_1 = x_0$, $A_2 = v_0 + \omega_n x_0$
under, $0 \leq \zeta < 1$	$\lambda \pm i\mu$, $\lambda = -\zeta\omega_n$, $\mu = \omega_n\sqrt{1-\zeta^2}$	$x = e^{\lambda t}(A_1 \cos \mu t + A_2 \sin \mu t)$; $A_1 = x_0$, $A_2 = (v_0 - \lambda x_0)/\mu$

SYMBOLS

m, b, k	mass, damping coefficient, spring stiffness	kg; N s/m; N/m
ω_n	undamped natural frequency	rad/s
ζ	damping ratio (1 = critical)	–
$\mu = \omega_d$	damped frequency	rad/s
A_1, A_2	constants from the initial conditions	m

Use: W8 [p. 16–26]: name the case from $b^2 - 4mk$ (> 0, $= 0$, < 0). For a unit-mass servo $\ddot{E} + k_v\dot{E} + k_p E = 0$: $\omega_n = \sqrt{k_p}$, $\zeta = k_v/(2\sqrt{k_p})$, critical $k_v = 2\sqrt{k_p}$; e.g. $\omega_n = 4$, $\zeta = 1$: $k_p = 16$, $k_v = 8$.

Examples. $m = 1, b = 2, k = 5$: $\omega_n = 2.236$, $\zeta = 0.447$, $s = -1 \pm 2i$; with $x_0 = 1$, $v_0 = 3$: $A_1 = 1$, $A_2 = (3+1)/2 = 2$. $m = 1, b = 3, k = 2$: $s = -1, -2$, $x_0 = 1, v_0 = 0$: $A_1 = 2, A_2 = -1$. $m = 1, b = 4, k = 4$: $\omega_n = 2$, $x = (1 + 2t)e^{-2t}$. [checked]

Step-response figures and Routh condition

$$M_p = \exp\!\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right), \qquad t_{\text{peak}} = \frac{\pi}{\omega_d}, \qquad t_s \approx \frac{4}{\zeta\omega_n}\ (2\%), \qquad \text{first order: } x = e^{-t/T_c}$$

Cubic $s^3 + a_2 s^2 + a_1 s + a_0$: stable (all roots in the left half-plane) iff $a_2, a_1, a_0 > 0$ **and** $a_2 a_1 > a_0$. Quadratic $s^2 + a_1 s + a_0$: stable iff $a_1, a_0 > 0$.

SYMBOLS

M_p	fractional overshoot of the step response above its final value (multiply by 100 for %)	–
t_{peak}, t_s	time of the first peak; time to stay within 2% of the final value (a rule of thumb)	s
T_c	time constant of a first-order response (63.2% of the way at $t = T_c$)	s
a_2, a_1, a_0	coefficients of the characteristic polynomial	–

Use: W8 PID: $s^3 + k_v s^2 + k_p s + k_i$ is stable iff $k_v k_p > k_i$; at equality the roots are $\pm j\sqrt{k_p}$ (marginal). M_p: $\zeta = 0.3$: 37.2%; 0.5: 16.3%; 0.707: 4.3%; ≥ 1: none.

Example. $m = 1, b = 2, k = 5$ step: $M_p = e^{-\pi \cdot 0.447/0.894} = e^{-1.571} = 0.208$ (20.8%), $t_{\text{peak}} = \pi/2 = 1.571$ s, $t_s \approx 4/(0.447 \cdot 2.236) = 4$ s. $s^3 + 3s^2 + 2s + c$: $ab = 6$, stable iff $0 < c < 6$; at $c = 6$ the roots are $-3, \pm j\sqrt{2}$; at $c = 7$ a pair has real part $+0.043$. $s^3 + 6s^2 + 11s + 6 = (s+1)(s+2)(s+3)$: $66 > 6$ stable. [checked]

11. Units, conversions and sign conventions

Conversions and unit check

$1^\circ = 0.017453$ rad	1 rad = 57.296°	π rad = 180°
1 rpm = $2\pi/60$ = 0.10472 rad/s	1 rad/s = 9.549 rpm	1 Hz = 2π rad/s
1000 mm = 1 m	1 N m = 1 J per rad	1 W = 1 N m/s

SYMBOLS

rpm	revolutions per minute	rev/min

Use: Convert before you substitute: wheel 120 rpm = 12.57 rad/s; 3000 rpm = 314.16 rad/s; 250 mm/s = 0.25 m/s; 10 rad/s = 95.5 rpm. Gearbox $N = 10$, motor 0.2 N m at 300 rad/s: load 2 N m at 30 rad/s (power unchanged). Quick check: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}}$ is N m (revolute) or N (prismatic). A Jacobian row is m (revolute joint) or dimensionless (prismatic joint). [checked]

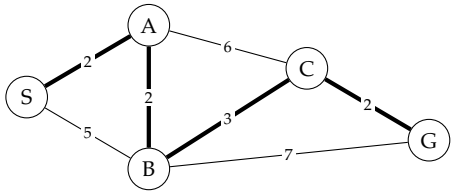
Sign and frame conventions used in this booklet (follow the slides)

Frames	right-handed, z up; ${}^A\mathbf{T}_B$ is the pose of $\{B\}$ in $\{A\}$ (maps ${}^B\mathbf{P} \mapsto {}^A\mathbf{P}$)
Positive rotation	anticlockwise when looking down the axis towards the origin (right-hand rule); angles in radians in formulas
Proximal DH	${}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})\, t_x(a_{i-1})\, \mathbf{R}_z(\theta_i)\, t_z(d_i)$
Joints	revolute: $q_i = \theta_i$ about z_i; prismatic: $q_i = d_i$ along z_i, positive when it extends
Generalised force	Q_i positive in the direction of positive q_i
Gravity	$\mathbf{g} = (0, 0, -9.81)$ m/s^2 in $\{0\}$; $P = mgh$ with h upward
Error	$\mathbf{E} = \mathbf{q}_d - \mathbf{q}$ (desired minus actual)
Wheeled robot	chassis $\{C\}$: u forward ($\hat{x}_C$), v left ($\hat{y}_C$), $\dot{\theta}$ positive anticlockwise; $\dot{\varphi}_i > 0$ rolls the wheel forward

BASICS

12. Graph basics (for W12)

Graph, path cost, tree; queue, stack, priority queue



[drawn for this booklet] graph with edge weights; thick = cheapest path S→G, cost $2+2+3+2=9$

$$G=(V,E), \qquad \text{path cost} = \sum_{e\in\text{path}} w_e, \qquad f(n)=g(n)+h(n)\ \ (A^*)$$

Tree: connected, no cycles, $|E|=|V|-1$.

structure	removes next	used by
queue (FIFO)	oldest item added	BFS (fewest edges)
stack (LIFO)	newest item added	DFS (dives down one branch)
priority queue	smallest key: $g(n)$ or $f(n)$	Dijkstra (g), A^* ($f=g+h$)

SYMBOLS

V, E	set of nodes (vertices); set of edges (links between two nodes)	–
w	edge weight (cost, e.g. distance)	m or as given
$g(n)$	cost of the cheapest known path start $\to n$	as w
$h(n)$	heuristic: estimate of the cost $n \to$ goal; *admissible* if never above the true cost	as w
$f(n)$	$g(n)+h(n)$	as w

Use: W12: neighbours, open (frontier) and closed lists; BFS finds the fewest edges, not the cheapest. On a grid, 4-connected steps cost 1 *and diagonal steps* $\sqrt{2}=1.414$ *(8-connected).*

Example (figure). Dijkstra: pop S ($g=0$): A= 2, B= 5. Pop A: B→ 4 (via A), C= 8. Pop B: C→ 7, G= 11. Pop C: G→ 9. Pop G: done, path S-A-B-C-G, cost 9. BFS would return S-B-G (two edges) with cost 12: not cheapest. [checked]

13. Notation used in this booklet

Frames, poses, joints, velocities

symbol	meaning	unit / week
$\{0\}$, $\{E\}$	base (world) frame; end-effector (tool) frame; $\{i\}$ is link frame i	W2
$\{C\}$, $\{O\}$	chassis frame (moves with the robot); world (odometry) frame of a mobile robot	W10
${}^A\mathbf{T}_B$	pose of $\{B\}$ in $\{A\}$, 4×4	m, –
${}^A\mathbf{R}_B$	orientation of $\{B\}$ in $\{A\}$, 3×3	–
${}^A\mathbf{P}$, ${}^A\mathbf{P}_B$	a point written in $\{A\}$; origin of $\{B\}$ in $\{A\}$	m
$\mathbf{q}$, $\dot{\mathbf{q}}$, $\ddot{\mathbf{q}}$	joint vector, rates, accelerations (rad or m)	W2–W9
$\mathbf{J}$	Jacobian, ${}^0\boldsymbol{\nu}={}^0\mathbf{J}\dot{\mathbf{q}}$	W6
$\boldsymbol{\nu}=(\boldsymbol{v},\boldsymbol{\omega})$	spatial velocity: linear (m/s) on top, angular (rad/s) below	W6

Dynamics, control, wheeled robots, planning

symbol	meaning	unit / week
$\mathbf{M}(\mathbf{q})$	joint-space mass (inertia) matrix, symmetric positive definite	W7
$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$	velocity terms (Coriolis and centrifugal)	W7
$\mathbf{g}(\mathbf{q})$	gravity torque vector	W7
$\mathbf{Q}$	joint torques (revolute, N m) or forces (prismatic, N): $\mathbf{Q}=\mathbf{M}\ddot{\mathbf{q}}+\mathbf{C}+\mathbf{g}$	W7–W9
$\mathbf{E}=\mathbf{q}_d-\mathbf{q}$	servo error; $\dot{\mathbf{E}}$ its rate; k_p, k_v, k_i gains	W8–W9
$\boldsymbol{\zeta}=(u,v,\dot{\theta})$	chassis velocity in $\{C\}$: forward, sideways, turning rate	m/s, rad/s; W10
$\dot{\varphi}_i$	rotation speed of wheel i; r wheel radius	rad/s; W10
β_i	drive direction of wheel i from $\hat{x}_C$ (bicycle: steering angle)	rad; W10
γ_i	free-roll direction of the rollers from the wheel-plane normal ($\gamma=0$ for a standard wheel)	rad; W10
$g(n)$, $h(n)$, $f(n)$	path cost so far, heuristic, $f=g+h$ for node n	W12

Comparison of the standard robots

Robot	joints	configuration	n	reachable workspace (no joint limits)	IK solutions (position)	singular poses ($\det \mathbf{J} = 0$)	treated in
1. planar RR	RR	planar, 2-link articulated	2	annulus $\lvert L_1 - L_2\rvert \le r \le L_1 + L_2$; full disc if $L_1 = L_2$	2 (elbow $\pm q_2$)	$q_2 = 0, \pi$: $\det \mathbf{J}_{xy} = L_1 L_2 \sin q_2$	W2 p. 33, W3 p. 45, W6 p. 78, W7 p. 91
2. planar RP	RP	polar (planar)	2	annulus $d_{\min} \le r \le d_{\max}$	1 (2 if $d_2 < 0$ is allowed)	$d_2 = 0$: $\det \mathbf{J}_{xy} = -d_2$	W2 p. 33, W3 p. 45, W6 p. 78, W7 p. 91
3. planar PR	PR	rail + rotary (planar)	2	"stadium": $D \times 2L$ rectangle + half-disc of radius L at each end	2 (q_2 and $\pi - q_2$)	$q_2 = \pm 90°$: $\det \mathbf{J}_{xy} = L_2 \cos q_2$	W2 p. 33, W3 p. 45, W6 p. 78, W7 p. 91
4. planar RRR	RRR	planar, 3-link	3	disc $r \le \sum L_i$ (hole if $2L_{\max} > \sum L_i$); dexterous: $L_3 + \lvert L_1 - L_2\rvert \le r \le L_1 + L_2 - L_3$	2 for (x, y, φ); ∞ for (x, y)	$q_2 = 0, \pi$ for (x, y, ω_z): $\det = L_1 L_2 \sin q_2$	W2 p. 33, W3 p. 45, W6 p. 78
5. Cartesian	PPP	gantry / Cartesian	3	box, edges = strokes	1	none: $\det \mathbf{J}_v = -1$	W1 p. 26
6. cylindrical	RPP	cylindrical	3	hollow cylinder sector: radius $d_{3\,\min}$ to $d_{3\,\max}$, height = stroke of d_2	1 (2 if $d_3 < 0$ is allowed)	$d_3 = 0$: $\det \mathbf{J}_v = d_3$	W1 p. 26
7. spherical	RRP	spherical (Stanford-like)	3	spherical shell $d_{3\,\min} \le \rho \le d_{3\,\max}$ about $(0, 0, H)$	2 (4 if $d_3 < 0$ is allowed)	$d_3 = 0$ or $\sin q_2 = 0$: $\det \mathbf{J}_v = -d_3^2 \sin q_2$	W3 p. 45, W6 p. 78
8. SCARA	RRPR (position part RRP)	SCARA	4 (3 for position)	flat ring $\lvert a_1 - a_2\rvert \le r \le a_1 + a_2$ times the stroke of d_3	2 (elbow); d_3, q_4 unique	$q_2 = 0, \pi$: $\det \mathbf{J}_v = -a_1 a_2 \sin q_2$	W1 p. 26, W2 p. 33, W3 p. 45
9. elbow arm	RRR	articulated / anthropomorphic (W13 Q1–Q3)	3	spherical shell $\lvert L_3 - L_4\rvert \le \rho \le L_3 + L_4$ about $(0, 0, L_1 + L_2)$	4 (shoulder × elbow)	$\sin \theta_3 = 0$ or $r = 0$	W2 p. 33, W3 p. 45, W6 p. 78, W7 p. 91
10. Puma 560	6R, spherical wrist	Puma 560 (zero-offset model and the course table)	6	wrist centre in a spherical shell about the shoulder	up to 8	elbow $\sin q_3 = 0$, shoulder (wrist centre on the waist axis), wrist $\sin q_5 = 0$	W1 p. 26, W2 p. 33, W3 p. 45, W6 p. 78
11. Franka Panda	7R	anthropomorphic, redundant	7	thick sphere about $(0, 0, 0.333)$, sampled 0.02–0.86 m	∞ (1-D self-motion)	rank $\mathbf{J} < 6$, e.g. wrist centre on the joint-1 axis	W2 p. 33, W3 p. 45 (numerical)
12. Furuta pendulum	RR (spatial: z, then x)	rotary pendulum	2	tip on a sphere of radius $\sqrt{L_1^2 + L_2^2}$ about $(0, 0, L_0)$, band $\lvert z - L_0\rvert \le L_2$	≤ 2 (only if the point is on the sphere)	$\sin \theta_2 = 0$	W2 p. 33

How to use this gallery [added]

PREMISE robot gallery **Given:** a named robot (RR, RP, PR, RRR, PPP, RPP, RRP, SCARA, elbow arm, Puma, Panda, Furuta) with its joint values $\mathbf{q}$, or a wanted tool position / pose
Find: its pose ${}^0\mathbf{T}_E(\mathbf{q})$, all IK branches, the full Jacobian ${}^0\mathbf{J}(\mathbf{q})$, its singular poses and its workspace, written out in whole matrices

Each entry has the same order: *card* (joints, variables, workspace, IK count, singular poses) → ETS beside proximal DH → the whole ${}^0\mathbf{T}_E$ → IK with every branch → the whole ${}^0\mathbf{J}$ with det and singular poses → a number check. Every matrix comes from one transform chain and is checked by script ([checked]); where a week part derives the same robot the matrices are copied from it. This part is a reference gallery, so everything here is [added] unless a slide page is quoted.

Conventions used in every entry

$$ {}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})\, t_x(a_{i-1})\, \mathbf{R}_z(\theta_i)\, t_z(d_i), \qquad {}^0\mathbf{T}_E = {}^0\mathbf{T}_1\, {}^1\mathbf{T}_2 \cdots {}^{n-1}\mathbf{T}_n $$

$$ {}^{i-1}\mathbf{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

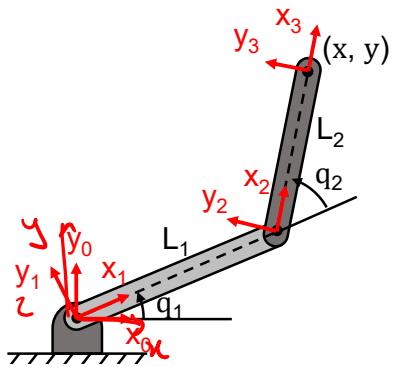
q_i	joint variable of joint i: θ_i (rad) if revolute, d_i (m) if prismatic	rad, m
L_i, a_i, H, L_0	link lengths, offsets, heights (fixed)	m
c_i, s_i, c_{12}, s_{123}	$\cos q_i$, $\sin q_i$, $\cos(q_1+q_2)$, $\sin(q_1+q_2+q_3)$	–
${}^0\mathbf{T}_E$	pose of the tool frame $\{E\}$ in the base frame $\{0\}$ (4×4; last column $= (x, y, z, 1)^\top$)	m
${}^0\mathbf{J}$	manipulator Jacobian in $\{0\}$ ($6 \times n$): rows $v_x, v_y, v_z, \omega_x, \omega_y, \omega_z$; ${}^0\boldsymbol{\nu} = {}^0\mathbf{J}\,\dot{\mathbf{q}}$	m, rad
$\mathbf{J}_v$, $\mathbf{J}_\omega$	top three rows (linear), bottom three rows (angular) of $\mathbf{J}$	–

Use: (times a constant tool transform if the question has one.) Right-handed frames, z up, z_i out of the page in plan views; radians in formulas, degrees only in numbers. Rows are $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$ [W2 slide 28]; ETS = walk the arm, base to tool [W2 slides 15–16]. Distal DH, POE, URDF: p. 33.

STEPS Which robot is it? (read the question)

1. Count the joints and write the letters in order from the base: R (turns), P (slides). The string is the name: RR, RP, PR, RRR, RRP, RPP, PPP, RRPR, 6R.
2. Planar (all axes parallel, $\perp$ the plane) or spatial? Planar RR/RP/PR/RRR live in the xy-plane; the rest are 3D.
3. Find the row of the comparison table (the table at the start of this part), then the entry. The card gives the workspace, the number of IK solutions and the singular poses at once.
4. If the question gives its own table, *use the question's table*; this gallery only shows the standard ones.
5. Close with the entry's number check: $\mathbf{q} = \mathbf{0}$ must match your drawing.

1. Planar RR: two revolute joints



[W06 lec p. 35] — RR arm with frames $\{0\}, \{1\}, \{2\}, \{3\}$ ($\{3\}$ = tool)

Joints	RR; planar; $n = 2$
Variables	$\mathbf{q} = (q_1, q_2)$; q_1 from x_0, q_2 relative to link 1; counter-clockwise +
Parameters	L_1, L_2
Workspace	annulus $\lvert L_1 - L_2 \rvert \le r \le L_1 + L_2$
IK	2 (elbow $\pm q_2$)
Singular	$q_2 = 0$ (stretched), π (folded)

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\, t_x(L_1)\, \mathbf{R}_z(q_2)\, t_x(L_2)$$

B. Proximal DH (3 rows, last is the tool)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	L_1	0	0	q_2
3 (tool)	L_2	0	0	0

FK: ${}^0\mathbf{T}_E$ of the RR arm [W2 slides 15–16]

$${}^0\mathbf{T}_E = \begin{bmatrix} c_{12} & -s_{12} & 0 & L_1 c_1 + L_2 c_{12} \\ s_{12} & c_{12} & 0 & L_1 s_1 + L_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SYMBOLS

L_1, L_2	link lengths	m
q_1, q_2	joint angles	rad

Use: tip $(x, y) = (L_1 c_1 + L_2 c_{12},\ L_1 s_1 + L_2 s_{12})$, *tool angle* $q_1 + q_2$ *(the running sum). Check:* $\mathbf{q} = \mathbf{0}$ *gives* $(L_1{+}L_2, 0)$; $L = (0.4, 0.3)$ *m,* $\mathbf{q} = (40°, 50°)$ *gives* $(0.3064,\ 0.5571)$ *m.* [checked]

IK, both branches [W3 slides 25–31]

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2 L_1 L_2}, \qquad q_2 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2 q_2},\ \cos q_2\right)$$

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin q_2,\ L_1 + L_2 \cos q_2)$$

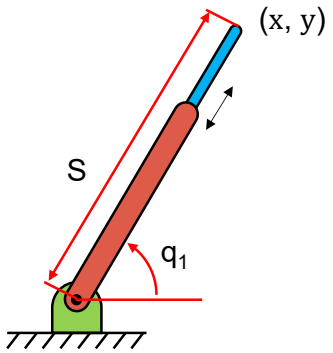
Use: Two solutions inside the annulus, one on its edge ($\sin q_2 = 0$)*, none outside* ($\lvert \cos q_2 \rvert > 1$)*. Example:* $L = (0.4, 0.3)$*, target* $(0.3064, 0.5571)$*:* $(q_1, q_2) = (40°, 50°)$ *or* $(82.38°, -50°)$*; both reproduce the target.* [checked]

Jacobian ${}^0\mathbf{J}$ [W6 slides 33–39, 45–46]

$${}^0\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \det \mathbf{J}_{xy} = L_1 L_2 \sin q_2$$

Use: singular at $q_2 = 0, \pi$*: the two columns are parallel, rank 1, and the tool cannot move along the arm; joint rates* $\to \infty$ *for any such request. Independent of* q_1*. Dynamics of this arm: W7 p. 91.* [checked]

2. Planar RP: polar arm



[W03 lec p. 23] — RP arm: revolute q_1, then a prismatic extension along the link (slide: S)

Joints	RP; planar; $n = 2$
Variables	q_1 (rad), d_2 (m; the slide calls it S)
Parameters	none (the stroke of d_2)
Workspace	annulus $d_{2\min} \le r \le d_{2\max}$
IK	1; $(-d_2,\ q_1{+}\pi)$ if $d_2 < 0$ is allowed
Singular	$d_2 = 0$ (tool on the joint-1 axis)

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\, t_x(d_2)$$

B. Proximal DH (prismatic axis must be z_2)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	$q_1 - 90°$
2	0	$-90°$	d_2	0

plus the constant tool transform ${}^2\mathbf{T}_E$ below.

FK: ${}^0\mathbf{T}_E$ of the RP arm [W2 p. 33]

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 & -s_1 & 0 & d_2 c_1 \\ s_1 & c_1 & 0 & d_2 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0\mathbf{T}_2 = \begin{bmatrix} s_1 & 0 & c_1 & d_2 c_1 \\ -c_1 & 0 & s_1 & d_2 s_1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^2\mathbf{T}_E = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^0\mathbf{T}_2\,{}^2\mathbf{T}_E = {}^0\mathbf{T}_E $$

SYMBOLS

q_1	revolute angle from x_0	rad
d_2	extension along the link, from the joint-1 axis to the tip	m

Use: tip = $d_2(c_1, s_1)$: polar coordinates. Check: $q_1 = 35°$, $d_2 = 0.6$ m gives $(0.4915, 0.3441)$ m. [checked]

IK [W3 slides 23–24]

$$ d_2 = \sqrt{x^2 + y^2}, \qquad q_1 = \operatorname{atan2}(y, x) \qquad \left(\text{second solution if } d_2 < 0 \text{ allowed: } (-d_2,\ q_1 + \pi)\right) $$

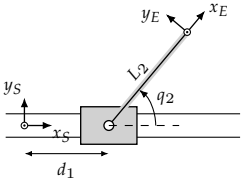
Use: atan2, not $\tan^{-1}(y/x)$. $(0.3, 0.4)$: $d_2 = 0.5$ m, $q_1 = 53.13°$; $(-0.3, 0.4)$: $q_1 = 126.87°$. Out of reach if d_2 is outside its stroke. [checked]

Jacobian ${}^0\mathbf{J}$ [W6 p. 78]

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_2 s_1 & c_1 \\ d_2 c_1 & s_1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_{xy} = -d_2 $$

Use: column 1 is tangential ($d_2\times$ unit tangent), column 2 radial. Singular only at $d_2 = 0$: column 1 vanishes and the tangential direction is lost. $q_1 = 40°$, $d_2 = 0.6$: $\mathbf{J}_{xy} = \begin{bmatrix} -0.3857 & 0.7660 \\ 0.4596 & 0.6428 \end{bmatrix}$. [checked]

3. Planar PR: rail, then rotary joint



[drawn for this booklet] z_S, z_E out of the page; slider along x_S

Joints	PR; planar; $n = 2$
Variables	d_1 (m) along x_S, q_2 (rad) from x_S
Parameters	L_2
Workspace	stadium: $D \times 2L_2$ rectangle, half-disc of radius L_2 at each end
IK	2: q_2 and $\pi - q_2$
Singular	$q_2 = \pm 90°$

ALTERNATIVE APPROACHES

A. ETS

$$ {}^S\mathbf{T}_E = t_x(d_1)\,\mathbf{R}_z(q_2)\,t_x(L_2) $$

B. Proximal DH (slide along x_S needs $z_0 = x_S$)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	90°	0	$q_2 + 90°$
3 (tool)	L_2	0	0	0

with ${}^S\mathbf{T}_0$ below.

FK: ${}^S\mathbf{T}_E$ of the PR arm [W2 p. 33]

$$ {}^S\mathbf{T}_E = \begin{bmatrix} c_2 & -s_2 & 0 & d_1 + L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^S\mathbf{T}_0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^S\mathbf{T}_0\ (\text{DH table product}) = {}^S\mathbf{T}_E $$

SYMBOLS

d_1	slider position along the rail	m
q_2	link angle from x_S	rad
L_2	link length	m

Use: tip $(d_1 + L_2 c_2,\ L_2 s_2)$. Check: $d_1 = 0.3$ m, $q_2 = 60°$, $L_2 = 0.4$ m gives $(0.5,\ 0.3464)$ m. A vertical slide (z_S) needs no extra frame: rows $(0, 0, d_1, 0), (0, 0, 0, q_2), (L_2, 0, 0, 0)$. [checked]

IK, both branches [added]

$$ \sin q_2 = \frac{y}{L_2}, \quad q_2 = \operatorname{atan2}\left(y,\ \pm\sqrt{L_2^2 - y^2}\right), \quad d_1 = x - L_2 \cos q_2 $$

Use: the slider cannot change y, so q_2 comes from y alone. None if $|y| > L_2$; one if $|y| = L_2$. $L_2 = 0.5$, target $(0.8, 0.3)$: $(d_1, q_2) = (0.4$ m, $36.87°)$ or $(1.2$ m, $143.13°)$. (W3 writes θ and L for q_2 and L_2.) [checked]

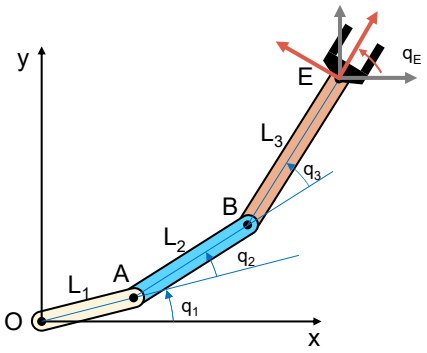
Jacobian ${}^S\mathbf{J}$ [W6 p. 78]

$$ {}^S\mathbf{J} = \begin{bmatrix} 1 & -L_2 s_2 \\ 0 & L_2 c_2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \det \mathbf{J}_{xy} = L_2 \cos q_2 $$

Use: singular at $q_2 = \pm 90°$: the link is perpendicular to the rail, both columns point along $\pm x$, and the y direction is lost. $L_2 = 0.5$, $q_2 = 30°$: det = 0.4330. [checked]

ROBOTS

4. Planar RRR: three revolute joints



[W02 lec p. 9] — planar RRR arm, joint angles each from the previous link

Joints	RRR; planar; $n = 3$
Variables	$\mathbf{q} = (q_1, q_2, q_3)$ (rad), each from the previous link
Parameters	L_1, L_2, L_3
Workspace	disc $r \le L_1{+}L_2{+}L_3$; dexterous annulus $L_3{+}\lvert L_1{-}L_2\rvert \le r \le L_1{+}L_2{-}L_3$
IK	2 for (x, y, φ); ∞ for (x, y)
Singular	$q_2 = 0, \pi$ (for (x, y, ω_z))

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\, t_x(L_1)\, \mathbf{R}_z(q_2) \cdot t_x(L_2)\, \mathbf{R}_z(q_3)\, t_x(L_3)$$

B. Proximal DH (4 rows, last is the tool)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	L_1	0	0	q_2
3	L_2	0	0	q_3
4 (tool)	L_3	0	0	0

FK: ${}^0\mathbf{T}_E$ of the RRR arm [W2 slides 6–9]

$${}^0\mathbf{T}_E = \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1c_1 + L_2c_{12} + L_3c_{123} \\ s_{123} & c_{123} & 0 & L_1s_1 + L_2s_{12} + L_3s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SYMBOLS

L_1, L_2, L_3	link lengths	m
q_1, q_2, q_3	joint angles	rad
$q_E = q_1{+}q_2{+}q_3$	tool angle, read from the matrix as $\operatorname{atan2}(T_{21}, T_{11})$	rad

Use: check: $\mathbf{q} = \mathbf{0}$ *gives* $(L_1{+}L_2{+}L_3, 0)$; $L = (0.5, 0.4, 0.2)$ *m,* $\mathbf{q} = (30°, 45°, -60°)$ *gives* $(0.7297,\ 0.6881)$ *m and* $q_E = 15°$ *(slide example).* *[checked]*

IK for the pose (x, y, φ), both branches [added]

$$\mathbf{p}_W = (x - L_3\cos\varphi,\ y - L_3\sin\varphi), \quad (q_1, q_2) = \text{RR solution for } \mathbf{p}_W \text{ (box 1)}, \quad q_3 = \varphi - q_1 - q_2$$

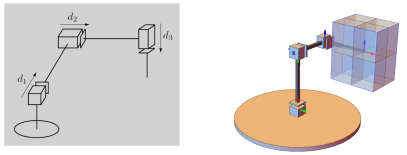
Use: two solutions; position only (x, y)*: infinitely many (one spare DoF).* $L = (1.0, 0.7, 0.3)$, $(x, y) = (1.2, 0.9)$, $\varphi = 60°$: $\mathbf{p}_W = (1.0500, 0.6402)$; $\mathbf{q} = (-3.32°,\ 89.09°,\ -25.77°)$ *or* $(66.06°,\ -89.09°,\ 83.02°)$. *[checked]*

Jacobian ${}^0\mathbf{J}$ [W6 p. 78]

$${}^0\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} - L_3s_{123} & -L_2s_{12} - L_3s_{123} & -L_3s_{123} \\ L_1c_1 + L_2c_{12} + L_3c_{123} & L_2c_{12} + L_3c_{123} & L_3c_{123} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Use: column j *sums every link from* j *outwards. Position only:* $\mathbf{J}_{xy}$ *is* 2×3 *(fat), rank 1 only when all links are collinear. Pose* (x, y, ω_z)*: the* 3×3 *matrix has* $\det = L_1L_2 \sin q_2$*, independent of* q_3*: singular when the wrist point is stretched or folded.* $L = (0.4, 0.3, 0.2)$, $\mathbf{q} = (20°, 40°, -30°)$: $\mathbf{J}_{xy} = \begin{bmatrix} -0.4966 & -0.3598 & -0.1000 \\ 0.6991 & 0.3232 & 0.1732 \end{bmatrix}$, $\det = 0.0771$. *[checked]*

5. Cartesian PPP: three prismatic joints



[W01 lec p. 56] — PPP (Cartesian) robot (slide)

Joints	PPP; $n = 3$
Variables	d_1 along z, d_2 along y, d_3 along x (m)
Parameters	none
Workspace	box: edges are the three strokes
IK	1: $(d_1, d_2, d_3) = (z, y, x)$
Singular	none ($\det \mathbf{J}_v = -1$)

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = t_z(d_1)\, t_y(d_2)\, t_x(d_3)$$

B. Proximal DH (each axis z_i is a slide)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	$-90°$	d_2	$90°$
3	0	$90°$	d_3	0

frame $\{3\}$ has z_3 along the last slide: ${}^0\mathbf{T}_3 = {}^0\mathbf{T}_E\, \mathbf{R}_y(90°)$.

FK: ${}^0\mathbf{T}_E$ of the PPP robot [W1 slide 56]

$${}^0\mathbf{T}_E = \begin{bmatrix} 1 & 0 & 0 & d_3 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^0\mathbf{T}_3 = \begin{bmatrix} 0 & 0 & 1 & d_3 \\ 0 & 1 & 0 & d_2 \\ -1 & 0 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SYMBOLS

d_1, d_2, d_3	strokes along z_0, y_0, x_0	m

Use: the orientation never changes: no joint turns the tool. The order of the three slides does not change the position. Check: all $d_i = 0$ *gives the origin;* $M = 6(4 - 3 - 1) + 3 = 3$ *(W1 G–K).* *[checked]*

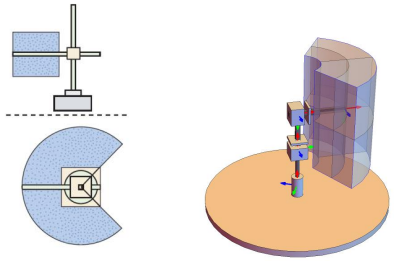
IK and Jacobian [W1 slide 56]

$(d_1, d_2, d_3) = (z,\ y,\ x)$ (always exactly one solution, if the point is inside the box)

$$ {}^0\mathbf{J} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = -1 $$

Use: a prismatic column is the unit axis of the slide and has no angular part; the three axes are orthogonal, so $\mathbf{J}_v$ is a permutation: never singular. [checked]

6. Cylindrical RPP: turn, lift, reach



[W01 lec p. 55] — RPP (cylindrical) robot (slide)

Joints	RPP; $n = 3$
Variables	q_1 (rad) about z_0, d_2 (m) up, d_3 (m) radial
Parameters	none
Workspace	hollow cylinder sector: inner and outer radius $d_{3\,\min}, d_{3\,\max}$, height = stroke of d_2
IK	1; $(q_1+\pi,\ d_2,\ -d_3)$ if $d_3 < 0$ allowed
Singular	$d_3 = 0$ (tool on the axis)

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\, t_z(d_2)\, t_x(d_3)$$

B. Proximal DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	0	0	d_2	$90°$
3	0	$90°$	d_3	0

with ${}^3\mathbf{T}_E$ below.

FK: ${}^0\mathbf{T}_E$ of the RPP robot [W1 slide 55]

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 & -s_1 & 0 & d_3 c_1 \\ s_1 & c_1 & 0 & d_3 s_1 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0\mathbf{T}_3 = \begin{bmatrix} -s_1 & 0 & c_1 & d_3 c_1 \\ c_1 & 0 & s_1 & d_3 s_1 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^3\mathbf{T}_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^0\mathbf{T}_3\, {}^3\mathbf{T}_E = {}^0\mathbf{T}_E $$

SYMBOLS

q_1	rotation of the whole arm about the vertical axis	rad
d_2	vertical slide (height of the tool)	m
d_3	radial slide (horizontal reach)	m

Use: cylindrical coordinates $(r, \varphi, z) = (d_3, q_1, d_2)$. Check: $q_1 = 40°$, $d_2 = 0.35$ m, $d_3 = 0.5$ m gives $(0.3830,\ 0.3214,\ 0.35)$ m. [checked]

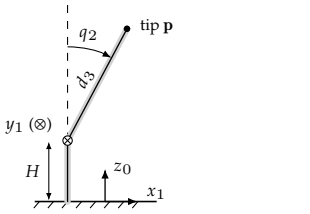
IK and Jacobian [W1 slide 55]

$$ q_1 = \operatorname{atan2}(y, x), \qquad d_2 = z, \qquad d_3 = \sqrt{x^2 + y^2} $$

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_3 s_1 & 0 & c_1 \\ d_3 c_1 & 0 & s_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = d_3 $$

Use: point $(0.3, 0.4, 0.6)$: $q_1 = 53.13°$, $d_2 = 0.6$ m, $d_3 = 0.5$ m. Singular only at $d_3 = 0$: column 1 vanishes (the tangential direction is lost) and q_1 is arbitrary. [checked]

7. Spherical RRP: turn, tilt, reach



[drawn for this booklet] side view in the arm's plane (turned q_1 about z_0); y_1 into the page

Joints	RRP; $n = 3$
Variables	q_1 about z_0, q_2 tilt of the boom from vertical (about y_1), d_3 along the boom
Parameters	H (height of joint 2)
Workspace	spherical shell $d_{3\,\min} \le \rho \le d_{3\,\max}$ about $(0, 0, H)$
IK	2: (q_1, q_2, d_3) and $(q_1+\pi, -q_2, d_3)$; 4 if $d_3 < 0$ allowed
Singular	$d_3 = 0$, or $\sin q_2 = 0$

ALTERNATIVE APPROACHES

A. ETS

$${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\, t_z(H)\, \mathbf{R}_y(q_2)\, t_z(d_3)$$

B. Proximal DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	H	q_1
2	0	$-90°$	0	q_2
3	0	$90°$	d_3	0

FK: ${}^0\mathbf{T}_E$ of the spherical RRP arm [W3 p. 45, W6 p. 78]

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & d_3 c_1 s_2 \\ s_1 c_2 & c_1 & s_1 s_2 & d_3 s_1 s_2 \\ -s_2 & 0 & c_2 & H + d_3 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

H	height of the joint-2 axis above $\{0\}$	m
q_1, q_2	turn about z_0; tilt of the boom from vertical about y_1 (positive tips the boom towards x_1)	rad
d_3	extension of the prismatic joint along the boom	m

Use: spherical coordinates about $(0, 0, H)$: radius d_3, polar angle q_2, azimuth q_1. Check: $\mathbf{q} = \mathbf{0}$ gives $(0, 0, H + d_3)$; $H = 0.5$, $(q_1, q_2, d_3) = (53.13°,\ 51.34°,\ 0.6403)$ gives $(0.3, 0.4, 0.9)$ m. [checked]

IK, both branches [W3 p. 45]

$$ q_1 = \operatorname{atan2}(y, x), \quad d_3 = \sqrt{r^2 + z'^2}, \quad q_2 = \operatorname{atan2}(r, z'), \qquad r = \sqrt{x^2 + y^2},\ z' = z - H $$

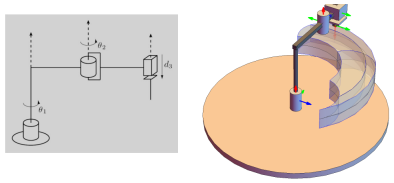
Use: second solution $(q_1 + \pi, -q_2, d_3)$; with $d_3 < 0$ allowed, also $(q_1, q_2 + \pi, -d_3)$ and $(q_1 + \pi, \pi - q_2, -d_3)$. $H = 0.5$, $(0.3, 0.4, 0.9)$: $(53.13°, 51.34°, 0.6403\ m)$ and $(233.13°, -51.34°, 0.6403\ m)$. [checked]

Jacobian ${}^0\mathbf{J}$ [W6 p. 78]

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_3 s_1 s_2 & d_3 c_1 c_2 & c_1 s_2 \\ d_3 c_1 s_2 & d_3 s_1 c_2 & s_1 s_2 \\ 0 & -d_3 s_2 & c_2 \\ 0 & -s_1 & 0 \\ 0 & c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = -d_3^2 \sin q_2 $$

Use: singular if $d_3 = 0$ (tool on the joint-2 axis) or $\sin q_2 = 0$ (boom vertical, tool on the z_0 axis, column 1 is zero). Boom 0.6403 m at $q_2 = 51.34°$, $q_1 = 53.13°$: $\det \mathbf{J}_v = -0.3202$. [checked]

8. SCARA: RRPR (position part RRP)



[W01 lec p. 54] — RRP (SCARA) robot (slide)

Joints	RRPR (RRP + tool yaw); $n = 4$
Variables	q_1, q_2 (rad) about vertical axes, d_3 (m) down, q_4 (rad) yaw of the tool
Parameters	H, a_1, a_2
Workspace	flat ring $\lvert a_1 - a_2 \rvert \le r \le a_1 + a_2$ times the stroke of d_3
IK	2 (elbow); d_3, q_4 follow
Singular	$q_2 = 0, \pi$

ALTERNATIVE APPROACHES

A. ETS (tool z_E up, q_4 about $+z$)

$$ {}^0\mathbf{T}_E = t_z(H)\,\mathbf{R}_z(q_1)\,t_x(a_1)\,\mathbf{R}_z(q_2) \cdot t_x(a_2)\,t_z(-d_3)\,\mathbf{R}_z(q_4) $$

B. Proximal DH (4 rows)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	H	q_1
2	a_1	0	0	q_2
3	a_2	180°	d_3	0
4	0	180°	0	q_4

FK: ${}^0\mathbf{T}_E$ of the SCARA robot [W2 p. 33; W1 slide 54]

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{124} & -s_{124} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{124} & c_{124} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & H - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

RRP only (3 rows, tool z down, W2): $$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{12} & s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & -c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & -1 & H - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

H	column height of the first joint	m
a_1, a_2	the two horizontal link lengths	m
d_3	downward stroke of the slide (positive lowers the tool)	m
q_4	tool yaw about the vertical, positive about $+z$	rad

Use: all revolute axes are vertical, so the angles add: yaw $\varphi = q_1 + q_2 + q_4$. The 180° twist in row 3 turns z_3 down so a positive d_3 lowers the tool; row 4 turns it back up. Check: $\mathbf{q} = \mathbf{0}$ gives $(a_1 + a_2, 0, H - d_3)$; $H = 0.4$, $a = (0.35, 0.25)$, $q = (40°, -70°, d_3 = 0.12, 25°)$ gives $(0.4846, 0.1000, 0.28)$ and yaw $-5°$ (W2 practice). [checked]

IK, both elbows [W3 p. 45]

$$ d_3 = H - z, \quad (q_1, q_2) = \text{RR solution for } (x, y) \text{ (box 1, } L_i \to a_i), \quad q_4 = \varphi - q_1 - q_2 $$

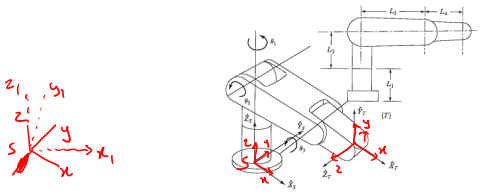
Use: two solutions (elbow $\pm q_2$); on the edge $r = a_1 + a_2$ one. $H = 0.40$, $a = (0.35, 0.25)$, target $(0.30, 0.25, 0.28)$ m, $\varphi = 60°$: $(q_1, q_2, d_3, q_4) = (0.83°, 100.70°, 0.12\ m, -41.53°)$ or $(78.79°, -100.70°, 0.12\ m, 81.92°)$. [checked]

Jacobian ${}^0\mathbf{J}$ [W6 p. 78]

$$ {}^0\mathbf{J} = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} & 0 & 0 \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}, \qquad \det \mathbf{J}_v = -a_1 a_2 \sin q_2 $$

Use: the slide column is $-\hat{z}$ because d_3 is measured downwards; the yaw column is ω_z only. The 4×4 block $(v_x, v_y, v_z, \omega_z)$ has the same determinant, so the SCARA is singular only at $q_2 = 0, \pi$ (arm stretched or folded). [checked]

9. Elbow arm: articulated RRR (W13 Q1–Q3)



[W13 lec p. 4] — W13 Q1 arm with the lecturer's frames $\{S\}, \{1\}, \{2\}, \{3\}, \{T\}$

Joints	RRR; spatial; $n = 3$
Variables	θ_1 about z_S; θ_2, θ_3 about horizontal axes, positive *lifts* the arm
Parameters	L_1, L_2 (shoulder height $L_1 + L_2$), L_3, L_4
Workspace	shell $\lvert L_3 - L_4 \rvert \le \rho \le L_3 + L_4$ about $(0, 0, L_1 + L_2)$ 4: shoulder × elbow
IK	
Singular	$\sin \theta_3 = 0$ (boundary); $r = 0$ (tool above the base axis)

ALTERNATIVE APPROACHES

A. ETS (the lecturer's, W13 p. 4)

$$ {}^S\mathbf{T}_T = \mathbf{R}_z(\theta_1)\,t_z(L_1 + L_2) \cdot \mathbf{R}_y(-\theta_2)\,t_x(L_3)\,\mathbf{R}_y(-\theta_3) \cdot t_x(L_4)\,\mathbf{R}_x(90°) $$

B. Proximal DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	$L_1 + L_2$	θ_1
2	0	90°	0	θ_2
3	L_3	0	0	θ_3
4 (tool)	L_4	0	0	0

the table gives the ETS exactly (no extra constant).

FK: ${}^{S}\mathbf{T}_{T}$ of the elbow arm [W13 slides 4–5]

$$ {}^{S}\mathbf{T}_{T} = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1(L_3c_2 + L_4c_{23}) \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1(L_3c_2 + L_4c_{23}) \\ s_{23} & c_{23} & 0 & L_1 + L_2 + L_3s_2 + L_4s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

θ_1	rotation about the vertical z_S	rad
θ_2, θ_3	shoulder and elbow, positive lifts the arm (rotation about $-y$)	rad
c_{23}	$\cos(\theta_2 + \theta_3)$	–

Use: same matrix as the W2 part (p. 33). $r = L_3c_2 + L_4c_{23}$ is the horizontal reach; $L_1{+}L_2{+}L_3s_2{+}L_4s_{23}$ the height. Check: $\boldsymbol{\theta} = \mathbf{0}$ gives $(L_3{+}L_4, 0, L_1{+}L_2)$ and $x_T = x_S$, $y_T = z_S$, $z_T = -y_S$; $L = (0.6, 0.5, 0.4, 0.3)$, $\boldsymbol{\theta} = (30°, 20°, 40°)$ gives $(0.4554,\ 0.2629,\ 1.4966)$ m. [checked]

IK, all four branches [W13 slide 7; W3 p. 54]

$$ \theta_1 = \operatorname{atan2}(y, x)\ (r = +\sqrt{x^2 + y^2}) \ \text{ or } \ \theta_1 + \pi\ (r = -\sqrt{x^2 + y^2}), \qquad z' = z - L_1 - L_2 $$

$$ \cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2L_3L_4}, \qquad \theta_3 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3\right) $$

$$ \theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4\sin\theta_3,\ L_3 + L_4\cos\theta_3) $$

Use: 4 solutions: (θ_1 or $\theta_1{+}\pi$) × ($\pm\theta_3$); none outside the shell ($|\cos\theta_3| > 1$). $L = (0.6, 0.5, 0.4, 0.3)$, $\mathbf{p} = (0.4554, 0.2629, 1.4966)$: $(30, 20, 40)$, $(30, 54.047, -40)$, $(-150, 125.953, 40)$, $(-150, 160, -40)$ degrees; each reproduces $\mathbf{p}$. [checked]

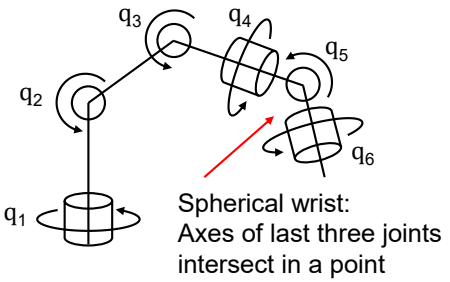
Jacobian ${}^{S}\mathbf{J}$ [W6 p. 78]

$$ {}^{S}\mathbf{J} = \begin{bmatrix} -s_1(L_3c_2 + L_4c_{23}) & -c_1(L_3s_2 + L_4s_{23}) & -c_1L_4s_{23} \\ c_1(L_3c_2 + L_4c_{23}) & -s_1(L_3s_2 + L_4s_{23}) & -s_1L_4s_{23} \\ 0 & L_3c_2 + L_4c_{23} & L_4c_{23} \\ 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix} $$

$$ \det\mathbf{J}_v = -L_3L_4\sin\theta_3\,(L_3c_2 + L_4c_{23}) $$

Use: angular columns are the joint axes in $\{S\}$: $\hat{z}_S$, then $(s_1, -c_1, 0)$ twice. Singular when $\sin\theta_3 = 0$ (arm stretched or folded: workspace boundary) or $r = L_3c_2 + L_4c_{23} = 0$ (tool on the vertical axis through the shoulder: interior). $L_3 = 0.4$, $L_4 = 0.3$, $\theta_2 = 60°$: $r = 0$ at $\theta_3 = 71.81°$. Equations of motion: W7 p. 91. [checked]

10. Puma 560: six revolute joints, spherical wrist



[W03 lec p. 32] — 6-DoF arm whose last three axes meet in a point (spherical wrist)

Joints	6R; $n = 6$; axes 4, 5, 6 meet at the wrist centre W
Variables	$q_1, \ldots, q_6$ (rad)
Parameters	zero-offset model: L_b, L_u, L_f; course table: a_2, a_3, d_3, d_4
Workspace	W in a spherical shell about the shoulder: inner/outer radius $\lvert L_u - L_f\rvert, L_u + L_f$ (zero-offset); 0.150 to 0.877 m (course table)
IK	up to 8: 2 (q_1) ×2 (q_3) ×2 (wrist flip)
Singular	elbow $\sin q_3 = 0$; shoulder (W on the waist axis); wrist $\sin q_5 = 0$

10a. Zero-offset model (W2 notebook Q5 / W6 notebook Q3)

ALTERNATIVE APPROACHES

A. ETS (home pose: arm straight up)

${}^{0}\mathbf{T}_{E} = t_z(L_b)\,\mathbf{R}_z(q_1)\,\mathbf{R}_y(q_2)\,t_z(L_u)$
$\cdot\,\mathbf{R}_y(q_3)\,t_z(L_f)\,\mathbf{R}_z(q_4)\,\mathbf{R}_y(q_5)\,\mathbf{R}_z(q_6)$
$\{E\}$ sits at the wrist centre.

B. Proximal DH (table of the W2 part)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_b	q_1
2	0	−90°	0	q_2−90°
3	L_u	0	0	q_3+90°
4	0	90°	L_f	q_4
5	0	−90°	0	q_5
6	0	90°	0	q_6

FK: ${}^{0}\mathbf{T}_{E}$ of the zero-offset Puma in block form [W2 p. 33]

$$ {}^{0}\mathbf{T}_{E} = \begin{bmatrix} {}^{0}\mathbf{R}_3\,{}^{3}\mathbf{R}_6 & \mathbf{p}_W \\ \mathbf{0}^{\top} & 1 \end{bmatrix} $$

$$ {}^{0}\mathbf{R}_3 = \mathbf{R}_z(q_1)\mathbf{R}_y(q_2{+}q_3) = \begin{bmatrix} c_1c_{23} & -s_1 & c_1s_{23} \\ s_1c_{23} & c_1 & s_1s_{23} \\ -s_{23} & 0 & c_{23} \end{bmatrix} $$

$$ {}^{3}\mathbf{R}_6 = \mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\mathbf{R}_z(q_6) = \begin{bmatrix} c_4c_5c_6 - s_4s_6 & -c_4c_5s_6 - s_4c_6 & c_4s_5 \\ s_4c_5c_6 + c_4s_6 & -s_4c_5s_6 + c_4c_6 & s_4s_5 \\ -s_5c_6 & s_5s_6 & c_5 \end{bmatrix} $$

ROBOTS

Wrist centre $\mathbf{p}_W$ = tool origin of the zero-offset Puma

$$\mathbf{p}_W = \begin{bmatrix} c_1(L_u s_2 + L_f s_{23}) \\ s_1(L_u s_2 + L_f s_{23}) \\ L_b + L_u c_2 + L_f c_{23} \end{bmatrix}$$

SYMBOLS

L_b	height of the shoulder above $\{0\}$	m
L_u, L_f	upper arm, forearm (shoulder to elbow, elbow to wrist centre)	m
q_1	waist; q_2, q_3 shoulder, elbow (about y, positive tips the arm towards $+x$); q_4, q_5, q_6 wrist (z, y, z)	rad
$\mathbf{p}_W$	wrist centre = tool origin, independent of q_4, q_5, q_6	m

Use: the rotation of the tool is the product ${}^0\mathbf{R}_3\,{}^3\mathbf{R}_6$ (multiply numerically; entries expand to long sums, so write the factors). Check: $\mathbf{q} = \mathbf{0}$ gives $\mathbf{I}_3$ and $\mathbf{p}_W = (0, 0, L_b{+}L_u{+}L_f)$; $L = (1.0, 0.5, 0.3)$, $\mathbf{q} = (30°, 20°, -40°, 0, 0, 0)$ gives $\mathbf{p}_W = (0.0592,\ 0.0342,\ 1.7518)$ m. [checked]

IK: decoupled, 8 branches [W3 slides 32–38]

$$\rho = \pm\sqrt{x_W^2 + y_W^2}, \quad q_1 = \operatorname{atan2}(y_W, x_W)\ (\rho > 0) \text{ or } q_1 + \pi\ (\rho < 0), \quad z' = z_W - L_b$$

$$\cos q_3 = \frac{\rho^2 + z'^2 - L_u^2 - L_f^2}{2L_u L_f}, \qquad q_3 = \operatorname{atan2}(\pm\sqrt{1 - \cos^2 q_3}, \cos q_3)$$

$$q_2 = \operatorname{atan2}(\rho, z') - \operatorname{atan2}(L_f \sin q_3,\ L_u + L_f \cos q_3)$$

$${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\,{}^0\mathbf{R}_E, \qquad q_5 = \operatorname{atan2}\left(\pm\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\right)$$

$$q_4 = \operatorname{atan2}(\pm r_{23}, \pm r_{13}), \qquad q_6 = \operatorname{atan2}(\pm r_{32}, \mp r_{31})$$

Use: angles are measured from the vertical here ($q_2 = 0$: upper arm straight up). Upper signs together, lower signs together (wrist flip: $q_5 \to -q_5$, $q_4 \to q_4 + \pi$, $q_6 \to q_6 + \pi$). If $q_5 = 0$ the wrist is singular: only $q_4 + q_6$ is fixed. Example $L = (1.0, 0.5, 0.3)$, $\mathbf{q} = (30, 20, -40, 25, 70, -35)°$; all 8 solutions (degrees), each reproduces the whole pose:

q_1	q_2	q_3	q_4	q_5	q_6
30	*20*	−40	*25*	*70*	−35
30	*20*	−40	−155	−70	*145*
30	−9.602	*40*	*54.876*	*29.048*	−77.116
30	−9.602	*40*	−125.124	−29.048	*102.884* [checked]
210	−20	*40*	−155	*70*	−35
210	−20	*40*	*25*	−70	*145*
210	*9.602*	−40	−125.124	*29.048*	−77.116
210	*9.602*	−40	*54.876*	−29.048	*102.884*

Jacobian ${}^0\mathbf{J}$ of the zero-offset Puma (6×6, three whole blocks) [W6 notebook Q3]

$${}^0\mathbf{J} = \begin{bmatrix} \mathbf{J}_v & \mathbf{0}_{3\times3} \\ \mathbf{J}_{\omega,1\text{–}3} & {}^0\mathbf{R}_3\mathbf{W} \end{bmatrix}$$

$$\mathbf{J}_v = \begin{bmatrix} -s_1(L_u s_2 + L_f s_{23}) & c_1(L_u c_2 + L_f c_{23}) & c_1 L_f c_{23} \\ c_1(L_u s_2 + L_f s_{23}) & s_1(L_u c_2 + L_f c_{23}) & s_1 L_f c_{23} \\ 0 & -(L_u s_2 + L_f s_{23}) & -L_f s_{23} \end{bmatrix}$$

$$\mathbf{J}_{\omega,1\text{–}3} = \begin{bmatrix} 0 & -s_1 & -s_1 \\ 0 & c_1 & c_1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{W} = \begin{bmatrix} 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \\ 1 & 0 & c_5 \end{bmatrix} \quad (\text{columns } \hat{z},\ \mathbf{R}_z(q_4)\hat{y},\ \mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\hat{z} \text{ in } \{3\})$$

$$\det \mathbf{J} = -L_u L_f \sin q_3\,(L_u s_2 + L_f s_{23}) \sin q_5$$

Use: the tool origin is the wrist centre, so columns 4–6 have no linear part and $\mathbf{J}$ is block lower-triangular: $\det \mathbf{J} = \det \mathbf{J}_v \cdot \det({}^0\mathbf{R}_3\mathbf{W})$ with $\det \mathbf{J}_v = L_u L_f \sin q_3 (L_u s_2 + L_f s_{23})$ and $\det \mathbf{W} = -\sin q_5$. Three singular families: elbow $\sin q_3 = 0$ (boundary), shoulder $L_u s_2 + L_f s_{23} = 0$ (W on the waist axis; $L_u = 0.5$, $L_f = 0.3$, $q_2 = 22.92°$: $q_3 = -63.39°$), wrist $\sin q_5 = 0$ (axes 4 and 6 aligned; rank 5). [checked]

10b. Puma 560 with the course table (W3 practical `puma560_fk_proximal_dh.py`)

ALTERNATIVE APPROACHES

A. ETS: each row expanded

${}^0\mathbf{T}_E = \prod_{i=1}^{6} \mathbf{R}_x(\alpha_{i-1})\,t_x(a_{i-1})\,\mathbf{R}_z(q_i)\,t_z(d_i)$, with the numbers of the table; no tool offset, $\{E\} = \{6\}$.

B. Proximal DH (metres, radians; $\theta_i = q_i$)

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	0	−90°	0	q_2
3	0.4318	0	0.15005	q_3
4	−0.0203	−90°	0.4318	q_4
5	0	90°	0	q_5
6	0	−90°	0	q_6

ROBOTS

FK and IK with the course table [added]

$$ {}^0\mathbf{T}_E(\mathbf{q}=\mathbf{0}) = \begin{bmatrix} 1 & 0 & 0 & 0.4115 \\ 0 & -1 & 0 & 0.1501 \\ 0 & 0 & -1 & -0.4318 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0\mathbf{T}_E(0, 45^\circ, -45^\circ, 0, 45^\circ, 0) = \begin{bmatrix} 0.7071 & 0 & -0.7071 & 0.2850 \\ 0 & -1 & 0 & 0.1501 \\ -0.7071 & 0 & -0.7071 & -0.7371 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ \mathbf{p}_W = \begin{bmatrix} c_1(a_2c_2 + a_3c_{23} - d_4s_{23}) - d_3s_1 \\ s_1(a_2c_2 + a_3c_{23} - d_4s_{23}) + d_3c_1 \\ -a_2s_2 - a_3s_{23} - d_4c_{23} \end{bmatrix}, \qquad \mathbf{p}_W = \text{origin of } \{4\} = \{5\} = \{6\} $$

SYMBOLS

a_2, a_3	table entries $a_2 = 0.4318$, $a_3 = -0.0203$	m
d_3, d_4	table entries $d_3 = 0.15005$, $d_4 = 0.4318$	m
$\mathbf{p}_W$	wrist centre, depends on q_1, q_2, q_3 only	m

Use: $\mathbf{p}_W$ does not depend on q_4, q_5, q_6 (the three wrist axes meet at one point). Check: $\mathbf{q} = \mathbf{0}$ puts the tool at $(0.4115, 0.1501, -0.4318)$ m pointing down. [checked]

STEPS IK of the course-table Puma 560 (derived [added])

1. $K = a_2^2 + a_3^2 + d_3^2 + d_4^2$; $|\mathbf{p}_W|^2 = K + 2a_2(a_3\cos q_3 - d_4 \sin q_3)$, so $A\cos q_3 + B \sin q_3 = C$ with $A = 2a_2a_3$, $B = -2a_2d_4$, $C = |\mathbf{p}_W|^2 - K$. Then $q_3 = \text{atan2}\left(C, \pm\sqrt{A^2 + B^2 - C^2}\right) - \text{atan2}(A, B)$ (the W3 trigonometric tool): two values.
2. $\rho = \pm\sqrt{x_W^2 + y_W^2 - d_3^2}$ (the offset d_3 separates the "left" and "right" shoulder), $q_1 = \text{atan2}(y_W, x_W) - \text{atan2}(d_3, \rho)$.
3. $m = a_2 + a_3\cos q_3 - d_4 \sin q_3$, $n = a_3 \sin q_3 + d_4 \cos q_3$, $q_2 = \text{atan2}(-z_W, \rho) - \text{atan2}(n, m)$.
4. Wrist: ${}^3\mathbf{R}_6 = \mathbf{R}_y(q_4)\mathbf{R}_z(q_5)\mathbf{R}_y(q_6)\,\mathbf{R}_x(-90^\circ)$ for this table. Compute ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top {}^0\mathbf{R}_E$ and $\mathbf{M} = {}^3\mathbf{R}_6\,\mathbf{R}_x(90^\circ)$; then $\sin q_5 = \pm\sqrt{M_{12}^2 + M_{32}^2}$, $q_5 = \text{atan2}(\sin q_5, M_{22})$, $q_4 = \text{atan2}\left(M_{32}/\sin q_5,\ -M_{12}/\sin q_5\right)$, $q_6 = \text{atan2}\left(M_{23}/\sin q_5,\ M_{21}/\sin q_5\right)$.
5. $2 \times 2 \times 2 = 8$ solutions (two q_3, two shoulders, two wrist signs). Example $\mathbf{q} = (0.4, 0.6, -0.5, 0.5, 0.9, -0.7)$ rad: all 8 reproduce the whole pose. The wrist centre lies in the shell $0.150 \le |\mathbf{p}_W| \le 0.877$ m about the origin of $\{0\}$. [checked]

SLIDE CHECK W3 practical files: two different Puma tables

`puma560_fk_proximal_dh.py` (table above) and `puma560_fk_distal_ets.py` (standard rows $(\theta, d, a, \alpha) = (q_1, 0, 0, 90^\circ), (q_2, 0, 0.4318, 0), (q_3, 0.15005, 0.0203, -90^\circ), \ldots$) do *not* give the same pose: at $\mathbf{q} = \mathbf{0}$ the first gives $(0.4115, 0.1501, -0.4318)$ m with the tool pointing down, the second $(0.4521, -0.1500, 0.4318)$ m with the tool axes parallel to the base. The sign of a_3 and of α_1 differ, so the second is not the first rewritten in distal form. Never assume a table: use the one in the question. [checked]

11. Franka Panda: seven revolute joints (redundant)



[W02 lec p. 10] — Panda from Franka Emika (photo, W2 p. 10)

Joints	7R; $n = 7$ (one more than the six needed for a pose)
Variables	$q_1, \ldots, q_7$ (rad)
Parameters	table below (m); flange offset 0.107 m
Workspace	thick spherical region about the shoulder point $(0, 0, 0.333)$; sampled flange distance 0.02 to 0.86 m
IK	∞: a 1-D family (self-motion) for every reachable pose
Singular	rank $\mathbf{J} < 6$ on special sets (below)

ALTERNATIVE APPROACHES

A. ETS: the same rows, expanded

${}^0\mathbf{T}_E = \prod_{i=1}^{7} \mathbf{R}_x(\alpha_{i-1})\,t_x(a_{i-1})\,\mathbf{R}_z(q_i)\,t_z(d_i) \cdot t_z(0.107)$ (flange).

The notebook's printed Panda ETS is a different arm (see W2 part, slide check).

B. Franka's published proximal (Craig) DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0.333	q_1
2	0	-90°	0	q_2
3	0	90°	0.316	q_3
4	0.0825	90°	0	q_4
5	-0.0825	-90°	0.384	q_5
6	0	90°	0	q_6
7	0.088	90°	0	q_7

a_{i-1}, d_i in m; flange: $t_z(0.107)$.

FK, Jacobian and what is special [W2 p. 33; W3 slides 40–52]

$$ {}^0\mathbf{T}_E = \left[\prod_{i=1}^{7} {}^{i-1}\mathbf{T}_i(q_i)\right] t_z(0.107), \qquad {}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})\,t_x(a_{i-1})\,\mathbf{R}_z(q_i)\,t_z(d_i) $$

$$ {}^0\mathbf{T}_E(\mathbf{q}=\mathbf{0}) = \begin{bmatrix} 1 & 0 & 0 & 0.088 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0.926 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

$\mathbf{z}_j$, $\mathbf{o}_j$	axis (third column of ${}^0\mathbf{T}_j$) and origin of proximal frame $\{j\}$, in $\{0\}$	–, m
$\mathbf{p}_E$	flange position (last column of ${}^0\mathbf{T}_E$)	m

Use: all joints revolute, so the Jacobian columns are ${}^0\mathbf{J}_{\cdot,j} = \left[\mathbf{z}_j \times (\mathbf{p}_E - \mathbf{o}_j);\ \mathbf{z}_j\right]$, $j = 1 \ldots 7$, a 6×7 matrix (fat: more joints than pose components). The flange points down at $\mathbf{q} = \mathbf{0}$. Matches the toolbox Panda flange. [checked]

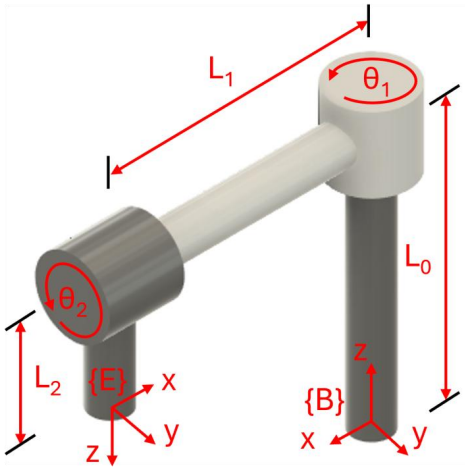
ROBOTS

Jacobian at $\mathbf{q} = 0$ (whole 6×7 matrix) and singular poses [added]

$$
{}^{0}\mathbf{J}(\mathbf{0}) = \begin{bmatrix}
0 & 0.5930 & 0 & -0.2770 & 0 & 0.1070 & 0 \\
0.0880 & 0 & 0.0880 & 0 & 0.0880 & 0 & 0 \\
0 & -0.0880 & 0 & 0.0055 & 0 & 0.0880 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & -1 & 0 & -1 & 0 \\
1 & 0 & 1 & 0 & 1 & 0 & -1
\end{bmatrix}
$$

Use: rank 5 here: the arm is straight up, all joint axes lie in the yz-plane, so the ω_x row is zero. In general $\operatorname{rank}\mathbf{J} = 6$ *(generic poses, 30 random tests) and there is a one-dimensional null space* $\mathbf{n}$ *with* $\mathbf{Jn} = \mathbf{0}$ ***(self-motion**: the elbow swivels while the flange stands still), so an exact IK has* ∞ *solutions: pick one by* $\dot{\mathbf{q}} = \mathbf{J}^{+}\boldsymbol{\nu} + (\mathbf{I} - \mathbf{J}^{+}\mathbf{J})\dot{\mathbf{q}}_0$ *(W3 pseudo-inverse, W6). Singular* ($\operatorname{rank}\mathbf{J} = 5$)*: $q_2 = q_4 = 0$ (any other joints), and every pose with the wrist centre (origin of $\{5\}$) on the joint-1 axis (shoulder singularity); a stretched elbow alone is not singular because the flange offset gives the wrist room to compensate. No closed-form IK is in the unit: numerical (W3 pp. 40–52).* [checked]

12. Furuta (rotary) pendulum: RR in 3D



[W02 lec p. 17] — Furuta pendulum: post L_0, arm L_1 turned θ_1 about z, pendulum L_2 turned θ_2 about the arm

Joints	RR (first axis vertical, second along the arm); $n = 2$
Variables	θ_1 about z_B; θ_2 about the arm's own x
Parameters	L_0, L_1, L_2
Workspace	tip on the sphere $x^2+y^2+(z-L_0)^2 = L_1^2+L_2^2$, band $\lvert z-L_0\rvert \le L_2 \le 2$; none unless the point is on the sphere
IK	
Singular	$\sin\theta_2 = 0$ (pendulum vertical)

ALTERNATIVE APPROACHES

A. ETS [W2 slide 19]

$$
{}^{B}\mathbf{T}_E = t_z(L_0)\,\mathbf{R}_z(\theta_1)\,t_x(L_1) \cdot \mathbf{R}_x(\theta_2)\,t_z(-L_2)\,\mathbf{R}_y(\pi)
$$

B. Proximal DH [W2 slide 34]

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_0	q_1+90°
2	0	90°	L_1	q_2
3	0	90°	L_2	0

plus ${}^{3}\mathbf{T}_E = \mathbf{R}_z(-90°)$.

FK: ${}^{B}\mathbf{T}_E$ of the Furuta pendulum [W2 slides 17–19]

$$
{}^{B}\mathbf{T}_E = \begin{bmatrix}
-c_1 & -s_1c_2 & -s_1s_2 & L_1c_1 - L_2s_1s_2 \\
-s_1 & c_1c_2 & c_1s_2 & L_1s_1 + L_2c_1s_2 \\
0 & s_2 & -c_2 & L_0 - L_2c_2 \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

SYMBOLS

L_0	post height	m
L_1	horizontal arm length	m
L_2	pendulum length (along $-z_E$)	m
θ_1, θ_2	arm angle about the post, pendulum angle about the arm	rad

Use: the table's ${}^{0}\mathbf{T}_3$ *differs from this matrix only by the constant* ${}^{3}\mathbf{T}_E$ *([W2 slide 32]: "{3} is not {E}"). Check:* $\boldsymbol{\theta} = \mathbf{0}$ *gives tip* $(L_1, 0, L_0 - L_2)$ *and* $z_E = -z_B$*;* $L = (0.5, 0.4, 0.3)$*,* $\boldsymbol{\theta} = (60°, 45°)$ *gives* $(0.0163,\ 0.4525,\ 0.2879)$ *m.* [checked]

IK, both branches [added]

$$
\cos\theta_2 = \frac{L_0 - z}{L_2}, \qquad \theta_2 = \operatorname{atan2}\left(\pm\sqrt{1-\cos^2\theta_2}, \cos\theta_2\right)
$$

$$
\theta_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2\sin\theta_2,\ L_1)
$$

Use: the tip (x, y) *is the vector* $(L_1,\ L_2\sin\theta_2)$ *turned by* θ_1*. A point is reachable only if it lies on the sphere* $x^2 + y^2 + (z - L_0)^2 = L_1^2 + L_2^2$ *(two unknowns cannot reach a 3-D region). Example: tip* $(0.0163, 0.4525, 0.2879)$*:* $(\theta_1, \theta_2) = (60°, 45°)$ *or* $(115.88°, -45°)$*.* [checked]

Jacobian ${}^{B}\mathbf{J}$ [added]

$$
{}^{B}\mathbf{J} = \begin{bmatrix}
-L_1s_1 - L_2c_1s_2 & -L_2s_1c_2 \\
L_1c_1 - L_2s_1s_2 & L_2c_1c_2 \\
0 & L_2s_2 \\
0 & c_1 \\
0 & s_1 \\
1 & 0
\end{bmatrix}, \qquad \left|\mathbf{J}_{v,1} \times \mathbf{J}_{v,2}\right|^2 = L_2^2(L_1^2 + L_2^2)\sin^2\theta_2
$$

Use: angular part: $\boldsymbol{\omega} = \dot{\theta}_1\hat{z}_B + \dot{\theta}_2\hat{x}_1$ *with* $\hat{x}_1 = (c_1, s_1, 0)$*. The two linear columns are parallel iff* $\sin\theta_2 = 0$ *(pendulum hanging straight down or standing straight up): then both are horizontal and tangential, so the radial and vertical directions are lost.* [checked]

Mobile robots at a glance

Robot	wheels	holonomic?	δ_m	δ_s	δ_M	ICR (in $\{C\}$)	treated in
differential drive	2 fixed standard (driven, at $\pm a$) + castor	no	2	0	2	on the axle line: $y_{ICR} = u/\dot\theta = a\,\dfrac{\dot\varphi_1+\dot\varphi_2}{\dot\varphi_2-\dot\varphi_1}$; $v=0$	W10 p. 122 (slides 43–47, 61)
unicycle model (added)	kinematic model of a diff. drive: inputs $u,\ \omega=\dot\theta$	no	2	0	2	$R = u/\omega$ on the axle line; $(\dot x,\dot y,\dot\theta) = (u\cos\theta, u\sin\theta, \omega)$	W10 p. 122
bicycle	rear fixed (driven) + front steered wheel at $(L,0)$	no	1	1	2	on the rear-axle line at $R = L/\tan\beta$; $\dot\theta = u\tan\beta/L$	W10 p. 122 (slide 69)
car (Ackermann)	2 rear fixed + 2 front steered	no	1	1	2	rear-axle line; $\tan\beta_{in} = \dfrac{L}{R-t/2}$, $\tan\beta_{out} = \dfrac{L}{R+t/2}$	W10 p. 122 (slides 59–61)
tricycle	2 rear fixed + 1 front wheel driven and steered at $(b,0)$	no	1	1	2	rear-axle line, $R = b/\tan\beta_1$; $\dot\varphi_1 = \frac{1}{r}\sqrt{u^2+b^2\dot\theta^2}$, $\beta_1 = \operatorname{atan2}(b\dot\theta, u)$	W10 p. 122 (slides 56–58)
3-wheel omni (Swedish triangle)	3 omni wheels at 120°, distance d from C	yes	3	0	3	anywhere: $(-v/\dot\theta,\ u/\dot\theta)$ for any $(u,v,\dot\theta)$	W10 p. 122 (slides 53–54, W13 Q10)
4-wheel mecanum / Swedish	4 wheels, roller angles $\gamma_i = \pm 45°$ at $(\pm\ell, \pm w)$	yes	3	0	3	anywhere (rank 3, 4 wheels for 3 unknowns: least squares)	W10 p. 122 (slide 55)
skid steer	4 standard wheels, left pair and right pair	no (slide model: $v=0$)	2 (model)	0	2	as the diff. drive; but 4 no-slide rows have rank 2 (also $\dot\theta=0$): turning needs slip	W10 p. 122 (slides 48–52)

Rules (W10). $\delta_M = \delta_m + \delta_s$; $\delta_m = 3 - \operatorname{rank}\mathbf{C}_1$ ($\mathbf{C}_1$ = no-slide rows of the standard wheels); a standard wheel gives a no-sliding row ⇒ non-holonomic; no standard wheel (omni, mecanum) ⇒ holonomic, $\delta_m = 3$. Wheel rows and every matrix are on p. 122. Table values follow the δ_m, δ_s table of slide 64 and the derived rows of the W10 part. [added]

What W1 asks, and where to look [added]

W1 is *counting and definitions*: no frames, no matrices. Every number comes from the Grübler–Kutzbach (G–K) count. Write $\{0\}$ for the ground (link 0).

If the question says…

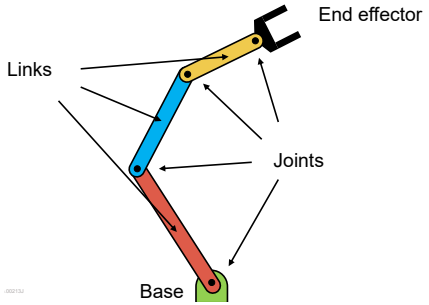
"find the mobility / DoF of this mechanism" ..... recipe, p. 28
"how many actuators / motors does it need" ..... *M* (slide p.39); p. 27
"planar parallel manipulator", "3-RPR" ..... W13 Q6, p. 32
"delta", "Stewart", "spatial parallel" ..... p. 29, W13 Q7 p. 32
"G–K gives 0 but it moves", "anything odd?" ..... p. 30
"fully / under-actuated / redundant" ..... p. 27
"reachable vs dexterous workspace" ..... p. 30
"serial vs parallel; open vs closed" ..... p. 26
"name the joint, give its DoF" ..... p. 27

Kinematics and dynamics [p. 18–19]

term	meaning (slide wording)
Kinematics [p. 18]	study of motion *without* the forces that cause it: only the geometry of motion (Ampère, 1834).
Dynamics [p. 19]	study of the forces required to cause motion: why objects move or do not.
Statics	dynamics of bodies *at rest / in equilibrium* (no acceleration).
Kinetics	dynamics of bodies *under acceleration*.

Use: definition questions: do not swap "kinetics" with "kinematics". In this unit: kinematics = W1–W6 (and W10), statics = $\mathbf{J}^\top$ forces (W6), kinetics = Lagrange / Newton–Euler (W7–W9) [added].

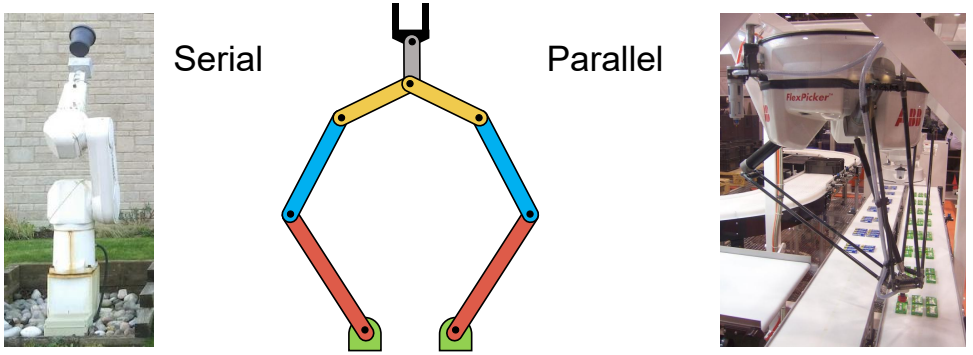
Mechanism, kinematic chain, link, joint [p. 21–23]



[W01 lec p. 22] — a kinematic chain: base, links (three bars), joints (four pins), end effector (gripper)

term	meaning
Mechanism [p. 21]	"a mechanical system that transfers/modifies motion and/or forces from one or more sources to one or more outputs".
Kinematic chain [p. 22]	"a system of rigid bodies connected by joints". A mechanism is a collection of one or more chains connecting the base to the end effector.
Link [p. 23]	a rigid body (a bar, a platform, a cylinder or a rod).
Joint	a connection that allows *relative* motion between two links.
Base	the link rigidly fixed to the ground: link 0.
End effector	the link that interacts with the world (gripper, tool).

Serial vs parallel, open vs closed chains [p. 24–26]



[W01 lec p. 24] — left: serial arm; centre: closed-chain gripper linkage; right: ABB FlexPicker, a parallel (delta) robot

	Open / serial	**Closed / parallel**
Definition [p. 25]	chain that is *not* closed; *serial* if every link except the first and last connects to two other links	chain that forms a closed loop; *parallel*: end effector joined to the base by several chains [added]
Loops ℓ [added]	0	≥ 1
Count	$N = J + 1$, so $M = \sum F_i$	$N - J - 1 < 0$ (loops remove DoF)
Examples	industrial arm, PUMA, SCARA	four-bar, gripper linkage, delta, Stewart
Actuators [p. 26]	on the joints, so the motors ride on the arm	can sit on the base: lighter moving parts
Load [p. 26]	one chain carries all	several legs share: larger load (mechanical advantage)
Motion [p. 26]	–	interesting motion paths (Strandbeest, Cyber Tiger)
Trade-off [added]	big workspace, easy FK	small workspace, easy IK, harder FK (W4)

Examples: a 6-joint arm: open and serial. A four-bar: closed and *not* serial (no "first" or "last" link). Two fingers on one palm: open but not serial [added]. Why a delta for fast pick-and-place: base-mounted motors, light rods, shared load.

TRAP
"Parallel" and "closed" overlap but are not synonyms: a four-bar is closed and not a parallel *robot*. The slides define only open, closed and serial (p.25); "parallel" is shown by pictures (p.24).

Degrees of freedom of a rigid body [p. 27–33]

PREMISE DoF of a body **Given:** a body or point and the space it moves in (plane or 3D); link geometry is known
Find: the number of independent parameters that fix its configuration

Degrees of freedom (DoF) [p. 27]
DoF = the number of independent parameters that define the configuration of a mechanism.

SYMBOLS

D	DoF of a free rigid body: $D = 3$ in a plane, $D = 6$ in 3D	–

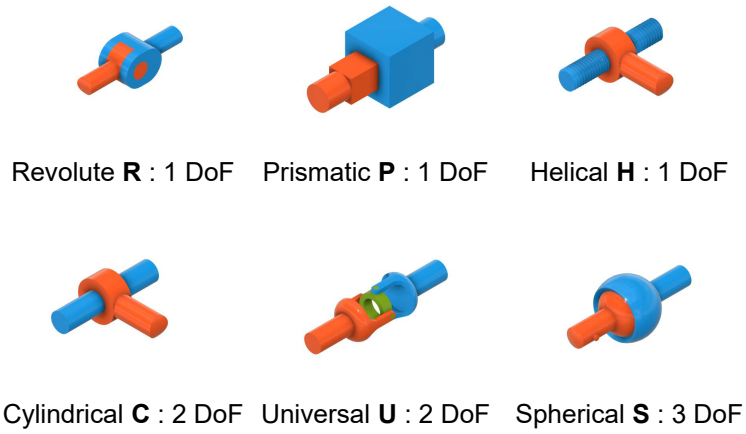
Use: DoF does not depend on the link lengths (notebook): we assume the geometry is known and ask how many numbers fix the pose.

system [p. 28–33]	DoF	parameters
point in 2D [p. 28–29]	2	x, y (a point has no orientation)
pendulum in 2D [p. 30–31]	1	angle θ about the pivot (x, y are tied: $x^2 + y^2 = L^2$)
rigid body in 2D (nb Q1)	3	x, y, θ
rigid body in 3D [p. 32–33]	6	x, y, z and rotations about x, y, z

DERIVATION why 3 and 6 [added]
Plane. A body is fixed by two of its points, $2 \times 2 = 4$ coordinates, but their distance is fixed: $4 - 1 = 3$. *Space.* Three non-collinear points, $3 \times 3 = 9$ coordinates, with three fixed distances: $9 - 3 = 6$. Each independent rigid constraint removes one DoF. *Pendulum:* the point (x, y) has 2 coordinates and one constraint $x^2 + y^2 = L^2$: $2 - 1 = 1$.

TRAP
A rigid body in 2D has 3 DoF, a *point* only 2. Joint DoF F is not the mechanism's mobility M.

Joint types and the constraints they add [p. 34–36]



[W01 lec p. 34] — the six joints on slide 34 (top row R, P, H; bottom row C, U, S)

joint [p. 34]	sym.	F	cuts 3D	cuts 2D	what it allows
Revolute (pin, hinge)	R	1	5	2	one rotation
Prismatic (slider)	P	1	5	2	one translation
Helical (screw)	H	1	5	–	rotation and translation tied by the pitch
Cylindrical	C	2	4	–	rotation about and translation along one axis
Universal	U	2	4	–	two perpendicular rotations
Spherical (ball)	S	3	3	–	three rotations about a point

Added: a planar joint E (3 DoF in 3D) is not on the slides. "Cuts" = $D - F$ is the number of relative freedoms the joint removes (added columns). Planar mechanisms in this unit use only R and P ($F = 1$, cuts 2).

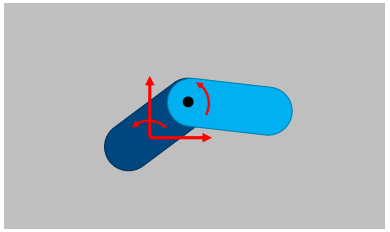
Constraints removed by a joint [p. 35]

$$\text{constraints removed} = D - F$$

SYMBOLS

D	freedoms of one free body: 3 (plane), 6 (3D)	–
F	DoF the joint allows between its two links	–

Use: start with the links free (each D), then every joint takes away $D - F$.



[W01 lec p. 36] — two planar rigid bodies joined by a revolute joint: 4 DoF

Example [p. 36]. Free: $3 + 3 = 6$ DoF, coordinates $(x_1, y_1, \theta_1, x_2, y_2, \theta_2)$. The pin forces one point of body 1 to coincide with one point of body 2: two equations (x and y). $6 - 2 = 4$ DoF: body 1 free (3) plus only the rotation of body 2 (1). [checked]

The same pin on a pendulum: the mass is a point (2 DoF); the revolute joint cuts it to 1.

Controllable vs total DoF; actuation [p. 37–38]

PREMISE actuation **Given:** the number of actuators and the mobility M (total DoF)
Find: fully actuated, under-actuated or redundant; and how many actuators are needed

A DoF is **controllable** if an actuator drives it (a motor on the joint); an unactuated DoF is uncontrollable [p. 37].

controllable vs total [p. 37]	name
controllable = total	**fully actuated**
controllable < total	**under-actuated**: fewer actuators than joints. Low cost, simpler design, lower weight [p. 38]
controllable > total	**redundant** (slide wording)

SLIDE CHECK redundant
Slide 37 says "controllable > total ⇒ redundant", yet notebook cell 8 calls a 7-DoF arm redundant (7 motors, 7 DoF: equal, so "fully actuated" by the slide's rule). The standard meaning [added]is *kinematic redundancy*: more joint DoF than the task needs (7 > 6 for a 3D pose; 3 > 2 for a planar 3R placing a point). In an answer state which meaning you use.

example	motors	M	name
6R arm [added]	6	6	fully actuated
same arm, 4 motors	4	6	under-actuated
planar 3-RPR, 3 prismatic motors	3	3	fully actuated
planar 4-RPR, 4 prismatic motors [added]	4	3	redundant (slide wording)
Stewart, 6 linear motors	6	6	fully actuated
delta, 3 base motors	3	3 (not 9)	fully actuated

TRAP
Using the plain G–K value 9 for a delta would make it "under-actuated"; its real mobility is 3 (p. 29). [checked]

Mobility: the Grübler–Kutzbach equation [p. 39–41]

PREMISE mobility **Given:** a drawing of the mechanism: every link (ground included), every joint with its type, and whether the motion is planar or spatial
Find: the mobility M (= DoF = the minimum number of actuators [p. 39])

G–K, first form [p. 40]

$$M_{3D} = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - P_5, \qquad M_{2D} = 3(N-1) - 2P_1 - P_2$$

SYMBOLS

M	mobility (DoF) of the mechanism	–
N	number of links, *including* the grounded link	–
P_i	number of joints (link pairs) with i DoF	–

Use: quick count when every joint is R or P (planar: P_1 only). The coefficient of P_i is $D - i$, the constraints one such joint removes.

G–K, second form [p. 41]: used in W13 Q6, Q7

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i = D(N-1) - DJ + \sum_{i=1}^{J} F_i$$

SYMBOLS

M	mobility / DoF of the mechanism	–
D	3 for a planar mechanism, 6 for a spatial one	–
N	number of links, ground included	–
J	number of joints	–
F_i	DoF of joint i ($F = 1$ for R, P, H; 2 for C, U; 3 for S)	–

Use: the general form: needs one F_i per joint, so it handles C, U, S. Equivalent to the textbook form $M = m(N - 1 - J) + \sum f_i$ (Lynch & Park, ch. 2) [added].

DERIVATION of form 2 and its link to form 1 [p. 41]
The $N-1$ moving links are free: $D(N-1)$ freedoms. Each joint removes the D relative freedoms and gives back the F_i it allows, so it nets $-(D - F_i)$:

$$M = D(N-1) - \sum_{i=1}^{J}(D - F_i) = D(N-1) - DJ + \sum_{i=1}^{J} F_i = D(N - J - 1) + \sum F_i.$$

Group the joints by their DoF i: $\sum_i (D - F_i) = \sum_i (D - i)P_i$, which is form 1 ($D - i = 5, 4, 3, 2, 1$ in 3D; 2, 1 in 2D). [checked]

DERIVATION loop view: $M = \sum F_i - D\,\ell$ [added]
Let $\ell = J - N + 1$ be the number of independent loops ($\ell = 0$ for an open chain). Then $N - J - 1 = -\ell$ and

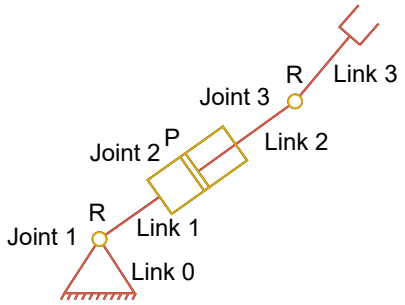
$$M = \sum_{i=1}^{J} F_i - D\,\ell.$$

Each loop removes D. Serial: $M = \sum F_i$. Four-bar: $4 - 3 \cdot 1 = 1$. 3-RPR: $9 - 3 \cdot 2 = 3$. Stewart: $36 - 6 \cdot 5 = 6$. [checked]

STEPS Mobility count (W13 Q6, Q7)
1. D: all joint axes perpendicular to the page ⇒ $D = 3$. Any S, U, C, or "still 3D" ⇒ $D = 6$.
2. **Links:** number them, **ground** = 0 (all ground pivots are *one* link). Between two joints in series there is one link: so an R–P–R leg has two links (cylinder, rod).
3. **Joints:** list each with its F; get J and $\sum F_i$.
4. Write $M = D(N - J - 1) + \sum F_i$, substitute, show the line. Cross-check with form 1 (planar).
5. **Check:** $M \geq 1$ for a machine; for a serial chain $M = \sum F_i$; if $M \leq 0$ ask whether constraints are dependent (p. 30).

Worked examples from the slides [p. 42–47]

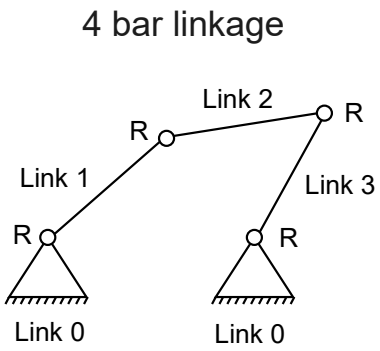
Planar serial manipulator RPR [p. 42–43]



[W01 lec p. 42] — planar serial RPR: links 0 to 3, joints R, P, R

WORKED RPR, $D = 3$ planar
links 0 ground, 1 (to the cylinder), 2 (rod), 3 (tool link): $N = 4$
joints J1 R, J2 P, J3 R: all $F = 1$, $J = 3$, $P_1 = 3$, $P_2 = 0$
Form 1: $M = 3(4-1) - 2 \cdot 3 - 1 \cdot 0 = 3$. Form 2: $M = 3(4 - 3 - 1) + (1 + 1 + 1) = 3$. [checked]
Serial: $M = \sum F_i = 3$: three motors set the tip (x, y) and its angle.

Planar four-bar [p. 44–45]



[W01 lec p. 44] — planar closed kinematic chain: four-bar, links 0 to 3, four R

WORKED four-bar, $D = 3$

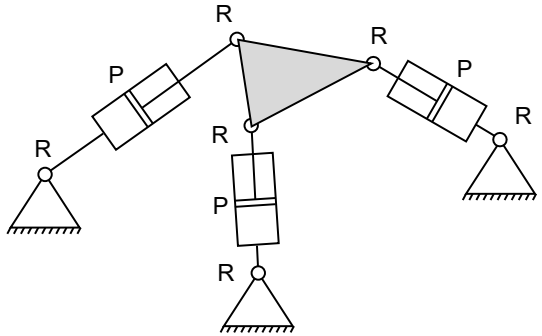
links 0 ground (drawn twice, still **one** link), 1 crank, 2 coupler, 3 rocker: $N = 4$

joints four R: $J = 4$, $P_1 = 4$, $P_2 = 0$

Form 1: $M = 3(4-1) - 2 \cdot 4 - 1 \cdot 0 = 1$. Form 2: $M = 3(4-4-1) + 4 = 1$. [checked]

Not 3: the loop gives 8 constraints on $3 \cdot 3 = 9$ freedoms. One motor on the crank drives it.

Planar parallel manipulator, 3-RPR [p. 46–47]; W13 Q6



[W01 lec p. 46] — planar parallel manipulator: triangular platform on three R–P–R legs

WORKED 3-RPR, $D = 3$

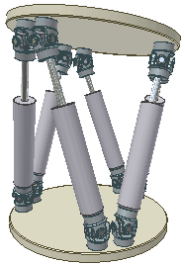
links ground, platform, and per leg cylinder + rod: $N = 1 + 1 + 3 \cdot 2 = 8$

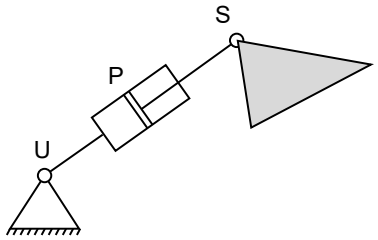
joints per leg R, P, R: $J = 9$, all $F = 1$, $P_1 = 9$, $P_2 = 0$

Form 1: $M = 3(8-1) - 2 \cdot 9 - 1 \cdot 0 = 3$. Form 2: $M = 3(8-9-1) + 9 = 3$. [checked]

Three motors (one prismatic joint per leg) place the platform's x, y and angle.

Spatial mechanisms: Stewart platform [p. 48], delta (W13 Q7)





[W01 lec p. 48] — Stewart platform (left) and one leg: ground–U–P–S–platform (right). Slide asks "what is the mobility of this system?"

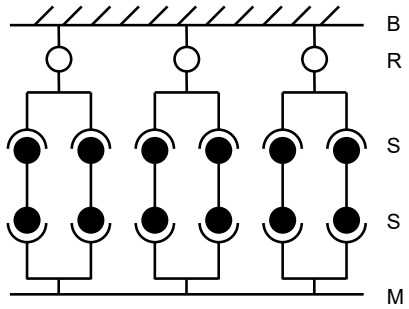
WORKED Stewart platform, $D = 6$ [added]: the slide gives no answer

links ground, platform, per leg cylinder + rod: $N = 1 + 1 + 6 \cdot 2 = 14$

joints per leg U ($F = 2$), P (1), S (3): $J = 18$, $\sum F_i = 6(2 + 1 + 3) = 36$

$$M = 6(14 - 18 - 1) + 36 = -30 + 36 = 6.$$

Six DoF, six actuators: fully actuated. [checked] One leg alone ($N = 4$, $J = 3$): $6(4-3-1) + 6 = 6$, because a serial chain has $M = \sum F_i$; that is the leg's DoF, not the platform's. Six SPS legs would give 12: each leg can spin about its own axis; the U at the base blocks that spin. [checked]



[W13 lec p. 15] — W13 p.15: delta mechanism: base B, one R per leg, two parallelogram rods per leg with S at each end, moving platform M

Extended G–K (W13 p.16; full treatment in W4)

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

SYMBOLS

R_C	number of over-constraints: $R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$ where u_k are the unique DoFs of closed loop k (as printed on the slide)	–
R_M	local mobility: DoF that do not move the output	–

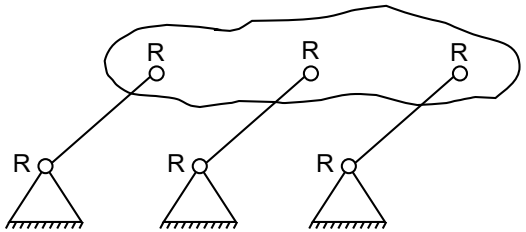
Use: plain G–K over-counts when parts can move without effect (spinning rods) and under-counts when constraints repeat (p. 30).

WORKED delta mechanism, W13 Q7
$D = 6$ ("the mechanism is still 3D despite this depiction", p.14). Links: base, platform, per leg 1 upper arm + 2 rods: $N = 2{+}3{\cdot}3 = 11$. Joints: per leg 1 R + 4 S: $J = 3{\cdot}5 = 15$; $\sum F_i = 3(1{+}4{\cdot}3) = 39$.

$$M = 6(11 - 15 - 1) + 39 = \mathbf{9} \quad \text{(plain)}, \qquad M = 9 + R_C - R_M = 9 + 0 - 6 = \mathbf{3}.$$

$R_M = 6$: each of the six rods has a ball joint at both ends and can spin about its own axis without moving anything else. The platform really translates in x, y, z with 3 motors. [checked]
Write both: plain 9, then "subtract the six rod spins: 3". If the question quotes only the plain equation, 9 is what it asks for.

Where G–K fails [p. 49–51]



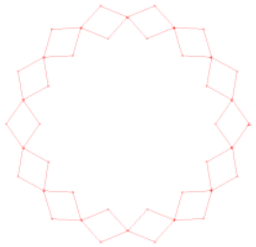
[W01 lec p. 49] — slide 49: a body hung by three equal parallel legs, R at each end. G–K gives 0, "but the system can move: it has 1 degree of freedom!"

WORKED the slide's system, $D = 3$
links ground, three legs, body: $N = 5$
joints six R: $J = 6$, $P_1 = 6$, $P_2 = 0$
$M = 3(5 - 1) - 2 \cdot 6 - 1 \cdot 0 = \mathbf{0}$ (form 2: $3(5 - 6 - 1) + 6 = 0$).
But the body moves on a circular arc without turning: **1 DoF**. Delete one middle leg: $N = 4$, $J = 4$, $M = 3(4 - 1) - 2 \cdot 4 = 1$ (a parallelogram four-bar). [checked]

Why (slide p.49): G–K "assumes that the constraints on motion are independent". One middle link can be removed without changing what the mechanism does: it is *redundant* and should not be counted.

DERIVATION why the count is 0: the closure equations [added]
Unknowns: three leg angles and the body's x, y, θ: 6. Each leg ties the body to the ground: 2 equations, 6 in all. G–K treats all 6 as independent: $6 - 6 = 0$. For equal, parallel legs spaced like the ground pivots, the third leg's two equations repeat what the first two already say: the constraint Jacobian has rank 5, not 6, so $6 - 5 = 1$ free motion. With unequal legs (16 random closed assemblies tested) the rank is 6: the body locks and 0 is right. G–K sees only N, J, F_i, never the lengths, so it cannot tell the two cases apart. [checked]

STEPS "Did you notice anything odd?" (nb Q4, Q5)
1. Do the count and show it: $M = 0$.
2. State what you see: the mechanism moves (1 DoF).
3. Reason (slide words): G–K assumes the constraints are independent; one leg is redundant.
4. Fix: remove one leg, recount: $N = 4$, $J = 4$, $M = 1$.
5. Caveat: it relies on equal, parallel legs; with unequal legs it locks.



[W01 lec p. 51] — Hoberman mechanism (p.51): 24 links, 36 revolute joints

Hoberman [p. 51]: the slide only states "24 links and 36 revolute joints". Counting one link as the frame [added]: $3(24 - 36 - 1) + 36 = -3$ ($D = 3$), $6(24 - 36 - 1) + 36 = -42$ ($D = 6$). Both are negative, yet it expands and contracts. [checked]

TRAP
False: "$M = 0$ (or $M < 0$) means it cannot move" and "G–K always gives the true DoF". It is exact only for independent constraints; it can under-count (p.49) or over-count (delta spins).

Workspace: reachable and dexterous [p. 52–57]

PREMISE workspace **Given:** link lengths and joint limits of the arm
Find: the region the end effector can reach (reachable) and the region it can reach in every orientation (dexterous)

Two factors set the workspace [p. 52]: (1) the **geometry** (link lengths); (2) the **constraints on the joints** (limits on angle or stroke).

Reachable and dexterous workspace [p. 57]

$$W_{\text{Dexterous}} \subseteq W_{\text{Reachable}} \qquad (W_R \supseteq W_D)$$

SYMBOLS

W_R	reachable workspace: points the end effector reaches in *one* orientation	m (region)
W_D	dexterous workspace: points reached in *all* possible orientations	m (region)

Use: answer "$W_R \supseteq W_D$" (nb Q7). $W_R \subseteq W_D$ is backwards; the strict $\subset$ or $\supset$ are traps: the sets can be equal (both empty).

Planar 2R and 3R reach [added]

$$\text{2R}: \ |L_1 - L_2| \le r \le L_1 + L_2 \qquad \text{3R (equal links } L): \ W_R: \ r \le 3L, \quad W_D: \ r \le L$$

SYMBOLS

r	distance of the tool from the base axis	m
L_1, L_2, L	link lengths	m

Use: 2R: fully folded to fully stretched. $L = (0.5, 0.3)$: annulus $0.2 \le r \le 0.8$ m. 3R with $L = 0.3$ m: W_R disc 0.9 m, W_D disc 0.3 m. [checked]

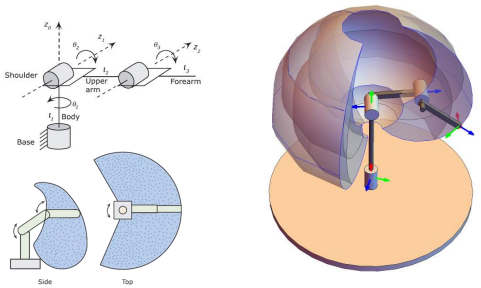
DERIVATION 3R, equal links: why W_D is the disc $r \le L$ [added]
Wrist point $\mathbf{w} = \mathbf{p} - L(\cos\varphi, \sin\varphi)$ for tool angle φ at tool point $\mathbf{p}$. The first two links reach $\mathbf{w}$ iff $0 \le |\mathbf{w}| \le 2L$ (equal links: no hole). Every φ must work: the circle of radius L about $\mathbf{p}$ lies in $|\mathbf{w}| \le 2L$ iff $r + L \le 2L$, i.e. $r \le L$. Reachable: some φ works, $r - L \le 2L$, i.e. $r \le 3L$. [checked]

STEPS Describe or sketch a workspace

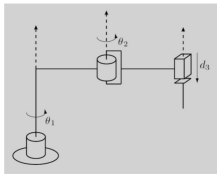
1. Centre the sketch on the first joint axis; name the frame.
2. Fold the arm: smallest tool distance. Stretch it: largest.
3. Sweep each joint that spins the arm about the centre: annulus → ring (plane) or shell (3D); a P adds a stroke (thickness/height).
4. Joint limits cut wedges out of the ring or shell.
5. State W_R and W_D; if the DoF are fewer than the pose (3 planar, 6 spatial), W_D is empty.

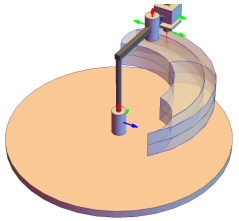
Dexterous is empty for any 2R arm: 2 DoF cannot set a point *and* an angle. At $(0.5, 0.3)$ an $L = (0.5, 0.3)$ arm reaches only the tool angles $90°$ and $-28.07°$. [checked]

Robot configurations and their workspaces [p. 53–56]

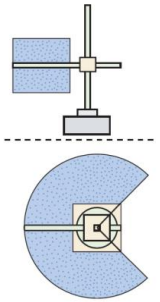


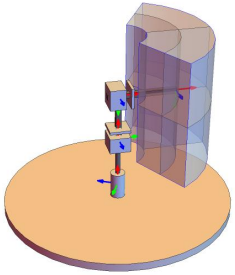
[W01 lec p. 53] — RRR (PUMA): shell



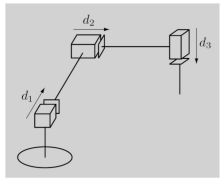


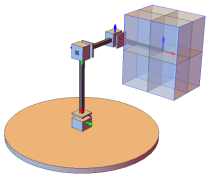
[W01 lec p. 54] — RRP (SCARA): thick ring





[W01 lec p. 55] — RPP (cylindrical): thick-walled sector





[W01 lec p. 56] — PPP (Cartesian): box

arm [p. 53–56]	joints	workspace shape (as drawn; added reading)
PUMA (Programmable Universal Machine for Assembly)	RRR	thick spherical shell about the shoulder; a wedge is cut out by the waist limit
SCARA (Selective Compliant Articulated Robot for Assembly)	RRP	flat ring from the two vertical-axis R joints, made thick by the vertical stroke; a gap from the limits
cylindrical (PlateCrane EX)	RPP	thick-walled cylinder sector: R gives the angle, one P the height, one P the radius range (inner hole = minimum extension)
Cartesian (Yamaha)	PPP	box whose edges are the three strokes
spherical [added]	RRP	thick spherical shell: radius range = stroke of the P

The slide's PUMA is the three-axis RRR (body, shoulder, elbow). Each of these has $M = 6(4 - 3 - 1) + 3 = 3$ in 3D: position only. Figures by A. Becker et al., "Robot Manipulator Workspaces" (as credited on the slides). [checked]

Manipulation problems: what comes next [p. 58–62]

problem [p. 59]	meaning	week
forward kinematics	joint coordinates → end-effector pose	W2
inverse kinematics	tool pose → joint coordinates	W3
velocity kinematics	joint rates ↔ tool velocity	W6
dynamics	torque to accelerate the tip; force at the tip	W7–W9
trajectory generation	path in joint or tool space	W5

Pose = position + orientation [p. 60]: 3 + 3 numbers, written as a 4×4 homogeneous matrix [p. 62]. Frames [p. 61]: {B} base, {W} wrist, {T} tool, {S} station, {G} goal, {C} camera. A pose has 6 numbers, so a 6-DoF arm is the minimum for arbitrary tool poses [added].

Simple scenarios: classify, count, actuate [added]

G–K value	reading
$M \geq 1$	a mechanism, M motors (if constraints are independent)
$M = 0$	a rigid structure, *or* dependent constraints (p.49): check the geometry
$M < 0$	over-constrained, *or* dependent constraints (Hoberman): recount first

Task DoF vs M ("how many actuators does it need?"): let m be the number the task needs: $m = 2$ (point in a plane), 3 (planar pose), 3 (point in 3D), 6 (3D pose). $M < m$: cannot do the task (2R cannot set a tool angle). $M = m$: exactly enough, fully actuated. $M > m$: kinematically redundant, $M - m$ spare DoF (7R for a 6-DoF pose).

Examples (all [checked]): planar 3R positioning a point: $M = 3$, $m = 2$: one spare DoF. Delta: $M = 3$, $m = 3$ (translation): 3 motors. 3-RPR: $M = 3$, $m = 3$ (planar pose): 3 motors. 3-RPS platform ($D = 6$, $N = 8$, $J = 9$, $\sum F = 15$): $6(8 - 9 - 1) + 15 = 3$.

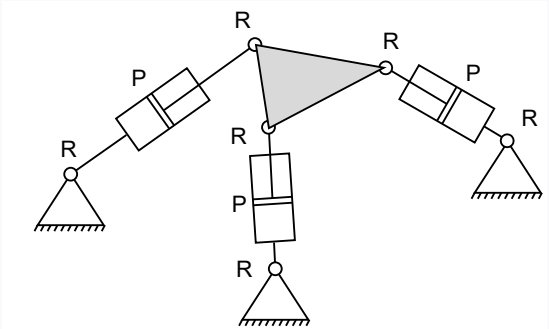
Notebook exercises Q1–Q7, solved

Q	answer
1	Rigid body in 2D: **3** DoF (two translations, one rotation); a point: 2 (no orientation).
2	Four-bar: $N = 4$, $P_1 = 4$: $3(4 - 1) - 2 \cdot 4 = 1$.
3	Planar parallel: $N = 8$, $P_1 = 9$: $3(8 - 1) - 2 \cdot 9 = 3$.
4	"Planar mechanism shown" (no answer in the notebook; figure assumed the p.49 system): $N = 5$, $P_1 = 6$: 0.
5	"Anything odd?": the count is 0 but it moves (1 DoF): dependent constraints, p. 30.
6	Workspaces: table of robot configurations (PUMA shell, SCARA ring, cylindrical sector, Cartesian box).
7	$W_R \supseteq W_D$ (option 3).

How this is examined: worked solutions

Marks are not published; indicative, about 1 mark per step. Marks come from: the right D; N with the ground; every F_i listed; the substitution shown; the answer; one sentence on why plain G–K and the real DoF may differ.

EXAM-STYLE QUESTION W13 Q6: mobility of a planar parallel manipulator [marks not published]



[W13 lec p. 13] — W13 pp.12–13: the same figure; on p.12 the lecturer numbers the links 1 ground, 2 platform, 3–4 left leg, 5–6 middle leg, 7–8 right leg

Find the mobility of the mechanism drawn (planar, three R–P–R legs).

Solution

$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$ (G–K). Planar, so $D = 3$.
Links: 1 ground, 2 platform, and each prismatic joint joins a cylinder to a rod (two links per leg: 3,4; 5,6; 7,8): $N = 8$.
Joints: per leg R, P, R: $J = 9$, each $F_i = 1$, $\sum F_i = 9$.

$$M = 3(8 - 9 - 1) + 9 = -6 + 9 = \boxed{3} \qquad \text{(form 1: } 3(8-1) - 2 \cdot 9 = 3\text{)}$$

Three DoF: platform x, y and rotation, driven by the three prismatic actuators. Check: $N = 7$ (ground forgotten) would give 0. [checked]

EXAM-STYLE QUESTION W13 Q7: mobility of a delta mechanism [marks not published]

A delta robot has a base, a moving platform and three legs. Each leg: one R at the base driving an upper arm, then two rods forming a parallelogram, each rod with a spherical joint at both ends. Find its mobility with (a) plain G–K, (b) the extended equation, given $R_C = 0$, $R_M = 6$.

Solution

3D, $D = 6$. $N = 2 + 3 \cdot 3 = 11$ (base, platform, 3 upper arms, 6 rods). $J = 3 \cdot (1 + 4) = 15$; $\sum F_i = 3(1 + 4 \cdot 3) = 39$.
(a) $M = 6(11 - 15 - 1) + 39 = \boxed{9}$.
(b) $M = 9 + 0 - 6 = \boxed{3}$. Six rods can spin about their own axes (S at both ends): $R_M = 6$. The platform translates in x, y, z; 3 motors. Fully actuated. [checked]

EXAM-STYLE QUESTION Derive and count a Watt six-bar [marks assumed 6]

(a) Derive $M = D(N - J - 1) + \sum F_i$. (b) A planar Watt six-bar has 6 links (ground included) and 7 revolute joints: find M. (c) Which slip gives -5?

Solution

(a) $N - 1$ moving links, free: $D(N - 1)$. Joint i allows F_i relative freedoms, so removes $D - F_i$. $M = D(N-1) - \sum(D - F_i) = D(N-1) - DJ + \sum F_i = D(N - J - 1) + \sum F_i$.
(b) $D = 3$, $N = 6$, $J = 7$, $\sum F_i = 7$: $M = 3(6 - 7 - 1) + 7 = \boxed{1}$. Form 1: $3(6-1) - 2 \cdot 7 = 1$.
(c) $D = 6$ on a planar drawing: $6(6 - 7 - 1) + 7 = -5$. (No ground, $N = 5$: -2.) [checked]

EXAM-STYLE QUESTION Odd mechanism and actuation (Mock 1 Q10 style) [marks assumed 4]

(a) Define fully actuated, under-actuated, redundant. (b) A body is hung from the ground by three equal parallel links (R at both ends). G–K gives $M = 0$ but it moves. Explain.

Solution

(a) By the slide: controllable DoF =, <, > total DoF. Example: 6R arm, 6 motors: fully; 4 motors: under (cheaper, lighter); 4-RPR platform ($M = 3$) with 4 motors: "redundant" by the slide's wording (kinematic redundancy: 7R arm for a 6-DoF pose; say which).
(b) $N = 5$, $J = 6$: $M = 3(5 - 1) - 2 \cdot 6 = \boxed{0}$. G–K assumes independent constraints; with equal parallel links the third repeats the others, one leg is redundant. Remove it: $N = 4$, $J = 4$, $M = \boxed{1}$ (the true value). [checked]

EXAM-STYLE QUESTION Workspace of a planar 3R arm [marks assumed 6]

A planar 3R arm has $L_1 = 0.5$, $L_2 = 0.3$, $L_3 = 0.2$ m, no joint limits. Find the reachable and dexterous workspaces.

Solution

Tool at distance r with direction φ: wrist $\mathbf{w} = \mathbf{p} - L_3(\cos\varphi, \sin\varphi)$ must lie in the 2R annulus $h \le |\mathbf{w}| \le H$, $h = |L_1 - L_2| = 0.2$, $H = L_1 + L_2 = 0.8$.
Reachable (some φ): $|\mathbf{w}|$ ranges over $[\,|r - L_3|,\ r + L_3\,]$; need it to meet $[h, H]$: $r \le H + L_3 = \boxed{1.0}$ m, a full disc.
Dexterous (every φ): the whole interval inside $[h, H]$: $r + L_3 \le H \Rightarrow r \le 0.6$; $|r - L_3| \ge h \Rightarrow r \ge L_3 + h = 0.4$ (the base point also works). Ring $\boxed{0.4 \le r \le 0.6}$ m. Sanity: equal links give $h = 0$ and the slides' $r \le L$. [checked]

EXAM-STYLE QUESTION Spot the errors [marks assumed 4]

"Four-bar: $D = 6$, $N = 4$, $J = 4$, so $M = -2$: a negative mobility means it cannot be built. Delta, 3 motors: $M = 9$, so under-actuated." Find the errors.

Solution

(1) Planar: $D = 3$, $M = 3(4 - 4 - 1) + 4 = 1$. (2) A negative or zero value does not prove a mechanism cannot move (p.49, Hoberman): it warns of dependent constraints. (3) 9 is the plain value; the delta's real $M = 3$ ($R_M = 6$), so 3 motors make it fully actuated. [checked]

What forward kinematics asks [p. 5, 10–11]

PREMISE forward kinematics **Given:** the joint variables $\mathbf{q} = (q_1, \ldots, q_n)^\top$ (angle θ_i in rad for a revolute joint, distance d_i in m for a prismatic joint) and the fixed link dimensions
Find: the pose of the end effector (tool) frame $\{E\}$ in the base frame $\{0\}$: position $({}^0x, {}^0y, {}^0z)$ and orientation ${}^0\mathbf{R}_E$, together as ${}^0\mathbf{T}_E(\mathbf{q})$

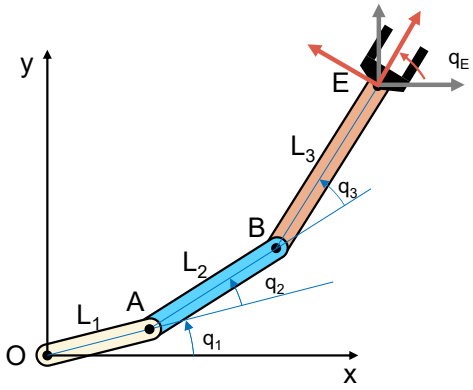
Forward kinematics is a *function* $\xi_n = \mathcal{K}(\mathbf{q})$: each joint vector gives exactly one pose. (The opposite question, W3, can have zero, one or many answers.) Three ways to write that function are on the slides, and this page gives all three:

1. trigonometry on the picture (planar arms only, [p. 6–9]);
2. the **elementary transform sequence (ETS)**: walk base → tool and multiply the moves ([p. 15–19]);
3. the **proximal (modified, Craig) DH table**: four numbers per joint ([p. 20–36]).

STEPS Answer layout for any FK question (marks come from method)

1. Draw the arm in its zero pose. Name the frames $\{S\}/\{0\}$, $\{E\}/\{T\}$, mark each joint axis and the *drawn* positive direction of each angle.
2. State the general rule you use: ${}^0\mathbf{T}_E = {}^0\mathbf{T}_1\,{}^1\mathbf{T}_2 \cdots {}^{n-1}\mathbf{T}_E$ (inner indices cancel).
3. Write the ETS (or DH table), one line per joint, then the matrix of every intermediate transform if the question says "indicate intermediate frames".
4. Multiply (at least the position column, then the rotation if asked). Keep symbols; put numbers in last, with units.
5. Box ${}^0\mathbf{T}_E$. Sanity check at $\mathbf{q} = \mathbf{0}$ against the drawing (and at one more pose with non-zero angles).

Planar RRR by trigonometry [p. 6–9]



[W02 lec p. 9] — planar RRR arm: joint angles q_1, q_2, q_3 each measured from the previous link, links L_1, L_2, L_3, tool frame at E

PREMISE **Given:** L_1, L_2, L_3 and the joint angles q_1, q_2, q_3 of a planar RRR arm in the xy-plane (frame $\{O\}$ at joint 1)
Find: the position (x, y) of E and its orientation angle q_E (measured from the x-axis)

Planar RRR forward kinematics [p. 7–9]

$$x = L_1\cos q_1 + L_2\cos(q_1+q_2) + L_3\cos(q_1+q_2+q_3)$$
$$y = L_1\sin q_1 + L_2\sin(q_1+q_2) + L_3\sin(q_1+q_2+q_3), \qquad q_E = q_1+q_2+q_3$$

SYMBOLS

L_k	length of link k (from joint k to joint k+1, or to E)	m
q_k	angle of joint k, measured from the extension of link $k-1$; counter-clockwise positive	rad
x, y	position of the tool point E in $\{O\}$	m
q_E	orientation of the end effector, angle of its x-axis from the x-axis of $\{O\}$	rad

Use: any planar serial chain of revolute joints. Angles add because each joint turns everything beyond it ($\phi_k = q_1 + \cdots + q_k$ is the absolute direction of link k).

WORKED planar RRR, numbers
$L = (0.5,\ 0.4,\ 0.2)$ m, $q = (30°, 45°, -60°)$. Absolute link directions $\phi = (30°,\ 75°,\ 15°)$.
Joint 2 at $A = (0.5\cos 30°,\ 0.5\sin 30°) = (0.4330,\ 0.2500)$.
Joint 3 at $B = A + 0.4(\cos 75°, \sin 75°) = (0.5365,\ 0.6364)$.
Tip $E = B + 0.2(\cos 15°, \sin 15°) = (0.7297,\ 0.6881)$ m, $q_E = 15°$. [checked]
Check at $\mathbf{q} = \mathbf{0}$: $E = (L_1{+}L_2{+}L_3,\ 0)$, arm straight along x.

TRAP
Using the joint angles themselves ($\cos 45°$ for link 2) instead of the running sums ϕ_k. At $\mathbf{q} = \mathbf{0}$ both agree, so test a pose with non-zero angles. Also: $q_E = q_1 + q_2 + q_3$, not q_3.

Frames, position vectors, rotation matrices [p. 10–12]

For spatial arms (UR5, Panda, YuMi [p. 10]) trigonometry on a drawing is hopeless, so the problem is set up with *frames*: a frame $\{A\}$ is an origin plus three orthogonal unit axes $\hat{x}_A, \hat{y}_A, \hat{z}_A$ (right-handed, z up in this unit). A point is a vector *written in a frame*: ${}^A\mathbf{P}$ means "the point P, expressed in $\{A\}$". Always write the left superscript; a vector without a frame is meaningless.

Rotation matrix ${}^A\mathbf{R}_B$ and the elementary rotations [added]

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix},\ \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix},\ \mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A\mathbf{R}_B = \begin{bmatrix} {}^A\hat{x}_B & {}^A\hat{y}_B & {}^A\hat{z}_B \end{bmatrix} \in SO(3), \qquad {}^A\mathbf{R}_B^\top\,{}^A\mathbf{R}_B = \mathbf{I}, \qquad \det {}^A\mathbf{R}_B = +1,$$
$${}^B\mathbf{R}_A = {}^A\mathbf{R}_B^{-1} = {}^A\mathbf{R}_B^\top$$

SYMBOLS

θ	angle of rotation about the axis in the subscript, right-hand rule (thumb along the axis, fingers curl the positive way)	rad
${}^A\hat{x}_B$	the x-axis of $\{B\}$ written in $\{A\}$ (a unit column vector); likewise $\hat{y}_B, \hat{z}_B$	–
$SO(3)$	the set of 3×3 rotation matrices (orthonormal columns, determinant +1)	–

Use: reading a pose off a figure: write the three axes of $\{B\}$ as columns, each in the axes of $\{A\}$. Check: columns are unit length, mutually orthogonal, and $\hat{x}_B \times \hat{y}_B = \hat{z}_B$.

WORKED reading a rotation matrix, [added]

$\{B\}$ has $\hat{x}_B = \hat{y}_A$, $\hat{y}_B = \hat{z}_A$, $\hat{z}_B = \hat{x}_A$. Columns: ${}^A\mathbf{R}_B = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Check $\hat{x}_B \times \hat{y}_B = \hat{y}_A \times \hat{z}_A = \hat{x}_A = \hat{z}_B$ ✓. As elementary rotations (any valid factorisation earns the marks): $\mathbf{R}_y(90°)\mathbf{R}_z(90°)$. [checked]

Homogeneous transformation matrix [p. 12]

PREMISE **Given:** the pose of frame $\{B\}$ relative to $\{A\}$: a rotation ${}^A\mathbf{R}_B$ and the position ${}^A\mathbf{t}_B$ of the origin of $\{B\}$ in $\{A\}$

Find: the single 4×4 matrix ${}^A\mathbf{T}_B \in SE(3)$; the coordinates ${}^A\mathbf{P}$ of a point known as ${}^B\mathbf{P}$; the inverse ${}^B\mathbf{T}_A$

Homogeneous transform, point mapping and the full inverse [p. 12]

$$ {}^A\mathbf{T}_B = \begin{bmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{t}_{B\text{-origin}} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \qquad \begin{bmatrix} {}^A\mathbf{P} \\ 1 \end{bmatrix} = {}^A\mathbf{T}_B \begin{bmatrix} {}^B\mathbf{P} \\ 1 \end{bmatrix} \Leftrightarrow {}^A\mathbf{P} = {}^A\mathbf{R}_B\,{}^B\mathbf{P} + {}^A\mathbf{t}_{B\text{-origin}} $$

$$ {}^B\mathbf{T}_A = ({}^A\mathbf{T}_B)^{-1} = \begin{bmatrix} {}^A\mathbf{R}_B^\top & -{}^A\mathbf{R}_B^\top\,{}^A\mathbf{t}_{B\text{-origin}} \\ \mathbf{0}^\top & 1 \end{bmatrix} $$

SYMBOLS

${}^A\mathbf{T}_B$	pose of $\{B\}$ in $\{A\}$ (maps coordinates in B to coordinates in A)	–
${}^A\mathbf{t}_{B\text{-origin}}$	position of the origin of $\{B\}$, written in $\{A\}$	m
${}^B\mathbf{P}$, ${}^A\mathbf{P}$	the same point written in $\{B\}$ and in $\{A\}$	m
$\mathbf{0}^\top$	row $(0\ 0\ 0)$	–

Use: moving a point or a frame between frames; the bottom row $(0\ 0\ 0\ 1)$ makes translation and rotation one multiplication. The inverse is not $-\mathbf{t}$ in the last column: transpose $\mathbf{R}$ and rotate the translation back, $-\mathbf{R}^\top\mathbf{t}$.

WORKED point mapping and inverse

${}^A\mathbf{R}_B = \mathbf{R}_z(90°)$, ${}^A\mathbf{t} = (2, 1, 3)^\top$ m. A point ${}^B\mathbf{P} = (1,\ 0.5,\ 0)^\top$:

$$ {}^A\mathbf{P} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 2 \\ 3 \end{bmatrix} $$

Inverse: ${}^A\mathbf{R}_B^\top = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $-{}^A\mathbf{R}_B^\top\,{}^A\mathbf{t} = -(1, -2, 3)^\top = (-1,\ 2,\ -3)^\top$:

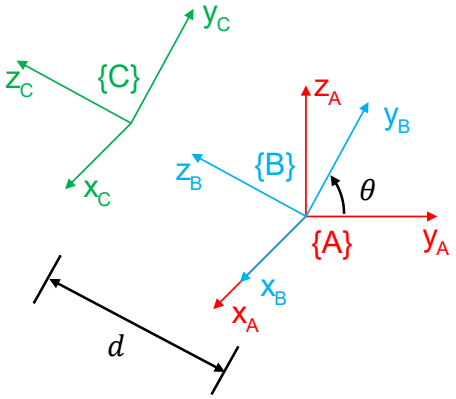
$$ {}^B\mathbf{T}_A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Check: ${}^B\mathbf{T}_A(1.5, 2, 3, 1)^\top = (1,\ 0.5,\ 0,\ 1)^\top$ ✓. [checked]

TRAP

Writing ${}^B\mathbf{T}_A$ with $-\mathbf{t} = (-2, -1, -3)$ in the last column (here the correct entry is $(-1, 2, -3)$). If the rotation is the identity the two agree, which hides the mistake.

Compound transformations [p. 13–14]



[W02 lec p. 13] — C is reached from A through B: rotate by θ about x_A, then translate d along z_B

Compounding [p. 13]

$$ {}^A\mathbf{T}_C = {}^A\mathbf{T}_B\,{}^B\mathbf{T}_C, \qquad {}^A\mathbf{T}_B = \mathbf{R}_x(\theta), \qquad {}^B\mathbf{T}_C = t_z(d) \Rightarrow {}^A\mathbf{T}_C = \mathbf{R}_x(\theta)\,t_z(d) $$

SYMBOLS

$\mathbf{R}_x(\theta)$	4×4 pure rotation about the x-axis of the frame it is applied in	rad
$t_z(d)$	4×4 pure translation d along that frame's z-axis	m

Use: chaining frames. The inner letters must cancel (B next to B). Slide's caution: "note the order in which the matrices are multiplied".

WORKED slide $\{A\}, \{B\}, \{C\}$ with $\theta = 30°$, $d = 2$ m

$$ {}^A\mathbf{T}_C = \mathbf{R}_x(30°)\,t_z(2) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.866 & -0.5 & -1 \\ 0 & 0.5 & 0.866 & 1.732 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

The origin of $\{C\}$ is $(0,\ -d\sin\theta,\ d\cos\theta) = (0, -1, 1.732)$ in $\{A\}$: you moved 2 m along the *tilted* z_B. [checked]

Reversed order $t_z(2)\,\mathbf{R}_x(30°)$: origin $(0, 0, 2)$, the translation happens along the *fixed* z_A first. [checked]

ALTERNATIVE APPROACHES

Read the product left to right: CURRENT frame

Each new factor on the *right* is a move made *in the frame you have just reached* (about its axes, along its axes). This is how a robot is built: joint i turns about *its own* axis.

$\mathbf{M} = \mathbf{R}_z(90°)\,t_x(1)$: turn 90° about z, then slide 1 m along the *new* x (which is the old y).

Read the product right to left: FIXED frame

Each new factor on the *left* is a move made *about the original (fixed) axes*. The same product read from the right: apply the last factor first.

$\mathbf{M} = \mathbf{R}_z(90°)\,t_x(1)$: first slide 1 m along the fixed x, then rotate the whole thing 90° about the fixed origin; the point $(1, 0)$ is carried to $(0, 1)$.

WORKED both readings give the same matrix; the order is what matters

$$\mathbf{M} = \mathbf{R}_z(90^\circ)t_x(1) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{origin } (0,1,0)$$

$$\mathbf{N} = t_x(1)\mathbf{R}_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{origin } (1,0,0)$$

Same rotation, different position. A link of length 1 turned by its own joint is $\mathbf{M}$ (turn, then move along the turned axis): $\mathbf{R}_z(q)\,t_x(L)$, the pattern of every planar arm. [checked]

Elementary Transform Sequence (ETS) [p. 15–16]

PREMISE ETS **Given:** the arm drawn in a pose where each joint is displaced from its rest position ([p. 18]), base frame $\{B\}$ and tool frame $\{E\}$ drawn

Find: the sequence $\{E_1, E_2, \dots\}$ of elementary transforms met on the way from $\{B\}$ to $\{E\}$ and ${}^B\mathbf{T}_E = E_1E_2\cdots E_m$

The six elementary transforms and the ETS product [p. 16]

$$t_x(\eta),\ t_y(\eta),\ t_z(\eta):\ \mathbf{T} = \begin{bmatrix} \mathbf{I}_3 & \eta\,\hat{\mathbf{e}} \\ \mathbf{0}^\top & 1 \end{bmatrix}\ (\hat{\mathbf{e}} = \hat{x}, \hat{y}, \hat{z}), \qquad \mathbf{R}_x(\eta),\ \mathbf{R}_y(\eta),\ \mathbf{R}_z(\eta):\ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

$${}^B\mathbf{T}_E = E_1(\eta_1)\,E_2(\eta_2)\cdots E_m(\eta_m), \qquad \eta_i = \begin{cases} c_i & \text{constant (offset, fixed rotation)} \\ q_j & \text{joint variable: } \theta_j \text{ revolute, } d_j \text{ prismatic} \end{cases}$$

SYMBOLS

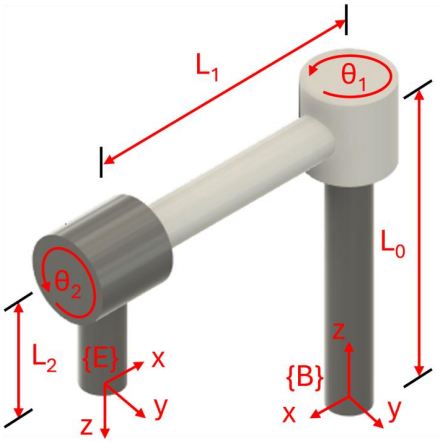
E_i	one elementary transform (a translation or a rotation along/about one axis of the *current* frame)	–
η_i	its parameter: a length (m) or an angle (rad)	m or rad
q_j	the j-th joint variable (changes with time)	rad or m
$\hat{\mathbf{e}}$	unit vector of the axis translated along	–

Use: the quickest FK method for exam arms: no table, no rules; you only need the drawing. The matrices $\mathbf{R}_x, \mathbf{R}_y, \mathbf{R}_z$ are the 3×3 ones above.

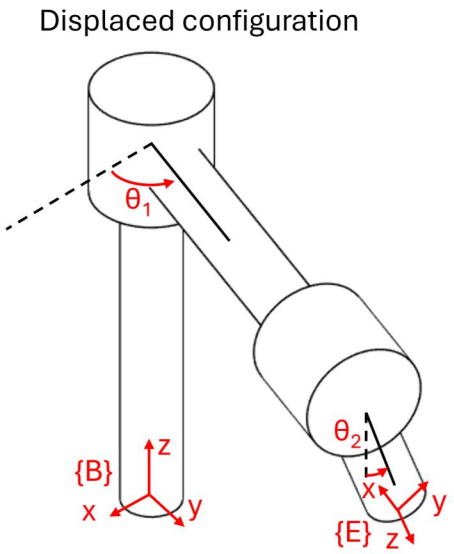
STEPS Write an ETS from a drawing [p. 18]

1. Redraw the arm with every joint displaced from rest, so each joint variable is visible.
2. Start at $\{B\}$. Move to the first joint: t_x, t_y, t_z with the signed distance along the *current* axes.
3. The joint itself: $\mathbf{R}_{axis}(\pm q)$ about the joint axis, expressed in the current frame's axes. A minus sign if the drawn positive arrow is opposite to the right-hand rule.
4. Continue joint by joint (translation along the link, then the next rotation). Joint axes that coincide need no translation between them.
5. End with the constant rotation that makes the last frame equal to the drawn $\{E\}$.
6. Group per joint: ${}^{i-1}\mathbf{T}_i$ (these are the "intermediate frames"), then multiply.

Example: Furuta pendulum [p. 17–19]



[W02 lec p. 17] — Furuta pendulum: L_0 post, L_1 arm turned θ_1 about z, L_2 pendulum turned θ_2; frames $\{B\}$ and $\{E\}$ (z_E down)



[W02 lec p. 19] — displaced configuration used to read off the sequence

Furuta pendulum ETS and full result [p. 19]

$$ {}^{B}\mathbf{T}_{E} = t_z(L_0)\,\mathbf{R}_z(\theta_1)\,t_x(L_1)\,\mathbf{R}_x(\theta_2)\,t_z(-L_2)\,\mathbf{R}_y(\pi) $$

$$ {}^{B}\mathbf{T}_{1} = t_z(L_0)\mathbf{R}_z(\theta_1) = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{1}\mathbf{T}_{2} = t_x(L_1)\mathbf{R}_x(\theta_2) = \begin{bmatrix} 1 & 0 & 0 & L_1 \\ 0 & c_2 & -s_2 & 0 \\ 0 & s_2 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}\mathbf{T}_{E} = t_z(-L_2)\mathbf{R}_y(\pi) = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & -L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{B}\mathbf{T}_{E} = {}^{B}\mathbf{T}_{1}\,{}^{1}\mathbf{T}_{2}\,{}^{2}\mathbf{T}_{E} = \begin{bmatrix} -c_1 & -s_1c_2 & -s_1s_2 & L_1c_1 - L_2s_1s_2 \\ -s_1 & c_1c_2 & c_1s_2 & L_1s_1 + L_2c_1s_2 \\ 0 & s_2 & -c_2 & L_0 - L_2c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

SYMBOLS

L_0, L_1, L_2	post height, horizontal arm length, pendulum length	m
θ_1	rotation of the arm about the vertical post, z_B	rad
θ_2	rotation of the pendulum about the arm's own x-axis	rad
c_i, s_i	$\cos\theta_i,\ \sin\theta_i$	–

Use: two joints, three constants. $\mathbf{R}_y(\pi)$ at the end flips z to point down and x back, so $\{E\}$ is the drawn tool frame.

WORKED Furuta numbers

$L = (0.5,\ 0.4,\ 0.3)$ m, $\theta = (60°, 45°)$ ($c_1 = 0.5$, $s_1 = 0.8660$, $c_2 = s_2 = 0.7071$):

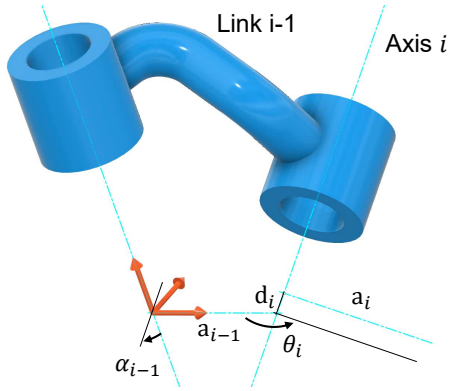
$$ {}^{B}\mathbf{T}_{E} = \begin{bmatrix} -0.5 & -0.6124 & -0.6124 & 0.0163 \\ -0.8660 & 0.3536 & 0.3536 & 0.4525 \\ 0 & 0.7071 & -0.7071 & 0.2879 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Check at $\theta = 0$: tip $(L_1, 0, L_0 - L_2) = (0.4, 0, 0.2)$, $z_E = -z_B$. [checked]

Workspace ([added]): $x^2 + y^2 + (z - L_0)^2 = L_1^2 + L_2^2$, so the tip lies on a sphere of radius $\sqrt{L_1^2 + L_2^2}$ about $(0, 0, L_0)$, band $|z - L_0| \le L_2$ (a spherical zone, not a torus). [checked]

Proximal DH notation: why a table, what the four numbers mean [p. 20–27]

Denavit–Hartenberg trades the freedom of the ETS (the joint axis may be any local axis) for a *systematic* recipe: fixed frame placement rules leave only four numbers per link. This unit uses **proximal DH (modified DH, Craig)**: frame $\{i\}$ sits on joint axis i, at the *near* (proximal) end of link i. Links are numbered $0 \ldots n$; link 0 is the fixed base; joint i joins link $i-1$ to link i ([p. 23]).



[W02 lec p. 28] — the four parameters between axis $i-1$ and axis i: a_{i-1}, α_{i-1} ([p. 24–25]), d_i ([p. 26]), θ_i ([p. 27])

The four proximal DH parameters [p. 24–27, 30]

SYMBOLS

a_{i-1}	**link length**: distance from z_{i-1} to z_i along x_{i-1} (length of the common normal); ≥ 0	m
α_{i-1}	**link twist**: angle from z_{i-1} to z_i about x_{i-1}, right-hand sense; signed	rad
d_i	**link offset**: signed distance from x_{i-1} to x_i along z_i; *variable* if joint i is prismatic	m
θ_i	**joint angle**: angle from x_{i-1} to x_i about z_i; *variable* if joint i is revolute	rad

Use: a_{i-1}, α_{i-1} describe the geometry behind joint i (axis $i-1 \to i$); d_i, θ_i describe motion along/about axis i. Row i of the table gives ${}^{i-1}\mathbf{T}_i$.

Proximal link transform [p. 28, 35]

$$ {}^{i-1}\mathbf{T}_{i} = \mathbf{R}_{x_{i-1}}(\alpha_{i-1})\,t_{x_{i-1}}(a_{i-1})\,\mathbf{R}_{z_i}(\theta_i)\,t_{z_i}(d_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{0}\mathbf{T}_{n} = {}^{0}\mathbf{T}_{1}\,{}^{1}\mathbf{T}_{2} \cdots {}^{n-1}\mathbf{T}_{n} $$

SYMBOLS

$c\theta_i, s\theta_i$	$\cos\theta_i,\ \sin\theta_i$	–
$c\alpha_{i-1}, s\alpha_{i-1}$	$\cos\alpha_{i-1},\ \sin\alpha_{i-1}$	–

Use: one row of the table in, one 4×4 out. Prismatic joint: d_i is the variable; revolute: θ_i is. The operator order is $\mathbf{R}_x\ t_x\ \mathbf{R}_z\ t_z$ (proximal); the distal form has a different order.

DERIVATION of the link transform: multiply the four moves in order

Frame $\{i{-}1\} \to \{i\}$ in four moves, each in the current frame: turn α_{i-1} about x, slide a_{i-1} along x, turn θ_i about z, slide d_i along z.

$$\mathbf{R}_x(\alpha)\,t_x(a) = \begin{bmatrix}1&0&0&0\\0&c\alpha&-s\alpha&0\\0&s\alpha&c\alpha&0\\0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&a\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&a\\0&c\alpha&-s\alpha&0\\0&s\alpha&c\alpha&0\\0&0&0&1\end{bmatrix}$$

(the x-translation is along the rotation axis, so it is unchanged.)

$$\mathbf{R}_z(\theta)\,t_z(d) = \begin{bmatrix}c\theta&-s\theta&0&0\\s\theta&c\theta&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&d\\0&0&0&1\end{bmatrix} = \begin{bmatrix}c\theta&-s\theta&0&0\\s\theta&c\theta&0&0\\0&0&1&d\\0&0&0&1\end{bmatrix}$$

Multiply the two results (rotation part first). Rows of $\mathbf{R}_x(\alpha)\mathbf{R}_z(\theta)$: $(c\theta, -s\theta, 0)$, $(c\alpha\,s\theta,\ c\alpha\,c\theta,\ -s\alpha)$, $(s\alpha\,s\theta,\ s\alpha\,c\theta,\ c\alpha)$. Translation $= \mathbf{R}_x(\alpha)(0,0,d)^\top + (a,0,0)^\top = (a,\ -s\alpha\,d,\ c\alpha\,d)^\top$. These are exactly the entries of the box above. [added]

Assigning frames to a mechanism [p. 29–30]

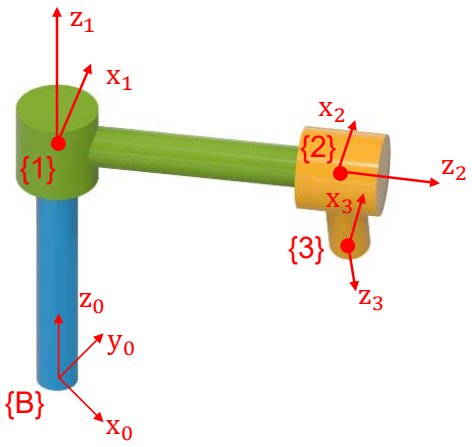
STEPS Proximal frame assignment [p. 29]

1. Identify the joint axes. For steps 2–5, look at axes i and $i{+}1$.
2. Find the common normal between them, or the point where they intersect. The origin of $\{i\}$ is where axis i meets that normal (if the axes intersect, that point is the origin).
3. z_i along joint axis i.
4. x_i along the common normal (towards axis $i{+}1$). If axes i and $i{+}1$ intersect, x_i is normal to the plane containing them (either sign: pick one).
5. $y_i = z_i \times x_i$ completes the right-handed frame.
6. $\{0\}$ coincides with $\{1\}$ when the first joint variable is zero. For $\{N\}$ choose the origin and x_N freely, so that as many parameters as possible become 0.
7. Read the table: a_{i-1} = distance $z_{i-1} \to z_i$ along x_{i-1}; α_{i-1} = angle $z_{i-1} \to z_i$ about x_{i-1}; d_i = distance $x_{i-1} \to x_i$ along z_i; θ_i = angle $x_{i-1} \to x_i$ about z_i.

TRAP

In proximal DH the frame $\{i\}$ is on axis i at the *near* end of link i, so ${}^0\mathbf{T}_n$ is the pose of the *last joint's* frame, **not** of the tool. A tool point beyond the last joint needs one extra row (planar arms) or a constant ${}^n\mathbf{T}_E$. Also: $a_{i-1} \geq 0$ is a distance; all the others are signed.

Building a table: Furuta pendulum [p. 31–36]



[W02 lec p. 32] — frames $\{0\}$ to $\{3\}$ assigned by the recipe; note $\{3\}$ is not $\{E\}$

WORKED Furuta DH table from the slide figure [p. 34]

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_0	$q_1 + 90°$
2	0	90°	L_1	q_2
3	0	90°	L_2	0

Row 1: $z_0 \parallel z_1$ so $\alpha_0 = 0$, $a_0 = 0$; $\{1\}$ sits L_0 above $\{0\}$ ($d_1 = L_0$); x_1 is turned 90° from x_0, so $\theta_1 = q_1 + 90°$. Row 2: axes 1, 2 are perpendicular ($\alpha_1 = 90°$) and intersect ($a_1 = 0$); $\{2\}$ is L_1 along z_2 ($d_2 = L_1$). Row 3: z_3 is perpendicular to z_2 ($\alpha_2 = 90°$), $\{3\}$ is L_2 along z_3 ($d_3 = L_2$), fixed ($\theta_3 = 0$).

Homework on slide 36: multiply the table and compare with the ETS.

$${}^0\mathbf{T}_3 = \begin{bmatrix}-s_1c_2 & c_1 & -s_1s_2 & L_1c_1 - L_2s_1s_2\\ c_1c_2 & s_1 & c_1s_2 & L_1s_1 + L_2c_1s_2\\ s_2 & 0 & -c_2 & L_0 - L_2c_2\\ 0 & 0 & 0 & 1\end{bmatrix}$$

The position column equals the ETS result and the rotation differs only by the fixed turn ${}^3\mathbf{T}_E = \mathbf{R}_z(-90°) = \begin{bmatrix}0&1&0&0\\-1&0&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$ ([p. 32]: "3 is not E"): ${}^0\mathbf{T}_3\,{}^3\mathbf{T}_E = {}^B\mathbf{T}_E$ for *all* q_1, q_2. [checked]

ALTERNATIVE APPROACHES

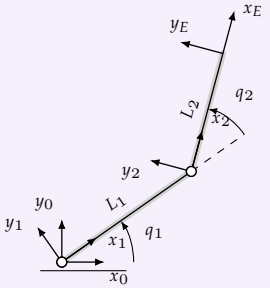
A. ETS (walk the arm)	B. Proximal DH table
${}^B\mathbf{T}_E = t_z(L_0)\mathbf{R}_z(\theta_1)t_x(L_1)$ $\mathbf{R}_x(\theta_2)\,t_z(-L_2)\mathbf{R}_y(\pi)$.	${}^B\mathbf{T}_E = {}^0\mathbf{T}_1\,{}^1\mathbf{T}_2\,{}^2\mathbf{T}_3\,{}^3\mathbf{T}_E$,
6 moves, no rules to remember, but you choose the axes yourself.	3 rows + $\mathbf{R}_z(-90°)$. 4 numbers per joint; rules fix the frames, so everyone gets the same table; but the frame may sit off the link and the tool needs a constant.

Simple robots: ETS and DH side by side

The exam favours simple arms. Each box has the picture with the frames drawn by the DH rules, the ETS, the proximal table, the *full* ${}^0\mathbf{T}_E$ and a number check. All tables are $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$; angles in degrees only in the numbers. Figures are *drawn for this booklet* (the slides have no figure of these arms); z_i is out of the page (circle with a dot) unless marked, and $\otimes$ means into the page. [added]

PREMISE simple arms **Given:** joint variables and link lengths of an RR, RP, PR, RRR or RRP arm
Find: ${}^0\mathbf{T}_E$ in full (position and orientation), from both the ETS and the DH table

WORKED 1. planar RR: ${}^0\mathbf{T}_E$ in full



[drawn] z_i out of page; $\{E\}$ is the tool row.

ETS: ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\,t_x(L_1)\,\mathbf{R}_z(q_2)\,t_x(L_2)$

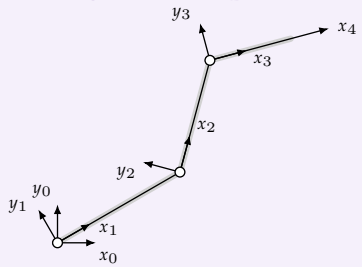
Proximal DH:

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	L_1	0	0	q_2
3 (tool)	L_2	0	0	0

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{12} & -s_{12} & 0 & L_1c_1 + L_2c_{12} \\ s_{12} & c_{12} & 0 & L_1s_1 + L_2s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{L=(0.4,0.3)\text{ m},\ q=(40°,50°)} \begin{bmatrix} 0 & -1 & 0 & 0.3064 \\ 1 & 0 & 0 & 0.5571 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$c_{12} = \cos(q_1 + q_2)$, $s_{12} = \sin(q_1 + q_2)$. Numbers: elbow at (0.3064, 0.2571); $q_1{+}q_2 = 90°$, so x_E points along $+y$. Check: $L_1 = L_2 = 1$, $q = (90°, -90°)$ gives tip $(1, 1)$. [checked]

WORKED 2. planar RRR: ${}^0\mathbf{T}_E$ in full (slide arm [p. 6–9])



[drawn] shown at $q = (30°, 45°, -60°)$; $\{4\} = \{E\}$ is the tool row.

ETS: ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)t_x(L_1)\mathbf{R}_z(q_2)$ $t_x(L_2)\mathbf{R}_z(q_3)t_x(L_3)$

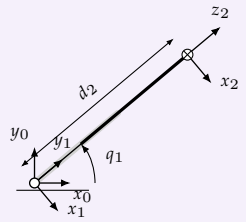
Proximal DH:

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	q_1
2	L_1	0	0	q_2
3	L_2	0	0	q_3
4 (tool)	L_3	0	0	0

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1c_1 + L_2c_{12} + L_3c_{123} \\ s_{123} & c_{123} & 0 & L_1s_1 + L_2s_{12} + L_3s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{as [p. 6–9]}} \begin{bmatrix} 0.9659 & -0.2588 & 0 & 0.7297 \\ 0.2588 & 0.9659 & 0 & 0.6881 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Numbers: $L = (0.5, 0.4, 0.2)$ m, $q = (30°, 45°, -60°)$, $q_1{+}q_2{+}q_3 = 15°$. The angle q_E is read from the matrix: $q_E = \text{atan2}(T_{21}, T_{11}) = 15°$. [checked]

WORKED 3. RP (polar): revolute then prismatic



[drawn] z_1 out of page; z_2 along the slider; y_2 into the page ($\otimes$); $\{E\}$: x_E along the slider.

ETS: ${}^0\mathbf{T}_E = \mathbf{R}_z(q_1)\,t_x(d_2)$

Proximal DH (prismatic axis must be a z-axis, so frame $\{1\}$ is turned $-90°$ and $\alpha_1 = -90°$):

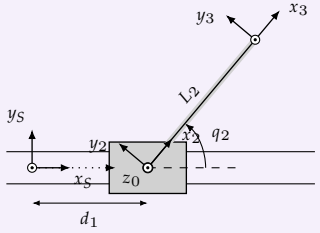
i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	$q_1 - 90°$
2	0	$-90°$	d_2	0

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_1 & -s_1 & 0 & d_2c_1 \\ s_1 & c_1 & 0 & d_2s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{q_1=35°,\ d_2=0.6\text{ m}} \begin{bmatrix} 0.8192 & -0.5736 & 0 & 0.4915 \\ 0.5736 & 0.8192 & 0 & 0.3441 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0\mathbf{T}_2 = \begin{bmatrix} s_1 & 0 & c_1 & d_2c_1 \\ -c_1 & 0 & s_1 & d_2s_1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^2\mathbf{T}_E = \mathbf{R}_x(90°)\mathbf{R}_z(90°) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Frame $\{2\}$ has the right position but z_2 along the slider; the constant ${}^2\mathbf{T}_E$ turns it into the tool frame of the ETS: ${}^0\mathbf{T}_2\,{}^2\mathbf{T}_E = {}^0\mathbf{T}_E$. Numbers: tip (0.4915, 0.3441) m. [checked]

WORKED 4. PR: prismatic then revolute (slider along x_S, vertical joint axis)



[drawn] z_S, z_2, z_3 out of page; $z_0 = x_S$ along the rail.

ETS: ${}^S\mathbf{T}_E = t_x(d_1)\,\mathbf{R}_z(q_2)\,t_x(L_2)$

Proximal DH (frame $\{0\}$ chosen with z_0 along the slide, $x_0 = -y_S$, $y_0 = -z_S$; ${}^S\mathbf{T}_0$ below):

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	0	90°	0	$q_2 + 90°$
3 (tool)	L_2	0	0	0

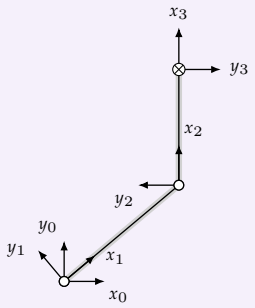
$$ {}^S\mathbf{T}_E = \begin{bmatrix} c_2 & -s_2 & 0 & d_1 + L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{d_1=0.3,\ q_2=60°,\ L_2=0.4} \begin{bmatrix} 0.5 & -0.866 & 0 & 0.5 \\ 0.866 & 0.5 & 0 & 0.3464 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^S\mathbf{T}_0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^0\mathbf{T}_3 = \begin{bmatrix} -s_2 & -c_2 & 0 & -L_2 s_2 \\ 0 & 0 & -1 & 0 \\ c_2 & -s_2 & 0 & L_2 c_2 + d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

${}^S\mathbf{T}_0\,{}^0\mathbf{T}_3 = {}^S\mathbf{T}_E$: the table plus the constant ${}^S\mathbf{T}_0 = \mathbf{R}_y(90°)\mathbf{R}_z(-90°)$ reproduces the ETS. Numbers: tip $(0.5,\ 0.3464)$ m. [checked]

Why this awkward table: a DH prismatic variable always runs along z_i and α is a rotation about x, so a slide along x_S is only possible if the base frame $\{0\}$ is itself chosen with $z_0 = x_S$. If the slide is vertical (z) the table is trivial: $(0, 0, d_1, 0)$, $(0, 0, 0, q_2)$, $(L_2, 0, 0, 0)$.

WORKED 5. spatial RRP (SCARA): ${}^0\mathbf{T}_E$ in full



[drawn] top view at $q = (40°, 50°)$: z_0, z_1, z_2 up (out of page), z_3 pointing *down* (⊗): d_3 is measured downwards.

ETS: ${}^0\mathbf{T}_E = t_z(H)\mathbf{R}_z(q_1)t_x(a_1)\,\mathbf{R}_z(q_2)\,t_x(a_2)\,t_z(-d_3)\,\mathbf{R}_x(180°)$

Proximal DH:

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	H	q_1
2	a_1	0	0	q_2
3	a_2	180°	d_3	0

$$ {}^0\mathbf{T}_E = \begin{bmatrix} c_{12} & s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & -c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & -1 & H - d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{H=0.5,\ a=(0.4,0.3)\text{ m} \\ q=(40°,50°),\ d_3=0.15\text{ m}}} \begin{bmatrix} 0 & 1 & 0 & 0.3064 \\ 1 & 0 & 0 & 0.5571 \\ 0 & 0 & -1 & 0.35 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$\mathbf{R}_x(180°)$ (equivalently $\alpha_2 = 180°$) makes the tool z-axis point down, so a positive d_3 lowers the tool: $z = H - d_3$. Same x, y as the RR example. [checked]

Puma 560 and Franka Panda tables [added]

The W2 deck shows photos only ([p. 10]); the Puma and Panda belong to the notebook (Q1, Q2, Q5). Neither table is on the slides.

ALTERNATIVE APPROACHES

Puma-like arm (notebook Q5), ETS

Home pose: shoulder and elbow axes along y, upper arm and forearm vertical; $L_b = 1.0$, $L_u = 0.5$, $L_f = 0.3$ m (own values, no offsets).

${}^0\mathbf{T}_E = t_z(L_b)\,\mathbf{R}_z(q_1)\,\mathbf{R}_y(q_2)\,t_z(L_u)$
$\mathbf{R}_y(q_3)\,t_z(L_f)\,\mathbf{R}_z(q_4)\,\mathbf{R}_y(q_5)\,\mathbf{R}_z(q_6)$

Coincident axes (shoulder 1–2, wrist 4–5–6) have *no* translation between the rotations.

the same arm, proximal DH

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	L_b	q_1
2	0	−90°	0	$q_2 - 90°$
3	L_u	0	0	$q_3 + 90°$
4	0	90°	L_f	q_4
5	0	−90°	0	q_5
6	0	90°	0	q_6

Frames 4, 5, 6 share one origin (the wrist centre).

WORKED Puma-like numbers

Home ($\mathbf{q} = \mathbf{0}$): ${}^0\mathbf{T}_E = \begin{bmatrix} \mathbf{I}_3 & (0,0,1.8)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}$, tool straight up. For $q = (30°, 20°, -40°, 0, 0, 0)$: wrist centre $\mathbf{p} = (c_1\rho,\ s_1\rho,\ L_b + L_u c_2 + L_f c_{23})$ with $\rho = L_u s_2 + L_f s_{23}$, i.e. $\mathbf{p} = (0.0592,\ 0.0342,\ 1.7518)$ m. Turning q_4, q_5, q_6 does not move $\mathbf{p}$ (spherical wrist: three axes meet at one point). The DH table above reproduces the ETS for all $\mathbf{q}$ (50 random poses). [checked]

Franka Panda, published proximal (Craig) DH [added]

i	1	2	3	4	5	6	7
a_{i-1} (m)	0	0	0	0.0825	−0.0825	0	0.088
α_{i-1}	0	−90°	90°	90°	−90°	90°	90°
d_i (m)	0.333	0	0.316	0	0.384	0	0
θ_i	q_1	q_2	q_3	q_4	q_5	q_6	q_7

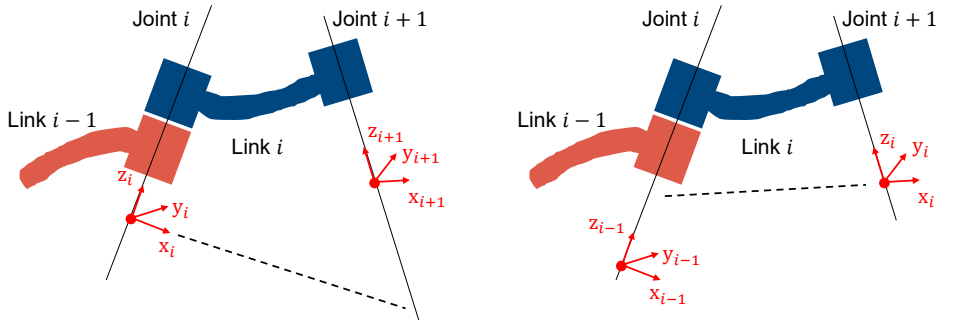
flange: constant $t_z(0.107)$. At $\mathbf{q} = \mathbf{0}$: ${}^0\mathbf{T}_{\text{flange}} = \begin{bmatrix} 1 & 0 & 0 & 0.088 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0.926 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (tool points down). [checked]

Use: each row is $\mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(q_i)t_z(d_i)$; the ETS is just those four moves written out for $i = 1 \ldots 7$ followed by $t_z(0.107)$. Matches the toolbox Panda flange position for random $\mathbf{q}$.

SLIDE CHECK notebook cell 42, Panda ETS

The notebook's printed ETS ($t_z(0.333)\,\mathbf{R}_z\,t_x(0.316)\,\mathbf{R}_x\,t_z(0.384)\ldots$) puts the flange at $(0.316,\ 0,\ 0.9945)$ m at $\mathbf{q} = \mathbf{0}$, but the Franka DH table above gives $(0.088,\ 0,\ 0.926)$ m. It is not the Panda. Use the DH rows (or the ETS built from them). Cell 35 also has a typo: `tz(-30)` should be `tz(-0.30)`, so its printed pose $(-18.17, 10.95, -20.71)$ is wrong. [checked]

Distal (standard) DH [p. 37–41]



[W02 lec p. 38] — the same two joints with proximal (left, frame $\{i\}$ on joint i, before the link) and distal (right, frame placed *after* the link, on the next joint) frames

ALTERNATIVE APPROACHES

Proximal (modified, Craig) [p. 40]

Frame $\{i\}$ on axis i (near end of link i). a_{i-1}: $z_{i-1} \to z_i$ along x_{i-1}; α_{i-1}: $z_{i-1} \to z_i$ about x_{i-1}; d_i: $x_{i-1} \to x_i$ along z_i; θ_i: $x_{i-1} \to x_i$ about z_i.

$^{i-1}\mathbf{T}_i = \mathbf{R}_{x_{i-1}}(\alpha_{i-1})\, t_{x_{i-1}}(a_{i-1})\, \mathbf{R}_{z_i}(\theta_i)\, t_{z_i}(d_i)$

Distal (standard) [p. 39–40]

Frame $\{i-1\}$ on axis i (far end of link $i-1$); joint i turns about z_{i-1}. a_i: $z_{i-1} \to z_i$ along x_i; α_i: $z_{i-1} \to z_i$ about x_i; d_i: $x_{i-1} \to x_i$ along z_{i-1}; θ_i: $x_{i-1} \to x_i$ about z_{i-1}.

$^{i-1}\mathbf{T}_i = \mathbf{R}_{z_{i-1}}(\theta_i)\, t_{z_{i-1}}(d_i)\, t_{x_i}(a_i)\, \mathbf{R}_{x_i}(\alpha_i)$

Distal link transform [p. 40]

$$^{i-1}\mathbf{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SYMBOLS

a_i, α_i	link length and twist of link i (axis $i \to i{+}1$) in the distal convention	m, rad
d_i, θ_i	offset and angle of joint i, along/about z_{i-1}	m, rad

Use: Corke's toolbox `DHRobot` with `RevoluteDH` is distal; `RevoluteMDH`/`PrismaticMDH` is proximal (this unit). Write "proximal (Craig) DH" on your script.

WORKED same arms, both conventions; conversion rule [added]

Rule: the distal row i has the same (θ_i, d_i) as the proximal row i, but takes (a_i, α_i) from the *next* proximal row: proximal (a_i, α_i) sits in row $i{+}1$, distal (a_i, α_i) in row i. Distal frame $\{n\}$ sits on the tool, so *no extra tool row* is needed.

Planar RR: distal rows $(\theta, d, a, \alpha) = (q_1, 0, L_1, 0)$, $(q_2, 0, L_2, 0)$ give $^0\mathbf{T}_2 = \mathbf{R}_z(q_1)t_x(L_1)\mathbf{R}_z(q_2)t_x(L_2)$ (2 rows; proximal needed 3). Planar RRR: $(q_1, 0, L_1, 0), (q_2, 0, L_2, 0), (q_3, 0, L_3, 0)$. SCARA: $(q_1, H, a_1, 0), (q_2, 0, a_2, 180°), (0, d_3, 0, 0)$. Furuta: $(q_1{+}90°, L_0, 0, 90°), (q_2, L_1, 0, 90°), (0, L_2, 0, 0)$. All four reproduce the proximal results above. [checked]

TRAP

Never mix conventions: a proximal table put into the distal formula describes a different robot. Reading the slide-p.40 numbers "$(a, \alpha, d, \theta) = (0.2, 90°, 0.05, -20°)$" with the wrong formula moves the origin of $\{2\}$ from $(0.1982, 0.0567, 0.30)$ to $(0.1970, 0.0347, 0.35)$ m (Practice Q&A, same numbers). [checked]

Other conventions, why 4 parameters, URDF [p. 41–47]

Variations [p. 41]**:** several DH variants exist (Springer Handbook of Robotics, geometric representation). Screw/POE and URDF are further alternatives; state which one you use.

Why four, not six? [p. 42–43] Two rigid bodies in 3D need 6 numbers (3 rotation, 3 translation). DH fixes two of them by *placing the frames by rule*: proximal: x_i intersects z_{i+1} and $x_i \perp z_{i+1}$ (distal: x_i intersects and is $\perp$ to z_{i-1}). $6 - 2 = 4$. Consequence: the link frame is not necessarily on the link.

A good kinematic model [p. 44]**:** (1) *completeness*: every spatial geometry between successive joints can be described; (2) *continuity*: small changes of geometry give small changes of the parameters (DH is not continuous for nearly parallel axes: the common normal jumps); (3) *minimality*: only the minimum number of parameters.

URDF [p. 45–47]**:** Unified Robot Description Format, an XML description of the kinematic structure. Each `<link>` is a body; each `<joint>` has a `type` (the slide uses `continuous` = a revolute joint without limits, and `fixed`; `revolute`, `prismatic` also exist [added]), a `parent`, a `child`, an `origin xyz` (a constant translation t_x, t_y, t_z from the parent frame; `rpy` sets a constant rotation [added]) and an `axis xyz` (which axis of the joint frame the joint variable turns about or slides along). The slide's planar example: joint `q1` (axis z, no origin) = $\mathbf{R}_z(q_1)$; joint `q2` (origin $x = 1$, axis z) = $t_x(1)\mathbf{R}_z(q_2)$; the `fixed` joint (origin $x = 1$) = $t_x(1)$. Product: $\mathbf{R}_z(q_1)\, t_x(1)\, \mathbf{R}_z(q_2)\, t_x(1)$.

Notebook exercises, solved

Q1/Q2 Panda by ETS. From the Franka rows $(\mathbf{R}_x(\alpha)t_x(a)\mathbf{R}_z(q)t_z(d)$ for $i = 1 \ldots 7)$:

$$^0\mathbf{T}_E = t_z(0.333)\mathbf{R}_z(q_1)\mathbf{R}_x(-90°)\mathbf{R}_z(q_2)\mathbf{R}_x(90°)t_z(0.316)\mathbf{R}_z(q_3)\mathbf{R}_x(90°)t_x(0.0825)\mathbf{R}_z(q_4)$$

$$\mathbf{R}_x(-90°)t_x(-0.0825)t_z(0.384)\mathbf{R}_z(q_5)\mathbf{R}_x(90°)\mathbf{R}_z(q_6)\mathbf{R}_x(90°)t_x(0.088)\mathbf{R}_z(q_7)\, t_z(0.107)$$

The relative poses $^{i-1}\mathbf{T}_i$ are the seven rows of the table above (each $\mathbf{R}_x t_x \mathbf{R}_z t_z$). Numeric check at $\mathbf{q} = \mathbf{0}$: flange $(0.088,\ 0,\ 0.926)$ m. [checked]

Q3 two frames. Read the picture: $\{B\}$ origin at $(3, 5, 2)$ in $\{A\}$; $x_B = -z_A$, $z_B = +y_A$, so $y_B = z_B \times x_B = -x_A$. Columns:

$$^A\mathbf{T}_B = \begin{bmatrix} 0 & -1 & 0 & 3 \\ 0 & 0 & 1 & 5 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^A\mathbf{T}_B = t(3, 5, 2)\, \mathbf{R}_y(90°)\, \mathbf{R}_x(-90°)$$ [checked]

Q4 ("write a method") on paper: the transform is the ordered product of the ETS list, $E_1 E_2 \cdots E_m$, each factor taken from the six elementary matrices above.

Q5 PUMA 560 by ETS. See the Puma-like box:

$$^0\mathbf{T}_E = t_z(L_b)\mathbf{R}_z(q_1)\mathbf{R}_y(q_2)\, t_z(L_u)\mathbf{R}_y(q_3)\, t_z(L_f)\mathbf{R}_z(q_4)\mathbf{R}_y(q_5)\mathbf{R}_z(q_6)$$

Adjacent frames: $^0\mathbf{T}_1 = t_z(L_b)\mathbf{R}_z(q_1)$, $^1\mathbf{T}_2 = \mathbf{R}_y(q_2)$, $^2\mathbf{T}_3 = t_z(L_u)\mathbf{R}_y(q_3)$, $^3\mathbf{T}_4 = t_z(L_f)\mathbf{R}_z(q_4)$, $^4\mathbf{T}_5 = \mathbf{R}_y(q_5)$, $^5\mathbf{T}_6 = \mathbf{R}_z(q_6)$. *Where joints coincide* (1–2, 4–5–6) write the rotations back to back with no translation: the joints share a point. [checked]

Q6 workspaces (by hand). Furuta: spherical zone $\rho^2 + (z - L_0)^2 = L_1^2 + L_2^2$, $|z - L_0| \le L_2$. Puma-like wrist centre: a vertical-plane 2R annulus (radii $|L_u - L_f|$ to $L_u + L_f$) about the shoulder $(0, 0, L_b)$, spun by q_1: a spherical shell, inner radius 0.2 m, outer radius 0.8 m, centre $(0, 0, 1.0)$ m (ignoring joint limits and collisions). [checked]

SLIDE CHECK notebook Q3 filled-in answer, cell 46

`ty(5) tx(3) Ry(90) Rx(90)` misses $t_z(2)$ and the last rotation must be $\mathbf{R}_x(-90°)$: that sequence gives

$$\begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & 5 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which has origin $(3, 5, 0)$ and both $y_B = +x_A$ and $z_B = -y_A$ flipped against the picture. Compare with the picture before trusting a printed answer. [checked]

Reading a pose from two drawn frames

PREMISE drawn frames **Given:** a picture of two frames $\{A\}$, $\{B\}$ (or a numeric ${}^A\mathbf{T}_B$)
Find: ${}^A\mathbf{T}_B$, or a description of $\{B\}$ in $\{A\}$, and the matching ETS

STEPS Transform between two drawn frames, and the reverse

1. Write the *origin* of $\{B\}$ in $\{A\}$ coordinates: it is the last column.
2. For each axis of $\{B\}$ ask "which axis of $\{A\}$ does it point along, with which sign?". Write $\hat{x}_B, \hat{y}_B, \hat{z}_B$ as the three *columns*.
3. Check: each column has length 1, $\hat{x}_B \times \hat{y}_B = \hat{z}_B$ (if you get $-\hat{z}_B$ you mis-read one axis).
4. ETS (optional): ${}^A\mathbf{T}_B = t(x, y, z)\,\mathbf{R}_a(\cdot)\,\mathbf{R}_b(\cdot)$; test the guess by multiplying the two 3×3 rotations and comparing the columns.
5. Reverse (matrix given): the columns tell you where $\{B\}$'s axes point; the last column is its origin.

How this is examined: worked solutions

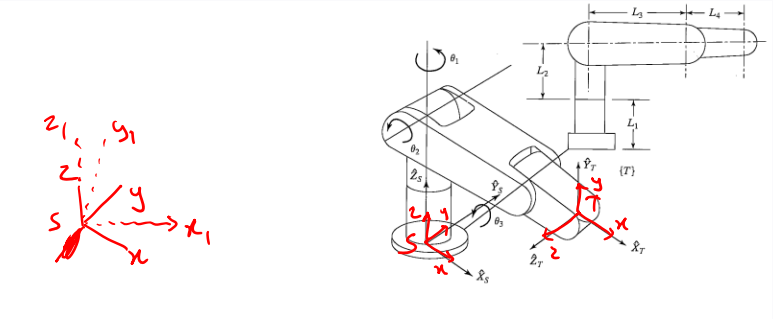
Scenarios: "given the joint angles, where is the tool and how is it oriented?" (W13 Q1; SCARA, elbow arm); "write the transform between two drawn frames" (notebook Q3, below); "find the frame given the transform" (below); "spot the error" (proximal vs distal, DH frame ≠ tool, order of multiplication). Marks are not published; "~1 mark per step" is an assumption.

TRAP W2: the five most common marks lost

(1) Order of multiplication (${}^A\mathbf{T}_B\,{}^B\mathbf{T}_C$, never reversed). (2) Treating the last DH frame as the tool. (3) Mixing proximal and distal formulas. (4) Wrong sign of a rotation because the drawn arrow is opposite to the right-hand rule. (5) Inverting ${}^A\mathbf{T}_B$ by negating $\mathbf{t}$ instead of $-\mathbf{R}^\top\mathbf{t}$. Calculator in degrees for degree data, radians for formulas.

EXAM-STYLE QUESTION W13 Q1: FK of an RRR arm by ETS [p. 4–5 of W13] [marks not published; indicative]

Write down the transformation matrix relating the tool frame $\{T\}$ to the station frame $\{S\}$ for the RRR manipulator below. Indicate intermediate frames. (Link data: L_1, L_2 vertical, L_3 upper arm, L_4 forearm; $L = (0.6, 0.5, 0.4, 0.3)$ m for the numbers.)



[W13 lec p. 4] — W13 Q1 arm with the lecturer's frames and ETS

Solution

Frames. $\{S\}$ on the base, z_S up along joint 1. $\{1\}$: $\{S\}$ turned θ_1 about z_S. $\{2\}$: shoulder, height $L_1 + L_2$ above $\{S\}$, with y_2 along the axis of joint 2. $\{3\}$: elbow, L_3 along the upper arm. $\{T\}$: tip, L_4 further; at $\theta = \mathbf{0}$ x_T points along the forearm and y_T up.
General rule: ${}^S\mathbf{T}_T = {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2\,{}^2\mathbf{T}_3\,{}^3\mathbf{T}_T$ (the lecturer's ETS, W13 p.4). The arrows of joints 2 and 3 lift the arm, which is *negative* about $+y$, hence the minus signs:

$${}^S\mathbf{T}_T = \mathbf{R}_z(\theta_1)\; t_z(L_1{+}L_2)\; \mathbf{R}_y(-\theta_2)\; t_x(L_3)\; \mathbf{R}_y(-\theta_3)\; t_x(L_4)\; \mathbf{R}_x(90°)$$

Intermediate frames ($c_i = \cos\theta_i$, $s_i = \sin\theta_i$):

$${}^S\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^1\mathbf{T}_2 = \begin{bmatrix} c_2 & 0 & -s_2 & 0 \\ 0 & 1 & 0 & 0 \\ s_2 & 0 & c_2 & L_1{+}L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2\mathbf{T}_3 = \begin{bmatrix} c_3 & 0 & -s_3 & L_3 \\ 0 & 1 & 0 & 0 \\ s_3 & 0 & c_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^3\mathbf{T}_T = \begin{bmatrix} 1 & 0 & 0 & L_4 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

EXAM-STYLE QUESTION W13 Q1 (cont.): multiplication, result, check

Multiply step by step ($c_{23} = \cos(\theta_2+\theta_3)$, $s_{23} = \sin(\theta_2+\theta_3)$, using $c_2c_3 - s_2s_3 = c_{23}$, $s_2c_3 + c_2s_3 = s_{23}$):

$$ {}^S\mathbf{T}_2 = {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2 = \begin{bmatrix} c_1c_2 & -s_1 & -c_1s_2 & 0 \\ s_1c_2 & c_1 & -s_1s_2 & 0 \\ s_2 & 0 & c_2 & L_1{+}L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^S\mathbf{T}_3 = {}^S\mathbf{T}_2\,{}^2\mathbf{T}_3 = \begin{bmatrix} c_1c_{23} & -s_1 & -c_1s_{23} & c_1c_2L_3 \\ s_1c_{23} & c_1 & -s_1s_{23} & s_1c_2L_3 \\ s_{23} & 0 & c_{23} & L_1{+}L_2{+}s_2L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Last factor: $\mathbf{R}_x(90^\circ)$ sends column 2 of the rotation to +column 3 and column 3 to −column 2, and $t_x(L_4)$ adds $L_4\times$ column 1 to the position. With $r = L_3c_2 + L_4c_{23}$:

$$ {}^S\mathbf{T}_T = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1\,r \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1\,r \\ s_{23} & c_{23} & 0 & L_1{+}L_2{+}L_3s_2{+}L_4s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Check at $\theta = 0$: tip $(L_3{+}L_4,\ 0,\ L_1{+}L_2)$ and $x_T = x_S$, $y_T = z_S$, $z_T = -y_S$, as drawn. At $\theta_2 = 90^\circ$, $\theta_3 = 0$: tip $(0, 0, L_1{+}L_2{+}L_3{+}L_4)$, arm straight up (as $\mathbf{R}_y(-\theta_2)$ predicts).
Numbers $\theta = (30^\circ, 20^\circ, 40^\circ)$: frame $\{2\}$ at $(0, 0, 1.1)$, frame $\{3\}$ at $(0.3255,\ 0.1879,\ 1.2368)$ m,

$$ {}^S\mathbf{T}_T = \begin{bmatrix} 0.4330 & -0.7500 & 0.5000 & 0.4554 \\ 0.2500 & -0.4330 & -0.8660 & 0.2629 \\ 0.8660 & 0.5000 & 0 & 1.4966 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$ [checked]

Marks lost by: $\mathbf{R}_y(+\theta_2)$ (arm dips when θ_2 grows), dropping the final $\mathbf{R}_x(90^\circ)$ (position right, orientation wrong), giving only the product with no intermediate frames when the question asks for them.

EXAM-STYLE QUESTION Mock 1 Q1 (SCARA arm) [mock marks: 6, assumed]

A SCARA arm stands on a column of height $H = 0.40$ m. Joint 1 turns about the vertical column axis; a horizontal link $a_1 = 0.35$ m leads to joint 2 (also vertical); a second horizontal link $a_2 = 0.25$ m carries joint 3, a prismatic joint that extends the tool *downwards* by d_3. $\{T\}$ has z_T down and x_T along link 2. (a) Write the ETS for ${}^0\mathbf{T}_T$ and a proximal DH table. (b) Evaluate ${}^0\mathbf{T}_T$ at $\theta_1 = 30^\circ$, $\theta_2 = -60^\circ$, $d_3 = 0.10$ m.

Solution
(a) ETS: ${}^0\mathbf{T}_T = t_z(H)\,\mathbf{R}_z(\theta_1)\,t_x(a_1)\,\mathbf{R}_z(\theta_2)\,t_x(a_2)\,t_z(-d_3)\,\mathbf{R}_x(180^\circ)$. Proximal rows $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: $(0, 0, H, \theta_1)$, $(a_1, 0, 0, \theta_2)$, $(a_2, 180^\circ, d_3, 0)$. The 180° twist turns z_3 down so that a positive d_3 lowers the tool.
(b) General result (see the SCARA box): $x = a_1c_1 + a_2c_{12}$, $y = a_1s_1 + a_2s_{12}$, $z = H - d_3$, rotation $\mathbf{R}_z(\theta_1+\theta_2)\mathbf{R}_x(180^\circ)$. With $\theta_1+\theta_2 = -30^\circ$: $x = 0.35(0.8660) + 0.25(0.8660) = 0.5196$, $y = 0.35(0.5) + 0.25(-0.5) = 0.0500$, $z = 0.40 - 0.10 = 0.30$ m.

$$ {}^0\mathbf{T}_T = \begin{bmatrix} 0.8660 & -0.5000 & 0 & 0.5196 \\ -0.5000 & -0.8660 & 0 & 0.0500 \\ 0 & 0 & -1 & 0.3000 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$ [checked]

Check: at all joints zero the tool is at $(a_1{+}a_2, 0, H - d_3)$ pointing down.

EXAM-STYLE QUESTION Practice Q&A (W2): SCARA with a tool yaw joint [marks not published]

A post of height H; revolute q_1 about the vertical; link L_1; revolute q_2 about the vertical; link L_2; a vertical slider that extends the tool downwards by d_3; a revolute q_4 about the vertical at the tool. (a) Write the ETS from $\{S\}$ to the tool $\{T\}$ and name each intermediate frame. (b) Multiply out position and orientation. (c) Evaluate for $H = 0.40$, $L_1 = 0.35$, $L_2 = 0.25$ m, $q_1 = 40^\circ$, $q_2 = -70^\circ$, $d_3 = 0.12$ m, $q_4 = 25^\circ$.

Solution
(a) ${}^S\mathbf{T}_T = t_z(H)\mathbf{R}_z(q_1)\,t_x(L_1)\mathbf{R}_z(q_2)\,t_x(L_2)t_z(-d_3)\,\mathbf{R}_z(q_4)$; ${}^S\mathbf{T}_1 = t_z(H)\mathbf{R}_z(q_1)$ (shoulder), ${}^1\mathbf{T}_2 = t_x(L_1)\mathbf{R}_z(q_2)$ (elbow), ${}^2\mathbf{T}_3 = t_x(L_2)t_z(-d_3)$ (tool carrier), ${}^3\mathbf{T}_T = \mathbf{R}_z(q_4)$.
(b) All three revolute axes are vertical, so the angles add: $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$, $z = H - d_3$, orientation $\mathbf{R}_z(q_1+q_2+q_4)$. At zero angles: $(L_1{+}L_2, 0, H - d_3)$.
(c) $q_1 + q_2 = -30^\circ$: $x = 0.35(0.7660) + 0.25(0.8660) = 0.4846$, $y = 0.35(0.6428) + 0.25(-0.5) = 0.1000$, $z = 0.40 - 0.12 = 0.28$ m, yaw $= 40 - 70 + 25 = -5^\circ$.

$$ {}^S\mathbf{T}_T = \begin{bmatrix} 0.9962 & 0.0872 & 0 & 0.4846 \\ -0.0872 & 0.9962 & 0 & 0.1000 \\ 0 & 0 & 1 & 0.28 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$ [checked]

Common wrong answer: $z = H + d_3$ (the tool would sit above the post top).

EXAM-STYLE QUESTION Practice Q&A (W2): proximal DH table for an elbow arm [marks not published]

A vertical post of height L_0; joint 1 turns about the vertical; joint 2 is a horizontal axis through the top of the post; the upper arm L_2, then joint 3 (parallel horizontal axis, the elbow) and the forearm L_3 to the tool point. At $q_2 = q_3 = 0$ both links point along $+x_1$ and positive q_2 raises the arm. (a) Draw the frames and fill in the proximal table including the tool row. (b) Write the ETS. (c) Find the tool point for $L_0 = 0.45$, $L_2 = 0.40$, $L_3 = 0.30$ m, $q = (35°, 50°, -30°)$.

Solution

(a) Axes 1 and 2 are perpendicular, so $\alpha_1 = 90°$; axes 2 and 3 are parallel, so $\alpha_2 = 0$. Rows $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: 1: $(0, 0, L_0, q_1)$; 2: $(0, 90°, 0, q_2)$; 3: $(L_2, 0, 0, q_3)$; tool: $(L_3, 0, 0, 0)$. Frame $\{i\}$ sits on axis i, and $a_2 = L_2$ is the distance from axis 2 to axis 3.
(b) ${}^0\mathbf{T}_E = t_z(L_0)\mathbf{R}_z(q_1)\,\mathbf{R}_x(90°)\mathbf{R}_z(q_2)\,t_x(L_2)\mathbf{R}_z(q_3)\,t_x(L_3)$.
(c) After $\mathbf{R}_x(90°)$, $z_2 = -y_1$, so a positive q_2 swings x_2 towards z_1 (up). In the arm's plane: $r = L_2c_2 + L_3c_{23} = 0.40(0.6428) + 0.30(0.9397) = 0.5390$ m; $z = L_0 + L_2s_2 + L_3s_{23} = 0.45+0.3064+0.1026 = 0.8590$ m; then spin by q_1: $x = r\cos 35° = 0.4415$, $y = r\sin 35° = 0.3092$.

$$\boxed{\mathbf{p} = (0.4415,\ 0.3092,\ 0.8590)\ \text{m}} \quad \text{[checked]}$$

Wrong answer to avoid: omitting the 90° twist (all rotations vertical, z stuck at 0.45 m).

EXAM-STYLE QUESTION Practice Q&A (W2): a camera on the wrist [marks not published]

A camera $\{C\}$ is fixed to the wrist $\{W\}$ of a robot with base $\{B\}$. ${}^B\mathbf{T}_W = t(0.5, 0.2, 0.6)\,\mathbf{R}_z(90°)$ and ${}^W\mathbf{T}_C = t(0.05, 0, 0.1)\,\mathbf{R}_y(90°)$ (m). The camera sees an object 0.8 m straight ahead along its own z_C axis. Find the object's position in $\{B\}$.

Solution

${}^B\mathbf{P} = {}^B\mathbf{T}_W\,{}^W\mathbf{T}_C\,{}^C\mathbf{P}$ (inner subscripts cancel), ${}^C\mathbf{P} = (0, 0, 0.8)^\top$. In $\{W\}$: $\mathbf{R}_y(90°)(0, 0, 0.8)^\top = (0.8, 0, 0)^\top$ ($\mathbf{R}_y(90°)$ sends z to $+x$), plus $(0.05, 0, 0.1)$ gives ${}^W\mathbf{P} = (0.85,\ 0,\ 0.10)$. In $\{B\}$: $\mathbf{R}_z(90°)(0.85, 0, 0.10)^\top = (0,\ 0.85,\ 0.10)$, plus $(0.5, 0.2, 0.6)$:

$$\boxed{{}^B\mathbf{P} = (0.50,\ 1.05,\ 0.70)\ \text{m}} \quad \text{[checked]}$$

The camera origin itself is at $(0.5, 0.25, 0.7)$. Wrong order ${}^W\mathbf{T}_C\,{}^B\mathbf{T}_W$ gives $(1.45, 0.2, -0.4)$: the subscripts do not cancel, which is the warning sign.

EXAM-STYLE QUESTION Practice Q&A (W2): reading a proximal DH table [marks not published]

Two rows of a proximal DH table $(a_{i-1}, \alpha_{i-1}, d_i, \theta_i)$: row 1 (0, 0, 0.3 m, 30°); row 2 (0.2 m, 90°, 0.05 m, −20°). Using ${}^{i-1}\mathbf{T}_i = \mathbf{R}_x(\alpha_{i-1})t_x(a_{i-1})\mathbf{R}_z(\theta_i)t_z(d_i)$, find the origin of $\{2\}$ in $\{0\}$.

Solution

Origin of $\{2\}$ in $\{1\}$: $\mathbf{R}_x(90°)\left[(0.2, 0, 0)^\top + \mathbf{R}_z(\theta_2)(0, 0, 0.05)^\top\right] = \mathbf{R}_x(90°)(0.2, 0, 0.05)^\top = (0.2,\ -0.05,\ 0)^\top$. ($\theta_2$ turns the frame about its own z_2 axis through the origin, so it does not move the origin.) In $\{0\}$: rotate by $\mathbf{R}_z(30°)$ and add $(0, 0, 0.3)$: $x = 0.2\cos 30° + 0.05\sin 30° = 0.1982$, $y = 0.2\sin 30° - 0.05\cos 30° = 0.0567$, $z = 0.3$.

$$\boxed{(0.1982,\ 0.0567,\ 0.30)\ \text{m}} \quad \text{[checked]}$$

The distal formula with the same numbers would give $(0.1970, 0.0347, 0.35)$: state "proximal (Craig)" on your script.

EXAM-STYLE QUESTION Find the frame from the transform [added] [marks not published]

The pose of $\{B\}$ in $\{A\}$ is ${}^A\mathbf{T}_B = \begin{bmatrix} 0 & 0 & 1 & 0.4 \\ 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. (a) State where the origin of $\{B\}$ is and which way its three axes point in $\{A\}$; verify that it is a valid pose. (b) Write it as an ETS. (c) A point is at ${}^B\mathbf{P} = (0, 0, 0.1)^\top$ m; find ${}^A\mathbf{P}$. (d) Find ${}^B\mathbf{T}_A$.

Solution

(a) Origin $(0.4, 0.2, 0.5)$ m. Columns: $\hat{x}_B = (0, 1, 0) = \hat{y}_A$, $\hat{y}_B = (0, 0, 1) = \hat{z}_A$, $\hat{z}_B = (1, 0, 0) = \hat{x}_A$. Unit length, orthogonal, $\hat{x}_B \times \hat{y}_B = \hat{y}_A \times \hat{z}_A = \hat{x}_A = \hat{z}_B$ so $\det = +1$.
(b) $t(0.4, 0.2, 0.5)\,\mathbf{R}_y(90°)\mathbf{R}_z(90°)$ (equally $\mathbf{R}_x(90°)\mathbf{R}_y(90°)$): $\mathbf{R}_y(90°)\mathbf{R}_z(90°)$ has columns $(0, 1, 0), (0, 0, 1), (1, 0, 0)$ ✓.
(c) ${}^A\mathbf{P} = \mathbf{R}\,(0, 0, 0.1)^\top + (0.4, 0.2, 0.5)^\top = (0.1, 0, 0)^\top + (0.4, 0.2, 0.5)^\top = (0.5,\ 0.2,\ 0.5)$ m.
(d) $\mathbf{R}^\top = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $-\mathbf{R}^\top\mathbf{t} = -(0.2, 0.5, 0.4)^\top$:

$$\boxed{{}^B\mathbf{T}_A = \begin{bmatrix} 0 & 1 & 0 & -0.2 \\ 0 & 0 & 1 & -0.5 \\ 1 & 0 & 0 & -0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}} \quad \text{[checked]}$$

Check: ${}^B\mathbf{T}_A(0.5, 0.2, 0.5, 1)^\top = (0, 0, 0.1, 1)^\top$.

EXAM-STYLE QUESTION Explain, 2–4 sentences [marks not published]

Why does DH need only four numbers per link where a general transform needs six, and what is the price?

Solution

Two bodies in 3D differ by 3 rotations and 3 translations, but DH *chooses* the frames by rule (x_i perpendicular to and intersecting the next axis), which removes two freedoms: $6 - 2 = 4$. The price is that the frame need not lie on the link (the tool or end of the link needs an extra constant transform), and the common normal jumps for nearly parallel axes, so the table is not continuous in the geometry. [p. 42–44]

EXAM-STYLE QUESTION Spot the error: the DH frame is not the tool [marks not published]

A planar 2R arm ($L_1 = 0.5$, $L_2 = 0.3$ m, $q_1 = 30°$, $q_2 = 45°$) has proximal rows $(0, 0, 0, q_1)$ and $(L_1, 0, 0, q_2)$. A student multiplies the two matrices and says: "the tip is the origin of ${}^0\mathbf{T}_2$, at $(0.433,\ 0.250)$ m". What is wrong and where is the tip?

Solution

In proximal DH, frame $\{2\}$ sits on the elbow axis, at the *near* end of link 2. So ${}^0\mathbf{T}_2$ gives the elbow $(0.5\cos 30°, 0.5\sin 30°) = (0.433, 0.250)$. Link 2 (length L_2) needs a third row $(L_2, 0, 0, 0)$, i.e. $t_x(L_2)$: $x = 0.433 + 0.3\cos 75° = 0.5107$, $y = 0.250 + 0.3\sin 75° = 0.5398$.

$$\boxed{\text{tip} = (0.5107,\ 0.5398)\ \text{m}}$$ [checked]

Same trap as the slide's "3 is not E" ([p. 32]).

Which recipe for which robot? [added]

Robot (given → find)	Method	Solutions
RP: $(x, y) \to S, q_1$	polar coordinates	1 (2 if $S < 0$ allowed)
PR: $(x, y) \to d_1, \theta$	$\sin\theta = y/L$, then $d_1 = x - L\cos\theta$	2
RR: $(x, y) \to q_1, q_2$	cosine rule, ± sine, atan2	2
RRR planar: (x, y, φ)	wrist point, then RR, $q_3 = \varphi - q_1 - q_2$	2
RRP: (x, y, φ) or (x, y, z)	quadratic in d_3, or polar in 3D	2
RRR in 3D (W13 Q3)	top view → θ_1, side view → planar RR	4
6-DoF, spherical wrist	wrist centre; q_1–q_3; then q_4–q_6	up to 8
no closed form	Newton–Raphson with $\mathbf{J}^\dagger$	depends on start

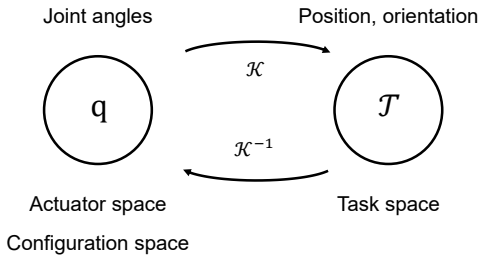
If the question says…

"derive the IK for (x_T, y_T, z_T)" ..... top view, then side view (p. 54)
"sketch the workspace" ..... outer and inner radius, sweep (p. 46)
"how many solutions" ..... count branches (table, §Multiple solutions)
"is the point reachable" ..... test $|\cos q_2| \le 1$ (§Existence)
"given a pose, find six joints" ..... wrist centre first (§Spherical wrist)

The IK problem: premise [p. 7–11]

PREMISE inverse kinematics **Given:** a desired tool pose ξ^* (written ${}^0\mathbf{T}_E$, or x, y, z and roll–pitch–yaw, or position and quaternion [p. 9]), expressed in the base frame $\{0\}$; the link constants (L_i, H, …)

Find: the joint vector $\mathbf{q}$ (angles in rad for R joints, extensions in m for P joints) with $\mathcal{K}(\mathbf{q}) = \xi^*$



[W03 lec p. 7] — FK $\mathcal{K}$ maps joint (configuration, actuator) space to task space; IK $\mathcal{K}^{-1}$ goes back

The two maps [p. 7–11]

$$\xi_n = \mathcal{K}(\mathbf{q}) \ \text{(FK)}, \qquad \mathbf{q} = \mathcal{K}^{-1}(\xi_n) \ \text{(IK)}$$

SYMBOLS

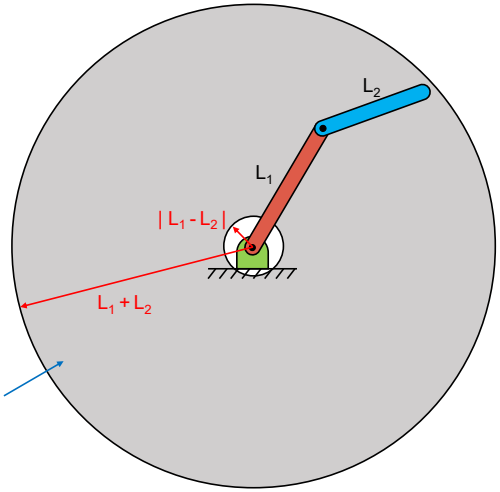
$\mathbf{q}$	joint vector, $q_i \in \mathbb{S}^1 = [0, 2\pi)$ for a revolute joint, extension for a prismatic joint	rad, m
$\mathcal{C}$	configuration (joint) space: the set of all $\mathbf{q}$ (two R joints: $\mathbb{S}^1 \times \mathbb{S}^1$) [p. 8]	–
$\mathcal{T}$	task space: the set of all tool poses ξ_n [p. 9]	–
$\mathcal{K}$	forward kinematics: many-to-one and nonlinear	–

Use: FK is always a formula; IK is an equation to solve. Because $\mathcal{K}$ is nonlinear, IK can have 0, 1, several or infinitely many solutions [p. 12].

The three key questions [p. 12]: *does a solution exist? how many? which method?*

Does a solution exist? Workspace [p. 12–16]

Reachable workspace W_R: points the tool can reach in *at least one* orientation. **Dexterous** workspace W_D: points it can reach in *every* orientation [p. 13]. A target given as a full pose must lie in W_D; a target given as a position only must lie in W_R. Always $W_D \subseteq W_R$.



[W03 lec p. 15] — planar RR, $L_1 = 1$, $L_2 = 0.7$ m, all joint angles free: reachable workspace = annulus between $|L_1 - L_2|$ and $L_1 + L_2$; dexterous workspace is empty

Reach condition for two links [p. 15]

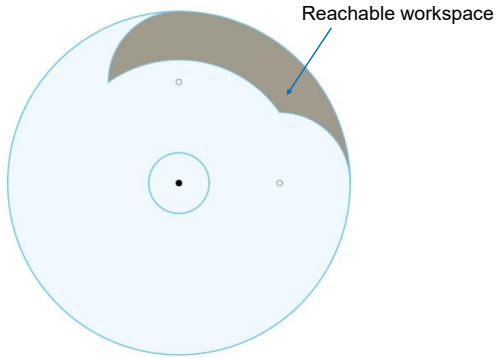
$$|L_1 - L_2| \ \le \ r \ \le \ L_1 + L_2, \qquad r = \sqrt{x^2 + y^2}$$

SYMBOLS

r	distance from the base axis to the target	m
L_1, L_2	link lengths	m

Use: test reachability before solving. Fully stretched gives $r = L_1 + L_2$ (outer edge); fully folded gives $r = |L_1 - L_2|$ (inner edge). Equivalent to $|\cos q_2| \le 1$.

Example (slide numbers): $L = (1.0, 0.7)$ m gives the annulus $0.3 \le r \le 1.7$ m. Is $(0, 0.2)$ reachable? $r = 0.2 < 0.3$, and $\cos q_2 = (0.04 - 1.49)/1.4 = -1.036 < -1$: no. [checked]



[W03 lec p. 16] — same arm with $q_1, q_2 \in [0, \pi/2]$: the reachable workspace shrinks to a crescent; dexterous workspace still empty

Joint limits [p. 16]. The boundary of the crescent is made of four arcs: the stretched arc $q_2 = 0$

(radius 1.7 m), the arc at the other q_2 limit (radius $\sqrt{1^2+0.7^2} = 1.221$ m at $q_2 = \pi/2$), and the two circles of radius $L_2 = 0.7$ m traced by the forearm about the elbow with q_1 held at 0 and at $\pi/2$ (the second reaches $x = -0.7$ m). [checked]

STEPS Sketch a workspace (W13 Q2) [added]

1. Name the frame and the point the sketch is centred on (the shoulder, not the base, if joint 1 is a pure lift).
2. Fold the arm: smallest distance from that point to the tool. Stretch it: largest distance. (Two links: $|L_a - L_b|$ and $L_a + L_b$.)
3. Sweep: every free joint that spins the arm about an axis through the centre turns the 2D annulus into a ring (planar) or a shell (3D).
4. Draw the outline and a second view (side and top), both radii and the height of the centre labelled to scale.
5. Say: reachable region, dexterous region (usually empty), what a joint limit would remove, and the one-line reason (cosine rule).

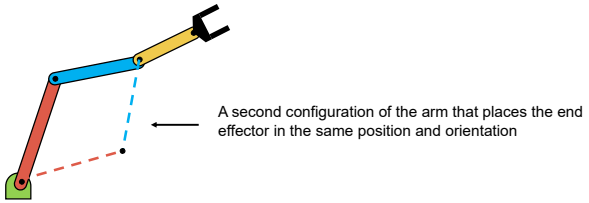
Workspaces of the simple robots (no joint limits) [added]

robot	reachable workspace W_R
RP, $S \in [S_{\min}, S_{\max}]$	annulus $S_{\min} \le r \le S_{\max}$ about the base
PR, $d_1 \in [0, D]$, link L	"stadium": rectangle $D \times 2L$ plus a half-disc of radius L at each end
RR	annulus $\lvert L_1 - L_2\rvert \le r \le L_1 + L_2$; if $L_1 = L_2$ a full disc
RRR planar	disc of radius $L_1 + L_2 + L_3$; hole of radius $2L_{\max} - \sum L_i$ if that is positive
RRR in 3D (Q2)	spherical shell about the shoulder, $\lvert L_3 - L_4\rvert \le \rho \le L_3 + L_4$
RRP (spherical)	spherical shell about the joint-2 axis point, $d_{3\,\min} \le \rho \le d_{3\,\max}$

Use: dexterous workspaces: RR, RP, PR, 3D RRR: empty (position fixes orientation). Planar RRR: tool orientation is free, so the wrist point $\mathbf{p} - L_3[\cos\varphi, \sin\varphi]^\top$ must stay in the RR annulus for every φ, which shrinks the annulus by L_3 on both sides: $L_3 + \lvert L_1 - L_2\rvert \le r \le L_1 + L_2 - L_3$ (for $r \ge L_3$).

Example: planar RRR, $L = (1.0, 0.7, 0.3)$: reachable disc $r \le 2.0$ m; dexterous annulus $0.6 \le r \le 1.4$ m. [checked]

Multiple solutions and how to choose [p. 17–20]



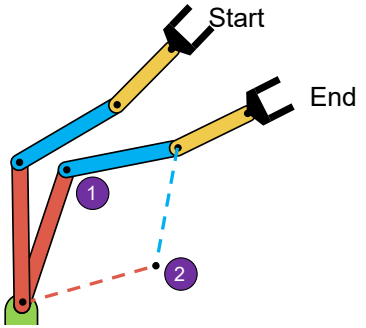
[W03 lec p. 17] — the dashed arm is a second configuration that puts the tool at the same position and orientation (elbow flip)

Number of solutions (interior of the workspace)

problem	number
RP, position	1; 2 if $S < 0$ allowed: $(-S, q_1 + \pi)$
PR, position	2: θ and $\pi - \theta$
RR, position	2 (elbow, $\pm q_2$)
RRR planar, pose (x, y, φ)	2
RRR planar, position only	infinitely many (1 DoF spare)
RRP spatial $(\theta_1, \theta_2, d_3 \ge 0)$	2
RRR in 3D, position (Q3)	4: shoulder × elbow
6-DoF spherical wrist, full pose	up to 8: $2 \times 2 \times 2$ [added]
general 6R serial arm	up to 16 [added](standard result, not on slides)

Use: on the workspace boundary two branches merge ($\sin q_2 = 0$): one solution. Outside: none. If an axis passes through the target ($x = y = 0$ for θ_1): infinitely many θ_1.

The slides state: a 6-DoF pose is 6 numbers and the arm has 6 joints, so there is no *continuous* redundancy [p. 33]; discrete solutions remain.



[W03 lec p. 18] — start and end poses: solution 1 is closer to the start than solution 2

Choosing one [p. 18–19]**:** closeness to the current configuration (absolute or weighted change of joint angles; number of joints to move; type of joints: prefer small distal joints over big proximal ones), or avoiding an obstacle (p. 19: only solution 1 is collision-free). Also apply joint limits *after* listing all branches.

How IK is solved [p. 20]**:** *closed form* (algebra or geometry; non-iterative; all solutions known) or *numerical* (iterative; slower; copes with long chains).

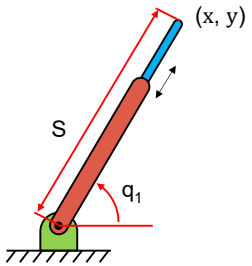
TRAP
Answering with one branch. Even if the question asks for "the" solution, list both, then say which you choose and why.

The analytic approach [p. 22]

STEPS Closed-form IK, slide recipe + exam layout

1. Sketch the robot, mark frame {0} and the positive direction of every joint.
2. Write the FK (position, and orientation if the pose is given).
3. Use geometry or algebra to isolate the joint variables: project to views (top, side); reduce to planar 2R triangles.
4. Solve with atan2; keep every branch (±).
5. State when no solution exists ($|\cos| > 1$) and how many solutions there are.
6. Check one branch by FK at a number.

Planar RP arm [p. 23–24]



[W03 lec p. 23] — planar RP arm: revolute q_1, then prismatic extension S along the link, tip at (x, y)

PREMISE RP **Given:** tip position (x, y) in {0}, m
Find: extension S (m) and angle q_1 (rad)

Polar coordinates [p. 24]

FK: $x = S\cos q_1,\ y = S\sin q_1$. Square and add; divide:

$$S = \sqrt{x^2 + y^2}, \qquad q_1 = \operatorname{atan2}(y, x)$$

SYMBOLS

S	extension of the prismatic joint, measured from the joint-1 axis	m
q_1	revolute angle from the x_0 axis, anticlockwise positive	rad
x, y	tip position in {0}	m

Use: the slide writes $q_1 = \tan^{-1}(y/x)$; use atan2 so the quadrant is right. If $S < 0$ is allowed, $(-S,\ q_1 + \pi)$ is a second solution.

Example: $(x, y) = (0.3, 0.4)$: $S = 0.5$ m, $q_1 = 53.130°$. $(-0.3, 0.4)$: $S = 0.5$, $q_1 = 126.870°$ ($\tan^{-1}$ would give $-53.13°$, wrong quadrant). FK check: $0.5(\cos 53.13°, \sin 53.13°) = (0.3, 0.4)$. [checked]
Reachability: $S_{\min} \le S \le S_{\max}$ (e.g. $0.2 \le S \le 0.8$ m: $(0.9, 0.5)$ needs $S = 1.030$ m, no).

PR arm (prismatic first, then revolute) [added]

PREMISE PR **Given:** tip (x, y) in {0}; slider d_1 along x_0; one revolute joint θ at the slider carrying a link of length L
Find: d_1 (m) and θ (rad)

FK: $x = d_1 + L\cos\theta,\ y = L\sin\theta$. The slider cannot change y, so θ comes from y alone:

$$\sin\theta = \frac{y}{L}, \quad \theta = \operatorname{atan2}\left(y,\ \pm\sqrt{L^2 - y^2}\right), \quad d_1 = x - L\cos\theta$$

Two solutions (θ and $\pi - \theta$); none if $|y| > L$; one if $|y| = L$.
Example: $L = 0.5$, target $(0.8, 0.3)$: $\theta = 36.870°$, $d_1 = 0.4$ m, or $\theta = 143.130°$, $d_1 = 1.2$ m. FK: $0.4 + 0.5\cos 36.87° = 0.8$ ✓, $0.5\sin 36.87° = 0.3$ ✓. [checked]

Planar RR arm: algebraic solution [p. 25–30]

PREMISE RR **Given:** tip position (x, y) in {0} (m), link lengths L_1, L_2
Find: joint angles q_1, q_2 (rad); q_1 from the x_0 axis, q_2 relative to link 1, both anticlockwise positive

DERIVATION of q_2 (square and add) [p. 26,28]

FK: $x = L_1\cos q_1 + L_2\cos(q_1 + q_2),\ y = L_1\sin q_1 + L_2\sin(q_1 + q_2)$. Square each and add:

$$\begin{aligned} x^2 + y^2 &= L_1^2(c_1^2 + s_1^2) + L_2^2(c_{12}^2 + s_{12}^2) + 2L_1L_2(c_1c_{12} + s_1s_{12}) \\ &= L_1^2 + L_2^2 + 2L_1L_2\cos\left((q_1 + q_2) - q_1\right) = L_1^2 + L_2^2 + 2L_1L_2\cos q_2 \end{aligned}$$

using $c_1c_{12} + s_1s_{12} = \cos(q_{12} - q_1)$. Solve for the cosine, then take $\sin q_2 = \pm\sqrt{1 - \cos^2 q_2}$.

Planar RR: q_2 [p. 28]

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}, \quad \sin q_2 = \pm\sqrt{1 - \cos^2 q_2}, \quad q_2 = \operatorname{atan2}(\sin q_2, \cos q_2)$$

SYMBOLS

x, y	tip position in {0}	m
L_1, L_2	link lengths	m
q_2	elbow angle, link 2 relative to link 1	rad
±	the two elbow branches (two solutions [p. 28])	–

Use: first line of every 2-link problem. If $|\cos q_2| > 1$ the point is out of reach (no solution).

DERIVATION of q_1: the point is a rotated vector [p. 30]

Expand the compound angles $c_{12} = c_1c_2 - s_1s_2,\ s_{12} = s_1c_2 + c_1s_2$ and collect in c_1, s_1:

$$\begin{aligned} x &= (L_1 + L_2c_2)\,c_1 - (L_2s_2)\,s_1, \\ y &= (L_2s_2)\,c_1 + (L_1 + L_2c_2)\,s_1. \end{aligned}$$

Write $k_1 = L_1 + L_2c_2$ and $k_2 = L_2s_2$ (known once q_2 is known). Then $(x, y)^\top$ is $(k_1, k_2)^\top$ rotated by q_1, so the polar angles subtract: $\operatorname{atan2}(y, x) = q_1 + \operatorname{atan2}(k_2, k_1)$. Hence

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2\sin q_2,\ L_1 + L_2\cos q_2)$$

One q_1 for each q_2 (the slide: "two solutions since q_2 has two solutions").

Planar RR: q_1 [p. 30]

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}\left(L_2 \sin q_2,\ L_1 + L_2 \cos q_2\right)$$

general tool [p. 30]: $A\cos\varphi + B\sin\varphi = C \Rightarrow \varphi = \operatorname{atan2}\left(C,\ \pm\sqrt{A^2 + B^2 - C^2}\right) - \operatorname{atan2}(A, B)$

SYMBOLS

A, B, C	known coefficients of the trigonometric equation	same unit
φ	unknown angle	rad
$\pm$	the two roots of the equation	–

Use: second line of the 2-link problem, one q_1 per branch. The general tool solves any $A\cos\varphi + B\sin\varphi = C$ (equivalent form [added]: $\varphi = \operatorname{atan2}(B, A) \pm \arccos\left(C/\sqrt{A^2 + B^2}\right)$).

SLIDE CHECK slides use $\tan^{-1}$, p. 24, 30; W13 p. 7

The slides write $q_1 = \tan^{-1}(y/x)$ and $\tan^{-1}(\cdot/\cdot)$ everywhere. $\tan^{-1}$ returns only $(-90°, 90°)$, so it cannot tell $(1, 1)$ from $(-1, -1)$. Use atan2 [p. 29]. The lecturer's own W13 Q3 sketch writes $\theta_1 = \tan^{-1}(y/x)$; atan2 is what gets the quadrant right.

atan2 [p. 29]

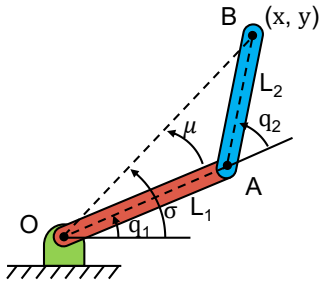
$$\operatorname{atan2}(y, x) = \begin{cases} \tan^{-1}(y/x) & x > 0 \\ \tan^{-1}(y/x) + \pi & x < 0,\ y \geq 0 \\ \tan^{-1}(y/x) - \pi & x < 0,\ y < 0 \\ +\pi/2 & x = 0,\ y > 0 \\ -\pi/2 & x = 0,\ y < 0 \\ \text{undefined} & x = y = 0 \end{cases} \qquad \text{range } [-\pi, \pi]$$

Use: first argument is the y-like (sine) quantity. $\operatorname{atan2}(-1, -1) = -135°$ *whereas* $\tan^{-1}(1) = 45°$. *Calculator:* `atan2(y,x)` *or Pol(x, y).*

TRAP spurious root

Solving only the x-equation $A\cos q_1 + B\sin q_1 = x$ gives two roots; just one also satisfies the y-equation. For the example below the roots are $7.576°$ and $-59.804°$, but $-59.804°$ gives $y = -0.8$. Use the atan2 pair, or check y.

Planar RR arm: geometric solution [p. 31]



[W03 lec p. 31] — triangle OAB: O base, A elbow, B tip (x, y); $\sigma = \operatorname{atan2}(y, x)$, μ = angle at O between OA and OB

DERIVATION triangle OAB [p. 31]

Sides: $|OA| = L_1$, $|AB| = L_2$, $|OB| = \sqrt{x^2 + y^2}$. The *interior* angle at A is $\pi - q_2$ (the joint angle q_2 is measured from the extension of link 1). Cosine rule:

$$x^2 + y^2 = L_1^2 + L_2^2 - 2L_1L_2\cos(\pi - q_2) = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$$

because $\cos(\pi - q_2) = -\cos q_2$. This is the algebraic result, so $\cos q_2$ is as in the box above. Sine rule (angle μ at O opposite L_2):

$$\frac{\sin\mu}{L_2} = \frac{\sin(\pi - q_2)}{\sqrt{x^2 + y^2}} = \frac{\sin q_2}{\sqrt{x^2 + y^2}}$$

With $\sigma = \operatorname{atan2}(y, x)$ the slide gives $q_1 = \sigma - \mu$. *Signs* [added]: μ has the sign of q_2; for $q_2 > 0$ the elbow lies on the clockwise side of OB and $q_1 = \sigma - \mu$; for $q_2 < 0$, $\mu < 0$ and $q_1 = \sigma + |\mu|$. Using $\mu = \operatorname{atan2}(L_2\sin q_2,\ L_1 + L_2\cos q_2)$ gives both automatically, and then $q_1 = \sigma - \mu$ equals the algebraic formula.

TRAP

Cosine rule with the joint angle: $x^2 + y^2 = L_1^2 + L_2^2 - 2L_1L_2\cos q_2$ is wrong. For the example below it gives $\pm 114.925° = 180° - 65.075°$ (the supplement). Always write "interior angle at the elbow $= \pi - q_2$". [checked]

ALTERNATIVE APPROACHES

A. Algebraic (square and add)

Needs only FK.
$\cos q_2$ from $x^2 + y^2$.
$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(k_2, k_1)$, $k_1 = L_1 + L_2c_2$, $k_2 = L_2s_2$.
Both branches: $\pm\sin q_2$.
Works for any number of links if you can reduce to $A\cos\varphi + B\sin\varphi = C$.

B. Geometric (triangle OAB)

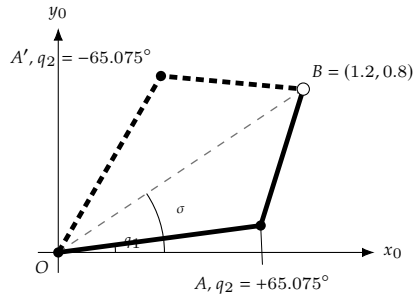
Needs a clean figure.
Cosine rule with **interior** angle $\pi - q_2$.
$q_1 = \sigma - \mu$ with μ from the sine rule.
Both branches: mirror the triangle about OB.
Fast for planar 2-link and 3D arms after the top/side view; full marks if the triangle is drawn and labelled.

STEPS IK of any 2-link planar sub-problem (W13 Q3) [added]

1. Write FK. Find the shoulder point and the target in the plane (subtract offsets and base height first).
2. $\cos q_2 = \dfrac{(\text{distance})^2 - L_1^2 - L_2^2}{2L_1L_2}$; check $|\cos q_2| \leq 1$.
3. $q_2 = \operatorname{atan2}(\pm\sqrt{1 - c_2^2}, c_2)$: two branches.
4. $q_1 = \operatorname{atan2}(\text{target direction}) - \operatorname{atan2}(L_2s_2,\ L_1 + L_2c_2)$ for each branch.
5. Check one branch by FK. Sanity: stretched arm $(L_1 + L_2, 0) \Rightarrow q = (0, 0)$.

WORKED RR, both elbows, IK→FK check

$L = (1.0, 0.7)$ m, target $(1.2, 0.8)$ m, $\{0\}$ at the base, degrees. $r^2 = 2.08$, $\cos q_2 = \dfrac{1.44 + 0.64 - 1 - 0.49}{1.4} = 0.421429$, $q_2 = \pm 65.075°$. $\sigma = \operatorname{atan2}(0.8, 1.2) = 33.690°$.
$q_2 = +65.075°$: $\mu = \operatorname{atan2}(0.7\sin 65.075°,\ 1 + 0.7\cos 65.075°) = \operatorname{atan2}(0.6348, 1.2950) = 26.114°$, $q_1 = 33.690 - 26.114 = \mathbf{7.576}°$.
$q_2 = -65.075°$: $\mu = -26.114°$, $q_1 = 33.690 + 26.114 = \mathbf{59.804}°$.
FK check ($q_2 > 0$): $x = \cos 7.576° + 0.7\cos 72.651° = 0.9913 + 0.2087 = 1.2000$ ✓, $y = \sin 7.576° + 0.7\sin 72.651° = 0.1318 + 0.6682 = 0.8000$ ✓. Other branch ($q_1 = 59.804°$, $q_1 + q_2 = -5.271°$): $x = 0.5030 + 0.6970 = 1.2000$ ✓, $y = 0.8643 - 0.0643 = 0.8000$ ✓. [checked]



[drawn for this booklet] both IK branches reach B; solid: $q = (7.576°, 65.075°)$, dashed: $(59.804°, -65.075°)$. For $q_2 > 0$ the elbow is on the clockwise side of OB.

Planar RRR with tool angle, RRP with tool angle [added]

PREMISE RRR **Given:** tool position (x, y) and tool angle φ (from x_0) in $\{0\}$; L_1, L_2, L_3
Find: $\theta_1, \theta_2, \theta_3$ (all relative, anticlockwise)

STEPS Planar RRR: position and tool angle (notebook Q10)

1. Wrist point $\mathbf{p}_W = (x - L_3\cos\varphi,\ y - L_3\sin\varphi)$ (end of link 2).
2. Solve the RR problem to $\mathbf{p}_W$: $\theta_2 = \operatorname{atan2}(\pm\sqrt{1-c_2^2}, c_2)$, θ_1 by the formula above.
3. $\theta_3 = \varphi - \theta_1 - \theta_2$ (the three relative angles add to the tool angle).
4. Two solutions. FK check with the pose (x, y, φ).

WORKED RRR
$L = (1.0, 0.7, 0.3)$, $(x, y) = (1.2, 0.9)$, $\varphi = 60°$: $\mathbf{p}_W = (1.2 - 0.15,\ 0.9 - 0.2598) = (1.0500, 0.6402)$; $\theta_2 = \pm 89.085°$; $\theta_1 = -3.319°$ or $66.061°$; $\theta_3 = -25.766°$ or $83.024°$. Both reproduce $(1.2, 0.9, 60°)$. Without φ the problem has infinitely many solutions. [checked]

PREMISE RRP, planar **Given:** tool position (x, y) and slider direction φ $(= \theta_1 + \theta_2)$ in $\{0\}$; first link L_1; slider extension $d_3 \geq 0$ along the second link
Find: θ_1, θ_2, d_3

FK: $\mathbf{p} = L_1(\cos\theta_1, \sin\theta_1) + d_3(\cos\varphi, \sin\varphi)$. Move the known slider term across and take the length of both sides: $|\mathbf{p} - d_3\mathbf{u}|^2 = L_1^2$ with $\mathbf{u} = (\cos\varphi, \sin\varphi)$, which expands to a quadratic in d_3:

$$d_3^2 - 2p\,d_3 + (x^2 + y^2 - L_1^2) = 0, \quad p = x\cos\varphi + y\sin\varphi, \quad d_3 = p \pm \sqrt{p^2 - (x^2 + y^2 - L_1^2)}$$

$$\theta_1 = \operatorname{atan2}(y - d_3\sin\varphi,\ x - d_3\cos\varphi), \qquad \theta_2 = \varphi - \theta_1$$

Real roots need $p^2 \geq x^2 + y^2 - L_1^2$; keep roots with $d_3 \geq 0$ (and within the stroke).
Example: $L_1 = 0.5$, $(x, y) = (0.7, 0.4)$, $\varphi = 30°$: $p = 0.8062$, discriminant 0.25, $d_3 = 1.3062$ or 0.3062 m; $(\theta_1, \theta_2) = (29.589°, 0.411°)$ for $d_3 = 0.3062$ (the other: $(-149.589°, 179.589°)$). Both give $(0.7, 0.4)$. [checked]

RRP arm in 3D (spherical type) [added]

PREMISE RRP, 3D **Given:** tip (x, y, z) in $\{0\}$; θ_1 about z_0; at height H a horizontal-axis joint tilts the boom by θ_2 from vertical; d_3 slides along the boom
Find: θ_1, θ_2, d_3

FK: $\mathbf{p} = (c_1 d_3 s_2,\ s_1 d_3 s_2,\ H + d_3 c_2)^\top$. Then $r = \sqrt{x^2 + y^2} = d_3 s_2$ and $z' = z - H = d_3 c_2$:

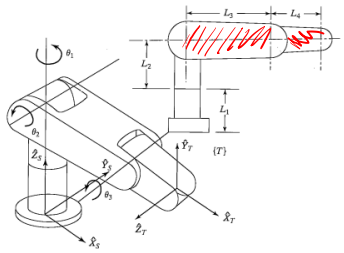
$$\theta_1 = \operatorname{atan2}(y, x), \quad d_3 = \sqrt{r^2 + z'^2}, \quad \theta_2 = \operatorname{atan2}(r, z')$$

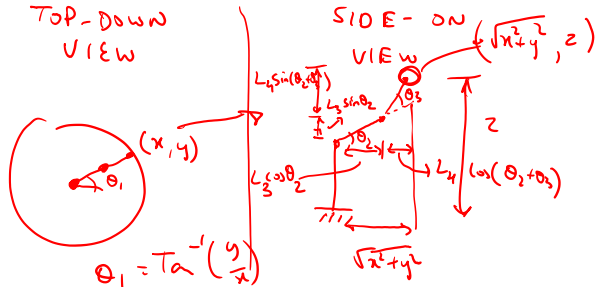
Second solution $(\theta_1 + \pi,\ -\theta_2,\ d_3)$; two more with $d_3 < 0$ if allowed. *Example:* $H = 0.5$, $\mathbf{p} = (0.3, 0.4, 0.9)$: $\theta_1 = 53.130°$, $r = 0.5$, $z' = 0.4$, $d_3 = 0.6403$ m, $\theta_2 = 51.340°$. FK: $(0.3, 0.4, 0.9)$ ✓. [checked]

3D RRR arm (W13 Q3): top view, side view [p. W13 7]

$z = L_1 + L_2 + L_3\sin\theta_2 + L_4\sin(\theta_2 + \theta_3)$

You are given the position of the end effector (x_T, y_T, z_T) $\sqrt{x^2+y^2} = L_3\cos\theta_2 + L_4\cos(\theta_2 + \theta_3)$





[W13 lec p. 7] — lecturer's solution sketch: top-down view gives θ_1; side-on view gives a planar RR problem in $(\sqrt{x^2 + y^2}, z)$

IK of the W13 RRR arm

FK: $x = c_1 r$, $y = s_1 r$, $z = L_1 + L_2 + L_3 s_2 + L_4 s_{23}$, with $r = L_3 c_2 + L_4 c_{23}$. Then

$$\theta_1 = \operatorname{atan2}(y, x)\ (r > 0) \ \text{ or } \ \theta_1 + \pi\ (r < 0)$$

$$\cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2L_3L_4}, \quad z' = z - L_1 - L_2, \quad \theta_3 = \operatorname{atan2}(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3)$$

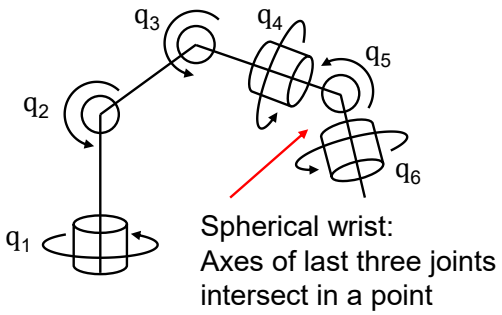
$$\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4\sin\theta_3,\ L_3 + L_4\cos\theta_3)$$

SYMBOLS

L_1, L_2	vertical offsets: base to shoulder height $L_1 + L_2$	m
L_3, L_4	upper arm and forearm (shoulder to elbow, elbow to tip)	m
θ_1	rotation about z_S	rad
θ_2, θ_3	rotations about horizontal axes, positive lifts the arm (about $-y$)	rad
r	horizontal radius of the tip, signed ($r = -\sqrt{x^2 + y^2}$ for the back-reaching branch)	m
z'	tip height above the shoulder	m

Use: any arm whose first axis is vertical and the rest lie in one plane. Four solutions: (θ_1 or $\theta_1 + \pi$) × ($\pm\theta_3$). Full derivation and numbers: exam question in this week's last section.

Six-DoF arm with a spherical wrist [p. 32–38]



[W03 lec p. 32] — 6-DoF arm; the last three axes meet in a point (spherical wrist)

PREMISE 6-DoF IK **Given:** desired tool pose ${}^0\mathbf{T}_E = \begin{bmatrix} \mathbf{R}_{3\times3} & \mathbf{T}_{3\times1} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}$ in $\{0\}$

Find: $q_1, \ldots, q_6$

Idea [p. 33–34]: a pose is 6 numbers and the arm has 6 joints. Because the three wrist joints rotate about *one point* (the wrist centre W), moving joints 4–6 cannot move W. So split the problem: **position of** $W \rightarrow q_1, q_2, q_3$; **orientation** $\rightarrow q_4, q_5, q_6$.

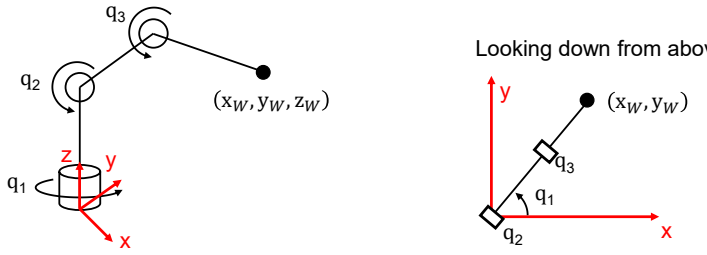
Wrist centre [added]; slide p. 34 gives no formula

$$ {}^0\mathbf{p}_W = {}^0\mathbf{p}_E - d_6\, {}^0\mathbf{R}_E \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} $$

SYMBOLS

${}^0\mathbf{p}_E$	tool position (last column of ${}^0\mathbf{T}_E$) in $\{0\}$	m
${}^0\mathbf{R}_E$	tool orientation in $\{0\}$; its third column is the approach axis $\hat{z}_E$	–
d_6	distance from the wrist centre to the tool point along $\hat{z}_E$	m
${}^0\mathbf{p}_W$	wrist centre (x_W, y_W, z_W) in $\{0\}$	m

Use: first step whenever a tool pose (not the wrist centre) is given. If the slide gives (x_W, y_W, z_W) directly, skip it.

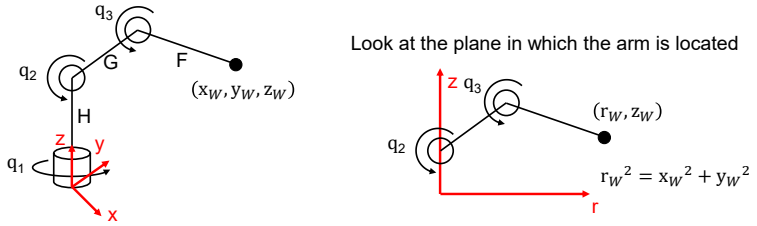


[W03 lec p. 35] — step 1: look down from above, $q_1 = \text{atan2}(y_W, x_W)$

DERIVATION the three position joints [p. 35–37]

Step 1 (top view). The arm lies in the vertical plane through z_0 and W: $q_1 = \text{atan2}(y_W, x_W)$. A second branch is $q_1 + \pi$ with the arm reaching backwards (then use $r_W \rightarrow -r_W$ below).
Step 2 (vertical plane). H = height of joint 2, G = link 2 (joint 2 to 3), F = link 3 (joint 3 to W), $r_W^2 = x_W^2 + y_W^2$. Triangle joint2–joint3–W: sides G, F and $\sqrt{r_W^2 + (z_W - H)^2}$. With q_3 the relative elbow angle (anticlockwise positive), the interior angle at joint 3 is $\pi - q_3$:

$$ r_W^2 + (z_W - H)^2 = G^2 + F^2 - 2GF\cos(\pi - q_3) = G^2 + F^2 + 2GF\cos q_3 $$



[W03 lec p. 36] — step 2: the plane that contains the arm, (r_W, z_W) with $r_W^2 = x_W^2 + y_W^2$

Position joints q_1, q_2, q_3 [p. 35–37], signed angles

$$ q_1 = \text{atan2}(y_W, x_W) $$

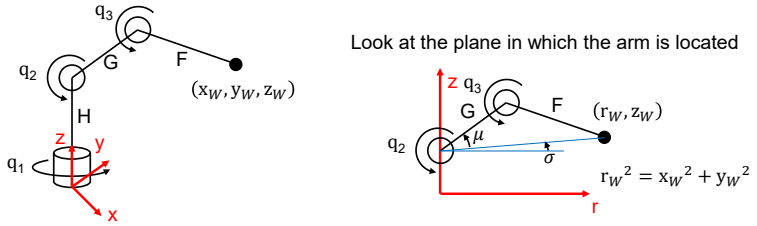
$$ \cos q_3 = \frac{r_W^2 + (z_W - H)^2 - F^2 - G^2}{2FG}, \quad q_3 = \text{atan2}\left(\pm\sqrt{1 - \cos^2 q_3},\ \cos q_3\right) $$

$$ \sigma = \text{atan2}(z_W - H,\ r_W), \quad \mu = \text{atan2}\left(F\sin q_3,\ G + F\cos q_3\right), \quad q_2 = \sigma - \mu $$

SYMBOLS

H	height of joint 2 above the base frame	m
G, F	link lengths joint 2 to 3, and joint 3 to the wrist centre	m
r_W	horizontal distance of W from the z_0 axis	m
σ	elevation of the line from joint 2 to W	rad
μ	angle at joint 2 between G and that line	rad

Use: after the wrist centre is known. Branches: q_1 or $q_1 + \pi$ (with $r_W \rightarrow -r_W$), and $\pm q_3$.



[W03 lec p. 37] — step 3: σ is the elevation of the line to W, μ the angle between it and link G

SLIDE CHECK elbow sign, pp. 36–37

Slide 36 writes $\cos(\pi - q_3)$ on the left of the equation. By the cosine rule the right-hand side equals $\cos q_3$ when q_3 is the relative elbow angle (as in the RR arm, and as μ on p. 37 assumes). Slide 37's $q_2 = \sigma + \mu$ is right only if q_3 is drawn as a clockwise bend (elbow up) and $\mu > 0$. With signed anticlockwise angles the formulas in the box above hold. In the exam, define your positive direction on the sketch and keep it. [checked]

WORKED wrist-arm position part, $\{0\}$ at the base

$H = 0.5$, $G = 0.4$, $F = 0.3$ m, $W = (0.4813, 0.4039, 0.5974)$ m. $q_1 = \text{atan2}(0.4039, 0.4813) = 40°$; $r_W = 0.6283$, $z_W - H = 0.0974$; $\cos q_3 = \dfrac{0.3948 + 0.0095 - 0.09 - 0.16}{0.24} = 0.6428$, $q_3 = \pm 50°$.

$q_3 = -50°$ (elbow up): $\sigma = 8.811°$, $\mu = \text{atan2}(0.3\sin(-50°),\ 0.4 + 0.3\cos 50°) = -21.189°$, $q_2 = \mathbf{30.0}°$.

$q_3 = +50°$ (elbow down): $\mu = +21.189°$, $q_2 = -12.378°$.

With $q_1 + \pi = 220°$ ($r_W \to -r_W$): $(q_2, q_3) = (150.0°, 50°)$ and $(192.378°, -50°)$. Four position solutions, all reproduce W by FK. With the wrist flip below: $4 \times 2 = 8$. [checked]

DERIVATION the wrist, step 4 [p. 38]

${}^0\mathbf{T}_6 = {}^0\mathbf{T}_3\,{}^3\mathbf{T}_6$. Joints 1–3 are now known, so ${}^0\mathbf{R}_3$ is known and

$$ {}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\,{}^0\mathbf{R}_6 \quad (\text{since } {}^0\mathbf{R}_3^{-1} = {}^0\mathbf{R}_3^\top) $$

${}^3\mathbf{R}_6$ is a product of three elementary rotations, "like the Euler angles of the previous lecture": read q_4, q_5, q_6 from it with compound-rotation algebra. For a wrist with axes z, y, z [added]: ${}^3\mathbf{R}_6 = R_z(q_4)R_y(q_5)R_z(q_6)$, so $r_{13} = c_4 s_5$, $r_{23} = s_4 s_5$, $r_{33} = c_5$, $r_{31} = -s_5 c_6$, $r_{32} = s_5 s_6$ (rows and columns of ${}^3\mathbf{R}_6$).

Z–Y–Z wrist angles [added]

$$ q_5 = \text{atan2}\left(\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\right), \quad q_4 = \text{atan2}(r_{23},\ r_{13}), \quad q_6 = \text{atan2}(r_{32},\ -r_{31}) $$

SYMBOLS

r_{ij}	element in row i, column j of ${}^3\mathbf{R}_6$	–
q_4, q_5, q_6	wrist joint angles	rad

Use: second solution (wrist flip): $q_5 \to -q_5$, $q_4 \to q_4 + \pi$, $q_6 \to q_6 + \pi$ (recompute with $-\sqrt{\cdot}$). If $q_5 = 0$ the wrist is singular: only $q_4 + q_6$ is fixed. A different wrist (e.g. x, y, z) needs its own extraction; derive it from the same product.

WORKED wrist angles from a numerical ${}^3\mathbf{R}_6$

$$ {}^3\mathbf{R}_6 = \begin{bmatrix} 0.4963 & -0.1684 & 0.8517 \\ -0.4014 & 0.8253 & 0.3971 \\ -0.7698 & -0.5390 & 0.3420 \end{bmatrix} $$

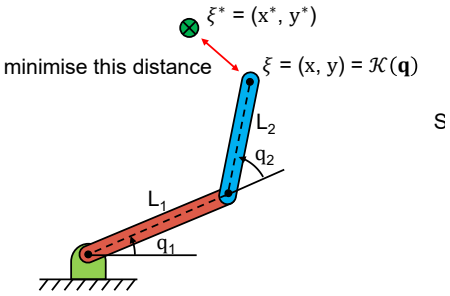
$q_5 = \text{atan2}(\sqrt{0.8517^2 + 0.3971^2},\ 0.3420) = \text{atan2}(0.9397, 0.3420) = 70°$; $q_4 = \text{atan2}(0.3971, 0.8517) = 25°$; $q_6 = \text{atan2}(-0.5390,\ 0.7698) = -35°$. Flip: $(-155°, -70°, 145°)$ gives the same matrix. [checked]

STEPS Decoupled IK of a 6-DoF arm with a spherical wrist (W13-style)

1. Wrist centre: ${}^0\mathbf{p}_W = {}^0\mathbf{p}_E - d_6\,{}^0\mathbf{R}_E[0\ 0\ 1]^\top$.
2. Top view: $q_1 = \text{atan2}(y_W, x_W)$ (also $q_1 + \pi$).
3. Vertical plane: $\cos q_3$, then $\pm q_3$; $q_2 = \sigma - \mu$.
4. ${}^3\mathbf{R}_6 = ({}^0\mathbf{R}_3)^\top\,{}^0\mathbf{R}_6$; read q_4, q_5, q_6 as Euler angles (both signs of q_5).
5. Count: 2 (q_1) ×2 (q_3) ×2 (wrist flip) = 8 for a PUMA-type arm. Check by FK.

Why a closed form exists [p. 55–56]: any 6-DoF serial chain of R and P joints is "solvable", but not always in closed form. A sufficient condition is that three neighbouring joint axes intersect at one point (the spherical wrist; Pieper's result [added]; three consecutive parallel axes also work [added]). The slides treat one generic 6-DoF arm; they do not give a PUMA 560 DH table in this week. The count 8 for a PUMA-type arm is the standard result [added].

Numerical IK [p. 40–52]



[W03 lec p. 42] — goal $\xi^* = (x^*, y^*)$; minimise the distance to $\xi = \mathcal{K}(\mathbf{q})$

Goal potential [p. 41–42]

$$ \mathbf{q}^* = \arg\min_{\mathbf{q}}\ G(\mathbf{q}), \qquad G(\mathbf{q}) = |\mathcal{K}(\mathbf{q}) \ominus \xi^*| = (x^* - x)^2 + (y^* - y)^2 \ \text{(position only)} $$

SYMBOLS

$G(\mathbf{q})$	goal potential: how far the tool is from the goal	m^2
$\ominus$	pose difference operator	–
$(x, y) = \mathcal{K}(\mathbf{q})$	current tool position	m

Use: adjust $\mathbf{q}$ until $\xi \to \xi^*$; the minimiser is the IK solution.

Newton–Raphson, scalar [p. 45–47]

$$ G(q) \approx G(q^0) + \frac{\partial G}{\partial q}(q^0)(q - q^0) \overset{G=0}{\Longrightarrow} q^{k+1} = q^k - \left(\frac{\partial G}{\partial q}(q^k)\right)^{-1} G(q^k), \quad \left|\frac{G(q^{k+1}) - G(q^k)}{G(q^k)}\right| \le \epsilon $$

SYMBOLS

$G(q)$	function whose root is wanted ($G : \mathbb{R} \to \mathbb{R}$)	–
q^k	k-th guess; q^0 the initial guess	rad
ϵ	stopping threshold on the relative change	–

Use: keep the first-order Taylor term only and solve for the zero. For IK use the residual form $G = \mathcal{F}(\mathbf{q}) - \mathbf{x}^*$ (p. 50), not the squared distance.

WORKED scalar Newton–Raphson [added]

One link, $L = 1$ m, wanted $x^* = 0.6$ m: $G(q) = \cos q - 0.6$, $G'(q) = -\sin q$. $q^0 = 1$ rad: $G = -0.0597$, $G' = -0.8415$, $q^1 = 1 - (-0.0597)/(-0.8415) = 0.9291$; $q^2 = 0.9273$; $q^3 = 0.9273 = \arccos 0.6$. [checked]

Jacobian update and pseudo-inverse [p. 48–52]

$$ \mathbf{e} = \mathbf{x}^* - \mathcal{F}(\mathbf{q}^i), \qquad \mathbf{q}^{i+1} = \mathbf{q}^i + \mathbf{J}^\dagger(\mathbf{q}^i)\,\mathbf{e}, \qquad \mathbf{J} = \frac{\partial \mathcal{F}}{\partial \mathbf{q}} \in \mathbb{R}^{m \times n} $$

SYMBOLS

$\mathbf{x}^* \in \mathbb{R}^m$	desired task vector	m
$\mathcal{F}(\mathbf{q})$	forward kinematics in task coordinates	m
$\mathbf{e}$	task-space error (goal minus current); stop when $\lvert\mathbf{e}\rvert \le \epsilon$	m
$\mathbf{J}$	Jacobian: m task rows, n joint columns	m/rad
$\mathbf{J}^\dagger$	(pseudo-)inverse of $\mathbf{J}$	rad/m

Use: one iteration = one FK, one $\mathbf{J}$, one solve. For a 2×2: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

W3

Which inverse? Decide from the shape of J [p. 51]

shape	$\mathbf{J}^\dagger$	meaning
square, $n = m$	$\mathbf{J}^{-1}$	exact step
fat, $n > m$ (redundant)	$\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$	infinitely many $\Delta\mathbf{q}$; picks the smallest
tall, $n < m$	$(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top$	no exact $\Delta\mathbf{q}$; least squares

Use: for planar 2R position IK $\mathbf{J}$ *is* 2×2: $\mathbf{J} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, $\det \mathbf{J} = L_1 L_2 \sin q_2$ *(singular when the arm is stretched or folded). Planar RRR with position only is fat* (2×3).

STEPS One Newton–Raphson iteration for IK

1. FK at $\mathbf{q}^i$; $\mathbf{e} = \mathbf{x}^* - \mathcal{F}(\mathbf{q}^i)$; if $|\mathbf{e}| \le \epsilon$ stop.
2. Write $\mathbf{J}(\mathbf{q}^i)$ in full (radians).
3. $\Delta\mathbf{q} = \mathbf{J}^\dagger\mathbf{e}$ (use the table for the shape).
4. $\mathbf{q}^{i+1} = \mathbf{q}^i + \Delta\mathbf{q}$; recompute $|\mathbf{e}|$ to confirm it fell.

WORKED one Newton step, RR, $\{0\}$ at base

$L = (1.0, 0.7)$, target $(1.2, 0.8)$, $\mathbf{q}^0 = (0.5, 0.5)$ rad. FK $= (1.2558, 1.0685)$, $\mathbf{e} = (-0.0558, -0.2685)$.

$\mathbf{J} = \begin{bmatrix} -1.0685 & -0.5890 \\ 1.2558 & 0.3782 \end{bmatrix}$, $\det \mathbf{J} = 0.3355$ ($= L_1 L_2 \sin 0.5$ ✓).

$\mathbf{J}^{-1} = \frac{1}{0.3355}\begin{bmatrix} 0.3782 & 0.5890 \\ -1.2558 & -1.0685 \end{bmatrix}$, $\Delta\mathbf{q} = (-0.5341,\ 1.0635)$ rad.

$\mathbf{q}^1 = (-0.0341,\ 1.5635)$ rad; $|\mathbf{e}|$: $0.2742 \to 0.2181$. It converges to $(7.576°, 65.075°)$ in 5 iterations. [checked]

WORKED fat Jacobian: planar RRR, position only [added]

$L = (1.0, 0.7, 0.3)$, $\mathbf{q} = (0.3, 0.5, 0.4)$ rad, want $\mathbf{e} = (0.05, -0.02)$ m. $\mathbf{J} = \begin{bmatrix} -1.0773 & -0.7818 & -0.2796 \\ 1.5517 & 0.5964 & 0.1087 \end{bmatrix}$, $\mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 1.8499 & -2.1683 \\ -2.1683 & 2.7754 \end{bmatrix}$ (det 0.4326). $\Delta\mathbf{q} = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\mathbf{e} = (0.0186, -0.0740, -0.0437)$ rad, $|\Delta\mathbf{q}| = 0.0879$; check $\mathbf{J}\Delta\mathbf{q} = (0.05, -0.02)$ ✓. [checked]

TRAP

$\mathbf{J}^\top$ instead of $\mathbf{J}^{-1}$ is gradient descent: a different direction, needs a step size. For tall $\mathbf{J}$, $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}$ is wrong ($\mathbf{J}\mathbf{J}^\top$ is singular). Degrees in $\mathbf{J}$ or in the update: radians only.

Why numerical IK can fail (name three; notebook Q4–Q6, p. 44): (1) $G = 0$ is impossible if ξ^* is outside the workspace (the best it can do is the nearest reachable point; for $(2, 0.5)$ with $L = (1, 0.7)$ the minimiser stops 0.362 m short); (2) it can stall in a local minimum, or blow up where $\mathbf{J}$ is singular ($\det \mathbf{J} = 0$ at $q_2 = 0$); (3) the answer depends on the initial guess; a redundant arm has infinitely many. [checked]

Closed form versus numerical [p. 54]

	closed form	numerical
speed	potentially faster, no iteration	iterative, slower
solutions	all known a priori	one near the start guess
many joints	harder	suits long chains
no exact answer	none	a close-enough answer

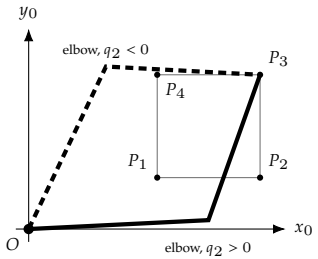
SLIDE CHECK p. 54

"a unique solution to all manipulator positions can be determined a priori" contradicts pp. 17 and 20 (several solutions). Read it as "a definite, complete set of solutions".

Simple scenarios: corners of a square, edge cases [added]

PREMISE square **Given:** planar RR, $L = (0.7, 0.6)$ m; the tool must visit the corners $(0.5, 0.2)$, $(0.9, 0.2)$, $(0.9, 0.6)$, $(0.5, 0.6)$ m in $\{0\}$
Find: (q_1, q_2) at each corner; one branch for the whole path

corner	$\cos q_2$	branch $q_2 > 0$	branch $q_2 < 0$
$P_1 = (0.5, 0.2)$	-0.6667	$(-34.34°,\ 131.81°)$	$(77.95°,\ -131.81°)$
$P_2 = (0.9, 0.2)$	0.0000	$(-28.07°,\ 90.00°)$	$(53.13°,\ -90.00°)$
$P_3 = (0.9, 0.6)$	0.3810	$(2.83°,\ 67.61°)$	$(64.55°,\ -67.61°)$
$P_4 = (0.5, 0.6)$	-0.2857	$(2.79°,\ 106.60°)$	$(97.60°,\ -106.60°)$



[drawn for this booklet] the arm at corner P_3 in both branches; r along the whole square stays between 0.538 and 1.082 m, inside $[0.1,\ 1.3]$ m.

Along the whole square, branch $q_2 > 0$ has $67.6° \le q_2 \le 131.8°$ and branch $q_2 < 0$ the mirror image: q_2 never reaches 0 or $\pm 180°$, so neither branch meets a singularity or the boundary, and the arm can follow the square without switching branches. **Choose** the branch by the start configuration (smallest joint change), joint limits (e.g. $q_2 \in [0, 150°]$ rules out the second column) or an obstacle. [checked]

Target out of reach / on the boundary / interior

target	result (RR, $L_1 = L_2 = 0.5$ m; RR slide arm)
$r > L_1 + L_2$ or $r < \lvert L_1 - L_2\rvert$	$\lvert\cos q_2\rvert > 1$: no solution (e.g. $(2, 0.5)$ for $L = (1, 0.7)$)
$r = L_1 + L_2$ (stretched)	$\cos q_2 = 1$: one solution, $q_2 = 0$ ($(1, 0) \to (0, 0)$)
$r = \lvert L_1 - L_2\rvert$ (folded)	$\cos q_2 = -1$: one solution, $q_2 = \pm 180°$
interior	two solutions
$L_1 = L_2$, target = base	$\cos q_2 = -1$ and q_1 arbitrary: infinitely many

Why it matters (explain, 2–4 sentences): at the boundary the two branches coincide and the Jacobian is singular ($\det \mathbf{J} = L_1 L_2 \sin q_2 = 0$), so a small move of the target needs a very large joint move; in a numerical solver this is where Newton steps blow up.

Practical (optional): `rrp_ik.py`

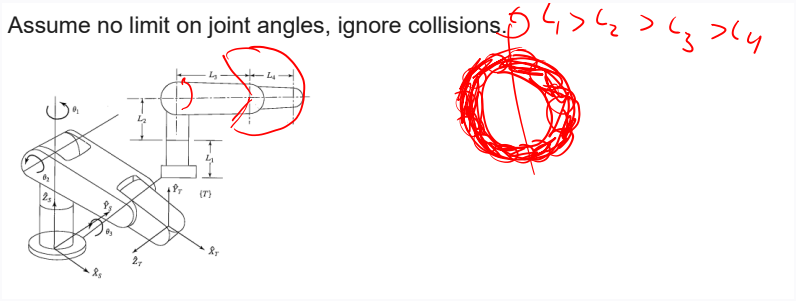
Despite its name the file is a planar **2R** arm with $L_1 = L_2 = 1$: analytic IK with both elbows, the 2×2 Jacobian above, and a Newton–Raphson loop. $\cos q_2$ is clipped to $[-1, 1]$, so an unreachable target silently returns the nearest boundary pose. The notebook builds the same arm with the

Robotics Toolbox and compares the numerical solution (Levenberg–Marquardt) to the analytic one; the W3 folder also has PUMA 560 FK (proximal DH and distal ETS), which pairs with W2. Notebook Q12: a 3-DoF arm (θ_1, horizontal θ_2 offset by L_2, slider d_3): θ_1 faces the point or $+\pi$, and in its plane $|d_3| = \sqrt{\rho^2 - L_2^2}$, so up to 4 solutions (2 if $d_3 \geq 0$); state the assumption (our reading of the figure) [added].

How this is examined: worked solutions

EXAM-STYLE QUESTION W13 Q2 [marks not published; ~1 mark per step]

Sketch the workspace of the manipulator (RRR arm of Q1: rotation θ_1 about z_S, then two joints about horizontal axes, links L_3, L_4 above a shoulder at height $L_1 + L_2$). Assume no limit on joint angles, ignore collisions.



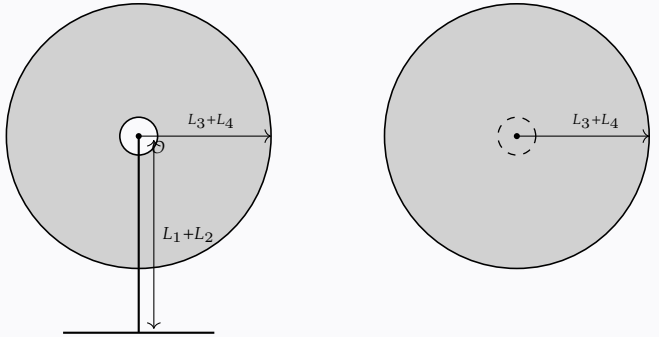
[W13 lec p. 6] — the question and the lecturer's "donut": a ring in the side view

Solution

Frames. $\{S\}$ at the base, z_S up. Shoulder point $O = (0, 0, L_1 + L_2)$.

Reasoning. Joints 2 and 3 are a planar RR arm (links L_3, L_4) in a vertical plane through z_S, pivoting about O. By the cosine rule the tip is at distance $\rho = \sqrt{L_3^2 + L_4^2 + 2L_3L_4\cos\theta_3}$ from O; with θ_3 free, ρ takes every value from $|L_3 - L_4|$ (folded) to $L_3 + L_4$ (stretched). With θ_2 free, the tip sweeps a full annulus in that plane. Joint 1 spins the plane about z_S, which passes through O.

Result. The workspace is a **hollow sphere (spherical shell) centred on** O, outer radius $L_3 + L_4$, inner radius $|L_3 - L_4|$; L_1, L_2 only lift the centre. For $L_3 = 0.4$, $L_4 = 0.3$ m: $0.1 \leq \rho \leq 0.7$ m about O. If $L_3 = L_4$ the hole closes: a solid ball. The dexterous workspace is empty (a position fixes the tool direction up to the branch).



side view ($x_S z_S$ plane) top view (disc)

[drawn for this booklet] side view: ring about O (hole radius $|L_3 - L_4|$, to scale; height of O not to scale). Top view: full disc of radius $L_3 + L_4$ (hole not visible from above: those points are reached at other heights).

Check at a simple pose: $\theta_2 = \theta_3 = 0$, arm horizontal, tip at $(L_3 + L_4, 0, L_1 + L_2)$, on the outer sphere ✓; folded $\theta_3 = \pi$: $\rho = |L_3 - L_4|$ ✓ (500 random poses give $\rho \in [0.1, 0.7]$). [checked]

EXAM-STYLE QUESTION W13 Q3 [marks not published]

Derive the inverse kinematics of the RRR arm of Q1. You are given the position of the end effector (x_T, y_T, z_T).

Solution

FK (from Q1). ${}^S\mathbf{p}_T = (c_1 r,\ s_1 r,\ L_1 + L_2 + L_3 s_2 + L_4 s_{23})$, $r = L_3 c_2 + L_4 c_{23}$; $c_{23} = \cos(\theta_2 + \theta_3)$.

Top-down view gives θ_1: $\theta_1 = \operatorname{atan2}(y_T, x_T)$, $r = +\sqrt{x_T^2 + y_T^2}$ (arm faces the point). Branch 2: $\theta_1 + \pi$ with $r = -\sqrt{x_T^2 + y_T^2}$ (arm reaches back over the base axis). atan2, not the $\tan^{-1}(y/x)$ of the slide.

Side-on view: planar RR in the plane (r, z'), $z' = z_T - L_1 - L_2$:

$$r = L_3 c_2 + L_4 c_{23}, \qquad z' = L_3 s_2 + L_4 s_{23}$$

Square and add (the cross terms give $2L_3L_4(c_2c_{23} + s_2s_{23}) = 2L_3L_4 \cos\theta_3$):

$$r^2 + z'^2 = L_3^2 + L_4^2 + 2L_3L_4 \cos\theta_3 \ \Rightarrow\ \cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2L_3L_4},$$

$$\theta_3 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3\right)$$

Expand c_{23}, s_{23} and write $r = (L_3 + L_4c_3)c_2 - (L_4s_3)s_2$, $z' = (L_4s_3)c_2 + (L_3 + L_4c_3)s_2$; the point (r, z') is $(L_3 + L_4c_3,\ L_4s_3)$ rotated by θ_2:

$$\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4 \sin\theta_3,\ L_3 + L_4 \cos\theta_3)$$

Number and existence. 2 (shoulder) $\times$ 2 (elbow) = 4 for an interior point. No solution if $|\cos\theta_3| > 1$, i.e. the point lies outside the shell of Q2 (outside $|L_3 - L_4| \le \rho \le L_3 + L_4$ about the shoulder). On the shell surface the two elbow branches merge. If $x_T = y_T = 0$, θ_1 is arbitrary.

Numbers. $L = (0.6, 0.5, 0.4, 0.3)$ m, $\mathbf{p}_T = (0.4554, 0.2629, 1.4966)$ m. $\theta_1 = \operatorname{atan2}(0.2629, 0.4554) = 30°$, $r = 0.5259$, $z' = 1.4966 - 1.1 = 0.3966$; $\cos\theta_3 = \dfrac{0.2766 + 0.1573 - 0.16 - 0.09}{0.24} = 0.7660$, $\theta_3 = \pm 40°$; $\sigma = \operatorname{atan2}(0.3966, 0.5259) = 37.024°$, $\mu = \pm 17.024°$. Solutions $(\theta_1, \theta_2, \theta_3)$ in degrees:

$$(30,\ 20,\ 40),\quad (30,\ 54.047,\ -40),\quad (-150,\ 125.953,\ 40),\quad (-150,\ 160,\ -40)$$

Each reproduces $\mathbf{p}_T$ by FK; the first is the Q1 pose, so the check closes. Reach test: a target $(0.5, 0.5, 1.8)$ is 0.995 m from the shoulder > 0.7: no solution. [checked]

Sanity: $\theta_2 = \theta_3 = 0$ gives $r = L_3 + L_4$, $z = L_1 + L_2$ (stretched, on the outer sphere) ✓.

EXAM-STYLE QUESTION SCARA-type: RRP with tool yaw [marks assumed 6]

A SCARA arm (column height $H = 0.40$ m, horizontal links $L_1 = 0.35$, $L_2 = 0.25$ m, vertical axes for q_1, q_2, q_4, slider d_3 down) must put the tool at $(0.30, 0.25, 0.28)$ m with yaw $\varphi = 60°$ in $\{0\}$. (a) Derive q_1, q_2, d_3, q_4 with every branch and the existence condition. (b) Evaluate. (c) How many solutions?

Solution

(a) FK: $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$, $z = H - d_3$, $\varphi = q_1 + q_2 + q_4$. Hence $d_3 = H - z$, the RR formulas for q_1, q_2 and $q_4 = \varphi - q_1 - q_2$. Existence: $|L_1 - L_2| \le r \le L_1 + L_2$ and $0 \le d_3 \le$ stroke.

(b) $r^2 = 0.1525$, $r = 0.3905 \in [0.10, 0.60]$ ✓. $\cos q_2 = \dfrac{0.1525 - 0.1225 - 0.0625}{0.175} = -0.1857$, $q_2 = \pm 100.70°$. $\operatorname{atan2}(0.25, 0.30) = 39.806°$.

$q_2 = +100.70°$: $\mu = \operatorname{atan2}(0.2457, 0.3036) = 38.980°$, $q_1 = 0.826°$, $q_4 = 60 - 0.826 - 100.703 = -41.53°$.

$q_2 = -100.70°$: $q_1 = 78.785°$, $q_4 = 81.917°$. $d_3 = 0.40 - 0.28 =$ **0.12** m in both.

(c) Two (elbow); d_3 and q_4 follow uniquely. One on the outer boundary, none beyond. Both reproduce the target by FK. [checked]

EXAM-STYLE QUESTION RRP in 3D [marks assumed 6]

A base joint rotates about z_0. At height $H = 0.5$ m a horizontal-axis joint tilts a boom by θ_2 from the vertical; a prismatic joint extends the tool by d_3 along the boom: $\mathbf{p} = (c_1 d_3 s_2,\ s_1 d_3 s_2,\ H + d_3 c_2)$. Derive the IK and evaluate it for $\mathbf{p} = (0.3, 0.4, 0.9)$ m. How many solutions?

Solution

$r = \sqrt{x^2 + y^2} = d_3 s_2$, $z' = z - H = d_3 c_2$. So $\theta_1 = \operatorname{atan2}(y, x)$, $d_3 = \sqrt{r^2 + z'^2}$, $\theta_2 = \operatorname{atan2}(r, z')$. Numbers: $\theta_1 = 53.130°$, $r = 0.5$, $z' = 0.4$, $d_3 = 0.6403$ m, $\theta_2 = 51.340°$. Second solution $(\theta_1 + 180°, -\theta_2, d_3)$. With $d_3 > 0$ there are exactly two; with $d_3 < 0$ allowed, two more $(\theta_1, \theta_2 + \pi, -d_3)$ and $(\theta_1 + \pi, \pi - \theta_2, -d_3)$. FK check of both: $(0.3, 0.4, 0.9)$ ✓. [checked]

EXAM-STYLE QUESTION Corners of a square [marks assumed 5]

A planar RR arm ($L_1 = 0.7$, $L_2 = 0.6$ m, base at the origin) must trace the square with corners $(0.5, 0.2)$, $(0.9, 0.2)$, $(0.9, 0.6)$, $(0.5, 0.6)$ m. (a) Find the joint angles at each corner for both elbow branches. (b) Is the square reachable? (c) Can the arm follow the square without changing branch? (d) Which branch would you choose if $q_2 \in [0, 150°]$?

Solution

Formulas: $\cos q_2 = \dfrac{x^2 + y^2 - 0.85}{0.84}$ ($L_1^2 + L_2^2 = 0.49 + 0.36 = 0.85$, $2L_1L_2 = 0.84$), $q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(0.6 \sin q_2,\ 0.7 + 0.6 \cos q_2)$.

(a) P_1: $r^2 = 0.29$, $\cos q_2 = -0.6667$, $q_2 = \pm 131.81°$, $q_1 = -34.34°$ or $77.95°$. P_2: $r^2 = 0.85$, $\cos q_2 = 0$, $q_2 = \pm 90°$, $q_1 = -28.07°$ or $53.13°$. P_3: $r^2 = 1.17$, $\cos q_2 = 0.381$, $q_2 = \pm 67.61°$, $q_1 = 2.83°$ or $64.55°$. P_4: $r^2 = 0.61$, $\cos q_2 = -0.2857$, $q_2 = \pm 106.60°$, $q_1 = 2.79°$ or $97.60°$ (table above).

(b) r ranges 0.538 to 1.082 m over the whole boundary of the square, inside $[0.1, 1.3]$ m: all points reachable.

(c) Yes: $\cos q_2 \in [-0.667, 0.381]$, so q_2 stays in $67.6°$–$131.8°$ (one branch) and never reaches 0 or $\pm 180°$; a branch can change only through a singular pose.

(d) Branch $q_2 > 0$, the only one inside the limits. [checked]

EXAM-STYLE QUESTION Joint limits choose the branch [marks assumed 4]

Planar RR, $L_1 = 0.6$, $L_2 = 0.45$ m; the elbow can only bend one way: $q_2 \in [0°, 150°]$. The tip must reach $(0.70, -0.30)$ m. Find the allowed (q_1, q_2) and say why the other solution is rejected.

Solution

$\cos q_2 = \dfrac{0.58 - 0.36 - 0.2025}{0.54} = 0.03241$, $q_2 = \pm 88.143°$. $\operatorname{atan2}(-0.3, 0.7) = -23.20°$. For $q_2 = +88.143°$: $\mu = \operatorname{atan2}(0.4498, 0.6146) = 36.20°$, $q_1 = -23.20 - 36.20 = -59.40°$. For $q_2 = -88.143°$: $q_1 = -23.20 + 36.20 = 13.00°$. Only $q_2 = +88.143° \in [0, 150°]$, so $(-59.40°,\ 88.14°)$; the other violates $q_2 \geq 0$. Rule: list all branches, then apply limits (slide 18: closeness, joint limits, obstacles). Both reproduce the target by FK. [checked]

EXAM-STYLE QUESTION Spot the errors [marks assumed 4]

A student solves a planar RR arm ($L_1 = 0.5$, $L_2 = 0.4$ m) for the target $(-0.45, 0.30)$ m: "$\cos q_2 = -0.29375$, so $q_2 = \cos^{-1}(-0.29375) = 107.08°$. Then $q_1 = \tan^{-1}(y/x) - \operatorname{atan2}(L_2 \sin q_2, L_1 + L_2 \cos q_2) = -78.68°$. Unique solution." Find the errors and give the correct q_1 for $q_2 = +107.08°$.

Solution

Error 1: $\tan^{-1}(0.30/-0.45) = -33.69°$ cannot tell the second quadrant from the fourth; $\operatorname{atan2}(0.30, -0.45) = 146.31°$. Error 2: $\cos^{-1}$ returns one sign only; the other branch is $q_2 = -107.08°$, so the solution is not unique. The student's $(-78.68°, 107.08°)$ puts the tip at $(0.45, -0.30)$, the point reflected through the base. Correct: $\mu = \operatorname{atan2}(0.4 \sin 107.08°,\ 0.5 + 0.4 \cos 107.08°) = 44.99°$, so $q_1 = 146.31 - 44.99 = \mathbf{101.32}°$; other branch $q_2 = -107.08°$, $q_1 = 146.31 + 44.99 = 191.30°$. Both check by FK. [checked]

EXAM-STYLE QUESTION One Newton step on a 2R arm [marks assumed 4]

$L_1 = 0.6$, $L_2 = 0.45$ m, start $\mathbf{q}^0 = (0.3, 0.8)$ rad. Take one Newton–Raphson step towards $(0.70, 0.30)$ m: give $\Delta\mathbf{q}$ and the distance left.

Solution

FK at $\mathbf{q}^0 = (0.7773, 0.5784)$, $\mathbf{e} = (-0.0773, -0.2784)$. $\mathbf{J} = \begin{bmatrix} -0.5784 & -0.4010 \\ 0.7773 & 0.2041 \end{bmatrix}$, $\det \mathbf{J} = L_1 L_2 \sin 0.8 = 0.27 \sin 0.8 = 0.1937$. $\mathbf{J}^{-1} = \frac{1}{0.1937} \begin{bmatrix} 0.2041 & 0.4010 \\ -0.7773 & -0.5784 \end{bmatrix}$.

$\Delta q_1 = [0.2041(-0.0773) + 0.4010(-0.2784)]/0.1937 = -0.6578$ and $\Delta q_2 = [-0.7773(-0.0773) - 0.5784(-0.2784)]/0.1937 = 1.1415$ rad. New angles $(-0.358, 1.941)$ rad; FK there is 0.156 m from the target (was 0.289 m). The step is large because $\mathbf{J}$ is only a local linear map: repeat. $\mathbf{J}^\top \mathbf{e} = (-0.1717, -0.0258)$ would be a gradient step (wrong method). [checked]

Serial vs parallel manipulators [p. 9–19, 74]

Definition [p. 12]: "the end effector is connected to the base by multiple parallel kinematic chains of links". It can be planar or spatial. Each actuator may have a wide range, but the end-effector motion is constrained by the other chains. The number of joints is usually larger than the DoF, so **several joints are passive** (not driven).

Examples on the slides: Gough tyre-testing machine (early 1950s) [p. 13], Stewart flight simulator (1965) [p. 14], ABB FlexPicker delta robot (Clavel, EPFL) [p. 15], surgical and spine hexapods [p. 16], a multi-finger grasp (point contact without slip acts as a spherical joint) [p. 17].

Comparison table [p. 74] (also pp. 9, 19, 75)

	Serial	Parallel
Kinematic chains	open	closed
Joints	all active	active + passive
Forward kinematics (FK)	simple	**complex**
Inverse kinematics (IK)	complex	**simple**
Joint errors	accumulate	do not accumulate
Work volume	large	small (restricted)
Payload-to-weight	low	very high
Inertia of moving parts	high	low

Use: any "choose serial or parallel and justify" question: quote the rows that matter for the task, then add the cost (small workspace, complex kinematics [p. 19]).

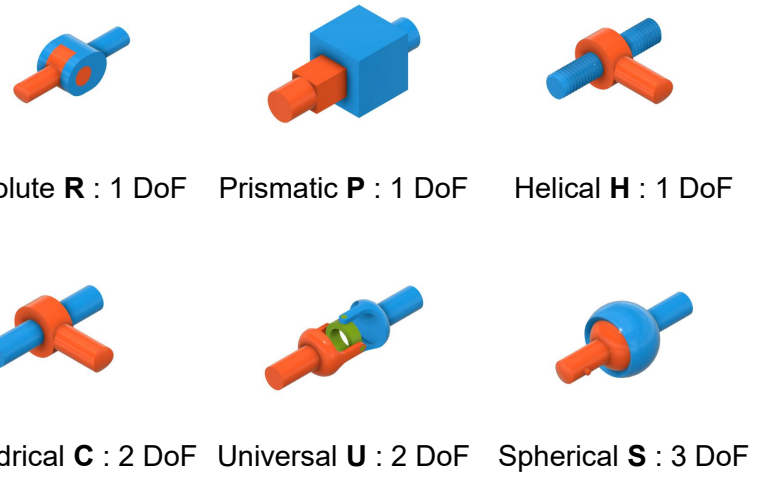
Payload example [p. 18]: a Stewart-based simulator of mass 1.8 t carries a payload of 10.0 t (a KUKA-arm-based simulator: mass 4.7 t, payload 1.3 t), numbers as printed on the slide.

WORKED which architecture? [added]

Task	Choose	Reason
150 light chocolates per min	delta	low moving mass, fast
Painting arm reaching 3 m into a car	serial	large work volume needed
2-tonne cockpit on a motion platform	Stewart	legs share the load
Positioning to a few μm	parallel	stiff, errors not accumulated

TRAP
Saying "parallel FK is easier" (wrong way round). Parallel: IK easy, FK hard. Serial: FK easy, IK hard.

Joint types and their degrees of freedom [p. 25; joint table W1 p. 34]



[W01 lec p. 34] — the six joint types the unit uses (DoF in the labels)

Joint DoF F_i used in Grübler–Kutzbach

Joint	Symbol	F_i	removes (3D)	appears in
Revolute	R	1	5	every mechanism here
Prismatic	P	1	5	3-PRR, Stewart
Helical	H	1	5	–
Cylindrical	C	2	4	–
Universal	U	2	4	simplest delta, Stewart
Spherical	S	3	3	common delta, Stewart

Use: each joint removes $D - F_i$ of the relative DoF of the two bodies it joins ($D = 6$ in 3D, $D = 3$ in the plane). In a planar mechanism only R and P occur, and each has $F = 1$ (removes 2).

Reading a drawing [added]: a U joint is two perpendicular R axes meeting at a point; an S joint is three R axes meeting at a point (so it can spin about the rod it holds). A prismatic pair joins *two* bodies (cylinder and piston). Three bodies pinned at one point count as *two* joints.

Workspace of parallel robots [p. 20–23]

Four workspaces [p. 21]

Reachable	points the end effector can reach, orientation arbitrary
Dexterous	points it can reach in *all* its possible orientations
Constant-orientation	points reachable with one given orientation
Orientation	all possible orientations at one given location

Use: say which workspace the question means. Dexterous $\subseteq$ constant-orientation $\subseteq$ reachable.

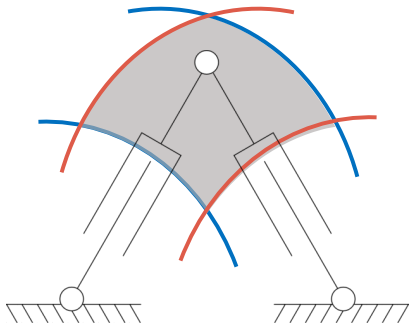
ALTERNATIVE APPROACHES

A. Discretisation [p. 22]

Make a grid by moving each actuator a small step; the boundary of the set of reached nodes is the workspace. Slow; accuracy is limited by the step size.

B. From geometry [p. 23]

Use the minimum and maximum of each actuator's limits to define the volume each chain allows; **intersect** them. Take all physical constraints into account.

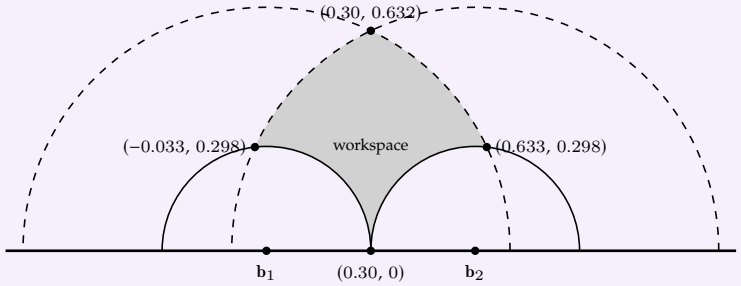


[W04 lec p. 23] — a planar two-leg robot: each leg's limits give an arc (blue/red); the shaded overlap is the workspace

WORKED sketch the workspace of a two-leg planar robot [added]

Leg base pivots $\mathbf{b}_1 = (0, 0)$, $\mathbf{b}_2 = (0.6, 0)$ m (platform = a point). Each prismatic leg has length $0.3 \le \ell \le 0.7$ m, so the point must lie in an annulus about each pivot:

$$0.3 \le \|\mathbf{p} - \mathbf{b}_1\| \le 0.7 \quad \text{and} \quad 0.3 \le \|\mathbf{p} - \mathbf{b}_2\| \le 0.7.$$



[drawn for this booklet] solid arcs: $\ell_{min} = 0.3$; dashed arcs: $\ell_{max} = 0.7$; mirror image below the axis if leg collisions are ignored

Corner points (circle intersections): top (0.30, 0.632) where both legs are at ℓ_{max}; the side corners (0.633, 0.298) and (−0.033, 0.298) where one leg is at ℓ_{max} and the other at ℓ_{min}; and (0.30, 0), the single point where both legs are at ℓ_{min}. Area for $y \ge 0$: about 0.175 m^2. [checked]

Mobility: Grübler–Kutzbach [p. 24–34]

PREMISE mobility **Given:** a drawn mechanism: its rigid bodies (including the ground) and its joints with their types

Find: the number of degrees of freedom M (mobility)

Grübler–Kutzbach (G–K) [p. 25]

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i$$

SYMBOLS

M	mobility = DoF of the mechanism	–
D	6 for a 3D mechanism, 3 for a planar (2D) one	–
N	number of links (rigid bodies), **ground included** ("one of which is grounded")	–
J	number of joints	–
F_i	DoF of joint i (R 1, P 1, U 2, S 3)	–

Use: first estimate for every mechanism. If M disagrees with what you can see moving, use the extended equation (next section).

DERIVATION of G–K [added]

Each of the $N-1$ moving bodies is free: $D(N-1)$ DoF. A joint of freedom F_i lets the two bodies it joins move relatively with only F_i of their D relative DoF, so it removes $D - F_i$ DoF:

$$M = D(N-1) - \sum_{i=1}^{J}(D - F_i) = D(N-1) - DJ + \sum_i F_i = D(N - J - 1) + \sum_i F_i.$$

Check: a pendulum ($N = 2$, one R): $3(2 - 1 - 1) + 1 = 1$.

STEPS count the DoF of a drawn mechanism

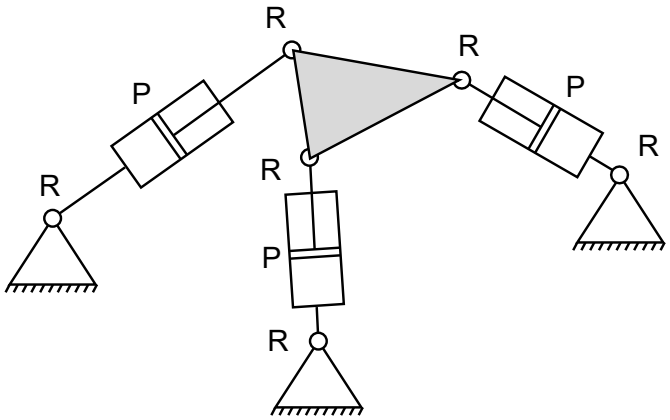
1. Planar or spatial? Planar: $D = 3$ (all axes parallel). The slides stress "the mechanism is still 3D despite this depiction" for the delta [p. 30]: $D = 6$.
2. Number **every rigid body, ground** = 1, on the figure. A prismatic joint is two bodies (cylinder, piston).
3. Table of joints: name the two bodies each joins and write F_i. A pin joining three bodies is two joints.
4. Substitute: $M = D(N - J - 1) + \sum F_i$.
5. Sanity check: does M match the motion you can see? $M \le 0$ or an implausible M means look for local mobility R_M or repeated constraints R_C.

What the sign of M means [added]

$M > 0$	mechanism: it moves (that many DoF)
$M = 0$	structure: rigid
$M < 0$	more constraints than the count allows (over-constrained), **or** the plain count is wrong: check R_C, R_M

Use: "is this a structure, a mechanism or over-constrained?". G–K assumes every constraint is independent. Equal parallel links break that: three equal parallel links hanging a planar body ($N = 5$, $J = 6$) give $M = 0$ but it really moves ($M = 1$, see extended G–K).

Example: planar 3-PRR [p. 26–27]



[W04 lec p. 27] — three R–P–R legs carry a triangular platform

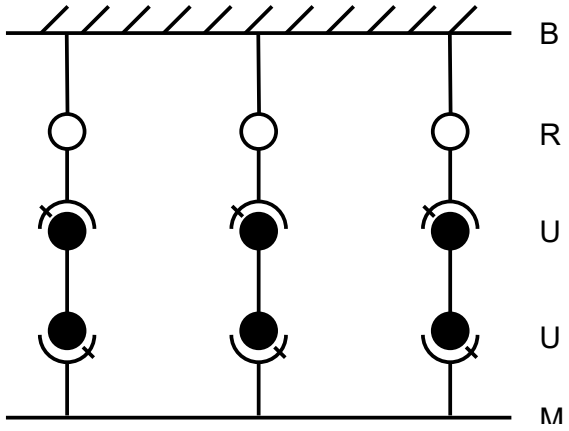
WORKED 3-PRR, planar, $D = 3$ [p. 27]; W13 Q6 [p. 12]

Bodies	ground; platform; 3 cylinders; 3 pistons	$N = 8$
Joints	3 R (ground–cylinder); 3 P (cylinder–piston); 3 R (piston–platform)	$J = 9$
F_i	all 1	$\sum F_i = 9$

$$M = 3(8 - 9 - 1) + 9 = 3$$

The platform has 3 DoF (two positions and one rotation) driven by the 3 prismatic actuators. [checked]

Example: the simplest delta [p. 29–31]



[W04 lec p. 30] — simplest delta, schematic: base B, active R, then U, link, U to the moving platform M

WORKED simplest delta, $D = 6$ [p. 30]

Bodies	base; 3 upper arms; 3 rods; platform	$N = 8$
Joints	3 R (base–arm, active); 3 U (arm–rod); 3 U (rod–platform)	$J = 9$
$\sum F_i$	$3\,(1 + 2 + 2)$	15

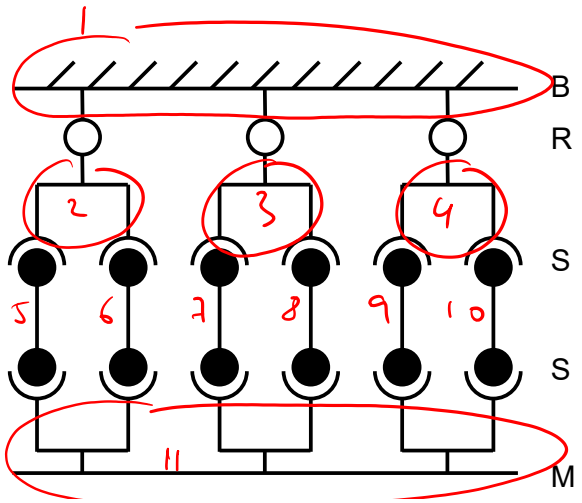
$$M = 6(8 - 9 - 1) + (1 + 2 + 2) \times 3 = -12 + 15 = 3$$

Three DoF, equal to the 3 motors. [checked] The slide notes that it is never built [p. 31]: not stiff enough, play-free universal joints are hard to find, and play lowers the usable frequency.

SLIDE CHECK slide 29

A handwritten note on the slide reads $M = D(N - J + 1) + \sum F_i$. The printed equation [p. 25] and every worked count (pp. 27, 30, 33) use $D(N - J - 1)$. With the +1 the simplest delta would give $6(8 - 9 + 1) + 15 = 15$, not 3. Use -1. [checked]

Example: the common delta [p. 32–34]



[W04 lec p. 33] — common delta: R, then a parallelogram of two bars with four S joints per leg (lecturer's link numbers 1–11)

WORKED common delta (parallelogram legs), $D = 6$ [p. 33]; W13 Q7 [p. 14]

Bodies	base (1); 3 upper arms (2–4); 6 parallelogram bars (5–10); platform (11)	$N = 11$
Joints	3 R (active); $3 \times 4 = 12$ S	$J = 15$
$\sum F_i$	$3\,(1 + 4 \times 3)$	39

$$M = 6(11 - 15 - 1) + (1 + 4 \times 3) \times 3 = -30 + 39 = 9$$

But the platform only translates (3 DoF): the plain count over-counts [p. 34]. The 6 *supplementary* DoF are internal: each of the 6 bars, with a ball joint at each end, can spin about its own axis without moving anything else. This is fixed by the extended equation. [checked]

Counts for the mechanisms in this week

Mechanism	D	N	J	$\sum F$	plain M
3-PRR [p. 27]	3	8	9	9	3
Simplest delta [p. 30]	6	8	9	15	3
Common delta [p. 33]	6	11	15	39	9
4RRR [p. 38]	6	10	12	12	−6
Four-bar [added]	3	4	4	4	1
Slider-crank [added]	3	4	4	4	1
Planar five-bar [added]	3	5	5	5	2
Gough–Stewart 6-UPS [added]	6	14	18	36	6

Use: Stewart: ground, platform, 6 legs of 2 bodies: $N = 2 + 12 = 14$; joints 6 U + 6 P + 6 S: $J = 18$; $\sum F = 6(2 + 1 + 3) = 36$; $M = 6(14 - 18 - 1) + 36 = 6$. [checked]

TRAP

Using $D = 6$ for a planar mechanism: slider-crank (ground, crank, rod, slider; 3 R + 1 P) gives $6(4 - 4 - 1) + 4 = -2$ ("a structure"). Planar: $3(4 - 4 - 1) + 4 = 1$. Forgetting the ground as a link, or counting a prismatic joint as one body, are the other usual slips. [checked]

Extended Grübler–Kutzbach [p. 35–43]

Extended G–K [p. 35]

$$M = D\,(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M, \qquad R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$$

SYMBOLS

R_C	number of **over-constraints**: constraints repeated by several joints, so removed twice by the plain count (add back)	–
R_M	**local mobility**: DoF internal to the mechanism that do not move the output (subtract)	–
L	number of closed loops considered	–
u_k	unique DoF of the platform allowed by loop k (a DoF shared by two loops is counted once in the union)	–

Use: whenever plain M is implausible (delta 9; 4RRR −6). Write which correction applies and why.

STEPS extended count

1. Plain G–K first (record the number).
2. Local mobility: is there a body that can move without moving the output? A rod with a ball joint at both ends spins about its own axis: each such rod adds 1 to R_M.
3. Over-constraints: take the loops one at a time; list the platform DoF each allows; $R_C = \sum F_i - \sum u_k$ (union, T_z shared counts once).
4. M = plain + $R_C - R_M$; check it against the motion you expect (delta: 3 translations).

WORKED common delta, extended [p. 36]; W13 Q7 [p. 16]

Plain $M = 6(11 - 15 - 1) + 39 = 9$. Six bars with S joints at both ends: $R_M = 6$. No repeated constraints: $R_C = 0$.

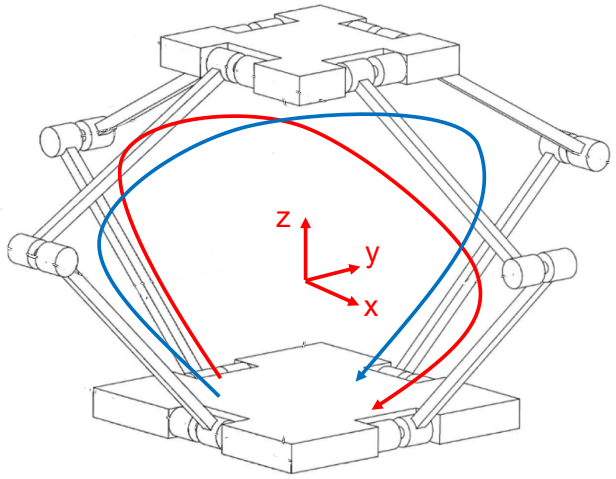
$$M = 6(11 - 15 - 1) + (1 + 4 \times 3) \times 3 + 0 - 6 = \mathbf{3}$$

Explain (2–4 sentences): plain G–K treats every DoF as moving the output; the bar spins do not, so subtract them. The platform of the delta only translates in x, y, z: 3 DoF. [checked]

SLIDE CHECK slide 35–36 and 42

The slides define $R_C = \sum_i F_i - \sum_k u_k$ and use it for the 4RRR example (7). For the delta they only state $R_C = 0$; the literal formula would need $\sum_k u_k = 39$, which is not how u_k is counted (a few platform DoF per loop). So: do not apply the u_k formula to the delta; argue "no repeated constraint". For 4RRR, follow the slide. The slides do not say how the loops are chosen (only two are shown for four legs). Ambiguity, not a proven error.

Example: parallel 4RRR [p. 37–43]



[W04 lec p. 41] — 4RRR: two square plates, four RR legs; the two loops (red, blue) each allow three platform DoF

WORKED 4RRR, $D = 6$ [p. 38–43]

Bodies: base, moving platform, 4 legs × 2 links: $N = 10$. Joints: 4 legs × 3 R: $J = 12$, $\sum F_i = 12$.

$$M = 6(10 - 12 - 1) + 12 + R_C - R_M = -6 + R_C - R_M$$

Local mobility: none, $R_M = 0$ [p. 42]. Loop 1 (a pair of legs) allows platform motion $\{T_y, T_z, R_x\}$ [p. 39]; loop 2 allows $\{T_x, T_z, R_y\}$ [p. 40]. Unique DoF: union = $\{T_x, T_y, T_z, R_x, R_y\}$, so $\sum_k u_k = 5$ (T_z once) [p. 41].

$$R_C = \sum F_i - \sum u_k = 12 - 5 = 7, \qquad M = -6 + 7 - 0 = \mathbf{1}$$

"The mechanism indeed has only 1 DoF" [p. 43] (the platform only slides along z). [checked]

WORKED 3-RPS and 3-SPS platforms [added]; from the mock exam

$D = 6$, $N = 2 + 3 \times 2 = 8$, $J = 9$ for both.
3-RPS (R at the ground, P, S at the platform): $\sum F = 3(1 + 1 + 3) = 15$, $M = 6(8 - 9 - 1) + 15 = 3$; the R axis stops the leg spinning, $R_M = 0$: three DoF.
3-SPS (S at both ends): $\sum F = 3(3 + 1 + 3) = 21$, plain $M = -12 + 21 = 9$; each leg spins about itself, $R_M = 3$: $M = 9 + 0 - 3 = 6$ (6 DoF but only 3 motors: under-actuated). [checked]

Plain vs extended, all examples

Mechanism	plain	R_C	R_M	M	reason
Simplest delta	3	0	0	3	U ends stop spin
Common delta	9	0	6	3	bars spin
4RRR	−6	7	0	1	repeated constraints
3-SPS [added]	9	0	3	6	legs spin

Configuration space: active and passive joints [p. 45–46]

Configuration [p. 45]

$$\mathbf{q} = (\boldsymbol{\theta}, \boldsymbol{\varphi}), \qquad \boldsymbol{\theta} = \{\theta_i \mid \theta_i \in [0, 2\pi),\ i \in 1..n\}, \qquad \boldsymbol{\varphi} = \{\varphi_i,\ i \in 1..m\}$$

SYMBOLS

$\boldsymbol{\theta}$	the n **active** (motor) joint variables	rad or m
$\boldsymbol{\varphi}$	the m **passive** joint variables	rad or m

Use: FK needs the m passive variables from the n active ones; for that you need m independent equations (loop closure).

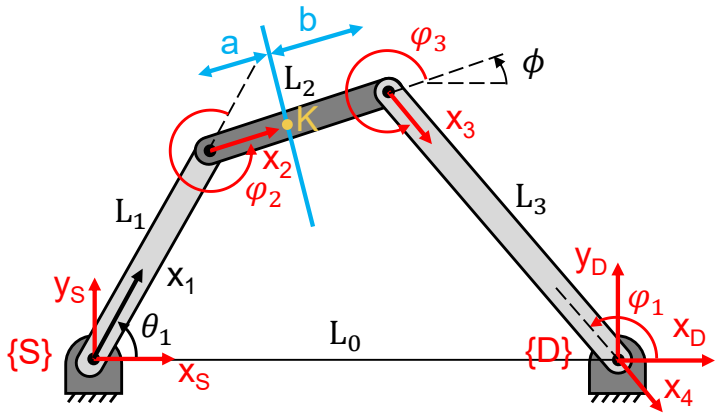
Mechanism	active n	passive m
Four-bar (break link 2 at K) [p. 54]	1 (θ_1)	3 ($\varphi_1, \varphi_2, \varphi_3$)
Four-bar (break both ends) [p. 57]	1	1 (φ_1)
3-PRR [added]	3 (P)	6 (R)
Gough–Stewart [p. 58]	6 (P)	30 (6 legs × (S 3 + U 2))

Loop-closure constraint equations [p. 47–58]

PREMISE loop closure **Given:** the geometry and the active joint values $\boldsymbol{\theta}$
Find: m independent equations $\eta_i(\mathbf{q}) = 0$ that tie the passive joints $\boldsymbol{\varphi}$ to $\boldsymbol{\theta}$

STEPS loop closure [p. 49]

1. Break the manipulator into two or more serial chains (choose the break giving the *fewest passive unknowns* m, hence the fewest equations).
2. For each chain, write the position and orientation of the **break point**.
3. Rejoin: equate the two descriptions (the joint constraint at the break).
4. Count: n active, m passive, m equations.



[W04 lec p. 71] — planar four-bar: frames S and D, L_0 between the pivots, K on link 2 (a, b = distances from K to the ends of link 2), output orientation ϕ (slide p. 54 breaks the loop at K)

Setup [p. 50]: one loop; frames {S} (left base pivot), {1}, {2}, {3}, {4}, {D} (right base pivot, at $(L_0, 0)$ in S). θ_1 is active; $\varphi_1, \varphi_2, \varphi_3$ passive. All angles are absolute, counter-clockwise, in radians.

Four-bar, break link 2 at K [p. 54]

$$^{S}x_K = L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2), \qquad ^{D}x_K = L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3)$$
$$^{S}y_K = L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2), \qquad ^{D}y_K = L_3\sin\varphi_1 + b\sin(\varphi_1+\varphi_3)$$

Impose that the broken link is rigid (the same K, $^{S}x_K = L_0 + {}^{D}x_K$):

$$L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2) = L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3)$$
$$L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2) = L_3\sin\varphi_1 + b\sin(\varphi_1+\varphi_3)$$
$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

SYMBOLS

L_0	distance S to D (ground link)	m
L_1, L_3	crank and rocker lengths	m
$L_2 = a + b$	coupler length; a, b from K to its two ends	m
θ_1	active crank angle at S, from x_S	rad
φ_1	passive angle at D, from x_D to link 3	rad
φ_2	passive angle at the crank–coupler joint	rad
φ_3	passive angle at the coupler–rocker joint	rad

Use: 3 independent equations for $\varphi_1, \varphi_2, \varphi_3$; $n = 1$, $m = 3$, $J = 4$ [p. 54].

DERIVATION of the three equations [added]

The coupler's direction seen from S is $\theta_1 + \varphi_2$. Seen from D, the vector from the joint B to K points back along the coupler, so $\varphi_1 + \varphi_3$ is the coupler direction plus π. Hence $\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi \pmod{2\pi}$. The first two equations are K reached from the left (crank, then a along the coupler) equals K reached from the right (L_0, rocker, then b back along the coupler). Verified at $\theta_1 = 70°$ for both assembly modes. [checked]

Freudenstein-type equation, break on both sides [p. 57]

U on the end of link 1, V on the end of link 3: ${}^{S}x_U = L_1\cos\theta_1$, ${}^{S}y_U = L_1\sin\theta_1$; ${}^{D}x_V = L_3\cos\varphi_1$, ${}^{D}y_V = L_3\sin\varphi_1$. Impose that link 2 has length L_2:

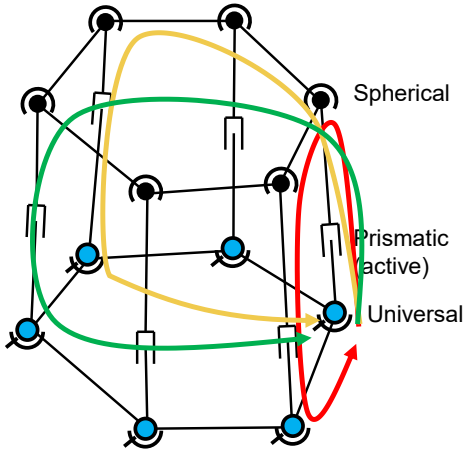
$$\eta_1(\theta_1,\varphi_1) = (L_1\cos\theta_1 - L_0 - L_3\cos\varphi_1)^2 + (L_1\sin\theta_1 - L_3\sin\varphi_1)^2 - L_2^2 = 0$$

Use: only one equation, $n=1$, $m=1$, $J=4$. Given θ_1 it is quadratic-in-trig in φ_1: two solutions, the two assembly modes. The slide calls it "the Freudenstein equation".

ALTERNATIVE APPROACHES

A. Break link 2 at K [p. 52–54]
Two RR serial chains meeting at K. Equations: 3 (position x, y and orientation). Unknowns $\varphi_1, \varphi_2, \varphi_3$.
Good when you also want K and the coupler orientation (IK, W13 Q5).

B. Break at both ends [p. 55–57]
Cut link 2 itself: chains S→U and D→V. Equation: 1 (length of link 2). Unknown φ_1 only.
Good for FK: one equation, two roots, then the other angles follow.



[W04 lec p. 58] — Gough–Stewart: 6 active (prismatic) and 30 passive parameters; several loops (coloured) are possible

Gough–Stewart [p. 58]: 6 active and 30 passive parameters. Many loops are possible; each potentially has 6 independent equations; "deciding which 30 of them to choose is not trivial". That is why it is not solved by hand in general.

Forward kinematics (hard) [p. 59–67]

PREMISE parallel FK **Given:** the geometry and the active joint values θ_i (θ_1 for the four-bar)
Find: the passive joints φ_i, then the pose (position and orientation) of the end effector

Steps [p. 60]: (1) write m loop-closure equations in the m passive and n active variables and solve this non-linear system for φ_i; (2) compute the output pose from θ_i, φ_i. "There is no general method like in serial manipulators." More complex mechanisms can have multiple solutions, some with imaginary passive joints; numerical methods are common [p. 67].

ALTERNATIVE APPROACHES

A. Slide route (algebra) [p. 61–66]
Eliminate angles with squaring and adding, giving $A\cos\delta + B\sin\delta + C = 0$ twice; solve with Cramer's rule or the half-angle substitution; then $\varphi_2 = \delta - \theta_1$, φ_1 from Freudenstein, $\varphi_3 = \theta_1 + \varphi_2 - \varphi_1 - \pi$.

B. Geometry (circle intersection) [added]
A is known from θ_1; B lies on the circle L_2 about A *and* on the circle L_3 about D. Two intersections = two assembly modes. Fastest by hand; use this in an exam unless told otherwise.

DERIVATION slide route, with all steps [p. 61–65]

Let $x \equiv {}^{S}x_K$, $y \equiv {}^{S}y_K$, $\delta = \theta_1 + \varphi_2$. Since $\varphi_1 + \varphi_3 = \delta - \pi$, $\cos(\varphi_1+\varphi_3) = -\cos\delta$ and the right-chain equations become

$$x - L_0 = L_3\cos\varphi_1 - b\cos\delta, \qquad y = L_3\sin\varphi_1 - b\sin\delta.$$

Isolate $L_3\cos\varphi_1 = (x-L_0) + b\cos\delta$ and $L_3\sin\varphi_1 = y + b\sin\delta$; square and add (this removes φ_1):

$$L_3^2 = (x-L_0)^2 + y^2 + b^2 + 2b\big[(x-L_0)\cos\delta + y\sin\delta\big]$$

$$\Rightarrow\ A_1\cos\delta + B_1\sin\delta + C_1 = 0, \quad A_1 = x - L_0,\ B_1 = y,\ C_1 = \tfrac{1}{2b}\big[(x-L_0)^2 + y^2 + b^2 - L_3^2\big].$$

The left chain gives $x - a\cos\delta = L_1\cos\theta_1$, $y - a\sin\delta = L_1\sin\theta_1$; square and add (removes θ_1):

$$A_2\cos\delta + B_2\sin\delta + C_2 = 0, \quad A_2 = x,\ B_2 = y,\ C_2 = \tfrac{1}{2a}\big[L_1^2 - a^2 - x^2 - y^2\big].$$

DERIVATION solving the two equations for δ [p. 65]

(i) Cramer's rule. Treat $\cos\delta, \sin\delta$ as unknowns of a linear system with determinant $\Delta = A_1B_2 - A_2B_1$:

$$\cos\delta = \frac{B_1C_2 - B_2C_1}{\Delta}, \quad \sin\delta = \frac{A_2C_1 - A_1C_2}{\Delta}, \quad \delta = \operatorname{atan2}(\sin\delta, \cos\delta).$$

Using $\cos^2 + \sin^2 = 1$ gives the slide's consistency condition $\Delta^2 = (A_1C_2 - A_2C_1)^2 + (B_1C_2 - B_2C_1)^2$.

(ii) Tangent half-angle $t = \tan(\delta/2)$: $\cos\delta = \frac{1-t^2}{1+t^2}$, $\sin\delta = \frac{2t}{1+t^2}$. Each equation becomes

$$(C_i - A_i)\,t^2 + 2B_i\,t + (A_i + C_i) = 0, \quad i = 1, 2.$$

Eliminate t^2: $(C_2 - A_2)\cdot$eq. 1 $-(C_1 - A_1)\cdot$eq. 2 is linear in t:

$$2t\big[B_1(C_2 - A_2) - B_2(C_1 - A_1)\big] + (A_1 + C_1)(C_2 - A_2) - (A_2 + C_2)(C_1 - A_1) = 0.$$

The constant part expands to $2(A_1C_2 - A_2C_1)$ and the bracket to $(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)$, so

$$t = \frac{A_2C_1 - A_1C_2}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}, \qquad \delta = 2\tan^{-1} t.$$

(sympy confirmed this elimination. [checked])

W4

SLIDE CHECK slide 65

Printed: $\delta = \tan^{-1}\dfrac{A_1C_2 - A_2C_1}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}$. The half-angle derivation gives the *opposite sign in the numerator and a factor 2*: $\delta = 2\tan^{-1}\dfrac{A_2C_1 - A_1C_2}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}$. Test (mode 1 below): slide form $0.300°$, true $\delta = -0.600°$. Use the Cramer/atan2 form (no half-angle ambiguity at $\delta = \pm\pi$). [checked]

Also: the slide's "simplifies to" quartic in x, y [p. 65] holds for every point K of the coupler (it is the coupler-curve equation [added]); it contains no θ_1, so it does not by itself give δ. And the slide treats (x, y) as known; for FK from θ_1 alone use route B.

Geometric FK of the four-bar, route B [added]

$$\mathbf{A} = L_1\begin{bmatrix}\cos\theta_1\\ \sin\theta_1\end{bmatrix},\ \mathbf{D} = \begin{bmatrix}L_0\\ 0\end{bmatrix},\ d = \|\mathbf{D}-\mathbf{A}\|,\ d^2 = L_0^2 + L_1^2 - 2L_0L_1\cos\theta_1$$

$$x_a = \frac{L_2^2 - L_3^2 + d^2}{2d},\ h = \sqrt{L_2^2 - x_a^2},\ \hat{\mathbf{u}} = \frac{\mathbf{D}-\mathbf{A}}{d},\ \hat{\mathbf{n}} = \begin{bmatrix}-u_y\\ u_x\end{bmatrix}$$

$$\mathbf{B} = \mathbf{A} + x_a\hat{\mathbf{u}} \pm h\,\hat{\mathbf{n}},\quad \phi = \operatorname{atan2}(B_y - A_y, B_x - A_x),\quad \varphi_2 = \phi - \theta_1,$$

$$\mathbf{K} = \mathbf{A} + a(\cos\phi, \sin\phi),\quad \varphi_1 = \operatorname{atan2}(B_y,\ B_x - L_0),\quad \varphi_3 = \phi + \pi - \varphi_1 \pmod{2\pi}$$

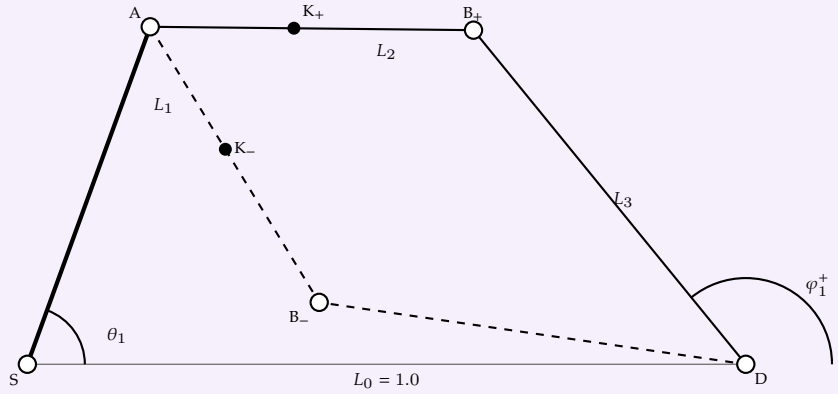
SYMBOLS

A, **B**, **D**	crank–coupler joint, coupler–rocker joint, rocker pivot, all in S	m
x_a, h	distance of B's foot from A along AD, and its height off AD (cosine rule)	m
$\pm$	+: B on the left of $\mathbf{A}\to\mathbf{D}$ ("above"); −: right ("below"): the two assembly modes	–
ϕ	orientation of the coupler (link 2), from x_S	rad

Use: FK of any four-bar from θ_1. It needs $\lvert L_2 - L_3\rvert \le d \le L_2 + L_3$, else the linkage cannot assemble at that θ_1.

WORKED four-bar FK, both assembly modes; closure checked

$L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$, $a = 0.2$, $b = 0.25$ m ($L_2 = 0.45$ m); crank $\theta_1 = 70°$ (angles in degrees here; radians in the formulas).



[drawn for this booklet] solid: mode + (B above AD); dashed: mode − (B below). Frame S at the left pivot, x to the right, y up.

Step 1 $\mathbf{A} = 0.5(\cos 70°, \sin 70°) = (0.1710, 0.4698)$; $d^2 = 1 + 0.25 - 2(0.5)\cos 70° = 0.9080$, $d = 0.9529$; $\hat{\mathbf{u}} = (0.8700, -0.4931)$, $\hat{\mathbf{n}} = (0.4931, 0.8700)$.

Step 2 $x_a = \dfrac{0.2025 - 0.36 + 0.9080}{2(0.9529)} = \dfrac{0.7505}{1.9058} = 0.3938$; $h = \sqrt{0.2025 - 0.1551} = 0.2178$.

Step 3 (B, K and the angles)

	mode +	mode −
B (m)	(0.6210, 0.4651)	(0.4062, 0.0862)
ϕ	−0.600°	−58.486°
$\varphi_2 = \phi - \theta_1$	289.400°	231.514°
φ_1	129.175°	171.739°
$\varphi_3 = \phi + 180° - \varphi_1$	50.225°	309.775°
K (m)	(0.3710, 0.4678)	(0.2756, 0.2993)

Closure check (mode +): left side $L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2) = 0.1710 + 0.2\cos(359.4°) = 0.3710$; right side $L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3) = 1 - 0.3790 + 0.25\cos 179.4° = 0.3710$. Second equation: $0.4698 + 0.2\sin 359.4° = 0.4678 = 0.6\sin 129.175° + 0.25\sin 179.4° = 0.4678$. Orientation: $\theta_1 + \varphi_2 = 359.4°$, $\varphi_1 + \varphi_3 + \pi = 359.4°$ (179.4 + 180). Also $|\mathbf{AB}| = 0.45$, $|\mathbf{DB}| = 0.60$ in both modes. [checked]

Solving Freudenstein for φ_1 [p. 57, 66]

With $u = L_1\cos\theta_1 - L_0$, $v = L_1\sin\theta_1$ ($u^2 + v^2 = d^2$): $\eta_1 = 0$ becomes $u\cos\varphi_1 + v\sin\varphi_1 = k$,

$$k = \frac{u^2 + v^2 + L_3^2 - L_2^2}{2L_3},\qquad \varphi_1 = \operatorname{atan2}(v, u) \pm \arccos\frac{k}{d}$$

Use: $|k/d| > 1$ means no assembly. Then $\varphi_2 = \delta - \theta_1$ and $\varphi_3 = \theta_1 + \varphi_2 - \varphi_1 - \pi$ [p. 66].

Numbers ($\theta_1 = 70°$): $u = -0.8290$, $v = 0.4698$, $k = 0.8879$, $d = 0.9529$, $\arccos(0.9318) = 21.282°$, $\operatorname{atan2}(v, u) = 150.457°$: $\varphi_1 = 129.175°$ (mode +) or $171.739°$ (mode −). [checked]

W4

Assembly range and crank test [added]

STEPS can the crank reach angle θ_1? does it turn fully?

1. The circles of radius L_2 about A and L_3 about D meet only if $|L_2 - L_3| \le d \le L_2 + L_3$, with $d^2 = L_0^2 + L_1^2 - 2L_0L_1\cos\theta_1$.
2. Solve for θ_1: $\dfrac{L_0^2 + L_1^2 - (L_2+L_3)^2}{2L_0L_1} \le \cos\theta_1 \le \dfrac{L_0^2 + L_1^2 - (L_2-L_3)^2}{2L_0L_1}$.
3. If the interval for $\cos\theta_1$ contains $[-1, 1]$, the crank turns fully; otherwise it rocks between the limits (the two modes merge there: a dead-centre position).
4. Shortcut (Grashof, [added]): if shortest + longest ≤ the other two, a link next to the shortest can turn fully.

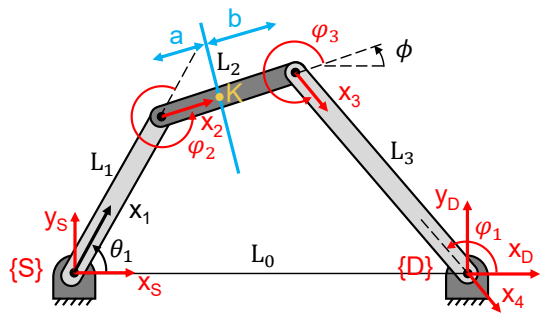
WORKED crank range for the example above

$L_2 + L_3 = 1.05$, $|L_2 - L_3| = 0.15$. Lower bound $\cos\theta_1 \ge \dfrac{1 + 0.25 - 1.1025}{1} = 0.1475$, so $|\theta_1| \le 81.52°$; upper bound $1.2275 > 1$, no limit. Non-Grashof ($0.45 + 1.0 = 1.45 > 0.5 + 0.6 = 1.1$): the crank only rocks. At $81.52°$, $d = 1.05 = L_2 + L_3$: coupler and rocker are collinear and the two modes meet. Mock-exam four-bar ($L_0 = 0.8$, $L_1 = 0.3$, $L_2 = 0.7$, $L_3 = 0.5$): $0.3 + 0.8 = 1.1 \le 0.5 + 0.7 = 1.2$: Grashof, the crank turns fully. [checked]

Inverse kinematics (easy) [p. 68–73]

PREMISE four-bar IK, W13 Q5 **Given:** position K= (x, y) in S of a point on the output link, and its orientation ϕ; known $L_0, L_1, L_2 = a + b, L_3$
Find: $\theta_1, \varphi_1, \varphi_2, \varphi_3$

Why easy [p. 69]: with the pose of the output link given, each leg is a serial chain whose end pose is known: solve each leg separately, no loop closure. Each leg must not be redundant, else the answer is not unique.



[W04 lec p. 71] — the same four-bar; K, a, b, $L_2 = a + b$ and the output orientation ϕ

DERIVATION four-bar IK [p. 70–72]

Left chain S→K [p. 71]: $x = L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2)$, $y = L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2)$, $\phi = \theta_1+\varphi_2-2\pi$ (so $\theta_1 + \varphi_2 = \phi \bmod 2\pi$). Substitute $\theta_1 + \varphi_2 = \phi$:

$$L_1\cos\theta_1 = x - a\cos\phi, \quad L_1\sin\theta_1 = y - a\sin\phi$$
$$\Rightarrow\ \theta_1 = \operatorname{atan2}(y - a\sin\phi,\ x - a\cos\phi), \quad \varphi_2 = \phi - \theta_1\ (+2\pi).$$

Right chain D→K [p. 72]: $x = L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$, $y = L_3\sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$, $\phi = \varphi_1 + \varphi_3 - \pi$. Substitute $\varphi_1 + \varphi_3 = \phi + \pi$, so $\cos(\varphi_1 + \varphi_3) = -\cos\phi$:

$$L_3\cos\varphi_1 = x - L_0 + b\cos\phi, \quad L_3\sin\varphi_1 = y + b\sin\phi$$
$$\Rightarrow\ \varphi_1 = \operatorname{atan2}(y + b\sin\phi,\ x + b\cos\phi - L_0), \quad \varphi_3 = \phi + \pi - \varphi_1.$$

(The slides leave "solve for φ_1 and φ_3" as an exercise.)

STEPS four-bar IK from K and ϕ (W13 Q5)

1. Coupler ends: $\mathbf{A} = \mathbf{K} - a(\cos\phi, \sin\phi)$, $\mathbf{B} = \mathbf{K} + b(\cos\phi, \sin\phi)$.
2. $\theta_1 = \operatorname{atan2}(y - a\sin\phi,\ x - a\cos\phi)$.
3. $\varphi_2 = \phi - \theta_1$ (add 2π if a positive angle is wanted).
4. $\varphi_1 = \operatorname{atan2}(y + b\sin\phi,\ x + b\cos\phi - L_0)$.
5. $\varphi_3 = \phi + \pi - \varphi_1 \pmod{2\pi}$.
6. Check: $|\mathbf{A}| = L_1$ and $|\mathbf{B} - \mathbf{D}| = L_3$ (K and ϕ over-specify a 1-DoF linkage); then redo the left-hand loop equation.

SLIDE CHECK W13 p. 11, Q5 continued

The slide prints $\varphi_1 - \phi = \varphi_3 - \pi$, i.e. $\varphi_3 = \varphi_1 - \phi + \pi$. With the position equations on the same slide this is only true when $\varphi_1 = \phi \pmod{\pi}$. W4 p. 72 prints $\phi = \varphi_1 + \varphi_3 - \pi$, giving $\varphi_3 = \phi + \pi - \varphi_1$, which is right. Test below: the printed form puts K at $(0.669,\ 0.711)$ instead of $(0.371,\ 0.468)$. If you use the slide's form, say so; the safe answer is $\varphi_1 + \varphi_3 = \phi + \pi$. [checked]

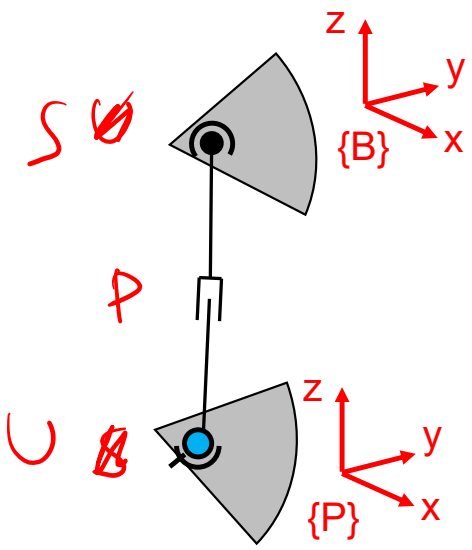
WORKED W13 Q5 numbers

$L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$, $a = 0.2$, $b = 0.25$ m. Given K= $(0.3710,\ 0.4678)$ m, $\phi = -0.600°$.
$\theta_1 = \operatorname{atan2}(0.4678 + 0.00209,\ 0.3710 - 0.2000) = \operatorname{atan2}(0.4699, 0.1710) = 70.0°$.
$\varphi_2 = -0.6° - 70.0° = -70.6° \equiv 289.4°$.
$\varphi_1 = \operatorname{atan2}(0.4678 - 0.00262,\ 0.3710 + 0.2500 - 1.0) = \operatorname{atan2}(0.4652, -0.3790) = 129.17°$.
$\varphi_3 = -0.6° + 180° - 129.175° = 50.225°$. Check: $\theta_1 + \varphi_2 = 359.4° = \varphi_1 + \varphi_3 + 180°$. This is the mode-+ pose of the previous example (the same K, ϕ as in the table). [checked]

TRAP

Radians vs degrees on the calculator; atan2 (not atan) because $x + b\cos\phi - L_0 < 0$ here, which puts φ_1 in the second quadrant; using the printed φ_3 relation; forgetting that FK has two solutions but IK has one per leg.

Gough–Stewart leg IK (homework, [p. 73])



[W04 lec p. 73] — one leg between B and P: S at the base, active P, U at the platform

PREMISE Gough–Stewart IK **Given:** ${}^{B}\mathbf{T}_{P} = [\mathbf{R}\,|\,\mathbf{t}]$ (platform pose in the base frame), base joint points $\mathbf{b}_i$ in B, platform joint points $\mathbf{p}_i$ in P
Find: for each leg i: the prismatic length d_i and the passive S and U angles

Leg vector, length, direction [p. 73]; formulas added

$$ {}^{B}\mathbf{l}_i = \mathbf{t} + \mathbf{R}\,\mathbf{p}_i - \mathbf{b}_i, \qquad d_i = \|{}^{B}\mathbf{l}_i\|, \qquad {}^{B}\hat{\mathbf{n}}_i = {}^{B}\mathbf{l}_i / d_i, \qquad {}^{P}\hat{\mathbf{n}}_i = \mathbf{R}^{\mathsf{T}}\,{}^{B}\hat{\mathbf{n}}_i $$

$$ \psi_i = \operatorname{atan2}(n_y, n_x), \qquad \varepsilon_i = \arcsin n_z \quad (\text{of } {}^{B}\hat{\mathbf{n}}_i : \text{ S angles at the base}) $$

SYMBOLS

$\mathbf{b}_i$	base attachment point of leg i, in B	m
$\mathbf{p}_i$	platform attachment point, in P	m
$\mathbf{t}, \mathbf{R}$	position and rotation of P in B	m, –
d_i	active prismatic length of leg i	m
ψ_i, ε_i	azimuth and elevation of the leg, the S joint's two non-spin angles	rad

Use: one answer per leg. The S spin about the leg axis is free (does nothing). The U angles at the platform are the azimuth/elevation of ${}^{P}\hat{\mathbf{n}}_i$ about the platform axes: the slide fixes no U-axis convention, so state yours.

WORKED Stewart leg lengths, hexagon geometry [added]
Base points on a 1.0 m circle at $-10°, 10°, 110°, 130°, 230°, 250°$; platform points on a 0.5 m circle at $-50°, 50°, 70°, 170°, 190°, 290°$.
Home (level, $\mathbf{t} = (0,0,1)$): leg 1, $\mathbf{b}_1 = (0.9848, -0.1736, 0)$, $\mathbf{p}_1 = (0.3214, -0.3830, 0)$: ${}^{B}\mathbf{l}_1 = (-0.6634, -0.2094, 1.0)$, $d_1 = \sqrt{0.4401 + 0.0438 + 1} = 1.2182$ m (all six equal); azimuth $-162.48°$, elevation $55.17°$.
Posed $\mathbf{t} = (0.05, -0.02, 1.0)$, $\mathbf{R} = R_z(5°)R_y(3°)$, leg 1: ${}^{B}\mathbf{l}_1 = (-0.5817, -0.1999, 0.9832)$, $d_1 = 1.1597$ m, ${}^{B}\hat{\mathbf{n}}_1 = (-0.5016, -0.1724, 0.8478)$: $\psi_1 = -161.03°$, $\varepsilon_1 = 57.97°$; ${}^{P}\hat{\mathbf{n}}_1 = (-0.5584, -0.1280, 0.8197)$ (azimuth $-167.09°$, elevation $55.05°$).
All six legs (m): 1.1597, 1.1974, 1.2170, 1.2812, 1.2152, 1.2504. [checked]

Delta robot IK [added]

PREMISE delta IK **Given:** platform centre $\mathbf{p} = (p_x, p_y, p_z)$ in the base frame B (z up, $p_z < 0$); base radius R_b, platform radius R_p, upper arm L_a, rod L_r
Find: the three motor angles $\theta_1, \theta_2, \theta_3$

Idea: the parallelogram keeps the platform level, so the rod joint of leg i is at $\mathbf{p} + R_p\hat{\mathbf{u}}_i$ and each leg is a small problem like the four-bar: solve every leg alone (this is the loop closure: each rod must stay L_r long).

DERIVATION leg i, azimuth $\gamma_i = 0°, 120°, 240°$
$\hat{\mathbf{u}}_i = (\cos\gamma_i, \sin\gamma_i, 0)$. Motor angle θ_i is measured *down* from horizontal, so the elbow is $\mathbf{E}_i = R_b\hat{\mathbf{u}}_i + L_a(\cos\theta_i\,\hat{\mathbf{u}}_i - \sin\theta_i\,\hat{\mathbf{z}})$. The rod joint is $\mathbf{p} + R_p\hat{\mathbf{u}}_i$. Let $\mathbf{d}_i = \mathbf{p} + (R_p - R_b)\hat{\mathbf{u}}_i$. Rod length: $\|L_a(\cos\theta_i\hat{\mathbf{u}}_i - \sin\theta_i\hat{\mathbf{z}}) - \mathbf{d}_i\| = L_r$. Expand:

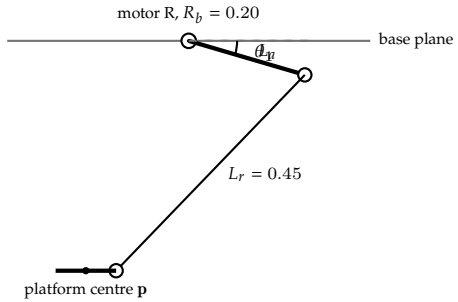
$$ L_a^2 + \|\mathbf{d}_i\|^2 - 2L_a\big(d_u\cos\theta_i - d_z\sin\theta_i\big) = L_r^2, \quad d_u = \mathbf{d}_i\cdot\hat{\mathbf{u}}_i,\ d_z = \mathbf{d}_i\cdot\hat{\mathbf{z}}. $$

With $A = d_u$, $B = -d_z$, $K = \dfrac{\|\mathbf{d}_i\|^2 + L_a^2 - L_r^2}{2L_a}$: $A\cos\theta_i + B\sin\theta_i = K$, and with $\rho = \sqrt{A^2 + B^2}$

$$ \theta_i = \operatorname{atan2}(B, A) - \arccos(K/\rho) \quad \text{(physical root)}, \qquad |K| \le \rho \text{ or the point is out of reach.} $$

Which root is physical
The other root, $\operatorname{atan2}(B,A) + \arccos(K/\rho)$, folds the arm back above the base (elbow inside the base circle, $z > 0$). Choose the root with the elbow **outside the base circle and below the base**: $\|\mathbf{E}_{i,xy}\| > R_b$, $E_{i,z} < 0$.
Use: state this choice; otherwise two answers per leg.

WORKED delta IK numbers
$R_b = 0.20$, $R_p = 0.05$, $L_a = 0.20$, $L_r = 0.45$ m; $\mathbf{p} = (0.03, -0.02, -0.38)$ m.
Leg 1 ($\gamma = 0$): $\mathbf{d}_1 = (0.03 - 0.15, -0.02, -0.38) = (-0.12, -0.02, -0.38)$, $\|\mathbf{d}_1\|^2 = 0.1592$, $K = \dfrac{0.1592 + 0.04 - 0.2025}{0.4} = -0.00825$, $A = -0.12$, $B = 0.38$, $\rho = 0.3985$. $\operatorname{atan2}(0.38, -0.12) = 107.526°$, $\arccos(-0.0207) = 91.186°$: $\theta_1 = 16.339°$ (other root $198.712°$, unphysical).
Legs 2, 3 the same way: $\theta_2 = 30.871°$, $\theta_3 = 22.996°$. Every rod comes out 0.45 m. At home $\mathbf{p} = (0, 0, -0.4)$: $\theta_i = 27.277°$. [checked]



[drawn for this booklet] delta leg 1 in the vertical plane through its arm (the small tangential offset -0.02 m of the rod is not visible in this projection)

Planar five-bar IK (2 DoF) [added]

Mobility $3(5-5-1)+5=2$: two actuators (the two base joints) position the tip $\mathbf{P}$ in the plane.

PREMISE five-bar IK **Given:** tip $\mathbf{P}=(x,y)$ in S; base pivots $\mathbf{O}_1,\mathbf{O}_2$; proximal links L_p, distal links L_d

Find: base angles θ_1,θ_2 (absolute, from $+x$)

DERIVATION each side is a planar 2R problem

For pivot $\mathbf{O}_j$ let $\mathbf{v}_j=\mathbf{P}-\mathbf{O}_j$, $r_j=\|\mathbf{v}_j\|$. The elbow $\mathbf{E}_j$ is at distance L_p from $\mathbf{O}_j$ and L_d from $\mathbf{P}$. Cosine rule in the triangle $\mathbf{O}_j\mathbf{E}_j\mathbf{P}$:

$$L_d^2=r_j^2+L_p^2-2r_jL_p\cos\beta_j \;\Rightarrow\; \cos\beta_j=\frac{r_j^2+L_p^2-L_d^2}{2r_jL_p},\qquad \theta_j=\operatorname{atan2}(v_{j,y},v_{j,x})\pm\beta_j.$$

Reachable on side j only if $|L_p-L_d|\le r_j\le L_p+L_d$. Two roots per side: 4 configurations in total; the usual working one is "elbows outward": $+\beta_1$ for the left pivot, $-\beta_2$ for the right.

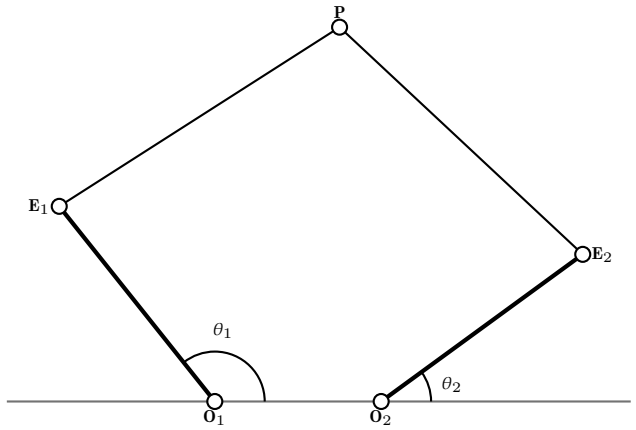
WORKED five-bar numbers

$\mathbf{O}_1=(-0.1,0)$, $\mathbf{O}_2=(0.1,0)$, $L_p=0.3$, $L_d=0.4$ m; $\mathbf{P}=(0.05,\,0.45)$.

Left: $\mathbf{v}_1=(0.15,0.45)$, $r_1=0.4743$, $\cos\beta_1=\dfrac{0.2250+0.09-0.16}{0.2846}=0.5446$, $\beta_1=57.00°$, atan2 $=71.565°$: $\theta_1=128.567°$, $\mathbf{E}_1=(-0.287,\,0.235)$.

Right: $\mathbf{v}_2=(-0.05,0.45)$, $r_2=0.4528$, $\cos\beta_2=0.4969$, $\beta_2=60.20°$, atan2 $=96.340°$: $\theta_2=96.340°-60.202°=36.138°$, $\mathbf{E}_2=(0.342,\,0.177)$.

Check: $\|\mathbf{P}-\mathbf{E}_1\|=\|\mathbf{P}-\mathbf{E}_2\|=0.4$. The other three configurations: $(14.563°,36.138°)$, $(128.567°,156.542°)$, $(14.563°,156.542°)$. FK from θ_1,θ_2: intersect the two radius-L_d circles about $\mathbf{E}_1,\mathbf{E}_2$: two points, $(0.05,0.45)$ and $(0.005,-0.039)$. [checked]



[drawn for this booklet] five-bar, elbows outward, $\theta_1=128.57°$, $\theta_2=36.14°$

Slider-crank (4 bodies, planar) [added]

Ground, crank L_1, rod L_2, slider on the ground line through the crank pivot; 3 R + 1 P. $M=3(4-4-1)+4=1$.

Slider position

$$x=L_1\cos\theta+\sqrt{L_2^2-L_1^2\sin^2\theta},\qquad \text{rod angle }\beta=-\arcsin\frac{L_1\sin\theta}{L_2}$$

SYMBOLS

θ	crank angle from the slider line	rad
L_1, L_2	crank and rod lengths ($L_2>L_1$)	m
x	slider distance from the crank pivot	m

Use: vertical closure: crank-tip height $L_1\sin\theta=-L_2\sin\beta$ (the slider is on the axis); horizontal closure gives x. IK: given x, $\cos\theta$ from the cosine rule.

Numbers $L_1=0.1$, $L_2=0.3$ m: $x(0°)=0.4000$, $x(60°)=0.3372$ ($\beta=-16.78°$, rod length check $=0.3$), $x(90°)=0.2828$, $x(180°)=0.2000$. Stroke $=2L_1=0.2$ m. [checked]

Singularities of parallel robots: what the slides say

The slides (W4) do **not** cover singularities of parallel robots; velocity kinematics and the Jacobian start in W6 [p. 76]. Assembly modes are only implied ("multiple solutions may exist" [p. 67]). [added] What can be said from this week's maths: the four-bar's two modes merge when $d=L_2+L_3$ or $d=|L_2-L_3|$ (circles tangent, $\cos\beta=\pm1$): there the linkage sits at a dead-centre and a small crank motion needs unbounded rocker speed, or the rocker cannot be driven back. A delta or five-bar leg is singular when it is fully stretched or folded ($r_j=L_p+L_d$ or $|L_p-L_d|$). Cross-reference the W6 Jacobian pages for the formal condition $\det\mathbf{J}=0$.

How this is examined: worked solutions

EXAM-STYLE QUESTION W13 Q5: four-bar IK [marks not published]

A planar four-bar has $L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$ m; the output link (coupler) has $a = 0.2$, $b = 0.25$ m. Point K on the output link is at $(0.3710,\ 0.4678)$ m in S, and the link is oriented at $\phi = -0.600°$. Determine $\theta_1, \varphi_1, \varphi_2, \varphi_3$.

Solution

Frames: S at the left pivot, D at $(L_0, 0)$, angles counter-clockwise from $+x$. Write each chain's K equations: $x = L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2)$, $y = L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2)$, $\theta_1+\varphi_2 = \phi \pmod{2\pi}$. [setup]

Substitute: $L_1\cos\theta_1 = x - a\cos\phi$, $L_1\sin\theta_1 = y - a\sin\phi$: $\theta_1 = \operatorname{atan2}(0.4699, 0.1710) = \boxed{70.0°}$; $\varphi_2 = \phi - \theta_1 = \boxed{289.4°}$ $(\equiv -70.6°)$. [left chain]

Right chain: $x = L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3)$ with $\varphi_1+\varphi_3 = \phi+\pi$: $L_3\cos\varphi_1 = x - L_0 + b\cos\phi = -0.3790$, $L_3\sin\varphi_1 = y + b\sin\phi = 0.4652$: $\varphi_1 = \operatorname{atan2}(0.4652, -0.3790) = \boxed{129.2°}$; $\varphi_3 = \phi + \pi - \varphi_1 = \boxed{50.2°}$. [right chain]

Check: $\|(-0.379, 0.4652)\| = 0.600 = L_3$; $\theta_1+\varphi_2 = 359.4° = \varphi_1+\varphi_3+180°$. Say: IK is easy because each leg is solved alone with atan2; one answer per leg. Using the slide's printed $\varphi_3 = \varphi_1 - \phi + \pi$ gives 309.8° and K misses by 0.4 m. [check]

EXAM-STYLE QUESTION W13 Q6: planar mobility [marks not published]

A planar platform is carried by three R–P–R legs (slide figure). Find its mobility with G–K and say whether it is a structure, a mechanism or over-constrained.

Solution

Planar: $D = 3$. Bodies: ground 1; platform 1; each leg is cylinder + piston: $3\times2 = 6$. $N = 8$. Joints: 3 R at the ground, 3 P, 3 R at the platform: $J = 9$, all $F_i = 1$, $\sum F_i = 9$. [counts]

$$M = D(N-J-1) + \sum F_i = 3(8-9-1)+9 = \boxed{3}$$

$M > 0$: a mechanism with 3 DoF (platform x, y and rotation), driven by the 3 prismatic actuators, so it is fully actuated. Check: if each P were locked, $N = 5$ (ground, platform, 3 rigid legs) and $J = 6$ (all R): $3(5-6-1)+6 = 0$, a structure, as it should be. [interpretation]

EXAM-STYLE QUESTION W13 Q7: delta mobility, plain vs extended [marks not published]

A delta robot has a base, three R joints, each followed by a parallelogram of two parallel bars with spherical joints at both ends of every bar, and a moving platform. (a) Find the mobility with plain G–K. (b) The platform has only 3 translational DoF. Reconcile with the extended equation.

Solution

(a) 3D, $D = 6$. $N = 1+3+6+1 = 11$. $J = 3+3\times4 = 15$. $\sum F_i = 3(1+4\times3) = 39$.

$$M = 6(11-15-1)+39 = \boxed{9}$$

(b) Each of the 6 bars has an S joint at both ends, so it can spin about its own axis with nothing else moving: $R_M = 6$. No constraint is repeated: $R_C = 0$.

$$M = 6(11-15-1)+39+0-6 = \boxed{3}$$

Explain: plain G–K counts every DoF as moving the output; the 6 bar spins are internal, so they are subtracted. The result 3 equals the 3 motors and the 3 platform translations.

EXAM-STYLE QUESTION four-bar: assembly modes and crank test (mock-exam style) [marks assumed]

A four-bar has $L_0 = 0.8$ m, crank $L_1 = 0.3$ m (angle θ_1 at S$= (0,0)$), coupler $L_2 = 0.7$ m, rocker $L_3 = 0.5$ m pivoted at $\mathbf{D} = (0.8, 0)$. (a) At $\theta_1 = 60°$ find both positions of the coupler–rocker joint B and the coupler angle ϕ. (b) Can the crank turn fully? (c) Mobility?

Solution

(a) $\mathbf{A} = 0.3(\cos60°, \sin60°) = (0.1500,\ 0.2598)$. $d^2 = 0.64 + 0.09 - 2(0.8)(0.3)(0.5) = 0.49$, $d = 0.7000$ m. B is on the circles ($\mathbf{A}$, 0.7) and ($\mathbf{D}$, 0.5). [setup]

$x_a = \dfrac{0.49 - 0.25 + 0.49}{1.4} = 0.5214$, $h = \sqrt{0.49 - 0.2719} = 0.4670$, $\hat{\mathbf{u}} = (0.9286, -0.3712)$, $\hat{\mathbf{n}} = (0.3712, 0.9286)$. $\mathrm{B}_+ = \mathbf{A} + x_a\hat{\mathbf{u}} + h\hat{\mathbf{n}} = \boxed{(0.8075,\ 0.4999)}$, $\phi = \operatorname{atan2}(0.2401, 0.6575) = 20.06°$; $\mathrm{B}_- = \boxed{(0.4608,\ -0.3674)}$, $\phi = -63.64°$. [two modes]

Check: $|\mathbf{DB}_+| = \sqrt{0.0075^2 + 0.4999^2} = 0.5$ ✓; $\varphi_1 = 89.14°$ (+), $-132.71°$ (−).

(b) Grashof: shortest + longest $= 0.3 + 0.8 = 1.1 \le 0.5 + 0.7 = 1.2$ and the shortest link is the crank: it turns fully (crank-rocker). Equivalent: d runs from 0.5 to 1.1, inside $[\,|L_2 - L_3|,\ L_2 + L_3\,] = [0.2, 1.2]$. [full turn]

(c) $M = 3(4-4-1)+4 = \boxed{1}$. [mobility]

EXAM-STYLE QUESTION planar 3-RPR IK (practice question) [marks assumed]

Leg i runs from a fixed pivot $\mathbf{b}_i$ to a platform point $\mathbf{p}_i$. $\mathbf{b}_1 = (0,0)$, $\mathbf{b}_2 = (1.0, 0)$, $\mathbf{b}_3 = (0.5, 0.9)$ m. In the platform frame $\mathbf{a}_i = 0.2(\cos\gamma_i, \sin\gamma_i)$ m, $\gamma_i = 0°, 120°, 240°$. Find the three leg lengths for the pose $(x, y, \varphi) = (0.5,\ 0.35,\ 10°)$, and state why FK is harder.

Solution

Each leg is a serial chain whose end is known: $\mathbf{p}_i = \mathbf{t} + \mathbf{R}(\varphi)\mathbf{a}_i$, $\ell_i = \|\mathbf{p}_i - \mathbf{b}_i\|$. Attachment directions rotate to $\gamma_i + 10° = 10°, 130°, 250°$. [formula]

$\mathbf{p}_1 = (0.5, 0.35) + 0.2(\cos10°, \sin10°) = (0.6970,\ 0.3847)$; $\mathbf{p}_2 = (0.3714,\ 0.5032)$; $\mathbf{p}_3 = (0.4316,\ 0.1621)$.

$\mathbf{p}_i - \mathbf{b}_i$: $(0.6970, 0.3847)$, $(-0.6286, 0.5032)$, $(-0.0684, -0.7379)$. Lengths:

$$\boxed{\ell_1 = 0.7961,\ \ell_2 = 0.8052,\ \ell_3 = 0.7411 \text{ m}}$$

Common wrong answer: forgetting to rotate $\mathbf{a}_i$ ($\ell_1 = 0.7826$ m). FK needs three coupled nonlinear loop equations with up to six solutions (standard result, [added]), IK none. [answer]

EXAM-STYLE QUESTION spot the errors: the simplest delta [marks assumed]

A student counts the simplest delta (R–U–U legs): "$D = 6$, $N = 8$, $J = 9$, $\sum F_i = 3(1+2+2) = 15$, $M = 6(8-9+1)+15 = 15$. The rod can spin, so $R_M = 6$ and $M = 9$." Find both errors and the correct mobility.

Solution

Error 1: the equation is $D(N-J-\mathbf{1})$, not $N-J+1$ (slide 29's handwritten note is the source of the slip). $M = 6(8-9-1)+15 = -12+15 = \boxed{3}$. [sign]

Error 2: a rod with a universal joint at each end cannot spin on its own (a U joint transmits rotation about the rod axis), so $R_M = 0$ and the plain count 3 is already right, equal to the 3 motors and to the table on slide 30. The student invented a local spin to "fix" a number that was wrong because of Error 1. [R_M]

Contrast: the common delta (ball joints, parallelogram bars) has plain 9, $R_M = 6$, true 3.

Path, trajectory, teach mode [p. 6–10]

PREMISE trajectory generation **Given:** start and goal pose of the tool $\{T\}$ (and optional via points), a duration t_F *or* limits on speed and acceleration
Find: position $q_i(t)$, velocity $\dot{q}_i(t)$, acceleration $\ddot{q}_i(t)$ of *every* joint i, for $t_0 \le t \le t_F$

Slide definition [p. 7]: **a trajectory is a time sequence of position (velocity and acceleration) for every degree of freedom.**

	Path [added]	Trajectory [p. 7]
What	the curve the tool follows in space: a set of poses, *no time*	the same curve *with timing*: $q, \dot{q}, \ddot{q}$ against t
Example	straight line from A to B	line from A to B in 2 s, starting and stopping at rest

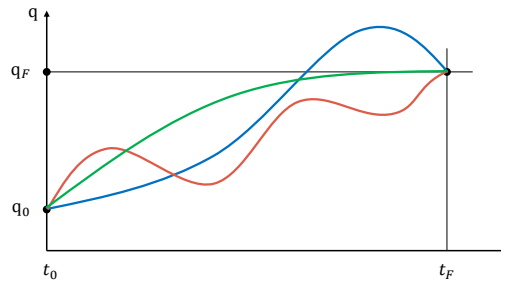
Teach mode [p. 9]: the operator jogs the robot (pendant, or holds the end effector); the joint positions are stored and replayed; *no kinematic calculation*. Good for a repeated task. **Generated trajectory** [p. 10]: given the initial and final pose of the tool, work out the motion in between. Whatever you plan, the motors always receive **joint** positions, velocities and accelerations [p. 68].

Joint space or Cartesian space? [p. 11–14, 42–47]

	Joint (configuration) space [p. 12]	Task (Cartesian) space [p. 43]
You give	IK of start and goal ⇒ joint values; one curve $q_i(t)$ per joint	the path of $\{T\}$ in $\{S\}$ (line, ellipse, …) as ${}^S\mathbf{T}_T(t)$
Cost	cheap: IK only at the two ends	heavier: IK at every sample, then fit and resample
Tool path	not controlled, in general curved [p. 46]	the shape you asked for
Use when	free motion A→B, speed matters	a prescribed path (line, ellipse), obstacles [p. 43]
Must check	joint limits at the ends	workspace, singularity and IK branch *all along* the path [p. 47]

Joint-space approach [p. 12–14]

Steps: (1) convert the initial and final poses to joint positions with IK; (2) generate $q_i(t)$ in between, one joint at a time. Many curves join the same two points [p. 13], so we must choose.



[W05 lec p. 13] — many curves $q(t)$ from q_0 at t_0 to q_F at t_F: which one?

Linear interpolation [p. 14, 31]

$$q(t) = q_0 + \frac{q_F - q_0}{t_F - t_0}(t - t_0), \qquad \dot{q} = \frac{q_F - q_0}{t_F - t_0} = \text{const}$$

SYMBOLS

q_0, q_F	start and goal joint value	rad, m
t_0, t_F	start and finish time	s

Use: simplest trajectory; but the velocity jumps from 0 to $\dot{q}$ at t_0 (infinite acceleration) and back at t_F. This is why we use cubics and blends.

Toolkit: constant-acceleration motion [added]

The three kinematics builders (from school physics)

$$s = ut + \tfrac{1}{2}at^2, \qquad v = u + at, \qquad v^2 = u^2 + 2as$$

SYMBOLS

s	distance moved (here Δq of a joint)	m or rad
u, v	speed at the start / at the end of the interval	per s
a	constant acceleration	per s^2
t	duration of the interval	s

Use: every parabolic blend. ***Graph facts:*** *slope of the q–t graph is the speed; slope of the speed graph is the acceleration;* ***area under the speed graph = distance moved****.*

WORKED from rest at a for t_B
$u = 0$: $v = at_B$ and $s = \frac{1}{2}at_B^2$ (a triangle of area $\frac{1}{2}t_B \cdot v$). Braking from v to 0 at a takes the same t_B and covers the same $\frac{1}{2}at_B^2$. A joint with $a = 300°/s^2$ accelerating for 0.2254 s reaches 67.6°/s having moved 7.62°. [checked]

Cubic polynomial [p. 15–16]

Order-N polynomial $q(t) = \sum_{k=0}^{N} a_k t^k$ has $N+1$ unknowns, so it needs $N+1$ conditions [p. 15]. "Start at rest, stop at the goal" gives **4** conditions (position and velocity at both ends), so the simplest function is a **cubic**.

Cubic trajectory, slide notation [p. 16], $t_0 = 0$

$$q = a_0 + a_1 t + a_2 t^2 + a_3 t^3, \quad \dot{q} = a_1 + 2a_2 t + 3a_3 t^2, \quad \ddot{q} = 2a_2 + 6a_3 t$$

Conditions: $q(0) = q_0$, $q(t_F) = q_F$, $\dot{q}(0) = \dot{q}_0$, $\dot{q}(t_F) = \dot{q}_F$.

SYMBOLS

a_0, a_1, a_2, a_3	the four unknown coefficients	a_k in unit/s^k
q_0, q_F	start and goal position	rad, m
$\dot{q}_0, \dot{q}_F$	start and goal velocity (0 for rest)	per s
t_F	duration (clock starts at $t = 0$ at the start)	s

Use: any move that must start and stop with given positions and speeds. The slide gives $\dot{q}$; the $\ddot{q}$ line is needed for plots and acceleration checks.

DERIVATION of the four coefficients

Let $\Delta q = q_F - q_0$. **Step 1 (start, $t = 0$):** $q(0) = a_0$ and $\dot{q}(0) = a_1$, so $a_0 = q_0$, $a_1 = \dot{q}_0$.
Step 2 (end, $t = t_F$): substitute a_0, a_1:

$$a_2 t_F^2 + a_3 t_F^3 = \Delta q - \dot{q}_0 t_F \quad (i), \qquad 2a_2 t_F + 3a_3 t_F^2 = \dot{q}_F - \dot{q}_0 \quad (ii)$$

Step 3 (eliminate a_2): from (i), $a_2 = (\Delta q - \dot{q}_0 t_F - a_3 t_F^3)/t_F^2$. Put it in (ii):

$$\frac{2(\Delta q - \dot{q}_0 t_F)}{t_F} - 2a_3 t_F^2 + 3a_3 t_F^2 = \dot{q}_F - \dot{q}_0 \;\Rightarrow\; a_3 t_F^2 = (\dot{q}_F + \dot{q}_0) - \frac{2\Delta q}{t_F}$$

Step 4: divide by t_F^2 for a_3; back-substitute for a_2:

$$\boxed{a_3 = \frac{-2\Delta q + (\dot{q}_0 + \dot{q}_F)t_F}{t_F^3}}, \qquad \boxed{a_2 = \frac{3\Delta q - (2\dot{q}_0 + \dot{q}_F)t_F}{t_F^2}}$$

Rest to rest ($\dot{q}_0 = \dot{q}_F = 0$): $a_2 = 3\Delta q/t_F^2$, $a_3 = -2\Delta q/t_F^3$. [checked]

Matrix form (the 4 conditions written as $\mathbf{Ma} = \mathbf{b}$) [p. 16], notebook

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & t_F & t_F^2 & t_F^3 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2t_F & 3t_F^2 \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_F \\ \dot{q}_0 \\ \dot{q}_F \end{bmatrix}, \qquad \mathbf{M}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{3}{t_F^2} & \frac{3}{t_F^2} & -\frac{2}{t_F} & -\frac{1}{t_F} \\ \frac{2}{t_F^3} & -\frac{2}{t_F^3} & \frac{1}{t_F^2} & \frac{1}{t_F^2} \end{bmatrix}$$

Rows of $\mathbf{M}$ in order: $q(0), q(t_F), \dot{q}(0), \dot{q}(t_F)$. $\mathbf{a} = \mathbf{M}^{-1}\mathbf{b}$ reproduces the closed forms above. [checked]
Use: if you forget the closed forms, write $\mathbf{M}$ from the four conditions and solve the 2×2 system; it is the same algebra.

Start time $t_0 \neq 0$ (added)

Use a local clock $\tau = t - t_0 \in [0, T]$ with $T = t_F - t_0$: $q(t) = a_0 + a_1\tau + a_2\tau^2 + a_3\tau^3$ and the coefficient formulas above with $t_F \to T$.
Use: via-point segments (each on its own clock) and any move that does not start at $t = 0$. Example: $10^\circ \to 50^\circ$ from $t_0 = 1$ to $t_F = 3$ has $T = 2$ and the same coefficients $(10, 0, 30, -10)$; $q(t{=}2) = 30^\circ$. [checked]

STEPS Cubic between two points

1. List the four conditions with units (rest $\Rightarrow \dot{q} = 0$); set the clock so the move starts at $\tau = 0$; $T = t_F - t_0$.
2. $a_0 = q_0$, $a_1 = \dot{q}_0$.
3. a_2 and a_3 from the boxed formulas (note T^2 and T^3 in the denominators).
4. Write q, $\dot{q}$, $\ddot{q}$ with numbers.
5. Check $q(T) = q_F$ and $\dot{q}(T) = \dot{q}_F$.
6. Peak speed (rest to rest): $1.5\,\Delta q/T$ at $T/2$; end accelerations $\pm 6\Delta q/T^2$.

TRAP

a_3 divides by T^3, not T^2 ($20^\circ \to 80^\circ$ in 3 s: $a_3 = -2(60)/27 = -4.44$, not -13.3). Units of a_k are unit/s^k. Degrees or radians, but not mixed. A cubic does *not* start or stop with zero acceleration: $\ddot{q} = \pm 6\Delta q/T^2$ at the ends.

Worked example: rest-to-rest cubic [p. 17–19]

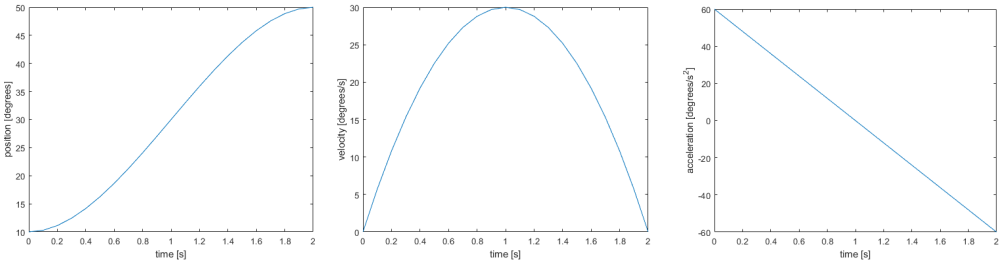
WORKED slide pp. 17–18: $10^\circ \to 50^\circ$ in 2 s, at rest at both ends

Conditions: $q(0) = 10^\circ$, $q(2) = 50^\circ$, $\dot{q}(0) = \dot{q}(2) = 0$. $\Delta q = 40^\circ$, $t_F = 2$.

$$a_0 = 10,\; a_1 = 0,\; a_2 = \frac{3(40)}{2^2} = 30,\; a_3 = \frac{-2(40)}{2^3} = -10.$$

$$q = 10 + 30t^2 - 10t^3\ [^\circ], \quad \dot{q} = 60t - 30t^2\ [^\circ/\text{s}], \quad \ddot{q} = 60 - 60t\ [^\circ/\text{s}^2]$$

Check: $q(2) = 10 + 120 - 80 = 50$ ✓; $\dot{q}(2) = 120 - 120 = 0$ ✓. Peak speed $\dot{q}(1) = 30^\circ/\text{s} = 1.5\Delta q/T$; $\ddot{q}$ runs from $+60$ to -60. [checked]



[W05 lec p. 19] — slide plots of this cubic: position (10 to 50), velocity (peak 30 at 1 s), acceleration (60 to -60)

WORKED extra: non-zero final speed, $20^\circ \to 80^\circ$ in 3 s, $\dot{q}_F = 10^\circ$/s

$\Delta q = 60$: $a_2 = \dfrac{3(60) - (0 + 10)(3)}{9} = 16.667$, $a_3 = \dfrac{-120 + (0 + 10)(3)}{27} = -3.333$. $q(3) = 20 + 150 - 90 = 80$ ✓, $\dot{q}(3) = 100 - 90 = 10$ ✓. [checked]

More than one joint [p. 20], notebook Q1

Each joint gets its *own* cubic with the *same* t_0, t_F ("multi-axis" trajectory), so all joints start and stop together. Example $\Delta q = (40, 90, -20)^\circ$, $T = 2$: $a_2 = (30, 67.5, -15)$, $a_3 = (-10, -22.5, 5)$, peak speeds $(30, 67.5, -15)^\circ/\text{s}$ at $t = 1$ s. [checked]

Via points [p. 21–28]

A **via point** is an extra point the robot must pass on the way to the goal [p. 21], e.g. $\{T\}_{via}$ between $\{T\}_{start}$ and $\{T\}_{final}$.

PREMISE via points **Given:** points q_k at times t_k ($k = 0, \ldots, n$), end velocities (usually 0)
Find: a cubic $q_k(t)$ for every segment $t_k \to t_{k+1}$, each on its own clock $\tau \in [0, T_k]$

Idea [p. 23]: a cubic between each pair of adjacent points. A cubic needs a velocity at both ends, so every via point needs a velocity [p. 24]. If each cubic is rest-to-rest the robot stops at every point: smooth, but wasteful [p. 25]. Three options [p. 26]:

1. the user specifies the via velocities;
2. the system computes them with a simple formula (in joint or task space);
3. the system chooses them so that the **acceleration is continuous**.

W5

Option 2: the simple rule [p. 27]

Chord slope of segment k: $s_k = \dfrac{q_{k+1}-q_k}{t_{k+1}-t_k}$. At a via point between segments $k-1$ and k:

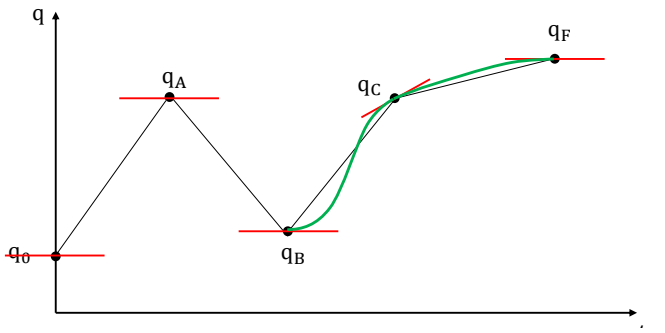
$$\dot{q}_k = \begin{cases} 0 & s_{k-1}s_k \le 0 \ \text{(slope changes sign)} \\ \tfrac{1}{2}(s_{k-1}+s_k) & \text{same sign (average of the two slopes)} \end{cases}$$

Start and end: $\dot{q} = 0$ (the figure on p. 27 shows this; state it as an assumption).

SYMBOLS

s_k	chord slope = average speed on segment k	per s
$\dot{q}_k$	speed prescribed at point k	per s

Use: "apply the slide rule to these points". Gives continuous q and $\dot{q}$; the acceleration can jump.



[W05 lec p. 27] — red tangents: zero speed at q_0, q_A, q_B, q_F (slope changes sign), sloped tangent at q_C (average); green cubic $q_B \to q_C$

STEPS Via points with the simple rule

1. Tabulate (t_k, q_k); segment times $T_k = t_{k+1} - t_k$, $\Delta_k = q_{k+1} - q_k$.
2. Slopes $s_k = \Delta_k / T_k$.
3. Via speeds by the rule; end speeds 0.
4. Each segment: cubic with $(q_0, q_F, \dot{q}_0, \dot{q}_F) = (q_k, q_{k+1}, \dot{q}_k, \dot{q}_{k+1})$ on its own clock.
5. Check $q_k(T_k) = q_{k+1}$ and equal speeds at each join.

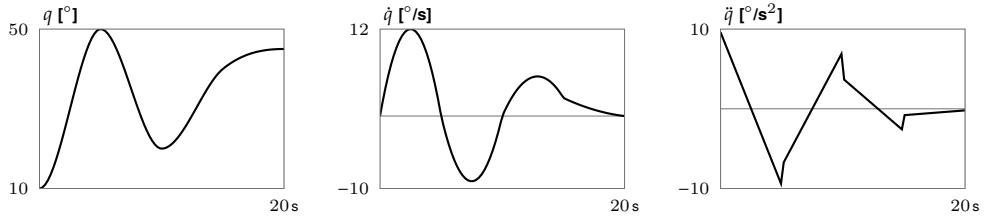
WORKED notebook Q2: $q = 10, 50, 20, 40, 45°$ **at** $t = 0, 5, 10, 15, 20$ **s**

Slopes $s = (+8, -6, +4, +1)°/\text{s}$. At q_A and q_B the sign changes, so 0; at q_C both slopes are positive: $\frac{1}{2}(4+1) = 2.5$. Speeds $(0, 0, 0, 2.5, 0)$. Each $T_k = 5$. Last segment: $\Delta = 5$, $\dot{q}_0 = 2.5$, $\dot{q}_F = 0$:

$$a_2 = \frac{15 - (5)(5)}{25} = -0.4, \quad a_3 = \frac{-10 + 2.5(5)}{125} = 0.02.$$

segment	a_0	a_1	a_2	a_3
$q_0 \to q_A$	10	0	4.8	−0.64
$q_A \to q_B$	50	0	−3.6	0.48
$q_B \to q_C$	20	0	1.9	−0.22
$q_C \to q_F$	40	2.5	−0.4	0.02

Check: last segment $q(5) = 40 + 12.5 - 10 + 2.5 = 45$ ✓. Acceleration jump at the joins (after − before): $+2.4$, -3.4, $+2.0°/\text{s}^2$: the rule keeps $q, \dot{q}$ continuous only. [checked]



[drawn for this booklet] notebook Q2: $q, \dot{q}, \ddot{q}$ plotted from the table above; $\ddot{q}$ jumps at $5, 10, 15$ s

What is continuous? (added)

scheme	q	$\dot{q}$	$\ddot{q}$
one cubic, rest to rest	yes	yes (0 at ends)	no: $\pm 6\Delta q/T^2$ at ends
cubics + simple rule	yes	yes	no: jumps
two cubics, option 3	yes	yes	**yes**
quintic (0 end acc.)	yes	yes	yes
LSPB (below)	yes	yes	no: steps $0 \leftrightarrow \pm a$

Option 3: continuous acceleration [p. 28], notebook Q3, W13 Q8

Slide p. 28: "single via point; given q_0, q_{via}, q_F, $\dot{q}_0 = \dot{q}_F = 0$ and a constant acceleration a_{via}". **Read "constant" as "continuous": $\ddot{q}_1 = \ddot{q}_2$ at the via point** (W13 Q8 says "continuous acceleration"). Prescribing a value a_{via} as well would be a 9th condition on 8 unknowns.

PREMISE Q8 **Given:** $\theta_0, \theta_v, \theta_g$; start and end at rest; segment times t_v (first) and t_F (second), each on its own clock

Find: the two cubics $\theta_1(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$, $\theta_2(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3$ (slide W13 p. 18 notation)

The eight conditions for eight unknowns $(a_0..a_3, b_0..b_3)$

1	$\theta_1(0) = \theta_0$	5	$\theta_2(t_F) = \theta_g$
2	$\dot{\theta}_1(0) = 0$	6	$\dot{\theta}_2(t_F) = 0$
3	$\theta_1(t_v) = \theta_v$	7	$\dot{\theta}_1(t_v) = \dot{\theta}_2(0)$ (speed continuous)
4	$\theta_2(0) = \theta_v$	8	$\ddot{\theta}_1(t_v) = \ddot{\theta}_2(0)$ (acceleration continuous)

Slide W13 p. 18 lists conditions 1, 3, 4, 5, 8; 2, 6, 7 are the rest and speed conditions that complete the count.

ALTERNATIVE APPROACHES

A. Brute force

Write 8 linear equations in a_k, b_k, e.g. $a_0 = \theta_0$, $a_1 = 0$, $b_0 = \theta_v$ from 1, 2, 4; five remain in a_2, a_3, b_1, b_2, b_3. Solve by substitution.

B. Via-speed shortcut (faster)

Call the unknown via speed $w = \dot{\theta}_1(t_v) = \dot{\theta}_2(0)$ (condition 7 holds by construction). Each segment is then an ordinary cubic with known end speeds; only condition 8 is left, and it fixes w.

DERIVATION of the via speed w and the coefficients, durations $T_1 = t_v, T_2 = t_F$

Let $d_1 = \theta_v - \theta_0$ and $d_2 = \theta_g - \theta_v$.

Segment 1 (end speeds 0 and w; use the cubic formulas with $\dot{q}_0 = 0, \dot{q}_F = w$):

$$a_0 = \theta_0,\ a_1 = 0,\ a_2 = \frac{3d_1 - wT_1}{T_1^2},\ a_3 = \frac{-2d_1 + wT_1}{T_1^3}$$

Segment 2 (end speeds w and 0):

$$b_0 = \theta_v,\ b_1 = w,\ b_2 = \frac{3d_2 - 2wT_2}{T_2^2},\ b_3 = \frac{-2d_2 + wT_2}{T_2^3}$$

Condition 8. $\ddot{\theta}_1(T_1) = 2a_2 + 6a_3T_1 = \dfrac{6d_1 - 2wT_1 - 12d_1 + 6wT_1}{T_1^2} = \dfrac{-6d_1 + 4wT_1}{T_1^2}$ and $\ddot{\theta}_2(0) = 2b_2 = \dfrac{6d_2 - 4wT_2}{T_2^2}$. Equate:

$$4w\left(\frac{1}{T_1} + \frac{1}{T_2}\right) = 6\left(\frac{d_1}{T_1^2} + \frac{d_2}{T_2^2}\right) \Rightarrow \boxed{w = \frac{3\,[\,T_1^2 d_2 + T_2^2 d_1\,]}{2T_1T_2(T_1 + T_2)}}$$

Equal durations $T_1 = T_2 = t_f$: $\boxed{w = \dfrac{3(\theta_g - \theta_0)}{4t_f}}$, independent of θ_v. [checked]

W13 Q8 answer, $T_1 = T_2 = t_f$ (symbolic)

$$a_0 = \theta_0,\quad a_1 = 0,\quad a_2 = \frac{12\theta_v - 3\theta_g - 9\theta_0}{4t_f^2},\quad a_3 = \frac{-8\theta_v + 3\theta_g + 5\theta_0}{4t_f^3},$$

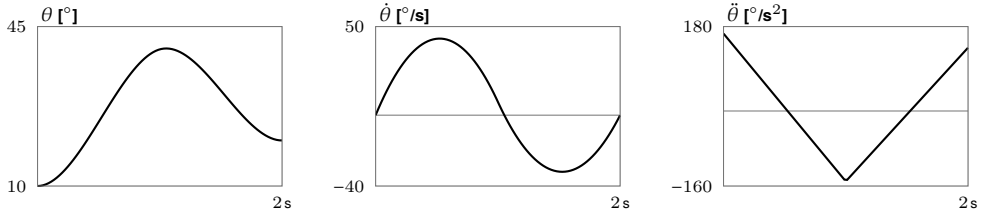
$$b_0 = \theta_v,\quad b_1 = \frac{3(\theta_g - \theta_0)}{4t_f},\quad b_2 = \frac{-12\theta_v + 6\theta_g + 6\theta_0}{4t_f^2},\quad b_3 = \frac{8\theta_v - 5\theta_g - 3\theta_0}{4t_f^3}$$

Shared acceleration at the via point: $\ddot{\theta} = 3(\theta_0 + \theta_g - 2\theta_v)/t_f^2$. [checked](sympy solves the 8×8 system and matches)

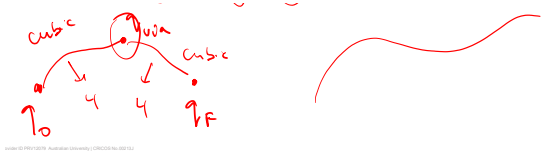
WORKED numbers: $\theta = 10°, 40°, 20°$; durations 1 s

$w = 3(10)/4 = 7.5°/\text{s}$. Segment 1: $10 + 0t + 82.5t^2 - 52.5t^3$; segment 2: $40 + 7.5t - 75t^2 + 47.5t^3$. Checks: $\theta_1(1) = 40$ ✓, $\dot{\theta}_1(1) = 165 - 157.5 = 7.5 = b_1$ ✓, $\ddot{\theta}_1(1) = 165 - 315 = -150 = 2b_2$ ✓, $\theta_2(1) = 40 + 7.5 - 75 + 47.5 = 20$ ✓, $\dot{\theta}_2(1) = 7.5 - 150 + 142.5 = 0$ ✓. [checked]

Unequal times $T_1 = 1, T_2 = 2$: $w = 3[1(-20) + 4(30)]/(2 \cdot 1 \cdot 2 \cdot 3) = 25°/\text{s}$; segment 1: $10 + 65t^2 - 35t^3$; segment 2: $40 + 25t - 40t^2 + 11.25t^3$; shared $\ddot{\theta} = -80$. [checked]



[drawn for this booklet] Q8 numbers, two cubics ($t = 0$ to 1 s segment 1, 1 to 2 s segment 2): the acceleration is a single straight line through the via point; the speed has a smooth minimum after it



[W05 lec p. 28] — the lecturer's sketch: two cubics meeting at q_{via}; $\dot{q}_{via}$ circled as the unknown

STEPS Two cubics through a via point, continuous acceleration

1. Write θ_1, θ_2 on their own clocks; list the 8 conditions (most marks are here).
2. $a_0 = \theta_0, a_1 = 0, b_0 = \theta_v$.
3. Call the via speed w; write both cubics with end speeds $(0, w)$ and $(w, 0)$.
4. Equate $\ddot{\theta}_1(T_1) = \ddot{\theta}_2(0)$; solve w.
5. Substitute w; check positions $\theta_1(T_1) = \theta_v$, $\theta_2(T_2) = \theta_g$ and equal accelerations.

Many via points: spline (added)

For $n+1$ points with $h_i = t_{i+1} - t_i$, chord slopes s_i, and end speeds 0, the interior speeds w_i satisfy

$$\frac{w_{i-1}}{h_{i-1}} + 2\left(\frac{1}{h_{i-1}} + \frac{1}{h_i}\right)w_i + \frac{w_{i+1}}{h_i} = 3\left(\frac{s_{i-1}}{h_{i-1}} + \frac{s_i}{h_i}\right)$$

For the Q2 data this gives $0, 2.304, -3.214, 4.554, 0°/\text{s}$ (the simple rule gave $0, 0, 0, 2.5, 0$). For one via point it reduces to $w = 3(s_1 + s_2)/4$ with equal times. [checked]

Use: only if asked for continuous acceleration through many points; with 8 conditions you solve it by hand for one via point.

Quintic, time scaling, why not polynomials [p. 29–30]

Quintic [p. 29]

$q = a_0 + a_1t + \cdots + a_5t^5$, $\dot{q} = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4$, $\ddot{q} = 2a_2 + 6a_3t + 12a_4t^2 + 20a_5t^3$

6 unknowns need 6 conditions: $q, \dot{q}, \ddot{q}$ at $t = 0$ and at t_F.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & t_F & t_F^2 & t_F^3 & t_F^4 & t_F^5 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2t_F & 3t_F^2 & 4t_F^3 & 5t_F^4 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6t_F & 12t_F^2 & 20t_F^3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_F \\ \dot{q}_0 \\ \dot{q}_F \\ \ddot{q}_0 \\ \ddot{q}_F \end{bmatrix}$$

Use: when acceleration must be continuous at the start and stop (less jerk). Linear: 2 conditions, cubic 4, quintic 6.

DERIVATION rest to rest, zero end accelerations: closed form [added]

$a_0 = q_0$; $\dot{q}(0) = 0 \Rightarrow a_1 = 0$; $\ddot{q}(0) = 0 \Rightarrow a_2 = 0$. With $u = t_F$ and $\Delta = q_F - q_0$ the end conditions are

$$a_3u^3 + a_4u^4 + a_5u^5 = \Delta, \quad 3a_3u^2 + 4a_4u^3 + 5a_5u^4 = 0, \quad 6a_3u + 12a_4u^2 + 20a_5u^3 = 0.$$

From the third: $a_3 = -2a_4u - \frac{10}{3}a_5u^2$. In the second: $-2a_4u^3 - 5a_5u^4 = 0 \Rightarrow a_4 = -\frac{5}{2}a_5u$, so $a_3 = \frac{5}{3}a_5u^2$. In the first: $a_5u^5(\frac{5}{3} - \frac{5}{2} + 1) = a_5u^5/6 = \Delta$.

$$\boxed{a_3 = \frac{10\Delta}{t_F^3}, \quad a_4 = -\frac{15\Delta}{t_F^4}, \quad a_5 = \frac{6\Delta}{t_F^5}}$$

Peak speed $\frac{15}{8}\Delta/t_F = 1.875\,\Delta/t_F$ at $t_F/2$; peak $|\ddot{q}| = \frac{10}{\sqrt{3}}\Delta/t_F^2 \approx 5.77\Delta/t_F^2$ at $t = \frac{1}{2}(1 \mp \frac{1}{\sqrt{3}})t_F$. [checked]

WORKED $0 \to 15°$ in 1 s

Quintic $(a_3, a_4, a_5) = (150, -225, 90)$: $q(1) = 15$, $\dot{q}(1) = 450 - 900 + 450 = 0$, $\ddot{q}(1) = 900 - 2700 + 1800 = 0$ ✓; peak 28.125°/s. Cubic: peak 22.5°/s, end accelerations ±90. [checked]

Time scaling: one shape for every joint (added)

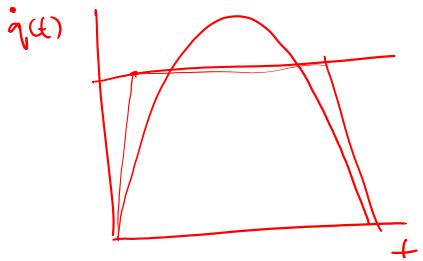
$$q(t) = q_0 + \Delta q\, s(\tau), \qquad \tau = \frac{t - t_0}{T} \in [0, 1]$$

$$s_{lin} = \tau, \qquad s_{cubic} = 3\tau^2 - 2\tau^3, \qquad s_{quintic} = 10\tau^3 - 15\tau^4 + 6\tau^5$$

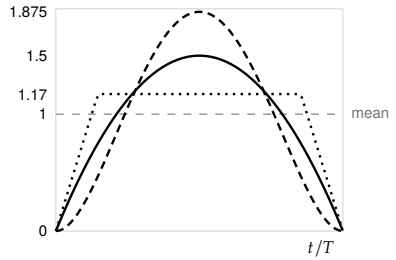
SYMBOLS

$s(\tau)$	dimensionless progress, $s(0) = 0$, $s(1) = 1$	–
τ	normalised time	–
T	duration $t_F - t_0$	s

Use: if all joints use the same $s(\tau)$ they finish together and the joint-space path is a straight line in **q***-space, traversed with a smooth speed. s_{cubic} reproduces $a_2 = 3\Delta q/T^2$, $a_3 = -2\Delta q/T^3$.*



[W05 lec p. 30] — slide p. 30 (lecturer's sketch): a polynomial's velocity is a hump, a trapezoid holds near-peak speed



solid: cubic (1.5) dashed: quintic (1.875) dotted: LSPB, $a = 2a_{min}$ (1.17)

[drawn for this booklet] speed $\dot{q}/(\Delta q/T)$ against t/T for the same move: the trapezoid (LSPB) has the lowest peak, the quintic the highest

	linear	cubic	quintic	LSPB (a=2a_{min})
conditions	2	4	6	3 phases
peak speed /($\Delta q/T$)	1	1.5	1.875	1.17
mean / peak speed	100 %	66.7 %	53.3 %	85.4 %
acceleration at ends	∞	$\pm 6\Delta q/T^2$	0	$\pm a = \pm 8\Delta q/T^2$

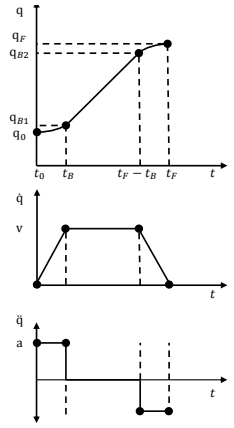
Why polynomials are inefficient [p. 30]**:** the speed is a hump that touches its peak for an instant, so for the same Δq and T it needs a higher peak speed (motor limit) than a trapezoid, which holds a constant speed. [checked]

Linear segment with parabolic blends (LSPB) [p. 31–37]

Problem with a straight line [p. 31]: its speed is constant, so starting and stopping needs infinite acceleration. **Fix** [p. 32]: blend with a parabola at each end; t_B is the duration for which blending is applied. Three phases: speed up at $+a$ for t_B, coast at v, slow down at $-a$ for t_B. The speed graph is a **trapezoid**.

PREMISE LSPB **Given:** $\Delta q = q_F - q_0$, duration t_F ($t_0 = 0$), blend acceleration magnitude a (motor limit)

Find: blend time t_B, coast speed v, and $q(t)$ in the three phases



Linear with parabolic blends

$$q_{B1} = q_0 + \frac{a\,(t_B - t_0)^2}{2}$$ ← Second order polynomial in time

$$q_{B2} = q_{B1} + v(t_F - 2t_B)$$ ← Linear in time

$$q_F = q_{B2} + vt_B - \frac{a\,(t_B)^2}{2}$$ ← Second order polynomial in time

$$t_B = \frac{v}{a}$$

[W05 lec p. 33] — slide pp. 33–34: $q(t)$ (parabola, line, parabola), $\dot{q}$ (trapezoid, v on the plateau), $\ddot{q}$ ($+a$, 0, $-a$) and the three position equations

Slide equations [p. 33–34], $t_0 = 0$

$$q_{B1} = q_0 + \tfrac{1}{2}at_B^2, \qquad q_{B2} = q_{B1} + v(t_F - 2t_B), \qquad q_F = q_{B2} + vt_B - \tfrac{1}{2}at_B^2, \qquad t_B = \frac{v}{a}$$

SYMBOLS

q_{B1}	position at the end of the first blend ($t = t_B$)	rad, deg
q_{B2}	position at the start of the last blend ($t = t_F - t_B$)	rad, deg
v	coast speed (plateau of the trapezoid)	per s
a	blend acceleration, magnitude ($a > 0$ if $q_F > q_0$; use $-a$ if $q_F < q_0$)	per s^2
t_B	duration of each blend	s

The slide writes $q_{B1} = q_0 + a(t_B - t_0)^2/2$; with $t_0 = 0$ this is the first equation. Eq. 1 and 3 are second-order in time, eq. 2 is linear in time.

DERIVATION blend time t_B and the existence condition [p. 36]

Area method. The distance moved is the area under the speed graph, a trapezoid of height v with parallel sides t_F and $t_F - 2t_B$:

$$\Delta q = \tfrac{1}{2}\big[t_F + (t_F - 2t_B)\big]v = v\,(t_F - t_B).$$

Slide method. Add the three equations: $q_F = q_0 + \frac{1}{2}at_B^2 + v(t_F - 2t_B) + vt_B - \frac{1}{2}at_B^2 = q_0 + v(t_F - t_B)$, the same result. Put $v = at_B$:

$$a\,t_B(t_F - t_B) = \Delta q \;\Rightarrow\; \boxed{a\,t_B^2 - a\,t_F\,t_B + \Delta q = 0} \quad [\text{p. 36}]$$

Solve:

$$t_B = \frac{t_F}{2} \pm \frac{\sqrt{a^2t_F^2 - 4a\,\Delta q}}{2a}$$

The root must be real: $a^2t_F^2 - 4a\Delta q \geq 0 \iff \boxed{a \geq \dfrac{4\,\Delta q}{t_F^2}}$ (slide: non-strict). The two blends cannot overlap, $2t_B \leq t_F$, so take the **minus** sign (the plus root is $> t_F/2$). At equality $t_B = t_F/2$: no coast, a triangle with peak $v = 2\Delta q/t_F$.

If v is given instead (toolbox `trapezoidal`): $t_B = t_F - \Delta q/v$ and $a = v/t_B$, which needs $\dfrac{\Delta q}{t_F} < v \leq \dfrac{2\Delta q}{t_F}$. [checked]

The three segments as functions of time, $t_0 = 0$ [added]

$$\begin{aligned}
0 \leq t \leq t_B: &\quad q = q_0 + \tfrac{1}{2}at^2, &\quad \dot q = at, &\quad \ddot q = +a\\
t_B \leq t \leq t_F - t_B: &\quad q = q_0 + v\,(t - \tfrac{1}{2}t_B), &\quad \dot q = v, &\quad \ddot q = 0\\
t_F - t_B \leq t \leq t_F: &\quad q = q_F - \tfrac{1}{2}a\,(t_F - t)^2, &\quad \dot q = a\,(t_F - t), &\quad \ddot q = -a
\end{aligned}$$

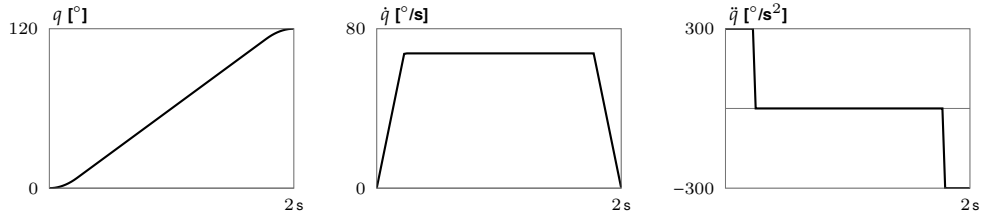
Use: plotting, and finding q at a given t. Check continuity at t_B: $\frac{1}{2}at_B^2 = vt_B/2$ ✓. The speed is continuous everywhere, the acceleration is not.

STEPS LSPB, given Δq, t_F, a

1. List Δq, $t_0 = 0$, t_F, a (units).
2. Existence: compute $4\Delta q/t_F^2$; need $a \geq$ it, else "not possible: larger a or longer t_F".
3. $t_B = \frac{t_F}{2} - \sqrt{a^2t_F^2 - 4a\Delta q}\,/(2a)$.
4. $v = a\,t_B$; coast time $t_F - 2t_B$.
5. Check: $v(t_F - t_B) = \Delta q$. Positions $q_{B1} = q_0 + \frac{1}{2}at_B^2$, $q_{B2} = q_F - \frac{1}{2}at_B^2$.

WORKED slide p. 37: $120°$ in 2 s, $a = 300°/s^2$, find the coast speed

Given $q_F - q_0 = 120$, $t_0 = 0$, $t_F = 2$, $a = 300$. **Step 1:** $4(120)/2^2 = 120 \leq 300$ ✓. **Step 2:** $t_B = 1 \pm \dfrac{\sqrt{300^2 \cdot 4 - 4 \cdot 300 \cdot 120}}{600} = 1 \pm \dfrac{464.76}{600} = 0.2254$ or 1.7746; reject $1.7746 > t_F/2$. **Step 3:** $v = at_B = 300(0.2254) = 67.62°/s$.
Check: $v(t_F - t_B) = 67.62(1.7746) = 120.0$ ✓; $q_{B1} = \frac{1}{2}(300)(0.2254)^2 = 7.62°$, $q_{B2} = 112.38°$. (The slide gives no final number.) [checked]
Variants: $a = 120$ (minimum): $t_B = 1$, $v = 120$, triangle. $a = 100$: discriminant $-8000 < 0$, impossible. $v = 70$ given: $t_B = 2 - 120/70 = 0.2857$ s, $a = 245°/s^2$. $a = 150$: $v = 82.9$. A bigger a gives a *lower* coast speed (it tends to the mean $60°/s$). [checked]



[drawn for this booklet] slide p. 37 example plotted from the three segment equations

TRAP

Strict ">" (the slide says ≥); taking the + root ($t_B = 1.7746$, $v = 532$); t_F instead of t_F^2 in the minimum a; reading a negative discriminant as a number; believing "bigger a gives a faster coast"; forgetting the sign of a when $q_F < q_0$.

SLIDE CHECK slides pp. 33–34

The slide's general-t_0 form $q_{B1} = q_0 + a(t_B - t_0)^2/2$ together with $q_{B2} = q_{B1} + v(t_F - 2t_B)$ and $t_B = v/a$ is only consistent for $t_0 = 0$ (it mixes t_B as a time and as a duration). Verified: $120°$ moved to $t_0 = 1$, $t_F = 3$ with the slide's quadratic $at_B^2 - a(t_F - t_0)t_B + (q_F - q_0 - at_0^2/2) = 0$ gives roots 2.049, -0.049, not 0.2254. **Use the duration $T = t_F - t_0$ and a local clock $\tau = t - t_0$.** All slide examples have $t_0 = 0$.

Fastest LSPB move under speed and acceleration limits [added]

Rest to rest, $|\dot{q}| \le v_{max}$, $|\ddot{q}| \le a_{max}$: from $\Delta = v(T - t_B)$ with $t_B = v/a$, $T = \dfrac{\Delta}{v} + \dfrac{v}{a}$, which is smallest for the largest allowed v.

$$T_{min} = \frac{\Delta}{v_{max}} + \frac{v_{max}}{a_{max}} \quad \text{if } \Delta \ge \frac{v_{max}^2}{a_{max}} \text{ (trapezoid)}$$

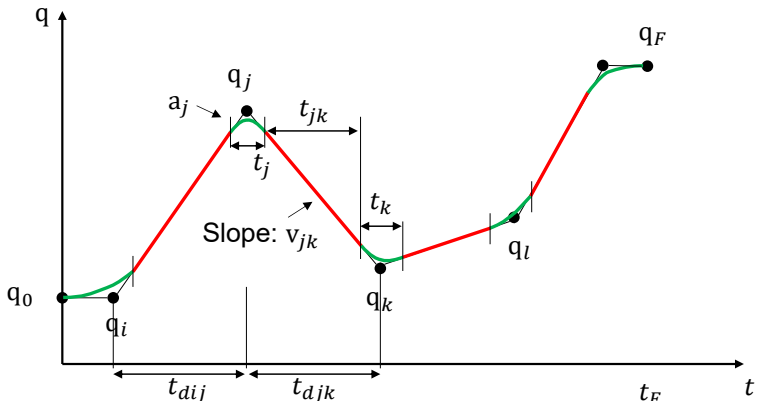
$$T_{min} = 2\sqrt{\frac{\Delta}{a_{max}}}, \quad v_{peak} = \sqrt{a_{max}\Delta} \text{ otherwise (triangle)}$$

SYMBOLS

Δ	distance to move, $\lvert q_F - q_0 \rvert$	deg, rad, m
v_{max}, a_{max}	speed and acceleration limits of the joint	per s, per s^2
T_{min}	shortest possible time	s

Use: "what is the minimum time?" Example $v_{max} = 60°/s$, $a_{max} = 120°/s^2$ (switch distance $v_{max}^2/a_{max} = 30°$): $\Delta = 90°$: $T = 1.5 + 0.5 = 2.0\,s$; $\Delta = 20°$: triangle, $T = 0.816\,s$, peak $49.0°/s$. A cubic with the same limits needs $T \ge \max(1.5\Delta/v_{max}, \sqrt{6\Delta/a_{max}}) = \max(2.25, 2.12) = 2.25\,s$ for $90°$. [checked]

Several segments with blends [p. 38–41]



[W05 lec p. 39] — slide p. 39: path through q_i, q_j, q_k, q_l with blends (green) at the corners and linear parts (red)

Symbols [p. 39]: t_{djk} total duration $q_j \to q_k$; t_{jk} duration of the linear part; t_j duration of the blend at q_j; v_{jk} slope of the linear part; a_j blend acceleration at q_j. **Given** [p. 40]: the via points q_k, the durations t_{djk}, the magnitudes $|a_j|$. **Find**: the blend times. The path passes *near* the via points (a rounded corner), not through them: hitting each exactly and stopping at the ends is over-constrained. The slides refer to Craig §7.3 for the solution and to Corke §3.3.3 (`mstraj`) [p. 41].

Blend times for a path $\theta_1, \theta_2, \ldots, \theta_n$ (Craig §7.3, textbook, not on the slides) [added]

First segment (starts at rest): $\ddot{\theta}_1 = \text{sgn}(\theta_2 - \theta_1)|\ddot{\theta}_1|$,

$$t_1 = t_{d12} - \sqrt{t_{d12}^2 - \frac{2(\theta_2 - \theta_1)}{\ddot{\theta}_1}}, \qquad v_{12} = \frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1}$$

Interior point j: $v_{jk} = \dfrac{\theta_k - \theta_j}{t_{djk}}$, $\ddot{\theta}_j = \text{sgn}(v_{jk} - v_{ij})|\ddot{\theta}_j|$, $t_j = \dfrac{v_{jk} - v_{ij}}{\ddot{\theta}_j}$.

Last segment (stops at rest): $\ddot{\theta}_n = \text{sgn}(\theta_{n-1} - \theta_n)|\ddot{\theta}_n|$,

$$t_n = t_{d,n-1\,n} - \sqrt{t_{d,n-1\,n}^2 + \frac{2(\theta_n - \theta_{n-1})}{\ddot{\theta}_n}}, \qquad v_{n-1\,n} = \frac{\theta_n - \theta_{n-1}}{t_{d,n-1\,n} - \frac{1}{2}t_n}$$

Linear parts: first $t_{12} = t_{d12} - t_1 - \frac{1}{2}t_2$; interior $t_{jk} = t_{djk} - \frac{1}{2}t_j - \frac{1}{2}t_k$; last $t_{n-1\,n} = t_{d,n-1\,n} - \frac{1}{2}t_{n-1} - t_n$. Corner miss = $|\ddot{\theta}_j|t_j^2/8$.

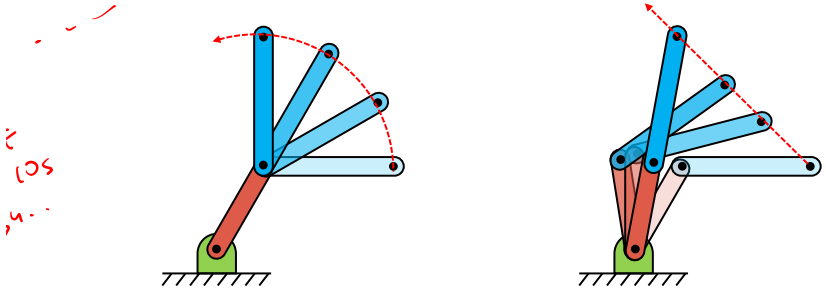
Use: why these: $t_j = \Delta v/a$ is $v = u + at$; the first-segment formulas come from the area rule $\Delta\theta = v_{12}(t_{d12} - \frac{1}{2}t_1)$ with $v_{12} = \ddot{\theta}_1 t_1$; the corner miss is $\frac{1}{2}a(t_j/2)^2$ from $s = \frac{1}{2}at^2$ over half a blend.

WORKED built from these formulas: $\theta = 0°, 40°, 10°$; $t_d = 2, 2$ s; $|\ddot{\theta}| = 50°/s^2$ at all three

$t_1 = 2 - \sqrt{4 - 80/50} = 0.4508$ s, $v_{12} = 40/(2 - 0.2254) = 22.54$. Last: $\ddot{\theta}_3 = \text{sgn}(40 - 10)\,50 = +50$ (the joint moves down and must slow to rest), $t_3 = 2 - \sqrt{4 + 2(10 - 40)/50} = 2 - \sqrt{2.8} = 0.3267$ s, $v_{23} = -30/(2 - 0.1633) = -16.33$. Interior: $\ddot{\theta}_2 = -50$, $t_2 = (-16.33 - 22.54)/(-50) = 0.7775$ s. Linear parts $t_{12} = 2 - 0.4508 - 0.3887 = 1.160$ s, $t_{23} = 2 - 0.3887 - 0.3267 = 1.285$ s. The path misses $40°$ by $50(0.7775)^2/8 = 3.78°$ (it reaches $36.22°$). [checked]

Task-space trajectories [p. 42–47]

Preferred when [p. 43]: a specific path (straight line, ellipse) is wanted for the end effector, or obstacles must be avoided. It costs more computation than joint-based generation. **Frames** [p. 44–45]: $\{B\}$ base, $\{W\}$ wrist, $\{T\}$ tool, $\{S\}$ station, $\{G\}$ goal, $\{C\}$ camera. A common choice: describe the trajectory as the motion of the tool $\{T\}$ in the station frame $\{S\}$, i.e. ${}^S\mathbf{T}_T(t)$. Benefits: the description is decoupled from the manipulator (same description for different robots or tools), and it adapts when $\{S\}$ moves (conveyor belt). For IK use ${}^B\mathbf{T}_T(t) = {}^B\mathbf{T}_S\,{}^S\mathbf{T}_T(t)$ [added].



Linear interpolation in joint space — Straight line path in task space

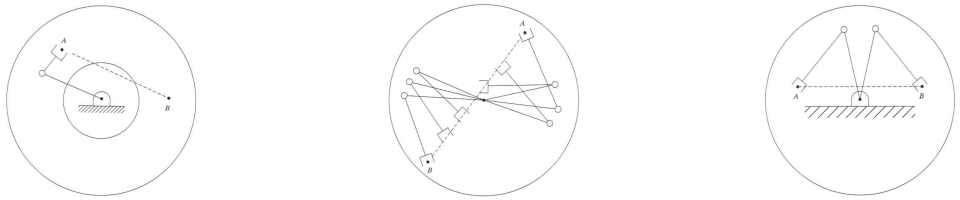
[W05 lec p. 46] — slide p. 46: linear interpolation in joint space (tool sweeps an arc) vs a straight-line path in task space (links reconfigure)

Joint-linear ≠ Cartesian-linear. RR arm L_1=4, L_2=3 m, tool from $(0, 4.75)$ to $(4.75, 0)$: joint-linear

W5

goes from $(51.07°, 95.83°)$ to $(-38.93°, 95.83°)$; q_2 stays constant, so the tool stays $4.75\,\text{m}$ from the base (an arc). Its midpoint is $(3.359, 3.359)$ against the straight-line midpoint $(2.375, 2.375)$: $1.39\,\text{m}$ off. [checked]

Three problems with Cartesian straight lines [p. 47]



Intermediate points may lie outside the workspace

The path may pass through a singularity

Start and goal are reachable in different configurations

[W05 lec p. 47] — slide p. 47: (left) the line crosses the unreachable hole near the base; (middle) it passes through a singularity; (right) A and B are reachable only in different configurations

STEPS Check a Cartesian line before planning

1. **Workspace.** Every point must satisfy $|L_1 - L_2| \le r \le L_1 + L_2$ (RR). Example L_1=1, L_2=0.7 (inner radius 0.3): line $y = 0.2$, $x \in [-1, 1]$: ends $r = 1.02$ (reachable), middle $r = 0.2 < 0.3$ (hole). [checked]
2. **Singularity.** $\det \mathbf{J} = L_1 L_2 \sin q_2$ (= 0 stretched or folded); joint rates $\propto 1/\sin q_2$. L_1=L_2=1, tool at $1\,\text{m/s}$ along x on y=0: largest joint rate $1.15\,\text{rad/s}$ at x=1, 3.20 at 1.9, 10.0 at 1.99, 31.6 at 1.999. [checked]
3. **Configuration.** Same IK branch at A and B? If not, the arm must jump or pass a singularity.

Fixes: plan in joint space around the problem, move the base or the path, or add a via point.

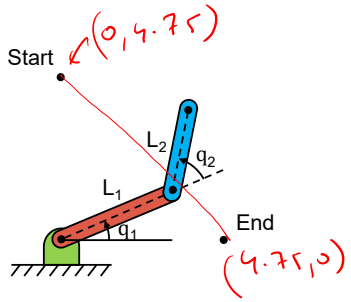
Task-space recipe and the planar RR example [p. 48–64]

Given [p. 48]: the tool trajectory in Cartesian coordinates. **Aim**: joint positions to send to the motors. **Method:** IK at every sample, fit a curve over the joint values, resample from the curve.

STEPS Task-space trajectory (slide steps 1–6)

1. Formulate the path as a function ($y = mx + c$ for a line).
2. Sample it: split t_F into equal sections, get (x_i, y_i).
3. IK at every sample, **same branch** throughout.
4. Fit a curve (cubic, or quintic if one cubic is not enough [p. 60]) to each joint's samples.
5. Resample the fit at the controller rate (slide: 50 points).
6. FK of the resampled joints to see the tool path actually drawn.

Fit first: numerical IK output is not smooth.



[W05 lec p. 49]

Slide p. 49: planar RR, $L_1 = 4\,\text{m}$, $L_2 = 3\,\text{m}$; start $(0, L_1 + 0.25L_2) = (0, 4.75)$, end $(4.75, 0)$; constant speed $3\,\text{m/s}$; tool on a straight line. Goal: joint coordinates for the motion.

Step 1 [p. 50–51]: line through $(0, 4.75)$ and $(4.75, 0)$: $y = -x + 4.75$ ($m = -1$). Length $\sqrt{4.75^2 + 4.75^2} = 4.75\sqrt{2} = 6.7175\,\text{m}$; time $6.7175/3 = 2.2392\,\text{s}$.

Step 2 [p. 52–54]: constant speed v along $y = mx + c$: the unit direction is $(1, m)/\sqrt{1+m^2}$, so

$$x(t) = x_0 + \frac{vt}{\sqrt{1+m^2}}, \qquad y(t) = y_0 + \frac{vmt}{\sqrt{1+m^2}} \;\Rightarrow\; x(t_i) = \frac{3t_i}{\sqrt{2}},\;\; y(t_i) = 4.75 - \frac{3t_i}{\sqrt{2}}$$

10 sections, $t_i = i\,t_F/10 = 0.2239\,i$ (slide: $0.224\,i$).

Planar RR: FK and IK [p. 56], week 3

$$x = L_1 c_1 + L_2 c_{12}, \qquad y = L_1 s_1 + L_2 s_{12}$$

$$\cos q_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}, \quad \sin q_2 = \pm\sqrt{1 - \cos^2 q_2}, \quad q_2 = \operatorname{atan2}(\sin q_2, \cos q_2)$$

$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin q_2,\; L_1 + L_2 \cos q_2)$$

SYMBOLS

q_1	angle of link 1 from the x-axis of the base frame $\{B\}$	rad
q_2	angle of link 2 relative to link 1	rad
L_1, L_2	link lengths	m
(x, y)	tool position in $\{B\}$	m
$\pm$	+: $q_2 > 0$ (elbow one way), −: mirrored arm	–

Use: step 3 of the recipe. Slide note: if the speed is not constant, get the joint velocity at each point from the Jacobian, $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}}$.

Step 3 (IK, branch $q_2 > 0$; computed, the slide table is not transcribed):

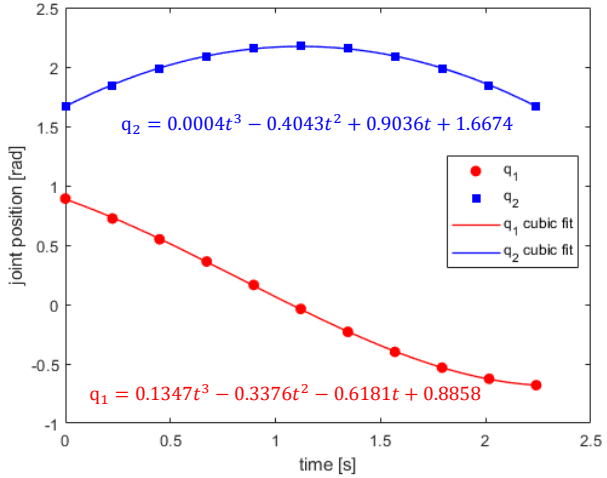
i	t_i [s]	x [m]	y [m]	q_1 [°]	q_2 [°]
0	0	0	4.750	51.07	95.83
1	0.224	0.475	4.275	41.48	105.71
2	0.448	0.950	3.800	31.44	113.73
3	0.672	1.425	3.325	20.75	119.76
4	0.896	1.900	2.850	9.44	123.56
5	1.120	2.375	2.375	−2.13	124.86
6	1.344	2.850	1.900	−13.18	123.56
7	1.567	3.325	1.425	−22.85	119.76
8	1.791	3.800	0.950	−30.48	113.73
9	2.015	4.275	0.475	−35.84	105.71
10	2.239	4.750	0	−38.93	95.83

[checked]

In radians the start is $(0.8914, 1.6725)$ and the end $(-0.6794, 1.6725)$. q_2 is symmetric because the

path is symmetric about the line $y = x$.

Step 4 [p. 58–60]: fit a cubic per joint.



[W05 lec p. 59] — slide p. 59: joint samples and cubic fits for q_1 (red) and q_2 (blue), radians

Fitting the samples by least squares gives $q_1 = 0.1346t^3 - 0.3375t^2 - 0.6182t + 0.8858$ (slide $0.1347, -0.3376, -0.6181$) and $q_2 = 0.0000\,t^3 - 0.4032t^2 + 0.9029t + 1.6674$. Largest fit error $0.46°$; the *fitted* tool path wanders up to 37 mm from the line, a quintic fit only 7.7 mm ([p. 60]: a single cubic may not be enough). [checked]

SLIDE CHECK slide p. 59

The slide prints $q_2 = 0.0004t^3 - 0.4043t^2 + 0.9036t + 1.6674$. Since q_2 is equal at both ends and the samples are symmetric, the cubic term must vanish; a least-squares fit gives $0.0000t^3 - 0.4032t^2 + 0.9029t$. q_1 agrees. Likely rounding of t_i; the method matters, not the last digits.

Steps 5–6 [p. 61–64]: sample each fit at 50 points, then FK $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$ gives the tool path actually followed.

Notebook Q4: the other IK solution ($q_2 < 0$): start $(128.93°, -95.83°)$, end $(38.93°, -95.83°)$; FK gives the *same* tool path with the arm mirrored. Do not switch branch mid-path. [checked]

WORKED velocity with the Jacobian, 3 m/s at the midpoint $(2.375, 2.375)$ (W6 preview)

$\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$ and $\dot{\mathbf{x}} = (3, -3)/\sqrt{2}$ (direction of the line). $\det \mathbf{J} = L_1L_2 \sin q_2 = 12(0.8205) = 9.846$ and $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}} = (-0.893,\ 0)$ rad/s. Sense check: the midpoint is the point of the line nearest the base ($r = 3.359$ is momentarily constant), so $\dot{q}_2 = 0$, and $\dot{q}_1 = -3/3.359$. [checked]

Poses and summary [p. 65–68]

Interpolating a pose: positions and angles can be interpolated directly, but **translation or rotation matrices cannot be interpolated entry by entry** (the result is not a rotation matrix); use `ctraj` [p. 65]. Summary [p. 68]: start and goal are usually given in Cartesian space as the pose of $\{T\}$ with respect to $\{S\}$; the path in between may be generated in joint or task space; the motor commands are *always* joint positions, velocities and accelerations.

Simple scenarios (added)

WORKED RR tool on a straight line: joint angles at $t = T/2$ (slide example)

$L_1 = 4$, $L_2 = 3$, line from $(0, 4.75)$ to $(4.75, 0)$ at 3 m/s: $T = 2.2392$ s, $t = T/2 = 1.1196$ s, point $(2.375, 2.375)$ (halfway along the line).

$\cos q_2 = \dfrac{2(2.375)^2 - 25}{24} = -0.5716$, $\sin q_2 = \pm 0.8205$. Branch +: $q_2 = 124.86°$; $q_1 = \operatorname{atan2}(2.375, 2.375) - \operatorname{atan2}(3(0.8205),\ 4 + 3(-0.5716)) = 45° - \operatorname{atan2}(2.4616, 2.2852) = 45° - 47.13° = -2.13°$. Branch −: $(92.13°, -124.86°)$. FK check on branch +: $x = 4c_1 + 3c_{12} = 3.9972 - 1.6222 = 2.3750$, $y = 4s_1 + 3s_{12} = -0.1486 + 2.5236 = 2.3750$ ✓. [checked]

STEPS When does a via point need a velocity? Checking continuity

1. A cubic needs a speed at *both* ends. At a via point two cubics meet, so both must be given the *same* speed there (option 1, 2 or 3).
2. Speed 0 means the robot stops: only choose it if the path reverses direction (slope changes sign) or the problem says stop.
3. To **check a spline**: (a) positions at every join; (b) speeds equal at every join; (c) accelerations equal only if the question asks for it (option 3, quintic). Evaluate each cubic and its derivatives at $\tau = T_k$ and at $\tau = 0$ of the next.

Reading map and quick answers [added]

Suggested, not from the unit: Craig *Introduction to Robotics* ch. 7 (path generation: cubics, via points, linear with parabolic blends, Cartesian paths); Lynch & Park ch. 9 (time-optimal time scaling, polynomial and trapezoidal profiles); Corke *Robotics, Vision and Control* §3.3 (`jtraj`, `trapezoidal`, `mstraj`, `ctraj`), which the unit's slides cite.

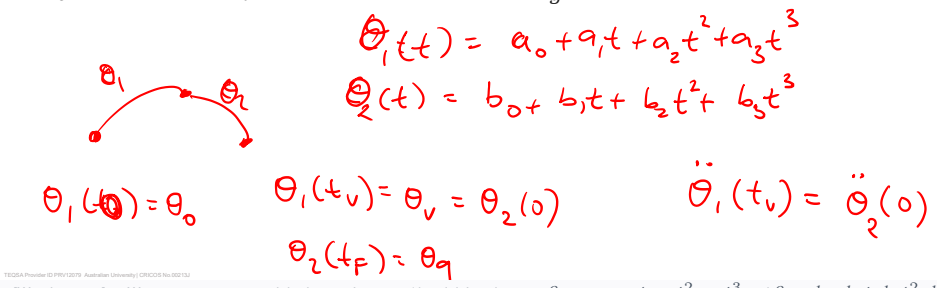
Quick numbers. Rest to rest, Δq in T: cubic $a_2 = 3\Delta q/T^2$, $a_3 = -2\Delta q/T^3$, peak $1.5\Delta q/T$; quintic peak $1.875\Delta q/T$; LSPB minimum $a = 4\Delta q/T^2$ (then $t_B = T/2$, peak $2\Delta q/T$). $90°$ in 1.5 s: $a_{min} = 160°/\text{s}^2$, $t_B = 0.75$ s, triangle peak $120°/\text{s}$. [checked]

How this is examined: worked solutions

Marks are not published; the steps below are where the method marks are likely (about one per step: conditions, formula, substitution, units, check).

EXAM-STYLE QUESTION W13 Q8, symbolic via-point cubics [marks not published]

Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 the via point θ_v is and the goal point is θ_g.



[W13 lec p. 18] — W13 p. 18: question and the lecturer's set-up (the slide's cubics are $\theta_1 = a_0+a_1t+a_2t^2+a_3t^3$ and $\theta_2 = b_0+b_1t+b_2t^2+b_3t^3$)

Solution

Assume (state it): start and end at rest, each segment on its own clock, equal durations t_f; frames: joint angle θ in rad (or deg).

Conditions (8 for 8 unknowns): $\theta_1(0) = \theta_0$, $\dot\theta_1(0) = 0$, $\theta_1(t_f) = \theta_v$, $\theta_2(0) = \theta_v$, $\theta_2(t_f) = \theta_g$, $\dot\theta_2(t_f) = 0$, $\dot\theta_1(t_f) = \dot\theta_2(0)$, $\ddot\theta_1(t_f) = \ddot\theta_2(0)$.

Step 1. $a_0 = \theta_0$, $a_1 = 0$, $b_0 = \theta_v$. Let $w = \dot\theta_2(0) = \dot\theta_1(t_f)$.

Step 2. Segment 1 has end speeds $(0, w)$ and $\Delta = \theta_v - \theta_0$: $a_2 = \dfrac{3\Delta - wt_f}{t_f^2}$, $a_3 = \dfrac{-2\Delta + wt_f}{t_f^3}$.

Segment 2 has $(w, 0)$ and $\Delta' = \theta_g - \theta_v$: $b_1 = w$, $b_2 = \dfrac{3\Delta' - 2wt_f}{t_f^2}$, $b_3 = \dfrac{-2\Delta' + wt_f}{t_f^3}$.

Step 3. $\ddot\theta_1(t_f) = 2a_2 + 6a_3t_f = \dfrac{-6\Delta + 4wt_f}{t_f^2}$, $\ddot\theta_2(0) = 2b_2 = \dfrac{6\Delta' - 4wt_f}{t_f^2}$. Equal: $8wt_f = 6(\Delta + \Delta') \Rightarrow \boxed{w = \dfrac{3(\theta_g - \theta_0)}{4t_f}}$.

Step 4. Substitute w:

$$\boxed{\begin{aligned} a_2 &= \tfrac{12\theta_v - 3\theta_g - 9\theta_0}{4t_f^2}, & a_3 &= \tfrac{-8\theta_v + 3\theta_g + 5\theta_0}{4t_f^3}, \\ b_1 &= \tfrac{3(\theta_g - \theta_0)}{4t_f}, & b_2 &= \tfrac{-12\theta_v + 6\theta_g + 6\theta_0}{4t_f^2}, & b_3 &= \tfrac{8\theta_v - 5\theta_g - 3\theta_0}{4t_f^3} \end{aligned}}$$

Check with $\theta = (10, 40, 20)°$, $t_f = 1$ s: $w = 7.5°/\text{s}$, $\theta_1 = 10 + 82.5t^2 - 52.5t^3$, $\theta_2 = 40 + 7.5t - 75t^2 + 47.5t^3$; $\theta_1(1) = 40$, $\theta_2(1) = 20$, $\ddot\theta_1(1) = -150 = \ddot\theta_2(0)$ ✓. Variant $(0, 40, 30)°$, $t_f = 1$: $w = 22.5$; segment 1 $97.5t^2 - 57.5t^3$, segment 2 $40 + 22.5t - 75t^2 + 42.5t^3$; peak speed in segment 1 $55.11°/\text{s}$ at $t = 0.565$ s. [checked]

Common slips: one cubic for both segments; forgetting $a_1 = 0$; using total time instead of t_f per segment; dropping the acceleration condition (then w is undetermined).

EXAM-STYLE QUESTION W13 Q9, cubic for a prismatic joint [marks not published]

A prismatic joint, initially at rest, is programmed to move from an initial position d=0.02m to a resting goal position for d=0.15m in 0.8 seconds. Calculate the coefficients of a cubic spline that accomplishes this motion.

[W13 lec p. 19] — W13 p. 19 (question text)

Solution

Prismatic joint, so $q = d$ in m. Conditions: $d(0) = 0.02$, $d(0.8) = 0.15$, $\dot d(0) = 0$, $\dot d(0.8) = 0$. $\Delta d = 0.13$ m, $t_F = 0.8$ s.

$a_0 = 0.02$ m, $a_1 = 0$, $a_2 = \dfrac{3(0.13)}{0.8^2} = 0.609375\,\text{m/s}^2$, $a_3 = \dfrac{-2(0.13)}{0.8^3} = -0.5078125\,\text{m/s}^3$.

$$\boxed{d(t) = 0.02 + 0.609375\,t^2 - 0.5078125\,t^3 \text{ m}} \qquad \dot d = 1.21875t - 1.5234375t^2 \text{ m/s}$$

Check: $d(0.8) = 0.02 + 0.39 - 0.26 = 0.15$ ✓; $\dot d(0.8) = 0.975 - 0.975 = 0$ ✓. Peak speed $1.5\Delta d/t_F = 0.24375$ m/s at $t = 0.4$ s ($d = 0.085$ m); end accelerations $\pm 1.21875\,\text{m/s}^2$. [checked]

Common slips: a_3 divided by t_F^2; cm/m mixed; forgetting $a_0 = 0.02$.

EXAM-STYLE QUESTION Via points, simple rule [marks not published]

A joint must pass $10°, 30°, 70°, 80°, 60°$ at $t = 0, 2, 5, 9, 12$ s. Use the slide rule (zero at the ends and where the chord slope changes sign, else the average of the two slopes). (a) The speed at $t = 5$ s. (b) a_2, a_3 of the cubic for $5 \to 9$ s on its own clock. (c) The jump in acceleration at $t = 5$ s.

Solution

Chord slopes: $10,\ 13.333,\ 2.5,\ -6.667°/\text{s}$.

(a) At $t = 5$ both neighbours are positive: $\dot q = \frac{1}{2}(13.333 + 2.5) = 7.917°/\text{s}$. (At $t = 2$: 11.667; at $t = 9$ the slope changes sign: 0.)

(b) Segment $5 \to 9$: $T = 4$, $\Delta = 10$, $\dot q_0 = 7.917$, $\dot q_F = 0$: $a_0 = 70$, $a_1 = 7.917$, $a_2 = \dfrac{30 - (15.833)(4)}{16} = -2.0833$, $a_3 = \dfrac{-20 + 7.917(4)}{64} = 0.18229$. Check $q(4) = 70 + 31.667 - 33.333 + 11.667 = 80$ ✓, $\dot q(4) = 7.917 - 16.667 + 8.750 = 0$ ✓.

(c) Segment $2 \to 5$ ($T = 3$, $\Delta = 40$, $\dot q_0 = 11.667$, $\dot q_F = 7.917$): $a_2 = 2.9167$, $a_3 = -0.7870$; acceleration at its end $2(2.9167) + 6(-0.7870)(3) = -8.333$. Start of the next: $2a_2 = -4.167$. Jump $= \boxed{4.167°/\text{s}^2}$: the rule gives continuous $q, \dot q$ only. [checked]

EXAM-STYLE QUESTION LSPB: minimum acceleration, blend time, cruise speed [marks not published]

A joint moves $75°$ in $t_F = 1.5$ s with a linear segment and parabolic blends, $a = 200°/\text{s}^2$. (a) Minimum acceleration for which the move is possible. (b) Blend time and cruise speed. (c) Would $a = 130°/\text{s}^2$ work?

Solution

(a) $a_{min} = 4\Delta q/t_F^2 = 4(75)/2.25 = 133.33°/\text{s}^2$.

(b) $200 \geq 133.33$ ✓. $a^2t_F^2 - 4a\Delta q = 90000 - 60000 = 30000$, $\sqrt{} = 173.21$. $t_B = 0.75 - \dfrac{173.21}{400} = 0.317$ s (smaller root). $v = at_B = \boxed{63.40°/\text{s}}$; coast time $1.5 - 0.634 = 0.866$ s. Check: $v(t_F - t_B) = 63.40(1.183) = 75.0$ ✓.

(c) $130 < 133.33$: discriminant $130^2(2.25) - 4(130)(75) = -975 < 0$, no real t_B: not possible in 1.5 s with that acceleration (larger a or longer t_F). [checked]

W5

EXAM-STYLE QUESTION LSPB with a sketch [marks not published]

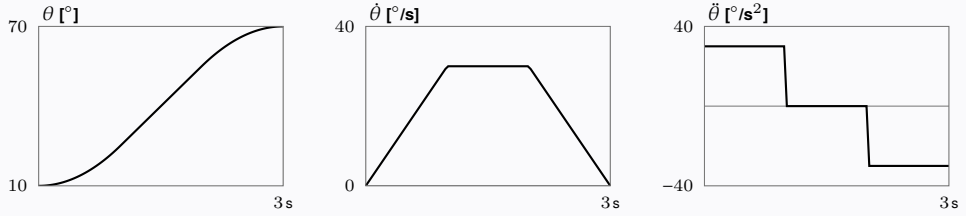
A joint moves from $10°$ to $70°$ in 3 s with blend acceleration $30°/s^2$. (a) Is the acceleration large enough? (b) Blend time, linear-segment speed, angle at the end of the first blend. (c) Sketch $\theta, \dot\theta, \ddot\theta$.

Solution

(a) $4(60)/3^2 = 26.67 \le 30$ ✓.

(b) $t_B = 1.5 - \dfrac{\sqrt{30^2 \cdot 9 - 4 \cdot 30 \cdot 60}}{60} = 1.5 - \dfrac{30}{60} = \boxed{1.0\text{ s}}$; $v = 30(1.0) = \boxed{30°/\text{s}}$; $\theta(t_B) = 10 + \frac{1}{2}(30)(1)^2 = \boxed{25°}$. Coast 1 s from $25°$ to $55°$; last blend $55° \to 70°$.

(c) Parabola, straight line (slope 30), parabola; speed trapezoid $0 \to 30 \to 0$; acceleration $+30$, 0, -30 (steps).



[drawn for this booklet] the sketch an examiner expects (axes labelled with the values $10, 25, 55, 70; 30; \pm 30$ and the times $1, 2, 3$ s)

[checked]

EXAM-STYLE QUESTION Planar RR tool on a circle [added] [marks not published]

A planar RR arm ($L_1 = 0.6$ m, $L_2 = 0.4$ m, base frame $\{B\}$ at the first joint) must move its tool once counter-clockwise round a circle of radius 0.2 m centred at $(0.5, 0.3)$ in $T = 4$ s at constant speed, starting at $(0.7, 0.3)$. (a) Show the path is reachable. (b) Joint angles at $t = T/2$ (branch $q_2 > 0$). (c) Joint speeds at $t = 0$.

Solution

Path: $x = 0.5 + 0.2\cos(2\pi t/4)$, $y = 0.3 + 0.2\sin(2\pi t/4)$; speed $\omega r = (2\pi/4)(0.2) = 0.314$ m/s.

(a) Distance of the centre from the base $\sqrt{0.5^2 + 0.3^2} = 0.583$, so path points lie 0.383 to 0.783 m from the base, inside $[|L_1 - L_2|,\ L_1 + L_2] = [0.2,\ 1.0]$ ✓; no hole or boundary crossing.

(b) $t = 2$: $(0.3, 0.3)$. $\cos q_2 = \dfrac{0.18 - 0.52}{0.48} = -0.7083$, $\sin q_2 = 0.7059$, $q_2 = 135.10°$. $q_1 = \text{atan2}(0.3, 0.3) - \text{atan2}(0.4(0.7059),\ 0.6 - 0.4(0.7083)) = 45° - \text{atan2}(0.2824, 0.3167) = 45° - 41.72° = \boxed{3.28°}$. Other samples ($q_2 > 0$): $t = 0$: $(-8.21°, 82.82°)$; $t = 1$: $(10.58°, 92.39°)$; $t = 3$: $(-29.95°, 122.80°)$.

(c) At $t = 0$: $\dot{\mathbf{x}} = (0, 0.3142)$ m/s. $\mathbf{J} = \begin{bmatrix} -0.3000 & -0.3857 \\ 0.7000 & 0.1061 \end{bmatrix}$ ($q_1 = -8.21°$, $q_{12} = 74.61°$), $\det \mathbf{J} = 0.2381$ (not singular), $\dot{\mathbf{q}} = \mathbf{J}^{-1}\dot{\mathbf{x}} = \boxed{(0.509,\ -0.396)\text{ rad/s}}$. [checked]

Check FK at $t = 2$ gives $(0.3, 0.3)$ ✓. If $\det \mathbf{J} \to 0$ along the path (arm stretched or folded) the joint speeds would blow up.

EXAM-STYLE QUESTION Spot the errors: via speed [marks not published]

Two cubics join $0°, 40°, 60°$ with $T_1 = T_2 = 2$ s, zero end speeds and continuous acceleration at the via point. A student writes: "the via speed is $3(q_F - q_0)/(4T)$ with $T = 4$ s, so $\dot q_{via} = 11.25°/\text{s}$. Segment 1 is the rest-to-rest cubic $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$." Find the errors and give the correct answer.

Solution

Error 1: T in $w = 3(q_F - q_0)/(4T)$ is the duration of *each* segment (2 s), not the total: $w = 3(60)/(4 \cdot 2) = \boxed{22.5°/\text{s}}$.

Error 2: segment 1 must *end* at $22.5°/\text{s}$, so it is not rest-to-rest; use the general cubic with $(\dot q_0, \dot q_F) = (0, 22.5)$, $\Delta = 40$: $a_2 = \dfrac{120 - 22.5(2)}{4} = 18.75$, $a_3 = \dfrac{-80 + 22.5(2)}{8} = -4.375$. Segment 2 ($22.5 \to 0$, $\Delta = 20$): $b_1 = 22.5$, $b_2 = -7.5$, $b_3 = 0.625$.

Check: $q_1(2) = 75 - 35 = 40$ ✓; $\dot q_1(2) = 2(18.75)(2) + 3(-4.375)(4) = 75 - 52.5 = 22.5$ ✓; $\ddot q_1(2) = 37.5 - 52.5 = -15 = 2b_2$ ✓; $q_2(2) = 40 + 45 - 30 + 5 = 60$ ✓. [checked]

The aim: joint rates → tool velocity [p. 6–7, 20–21]

PREMISE velocity kinematics **Given:** joint values $\mathbf{q}$ and joint rates $\dot{\mathbf{q}}$ of an n-joint arm
Find: tool linear velocity $\boldsymbol{v}$ (m/s) and angular velocity $\boldsymbol{\omega}$ (rad/s), both in frame $\{0\}$

Manipulator Jacobian [p. 20–21]

$$ {}^0\boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_v(\mathbf{q}) \\ \mathbf{J}_\omega(\mathbf{q}) \end{bmatrix} \dot{\mathbf{q}} = {}^0\mathbf{J}(\mathbf{q})\,\dot{\mathbf{q}}, \qquad \underbrace{{}^0\boldsymbol{\nu}}_{6\times 1} = \underbrace{{}^0\mathbf{J}}_{6\times n}\ \underbrace{\dot{\mathbf{q}}}_{n\times 1} $$

SYMBOLS

${}^0\boldsymbol{\nu}$	spatial velocity (twist) of the tool in $\{0\}$: linear part on top, angular part below	m/s; rad/s
$\boldsymbol{v}, \boldsymbol{\omega}$	linear velocity of the tool origin, angular velocity of the tool (3×1 each)	m/s; rad/s
$\mathbf{J}_v$, $\mathbf{J}_\omega$	the top three and bottom three rows of $\mathbf{J}$ ($3\times n$)	m (revolute joint); 1
${}^0\mathbf{J}(\mathbf{q})$	manipulator Jacobian in frame $\{0\}$; depends on the pose $\mathbf{q}$	–
$\dot{\mathbf{q}}$	joint rates: $\dot{\theta}_j$ (revolute) or $\dot{d}_j$ (prismatic)	rad/s; m/s
n	number of joints (6 for a general arm, 2 for RR)	–

Use: joint rates are known, tool velocity wanted. The left superscript 0 is the frame in which $\boldsymbol{v}, \boldsymbol{\omega}$ are written (slide 21: "pay attention to the frame"). $\mathbf{J}$ is linear in $\dot{\mathbf{q}}$ but changes with $\mathbf{q}$.

Which formula for which question (W6 map) [added]

the question gives …finds …	use
$\mathbf{q}, \dot{\mathbf{q}}$; tool velocity	$\boldsymbol{\nu} = \mathbf{J}\dot{\mathbf{q}}$
$\mathbf{q}$, wanted $\boldsymbol{\nu}$; joint rates	$\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$ (square), pseudo-inverse otherwise
"where is it singular?"	$\det \mathbf{J} = 0$ (rank drop)
"how well can it move in each direction?"	$\mathbf{J}\mathbf{J}^\top$: ellipsoid, m
tool force; joint torques	$\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F}_E$
joint torques; tool force	$\mathbf{F} = \mathbf{J}^{-\top}\boldsymbol{\tau}$ (square, non-singular)

Prerequisite: skew-symmetric matrix and vex [p. 9–12]

Skew matrix $[\mathrm{s}]_\times$ and vex operator $\vee_\times$ [p. 10]

$$ \mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \longmapsto \mathbf{S} = [\mathbf{s}]_\times = \begin{bmatrix} 0 & -s_3 & s_2 \\ s_3 & 0 & -s_1 \\ -s_2 & s_1 & 0 \end{bmatrix} \in so(3), \qquad \mathbf{s} = \vee_\times(\mathbf{S}) = \begin{bmatrix} S_{32} \\ S_{13} \\ S_{21} \end{bmatrix} $$

SYMBOLS

$\mathbf{s}$	any vector in $\mathbb{R}^3$ (for angular velocity: $\boldsymbol{\omega}$)	any; rad/s
$[\mathbf{s}]_\times$	the skew-symmetric 3×3 matrix of $\mathbf{s}$: $\mathbf{S}^\top = -\mathbf{S}$, zero diagonal	same as $\mathbf{s}$
$\vee_\times$	vex: reads $\mathbf{s}$ back from a skew matrix	–
S_{ij}	entry of $\mathbf{S}$ in row i, column j	–
$so(3)$	the set of all 3×3 skew-symmetric matrices	–

Use: write a cross product as a matrix product, so it can sit inside matrix algebra ($[\boldsymbol{\omega}]_\times\mathbf{R}$); vex reads $\boldsymbol{\omega}$ off $\dot{\mathbf{R}}\mathbf{R}^\top$.

Properties of $[\mathrm{a}]_\times$ (for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, $\mathbf{R}$ a rotation) [added]

$[\mathbf{a}]_\times^\top = -[\mathbf{a}]_\times$ — $[\mathbf{a}]_\times\mathbf{a} = \mathbf{0}$
$[\mathbf{a}]_\times\mathbf{b} = \mathbf{a}\times\mathbf{b} = -[\mathbf{b}]_\times\mathbf{a}$ — $[\mathbf{a}+\mathbf{b}]_\times = [\mathbf{a}]_\times + [\mathbf{b}]_\times$
$[\mathbf{a}]_\times^2 = \mathbf{a}\mathbf{a}^\top - |\mathbf{a}|^2\mathbf{I}$ — $\mathbf{R}[\mathbf{a}]_\times\mathbf{R}^\top = [\mathbf{R}\mathbf{a}]_\times$
$\det[\mathbf{a}]_\times = 0$ (rank 2) — $\vee_\times([\mathbf{a}]_\times) = \mathbf{a}$

Use: the sixth identity is why $\dot{\mathbf{R}}\mathbf{R}^\top$ is the same skew matrix in any frame; the second says a vector crossed with itself is zero. All verified numerically. [checked]

Augmented skew matrix (twist matrix) $\hat{\mathbf{S}}$ [p. 11–12]

$$ \hat{\mathbf{s}} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \end{bmatrix} \longmapsto \hat{\mathbf{S}} = [\hat{\mathbf{s}}]_\times = \begin{bmatrix} 0 & -s_6 & s_5 & s_1 \\ s_6 & 0 & -s_4 & s_2 \\ -s_5 & s_4 & 0 & s_3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \in se(3), \qquad \hat{\mathbf{s}} = \vee_\times(\hat{\mathbf{S}}) $$

SYMBOLS

s_1, s_2, s_3	linear (translation) part	m/s
s_4, s_5, s_6	angular part, the entries of $[\,\cdot\,]_\times$	rad/s
$se(3)$	the set of such 4×4 matrices	–

Use: the 4×4 derivative matrices $\hat{\mathbf{R}}_x, \hat{\mathbf{R}}_y, \hat{\mathbf{R}}_z$ and $\hat{\mathbf{t}}_x, \hat{\mathbf{t}}_y, \hat{\mathbf{t}}_z$ used below are special cases (unit axis, or unit slide). Same order as $\boldsymbol{\nu}$: linear first, angular second.

The cross product: determinant form and $[\mathrm{a}]_\times\mathrm{b}$ [p. W6 note]

Cross product $\omega \times \mathrm{P}$, written two ways

$$ \boldsymbol{\omega}\times\mathbf{P} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \omega_x & \omega_y & \omega_z \\ P_x & P_y & P_z \end{vmatrix} = \hat{\mathbf{i}}\begin{vmatrix} \omega_y & \omega_z \\ P_y & P_z \end{vmatrix} - \hat{\mathbf{j}}\begin{vmatrix} \omega_x & \omega_z \\ P_x & P_z \end{vmatrix} + \hat{\mathbf{k}}\begin{vmatrix} \omega_x & \omega_y \\ P_x & P_y \end{vmatrix} $$

$$ = (\omega_y P_z - \omega_z P_y)\hat{\mathbf{i}} - (\omega_x P_z - \omega_z P_x)\hat{\mathbf{j}} + (\omega_x P_y - \omega_y P_x)\hat{\mathbf{k}} $$

$$ [\boldsymbol{\omega}]_\times\mathbf{P} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 0\cdot P_x - \omega_z P_y + \omega_y P_z \\ \omega_z P_x + 0\cdot P_y - \omega_x P_z \\ -\omega_y P_x + \omega_x P_y + 0\cdot P_z \end{bmatrix} = \begin{bmatrix} \omega_y P_z - \omega_z P_y \\ \omega_z P_x - \omega_x P_z \\ \omega_x P_y - \omega_y P_x \end{bmatrix} $$

SYMBOLS

$\boldsymbol{\omega}$, $\mathbf{P}$	the two vectors (here: angular velocity, position vector), same frame	rad/s; m
$\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$	unit vectors along x, y, z of that frame	–
$\boldsymbol{\omega}\times\mathbf{P}$	velocity of a point at $\mathbf{P}$ on a body turning at $\boldsymbol{\omega}$: perpendicular to both, length $\lvert\boldsymbol{\omega}\rvert\lvert\mathbf{P}\rvert\sin\varphi$	m/s

Use: Craig's $\boldsymbol{\omega}\times\mathbf{P}$ is this ordinary cross product, not a product of two 3×1 columns. The $\hat{\mathbf{j}}$ term carries the cofactor sign −: its coefficient is $\omega_z P_x - \omega_x P_z$, equal to the middle row of $[\boldsymbol{\omega}]_\times\mathbf{P}$.

W6

WORKED $\boldsymbol{\omega} = (1,2,3)$, $\mathbf{P} = (4,5,6)$

$\hat{\mathbf{i}}$: $2\cdot 6 - 3\cdot 5 = -3$. $\hat{\mathbf{j}}$: $-(1\cdot 6 - 3\cdot 4) = -(6-12) = +6$. $\hat{\mathbf{k}}$: $1\cdot 5 - 2\cdot 4 = -3$. So $\boldsymbol{\omega}\times\mathbf{P} = (-3,6,-3)$.

Skew form: $[\boldsymbol{\omega}]_\times = \begin{bmatrix} 0 & -3 & 2 \\ 3 & 0 & -1 \\ -2 & 1 & 0 \end{bmatrix}$, times $(4,5,6)^\top$ gives $(-15+12,\ 12-6,\ -8+5) = (-3,6,-3)$. [checked]

Point on a spinning body: $\boldsymbol{\omega} = (0,0,2)\,\text{rad/s}$, $\mathbf{r} = (0.3,0.4,0)\,\text{m}$: $\mathbf{v} = (-0.8,0.6,0)\,\text{m/s}$; $|\mathbf{v}| = 2\times 0.5 = 1\,\text{m/s}$, perpendicular to $\mathbf{r}$. [checked]

SLIDE CHECK cross-product note, w06-cross

For two vectors $\omega = [\omega_x,\omega_y,\omega_z]^T$ and $P = [P_x,P_y,P_z]^T$, the cross product is

$$\omega \times P = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \omega_x & \omega_y & \omega_z \\ P_x & P_y & P_z \end{vmatrix}$$

Expanding this determinant gives:

$$\omega \times P = (\omega_y P_z - \omega_z P_y)\hat{\imath} + (\omega_x P_z - \omega_z P_x)\hat{\jmath} + (\omega_x P_y - \omega_y P_x)\hat{k}$$

[W06 cross p. 1] — the note's determinant and its expansion

The expansion on the note prints the $\hat{\mathbf{j}}$ term as $+(\omega_x P_z - \omega_z P_x)$. A determinant expanded along row 1 gives $-(\omega_x P_z - \omega_z P_x) = \omega_z P_x - \omega_x P_z$. With the example above the printed term gives -6, the right one $+6$. The note's own skew-matrix form is right (its second row is $\omega_z P_x - \omega_x P_z$), so as printed its two "equivalent" forms disagree. The note also drops Craig's rotation matrix and frame labels. Follow this page. [checked]

Derivative of a rotation matrix: $\dot{\mathbf{R}} = [\omega]_\times \mathbf{R}$ [p. 18–19]

Angular velocity from a rotation matrix

$$\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times \mathbf{R} \iff [\boldsymbol{\omega}]_\times = \dot{\mathbf{R}}\,\mathbf{R}^\top, \qquad \boldsymbol{\omega} = \vee_\times\left(\dot{\mathbf{R}}\,\mathbf{R}^\top\right)$$

SYMBOLS

$\mathbf{R}(t)$	rotation matrix of the body (3×3), here ${}^0\mathbf{R}_E$	–
$\dot{\mathbf{R}}$	its time derivative, entry by entry (3×3, *not* a vector)	1/s
$\boldsymbol{\omega}$	angular velocity of the body, in the frame $\mathbf{R}$ is measured from	rad/s

Use: the slide's step: $\dot{\mathbf{R}}$ is 3×3 but $\boldsymbol{\omega}$ is 3×1, so convert through the skew matrix. For a rotation θ about a fixed unit axis $\hat{\mathbf{n}}$: $\boldsymbol{\omega} = \dot{\theta}\,\hat{\mathbf{n}}$.

DERIVATION why $\dot{\mathbf{R}}\mathbf{R}^\top$ is skew [added]

$\mathbf{R}\mathbf{R}^\top = \mathbf{I}$. Differentiate with the product rule: $\dot{\mathbf{R}}\mathbf{R}^\top + \mathbf{R}\dot{\mathbf{R}}^\top = \mathbf{0}$. The second term is the transpose of the first, because $(\dot{\mathbf{R}}\mathbf{R}^\top)^\top = \mathbf{R}\dot{\mathbf{R}}^\top$. So $\mathbf{S} = \dot{\mathbf{R}}\mathbf{R}^\top$ obeys $\mathbf{S} + \mathbf{S}^\top = \mathbf{0}$: skew-symmetric. Name it $[\boldsymbol{\omega}]_\times$ and multiply by $\mathbf{R}$ on the right (using $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$): $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times\mathbf{R}$.

WORKED $\mathbf{R}_z(\theta)$, $\theta = 30°$, $\dot{\theta} = 2$ **rad/s**

$\mathbf{R}_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\dot{\mathbf{R}}_z = \dot{\theta}\begin{bmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Then

$$\dot{\mathbf{R}}_z\mathbf{R}_z^\top = \dot{\theta}\begin{bmatrix} -s & -c & 0 \\ c & -s & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} = \dot{\theta}\begin{bmatrix} 0 & -(s^2+c^2) & 0 \\ c^2+s^2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \dot{\theta}\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\vee_\times$: $(S_{32}, S_{13}, S_{21}) = (0,0,\dot{\theta})$, so $\boldsymbol{\omega} = (0,0,2)\,\text{rad/s}$ along z. Numbers: $\dot{\mathbf{R}}_z = \begin{bmatrix} -1 & -1.732 & 0 \\ 1.732 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $[\boldsymbol{\omega}]_\times\mathbf{R}_z$ is the same matrix. [checked]

Velocity of a point in a moving frame [added]

Point Q carried by frame $\{B\}$ (frames explicit)

$$^A\mathbf{v}_Q = {}^A\mathbf{v}_B + {}^A\boldsymbol{\omega}_B \times \left({}^A\mathbf{R}_B\,{}^B\mathbf{Q}\right) + {}^A\mathbf{R}_B\,{}^B\mathbf{v}_Q$$

SYMBOLS

${}^A\mathbf{v}_Q$	velocity of the point Q seen from $\{A\}$, written in $\{A\}$	m/s
${}^A\mathbf{v}_B$	velocity of the origin of $\{B\}$ in $\{A\}$	m/s
${}^A\boldsymbol{\omega}_B$	angular velocity of $\{B\}$ relative to $\{A\}$, in $\{A\}$ (${}^A\dot{\mathbf{R}}_B = [{}^A\boldsymbol{\omega}_B]_\times {}^A\mathbf{R}_B$)	rad/s
${}^B\mathbf{Q}$	position of Q in $\{B\}$ (constant if Q is fixed to the body)	m
${}^B\mathbf{v}_Q$	velocity of Q inside $\{B\}$; zero for a fixed point (drop the last term)	m/s
${}^A\mathbf{R}_B$	rotation of $\{B\}$ in $\{A\}$; turns a vector from $\{B\}$ into $\{A\}$	–

Use: the three terms: the frame's origin moves, the frame spins (lever arm ${}^A\mathbf{R}_B\,{}^B\mathbf{Q}$), the point slides inside the frame. This is the step Craig repeats link by link (velocity propagation below).

DERIVATION from the position [added]

${}^A\mathbf{Q} = {}^A\mathbf{p}_B + {}^A\mathbf{R}_B\,{}^B\mathbf{Q}$. Differentiate (product rule): ${}^A\dot{\mathbf{Q}} = {}^A\dot{\mathbf{p}}_B + {}^A\dot{\mathbf{R}}_B\,{}^B\mathbf{Q} + {}^A\mathbf{R}_B\,{}^B\dot{\mathbf{Q}}$. Insert $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times\mathbf{R}$: the middle term is $[{}^A\boldsymbol{\omega}_B]_\times {}^A\mathbf{R}_B\,{}^B\mathbf{Q} = {}^A\boldsymbol{\omega}_B \times ({}^A\mathbf{R}_B\,{}^B\mathbf{Q})$.

WORKED numbers

${}^A\mathbf{R}_B = \mathbf{R}_z(90°)$, ${}^A\mathbf{v}_B = (0.1,0,0)\,\text{m/s}$, ${}^A\boldsymbol{\omega}_B = (0,0,2)\,\text{rad/s}$, ${}^B\mathbf{Q} = (0.3,0,0)\,\text{m}$. Lever arm in $\{A\}$: $\mathbf{R}_z(90°)(0.3,0,0) = (0,0.3,0)$. $\boldsymbol{\omega}\times$ lever $= (0,0,2)\times(0,0.3,0) = (-0.6,0,0)$. Fixed point: ${}^A\mathbf{v}_Q = (0.1,0,0) + (-0.6,0,0) = (-0.5,0,0)\,\text{m/s}$. If Q also slides at ${}^B\mathbf{v}_Q = (0,0.05,0)$: add $\mathbf{R}_z(90°)(0,0.05,0) = (-0.05,0,0)$, total $(-0.55,0,0)\,\text{m/s}$. [checked](finite difference of the position agrees)

W6

ETS and the time derivative of the pose [p. 13–16]

Chain rule on the pose

$$ {}^0\mathbf{T}_E = \prod_{i=1}^{M} \mathbf{E}_i(\eta_i), \qquad \dot{\mathbf{T}} = \frac{d\mathbf{T}}{dt} = \frac{\partial \mathbf{T}}{\partial q_1}\dot{q}_1 + \cdots + \frac{\partial \mathbf{T}}{\partial q_n}\dot{q}_n, \qquad \frac{\partial \mathbf{T}}{\partial q_j} = \begin{bmatrix} \mathbf{J}_{R_j} & \mathbf{J}_{t_j} \\ \mathbf{0} & 0 \end{bmatrix} $$

so $\dot{\mathbf{R}} = \sum_j \mathbf{J}_{R_j}\dot{q}_j$ and $\dot{\mathbf{t}} = \sum_j \mathbf{J}_{t_j}\dot{q}_j$.

SYMBOLS

$\mathbf{E}_i(\eta_i)$	the ith elementary transform (4×4); η_i is a constant (link length, fixed twist) or a joint variable q_j	m or rad
M	number of elementary transforms in the sequence ($M \geq n$)	–
$\partial\mathbf{T}/\partial q_j$	4×4 matrix, bottom row zero	per unit q_j
$\mathbf{J}_{R_j}$	3×3 block: rate of change of $\mathbf{R}$ per unit q_j	–
$\mathbf{J}_{t_j}$	3×1 block: rate of change of the tool position per unit q_j	m per q_j

Use: the pose is a function of $\mathbf{q}$ *only, so every other time dependence passes through* $\dot{q}_j$.

Jacobian columns from $\partial\mathbf{T}/\partial q_j$ [p. 17–19, 25–26]

$$ \mathbf{J}_v = \begin{bmatrix} \mathbf{J}_{t_1} & \cdots & \mathbf{J}_{t_n} \end{bmatrix}, \quad \mathbf{J}_{t_j} = \tau\left(\frac{\partial \mathbf{T}}{\partial q_j}\right); \qquad \mathbf{J}_\omega = \begin{bmatrix} \mathbf{J}_{\omega_1} & \cdots & \mathbf{J}_{\omega_n} \end{bmatrix}, \quad \mathbf{J}_{\omega_j} = \vee_\times\left(\rho\left(\frac{\partial \mathbf{T}}{\partial q_j}\right)\rho(\mathbf{T})^\top\right) $$

SYMBOLS

$\tau(\mathbf{M})$	the 3×1 translation column of a 4×4 matrix (rows 1–3 of column 4)	–
$\rho(\mathbf{M})$	the 3×3 rotation block (rows 1–3, columns 1–3)	–
$\rho(\mathbf{T}) = \mathbf{R}$	rotation part of the pose itself	–

Use: two read-offs from the same 4×4 *matrix* $\partial\mathbf{T}/\partial q_j$*; stack* $\mathbf{J}_v$ *above* $\mathbf{J}_\omega$ *to get* ${}^0\mathbf{J}$ $(6\times n)$.

DERIVATION of the rotation part [p. 18–19]

$\dot{\mathbf{R}} = \mathbf{J}_{R_1}\dot{q}_1 + \cdots + \mathbf{J}_{R_n}\dot{q}_n$ and $\dot{\mathbf{R}} = [\boldsymbol{\omega}]_\times\mathbf{R}$, so $[\boldsymbol{\omega}]_\times\mathbf{R} = \sum_j \mathbf{J}_{R_j}\dot{q}_j$. Post-multiply every term by $\mathbf{R}^\top$ ($\mathbf{R}\mathbf{R}^\top = \mathbf{I}$): $[\boldsymbol{\omega}]_\times = \sum_j(\mathbf{J}_{R_j}\mathbf{R}^\top)\dot{q}_j$. Take $\vee_\times$ (it is linear): $\boldsymbol{\omega} = \sum_j \vee_\times(\mathbf{J}_{R_j}\mathbf{R}^\top)\dot{q}_j$. The translation part needs no trick: $\boldsymbol{v} = \dot{\mathbf{t}} = \sum_j \mathbf{J}_{t_j}\dot{q}_j$ (slide 17).

What the columns mean; a 6-DoF Jacobian [p. 22–24]

Reading a 6×6 Jacobian [p. 22–23]

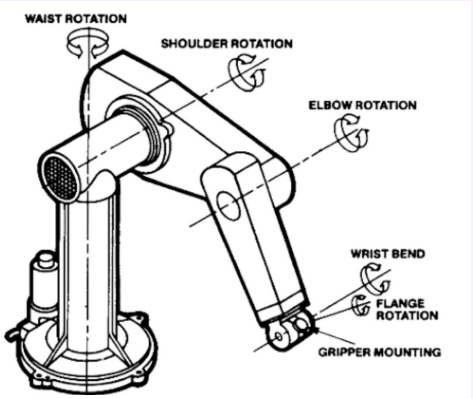
$$ \begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} j_{11} & j_{12} & j_{13} & j_{14} & j_{15} & j_{16} \\ j_{21} & j_{22} & j_{23} & j_{24} & j_{25} & j_{26} \\ j_{31} & j_{32} & j_{33} & j_{34} & j_{35} & j_{36} \\ j_{41} & j_{42} & j_{43} & j_{44} & j_{45} & j_{46} \\ j_{51} & j_{52} & j_{53} & j_{54} & j_{55} & j_{56} \\ j_{61} & j_{62} & j_{63} & j_{64} & j_{65} & j_{66} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix} $$

Use: column j *= the tool's spatial velocity when only joint* j *moves at unit rate (1 rad/s or 1 m/s); the effects of the joints add. Row* i *= how every joint feeds the* i*th component. Units: linear rows are m per rad for a revolute joint (m/s per rad/s), angular rows are dimensionless. 6-joint arm:* 6×6*; planar RR:* 6×2 *with rows* v_z, ω_x, ω_y *identically zero.*

SLIDE CHECK slides 22–23

The slide prints the last entry as j_{76}; the matrix has six rows, so it is j_{66} (typed above).

WORKED slide 24: PUMA 560, a numeric 6×6 Jacobian



PUMA 560

[W06 lec p. 24] — PUMA 560 joints: waist, shoulder, elbow, wrist bend, flange rotation

$$ {}^0\mathbf{J} = \begin{bmatrix} 0.13 & 0 & 0.306 & 0 & 0 & 0 \\ 0.664 & 0 & 0 & -0.038 & 0 & 0 \\ 0 & 0.664 & 0.359 & 0 & 0.053 & 0 \\ 0 & 0 & 0 & 0.707 & 0 & 1 \\ 0 & -1 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -0.707 & 0 & 0 \end{bmatrix} $$

The slide gives no pose, frame or units. It equals the toolbox PUMA 560 at the nominal pose $\mathbf{q} = (0, 45°, 180°, 0, 45°, 0)$ in $\{0\}$ (m/s and rad/s per rad/s): say "assumed nominal pose, frame $\{0\}$". [checked]

Rows 4–6 are the joint axes in $\{0\}$**:** column 1 $(0,0,1)$ (vertical waist axis); columns 2, 3, 5 all $(0,-1,0)$ (three parallel horizontal axes); column 4 $(0.707, 0, -0.707)$; column 6 $(1,0,0)$. The dot product of columns 4 and 6 is 0.707: the angle between axes 4 and 6 is $45° = q_5$. **Rows 1–3 of a revolute column** are $\hat{\mathbf{z}}_j \times (\mathbf{p}_E - \mathbf{o}_j)$; column 1 is $(-p_y, p_x, 0) = (0.13, 0.664, 0)$, so the tool is at $p_x = 0.664$, $p_y = -0.130$ m. $\det \approx -0.081$, rank 6: not singular. [checked]

W6

Derivative of one elementary transform [p. 27–32]

The six constant matrices and the product rule [p. 29, 31, 32]

$$\frac{\partial \mathbf{T}_{R_k}(\theta)}{\partial \theta} = \hat{\mathbf{R}}_k \, \mathbf{T}_{R_k}(\theta), \qquad \frac{\partial \mathbf{T}_{t_k}(d)}{\partial d} = \hat{\mathbf{t}}_k$$

$$\hat{\mathbf{R}}_x = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \hat{\mathbf{R}}_y = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \hat{\mathbf{R}}_z = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\mathbf{t}}_x = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \hat{\mathbf{t}}_y = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \hat{\mathbf{t}}_z = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial \mathbf{T}}{\partial q_j} = \Big(\prod_{i=1}^{\mu(j)-1} \mathbf{E}_i(\eta_i) \Big) \frac{\partial \mathbf{E}_{\mu(j)}(q_j)}{\partial q_j} \Big(\prod_{i=\mu(j)+1}^{M} \mathbf{E}_i(\eta_i) \Big)$$

SYMBOLS

$\hat{\mathbf{R}}_k$	4×4 skew matrix of the unit axis $k \in \{x,y,z\}$ (the augmented skew matrix with $\hat{\mathbf{s}} = (0,0,0,\hat{\mathbf{k}})$)	–
$\hat{\mathbf{t}}_k$	zeros with a single 1 in row k of column 4	–
$\mathbf{T}_{R_k}(\theta)$	rotation of θ about axis k as a 4×4 matrix	–
$\mu(j)$	position in the sequence of the one elementary transform that holds joint j	–

Use: freeze every factor except the one containing q_j, differentiate that one, multiply the pieces back in the same order.
Sign flip: if the joint's positive sense is opposite to the drawn axis (e.g. $R_y(-q)$ or $t_z(-d)$) the derivative changes sign: $\frac{d}{dq}R_y(-q) = -\hat{\mathbf{R}}_y \mathbf{T}_{R_y}(-q)$, $\frac{d}{dd}t_z(-d) = -\hat{\mathbf{t}}_z$.

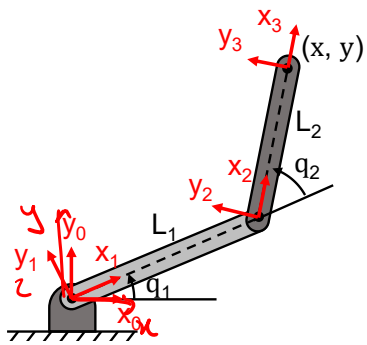
DERIVATION check of one derivative: $\mathbf{T}_{R_x}$ [added]

$\mathbf{T}_{R_x}(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c & -s & 0 \\ 0 & s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, so $\frac{\partial \mathbf{T}_{R_x}}{\partial \theta} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -s & -c & 0 \\ 0 & c & -s & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. And $\hat{\mathbf{R}}_x \mathbf{T}_{R_x}$: row 2 = −(row 3 of $\mathbf{T}_{R_x}$) $= (0,-s,-c,0)$; row 3 = row 2 of $\mathbf{T}_{R_x} = (0,c,-s,0)$; rows 1, 4 zero. Equal. Same for y and z (checked symbolically). [checked]

STEPS Jacobian of any arm by the slides' method [p. 25–26, 35–39]

1. Write the ETS from base to tool, ${}^0\mathbf{T}_E = \mathbf{E}_1 \cdots \mathbf{E}_M$; mark which factor holds each q_j ($\mu(j)$).
2. For joint j, replace *only* $\mathbf{E}_{\mu(j)}$ by its derivative (table above, sign flip if needed); keep all other factors.
3. Multiply out $\partial\mathbf{T}/\partial q_j$ (4×4).
4. Column j: top three entries $\tau(\partial\mathbf{T}/\partial q_j)$; bottom three $\vee_\times(\rho(\partial\mathbf{T}/\partial q_j)\,\rho(\mathbf{T})^\top)$.
5. Stack the columns into the $6\times n$ matrix ${}^0\mathbf{J}$. State the frame.
6. Sanity check at a pose you can see ($\mathbf{q} = \mathbf{0}$): the columns must be the velocities you get by turning one joint by hand.

Worked example: 2-link planar RR, two ways [p. 33–39]



[W06 lec p. 35] — RR arm with frames $\{0\},\{1\},\{2\},\{3\}$; q_1 from x_0, q_2 relative to link 1 (slide 35)

ALTERNATIVE APPROACHES

A. Direct differentiation [p. 33–34]
FK: $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$ ($c_{12} = \cos(q_1+q_2)$).
$\dot{x} = -L_1s_1\dot{q}_1 - L_2s_{12}(\dot{q}_1+\dot{q}_2)$
$\dot{y} = L_1c_1\dot{q}_1 + L_2c_{12}(\dot{q}_1+\dot{q}_2)$
collect $\dot{q}_1, \dot{q}_2$: the matrix of coefficients is $\mathbf{J}_{xy}$.
Quick for a planar arm; gives only the position rows; $\omega_z = \dot{q}_1 + \dot{q}_2$ is added by hand.

B. The ETS method [p. 35–39]
${}^0\mathbf{T}_3 = \mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2) = \mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4$, $\mu(1)=1$, $\mu(2)=3$.
$\partial\mathbf{T}/\partial q_1 = \hat{\mathbf{R}}_z\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$
$\partial\mathbf{T}/\partial q_2 = \mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\hat{\mathbf{R}}_z\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$
Works for any arm, 3D, any joint type; gives all six rows.

DERIVATION method B, every line (slides 36–39)

Step 1 ETS above. **Step 2** $\mu(1) = 1$, $\mu(2) = 3$ (q_2 is in $\mathbf{E}_3$). **Step 3.** The pose is $\mathbf{T} = \begin{bmatrix} c_{12} & -s_{12} & 0 & x \\ s_{12} & c_{12} & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$. Joint 1: $\hat{\mathbf{R}}_z\mathbf{T}$ (row 1 = −row 2 of $\mathbf{T}$, row 2 = row 1 of $\mathbf{T}$):

$$\frac{\partial \mathbf{T}}{\partial q_1} = \begin{bmatrix} -s_{12} & -c_{12} & 0 & -y \\ c_{12} & -s_{12} & 0 & x \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \frac{\partial \mathbf{T}}{\partial q_2} = \begin{bmatrix} -s_{12} & -c_{12} & 0 & -L_2s_{12} \\ c_{12} & -s_{12} & 0 & L_2c_{12} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(for q_2 the factors $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)$ sit in front, so only the part beyond joint 2 is turned: the translation is $L_2(-s_{12}, c_{12}, 0)$.) **Step 4.** Translation: $\mathbf{J}_{t_1} = (-y, x, 0)^\top$, $\mathbf{J}_{t_2} = L_2(-s_{12}, c_{12}, 0)^\top$. Rotation, column 1:

$$\rho\Big(\frac{\partial \mathbf{T}}{\partial q_1}\Big)\mathbf{R}^\top = \begin{bmatrix} -s_{12} & -c_{12} & 0 \\ c_{12} & -s_{12} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\vee_\times} (0,0,1)^\top$$

and the same product for q_2 gives $(0,0,1)^\top$. Stack:

Planar RR in frame $\{0\}$, full 6×2 [p. 34–39]

$$ {}^0\mathbf{J}(\mathbf{q}) = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{matrix} \leftarrow v_x \\ \leftarrow v_y \\ \leftarrow v_z \\ \leftarrow \omega_x \\ \leftarrow \omega_y \\ \leftarrow \omega_z \end{matrix}, \qquad \det \mathbf{J}_{xy} = L_1 L_2 \sin q_2 $$

SYMBOLS

L_1, L_2	link lengths	m
q_1	angle of link 1 from x_0	rad
q_2	angle of link 2 relative to link 1 (running sum q_1+q_2 is the absolute angle of link 2)	rad
s_{12}, c_{12}	$\sin(q_1+q_2)$, $\cos(q_1+q_2)$	–

Use: rows 3–5 are zero in the plane (slide 45: "ignore in 2D"); the 2×2 block $\mathbf{J}_{xy}$ and the row ω_z carry all the information.

WORKED notebook Question 1: $L_1=3$ m, $L_2=4$ m, $q_1=30°$, $q_2=15°$ (angles in degrees)

$q_1+q_2 = 45°$. $J_{11} = -3\sin 30° - 4\sin 45° = -1.5 - 2.8284 = -4.3284$; $J_{12} = -4\sin 45° = -2.8284$; $J_{21} = 3\cos 30° + 4\cos 45° = 2.5981 + 2.8284 = 5.4265$; $J_{22} = 4\cos 45° = 2.8284$.

$$ {}^0\mathbf{J} = \begin{bmatrix} -4.3284 & -2.8284 \\ 5.4265 & 2.8284 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} $$

$$ \det \mathbf{J}_{xy} = (-4.3284)(2.8284) - (-2.8284)(5.4265) = 3.1058 = 12\sin 15° $$

For $\dot{\mathbf{q}} = (0.2, -0.1)$ rad/s: $\boldsymbol{v} = (-0.5828, 0.8025)$ m/s, $\omega_z = 0.2 - 0.1 = 0.1$ rad/s. [checked]

Changing the frame of the Jacobian [p. 40, 45]

Express a Jacobian in another frame [p. 40]

$$ {}^A\mathbf{J}(\mathbf{q}) = \begin{bmatrix} {}^A\mathbf{R}_B & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & {}^A\mathbf{R}_B \end{bmatrix} {}^B\mathbf{J}(\mathbf{q}), \qquad {}^E\mathbf{R}_0 = ({}^0\mathbf{R}_E)^\top $$

SYMBOLS

${}^B\mathbf{J}$	the Jacobian written in $\{B\}$ ($6\times n$)	as $\mathbf{J}$
${}^A\mathbf{R}_B$	rotation of $\{B\}$ in $\{A\}$	–
$\mathbf{0}_{3\times3}$	zero block	–
${}^A\mathbf{J}$	the same Jacobian written in $\{A\}$	as $\mathbf{J}$

Use: rotate both halves (linear and angular) by the same matrix. To go from $\{0\}$ to the tool frame $\{E\}$ use ${}^E\mathbf{R}_0 = {}^0\mathbf{R}_E{}^\top$. A 3×3 matrix cannot multiply a $6\times n$ one. det *and the singular values of the $\mathbf{J}_v$ block do not change: singularities do not depend on the frame.*

WORKED RR of Question 1 in the tool frame $\{3\}$ [p. 45]

${}^3\mathbf{R}_0 = \mathbf{R}_z(-45°)$, so the six rows of ${}^3\mathbf{J}$ are ${}^3\mathbf{R}_0$ applied to rows 1–3 and to rows 4–6 of ${}^0\mathbf{J}$:

$$ {}^3\mathbf{J} = \begin{bmatrix} L_1 s_2 & 0 \\ L_1 c_2 + L_2 & L_2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.7765 & 0 \\ 6.8978 & 4 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} $$

(slide 45 prints the yellow 2×2 block; it depends on q_2 only). det $= 3.1058$, the same as in $\{0\}$. $\dot{\mathbf{q}} = (0.2, -0.1)$: ${}^3\boldsymbol{v} = (0.1553, 0.9796)$ m/s $= \mathbf{R}_z(45°)^\top(-0.5828, 0.8025)$; speed 0.9918 m/s in both frames; $\omega_z = 0.1$ in both. [checked]

Alternative: velocity propagation and the geometric rule [added]

Craig's outward velocity propagation (frames explicit)

Revolute joint $i{+}1$:

$$ {}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_{i}\mathbf{R}\,{}^i\boldsymbol{\omega}_i + \dot{\theta}_{i+1}\,{}^{i+1}\hat{\mathbf{Z}}_{i+1}, \qquad {}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}_{i}\mathbf{R}\left({}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{P}_{i+1}\right) $$

Prismatic joint $i{+}1$: ${}^{i+1}\boldsymbol{\omega}_{i+1} = {}^{i+1}_{i}\mathbf{R}\,{}^i\boldsymbol{\omega}_i$ and ${}^{i+1}\mathbf{v}_{i+1} = {}^{i+1}_{i}\mathbf{R}\left({}^i\mathbf{v}_i + {}^i\boldsymbol{\omega}_i \times {}^i\mathbf{P}_{i+1}\right) + \dot{d}_{i+1}\,{}^{i+1}\hat{\mathbf{Z}}_{i+1}$.
Start: ${}^0\boldsymbol{\omega}_0 = {}^0\mathbf{v}_0 = \mathbf{0}$.

SYMBOLS

${}^i\boldsymbol{\omega}_i$, ${}^i\mathbf{v}_i$	angular and linear velocity of frame $\{i\}$, written in $\{i\}$	rad/s; m/s
${}^{i+1}_{i}\mathbf{R}$	rotation of $\{i\}$ expressed in $\{i{+}1\}$ (the transpose of ${}^i\mathbf{R}_{i+1}$)	–
${}^i\mathbf{P}_{i+1}$	origin of $\{i{+}1\}$ in $\{i\}$	m
${}^{i+1}\hat{\mathbf{Z}}_{i+1}$	unit joint axis $(0,0,1)^\top$ of $\{i{+}1\}$	–
$\dot{\theta}_{i+1}, \dot{d}_{i+1}$	joint rate (revolute, prismatic)	rad/s; m/s

Use: the result is in the last frame; the cross product is the one on the cross-product note. Read the Jacobian from ${}^n\boldsymbol{\nu} = {}^n\mathbf{J}\dot{\mathbf{q}}$ (coefficients of each $\dot{q}_j$). Only if the question says "velocity propagation" (Craig). Rotate with the change-of-frame rule to get ${}^0\mathbf{J}$.

DERIVATION propagation for the planar RR [added]

Frames as on slide 35: $\{1\}$ rotated q_1 from $\{0\}$ at the same origin; $\{2\}$ at ${}^1\mathbf{P}_2 = (L_1, 0, 0)$, rotated q_2 from $\{1\}$; $\{3\}$ at ${}^2\mathbf{P}_3 = (L_2, 0, 0)$, no rotation. ${}^2_1\mathbf{R} = \mathbf{R}_z(-q_2) = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

$$ {}^1\boldsymbol{\omega}_1 = (0, 0, \dot{q}_1), \quad {}^1\mathbf{v}_1 = \mathbf{0} $$
$$ {}^2\boldsymbol{\omega}_2 = {}^2_1\mathbf{R}\,{}^1\boldsymbol{\omega}_1 + (0, 0, \dot{q}_2) = (0, 0, \dot{q}_1 + \dot{q}_2) $$
$$ {}^2\mathbf{v}_2 = {}^2_1\mathbf{R}\left(\mathbf{0} + (0, 0, \dot{q}_1) \times (L_1, 0, 0)\right) = {}^2_1\mathbf{R}\,(0, L_1\dot{q}_1, 0) = (L_1 s_2 \dot{q}_1,\ L_1 c_2 \dot{q}_1,\ 0) $$
$$ {}^3\mathbf{v}_3 = {}^2\mathbf{v}_2 + (0, 0, \dot{q}_1 + \dot{q}_2) \times (L_2, 0, 0) = \left(L_1 s_2 \dot{q}_1,\ L_1 c_2 \dot{q}_1 + L_2(\dot{q}_1 + \dot{q}_2),\ 0\right) $$

Coefficients of $\dot{q}_1, \dot{q}_2$: ${}^3\mathbf{J} = \begin{bmatrix} L_1 s_2 & 0 \\ L_1 c_2 + L_2 & L_2 \end{bmatrix}$, ω_z row $(1,1)$: identical to slide 45. [checked]

Geometric column rule (inspection method) [added]

All vectors in $\{0\}$: $\hat{\mathbf{z}}_j$ unit axis of joint j, $\mathbf{o}_j$ a point on that axis, $\mathbf{p}_E$ the tool point.

$$\text{revolute: } {}^0\mathbf{J}_{:,j} = \begin{bmatrix} \hat{\mathbf{z}}_j \times (\mathbf{p}_E - \mathbf{o}_j) \\ \hat{\mathbf{z}}_j \end{bmatrix}, \qquad \text{prismatic: } {}^0\mathbf{J}_{:,j} = \begin{bmatrix} \hat{\mathbf{z}}_j \\ \mathbf{0} \end{bmatrix}$$

Use: fastest way to write the $6 \times n$ matrix of a simple arm: a spinning joint moves the tool perpendicular to the lever arm $\mathbf{p}_E - \mathbf{o}_j$ at rate lever$\times\dot{\theta}$ and contributes its own axis to $\boldsymbol{\omega}$; a slider moves the tool along its axis and contributes no $\boldsymbol{\omega}$. Planar RR: column 1 = $(-y, x, 0; 0, 0, 1)$, column 2 = $(L_2(-s_{12}, c_{12}, 0); 0, 0, 1)$. [checked]

ALTERNATIVE APPROACHES

A. Slides: differentiate the ETS

One 4×4 matrix per joint, then read off τ and $\vee_\times$. Systematic, same steps for every robot, result in $\{0\}$ directly.

B. Craig propagation / geometric rule

Walk link by link (recursion) or write columns by inspection. Quicker for planar arms; propagation gives the result in the last frame, rotate to $\{0\}$ if asked. Both give the same $\mathbf{J}$ (verified).

SLIDE CHECK slides 21 vs 45; p. 32 indices

Slide 21 stresses frame $\{0\}$ while slide 45 prints the RR matrix in the tool frame $\{3\}$: both right, they differ by the rotation above. Slide 32 runs the first product to $\mathbf{E}_n(\eta_n)$; the ETS has $M \geq n$ factors, so it should end at $\mathbf{E}_M$ (as written in the box above).

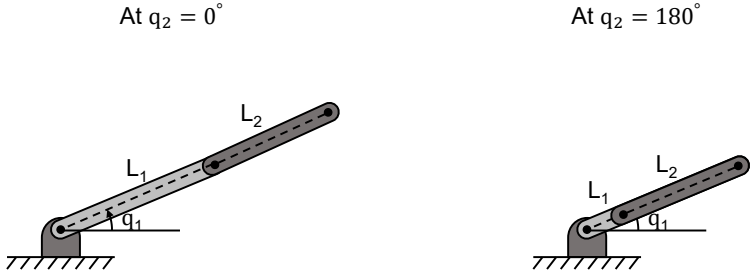
Jacobians of the simple robots (full matrices, ETS method) [added]

Each: ETS (base to tool), full $6 \times n$ matrix in $\{0\}$, det of the square block, singular poses and the lost direction. $\mathbf{J}$ columns were verified against finite differences of the FK. [checked]

RR (planar, two revolute joints)

PREMISE RR **Given:** L_1, L_2, joint angles q_1, q_2
Find: ${}^0\mathbf{J}$ and its singular poses

ETS $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$; ${}^0\mathbf{J}$ as in the box above, $\det \mathbf{J}_{xy} = L_1 L_2 \sin q_2 = 0$ at $q_2 = 0$ (stretched) and $q_2 = \pi$ (folded): rank 1, both columns perpendicular to the arm, lost direction *along* the arm.



[W06 lec p. 46] — slide 46: the two singular poses of the RR arm, $q_2 = 0°$ (stretched) and $q_2 = 180°$ (folded)

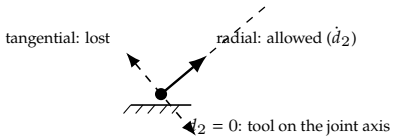
RP (polar: revolute, then prismatic along the link)

PREMISE RP **Given:** extension d_2 (m) along the link, base angle q_1
Find: ${}^0\mathbf{J}$, singular pose

ETS $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(d_2)$, tool $x = d_2 c_1$, $y = d_2 s_1$.

$${}^0\mathbf{J} = \begin{bmatrix} -d_2 s_1 & c_1 \\ d_2 c_1 & s_1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_{xy} = -d_2 s_1^2 - d_2 c_1^2 = -d_2$$

Column 1 is tangential ($d_2\times$ unit tangent), column 2 radial. Singular only at $d_2 = 0$ (tool on the joint axis): column 1 vanishes, the tangential direction is lost. Example $q_1 = 40°$, $d_2 = 0.6\,\text{m}$: $\mathbf{J}_{xy} = \begin{bmatrix} -0.3857 & 0.7660 \\ 0.4596 & 0.6428 \end{bmatrix}$, det $= -0.6$; tool moving tangentially at $0.3\,\text{m/s}$ needs $\dot{q}_1 = 0.5\,\text{rad/s}$, $\dot{d}_2 = 0$; radially at $0.3\,\text{m/s}$ needs $\dot{q}_1 = 0$, $\dot{d}_2 = 0.3\,\text{m/s}$. [checked]



[drawn for this booklet] RP singular pose $d_2 = 0$

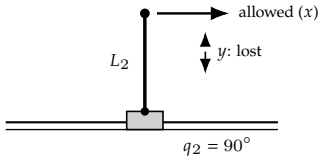
PR (prismatic along x_0, then revolute)

PREMISE PR **Given:** slide d_1 along x_0, link L_2, angle q_2 from x_0
Find: ${}^0\mathbf{J}$, singular pose

ETS $\mathbf{T}_{t_x}(d_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)$, tool $x = d_1 + L_2 c_2$, $y = L_2 s_2$.

$${}^0\mathbf{J} = \begin{bmatrix} 1 & -L_2 s_2 \\ 0 & L_2 c_2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \det \mathbf{J}_{xy} = L_2 \cos q_2$$

Singular at $q_2 = \pm 90°$: the link stands perpendicular to the rail, the revolute column becomes $(\mp L_2, 0)$ parallel to the prismatic column $(1, 0)$, and the y direction is lost. Example $L_2 = 0.5\,\text{m}$, $q_2 = 30°$: $\mathbf{J}_{xy} = \begin{bmatrix} 1 & -0.25 \\ 0 & 0.4330 \end{bmatrix}$, det $= 0.4330 = 0.5 \cos 30°$; at $q_2 = 90°$: $\begin{bmatrix} 1 & -0.5 \\ 0 & 0 \end{bmatrix}$, rank 1. [checked]



[drawn for this booklet] PR singular pose $q_2 = 90°$

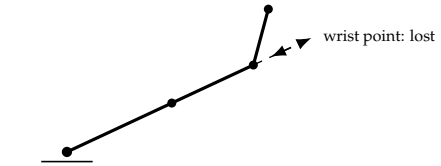
Planar RRR (three revolute joints, planar)

PREMISE RRR **Given:** L_1, L_2, L_3, joint angles q_1, q_2, q_3
Find: ${}^0\mathbf{J}$ (6×3), singular poses for position only and for (x, y, φ)

ETS $\mathbf{T}_{R_z}(q_1)\mathbf{T}_{t_x}(L_1)\mathbf{T}_{R_z}(q_2)\mathbf{T}_{t_x}(L_2)\mathbf{T}_{R_z}(q_3)\mathbf{T}_{t_x}(L_3)$; $s_{123} = \sin(q_1+q_2+q_3)$:

$$ {}^0\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} - L_3s_{123} & -L_2s_{12} - L_3s_{123} & -L_3s_{123} \\ L_1c_1 + L_2c_{12} + L_3c_{123} & L_2c_{12} + L_3c_{123} & L_3c_{123} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} $$

Column j sums the lengths of every link from j outwards. **(i)** Position only: $\mathbf{J}_{xy}$ is 2×3 (fat); rank 1 only if all columns are parallel, i.e. all three links collinear (q_2, q_3 multiples of π). **(ii)** Pose (x, y, ω_z): the 3×3 matrix has $\det = L_1L_2\sin q_2$ (symbolically verified), independent of q_3: singular when the wrist point (end of link 2) is stretched or folded. Example $L = (0.4, 0.3, 0.2)$ m, $\mathbf{q} = (20°, 40°, -30°)$: $\mathbf{J}_{xy} = \begin{bmatrix} -0.4966 & -0.3598 & -0.1000 \\ 0.6991 & 0.3232 & 0.1732 \end{bmatrix}$, $\det_{(x,y,\omega_z)} = 0.4{\cdot}0.3\sin 40° = 0.0771$. For $\dot{\mathbf{q}} = (0.3, -0.2, 0.5)$: $\boldsymbol{v} = (-0.1270, 0.2317)$ m/s, $\omega_z = 0.6$ rad/s. [checked]



[drawn for this booklet] planar RRR with $q_2 = 0$: (x, y, φ) singular

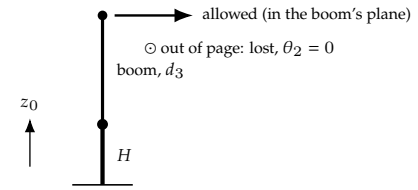
RRP (spatial, spherical type: θ_1 about z_0, θ_2 tilts the boom, d_3 slides along it)

PREMISE RRP **Given:** $H, \theta_1, \theta_2, d_3$ (the arm of the W3 part)
Find: ${}^0\mathbf{J}$ (6×3), $\det \mathbf{J}_v$, singular poses

ETS $\mathbf{T}_{R_z}(\theta_1)\mathbf{T}_{t_z}(H)\mathbf{T}_{R_y}(\theta_2)\mathbf{T}_{t_z}(d_3)$; tip $\mathbf{p} = (c_1d_3s_2,\ s_1d_3s_2,\ H + d_3c_2)^\top$. Axes in $\{0\}$: joint 1 $= (0, 0, 1)$, joint 2 $= \mathbf{R}_z(\theta_1)(0, 1, 0) = (-s_1, c_1, 0)$, joint 3 slides along (c_1s_2, s_1s_2, c_2).

$$ {}^0\mathbf{J} = \begin{bmatrix} -d_3s_1s_2 & d_3c_1c_2 & c_1s_2 \\ d_3c_1s_2 & d_3s_1c_2 & s_1s_2 \\ 0 & -d_3s_2 & c_2 \\ 0 & -s_1 & 0 \\ 0 & c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \det \mathbf{J}_v = -d_3^2\sin\theta_2 $$

Singular if $d_3 = 0$ (tool on the joint-2 axis: both rotations only turn the boom about the tool) or $\sin\theta_2 = 0$ (boom vertical: tool on the z_0 axis, column 1 is zero; the horizontal direction perpendicular to the boom's plane, $(-s_1, c_1, 0)$, is lost). Example $H = 0.5$ m, tip $(0.3, 0.4, 0.9)$ (so $\theta_1 = 53.13°$, $\theta_2 = 51.34°$, $d_3 = 0.6403$ m): $\mathbf{J}_v = \begin{bmatrix} -0.4 & 0.24 & 0.4685 \\ 0.3 & 0.32 & 0.6247 \\ 0 & -0.5 & 0.6247 \end{bmatrix}$, $\det = -0.3202$; for $\dot{\mathbf{q}} = (0.2, -0.1, 0.05)$: $\boldsymbol{v} = (-0.0806, 0.0592, 0.0812)$ m/s, $\boldsymbol{\omega} = (0.08, -0.06, 0.2)$ rad/s. [checked]



[drawn for this booklet] RRP with the boom vertical ($\theta_2 = 0$), side view

Offset RP arm (extra, W7–8 folder): slider perpendicular to the link

$x = l_1c_1 - d_2s_1$, $y = l_1s_1 + d_2c_1$: $\mathbf{J}_{xy} = \begin{bmatrix} -l_1s_1 - d_2c_1 & -s_1 \\ l_1c_1 - d_2s_1 & c_1 \end{bmatrix}$, $\det = -d_2$. Singular only at $d_2 = 0$, where column 1 $= l_1\times$ column 2: the radial direction is lost; null motion $\dot{q}_1 = -1/l_1$, $\dot{d}_2 = 1$. [checked]

Inverse velocity: tool velocity → joint rates [p. 41–42]

PREMISE inverse velocity **Given:** pose $\mathbf{q}$ and the wanted tool velocity ${}^0\boldsymbol{\nu}$ (or $\boldsymbol{v}$ only)
Find: joint rates $\dot{\mathbf{q}}$

Inverse Jacobian and pseudo-inverses [p. 41–42]

$$ \dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\,\boldsymbol{\nu}, \qquad \mathbf{J}_{RR}^{-1} = \frac{1}{L_1L_2\sin q_2}\begin{bmatrix} L_2c_{12} & L_2s_{12} \\ -L_1c_1 - L_2c_{12} & -L_1s_1 - L_2s_{12} \end{bmatrix} $$

$$ \text{tall } (m > n):\ \mathbf{J}^+ = (\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top \text{ (least squares)} $$

$$ \text{fat } (n > m):\ \mathbf{J}^+ = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1} \text{ (minimum norm)} $$

SYMBOLS

$\mathbf{J}^{-1}$	ordinary inverse; exists only if $\mathbf{J}$ is square and $\det\mathbf{J} \neq 0$	–
$\mathbf{J}^+$	pseudo-inverse for a non-square $\mathbf{J}$ (tall: more task components than joints; fat: more joints than components)	–
m, n	number of rows (task components), number of joints	–
$\boldsymbol{\nu}$	wanted tool velocity (the components you actually want to control)	m/s; rad/s

Use: the slide's three questions (p. 42): is $\mathbf{J}$ square? (if not, pseudo-inverse); is it non-singular? (if not, no inverse); is it invertible for all $\mathbf{q}$? (no: find the singular poses). Resolved-rate control: $\dot{\mathbf{q}}_k = \mathbf{J}^{-1}(\mathbf{q}_k)\boldsymbol{\nu}$, $\mathbf{q}_{k+1} = \mathbf{q}_k + \dot{\mathbf{q}}_k\Delta t$, recompute $\mathbf{J}$ every step (radians). Pseudo-inverses and damped least squares $\mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top + \lambda^2\mathbf{I})^{-1}$ are added (not on the slides).

WORKED notebook Question 4: $L = (3, 4)$ **m,** $\mathbf{q} = (30°, 60°)$, $\boldsymbol{\nu} = (0.2, -0.3, -0.2)$ **(m/s, m/s, rad/s)**

$$ \mathbf{J} = \begin{bmatrix} -5.5 & -4 \\ 2.5981 & 0 \\ 1 & 1 \end{bmatrix} \ (v_x, v_y, \omega_z \text{ rows}), \qquad \det\mathbf{J}_{xy} = 10.3923 = 12\sin 60° $$

Two joints, three components: over-determined, and $\omega_z = \dot{q}_1 + \dot{q}_2$ is not free. **Reading A** (follow the position): $v_y = 2.5981\dot{q}_1 \Rightarrow \dot{q}_1 = -0.1155$; $-5.5\dot{q}_1 - 4\dot{q}_2 = 0.2 \Rightarrow \dot{q}_2 = 0.1088$ rad/s, giving $\omega_z = -0.0067$. After $\Delta t = 0.1$ s: $\mathbf{q} = (29.338°, 60.623°)$. **Reading B** (least squares): $\dot{\mathbf{q}} = \mathbf{J}^+\boldsymbol{\nu} = (-0.1056, 0.0840)$, residual $(0.0446, 0.0258, 0.1784)$, $\mathbf{q} = (29.395°, 60.481°)$. State which reading you use. Over 5 s (50 steps) A ends with the tip at $(3.591, 3.997)$, B at $(3.795, 4.058)$; the ideal straight line ends at $(3.598, 4.000)$. [checked]

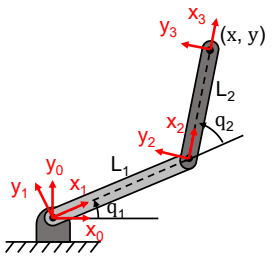
Singularities [p. 43–47]

Definition, two types, what is lost [p. 44]

Singular configuration: $\det \mathbf{J}(\mathbf{q}) = 0$ (non-square $\mathbf{J}$: rank drops, $\det \mathbf{J}\mathbf{J}^\top = 0$, or $\sigma_{\min} = 0$). Then there is a direction $\mathbf{u} \neq \mathbf{0}$ with $\mathbf{u}^\top \mathbf{J} = \mathbf{0}$ in which the tool cannot move, and $\mathbf{J}^{-1}$ (which contains $1/\det \mathbf{J}$) blows up.

type	meaning, example
workspace-*boundary*	arm stretched out or folded back on itself (RR: $q_2 = 0, \pi$)
workspace-*interior*	inside the workspace, joint axes line up (wrist $q_5 = 0$; tool on the base axis)

Use: RR: $\det \begin{bmatrix} L_1 s_{q_2} & 0 \\ L_1 c_{q_2} + L_2 & L_2 \end{bmatrix} = L_1 L_2 \sin q_2 = 0 \iff q_2 = 0^\circ$ *or* 180° *(slide 45). Independent of* q_1 *and of the frame.*



$${}^3J(q) = \begin{bmatrix} L_1 s q_2 & 0 \\ L_1 c q_2 + L_2 & L_2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Corresponds to v_z, ignore in 2D case
Corresponds to ω_x, ignore in 2D case
Corresponds to ω_y, ignore in 2D case

Compute the determinant

$$\begin{vmatrix} L_1 s q_2 & 0 \\ L_1 c q_2 + L_2 & L_2 \end{vmatrix} = L_1 L_2 s q_2$$

When is the determinant = 0?
At $q_2 = 0^\circ$ and at $q_2 = 180^\circ$

[W06 lec p. 45] — slide 45: determine the singularities of the RR arm

WORKED planar RR $L = (3, 4)$, $q_1 = 30^\circ$ at the two singular poses (notebook cell 46)

Stretched $q_2 = 0$: frame $\{0\}$ block $\begin{bmatrix} -3.5 & -2 \\ 6.0622 & 3.4641 \end{bmatrix}$, column 1 = 1.75× column 2, both along $(-0.5, 0.866)$ (perpendicular to the arm direction $(0.866, 0.5)$): rank 1. In the tool frame ${}^3\mathbf{J}_{xy} = \begin{bmatrix} 0 & 0 \\ 7 & 4 \end{bmatrix}$: the whole first row (the direction along the arm) is zero. Folded $q_2 = 180^\circ$: ${}^3\mathbf{J}_{xy} = \begin{bmatrix} 0 & 0 \\ 1 & 4 \end{bmatrix}$, rank 1. [checked]

WORKED 3D arms: where are they singular?

W13-type RRR (W2/W3 part): $\det \mathbf{J}_v = -L_3 L_4 \sin\theta_3 \,(L_3 \cos\theta_2 + L_4 \cos(\theta_2 + \theta_3))$: $\sin\theta_3 = 0$ boundary; reach $r = L_3 c_2 + L_4 c_{23} = 0$ interior (tool on the vertical axis through the shoulder). For $L_3 = 0.4$, $L_4 = 0.3$, $\theta_2 = 60^\circ$: $\theta_3 = 71.81^\circ$. **PUMA 560 (zero-offset), notebook Question 3:** $\det \mathbf{J} = -L_u L_f \sin q_3 \,(L_u \sin q_2 + L_f \sin(q_2 + q_3)) \sin q_5$: elbow $\sin q_3 = 0$ (boundary); shoulder $L_u \sin q_2 + L_f \sin(q_2 + q_3) = 0$ (wrist centre on the waist axis, interior; e.g. $L_u = 0.5$, $L_f = 0.3$, $q_2 = 22.92^\circ$, $q_3 = -63.39^\circ$); wrist $\sin q_5 = 0$ (axes 4 and 6 aligned, rank 5). The real PUMA has offsets: same three types at shifted poses (added). [checked]

STEPS Find and explain the singularities

1. Write $\mathbf{J}$ (or the square block that matters; say which rows).
2. Compute det and factorise (RR: $L_1 L_2 \sin q_2$).
3. Solve each factor for the angles; list every solution in the range.
4. Name each: boundary or interior; draw the pose (slide 46 style).
5. State what is lost ("the tool cannot move along …") and that rates $\to \infty$ for a velocity with a component there.
6. Sanity check: at the singular pose two columns are parallel (rank 1 for RR).

Why rates blow up near $\det \mathbf{J} = 0$ [added]

$\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$ and $\mathbf{J}^{-1} = \operatorname{adj}\mathbf{J}/\det\mathbf{J}$: for RR, $\dot{q}_1 = \dfrac{v_\perp}{L_1 \sin q_2}$ for a velocity $v_\perp$ across the arm, so $\dot{q}_1 \to \infty$ as $q_2 \to 0$. Tool speed along the arm is finite only if it is zero. $L = (3, 4)$, 0.1 m/s perpendicular to the arm:

q_2	90°	30°	15°	5°	1°
$\dot{q}_1$	0.0333	0.0667	0.1288	0.3825	1.9100
$\dot{q}_2$	−0.0333	−0.1100	−0.2221	−0.6682	−3.3422

growth 30° → 5°: $\sin 30^\circ / \sin 5^\circ = 5.7\times$. [checked]

Simple scenario questions: line, circle, rectangle [added]

All use the RR arm $L_1 = 0.5$ m, $L_2 = 0.4$ m, elbow-left branch ($q_2 > 0$), frame $\{0\}$, and the pose $\mathbf{q} = (30^\circ, 60^\circ)$ with tool at $\mathbf{p} = (0.4330, 0.6500)$ m:

$$\mathbf{J} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}, \quad \det \mathbf{J} = 0.1732 = 0.2 \sin 60^\circ, \quad \mathbf{J}^{-1} = \begin{bmatrix} 0 & 2.3094 \\ -2.5 & -3.7528 \end{bmatrix}$$

(a) Straight line

PREMISE line **Given:** tool speed $v = 0.2$ m/s along $+x_0$ at the pose above
Find: joint rates; time; how far can it go

$\dot{\mathbf{q}} = \mathbf{J}^{-1}(0.2, 0)^\top = (0, -0.5)$ rad/s: only joint 2 moves ($v_y = 0.433\dot{q}_1$ must be 0). Along $+y_0$ at 0.2 m/s: $\dot{\mathbf{q}} = (0.4619, -0.7506)$ rad/s. Time for 0.1 m: $0.1/0.2 = 0.5$ s. $\mathbf{J}$ changes along the line, so recompute at every step (resolved rate): after 0.5 s $\mathbf{q} = (32.03^\circ, 42.14^\circ)$, tip $(0.5330, 0.6500)$ (checked against IK). The line $y = 0.65$ ends at the reach limit $x = \sqrt{0.9^2 - 0.65^2} = 0.6225$ m, 0.1895 m away (0.947 s): rates grow as it nears the boundary.

x (m)	q_1, q_2 (°)	$\det \mathbf{J}$	$\dot{q}_1$	$\dot{q}_2$ (rad/s)
0.433	30.00, 60.00	0.1732	0	−0.500
0.533	32.03, 42.14	0.1342	0.163	−0.794
0.550	32.93, 38.05	0.1233	0.211	−0.892
0.600	37.81, 21.37	0.0729	0.562	−1.647
0.620	43.18, 7.14	0.0249	2.055	−4.990

(b) Circle at constant speed

PREMISE circle **Given:** circle of radius $\rho = 0.1$ m, speed $v = 0.2$ m/s, counter-clockwise, tool at its rightmost point at $t = 0$ (centre $\mathbf{c} = (0.3330, 0.6500)$)
Find: $\boldsymbol{\nu}(t)$, joint rates at sample points, period

STEPS circle

1. Angular rate $\Omega = v/\rho = 2$ rad/s; period $T = 2\pi/\Omega = \pi = 3.1416$ s.
2. $\mathbf{p}(t) = \mathbf{c} + \rho(\cos\Omega t, \sin\Omega t)$; differentiate: $\boldsymbol{v}(t) = v\,(-\sin\Omega t,\ \cos\Omega t) = 0.2(-\sin 2t, \cos 2t)$ m/s (check $|\boldsymbol{v}| = v$). ω_z is not controlled (two joints give two components).
3. Check the circle lies in the workspace and away from singularities: $|\mathbf{c}| = 0.7303$ m, farthest point 0.83 m < 0.9; smallest q_2 on the circle $= 45.7°$.
4. At the sample time: IK for $\mathbf{q}$ (branch $q_2 > 0$), build $\mathbf{J}(\mathbf{q})$, solve $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{v}$. Repeat at every time (the circle needs $\mathbf{J}$ updated).
5. Sanity: after one period $\mathbf{q}$ returns to $(30°, 60°)$ (integrated and checked).

Ωt	$\mathbf{p}$ (m)	$\boldsymbol{v}$ (m/s)	$\mathbf{q}$ (°)	$\dot{\mathbf{q}}$ (rad/s)
0	(0.433, 0.650)	(0, 0.2)	(30.0, 60.0)	(0.462, −0.751)
$\pi/2$	(0.333, 0.750)	(−0.2, 0)	(44.5, 48.8)	(0.031, 0.443)
π	(0.233, 0.650)	(0, −0.2)	(35.4, 80.4)	(−0.365, 0.659)
$3\pi/2$	(0.333, 0.550)	(0.2, 0)	(20.3, 89.5)	(−0.136, −0.333)

At $t = 0$: $\dot{\mathbf{q}} = \mathbf{J}^{-1}(0, 0.2)^{\top} = (0.2 \cdot 2.3094,\ 0.2 \cdot (-3.7528)) = (0.462, -0.751)$. Largest rate on the whole circle: 0.78 rad/s. [checked]

(c) Rectangle

PREMISE rectangle **Given:** corners $A(0.45, 0.45)$, $B(0.65, 0.45)$, $C(0.65, 0.55)$, $D(0.45, 0.55)$ m, speed $v = 0.1$ m/s along each side
Find: time per side, joint rates at a sample point of each side

Side times = length/v: AB 2 s, BC 1 s, CD 2 s, DA 1 s (total 6 s). On each side $\boldsymbol{v}$ is constant; $\mathbf{J}$ is not, so evaluate at a sample point (here the midpoint, via IK, $q_2 > 0$):

side	midpoint	$\boldsymbol{v}$	$\mathbf{q}$ (°)	$\dot{\mathbf{q}}$ (rad/s)
AB	(0.55, 0.45)	(0.1, 0)	(6.1, 76.3)	(0.027, −0.283)
BC	(0.65, 0.50)	(0, 0.1)	(16.0, 49.0)	(0.240, −0.331)
CD	(0.55, 0.55)	(−0.1, 0)	(18.3, 60.8)	(−0.043, 0.315)
DA	(0.45, 0.50)	(0, −0.1)	(11.8, 83.9)	(−0.200, 0.251)

Corner C is the farthest ($|\mathbf{C}| = 0.8515$ m, $q_2 = 38.0°$, $\det\mathbf{J} = 0.1233$): at the end of side BC (velocity $(0, 0.1)$) the rates are $(0.285, -0.446)$ rad/s, against $(0.240, -0.331)$ at its midpoint. At a corner $\boldsymbol{v}$ jumps from one side direction to the next: infinite acceleration, so the robot must stop or blend (W5). Resolved-rate integration of side AB from corner A reaches B exactly. [checked]

(d) Near a singularity: moving along the arm

Radial speed $\dot{r}$ outward: from $r^2 = L_1^2 + L_2^2 + 2L_1L_2\cos q_2$ differentiate, $2r\dot{r} = -2L_1L_2\sin q_2\,\dot{q}_2$, so

$$\dot{q}_2 = -\frac{r\,\dot{r}}{L_1 L_2 \sin q_2} \ \to\ \infty \ \text{ as } q_2 \to 0\ (r \to L_1 + L_2)$$

$\dot{r} = 0.1$ m/s:

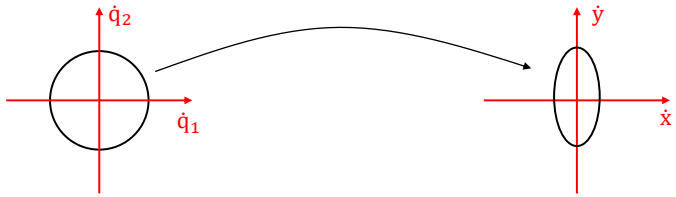
r (m)	0.60	0.80	0.85	0.89	0.899
$\dot{q}_2$ (rad/s)	−0.302	−0.489	−0.681	−1.504	−4.745

With a joint-rate limit of 2 rad/s at $r = 0.85$ m the largest allowed $\dot{r}$ is $2 \cdot 0.2 \sin 38.6° / 0.85 = 0.294$ m/s. [checked]

TRAP
Using one $\mathbf{J}$ (the starting pose) for a whole path. $\mathbf{J}$ depends on $\mathbf{q}$: the rates change along the circle or side. Also: forgetting that the end point must be inside the workspace (0.62 m on the line above, not 0.633).

Manipulability and the velocity ellipsoid [p. 49–55]

PREMISE manipulability **Given:** pose $\mathbf{q}$ (so $\mathbf{J}(\mathbf{q})$)
Find: shape of the set of tool velocities reachable with unit joint rates; how close to singular



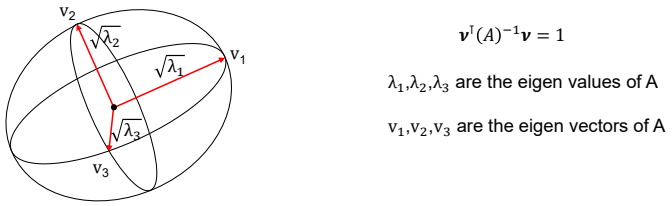
[W06 lec p. 51] — slide 51: unit circle of joint rates $(\dot{q}_1, \dot{q}_2)$ mapped to an ellipse of tool velocities $(\dot{x}, \dot{y})$

DERIVATION the ellipsoid equation [p. 51–53]

Joint rates of unit norm: $\dot{\mathbf{q}}^\top \dot{\mathbf{q}} = 1$ (the slide prints $\dot{\mathbf{q}}\dot{\mathbf{q}}^\top$, see below). For square, non-singular $\mathbf{J}$: $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$, so

$$\dot{\mathbf{q}}^\top \dot{\mathbf{q}} = \boldsymbol{\nu}^\top (\mathbf{J}^{-1})^\top \mathbf{J}^{-1} \boldsymbol{\nu} = \boldsymbol{\nu}^\top (\mathbf{J}\mathbf{J}^\top)^{-1} \boldsymbol{\nu} = 1,$$

because $(\mathbf{J}^{-1})^\top \mathbf{J}^{-1} = (\mathbf{J}\mathbf{J}^\top)^{-1}$. Unit ball ("$\le 1$") gives the solid ellipsoid. With $\mathbf{A} = \mathbf{J}\mathbf{J}^\top$ (symmetric), eigenvalues λ_i, unit eigenvectors $\mathbf{v}_i$, the semi-axes point along $\mathbf{v}_i$ and have length $\sqrt{\lambda_i}$ (slide 53). *Why:* for $\boldsymbol{\nu} = \sqrt{\lambda_i}\,\mathbf{v}_i$ we get $\boldsymbol{\nu}^\top \mathbf{A}^{-1} \boldsymbol{\nu} = \lambda_i \mathbf{v}_i^\top \mathbf{A}^{-1} \mathbf{v}_i = \lambda_i/\lambda_i = 1$. [added]



[W06 lec p. 53] — slide 53: $\boldsymbol{\nu}^\top \mathbf{A}^{-1} \boldsymbol{\nu} = 1$, semi-axes $\sqrt{\lambda_i}$ along the eigenvectors $\mathbf{v}_i$ of $\mathbf{A}$

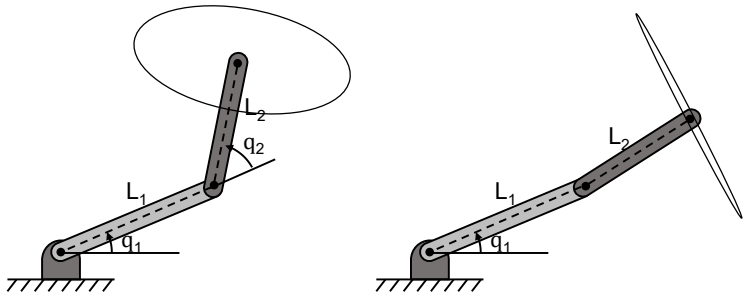
Velocity ellipsoid and manipulability measure [p. 52–55]

$$\boldsymbol{\nu}^\top \left(\mathbf{J}\mathbf{J}^\top\right)^{-1} \boldsymbol{\nu} \le 1, \qquad \mathbf{A} = \mathbf{J}\mathbf{J}^\top, \qquad m := \frac{\sqrt{\lambda_{\min}(\mathbf{A})}}{\sqrt{\lambda_{\max}(\mathbf{A})}} \in [0,1]$$

SYMBOLS

$\mathbf{A}$	$\mathbf{J}\mathbf{J}^\top$ ($m \times m$, symmetric, positive semi-definite)	m^2 (position rows)
λ_i, $\mathbf{v}_i$	eigenvalues and eigenvectors of $\mathbf{A}$	m^2; –
$\sqrt{\lambda_i} = \sigma_i$	semi-axis length = singular value of $\mathbf{J}$: tool speed per unit joint-rate norm along $\mathbf{v}_i$	m/s per rad/s
m	manipulability: 1 isotropic (equally easy in all directions), 0 singular (ellipse collapsed)	–
$w = \sqrt{\det \mathbf{J}\mathbf{J}^\top}$	Yoshikawa measure [added]; $= \lvert\det \mathbf{J}\rvert = \sigma_1\sigma_2\cdots$ (RR: $L_1 L_2 \lvert\sin q_2\rvert$): the "size" of the ellipsoid	m^2

Use: the robot moves fastest along the long axis and slowest along the short axis (slide 54). Some books define m as the reciprocal ($\lambda_{\max}/\lambda_{\min}$): state your definition. For a 2×2 $\mathbf{A}$: $\lambda = \frac{1}{2}\left(\operatorname{tr}\mathbf{A} \pm \sqrt{\operatorname{tr}^2\mathbf{A} - 4\det\mathbf{A}}\right)$, $\det\mathbf{A} = (\det\mathbf{J})^2$, $\operatorname{tr}\mathbf{A}$ = sum of squared column norms $= L_1^2 + 2L_2^2 + 2L_1L_2\cos q_2$ for the RR position block.



[W06 lec p. 54] — slide 54: ellipsoids at two configurations (fat far from a singularity, a needle close to one)

WORKED RR $L = (3,4)$, position block (slides' RR), singular values, m, w

q_2	σ_1	σ_2	m	w
15°	8.002	0.3881	0.0485	3.1058
60°	7.1328	1.4570	0.2043	10.3923
90°	6.0927	1.9696	0.3233	12.0000
150°	4.2711	1.4048	0.3289	6.0000

$\sigma_1\sigma_2 = w$ in every row. At 15°: $\operatorname{tr}\mathbf{A} = 9 + 32 + 24\cos 15° = 64.182$, $\det\mathbf{A} = 3.1058^2 = 9.646$, $\lambda = 64.032,\ 0.1506$. A planar 2R never reaches $m = 1$ (best 0.4267 at $q_2 \approx 125.8°$; equal links 0.618 at 131.8°); w peaks at 90°. [checked]

WORKED equal links $L_1 = L_2 = 0.5$ m, $q_2 = 110°$ (any q_1)

Tool-frame position block $\mathbf{J} = \begin{bmatrix} 0.5\sin 110° & 0 \\ 0.5\cos 110° + 0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.4698 & 0 \\ 0.3290 & 0.5 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} 0.2208 & 0.1546 \\ 0.1546 & 0.3582 \end{bmatrix}$, $\operatorname{tr}\mathbf{A} = 0.5790$, $\det\mathbf{A} = (0.25\sin 110°)^2 = 0.0552$, $\lambda = \frac{1}{2}(0.5790 \pm 0.3383) = 0.4587,\ 0.1203$. Semi-axes $\sigma_1 = 0.6772$, $\sigma_2 = 0.3469$ m/s per rad/s, $m = 0.3469/0.6772 = 0.5122$, $w = 0.2349 = L_1L_2\sin q_2$. Common wrong $m = \lambda_{\min}/\lambda_{\max} = 0.2623$ (forgot the roots). [checked]

WORKED $L = (0.5, 0.4)$ m, $\mathbf{q} = (30°, 60°)$: what to draw

$\mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 0.5825 & -0.2815 \\ -0.2815 & 0.1875 \end{bmatrix}$, $\lambda = 0.7288,\ 0.04116$. Semi-axes 0.8537 (long, along 152.5° from x_0) and 0.2029 (short, along 62.5°); $m = 0.2376$. Sketch a thin ellipse centred on the tool with its long axis tilted up-left, and its short axis nearly along the (vertical) forearm. [checked]

TRAP

Reporting λ where the semi-axis is $\sqrt{\lambda}$; m as $\lambda_{\min}$ alone; "near a singularity the ellipsoid is large so the arm is faster": false, one axis collapses ($\sigma_{\min} \to 0$), the tool cannot move that way.

SLIDE CHECK slide 51

The unit-norm condition is printed $\dot{\mathbf{q}}\dot{\mathbf{q}}^\top = 1$; for a column vector that is an $n \times n$ matrix. The scalar condition is $\dot{\mathbf{q}}^\top \dot{\mathbf{q}} = 1$ (used above).

Statics: joint torques and tool forces [p. 57–59]

PREMISE statics **Given:** tool wrench $\mathbf{F}_E$ (force and moment the tool applies) in the same frame as $\mathbf{J}$

Find: joint torques $\boldsymbol{\tau}$ that hold it (or the reverse)

$\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F}_E$ [p. 58]

$$\boldsymbol{\tau} = \mathbf{J}^\top(\mathbf{q})\,\mathbf{F}_E, \qquad \mathbf{F}_E = \begin{bmatrix} F_x & F_y & F_z & n_x & n_y & n_z \end{bmatrix}^\top, \qquad \boldsymbol{\tau}^\top \dot{\mathbf{q}} = \mathbf{F}_E^\top \boldsymbol{\nu}$$

SYMBOLS

$\mathbf{F}_E$	wrench at the tool: forces F_x, F_y, F_z then moments n_x, n_y, n_z (same order as $\boldsymbol{\nu}$)	N; N m
$\mathbf{J}^\top$	transpose of the $6 \times n$ Jacobian ($n \times 6$), same frame as $\mathbf{F}_E$	–
$\boldsymbol{\tau}$	joint torques/forces ($n \times 1$) the actuators supply	N m; N

Use: planar arm: $\boldsymbol{\tau} = \mathbf{J}_{xy}^\top \mathbf{F} + n_z(1, 1, \ldots)^\top$ *(the ω_z row). Tool force from torques:* $\mathbf{F} = \mathbf{J}^{-\top}\boldsymbol{\tau}$ *(square, non-singular). $\mathbf{J}^\top$ exists even when $\mathbf{J}^{-1}$ does not.* ***Convention (not given on the slide,*** *[added]****):*** *$\mathbf{F}_E$ is the wrench the tool applies to the environment and $\boldsymbol{\tau}$ is what the actuators supply to hold it; if $\mathbf{F}_E$ is the load acting on the tool, the signs flip. Always state frame and sign.*

DERIVATION by virtual work [added]

Power in the joints equals power at the tool (no losses, static): $\boldsymbol{\tau}^\top \dot{\mathbf{q}} = \mathbf{F}_E^\top \boldsymbol{\nu}$. Insert $\boldsymbol{\nu} = \mathbf{J}\dot{\mathbf{q}}$: $\boldsymbol{\tau}^\top \dot{\mathbf{q}} = \mathbf{F}_E^\top \mathbf{J}\dot{\mathbf{q}}$ for every $\dot{\mathbf{q}}$, so $\boldsymbol{\tau}^\top = \mathbf{F}_E^\top \mathbf{J}$, i.e. $\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F}_E$ (transpose both sides).

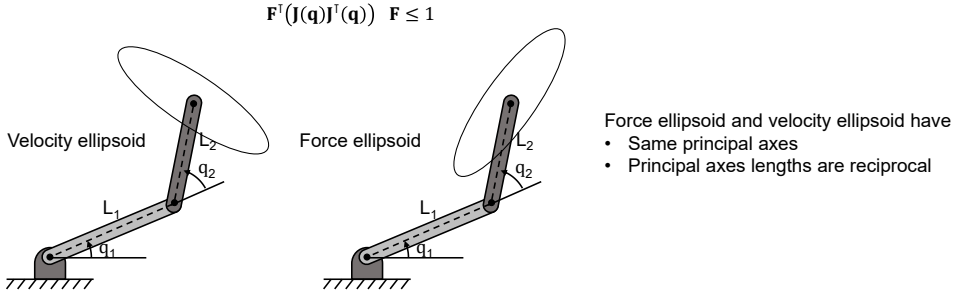
Force ellipsoid [p. 59]

Unit joint torques $\boldsymbol{\tau}^\top\boldsymbol{\tau} \le 1$ with $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$:

$$\mathbf{F}^\top \mathbf{J}\mathbf{J}^\top \mathbf{F} \le 1 \qquad \text{(no inverse; the velocity ellipsoid has } (\mathbf{J}\mathbf{J}^\top)^{-1})$$

Same principal axes (eigenvectors of $\mathbf{J}\mathbf{J}^\top$), semi-axes $1/\sqrt{\lambda_i}$ (reciprocal).

Use: where the tool can move fast it can push only weakly, and the reverse. At a singularity $\lambda_{\min} \to 0$: the force semi-axis $\to \infty$, the structure carries load along the lost direction with zero joint torque.



[W06 lec p. 59] — slide 59: velocity ellipsoid and force ellipsoid at the same pose: same axes, reciprocal lengths

WORKED $L = (0.5, 0.4)$ m, $\mathbf{q} = (30°, 60°)$; $\mathbf{J}_{xy} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$, $\mathbf{J}^\top = \begin{bmatrix} -0.65 & 0.4330 \\ -0.4 & 0 \end{bmatrix}$

Torques to push with a force. The tool presses down on a table, $\mathbf{F} = (0, -15)$ N: $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F} = (0.4330 \cdot (-15),\ 0) = (-6.495,\ 0)$ N m. $\tau_2 = 0$: $q_1 + q_2 = 90°$, link 2 is vertical, the vertical force passes through the elbow axis. Pushing $+x_0$ with 10 N: $\boldsymbol{\tau} = (-6.5, -4.0)$. For $\mathbf{F} = (10, -20)$ N: $\boldsymbol{\tau} = (-6.5 - 8.660,\ -4.0) = (-15.160, -4.000)$ N m (the wrong $\mathbf{JF}$ gives $(1.5, 4.330)$). A moment $n_z = 2$ N m adds $2(1, 1)$.

Force produced by torques. $\boldsymbol{\tau} = (2, 1)$ N m: solve $\mathbf{J}^\top\mathbf{F} = \boldsymbol{\tau}$: row 2: $-0.4F_x = 1 \Rightarrow F_x = -2.5$; row 1: $-0.65(-2.5) + 0.4330F_y = 2 \Rightarrow F_y = 0.866$. $\mathbf{F} = (-2.5, 0.866)$ N, $|\mathbf{F}| = 2.646$ N.

Stretched pose $\mathbf{q}$ $= (30°, 0)$: 50 N along the arm needs $\boldsymbol{\tau} = (0, 0)$; 50 N perpendicular needs $(45, 20)$ N m (= $50 \cdot 0.9, 50 \cdot 0.4$).

Frame: ${}^3\mathbf{F} = \mathbf{R}_z(45°)^\top \mathbf{F}$ with ${}^3\mathbf{J}^\top$ gives the same $\boldsymbol{\tau}$ (checked), so $\boldsymbol{\tau} = {}^0\mathbf{J}^{\top 0}\mathbf{F} = {}^3\mathbf{J}^{\top 3}\mathbf{F}$. Axes: velocity 0.2029, 0.8537; force 4.929, 1.171. [checked]

SLIDE CHECK slide 58

Slide 58 gives neither the frame of $\mathbf{F}_E$ nor the sign convention for $\boldsymbol{\tau}$. Write both in your answer (above).

STEPS Statics question

1. Sketch the arm and the force arrow; choose frame {0}; fix the sign convention (force *on the environment*).
2. Write $\mathbf{J}_{xy}$ in that frame (add the ω_z row if a moment is given).
3. $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$: each torque is a column of $\mathbf{J}$ dotted with $\mathbf{F}$.
4. Units N m. Check by moments: $\tau_1 = xF_y - yF_x$ about joint 1.
5. Singular pose: say which force directions need zero torque.

Summary of the week [p. 60]

Everything in one box

quantity	formula
tool velocity	${}^0\boldsymbol{\nu} = {}^0\mathbf{J}\dot{\mathbf{q}}$, column $j = \left[\tau(\partial\mathbf{T}/\partial q_j);\ \vee_\times(\rho(\partial\mathbf{T}/\partial q_j)\rho(\mathbf{T})^\top)\right]$
planar RR	$\mathbf{J}_{xy} = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$, ω_z row $(1, 1)$
frame change	${}^A\mathbf{J} = \mathrm{diag}({}^A\mathbf{R}_B, {}^A\mathbf{R}_B)\,{}^B\mathbf{J}$
joint rates	$\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu}$ or $\mathbf{J}^+\boldsymbol{\nu}$
singular	$\det \mathbf{J} = 0$; RR: $L_1 L_2 \sin q_2 = 0$
ellipsoid	velocity $\boldsymbol{\nu}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{\nu} \le 1$; force $\mathbf{F}^\top\mathbf{J}\mathbf{J}^\top\mathbf{F} \le 1$; $m = \sqrt{\lambda_{\min}/\lambda_{\max}}$
statics	$\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}_E$

How this is examined: worked solutions

No W13 sample question is on velocity kinematics (marks not published). The questions below are exam-style (marks assumed, about 1 per step), reusing the W6 practice and mock questions. Typical lost marks: running sums forgotten, frame mixed, $\mathbf{JF}$ for $\mathbf{J}^\top\mathbf{F}$, radians, one $\mathbf{J}$ for the whole path.

EXAM-STYLE QUESTION Q1: Jacobian, singularities, statics, rates [marks assumed 7]

Planar 2R arm, $L_1 = 0.5$ m, $L_2 = 0.4$ m. (a) Derive the 2×2 position Jacobian in $\{0\}$ and the singular configurations [3]. (b) At $\mathbf{q} = (30°, 60°)$ find the joint torques that hold a tip force $\mathbf{F} = (0, -10)$ N; explain τ_2 [2]. (c) Find $\dot{\mathbf{q}}$ for a tip velocity $(0.1, 0)$ m/s at that pose [2].

Solution

Setup. Frame $\{0\}$ at joint 1, x_0 horizontal; q_1 from x_0, q_2 relative to link 1; $\mathbf{F}$ is the force the tip applies to the environment.

(a) $x = L_1c_1 + L_2c_{12}$, $y = L_1s_1 + L_2s_{12}$. Differentiate: $\mathbf{J} = \begin{bmatrix} -L_1s_1 - L_2s_{12} & -L_2s_{12} \\ L_1c_1 + L_2c_{12} & L_2c_{12} \end{bmatrix}$. $\det = (-L_1s_1 - L_2s_{12})L_2c_{12} + L_2s_{12}(L_1c_1 + L_2c_{12}) = L_1L_2(s_{12}c_1 - c_{12}s_1) = L_1L_2 \sin q_2$. Zero at $q_2 = 0$ (stretched) and $q_2 = \pi$ (folded): workspace boundary, independent of q_1 [3].

(b) $q_1 + q_2 = 90°$: $\mathbf{J} = \begin{bmatrix} -0.5(0.5) - 0.4(1) & -0.4 \\ 0.5(0.866) + 0 & 0 \end{bmatrix} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$. $\boldsymbol{\tau} = \mathbf{J}^\top \mathbf{F} = \begin{bmatrix} -0.65 & 0.4330 \\ -0.4 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -10 \end{bmatrix} = \begin{bmatrix} -4.330 \\ 0 \end{bmatrix}$ N m [1]. $\tau_2 = 0$ because link 2 points straight up: the vertical force acts along link 2, with no moment about joint 2 [1]. (If $\mathbf{F}$ is a load on the tip the holding torques are $(+4.330, 0)$; state your reading.)

(c) $\det = 0.5 \cdot 0.4 \sin 60° = 0.1732$; $\dot{\mathbf{q}} = \mathbf{J}^{-1}\boldsymbol{\nu} = \frac{1}{0.1732} \begin{bmatrix} 0 & 0.4 \\ -0.433 & -0.65 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} = (0, -0.25)$ rad/s [2]. Check: $v_x = -0.4(-0.25) = 0.1$ ✓. Only joint 2 moves: link 2 is vertical, so turning it moves the tip horizontally. [checked]

EXAM-STYLE QUESTION Q2: spot the errors in a Jacobian [marks assumed 5]

A student gives the frame-$\{0\}$ Jacobian of a planar RR arm as $\begin{bmatrix} -L_1s_1 - L_2s_2 & -L_2s_2 \\ L_1c_1 + L_2c_2 & L_2c_2 \end{bmatrix}$, says "$\det = L_1L_2 \sin q_2$ only when $q_1 = 0$" and "$\omega_z = \dot{q}_2$". For $L_1 = 0.5$, $L_2 = 0.3$ m, $\mathbf{q} = (30°, 60°)$ give the correct $\mathbf{J}$ and mark the three claims.

Solution

Error 1 (running sum). In $\{0\}$ the second link's absolute angle is $q_1 + q_2 = 90°$, so every s_2, c_2 must be s_{12}, c_{12}. Correct: $J_{11} = -0.5(0.5) - 0.3(1) = -0.55$, $J_{12} = -0.3$, $J_{21} = 0.5(0.8660) + 0.3(0) = 0.4330$, $J_{22} = 0$. The student's formula gives $J_{11} = -0.5098$, $J_{21} = 0.5830$ [2].

Error 2. $\det \mathbf{J} = L_1L_2 \sin q_2$ for every q_1 (here $0.15 \sin 60° = 0.1299$): a singularity is a property of the arm's shape, and det is the same in every frame [2].

Error 3. Parallel axes: $\omega_z = \dot{q}_1 + \dot{q}_2$ [1]. [checked]

EXAM-STYLE QUESTION Q3: trace a circle [marks assumed 8]

RR arm $L_1 = 0.5$ m, $L_2 = 0.4$ m. The tool must trace a circle of radius 0.1 m, counter-clockwise, at constant speed 0.2 m/s, starting at its rightmost point with $\mathbf{q} = (30°, 60°)$. (a) Write $\boldsymbol{v}(t)$ and the time for one lap [3]. (b) Find $\dot{\mathbf{q}}$ at $t = 0$ [3]. (c) Find $\dot{\mathbf{q}}$ a quarter lap later [1]. (d) Why is $\dot{\mathbf{q}}$ not constant? [1]

Solution

Setup. Frame $\{0\}$; tool at $\mathbf{p}_0 = (0.4330, 0.6500)$ m, so the centre is $\mathbf{c} = (0.3330, 0.6500)$ m.

(a) $\Omega = v/\rho = 0.2/0.1 = 2$ rad/s. $\mathbf{p}(t) = \mathbf{c} + 0.1(\cos 2t, \sin 2t)$; $\boldsymbol{v} = \dot{\mathbf{p}} = 0.2(-\sin 2t, \cos 2t)$ m/s, $|\boldsymbol{v}| = 0.2$ ✓. One lap $T = 2\pi/\Omega = 3.14$ s [3].

(b) $t = 0$: $\boldsymbol{v} = (0, 0.2)$. $\mathbf{J} = \begin{bmatrix} -0.65 & -0.4 \\ 0.4330 & 0 \end{bmatrix}$, $\det = 0.1732$. Row 2: $0.4330\dot{q}_1 = 0.2 \Rightarrow \dot{q}_1 = 0.4619$. Row 1: $-0.65(0.4619) - 0.4\dot{q}_2 = 0 \Rightarrow \dot{q}_2 = -0.7506$. $\dot{\mathbf{q}} = (0.462, -0.751)$ rad/s [3].

(c) $t = \pi/4 = 0.785$ s: tip $(0.3330, 0.7500)$, $\boldsymbol{v} = (-0.2, 0)$. IK: $|\mathbf{p}|^2 = 0.3330^2 + 0.75^2 = 0.6734$, $\cos q_2 = (0.6734 - 0.41)/0.4 = 0.6585$, $q_2 = 48.8°$, $q_1 = \text{atan2}(0.75, 0.333) - \text{atan2}(0.4 \sin q_2, 0.5 + 0.4 \cos q_2) = 66.07° - 21.53° = 44.5°$. Solve $\mathbf{J}\dot{\mathbf{q}} = (-0.2, 0)$: $\dot{\mathbf{q}} = (0.031, 0.443)$ rad/s [1].

(d) $\mathbf{J}$ depends on $\mathbf{q}$, which changes around the circle: the rates must be recomputed at every point. Largest rate on the lap 0.78 rad/s; smallest $q_2 = 45.7°$, far from the singularity $q_2 = 0$. ω_z is not controlled (two joints) [1]. [checked]

EXAM-STYLE QUESTION Q4: planar 3R, redundant solve [marks assumed 8]

Planar 3R, $L = (0.4, 0.3, 0.2)$ m. (a) Derive the 2×3 position Jacobian in $\{0\}$ and the angular row [2]. (b) At $\mathbf{q} = (20°, 40°, -30°)$ find the tool velocity for $\dot{\mathbf{q}} = (0.3, -0.2, 0.5)$ rad/s [2]. (c) Find the smallest-norm $\dot{\mathbf{q}}$ for a tool velocity $(0.05, 0.02)$ m/s [3]. (d) When is $\mathbf{J}_{xy}$ rank-deficient? [1]

Solution

(a) With c_k the running sums: $\mathbf{J}_{xy} = \begin{bmatrix} -L_1s_1 - L_2s_{12} - L_3s_{123} & -L_2s_{12} - L_3s_{123} & -L_3s_{123} \\ L_1c_1 + L_2c_{12} + L_3c_{123} & L_2c_{12} + L_3c_{123} & L_3c_{123} \end{bmatrix}$, ω_z row $(1, 1, 1)$ [2].

(b) Running angles $20°, 60°, 30°$: $\mathbf{J}_{xy} = \begin{bmatrix} -0.4966 & -0.3598 & -0.1000 \\ 0.6991 & 0.3232 & 0.1732 \end{bmatrix}$. $v_x = -0.4966(0.3) - 0.3598(-0.2) - 0.1(0.5) = -0.1270$; $v_y = 0.6991(0.3) + 0.3232(-0.2) + 0.1732(0.5) = 0.2317$ m/s; $\omega_z = 0.3 - 0.2 + 0.5 = 0.6$ rad/s [2].

(c) Three joints, two components: $\mathbf{J}$ is fat. $\dot{\mathbf{q}} = \mathbf{J}^\top(\mathbf{J}\mathbf{J}^\top)^{-1}\boldsymbol{v}$. $\mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 0.3861 & -0.4808 \\ -0.4808 & 0.6232 \end{bmatrix}$, $\det = 0.00945$. Solve $(\mathbf{J}\mathbf{J}^\top)\mathbf{y} = (0.05, 0.02)$: $\mathbf{y} = (4.3169, 3.3626)$. $\dot{\mathbf{q}} = \mathbf{J}^\top\mathbf{y} = (0.2069, -0.4664, 0.1507)$ rad/s, $\|\dot{\mathbf{q}}\| = 0.5321$ rad/s [3]. Check $\mathbf{J}\dot{\mathbf{q}} = (0.05, 0.02)$ ✓.

(d) Rank 1 only when all columns are parallel, i.e. all links collinear (q_2, q_3 multiples of 180°). Unlike the 2R arm, one straight elbow is not singular for position (the third joint can still move the tip across the arm); (x, y, φ) has $\det = L_1L_2 \sin q_2$ [1]. [checked]

EXAM-STYLE QUESTION Q5: manipulability, and what a singularity does [marks assumed 7]

Planar RR, $L_1 = L_2 = 0.5$ m, $q_2 = 110°$. (a) Semi-axes of the velocity ellipse, $m = \sigma_{\min}/\sigma_{\max}$ and w [4]. (b) The arm is nearly stretched. Which tool velocity becomes impossible, what happens to joint rates for a wanted tool velocity, what happens to the force the tool can resist, and one remedy? [2] (c) At constant speed across the arm, by what factor does $\dot{q}_1$ grow when q_2 falls from $20°$ to $4°$? [1]

Solution

(a) Tool-frame block $\mathbf{J} = \begin{bmatrix} 0.4698 & 0 \\ 0.3290 & 0.5 \end{bmatrix}$ (q_1 drops out). $\mathbf{A} = \mathbf{J}\mathbf{J}^\top = \begin{bmatrix} 0.2208 & 0.1546 \\ 0.1546 & 0.3582 \end{bmatrix}$; $\mathrm{tr} = 0.5790$, $\det = (0.25\sin 110°)^2 = 0.0552$; $\lambda = \frac{1}{2}(0.5790 \pm \sqrt{0.5790^2 - 4(0.0552)}) = 0.4587,\ 0.1203$. Semi-axes $\sqrt{\lambda} = 0.6772,\ 0.3469$ m/s per rad/s; $m = 0.3469/0.6772 = 0.512$; $w = 0.6772 \times 0.3469 = 0.2349$ m^2 [4].
(b) Motion along the arm is impossible (all columns of $\mathbf{J}_v$ are perpendicular to the straight arm). $\mathbf{J}^{-1}$ contains $1/(L_1 L_2 \sin q_2)$: rates for a velocity with a component across/along the arm $\to \infty$ and exceed motor limits. Force semi-axis $1/\sigma \to \infty$: a force along the arm needs almost zero torque. Remedies: plan around the pose, damped least squares, or a redundant joint [2].
(c) $\dot{q}_1 = v_\perp/(L_1 \sin q_2)$: factor $\sin 20°/\sin 4° = 0.3420/0.0698 = 4.90$ [1]. [checked]

EXAM-STYLE QUESTION Q6: derive $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$, then use it [marks assumed 6]

(a) Use virtual work to show that joint torques producing a tip force $\mathbf{F}$ are $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ [2]. (b) RR arm $L_1 = 0.5, L_2 = 0.4$ m at $\mathbf{q} = (60°, -30°)$, tip force $\mathbf{F} = (-2, 6)$ N in $\{0\}$ (force the tip applies): find $\boldsymbol{\tau}$ and check by moments [4].

Solution

(a) A small joint motion $\delta\mathbf{q}$ moves the tip by $\delta\mathbf{x} = \mathbf{J}\delta\mathbf{q}$. Work by the actuators equals work against the tip force: $\boldsymbol{\tau}^\top\delta\mathbf{q} = \mathbf{F}^\top\delta\mathbf{x} = \mathbf{F}^\top\mathbf{J}\delta\mathbf{q}$ for every $\delta\mathbf{q}$, so $\boldsymbol{\tau}^\top = \mathbf{F}^\top\mathbf{J}$, $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ [2].
(b) $q_1 + q_2 = 30°$: $\mathbf{J} = \begin{bmatrix} -0.5(0.8660) - 0.4(0.5) & -0.4(0.5) \\ 0.5(0.5) + 0.4(0.8660) & 0.4(0.8660) \end{bmatrix} = \begin{bmatrix} -0.633 & -0.2 \\ 0.5964 & 0.3464 \end{bmatrix}$. $\boldsymbol{\tau} = \begin{bmatrix} -0.633 & 0.5964 \\ -0.2 & 0.3464 \end{bmatrix}\begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 1.266 + 3.578 \\ 0.4 + 2.078 \end{bmatrix} = (4.8445,\ 2.4785)$ N m [2].
Moments: tip $(0.5964, 0.633)$: about joint 1, $xF_y - yF_x = 0.5964(6) - 0.633(-2) = 4.8445$ ✓; elbow $(0.25, 0.433)$, tip relative to elbow $(0.3464, 0.2)$: $0.3464(6) - 0.2(-2) = 2.4785$ ✓ [2].
Wrong $\mathbf{JF} = (0.066, 0.886)$: not torques. [checked]

Which recipe? Lookup [added]

If the question says …	Go to …
"derive the equations of motion" (RR, RP, PR, 3D RRR)	Lagrange recipe, then the robot that matches (p. 93)
"what torque holds the arm still at $\mathbf{q}$"	$\mathbf{Q} = \mathbf{g}(\mathbf{q})$ (p. 95)
"what torque gives this acceleration / motion"	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$ (p. 95)
"what acceleration results from torque $\mathbf{Q}$"	forward dynamics, $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\dots)$ (p. 95)
"angular / linear acceleration of the centre of mass"	Newton–Euler outward pass (p. 96)
"inertia tensor of a box / rod in frame $\{A\}$"	integrals, then parallel axis (p. 91)
"compare Lagrange and Newton–Euler"	side-by-side table (p. 97)

Why dynamics: forward and inverse [p. 6–12]

Kinematics (W2–W6) says *where* and *how fast*. Dynamics says *how hard the motors must push*: torque (N m, revolute) or force (N, prismatic) as a function of position, velocity and acceleration. All equations below use the joint force/torque vector $\mathbf{Q}$ (the slides' symbol).

PREMISE inverse dynamics [p. 10] **Given:** joint positions $\mathbf{q}$, velocities $\dot{\mathbf{q}}$, accelerations $\ddot{\mathbf{q}}$ (a planned motion) and the robot's mass data
Find: joint torques/forces $\mathbf{Q}$. Use: *control* (computed torque, W9) and sizing motors

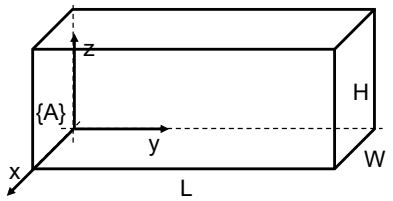
PREMISE forward dynamics [p. 9, 48] **Given:** joint torques/forces $\mathbf{Q}$, the present state $\mathbf{q}, \dot{\mathbf{q}}$ and the mass data
Find: the motion $\ddot{\mathbf{q}}$ (integrate to get $\dot{\mathbf{q}}, \mathbf{q}$). Use: *simulation*

What the actuator torque has to balance [p. 11]: (1) *dynamic* torques from motion: inertial (proportional to acceleration), centripetal (holds a body on a circle, directed to the centre), Coriolis (interaction of two rotating links); (2) *static* torques from an external load; (3) *gravity*; (4) *friction*; (5) other external forces.

TRAP
Mixing up the two directions. Inverse = motion in, torques out (needs no integration). Forward = torques in, motion out (needs $\mathbf{M}^{-1}$ and integration).

Rigid-body basics: mass, centre of mass, inertia tensor [p. 34–39]

A link is a rigid body of mass m. Its **centre of mass** (COM) is $\mathbf{p}_C = \frac{1}{m}\iiint \mathbf{r}\,\rho\,dv$. Frame $\{C\}$ is a frame with its origin at the COM and axes parallel to the link frame. The **inertia tensor** says how hard the body is to spin about each axis, and depends on the frame in which it is written [p. 39].



[W07 lec p. 37] — cuboid with frame $\{A\}$ at a corner: x spans W, y spans L, z spans H

Inertia tensor in frame $\{A\}$, uniform density ρ [p. 36]

$$ {}^{A}\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}, \quad I_{xx} = \iiint_V (y^2+z^2)\rho\,dv, \quad I_{xy} = -\iiint_V xy\,\rho\,dv $$
$$ I_{yy} = \iiint_V (x^2+z^2)\rho\,dv, \quad I_{yz} = -\iiint_V yz\,\rho\,dv $$
$$ I_{zz} = \iiint_V (x^2+y^2)\rho\,dv, \quad I_{xz} = -\iiint_V xz\,\rho\,dv $$

SYMBOLS

${}^{A}\mathbf{I}$	inertia tensor about the origin of $\{A\}$, written in $\{A\}$; symmetric	$\mathrm{kg\,m^2}$
I_{xx}, I_{yy}, I_{zz}	moments of inertia about the x, y, z axes of $\{A\}$	$\mathrm{kg\,m^2}$
I_{xy}, I_{yz}, I_{xz}	products of inertia (note the minus sign)	$\mathrm{kg\,m^2}$
ρ	mass density (uniform)	$\mathrm{kg\,m^{-3}}$
x, y, z	coordinates of the volume element $dv = dx\,dy\,dz$ in $\{A\}$	m

Use: whenever a link's rotation matters: Euler's equation, kinetic energy $\frac{1}{2}\boldsymbol{\omega}^\top\mathbf{I}\boldsymbol{\omega}$. Moments are always ≥ 0; products can have either sign.

DERIVATION cuboid, frame at a corner [p. 37–38]
$\rho = m/(WLH)$, limits $x \in [0, W]$, $y \in [0, L]$, $z \in [0, H]$.

$$ I_{xx} = \rho\int_0^W\int_0^L\int_0^H (y^2+z^2)\,dz\,dy\,dx = \rho W\left(\frac{L^3}{3}H + L\frac{H^3}{3}\right) = \frac{m(L^2+H^2)}{3} $$
$$ I_{xy} = -\rho\int_0^W x\,dx\int_0^L y\,dy\int_0^H dz = -\rho\,\frac{W^2}{2}\,\frac{L^2}{2}\,H = -\frac{m\,WL}{4} $$

The other entries follow by permuting $(x, y, z) \to (y, z, x)$ and $(W, L, H) \to (L, H, W)$.

Cuboid $W \times L \times H$ (x, y, z), mass m [p. 38–39]

$$ \text{corner } \{A\}: \; {}^{A}\mathbf{I} = \begin{bmatrix} \frac{m(L^2+H^2)}{3} & -\frac{m\,WL}{4} & -\frac{m\,HW}{4} \\ -\frac{m\,WL}{4} & \frac{m(W^2+H^2)}{3} & -\frac{m\,HL}{4} \\ -\frac{m\,HW}{4} & -\frac{m\,HL}{4} & \frac{m(L^2+W^2)}{3} \end{bmatrix} $$
$$ \text{centroid } \{C\}: \; {}^{C}\mathbf{I} = \begin{bmatrix} \frac{m(L^2+H^2)}{12} & 0 & 0 \\ 0 & \frac{m(W^2+H^2)}{12} & 0 \\ 0 & 0 & \frac{m(L^2+W^2)}{12} \end{bmatrix} $$

Use: the same body has a different tensor when the frame moves [p. 39]. At the centroid, symmetry makes all products zero.

SLIDE CHECK sign of I_{xy}, p. 37
Slide 37 prints $I_{xy} = \iiint xy\rho\,dv = m(WL)/4$ (no minus). Slide 36 defines $I_{xy} = -\iiint xy\rho\,dv$ and slide 38's matrix has $-m(WL)/4$. The matrix entry is $-mWL/4$.

Parallel-axis theorem [added]

$$ {}^{A}\mathbf{I} = {}^{C}\mathbf{I} + m\left(\mathbf{p}^{\top}\mathbf{p}\,\mathbf{1}_3 - \mathbf{p}\,\mathbf{p}^{\top}\right) $$

SYMBOLS

$\mathbf{p}$	position of the COM in $\{A\}$ (axes of $\{A\}$ parallel to those of $\{C\}$)	m
$\mathbf{1}_3$	3×3 identity	–

Use: moving the tensor from the centroid to another origin. Entrywise: $I_{xx} = {}^{C}I_{xx} + m(p_y^2 + p_z^2)$, $I_{xy} = {}^{C}I_{xy} - m\,p_x p_y$. Check with the slides: $\mathbf{p} = (\frac{W}{2}, \frac{L}{2}, \frac{H}{2})$ turns the centroid tensor into the corner tensor above.

Other bodies used for links [added]

body	moment of inertia
thin rod, length ℓ, about an axis $\perp$ rod through the centre	$m\ell^2/12$
same rod, about the end (parallel axis: $\frac{m\ell^2}{12} + m\frac{\ell^2}{4}$)	$m\ell^2/3$
point mass at distance ℓ	$m\ell^2$
solid cylinder, radius r, length h, axis z, centroid	$I_{zz} = \frac{1}{2}mr^2$, $I_{xx} = I_{yy} = \frac{m(3r^2+h^2)}{12}$

Use: a slender rod rotating in its plane about a joint at one end: $I = m\ell^2/3$. A uniform link of length ℓ has its COM at $\ell/2$.

Principal axes [added]

The axes along which ${}^{C}\mathbf{I}$ is diagonal. They are the eigenvectors of $\mathbf{I}$; the eigenvalues are the principal moments.

Use: if $\boldsymbol{\omega}$ is along a principal axis then $\mathbf{I}\boldsymbol{\omega} \parallel \boldsymbol{\omega}$ and $\boldsymbol{\omega}\times\mathbf{I}\boldsymbol{\omega} = \mathbf{0}$ (Euler's equation loses its gyroscopic term). Symmetry axes of a body are principal axes.

Example. $\mathbf{I} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ has eigenvalues $1, 3, 3$ with principal axes $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{2}}(1,1,0)$, $(0,0,1)$. For $\boldsymbol{\omega} = (1,1,0)$: $\mathbf{I}\boldsymbol{\omega} = (3,3,0)$, $\boldsymbol{\omega}\times\mathbf{I}\boldsymbol{\omega} = \mathbf{0}$. For $\boldsymbol{\omega} = (1,0,0)$: $\mathbf{I}\boldsymbol{\omega} = (2,1,0)$, $\boldsymbol{\omega}\times\mathbf{I}\boldsymbol{\omega} = (0,0,1) \neq \mathbf{0}$. [checked]

WORKED box $m = 6$ kg, $W = 0.2$, $L = 0.3$, $H = 0.1$ m

Centroid: $\operatorname{diag}\left(\frac{6}{12}(0.09+0.01), \frac{6}{12}(0.04+0.01), \frac{6}{12}(0.09+0.04)\right) = \operatorname{diag}(0.050, 0.025, 0.065)$.

Corner tensor by the slide formulas: $I_{xx} = \frac{6}{3}(0.09+0.01) = 0.200$, $I_{xy} = -\frac{6}{4}(0.2)(0.3) = -0.090$, $I_{xz} = -\frac{6}{4}(0.1)(0.2) = -0.030$, $I_{yz} = -\frac{6}{4}(0.1)(0.3) = -0.045$:

$$ {}^{A}\mathbf{I} = \begin{bmatrix} 0.200 & -0.090 & -0.030 \\ -0.090 & 0.100 & -0.045 \\ -0.030 & -0.045 & 0.260 \end{bmatrix} \text{ kg m}^2 $$

Parallel axis check on I_{xx}: $0.050 + 6(0.15^2 + 0.05^2) = 0.050 + 0.150 = 0.200$. [checked]

Newton's and Euler's equations for one rigid body [p. 34]

Newton and Euler [p. 34]

$$ \mathbf{F} = m\,\dot{\boldsymbol{v}}_C, \qquad \mathbf{N} = {}^{C}\mathbf{I}\,\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}\times {}^{C}\mathbf{I}\,\boldsymbol{\omega} $$

SYMBOLS

$\mathbf{F}$	net force that causes the COM acceleration	N
$\mathbf{N}$	net moment about the COM that causes the rotation	N m
$\dot{\boldsymbol{v}}_C$	linear acceleration of the COM	m s^{-2}
$\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}$	angular velocity, acceleration of the link	rad s^{-1}, rad s^{-2}
${}^{C}\mathbf{I}$	inertia tensor about the COM, in the same frame as $\boldsymbol{\omega}$	kg m^2

Use: all vectors in the same frame (the link frame $\{1\}$ in the examples). The term $\boldsymbol{\omega}\times\mathbf{I}\boldsymbol{\omega}$ is the gyroscopic moment; it vanishes for planar motion about a principal axis.

WORKED W8 practical Q1: single link, every step

PREMISE **Given:** $m = 1$ kg, ${}^{1}\mathbf{P}_C = (3,0,0)^{\top}$ m, ${}^{C}\mathbf{I} = \operatorname{diag}(1,2,3)$ kg m^2, $q(t) = bt + ct^2$, rest at $t = 0$

Find: $\dot{\boldsymbol{\omega}}$ and $\dot{\boldsymbol{v}}_C$ in $\{1\}$, then the joint torque

Joint axis is z_1: $\boldsymbol{\omega} = (0,0,\dot{q})^{\top} = (0,0,b+2ct)^{\top}$, $\dot{\boldsymbol{\omega}} = (0,0,2c)^{\top}$ rad/s^2.

$\dot{\boldsymbol{v}}_C = \dot{\boldsymbol{\omega}}\times\mathbf{P}_C + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{P}_C)$ (the joint origin does not accelerate).

$\dot{\boldsymbol{\omega}}\times\mathbf{P}_C = (0,0,2c)\times(3,0,0) = (0,\,6c,\,0)$ (tangential); $\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{P}_C) = -\dot{q}^2(3,0,0) = (-3\dot{q}^2,0,0)$ (centripetal, towards the axis).

$$ \boxed{\dot{\boldsymbol{v}}_C = \left(-3(b+2ct)^2,\; 6c,\; 0\right)^{\top} \text{ m/s}^2} $$

Then $\mathbf{F} = m\dot{\boldsymbol{v}}_C$; $\mathbf{N} = {}^{C}\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}\times{}^{C}\mathbf{I}\boldsymbol{\omega} = (0,0,3\cdot 2c) + \mathbf{0} = (0,0,6c)$ ($\boldsymbol{\omega} \parallel z$ is a principal axis). Joint moment: $n_z = N_z + (\mathbf{P}_C\times\mathbf{F})_z = 6c + 3(6c)$, so $\boxed{\tau = 24c}$ N m. Check: $(I_{zz} + m\,3^2)\ddot{q} = (3+9)(2c) = 24c$. Numbers $b = 1, c = 0.5, t = 1$: $\dot{q} = 2$, $\ddot{q} = 1$, $\dot{\boldsymbol{v}}_C = (-12,3,0)$, $\tau = 12$ N m. [checked]

Energy of a link: K and P [added] (extends p. 16–21)

Kinetic and potential energy of link i

$$ K_i = \tfrac{1}{2}m_i\,{}^{0}\mathbf{v}_{C_i}^{\top}\,{}^{0}\mathbf{v}_{C_i} + \tfrac{1}{2}\,{}^{i}\boldsymbol{\omega}_i^{\top}\,{}^{C_i}\mathbf{I}_i\,{}^{i}\boldsymbol{\omega}_i, \qquad P_i = -m_i\,\mathbf{g}^{\top}\mathbf{p}_{C_i} = m_i g\,h_{C_i} $$

SYMBOLS

${}^{0}\mathbf{v}_{C_i}$	velocity of the COM of link i in $\{0\}$ (differentiate its position)	m s^{-1}
${}^{i}\boldsymbol{\omega}_i$	angular velocity of link i and ${}^{C_i}\mathbf{I}_i$ its tensor, both in the link frame	rad s^{-1}, kg m^2
h_{C_i}	height of the COM above a chosen level (gravity along $-y$ or $-z$)	m
g	9.81	m s^{-2}

Use: point mass: drop the second term. Planar link rotating at $\dot{\theta}$ (total, i.e. $\dot{q}_1 + \dot{q}_2$ for link 2 of a 2R): second term $= \frac{1}{2}I_{zz}\dot{\theta}^2$. The choice of zero height is free: constants in P vanish when you differentiate.

Lagrange's method [p. 15–17]

Lagrangian and Euler–Lagrange equation [p. 16–17]

$$\mathcal{L}(\mathbf{q},\dot{\mathbf{q}}) = K(\mathbf{q},\dot{\mathbf{q}}) - P(\mathbf{q}), \qquad Q_i = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} \quad (i = 1,\dots,n)$$

SYMBOLS

$\mathcal{L}$	Lagrangian (scalar)	J
K, P	total kinetic and potential energy of the robot	J
$q_i, \dot{q}_i$	generalised coordinate = joint variable and its rate	rad or m; · s^{-1}
Q_i	generalised force at joint i: torque (revolute) or force (prismatic)	N m or N

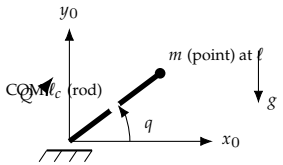
Use: write K and P once for the whole robot; internal reaction forces never appear [p. 14]. Gives closed-form equations: best for analysis and control design.

STEPS Equations of motion by Lagrange (W13 Q4)

1. Draw the robot, frame {0} (x right, y up), the joint variables q_i and the direction of gravity. State where each angle is measured from.
2. Position of every COM (point mass) in {0} from the geometry/FK; differentiate to get the velocities, then $v^2 = \dot{x}^2 + \dot{y}^2$.
3. Write K (translation + rotation) and $P = \sum m_i g h_i$ *in full*; $\mathcal{L} = K - P$.
4. For each joint: (a) $\partial\mathcal{L}/\partial\dot{q}_i$, (b) d/dt of it (product rule!), (c) $\partial\mathcal{L}/\partial q_i$, then $Q_i = (b) - (c)$.
5. Collect: coefficients of $\ddot{q}_j$ give $\mathbf{M}$; terms with $\dot{q}_j\dot{q}_k$ give $\mathbf{C}$; terms with no $\dot{q}$ give $\mathbf{g}$.
6. Check: $\mathbf{M}$ symmetric with $M_{ii} > 0$; $\mathbf{g}$ has no $\dot{q}$; at rest only $\mathbf{g}$ is left; units (N m); one pose with an easy answer (arm horizontal).

Time-saving form. If $K = \frac{1}{2}\dot{\mathbf{q}}^\top \mathbf{M}(\mathbf{q})\dot{\mathbf{q}}$ (always true), then $Q_i = \sum_j M_{ij}\ddot{q}_j + \sum_{j,k}\left(\frac{\partial M_{ij}}{\partial q_k} - \frac{1}{2}\frac{\partial M_{jk}}{\partial q_i}\right)\dot{q}_j\dot{q}_k + \partial P/\partial q_i$ [added]: read $\mathbf{M}$ off K first, then differentiate.

Single link: pendulum [added]



[drawn for this booklet] one revolute joint about z_0 (out of the page), q from x_0, gravity along $-y_0$.

DERIVATION point mass m at distance ℓ; then a rod

$x = \ell c_q$, $y = \ell s_q$, $v^2 = \ell^2\dot{q}^2$. $K = \frac{1}{2}m\ell^2\dot{q}^2$, $P = mg\ell s_q$, $\mathcal{L} = \frac{1}{2}m\ell^2\dot{q}^2 - mg\ell s_q$.
$\partial\mathcal{L}/\partial\dot{q} = m\ell^2\dot{q}$; $\frac{d}{dt}(\cdot) = m\ell^2\ddot{q}$; $\partial\mathcal{L}/\partial q = -mg\ell c_q$.

$$Q = m\ell^2\ddot{q} + mg\ell\cos q \qquad (M = m\ell^2,\ C = 0,\ g = mg\ell\cos q)$$

Rod (mass m, COM at ℓ_c, inertia I_c about the COM): $K = \frac{1}{2}(m\ell_c^2 + I_c)\dot{q}^2$, $P = mg\ell_c s_q$, so

$$Q = (I_c + m\ell_c^2)\ddot{q} + mg\ell_c\cos q$$

Uniform rod of length ℓ: $I_c + m\ell_c^2 = m\ell^2/12 + m\ell^2/4 = m\ell^2/3$.

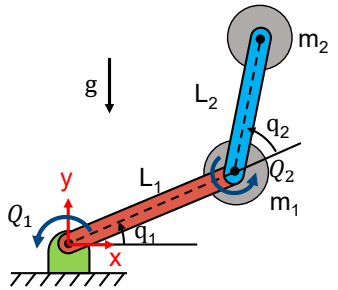
WORKED point mass and rod, with an energy check

Point mass $m = 2$ kg, $\ell = 0.5$ m. Hold at $q = 30°$: $Q = 2(9.81)(0.5)\cos 30° = 8.496$ N m. Horizontal ($q = 0$): 9.810 N m (maximum). Vertical up: 0.
Uniform rod $m = 2$ kg, $\ell = 0.6$ m: $I_c = 0.060$, $I_{end} = 0.240$ kg m^2, $mg\ell_c = 5.886$ N m. At $q = 30°$ with $\ddot{q} = 4$ rad/s^2: $Q = 0.240(4) + 5.886\cos 30° = 0.960 + 5.097 = 6.057$ N m.
Energy check: release from horizontal at rest ($q = 0$), $Q = 0$. At the bottom ($q = -90°$) the COM has dropped $\ell_c = 0.3$ m: $\frac{1}{2}I_{end}\dot{q}^2 = mg\ell_c \Rightarrow \dot{q} = \sqrt{2(5.886)/0.240} = 7.004$ rad/s; integrating the equation of motion numerically gives the same 7.004. [checked]

TRAP

Using I_c alone for a rod hinged at its end: the joint feels $I_c + m\ell_c^2$ (parallel axis). Using $m\ell^2$ (point-mass formula) for a rod: $m\ell^2$ is three times too large.

Planar two-link arm, point masses [p. 18–25]



[W07 lec p. 18] — the slides' 2R arm: point masses m_1 (end of L_1), m_2 (end of L_2); q_1 from x_0, q_2 relative to link 1; gravity along $-y$; Q_1, Q_2 at the joints

PREMISE RR, Lagrange **Given:** L_1, L_2, m_1, m_2 (point masses at the link tips), q_1, q_2 in {0} (x right, y up, g along $-y$)
Find: Q_1, Q_2 in terms of $q, \dot{q}, \ddot{q}$, then $\mathbf{M}$, $\mathbf{C}$, $\mathbf{g}$

DERIVATION 1. positions, velocities, energies [p. 19–22]

Mass 1: $x_1 = L_1c_1$, $y_1 = L_1s_1$; $\dot{x}_1 = -L_1s_1\dot{q}_1$, $\dot{y}_1 = L_1c_1\dot{q}_1$; $K_1 = \frac{1}{2}m_1L_1^2\dot{q}_1^2$; $P_1 = m_1gL_1s_1$.
Mass 2: $x_2 = L_1c_1 + L_2c_{12}$, $y_2 = L_1s_1 + L_2s_{12}$ ($c_{12} = \cos(q_1 + q_2)$);

$$\dot{x}_2 = -L_1s_1\dot{q}_1 - L_2s_{12}(\dot{q}_1 + \dot{q}_2), \qquad \dot{y}_2 = L_1c_1\dot{q}_1 + L_2c_{12}(\dot{q}_1 + \dot{q}_2)$$

Square and add (the cross term uses $s_1s_{12} + c_1c_{12} = \cos q_2$):

$$\dot{x}_2^2 + \dot{y}_2^2 = L_1^2\dot{q}_1^2 + L_2^2(\dot{q}_1 + \dot{q}_2)^2 + 2L_1L_2c_2\,\dot{q}_1(\dot{q}_1 + \dot{q}_2)$$

Collect by $\dot{q}_1^2$, $\dot{q}_1\dot{q}_2$, $\dot{q}_2^2$:

$$K_2 = \tfrac{1}{2}m_2\left[(L_1^2 + 2L_1L_2c_2 + L_2^2)\dot{q}_1^2 + 2(L_2^2 + L_1L_2c_2)\dot{q}_1\dot{q}_2 + L_2^2\dot{q}_2^2\right]$$

$P_2 = m_2g\,(L_1s_1 + L_2s_{12})$. Abbreviate

$$A = m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2), \quad B = m_2(L_1L_2c_2 + L_2^2), \quad D = m_2L_2^2$$

so $K = K_1 + K_2 = \frac{1}{2}A\dot{q}_1^2 + B\dot{q}_1\dot{q}_2 + \frac{1}{2}D\dot{q}_2^2$, $P = (m_1 + m_2)gL_1s_1 + m_2gL_2s_{12}$, $\mathcal{L} = K - P$.

W7

DERIVATION 2. Euler–Lagrange, joint 1 (every line)

$\frac{\partial \mathcal{L}}{\partial \dot{q}_1} = A\dot{q}_1 + B\dot{q}_2$. Only A and B depend on time (through c_2): $\dot{A} = -2m_2L_1L_2s_2\dot{q}_2$, $\dot{B} = -m_2L_1L_2s_2\dot{q}_2$. So

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_1} = A\ddot{q}_1 + B\ddot{q}_2 - 2m_2L_1L_2s_2\,\dot{q}_1\dot{q}_2 - m_2L_1L_2s_2\,\dot{q}_2^2$$

K does not contain q_1, so $\frac{\partial \mathcal{L}}{\partial q_1} = -\frac{\partial P}{\partial q_1} = -\big[(m_1+m_2)gL_1c_1 + m_2gL_2c_{12}\big]$.

$$Q_1 = A\ddot{q}_1 + B\ddot{q}_2 - m_2L_1L_2s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) + (m_1+m_2)gL_1c_1 + m_2gL_2c_{12}$$

DERIVATION 3. Euler–Lagrange, joint 2 (every line)

$\frac{\partial \mathcal{L}}{\partial \dot{q}_2} = B\dot{q}_1 + D\dot{q}_2$; $\frac{d}{dt}(\cdot) = B\ddot{q}_1 + D\ddot{q}_2 - m_2L_1L_2s_2\,\dot{q}_1\dot{q}_2$.

$\frac{\partial \mathcal{L}}{\partial q_2} = \frac{1}{2}\frac{\partial A}{\partial q_2}\dot{q}_1^2 + \frac{\partial B}{\partial q_2}\dot{q}_1\dot{q}_2 - \frac{\partial P}{\partial q_2} = -m_2L_1L_2s_2\dot{q}_1^2 - m_2L_1L_2s_2\dot{q}_1\dot{q}_2 - m_2gL_2c_{12}$.

Subtract: the two $m_2L_1L_2s_2\dot{q}_1\dot{q}_2$ terms cancel (so joint 2 has no Coriolis term):

$$Q_2 = B\ddot{q}_1 + D\ddot{q}_2 + m_2L_1L_2s_2\,\dot{q}_1^2 + m_2gL_2c_{12}$$

Result: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g}$ [p. 24–25]

$$\mathbf{M} = \begin{bmatrix} m_1L_1^2 + m_2(L_1^2 + 2L_1L_2c_2 + L_2^2) & m_2(L_1L_2c_2 + L_2^2) \\ m_2(L_1L_2c_2 + L_2^2) & m_2L_2^2 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -m_2L_1L_2s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ m_2L_1L_2s_2\,\dot{q}_1^2 \end{bmatrix}$$

$$\mathbf{g} = \begin{bmatrix} (m_1+m_2)gL_1c_1 + m_2gL_2c_{12} \\ m_2gL_2c_{12} \end{bmatrix}$$

Use: c_{12} in $\mathbf{g}$ because gravity on mass 2 acts at the absolute angle q_1+q_2. The slide matches these rows term for term.

WORKED numbers: $m = (2,1)$ kg, $L = (1.0, 0.8)$ m, $q = (30°, 45°)$, $\dot{q} = (1, -0.5)$ rad/s, $\ddot{q} = (0.5, 2)$ rad/s^2

$c_2 = 0.7071$, $s_2 = 0.7071$, $h = m_2L_1L_2s_2 = 0.5657$, $q_1+q_2 = 75°$.

$M_{11} = 2 + 1(1 + 2(0.8)(0.7071) + 0.64) = 4.7714$; $M_{12} = 0.8(0.7071) + 0.64 = 1.2057$; $M_{22} = 0.64$.

$C_1 = -0.5657\big(2(1)(-0.5) + 0.25\big) = +0.4243$; $C_2 = 0.5657(1)^2 = 0.5657$.

$g_1 = 3(9.81)(1)\cos 30° + 0.8(9.81)\cos 75° = 25.4885 + 2.0312 = 27.5183$; $g_2 = 2.0312$.

$$Q_1 = 4.7714(0.5) + 1.2057(2) + 0.4243 + 27.5183$$
$$= 2.3857 + 2.4114 + 0.4243 + 27.5183 = \boxed{32.740}\ \text{N m}$$
$$Q_2 = 1.2057(0.5) + 0.64(2) + 0.5657 + 2.0312$$
$$= 0.6028 + 1.28 + 0.5657 + 2.0312 = \boxed{4.480}\ \text{N m}$$

Energy check (power balance): $\dot{\mathbf{q}}^\top\mathbf{Q} = 1(32.740) - 0.5(4.480) = 30.500$ W must equal $\dot{K} + \dot{P}$. With $\dot{K} = \dot{\mathbf{q}}^\top\mathbf{M}\ddot{\mathbf{q}} + \frac{1}{2}\dot{\mathbf{q}}^\top\dot{\mathbf{M}}\dot{\mathbf{q}} = 3.997$ W and $\dot{P} = \dot{\mathbf{q}}^\top\mathbf{g} = 26.503$ W the sum is 30.500 W. [checked] Newton–Euler gives the same $\mathbf{Q}$ (p. 96). [checked]

TRAP

$\mathcal{L} = K + P$. Missing the cross term $2L_1L_2c_2$ in v_2^2. Writing g_2 with $\sin(q_1+q_2)$ (with q from the horizontal and gravity along $-y$ it is cos). Calling $\dot{q}_1\dot{q}_2$ centripetal (it is Coriolis). M_{12} without the L_2^2. Differentiating $\partial\mathcal{L}/\partial\dot{q}$ without the product rule on $\cos q_2 \cdot \dot{q}$.

Structure: M, C, g and their properties [p. 25–31]

General equation of motion [p. 25–26]

$$\underbrace{\mathbf{Q}}_{n\times1} = \underbrace{\mathbf{M}(\mathbf{q})}_{n\times n}\underbrace{\ddot{\mathbf{q}}}_{n\times1} + \underbrace{\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})}_{n\times1} + \underbrace{\mathbf{g}(\mathbf{q})}_{n\times1} \quad \big(+\ \mathbf{f}(\dot{\mathbf{q}}) + \mathbf{J}^\top(\mathbf{q})\mathbf{w}\big)$$

SYMBOLS

$\mathbf{Q}$	joint torques/forces	N m, N
$\mathbf{M}(\mathbf{q})$	mass (joint-space inertia) matrix	kg m^2 (R), kg (P)
$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$	velocity-product vector (centripetal + Coriolis)	N m
$\mathbf{g}(\mathbf{q})$	gravity torque vector; other potential energy (springs) also lands here	N m
$\mathbf{f}(\dot{\mathbf{q}})$	friction at the joints	N m
$\mathbf{J}^\top\mathbf{w}$	joint torques that balance an external wrench $\mathbf{w}$ at the tool (contact)	N m

Use: the slides' compact form is the first three terms; $\mathbf{f}$ and $\mathbf{J}^\top\mathbf{w}$ are added on p. 26. Forward dynamics: $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g} - \mathbf{f} - \mathbf{J}^\top\mathbf{w})$ *[p. 48].*

Split of C: centripetal and Coriolis [p. 30]

$$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) = C(\mathbf{q})[\dot{\mathbf{q}}^2] + B(\mathbf{q})[\dot{\mathbf{q}}\dot{\mathbf{q}}]$$

SYMBOLS

$C(\mathbf{q})$	$n \times n$ matrix of centripetal coefficients	–
$[\dot{\mathbf{q}}^2]$	the $n \times 1$ vector $(\dot{q}_1^2, \ldots, \dot{q}_n^2)^\top$	rad^{2}s^{-2}
$B(\mathbf{q})$	$n \times \frac{n(n-1)}{2}$ matrix of Coriolis coefficients	–
$[\dot{\mathbf{q}}\dot{\mathbf{q}}]$	the $\frac{n(n-1)}{2}$ products $\dot{q}_i\dot{q}_j$, $i<j$ ($n=3$: $\dot{q}_1\dot{q}_2, \dot{q}_1\dot{q}_3, \dot{q}_2\dot{q}_3$)	rad^{2}s^{-2}

Use: centripetal terms have squares of velocities; Coriolis terms have products. For the 2R arm with $h = m_2L_1L_2s_2$ (my split of the slide's vector, [added]):

$$\mathbf{C} = \underbrace{\begin{bmatrix} 0 & -h \\ h & 0 \end{bmatrix}\begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix}}_{\text{centripetal}} + \underbrace{\begin{bmatrix} -2h \\ 0 \end{bmatrix}[\dot{q}_1\dot{q}_2]}_{\text{Coriolis}}$$

Properties [p. 28] and [added]

- $\mathbf{M}(\mathbf{q}) = \mathbf{M}^\top$ and positive definite [p. 28]: $K = \frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}} > 0$ for any motion, so $\mathbf{M}^{-1}$ exists. Diagonal M_{jj} = inertia felt by joint j; M_{ij} couples $\ddot{q}_j$ into torque i.
- [added] $\mathbf{M}$ depends on $\mathbf{q}$ only; $\mathbf{g}$ on $\mathbf{q}$ only; $\mathbf{C}$ is quadratic in $\dot{\mathbf{q}}$ and vanishes at rest.
- [added] With $\mathbf{C} = \mathbf{C}_m(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}}$: $\dot{\mathbf{M}} - 2\mathbf{C}_m$ is skew-symmetric, so $\dot{\mathbf{q}}^\top\mathbf{C} = \frac{1}{2}\dot{\mathbf{q}}^\top\dot{\mathbf{M}}\dot{\mathbf{q}}$ and, with $\mathbf{Q} = \mathbf{0}$ and no friction, $K + P$ is constant.

2R: $\det\mathbf{M} = m_2L_1^2L_2^2(m_1 + m_2s_2^2) > 0$ always. $\mathbf{C}_m = \begin{bmatrix} -h\dot{q}_2 & -h(\dot{q}_1+\dot{q}_2) \\ h\dot{q}_1 & 0 \end{bmatrix}$, $\dot{\mathbf{M}} - 2\mathbf{C}_m = h(2\dot{q}_1 + \dot{q}_2)\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. [checked]

PUMA 560 numbers on the slides [p. 27, 29]

pose	$\mathbf{g}$ (joints 1–6, N m)
rest	$[0,\ 31.64,\ 6.035,\ 0,\ 0.02825,\ 0]$
arm stretched out	$[0,\ 46.01,\ 8.772,\ 0,\ 0.02825,\ 0]$
arm pointing up	$[0,\ -0.7752,\ 0.2489,\ 0,\ 0,\ 0]$

Friction on joint 1 at 1 rad/s from rest: -30.53 (other joints 0) [p. 29].

Use: gravity is usually the dominant term and is present when the robot is stationary; it is largest when the arm is stretched out (longest lever arm) and almost zero when it points straight up. The friction sign is the Toolbox's convention; the point is that it depends on $\dot{\mathbf{q}}$ only.

Scenarios: hold still, accelerate, forward dynamics [added]

situation	torque needed
held still ($\dot{\mathbf{q}} = \ddot{\mathbf{q}} = \mathbf{0}$)	$\mathbf{Q} = \mathbf{g}(\mathbf{q})$ [p. 27]
constant joint speed ($\ddot{\mathbf{q}} = \mathbf{0}$)	$\mathbf{Q} = \mathbf{C} + \mathbf{g}$
starts from rest ($\dot{\mathbf{q}} = \mathbf{0}$)	$\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}$
tool presses with wrench $\mathbf{w}$	add $\mathbf{J}^\top\mathbf{w}$ ($\mathbf{w}$ = force the tool applies; for a load on the tool the sign flips: state your reading)
torque given, find motion	$\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g})$

WORKED slide 2R at $q = (30°, 45°)$, $m = (2, 1)$, $L = (1, 0.8)$; M, g from the box above

Hold still: $\mathbf{Q} = \mathbf{g} = (27.518,\ 2.031)$ N m. Arm stretched along x_0 ($q = 0$): $\mathbf{g} = (3(9.81)(1) + 1(9.81)(0.8),\ 0.8(9.81)) = (37.278,\ 7.848)$ N m.

Start from rest with $\ddot{\mathbf{q}} = (0.5, 2)$: $\mathbf{M}\ddot{\mathbf{q}} = (4.797,\ 1.883)$, $\mathbf{Q} = (32.315,\ 3.914)$ N m.

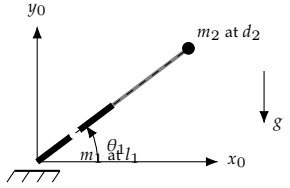
Forward dynamics, $\mathbf{Q} = \mathbf{0}$ *released at rest:* $\det\mathbf{M} = 1.600$, $\ddot{\mathbf{q}} = -\mathbf{M}^{-1}\mathbf{g} = -\frac{1}{1.6}\begin{bmatrix} 0.64 & -1.2057 \\ -1.2057 & 4.7714 \end{bmatrix}\begin{bmatrix} 27.518 \\ 2.031 \end{bmatrix} = (-9.477,\ 14.679)$ rad/s^2.

Pressing on a surface with $\mathbf{w} = (-5, 12)$ N (the force the tip applies to the surface, entered as $+\mathbf{J}^\top\mathbf{w}$ as in the slides' general form; for a load *on* the tip the sign flips): $\mathbf{J} = \begin{bmatrix} -1.2727 & -0.7727 \\ 1.0731 & 0.2071 \end{bmatrix}$, $\mathbf{J}^\top\mathbf{w} = (19.241,\ 6.348)$, $\mathbf{Q} = \mathbf{g} + \mathbf{J}^\top\mathbf{w} = (46.759,\ 8.380)$ N m. [checked]

TRAP

Using $\mathbf{Jw}$ instead of $\mathbf{J}^\top\mathbf{w}$ (a Jacobian maps joint rates to tip velocity; its transpose maps tip force to joint torque). Keeping $\mathbf{C}$ in a "hold still" answer. Forgetting that at rest $\mathbf{C} = \mathbf{0}$ (it is quadratic in $\dot{\mathbf{q}}$).

Planar RP arm [added](Craig example; unit practical code)



[drawn for this booklet] revolute θ_1 about z_0, then a prismatic d_2 along the arm (measured from the joint-1 axis); gravity along $-y_0$.

PREMISE RP, Lagrange **Given:** m_1, l_1, I_{zz1} (link 1: mass, COM distance, inertia about its COM), m_2, I_{zz2} (sliding link, COM at d_2), variables θ_1, d_2
Find: τ_1 (N m) and f_2 (N)

DERIVATION RP: positions, energies, Euler–Lagrange

COM of link 2: $(d_2c_1,\ d_2s_1)$, so $\dot{x} = \dot{d}_2c_1 - d_2s_1\dot{\theta}_1$, $\dot{y} = \dot{d}_2s_1 + d_2c_1\dot{\theta}_1$ and $v_2^2 = \dot{d}_2^2 + d_2^2\dot{\theta}_1^2$ (radial plus tangential).

$$K_1 = \tfrac{1}{2}(m_1l_1^2 + I_{zz1})\dot{\theta}_1^2, \quad K_2 = \tfrac{1}{2}m_2(\dot{d}_2^2 + d_2^2\dot{\theta}_1^2) + \tfrac{1}{2}I_{zz2}\dot{\theta}_1^2, \quad P = m_1gl_1s_1 + m_2gd_2s_1$$

θ_1: $\frac{\partial\mathcal{L}}{\partial\dot{\theta}_1} = (m_1l_1^2 + I_{zz1} + I_{zz2} + m_2d_2^2)\dot{\theta}_1$; $\frac{d}{dt}(\cdot) = (\dots)\ddot{\theta}_1 + 2m_2d_2\dot{d}_2\dot{\theta}_1$; $\frac{\partial\mathcal{L}}{\partial\theta_1} = -(m_1l_1 + m_2d_2)gc_1$.

d_2: $\frac{\partial\mathcal{L}}{\partial\dot{d}_2} = m_2\dot{d}_2$; $\frac{d}{dt}(\cdot) = m_2\ddot{d}_2$; $\frac{\partial\mathcal{L}}{\partial d_2} = m_2d_2\dot{\theta}_1^2 - m_2gs_1$.

Result: RP arm

$$\begin{bmatrix}\tau_1 \\ f_2\end{bmatrix} = \underbrace{\begin{bmatrix} m_1l_1^2 + I_{zz1} + I_{zz2} + m_2d_2^2 & 0 \\ 0 & m_2 \end{bmatrix}}_{\mathbf{M}}\begin{bmatrix}\ddot{\theta}_1 \\ \ddot{d}_2\end{bmatrix} + \underbrace{\begin{bmatrix} 2m_2d_2\dot{\theta}_1\dot{d}_2 \\ -m_2d_2\dot{\theta}_1^2 \end{bmatrix}}_{\mathbf{C}} + \underbrace{\begin{bmatrix} (m_1l_1 + m_2d_2)g\cos\theta_1 \\ m_2g\sin\theta_1 \end{bmatrix}}_{\mathbf{g}}$$

Use: $\mathbf{M}$ *is diagonal but* $\mathbf{C} \neq \mathbf{0}$*: $2m_2d_2\dot{\theta}_1\dot{d}_2$ is Coriolis (product), $-m_2d_2\dot{\theta}_1^2$ is centrifugal (square, pushes the slider outward so the actuator must pull in). M_{11} grows with d_2^2 as the arm extends. At $\theta_1 = 90°$ and no force, $\ddot{d}_2 = -g$ (free fall).*

WORKED numbers

$m = (2, 1)$ kg, $l_1 = 0.3$ m, $I_{zz} = (0.05, 0.02)$ kg m^2, $\theta_1 = 0.7$ rad, $d_2 = 0.6$ m, $\dot{\theta}_1 = 1.3$, $\dot{d}_2 = -0.8$, $\ddot{\theta}_1 = 2.0$, $\ddot{d}_2 = 1.5$.

$M_{11} = 2(0.09) + 0.05 + 0.02 + 1(0.36) = 0.61$. $\tau_1 = 0.61(2.0) + 2(1)(0.6)(1.3)(-0.8) + (2(0.3) + 0.6)(9.81)\cos 0.7 = 1.220 - 1.248 + 9.004 = \boxed{8.976}$ N m.

$f_2 = 1(1.5) - 1(0.6)(1.69) + 1(9.81)\sin 0.7 = 1.500 - 1.014 + 6.320 = \boxed{6.806}$ N.

Hold still at $\theta_1 = 30°$, $d_2 = 0.6$: $\tau_1 = 1.2(9.81)\cos 30° = 10.195$ N m, $f_2 = 9.81\sin 30° = 4.905$ N.

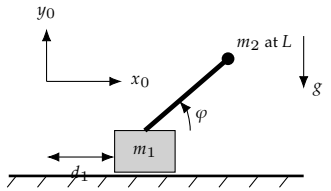
Energy check: $\dot{\mathbf{q}}^\top\mathbf{Q} = 1.3(8.976) - 0.8(6.806) = 6.224$ W $= \dot{K} + \dot{P}$. [checked]

TRAP

Taking $K_2 = \frac{1}{2}m_2\dot{d}_2^2$ (forgetting the sideways speed $d_2\dot{\theta}_1$); $\partial P/\partial d_2 = m_2g\cos\theta_1$ (the weight component along the arm is $\sin\theta_1$ for θ_1 from the horizontal); "M diagonal, so no Coriolis".

Prismatic joints: vertical slider and a PR arm [added]

A mass m on a vertical prismatic joint (z up): $K = \frac{1}{2}m\dot{z}^2$, $P = mgz$, so $f = m(\ddot{z} + g)$. Hold still: $f = mg$ ($m = 2$ kg: 19.62 N); accelerate up at 1.5 m/s^2: 22.62 N. [checked]



[drawn for this booklet] PR arm: horizontal slider d_1 (mass m_1) carrying a massless link of length L with tip mass m_2; φ from x_0.

DERIVATION PR arm: Lagrange

Tip: $x_2 = d_1 + Lc_\varphi$, $y_2 = Ls_\varphi$; $\dot{x}_2 = \dot{d}_1 - Ls_\varphi\dot{\varphi}$, $\dot{y}_2 = Lc_\varphi\dot{\varphi}$; $v_2^2 = \dot{d}_1^2 - 2Ls_\varphi\dot{d}_1\dot{\varphi} + L^2\dot{\varphi}^2$.

$$K = \tfrac{1}{2}(m_1+m_2)\dot{d}_1^2 - m_2Ls_\varphi\dot{d}_1\dot{\varphi} + \tfrac{1}{2}m_2L^2\dot{\varphi}^2, \quad P = m_2gLs_\varphi$$

d_1: $\dfrac{\partial\mathcal{L}}{\partial\dot{d}_1} = (m_1+m_2)\dot{d}_1 - m_2Ls_\varphi\dot{\varphi}$; $\dfrac{d}{dt}(\cdot) = (m_1+m_2)\ddot{d}_1 - m_2Ls_\varphi\ddot{\varphi} - m_2Lc_\varphi\dot{\varphi}^2$; $\dfrac{\partial\mathcal{L}}{\partial d_1} = 0$.

φ: $\dfrac{\partial\mathcal{L}}{\partial\dot{\varphi}} = m_2L^2\dot{\varphi} - m_2Ls_\varphi\dot{d}_1$; $\dfrac{d}{dt}(\cdot) = m_2L^2\ddot{\varphi} - m_2Ls_\varphi\ddot{d}_1 - m_2Lc_\varphi\dot{d}_1\dot{\varphi}$; $\dfrac{\partial\mathcal{L}}{\partial\varphi} = -m_2Lc_\varphi\dot{d}_1\dot{\varphi} - m_2gLc_\varphi$. The $\dot{d}_1\dot{\varphi}$ terms cancel.

Result: PR arm

$$\begin{bmatrix}F_1\\ \tau_2\end{bmatrix} = \underbrace{\begin{bmatrix}m_1+m_2 & -m_2Ls_\varphi\\ -m_2Ls_\varphi & m_2L^2\end{bmatrix}}_{\mathbf{M}}\begin{bmatrix}\ddot{d}_1\\ \ddot{\varphi}\end{bmatrix} + \underbrace{\begin{bmatrix}-m_2Lc_\varphi\dot{\varphi}^2\\ 0\end{bmatrix}}_{\mathbf{C}} + \underbrace{\begin{bmatrix}0\\ m_2gLc_\varphi\end{bmatrix}}_{\mathbf{g}}$$

Use: $\mathbf{M}$ *is not diagonal (acceleration of the cart pushes on the link) and depends on* φ *only. The only velocity term is centripetal; the cart feels no gravity because it moves horizontally.*

WORKED numbers

$m_1 = 3$, $m_2 = 1$ kg, $L = 0.5$ m, $\varphi = 40°$, $\dot{d}_1 = 1$, $\dot{\varphi} = 2$, $\ddot{d}_1 = 0.5$, $\ddot{\varphi} = 1$.

$F_1 = 4(0.5) - 0.5\sin 40°(1) - 0.5\cos 40°(4) = 2.000 - 0.321 - 1.532 = \boxed{0.147}$ N.

$\tau_2 = -0.5\sin 40°(0.5) + 0.25(1) + 9.81(0.5)\cos 40° = -0.161 + 0.250 + 3.757 = \boxed{3.847}$ N m.

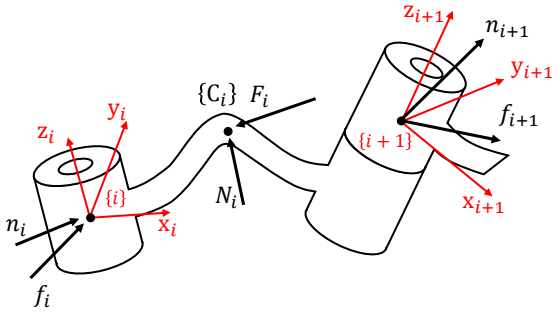
Hold still ($\varphi = 40°$): $F_1 = 0$, $\tau_2 = 3.757$ N m. Energy check $\dot{\mathbf{q}}^\top\mathbf{Q} = 7.840$ W $= \dot{K} + \dot{P}$. [checked]

Variant with the angle from the downward vertical (cart–pendulum, Practice Q7.1): $K = \tfrac{1}{2}(M+m)\dot{x}^2 + ml\dot{x}\dot{\varphi}\cos\varphi + \tfrac{1}{2}ml^2\dot{\varphi}^2$, $P = -mgl\cos\varphi$ give $F = (M+m)\ddot{x} + ml\ddot{\varphi}\cos\varphi - ml\dot{\varphi}^2\sin\varphi$, $\tau_\varphi = ml^2\ddot{\varphi} + ml\ddot{x}\cos\varphi + mgl\sin\varphi$; $M = 2, m = 1, l = 0.5, \varphi = 30°, \dot{\varphi} = 2, \ddot{\varphi} = 3, \ddot{x} = 1.5$: $F = 4.799$ N, $\tau_\varphi = 3.852$ N m. [checked]

Iterative Newton–Euler [p. 40–46]

Newton–Euler writes force and moment balance *link by link* [p. 14, 41]: an **outward** pass (base → tool) propagates $\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}, \dot{\boldsymbol{v}}$ and finds the force $\mathbf{F}_i$ and moment $\mathbf{N}_i$ each link needs [p. 43]; an **inward** pass (tool → base) balances the forces and moments between links to find the joint torques [p. 44–45].

PREMISE Newton–Euler (inverse dynamics) [p. 42] **Given:** $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$; the kinematics (frames $\{i\}$, origins ${}^{i-1}\mathbf{P}_i$, rotations ${}^{i-1}_{i}\mathbf{R}$); masses m_i, COM positions ${}^{i}\mathbf{P}_{C_i}$, tensors ${}^{C_i}\mathbf{I}_i$
Find: joint torques $\mathbf{Q}$



[W07 lec p. 45] — link i with its COM frame $\{C_i\}$: force $\mathbf{f}_i$ and moment $\mathbf{n}_i$ from link $i-1$ at the origin of $\{i\}$, $\mathbf{f}_{i+1}, \mathbf{n}_{i+1}$ from link $i+1$; $\mathbf{F}_i, \mathbf{N}_i$ are what the motion needs

Outward pass, $i = 0 \to n-1$, revolute joint $i{+}1$ (Craig form; joint axis $\hat{\mathbf{Z}} = (0,0,1)^\top$) [added]

With $\mathbf{R} = {}^{i}_{i+1}\mathbf{R}$ (rotation of $\{i{+}1\}$ in $\{i\}$) and $\mathbf{P} = {}^{i}\mathbf{P}_{i+1}$; all vectors on the left in $\{i{+}1\}$:

$$\begin{aligned}
{}^{i+1}\boldsymbol{\omega}_{i+1} &= \mathbf{R}^\top{}^{i}\boldsymbol{\omega}_i + \dot{q}_{i+1}\hat{\mathbf{Z}}\\
{}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} &= \mathbf{R}^\top{}^{i}\dot{\boldsymbol{\omega}}_i + \mathbf{R}^\top{}^{i}\boldsymbol{\omega}_i \times \dot{q}_{i+1}\hat{\mathbf{Z}} + \ddot{q}_{i+1}\hat{\mathbf{Z}}\\
{}^{i+1}\dot{\boldsymbol{v}}_{i+1} &= \mathbf{R}^\top\left({}^{i}\dot{\boldsymbol{\omega}}_i \times \mathbf{P} + {}^{i}\boldsymbol{\omega}_i \times ({}^{i}\boldsymbol{\omega}_i \times \mathbf{P}) + {}^{i}\dot{\boldsymbol{v}}_i\right)\\
\dot{\boldsymbol{v}}_C &= \dot{\boldsymbol{\omega}} \times \mathbf{P}_C + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{P}_C) + \dot{\boldsymbol{v}} \qquad \text{(all in the same frame } \{i{+}1\})\\
\mathbf{F}_{i+1} &= m_{i+1}\dot{\boldsymbol{v}}_C, \qquad \mathbf{N}_{i+1} = {}^{C}\mathbf{I}\,\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times {}^{C}\mathbf{I}\,\boldsymbol{\omega}
\end{aligned}$$

SYMBOLS

$\mathbf{P}_C$	COM of the link in its own frame	m
$\mathbf{R}^\top$	turns a vector from $\{i\}$ into $\{i{+}1\}$	–
$\dot{q}, \ddot{q}$	joint rate and acceleration	rad s^{-1}, rad s^{-2}

Use: start: ${}^{0}\boldsymbol{\omega}_0 = \dot{\boldsymbol{\omega}}_0 = \mathbf{0}$. *Gravity trick: set* ${}^{0}\dot{\boldsymbol{v}}_0 = (0, g, 0)^\top$ *(base accelerates up) instead of adding* $m\mathbf{g}$ *to each link.*

Prismatic joint $i{+}1$: the outward lines that change (full list) [added]

$$\begin{aligned}
{}^{i+1}\boldsymbol{\omega}_{i+1} &= \mathbf{R}^\top{}^{i}\boldsymbol{\omega}_i, \qquad {}^{i+1}\dot{\boldsymbol{\omega}}_{i+1} = \mathbf{R}^\top{}^{i}\dot{\boldsymbol{\omega}}_i\\
{}^{i+1}\dot{\boldsymbol{v}}_{i+1} &= \mathbf{R}^\top\left({}^{i}\dot{\boldsymbol{\omega}}_i \times \mathbf{P} + {}^{i}\boldsymbol{\omega}_i \times ({}^{i}\boldsymbol{\omega}_i \times \mathbf{P}) + {}^{i}\dot{\boldsymbol{v}}_i\right) + 2\,{}^{i+1}\boldsymbol{\omega}_{i+1} \times \dot{d}_{i+1}\hat{\mathbf{Z}} + \ddot{d}_{i+1}\hat{\mathbf{Z}}
\end{aligned}$$

Use: the extra term $2\boldsymbol{\omega} \times \dot{d}\hat{\mathbf{Z}}$ *is the Coriolis acceleration; the inward pass is unchanged except* $Q_i = \mathbf{f}_i \cdot \hat{\mathbf{Z}}$ *(a force).*

Inward pass, $i = n \to 1$, with $\mathbf{f}_{n+1} = \mathbf{n}_{n+1} = \mathbf{0}$ (free tool) [p. 44–45, Craig form]

$\mathbf{R}' = {}^{i}_{i+1}\mathbf{R}$; ${}^{i}\mathbf{P}_{i+1}$ origin of $\{i{+}1\}$ in $\{i\}$; all in $\{i\}$:

$$\begin{aligned}
\mathbf{f}_i &= \mathbf{R}'\mathbf{f}_{i+1} + \mathbf{F}_i\\
\mathbf{n}_i &= \mathbf{N}_i + \mathbf{R}'\mathbf{n}_{i+1} + \mathbf{P}_{C_i} \times \mathbf{F}_i + \mathbf{P}_{i+1} \times \mathbf{R}'\mathbf{f}_{i+1}\\
Q_i &= \mathbf{n}_i^\top\hat{\mathbf{Z}} \text{ (revolute)}, \qquad Q_i = \mathbf{f}_i^\top\hat{\mathbf{Z}} \text{ (prismatic)}
\end{aligned}$$

SYMBOLS

$\mathbf{f}_i, \mathbf{n}_i$	force, moment exerted on link i by link $i-1$, at the origin of $\{i\}$	N, N m
$\mathbf{F}_i, \mathbf{N}_i$	force, moment on the COM that the motion requires (outward pass)	N, N m
$\mathbf{P}_{C_i}$	COM in $\{i\}$	m

Use: the tool's contact wrench, if any, enters as $\mathbf{f}_{n+1}, \mathbf{n}_{n+1}$.

DERIVATION where the inward pass comes from: the slides' balances, then the form above [added]

Slide p. 45 (as printed): ${}^{i}\mathbf{F}_i = {}^{i}\mathbf{f}_i - {}^{i}\mathbf{T}_{i+1}{}^{i+1}\mathbf{f}_{i+1}$, ${}^{i}\mathbf{N}_i = {}^{i}\mathbf{n}_i - {}^{i}\mathbf{n}_{i+1} + (-{}^{i}\mathbf{P}_{C_i}) \times {}^{i}\mathbf{f}_i - ({}^{i}\mathbf{P}_{i+1} - {}^{i}\mathbf{P}_{C_i}) \times {}^{i}\mathbf{f}_{i+1}$. Replace the transform by the rotation $\mathbf{R}'$ (it is a rotation of vectors) and write ${}^{i}\mathbf{n}_{i+1} = \mathbf{R}'\mathbf{n}_{i+1}$, ${}^{i}\mathbf{f}_{i+1} = \mathbf{R}'\mathbf{f}_{i+1}$. Force: $\mathbf{f}_i = \mathbf{F}_i + \mathbf{R}'\mathbf{f}_{i+1}$. Moment: $\mathbf{n}_i = \mathbf{N}_i + \mathbf{R}'\mathbf{n}_{i+1} + \mathbf{P}_{C_i} \times \mathbf{f}_i + (\mathbf{P}_{i+1} - \mathbf{P}_{C_i}) \times \mathbf{R}'\mathbf{f}_{i+1}$. Insert $\mathbf{f}_i = \mathbf{F}_i + \mathbf{R}'\mathbf{f}_{i+1}$ into $\mathbf{P}_{C_i} \times \mathbf{f}_i$: the $\mathbf{P}_{C_i} \times \mathbf{R}'\mathbf{f}_{i+1}$ term cancels against $-\mathbf{P}_{C_i} \times \mathbf{R}'\mathbf{f}_{i+1}$, leaving the second line of the box.

SLIDE CHECK Newton–Euler, p. 45

The slide writes ${}^{i}\mathbf{T}_{i+1}$ in the force balance, where a rotation ${}^{i}_{i+1}\mathbf{R}$ is meant (a force is a vector), and in the moment balance ${}^{i}\mathbf{n}_{i+1}$ is not rotated into $\{i\}$ explicitly. Use the rotated form in the box. Also the slide's outward equations are given only in words (p. 43); the equations above are Craig's, as coded in the unit's practical script.

STEPS Newton–Euler by hand (small robots)

1. Frames $\{0\},\{1\},\ldots$ with joint axes along z; list ${}^{i}\mathbf{P}_{i+1}$, ${}^{i}_{i+1}\mathbf{R}$, ${}^{i}\mathbf{P}_{C_i}$, m_i, $\mathbf{I}_i$.
2. Initial values ${}^{0}\boldsymbol{\omega}_0 = {}^{0}\dot{\boldsymbol{\omega}}_0 = \mathbf{0}$, ${}^{0}\dot{\boldsymbol{v}}_0 = (0, g, 0)^{\top}$.
3. Outward, $i = 1 \ldots n$: $\boldsymbol{\omega}_i, \dot{\boldsymbol{\omega}}_i, \dot{\boldsymbol{v}}_i, \dot{\boldsymbol{v}}_{C_i}, \mathbf{F}_i, \mathbf{N}_i$ (point masses: $\mathbf{N}_i = \mathbf{0}$).
4. Inward, $i = n \ldots 1$: $\mathbf{f}_i, \mathbf{n}_i$.
5. $Q_i = \mathbf{n}_i \cdot \hat{\mathbf{Z}}$ (or $\mathbf{f}_i \cdot \hat{\mathbf{Z}}$ if prismatic).
6. Check against the static case ($\dot{\mathbf{q}} = \ddot{\mathbf{q}} = \mathbf{0}$): only $\mathbf{g}$ remains.

DERIVATION slide homework p. 46: Newton–Euler on the 2R arm with point masses (symbolic)

Frames $\{1\},\{2\}$ with z out of the page; ${}^{0}_{1}\mathbf{R} = R_z(q_1)$, ${}^{1}_{2}\mathbf{R} = R_z(q_2)$, ${}^{0}\mathbf{P}_1 = \mathbf{0}$, ${}^{1}\mathbf{P}_2 = (L_1,0,0)^{\top}$, ${}^{1}\mathbf{P}_{C_1} = (L_1,0,0)^{\top}$, ${}^{2}\mathbf{P}_{C_2} = (L_2,0,0)^{\top}$, $\mathbf{N}_i = \mathbf{0}$. Let $a = -L_1\dot{q}_1^2 + g s_1$, $b = L_1\ddot{q}_1 + g c_1$.

Outward. ${}^{1}\boldsymbol{\omega}_1 = (0,0,\dot{q}_1)$, ${}^{1}\dot{\boldsymbol{\omega}}_1 = (0,0,\ddot{q}_1)$, ${}^{1}\dot{\boldsymbol{v}}_1 = R_z(q_1)^{\top}(0,g,0) = (g s_1, g c_1, 0)$.
${}^{1}\dot{\boldsymbol{v}}_{C_1} = (0, L_1\ddot{q}_1, 0) + (-L_1\dot{q}_1^2, 0, 0) + (g s_1, g c_1, 0) = (a,\ b,\ 0)$; $\mathbf{F}_1 = m_1(a,b,0)$.
Origin of $\{2\}$: ${}^{1}(\dot{\boldsymbol{\omega}}_1 \times \mathbf{P}_2 + \boldsymbol{\omega}_1 \times (\boldsymbol{\omega}_1 \times \mathbf{P}_2) + \dot{\boldsymbol{v}}_1) = (a,b,0)$ (equal to $\dot{\boldsymbol{v}}_{C_1}$ because $\mathbf{P}_{C_1} = \mathbf{P}_2$), so ${}^{2}\dot{\boldsymbol{v}}_2 = R_z(q_2)^{\top}(a,b,0) = (c_2 a + s_2 b,\ -s_2 a + c_2 b,\ 0)$. ${}^{2}\boldsymbol{\omega}_2 = (0,0,\dot{q}_1+\dot{q}_2)$, ${}^{2}\dot{\boldsymbol{\omega}}_2 = (0,0,\ddot{q}_1+\ddot{q}_2)$.
${}^{2}\dot{\boldsymbol{v}}_{C_2} = \big(-L_2(\dot{q}_1+\dot{q}_2)^2,\ L_2(\ddot{q}_1+\ddot{q}_2), 0\big) + {}^{2}\dot{\boldsymbol{v}}_2$; $\mathbf{F}_2 = m_2\dot{\boldsymbol{v}}_{C_2}$.

Inward. $\mathbf{f}_2 = \mathbf{F}_2$; $\mathbf{n}_2 = \mathbf{P}_{C_2} \times \mathbf{F}_2 = (0,0,L_2F_{2y})$, so

$$Q_2 = L_2F_{2y} = m_2L_2\big[L_2(\ddot{q}_1+\ddot{q}_2) - s_2a + c_2b\big], \quad -s_2a + c_2b = L_1s_2\dot{q}_1^2 + L_1c_2\ddot{q}_1 + gc_{12}$$

$$Q_2 = m_2L_2^2(\ddot{q}_1+\ddot{q}_2) + m_2L_1L_2c_2\ddot{q}_1 + m_2L_1L_2s_2\dot{q}_1^2 + m_2gL_2c_{12} \quad \checkmark \text{ same as Lagrange}$$

Link 1: $\mathbf{f}_1 = R_z(q_2)\mathbf{f}_2 + \mathbf{F}_1$; $Q_1 = n_{1z} = Q_2 + L_1F_{1y} + L_1(s_2F_{2x} + c_2F_{2y})$ (the z component of $\mathbf{P}_{C_1}\times\mathbf{F}_1 + \mathbf{P}_2\times\mathbf{R}'\mathbf{f}_2$). With $F_{1y} = m_1 b$ and $s_2F_{2x} + c_2F_{2y} = m_2[b - L_2s_2(\dot{q}_1+\dot{q}_2)^2 + L_2c_2(\ddot{q}_1+\ddot{q}_2)]$, collecting $\ddot{q}_1, \ddot{q}_2$, velocity and g terms reproduces the Lagrange Q_1 exactly (symbolic check, [checked]).

WORKED Newton–Euler numbers, same state as the Lagrange example

$g = 9.81$, $q = (30°, 45°)$, $\dot{q} = (1, -0.5)$, $\ddot{q} = (0.5, 2)$, $m = (2,1)$, $L = (1, 0.8)$.
Link 1: $\boldsymbol{\omega}_1 = (0,0,1)$, $\dot{\boldsymbol{\omega}}_1 = (0,0,0.5)$, $\dot{\boldsymbol{v}}_1 = (4.905, 8.496, 0)$, $\dot{\boldsymbol{v}}_{C_1} = (3.905, 8.996, 0)$, $\mathbf{F}_1 = (7.810, 17.991, 0)$ N.
Link 2: $\boldsymbol{\omega}_2 = (0,0,0.5)$, $\dot{\boldsymbol{\omega}}_2 = (0,0,2.5)$, $\dot{\boldsymbol{v}}_2 = (9.122, 3.600, 0)$, $\dot{\boldsymbol{v}}_{C_2} = (8.922, 5.600, 0)$, $\mathbf{F}_2 = \mathbf{f}_2 = (8.922, 5.600, 0)$ N; $\mathbf{n}_2 = (0,0,0.8(5.600)) = (0,0,4.480)$, so $Q_2 = 4.480$.
$\mathbf{f}_1 = (10.159, 28.260, 0)$ N, $\mathbf{n}_1 = (0,0,32.740)$, so $Q_1 = 32.740$ N m. Identical to the Lagrange result. [checked]

Lagrange versus Newton–Euler [p. 12–14, 47]

ALTERNATIVE APPROACHES

A. Lagrange (energy)	B. Newton–Euler (forces)
Whole robot at once; internal reaction forces never appear [p. 14].	Link by link: outward pass (motion), inward pass (forces) [p. 41].
Needs: positions of the COMs, K, P and derivatives.	Needs: frames, rotations, vectors $\mathbf{P}_i, \mathbf{P}_{C_i}$, $\mathbf{I}_i$; numbers in, numbers out.
Gives *closed-form* $\mathbf{M}, \mathbf{C}, \mathbf{g}$.	Iterative, numerical; gives the reaction forces for free.
Best for analysis and control design (structure, symmetry, energy).	Best for real-time inverse dynamics, model-based control.
Cost $32n^4 + 86n^3 + 171n^2 + 53n - 128$ mult. ($O(n^4)$).	Cost $104n - 77$ mult. ($O(n)$).

Computational cost [p. 47]

	NE mult. $104n - 77$	Lagrange mult. $32n^4 + 86n^3 + 171n^2 + 53n - 128$	$n = 6$ (slide)
$n = 2$	131	1 862	–
$n = 3$	235	6 484	–
$n = 6$	547	66 394	additions 482 vs 51 456

Use: both give the same torques; the difference is the route and the cost. NE additions: $92n - 70$; Lagrange additions: $25n^4 + 66n^3 + 129n^2 + 42n - 96$. Source on the slide: Handbook of Industrial Robotics.

Forward dynamics and task space [p. 32, 48]

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1}(\mathbf{q})\big(\mathbf{Q} - \mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) - \mathbf{g}(\mathbf{q}) - \mathbf{f}(\dot{\mathbf{q}}) - \mathbf{J}^{\top}(\mathbf{q})\mathbf{w}\big), \qquad \mathcal{F} = \mathbf{M}_x\ddot{\mathbf{x}} + \mathbf{C}_x + \mathbf{g}_x, \quad \mathbf{Q} = \mathbf{J}^{\top}(\mathbf{q})\mathcal{F}$$

Use: first: simulate (integrate $\ddot{\mathbf{q}}$). Second: the same dynamics in Cartesian coordinates $\mathbf{x}$; $\mathcal{F}$ is a tip force/moment and $\mathbf{J}^{\top}$ turns it into joint torques (W6, W9).

How this is examined: worked solutions

Expect one derivation question (W13 Q4: "derive the equations of motion"; marks for COM positions, velocities, K, P, $\mathcal{L}$, one Euler–Lagrange line per joint, $\mathbf{M}, \mathbf{C}, \mathbf{g}$ identified) and short calculations (torque to hold, torque for a given motion, a tensor in a shifted frame, an acceleration from Newton–Euler) or explanations. Marks are not published; the grey bracketed tags below are indicative, about 1 mark per step. If time runs out: give $\mathbf{M}$ and $\mathbf{g}$ correctly and set up $\mathbf{C}$.

EXAM-STYLE QUESTION Torque to hold, to accelerate, and the resulting motion (2R, point masses) [indicative 6]

A planar 2R arm, point masses $m_1 = 1.5$ kg (tip of link 1, $L_1 = 0.9$ m) and $m_2 = 0.8$ kg (tip of link 2, $L_2 = 0.6$ m); q_1 from the horizontal, q_2 relative to link 1; gravity along $-y$. (a) Torque to hold the arm still at $q = (60°, -30°)$. (b) At $q = (60°, -30°)$, $\dot{q} = (0.8, 1.2)$, $\ddot{q} = (-1.0, 2.5)$ find Q_1, Q_2.

Solution

Use $\mathbf{M}, \mathbf{C}, \mathbf{g}$ from p. 95's box. $c_2 = 0.8660$, $s_2 = -0.5$, $q_1 + q_2 = 30°$, $h = m_2 L_1 L_2 s_2 = 0.8(0.9)(0.6)(-0.5) = -0.216$.

(a) $\mathbf{Q} = \mathbf{g}$: $g_1 = 2.3(9.81)(0.9)\cos 60° + 0.8(9.81)(0.6)\cos 30° = 10.153 + 4.078 = 14.231$; $g_2 = 4.078$. $\boxed{\mathbf{Q} = (14.231,\ 4.078)\ \text{N m}}$.

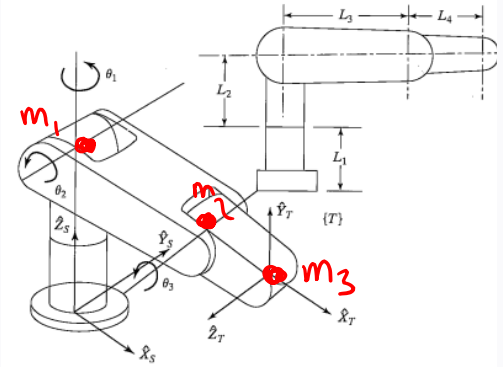
(b) $M_{11} = 1.5(0.81) + 0.8(0.81 + 0.9353 + 0.36) = 2.8992$; $M_{12} = 0.8(0.4677 + 0.36) = 0.6621$; $M_{22} = 0.288$. $\mathbf{M}\ddot{\mathbf{q}} = (-2.8992 + 1.6553,\ -0.6621 + 0.72) = (-1.2439,\ 0.0579)$.

$C_1 = -h(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) = 0.216(1.92 + 1.44) = 0.7258$; $C_2 = h\dot{q}_1^2 = -0.216(0.64) = -0.1382$.

$Q_1 = -1.2439 + 0.7258 + 14.231 = \boxed{13.713}$; $Q_2 = 0.0579 - 0.1382 + 4.078 = \boxed{3.998}$ N m. [checked]

Sanity: dropping $\mathbf{C}$ changes the answer by only a few per cent (12.99, 4.14), which is easy to miss; the moving joints make $\mathbf{C} \neq \mathbf{0}$.

EXAM-STYLE QUESTION W13 Q4 (a): derive the equations of motion, 3D RRR arm [marks not published; indicative, 1 per step]



[W13 lec p. 8] — W13 p. 8: RRR arm; the lecturer's red dots mark m_1 (shoulder), m_2 (elbow), m_3 (tool). θ_1 about z_S, θ_2, θ_3 in the vertical arm plane

"Mass, moments of inertia and positions of centre of mass are given. Derive the equations of motion."

Solution

Set-up (state it). Frame $\{S\}$: z_S up, gravity along $-z_S$. θ_1 turns the whole arm about z_S. The arm plane is vertical, at azimuth θ_1; shoulder at height $H = L_1 + L_2$ on the axis. θ_2: elevation of the upper arm (length L_3) from the horizontal; θ_3: forearm (length L_4) relative to the upper arm, so the forearm's absolute elevation is $\theta_2 + \theta_3$. (If the exam measures angles from the vertical, swap $c \leftrightarrow s$ in heights.) Point masses first (the red dots): m_2 at the elbow, m_3 at the tool; m_1 sits on the axis, so it only adds a constant to P and nothing to the equations. Use $\rho =$ horizontal distance from z_S and $h =$ height.

Positions. $\rho_2 = L_3 c_2$, $h_2 = H + L_3 s_2$; $\rho_3 = L_3 c_2 + L_4 c_{23}$, $h_3 = H + L_3 s_2 + L_4 s_{23}$.

Velocities. A point at (ρ, h) on a plane spinning at $\dot{\theta}_1$: $v^2 = \dot{\rho}^2 + \dot{h}^2 + \rho^2\dot{\theta}_1^2$. $\dot{\rho}_2^2 + \dot{h}_2^2 = L_3^2\dot{\theta}_2^2$. For mass 3: $(\dot{\rho}_3, \dot{h}_3) = L_3\dot{\theta}_2(-s_2, c_2) + L_4(\dot{\theta}_2 + \dot{\theta}_3)(-s_{23}, c_{23})$, so $\dot{\rho}_3^2 + \dot{h}_3^2 = L_3^2\dot{\theta}_2^2 + L_4^2(\dot{\theta}_2 + \dot{\theta}_3)^2 + 2L_3L_4c_3\,\dot{\theta}_2(\dot{\theta}_2 + \dot{\theta}_3)$ (using $s_2s_{23} + c_2c_{23} = c_3$).

Energies.

$$K = \tfrac{1}{2}(m_2\rho_2^2 + m_3\rho_3^2)\dot{\theta}_1^2 + \tfrac{1}{2}\big[(m_2 + m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3\big]\dot{\theta}_2^2 + m_3(L_4^2 + L_3L_4c_3)\dot{\theta}_2\dot{\theta}_3 + \tfrac{1}{2}m_3L_4^2\dot{\theta}_3^2$$

$$P = m_2g(H + L_3s_2) + m_3g(H + L_3s_2 + L_4s_{23}) + \text{const}, \qquad \mathcal{L} = K - P$$

Reading $\mathbf{M}$ off $K = \tfrac{1}{2}\dot{\boldsymbol{\theta}}^\top\mathbf{M}\dot{\boldsymbol{\theta}}$: $M_{11} = m_2\rho_2^2 + m_3\rho_3^2$; $M_{22} = (m_2 + m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3$; $M_{23} = m_3(L_4^2 + L_3L_4c_3)$; $M_{33} = m_3L_4^2$; $M_{12} = M_{13} = 0$.

EXAM-STYLE QUESTION W13 Q4 (b): Euler–Lagrange lines [cont.]

Joint 1. $\partial\mathcal{L}/\partial\dot\theta_1 = M_{11}\dot\theta_1$, $\frac{d}{dt} = M_{11}\ddot\theta_1 + \dot M_{11}\dot\theta_1$; $\partial\mathcal{L}/\partial\theta_1 = 0$ (nothing depends on θ_1). With $\dot\rho_2 = -L_3 s_2\dot\theta_2$, $\dot\rho_3 = -L_3 s_2\dot\theta_2 - L_4 s_{23}(\dot\theta_2+\dot\theta_3)$, $\dot M_{11} = 2m_2\rho_2\dot\rho_2 + 2m_3\rho_3\dot\rho_3$:

$$Q_1 = M_{11}\ddot\theta_1 - 2\dot\theta_1\Big[m_2\rho_2 L_3 s_2\dot\theta_2 + m_3\rho_3\big(L_3 s_2\dot\theta_2 + L_4 s_{23}(\dot\theta_2+\dot\theta_3)\big)\Big]$$

Joint 2. $\partial\mathcal{L}/\partial\dot\theta_2 = M_{22}\dot\theta_2 + M_{23}\dot\theta_3$; $\frac{d}{dt} = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 - m_3L_3L_4s_3\dot\theta_3(2\dot\theta_2+\dot\theta_3)$ (using $\dot M_{22} = -2m_3L_3L_4s_3\dot\theta_3$, $\dot M_{23} = -m_3L_3L_4s_3\dot\theta_3$). $\partial\mathcal{L}/\partial\theta_2 = \frac12\dot\theta_1^2\,\partial M_{11}/\partial\theta_2 - g\big[(m_2+m_3)L_3c_2 + m_3L_4c_{23}\big]$ with $\frac12\partial M_{11}/\partial\theta_2 = -\big[m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2+L_4s_{23})\big]$.

$$\begin{aligned}Q_2 = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 + \dot\theta_1^2\big[&m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2+L_4s_{23})\big]\\ &- m_3L_3L_4s_3(2\dot\theta_2\dot\theta_3+\dot\theta_3^2) + g\big[(m_2+m_3)L_3c_2 + m_3L_4c_{23}\big]\end{aligned}$$

Joint 3. $\partial\mathcal{L}/\partial\dot\theta_3 = M_{23}\dot\theta_2 + M_{33}\dot\theta_3$; $\frac{d}{dt} = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 - m_3L_3L_4s_3\dot\theta_3\dot\theta_2$. $\partial K/\partial\theta_3 = -m_3\rho_3L_4s_{23}\dot\theta_1^2 - m_3L_3L_4s_3(\dot\theta_2^2+\dot\theta_2\dot\theta_3)$; $\partial P/\partial\theta_3 = m_3gL_4c_{23}$. The $\dot\theta_2\dot\theta_3$ terms cancel:

$$Q_3 = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 + m_3\rho_3L_4s_{23}\dot\theta_1^2 + m_3L_3L_4s_3\dot\theta_2^2 + g\,m_3L_4c_{23}$$

EXAM-STYLE QUESTION W13 Q4 (c): M, C, g, general links, numbers [cont.]

$$\mathbf{M} = \begin{bmatrix}M_{11}&0&0\\0&M_{22}&M_{23}\\0&M_{23}&M_{33}\end{bmatrix},\quad \mathbf{g} = \begin{bmatrix}0\\g[(m_2+m_3)L_3c_2 + m_3L_4c_{23}]\\g\,m_3L_4c_{23}\end{bmatrix}$$

$$\mathbf{C} = (V_1,V_2,V_3)^\top = \text{the velocity terms of } Q_1,Q_2,Q_3$$

Centrifugal: the $\dot\theta_1^2$ terms in V_2, V_3 and $m_3L_3L_4s_3\dot\theta_2^2$ (squares). Coriolis: $\dot\theta_1\dot\theta_2$, $\dot\theta_1\dot\theta_3$ in V_1; $\dot\theta_2\dot\theta_3$ in V_2 (products). Physical reading: spinning about z_S flings the arm outwards; raising the arm while spinning changes the inertia about z_S, so joint 1 must push; gravity loads joints 2 and 3 only, most when the arm is horizontal.

With link inertias (slender links, COM at l_2 from the shoulder and l_3 from the elbow, I_2, I_3 about axes $\perp$ the link through the COM, ≈ 0 about the link axis, I_1 the column about z_S): replace $L_3 \to l_2$ in link 2's terms and $L_4 \to l_3$ in link 3's (keep L_3 where the elbow position enters ρ_3), $\rho_2 = l_2c_2$, $\rho_3 = L_3c_2 + l_3c_{23}$, and

$$M_{11} = I_1 + m_2\rho_2^2 + m_3\rho_3^2 + I_2c_2^2 + I_3c_{23}^2,\quad M_{22} = m_2l_2^2 + I_2 + m_3(L_3^2 + l_3^2 + 2L_3l_3c_3) + I_3$$

$$M_{23} = m_3(l_3^2 + L_3l_3c_3) + I_3,\quad M_{33} = m_3l_3^2 + I_3,\quad \mathbf{g} = \begin{bmatrix}0\\g[(m_2l_2+m_3L_3)c_2 + m_3l_3c_{23}]\\g\,m_3l_3c_{23}\end{bmatrix}$$

$$V_1 = -2\dot\theta_1\big[m_2\rho_2l_2s_2\dot\theta_2 + m_3\rho_3(L_3s_2\dot\theta_2 + l_3s_{23}(\dot\theta_2+\dot\theta_3)) + I_2c_2s_2\dot\theta_2 + I_3c_{23}s_{23}(\dot\theta_2+\dot\theta_3)\big]$$

$$V_2 = \dot\theta_1^2\big[m_2\rho_2l_2s_2 + m_3\rho_3(L_3s_2 + l_3s_{23}) + I_2c_2s_2 + I_3c_{23}s_{23}\big] - m_3L_3l_3s_3(2\dot\theta_2\dot\theta_3 + \dot\theta_3^2)$$

$$V_3 = \dot\theta_1^2\big[m_3\rho_3l_3s_{23} + I_3c_{23}s_{23}\big] + m_3L_3l_3s_3\dot\theta_2^2$$

(The red-dot case is $l_2 = L_3$, $l_3 = L_4$, $I_2 = I_3 = 0$.) Rotational energy added: $\frac12I_1\dot\theta_1^2 + \frac12I_2(\dot\theta_2^2 + c_2^2\dot\theta_1^2) + \frac12I_3((\dot\theta_2+\dot\theta_3)^2 + c_{23}^2\dot\theta_1^2)$.

Numbers (point masses): $m_2 = 2$, $m_3 = 1$ kg, $L_3 = 0.6$, $L_4 = 0.4$ m, $\boldsymbol\theta = (20°, 30°, -40°)$, $\dot{\boldsymbol\theta} = (1, 0.5, -0.5)$ rad/s, $\ddot{\boldsymbol\theta} = (0.2, 1, 0.5)$ rad/s^2. $\rho_2 = 0.5196$, $\rho_3 = 0.9135$ m, $c_3 = 0.7660$, $s_3 = -0.6428$, $c_{23} = 0.9848$, $s_{23} = -0.1736$.

$$\mathbf{M} = \begin{bmatrix}1.3746&0&0\\0&1.6077&0.3439\\0&0.3439&0.1600\end{bmatrix},\ \mathbf{C} = \begin{bmatrix}-0.5858\\0.4838\\-0.1020\end{bmatrix},\ \mathbf{g} = \begin{bmatrix}0\\19.157\\3.864\end{bmatrix}$$

$Q_1 = 1.3746(0.2) - 0.5858 = \boxed{-0.311}$; $Q_2 = 1.6077(1) + 0.3439(0.5) + 0.4838 + 19.157 = \boxed{21.420}$; $Q_3 = 0.3439(1) + 0.16(0.5) - 0.1020 + 3.864 = \boxed{4.186}$ N m. Hold still: $\mathbf{Q} = \mathbf{g} = (0, 19.157, 3.864)$; arm stretched out horizontally: $(0, 21.582, 3.924)$ N m (largest). [checked]

Checks. $\mathbf{M}$ symmetric, positive definite; $\mathbf{g}$ contains no $\dot\theta$; $\mathbf{C} = \mathbf{0}$ at rest; power $\dot{\boldsymbol\theta}^\top\mathbf{Q} = 8.306$ W $= \dot K + \dot P$; free swing conserves $K + P$ to 10^{-9} J over 3 s. [checked]

EXAM-STYLE QUESTION Uniform links: M and g of the 2R arm [indicative 8]

Planar 2R arm in a vertical plane (q_1 from the horizontal, q_2 relative, gravity $-y$) with uniform rods $m_1 = 3$ kg, $\ell_1 = 0.6$ m; $m_2 = 2$ kg, $\ell_2 = 0.4$ m (COM at mid-length, $I_c = m\ell^2/12$). (a) Write $\mathbf{M}(\mathbf{q})$ and $\mathbf{g}(\mathbf{q})$. (b) Momentarily at rest at $q = (45°, -30°)$ with $\ddot{\mathbf{q}} = (1, 2)$ rad/s^2: find the torques.

Solution

Link 1 turns about its end: $K_1 = \frac{1}{2}(I_1 + m_1\frac{\ell_1^2}{4})\dot{q}_1^2$. Link 2: COM at $\mathbf{p} = (\ell_1 c_1 + \frac{\ell_2}{2}c_{12},\ \ell_1 s_1 + \frac{\ell_2}{2}s_{12})$, $v_{C_2}^2 = \ell_1^2\dot{q}_1^2 + \frac{\ell_2^2}{4}(\dot{q}_1+\dot{q}_2)^2 + \ell_1\ell_2 c_2\dot{q}_1(\dot{q}_1+\dot{q}_2)$, spin $\dot{q}_1+\dot{q}_2$: $K_2 = \frac{1}{2}m_2 v_{C_2}^2 + \frac{1}{2}I_2(\dot{q}_1+\dot{q}_2)^2$. Reading $\mathbf{M}$ off K:

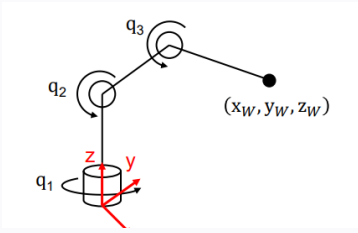
$$M_{11} = I_1 + \tfrac{m_1\ell_1^2}{4} + I_2 + m_2\big(\ell_1^2 + \tfrac{\ell_2^2}{4} + \ell_1\ell_2 c_2\big),\ M_{12} = I_2 + m_2\big(\tfrac{\ell_2^2}{4} + \tfrac{\ell_1\ell_2}{2}c_2\big),\ M_{22} = I_2 + \tfrac{m_2\ell_2^2}{4}$$

$$g_1 = \big(\tfrac{m_1\ell_1}{2} + m_2\ell_1\big)g c_1 + \tfrac{m_2\ell_2}{2}g c_{12}, \qquad g_2 = \tfrac{m_2\ell_2}{2}g\,c_{12}$$

(b) $I_1 = 0.0900$, $I_2 = 0.0267$; $c_2 = 0.8660$. $\mathbf{M} = \begin{bmatrix} 1.6024 & 0.3145 \\ 0.3145 & 0.1067 \end{bmatrix}$ kg m^2, $\mathbf{g} = (18.357,\ 3.790)$ N m. At rest $\mathbf{C} = \mathbf{0}$: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g} = (1.6024 + 0.6290 + 18.357,\ 0.3145 + 0.2133 + 3.790) =$ $\boxed{(20.589,\ 4.318)\ \text{N m}}$. [checked]

Check: $M_{22} = \frac{m_2\ell_2^2}{3} = 0.1067$ (rod about its end), as the formula gives.

EXAM-STYLE QUESTION W8 practical Q3: 3-link planar arm, torque along a cubic [indicative 8]



[W08 prac p. 1] — W8 practical Q3: planar 3R, $\theta_1, \theta_2, \theta_3$ relative

$L = (3, 2, 1)$ m. The arm starts at rest at tool pose $(5, 1, 90°)$ and must come to rest at $(-3, 3, 180°)$ in $T = 2$ s using cubics. Masses $3, 2, 1$ kg concentrated at the distal ends of the links (vertical plane, gravity $-y$, assumed). Find the joint angles, the cubics and the torques at $t = 0$.

Solution

Joint angles. Wrist point $= (x - L_3\cos\varphi,\ y - L_3\sin\varphi)$, $\cos\theta_2 = (w^2 - L_1^2 - L_2^2)/(2L_1L_2)$. Start: wrist $(5, 0)$, $\cos\theta_2 = (25 - 13)/12 = 1$, so $q_0 = (0°, 0°, 90°)$ (stretched, one solution). Goal: wrist $(-2, 3)$, $w^2 = 13$, $\cos\theta_2 = 0$. Branch $\theta_2 = +90°$: $\theta_1 = \text{atan2}(3,-2) - \text{atan2}(2,3) = 123.69° - 33.69° = 90°$, $\theta_3 = 180° - 90° - 90° = 0$: $q_f = (90°, 90°, 0°)$. (Other branch $(157.38°, -90°, 112.62°)$.) Choose the first: smaller joint travel.

Cubic (rest to rest): $\Delta = (90, 90, -90)°$, $a_2 = 3\Delta/T^2 = (67.5, 67.5, -67.5)°/\text{s}^2$, $a_3 = -2\Delta/T^3 = (-22.5, -22.5, 22.5)°/\text{s}^3$; $\ddot{q}(0) = 2a_2 = (135, 135, -135)°/\text{s}^2 = (2.356, 2.356, -2.356)$ rad/s^2.

Torque at $t = 0$ ($\dot{q} = 0$, so $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}$). Tips at $(3, 0)$, $(5, 0)$, $(5, 1)$, joint origins $(0, 0)$, $(3, 0)$, $(5, 0)$. Column j of a tip Jacobian is $\hat{z} \times (\mathbf{p}_{tip} - \mathbf{o}_j) = (-r_y, r_x)$ [added]:

tip	$\partial\mathbf{p}/\partial q_1$	$\partial\mathbf{p}/\partial q_2$	$\partial\mathbf{p}/\partial q_3$
1	$(0, 3)$	–	–
2	$(0, 5)$	$(0, 2)$	–
3	$(-1, 5)$	$(-1, 2)$	$(-1, 0)$

$\mathbf{M} = \sum m_i \mathbf{J}_i^{\top}\mathbf{J}_i$: $M_{11} = 3(9) + 2(25) + 1(1 + 25) = 103$, $M_{12} = 2(5)(2) + 1(1 + 10) = 31$, $M_{13} = 1$, $M_{22} = 2(4) + 1(1 + 4) = 13$, $M_{23} = 1$, $M_{33} = 1$. $\mathbf{g}_j = g\sum m_i(\mathbf{p}_i - \mathbf{o}_j)_x$: $g = (9.81(9 + 10 + 5),\ 9.81(4 + 2),\ 0) = (235.44,\ 58.86,\ 0)$.

$$\mathbf{Q}(0) = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2.356 \\ 2.356 \\ -2.356 \end{bmatrix} + \begin{bmatrix} 235.44 \\ 58.86 \\ 0 \end{bmatrix} = \begin{bmatrix} 313.37 \\ 101.32 \\ 2.356 \end{bmatrix} + \begin{bmatrix} 235.44 \\ 58.86 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 548.81 \\ 160.18 \\ 2.356 \end{bmatrix}}\ \text{N m}$$

Other instants (Lagrange, point masses): $t = 1$ s, $q = (45°, 45°, 45°)$, $\ddot{q} = 0$, $\dot{q} = (1.178, 1.178, -1.178)$ rad/s: $\mathbf{Q} = (70.82, 20.78, 5.08)$; $t = 2$ s, $q = (90°, 90°, 0)$, $\ddot{q} = -\ddot{q}(0)$: $\mathbf{Q} = (-268.95, -141.71, -21.59)$ N m. [checked]

EXAM-STYLE QUESTION RP arm, point mass on a massless arm [indicative 8]

An RP arm moves in a vertical plane. Joint 1 turns the arm by θ from the horizontal; joint 2 slides a point mass m along it to distance d. Arm massless, gravity $-y$. (a) Derive the equations of motion and identify $\mathbf{M}, \mathbf{C}, \mathbf{g}$. (b) $m = 2$ kg, $\theta = 30°$, $d = 0.6$ m, $\dot\theta = 0.5$, $\dot d = 0.2$, $\ddot\theta = 1.0$, $\ddot d = -0.3$: find τ_1, f_2.

Solution

(a) Position (dc_θ, ds_θ); $v^2 = \dot d^2 + d^2\dot\theta^2$. $K = \frac{1}{2}m(\dot d^2 + d^2\dot\theta^2)$, $P = mgds_\theta$. θ: $\frac{d}{dt}(md^2\dot\theta) = md^2\ddot\theta + 2md\dot d\dot\theta$, $\partial\mathcal{L}/\partial\theta = -mgdc_\theta$. d: $\frac{d}{dt}(m\dot d) = m\ddot d$, $\partial\mathcal{L}/\partial d = md\dot\theta^2 - mgs_\theta$.

$$\tau_1 = md^2\ddot\theta + 2md\dot d\dot\theta + mgd\cos\theta, \qquad f_2 = m\ddot d - md\dot\theta^2 + mg\sin\theta$$

$\mathbf{M} = \mathrm{diag}(md^2, m)$; $\mathbf{C} = (2md\dot d\dot\theta$ [Coriolis], $-md\dot\theta^2$ [centrifugal]); $\mathbf{g} = (mgd\cos\theta,\ mg\sin\theta)$.
(b) $\tau_1 = 2(0.36)(1) + 2(2)(0.6)(0.2)(0.5) + 2(9.81)(0.6)\cos 30° = 0.72 + 0.24 + 10.195 = \boxed{11.155}$ N m; $f_2 = 2(-0.3) - 2(0.6)(0.25) + 2(9.81)(0.5) = -0.6 - 0.3 + 9.81 = \boxed{8.910}$ N. [checked]

EXAM-STYLE QUESTION Spot the errors: a student's RP arm [indicative 5]

Revolute θ (from the horizontal), then prismatic d along the arm, point mass $m_2 = 1.5$ kg at d, gravity $-y$. The student writes: "$K = \frac{1}{2}m_2\dot d^2$; $P = m_2gd\sin\theta$; $f_2 = m_2\ddot d + m_2g\cos\theta$; $\tau_1 = m_2gd\cos\theta$; $\mathbf{M}$ is diagonal, so there is no Coriolis term." Find every error; give the correct f_2, τ_1 at $d = 0.4$, $\dot d = 0.3$, $\ddot d = 0.5$, $\theta = 0.9$ rad, $\dot\theta = 1.1$, $\ddot\theta = 0.6$.

Solution

(1) K misses the sideways speed: $K = \frac{1}{2}m_2(\dot d^2 + d^2\dot\theta^2)$. (2) $\partial P/\partial d = m_2g\sin\theta$, not cos. (3) Because of (1), τ_1 also has $m_2d^2\ddot\theta$ and $2m_2d\dot d\dot\theta$. (4) A diagonal $\mathbf{M}$ only removes *acceleration* coupling; the velocity products remain.
$f_2 = 1.5(0.5 - 0.4(1.21)) + 1.5(9.81)\sin 0.9 = 0.024 + 11.527 = \boxed{11.551}$ N; $\tau_1 = 1.5(0.16)(0.6) + 2(1.5)(0.4)(0.3)(1.1) + 1.5(9.81)(0.4)\cos 0.9 = 0.144 + 0.396 + 3.659 = \boxed{4.199}$ N m. (The student's formulas give 9.897 N and 3.659 N m.) [checked]

EXAM-STYLE QUESTION Inertia tensor about a shifted frame [indicative 4]

Uniform box, $m = 4$ kg, edges $W = 0.3$ (x), $L = 0.2$ (y), $H = 0.5$ m (z). Frame $\{A\}$ has parallel axes; the COM is at $\mathbf{p} = (0, 0.1, 0.25)$ m in $\{A\}$. Find ${}^A\mathbf{I}$.

Solution

Centroid: $\mathrm{diag}\big(\frac{4}{12}(0.04 + 0.25),\ \frac{4}{12}(0.09 + 0.25),\ \frac{4}{12}(0.04 + 0.09)\big) = \mathrm{diag}(0.09667,\ 0.11333,\ 0.04333)$. Parallel axis, $\mathbf{p}^\top\mathbf{p} = 0.0725$:

$${}^A\mathbf{I} = {}^C\mathbf{I} + 4\big(0.0725\,\mathbf{1}_3 - \mathbf{p}\mathbf{p}^\top\big) = \begin{bmatrix} 0.38667 & 0 & 0 \\ 0 & 0.36333 & -0.1 \\ 0 & -0.1 & 0.08333 \end{bmatrix} \mathrm{kg\,m^2}$$

$I_{yz} = 0 - 4(0.1)(0.25) = -0.1$: the off-diagonal term carries the minus sign. Common slip: using the centroid $I_{xx} = 0.09667$ after the frame has moved. [checked]

EXAM-STYLE QUESTION Short explanations [indicative 2 each]

(a) Forward vs inverse dynamics, one use each. **(b)** Why is $\mathbf{M}$ invertible? **(c)** Why is Newton–Euler used for real-time control, and when is Lagrange better? **(d)** Which terms are non-zero when the arm is held still? When it moves at constant joint speed? **(e)** Multiplications for $n = 5$.

Solution

(a) Inverse: $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \to \boldsymbol{\tau}$, used in model-based control (computed torque). Forward: $\boldsymbol{\tau} \to \ddot{\mathbf{q}}$, used to simulate. (b) $\mathbf{M}$ is symmetric positive definite because $K = \frac{1}{2}\dot{\mathbf{q}}^\top\mathbf{M}\dot{\mathbf{q}} > 0$ for every non-zero $\dot{\mathbf{q}}$; a positive-definite matrix has non-zero determinant. (c) NE visits each link twice with a fixed amount of work, so the cost is linear ($104n - 77$ multiplications) against $O(n^4)$ for the closed-form Lagrangian; Lagrange gives the symbolic structure $\mathbf{M}, \mathbf{C}, \mathbf{g}$ for analysis and controller design, and both give identical torques. (d) Held still: only $\mathbf{g}$. Constant speed: $\mathbf{C}$ and $\mathbf{g}$ ($\mathbf{M}\ddot{\mathbf{q}} = \mathbf{0}$). (e) NE $104(5) - 77 = 443$; Lagrange $32(625) + 86(125) + 171(25) + 53(5) - 128 = 35\,162$.
[checked]

Which controller for which situation [added]

Law	Control signal	What it fixes / what it cannot fix	Must know
P	$f = -k_p x$	stiffness only; no damping added, so a light plant rings; constant disturbance leaves an offset	x
PD	$f = -k_p x - k_v \dot{x}$	stiffness $k + k_p$ and damping $b + k_v$; critical damping needs m, b, k to pick the gains; offset under a constant disturbance remains	$x, \dot{x}$; m, b, k for tuning
Partitioned (model + servo)	$f = \alpha f' + \beta$, $f' = -k_v \dot{x} - k_p x$	cancels the plant: unit mass, so gains come from ω_n alone ($k_p = \omega_n^2$, $k_v = 2\omega_n$); regulation only	$x, \dot{x}$; the model m, b, k
Partitioned + feed-forward	$f' = \ddot{x}_d + k_v \dot{E} + k_p E$	trajectory following: $\ddot{E} + k_v \dot{E} + k_p E = 0$, error $\to 0$; needs a smooth path	model; $x_d, \dot{x}_d, \ddot{x}_d$
PID	$f' = \ddot{x}_d + k_v \dot{E} + k_p E + k_i \int E\, dt$	removes the offset of a constant disturbance; third order, stable only if $k_v k_p > k_i$	as above, plus $\int E$
PI (no k_v) [added]	$f' = k_p E + k_i \int E\, dt$	unstable on a unit mass: $s^3 + k_p s + k_i$ has no s^2 term, so the roots sum to 0	–

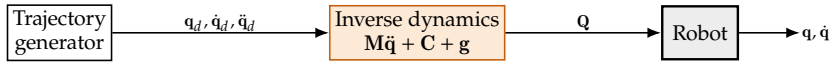
If the question says…

"classify / roots / sketch the response" ..... p. 103
"choose k_p, k_v for critical damping" ..... p. 106
"find the force f at this instant" ..... p. 106
"steady-state error / disturbance" ..... p. 108
"is the PID loop stable" ..... p. 108
"motor, gearbox, armature current" ..... p. 108

Open loop and closed loop [p. 8–12]

PREMISE control **Given:** a desired trajectory $\mathbf{q}_d(t), \dot{\mathbf{q}}_d(t), \ddot{\mathbf{q}}_d(t)$ from the trajectory generator (W5), and the robot dynamics $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q})$ (W7)
Find: the joint torques $\mathbf{Q}(t)$ (N m for revolute joints, N for prismatic) that make the real joints follow the trajectory, despite model errors and disturbances

(1) Open loop [p. 11]. Inverse dynamics turns the trajectory into torques and the robot is driven blind:

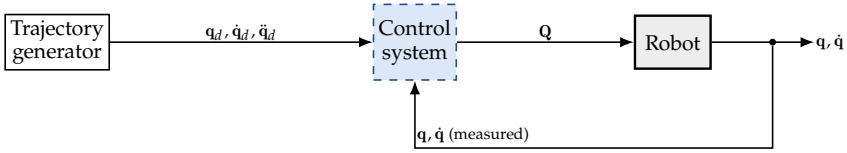


no feedback path: nothing measures or corrects an error

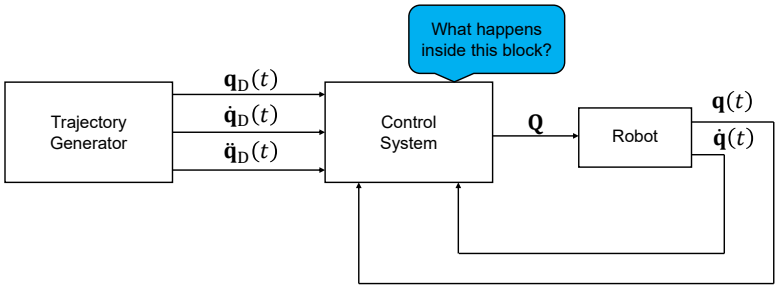
[drawn for this booklet] open-loop control. The slide (p. 11) labels the trajectory signals $\mathbf{q}(t), \dot{\mathbf{q}}(t), \ddot{\mathbf{q}}(t)$; p. 12 and p. 13 call them $\mathbf{q}_D, \mathbf{q}_d$ (same thing).

(2) Closed loop [p. 12]. The measured $\mathbf{q}, \dot{\mathbf{q}}$ are fed back to the control system:

"what happens inside this block?"



[drawn for this booklet] closed-loop control; the rest of this week fills in the control-system block.



[W08 lec p. 12] — the slide's closed-loop diagram: two feedback lines, $\mathbf{q}$ and $\dot{\mathbf{q}}$

Why open loop fails [p. 11, 12]

Exact model, exact start: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}}_d + \mathbf{C} + \mathbf{g}$ gives $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d$, so $\ddot{E} = 0$ (one joint, unit mass):

$$\ddot{E} = 0 \implies E(t) = E(0) + \dot{E}(0)\, t$$

Use: any start error, wrong model or push is never removed: it stays or grows linearly. Feedback is the only cure.

WORKED open loop with a start error [checked]
$E(0) = 0.1$ rad, $\dot{E}(0) = 0.02$ rad/s: $E(5) = 0.1 + 0.02(5) = 0.2$ rad (doubled). With $E(0) = 0.1$, $\dot{E}(0) = 0$ it stays 0.1 rad for ever.

Servo error and independent joint control [p. 13–14]

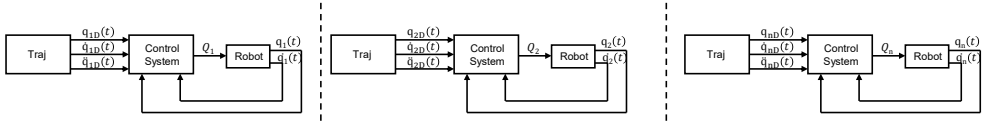
Servo error [p. 13]

$$E = \mathbf{q}_d - \mathbf{q}, \qquad \dot{E} = \dot{\mathbf{q}}_d - \dot{\mathbf{q}} \qquad \text{(one joint: } E = x_d - x,\ \dot{E} = \dot{x}_d - \dot{x}\text{)}$$

SYMBOLS

$\mathbf{q}_d$	desired joint positions	rad or m
$\mathbf{q}$	measured (actual) joint positions	rad or m
$E, \dot{E}$	position error and velocity error (desired − actual)	rad, rad/s

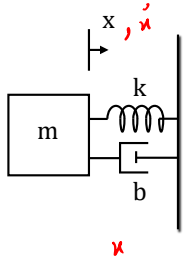
Use: the goal of a control system is $E \to 0$ quickly. Every later formula uses this sign (desired minus actual); with $E = q - q_d$ every sign flips.



[W08 lec p. 14] — independent joint control: the $n \times 1$ loop of p. 12 is split into n separate single-joint loops (each line is a single number)

Independent joint control [p. 14]. A robot has n coupled joints, but we treat each as its own single-input loop (inputs $q_{iD}, \dot{q}_{iD}, \ddot{q}_{iD}$, output torque Q_i). This ignores the coupling terms of $\mathbf{M}, \mathbf{C}$ (the other joints appear as disturbances); the rest of the week therefore uses *one joint modelled as a mass–spring–damper*.

Spring–mass–damper: characteristic equation [p. 16–17]



[W08 lec p. 16] — mass m on a spring k and damper b against a wall; x is the position of the block

PREMISE free response **Given:** m, b, k and the initial state $x(0), \dot{x}(0)$
Find: $x(t)$ for $t \geq 0$ for $m\ddot{x} + b\dot{x} + kx = 0$

Forces on the block: the spring pulls back with kx, the damper opposes the velocity with $b\dot{x}$, so $m\ddot{x} = -kx - b\dot{x}$ (the slide's hand sketch).

Characteristic equation and poles [p. 16, 17]

$$m\ddot{x} + b\dot{x} + kx = 0 \;\Rightarrow\; ms^2 + bs + k = 0, \qquad s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

SYMBOLS

m	mass (a joint: inertia I)	kg
b	damping coefficient (viscous friction)	N s/m
k	spring stiffness	N/m
$x, \dot{x}, \ddot{x}$	position, velocity, acceleration of the block	m, m/s, m/s^2
s_1, s_2	roots of the characteristic equation = the **poles**	s^{-1}

Use: the slide calls it a second-order, homogeneous, linear ODE with constant coefficients. The poles decide the whole shape of $x(t)$.

DERIVATION of the characteristic equation [added]

Try $x = e^{st}$. Then $\dot{x} = se^{st}$, $\ddot{x} = s^2e^{st}$. Substitute: $(ms^2 + bs + k)e^{st} = 0$. Since $e^{st} \neq 0$, the bracket must vanish: $ms^2 + bs + k = 0$. Its two roots s_1, s_2 each give a solution; the general solution is a combination, with two constants fixed by $x(0)$ and $\dot{x}(0)$.

The three root cases [p. 18–26]

Damping ratio and natural frequency [added]

$$\omega_n = \sqrt{\frac{k}{m}}, \qquad \zeta = \frac{b}{2\sqrt{mk}}, \qquad ms^2 + bs + k = m\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

SYMBOLS

ω_n	undamped natural frequency (oscillation rate if $b = 0$)	rad/s
ζ	damping ratio (1 = critical)	–

Use: $\zeta > 1$ over-damped, $\zeta = 1$ critical, $0 < \zeta < 1$ under-damped, $\zeta = 0$ undamped. Same test as $b^2 - 4mk$ (> 0, $= 0$, < 0). The slides never name ζ, ω_n; the exam may.

Case	Poles	$x(t)$ ([p. 18], [p. 21], [p. 24])
$b^2 - 4mk > 0$ over-damped [p. 18]	real, unequal s_1, s_2	$A_1e^{s_1t} + A_2e^{s_2t}$
$b^2 - 4mk < 0$ under-damped [p. 21]	complex $\lambda \pm i\mu$	$A_1e^{\lambda t}\cos\mu t + A_2e^{\lambda t}\sin\mu t$
$b^2 - 4mk = 0$ critically damped [p. 24]	real, equal $s = -\frac{b}{2m}$	$(A_1 + A_2t)\,e^{-\frac{b}{2m}t}$

SYMBOLS

λ	real part of the complex poles, $\lambda = -\frac{b}{2m}$ [added]	s^{-1}
μ	imaginary part, $\mu = \frac{\sqrt{4mk - b^2}}{2m}$ [added]	rad/s
A_1, A_2	constants from $x(0), \dot{x}(0)$	m

DERIVATION constants from the initial conditions [added]

Over: $x(0) = A_1 + A_2$, $\dot{x}(0) = s_1A_1 + s_2A_2$. Under: $x(0) = A_1$, $\dot{x}(0) = \lambda A_1 + \mu A_2$ (not μA_2 alone). Critical: $\dot{x} = (A_2 + sA_1 + sA_2t)e^{st}$, so $x(0) = A_1$, $\dot{x}(0) = sA_1 + A_2$. Solve the 2×2 system.

STEPS Classify and solve $x(t)$

1. Write $ms^2 + bs + k = 0$; compute $b^2 - 4mk$ (or ζ); name the case.
2. Find the poles ($\lambda \pm i\mu$ if complex); write the matching form of $x(t)$.
3. Substitute $x(0)$ and $\dot{x}(0)$: two equations, two unknowns A_1, A_2.
4. Check: the formula returns $x(0), \dot{x}(0)$ and $x \to 0$.

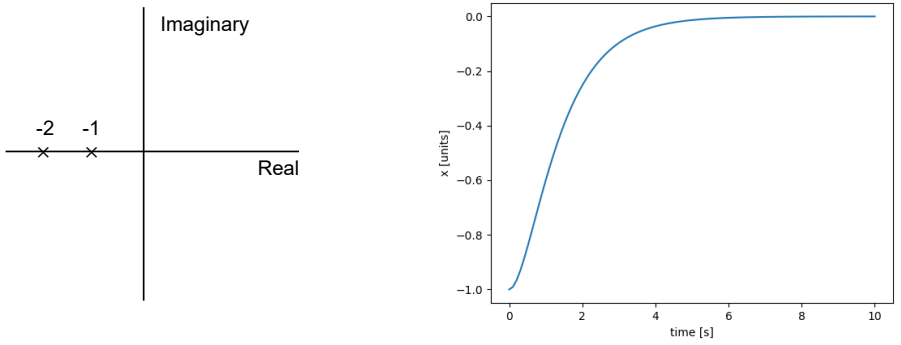
All three slide examples use $x(0) = -1$, $\dot{x}(0) = 0$:

WORKED over-damped, $m, b, k = 1, 3, 2$ [p. 19, 20]

$s^2 + 3s + 2 = 0$, $b^2 - 4mk = 9 - 8 = 1 > 0$. $s = \frac{-3\pm1}{2} = -1, -2$. $x = A_1e^{-t} + A_2e^{-2t}$.
$x(0) = A_1 + A_2 = -1$; $\dot{x}(0) = -A_1 - 2A_2 = 0 \Rightarrow A_1 = -2A_2$; so $-2A_2 + A_2 = -1$: $A_2 = 1$, $A_1 = -2$.

$$x(t) = -2e^{-t} + e^{-2t}, \qquad x(1) = -0.6004 \text{ m}$$

$\zeta = 1.061$, $\omega_n = \sqrt{2}$. Rises monotonically from -1 to 0, no overshoot. [checked]



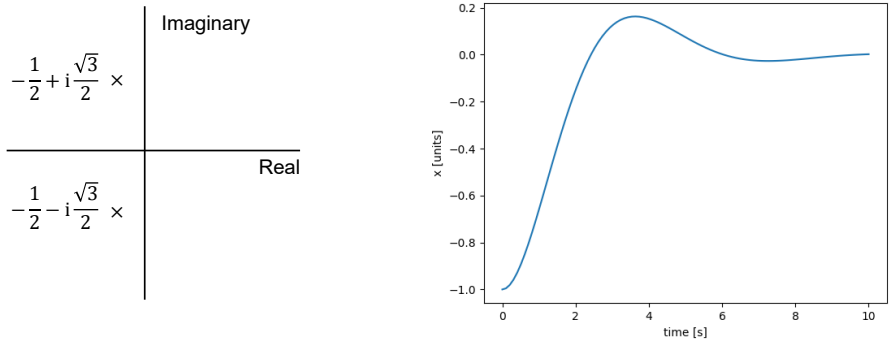
[W08 lec p. 20] — poles at $-1, -2$ (left) and the response $x(t)$ (right)

WORKED under-damped, $m, b, k = 1, 1, 1$ [p. 21–23]

$b^2 - 4mk = -3 < 0$: $s = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$, so $\lambda = -\frac{1}{2}$, $\mu = \frac{\sqrt{3}}{2}$. $x = A_1 e^{-t/2}\cos\frac{\sqrt{3}}{2}t + A_2 e^{-t/2}\sin\frac{\sqrt{3}}{2}t$.
$x(0) = A_1 = -1$; $\dot{x}(0) = \lambda A_1 + \mu A_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}A_2 = 0 \Rightarrow A_2 = -\frac{1}{\sqrt{3}} = -0.5774$.

$$x(t) = -e^{-t/2}\left[\cos\tfrac{\sqrt{3}}{2}t + \tfrac{1}{\sqrt{3}}\sin\tfrac{\sqrt{3}}{2}t\right]$$

$\zeta = 0.5$, $\omega_n = 1$. First peak $+0.1630$ at $t = 2\pi/\sqrt{3} = 3.628$ s ($= e^{-\pi/\sqrt{3}}$, i.e. 16.3% of the initial offset), then a trough -0.027 at 7.26 s. [checked]



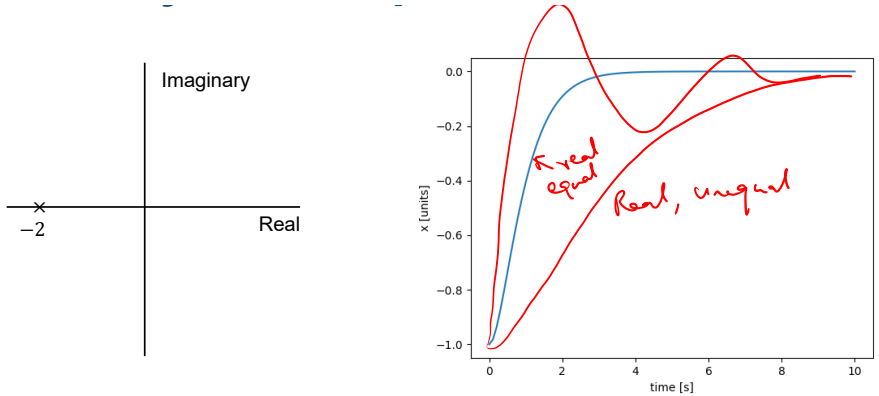
[W08 lec p. 23] — poles $-\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$ and the overshooting response

WORKED critically damped, $m, b, k = 1, 4, 4$ [p. 24–26]

$b^2 - 4mk = 0$: $s = -2$ (twice). $x = (A_1 + A_2 t)e^{-2t}$.
$x(0) = A_1 = -1$; $\dot{x}(0) = sA_1 + A_2 = 2 + A_2 = 0 \Rightarrow A_2 = -2$.

$$x(t) = (-1 - 2t)\,e^{-2t}, \qquad x(1) = -0.4060 \text{ m}$$

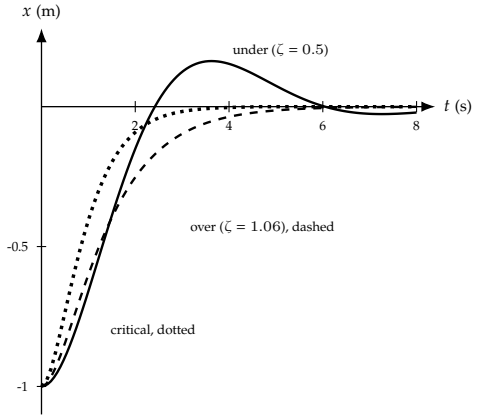
$\zeta = 1$, $\omega_n = 2$. Fastest approach to 0 with no overshoot. [checked]



[W08 lec p. 26] — all three overlaid on the slide. The red curves are a hand sketch, not to scale

SLIDE CHECK p.26 hand sketch

The red sketch on p. 26 draws "complex" with a large late bump and "real, unequal" as the slowest curve. The exact solutions above give: under-damped peak $+0.163$ then -0.027; 2% settling times critical 2.92 s < over 4.60 s < under 8.08 s (so here the under-damped one is the slowest). Only "critical is fastest without overshoot" is safe to claim in general. [checked]



[drawn for this booklet] the three slide examples plotted from the exact formulas above ($x(0) = -1$, $\dot{x}(0) = 0$)

Pole map (slides p. 20, 23, 26). Real poles $-1, -2$ (over), $-\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$ (under), double pole -2 (critical). The farther left, the faster the decay; complex poles oscillate, their real part sets the decay and their imaginary part the frequency.

TRAP

Common slips: $\lambda = -b/m$ (it is $-b/2m$); $A_2 = \dot{x}(0)/\mu$ (forget λA_1); calling $b^2 = 4mk$ "under-damped".

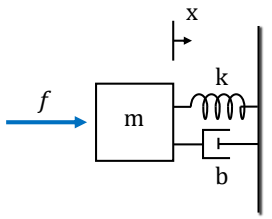
WORKED free response with an initial velocity (Practice Q8.2) [checked]

$m = 1, b = 2, k = 5, x(0) = 1$ m, $\dot{x}(0) = 3$ m/s. $b^2 - 4mk = -16 < 0$: under-damped; $s = -1 \pm 2i$, $\lambda = -1$, $\mu = 2$ ($\zeta = 0.447$). $A_1 = x(0) = 1$; $\dot{x}(0) = \lambda A_1 + \mu A_2 = -1 + 2A_2 = 3 \Rightarrow A_2 = 2$.

$$x(t) = e^{-t}(\cos 2t + 2\sin 2t), \qquad x(0.5) = 0.6065(0.5403 + 1.6829) = 1.3485 \text{ m}$$

Wrong: $A_2 = \dot{x}(0)/\mu = 1.5$.

Actuated system and PD control [p. 27–32]



[W08 lec p. 28] — the actuated mass–spring–damper: actuator force f added to the block, $m\ddot{x} + b\dot{x} + kx = f$

PREMISE PD regulation **Given:** $m\ddot{x} + b\dot{x} + kx = f$ with the actuator force f; $x, \dot{x}$ measured
Find: k_p, k_v so that the closed loop is critically damped (or has chosen poles)

Actuated plant and PD law [p. 28–32]

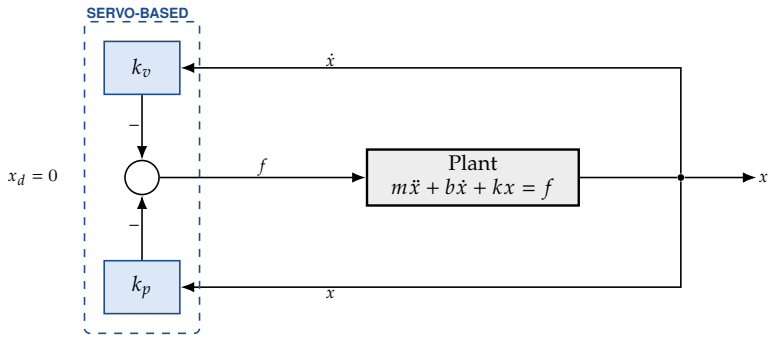
$$m\ddot{x} + b\dot{x} + kx = f, \qquad f = -k_p x - k_v \dot{x}$$
$$\implies m\ddot{x} + \underbrace{(b + k_v)}_{b'}\dot{x} + \underbrace{(k + k_p)}_{k'}x = 0$$

SYMBOLS

f	actuator force (a joint: torque Q)	N
k_p	position gain (virtual extra spring)	N/m
k_v	velocity gain (virtual extra damper)	N s/m
b', k'	closed-loop damping and stiffness (acute accent on the slide)	N s/m, N/m

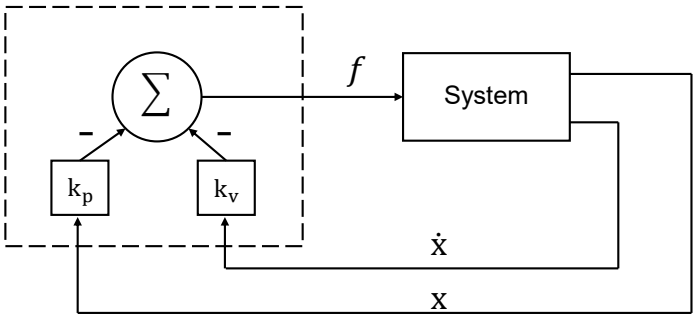
Use: critically damped if $b'^2 = 4mk'$, i.e. $(b + k_v)^2 = 4m(k + k_p)$. Always include the plant's own b and k.

DERIVATION closed-loop equation [added]
Substitute the law into the plant: $m\ddot{x} + b\dot{x} + kx = -k_p x - k_v \dot{x}$. Move the right side to the left: $m\ddot{x} + (b + k_v)\dot{x} + (k + k_p)x = 0$. This is the spring–mass–damper again, so the case test of the previous section applies with b', k'.



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] PD regulation of the mass–spring–damper (after the slide's p. 30): $f = -k_p x - k_v \dot{x}$; both signs are minus because the target is $x = 0$



[W08 lec p. 30] — the slide's block diagram of the simple control law (dashed box = controller; both inputs to the sum carry −)

WORKED critical damping of the raw plant [p. 32]
$m = 2$, $b = 3$, $k = 8$ (under-damped: $9 - 64 < 0$); choose $k_p = 10$. $k' = 8 + 10 = 18$; $b' = \sqrt{4mk'} = \sqrt{144} = 12$; so $k_v = b' - b = 9$ N s/m. Double pole $-b'/2m = -3$. [checked]
Trap: using $k_v = 2\sqrt{k_p} = 6.32$ (the unit-mass rule) leaves it under-damped (discriminant -57.05). That rule needs partitioning (next sections).

Control-law partitioning [p. 33–37]

PREMISE partitioning **Given:** the model m, b, k of the joint and the measured $x, \dot{x}$
Find: a force $f = \alpha f' + \beta$ so that $\ddot{x} = f'$ and a servo law f' with gains independent of m, b, k

Reasoning ("unit mass"). We split the controller in two. The *model-based* part uses what we know about the joint (m, b, k) to cancel the physics, so the joint behaves like a mass of 1 kg with no spring or damper: a pure double integrator $\ddot{x} = f'$. The *servo* part then controls this simple joint with the same recipe whatever the real m, b, k are. The α, β may be functions or constants chosen to make this true [p. 33–34].

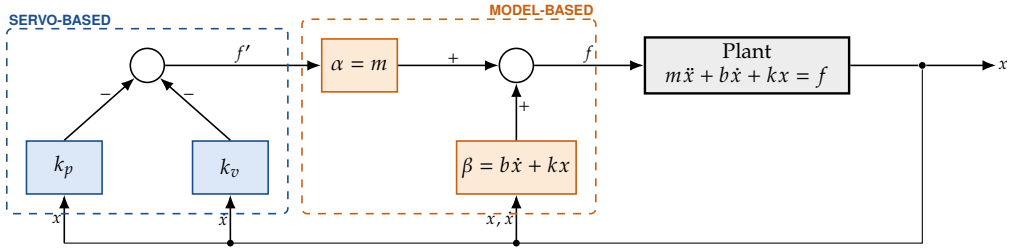
Partitioned law [p. 33–36]

$$f = \alpha f' + \beta, \qquad \alpha = m, \qquad \beta = b\dot{x} + kx \implies m\ddot{x} + b\dot{x} + kx = mf' + b\dot{x} + kx \implies \ddot{x} = f'$$
$$f' = -k_v\dot{x} - k_p x \implies \ddot{x} + k_v\dot{x} + k_p x = 0, \qquad s^2 + k_v s + k_p = 0$$
$$\textbf{critically damped: } k_v^2 = 4k_p \iff k_v = 2\sqrt{k_p}$$

SYMBOLS

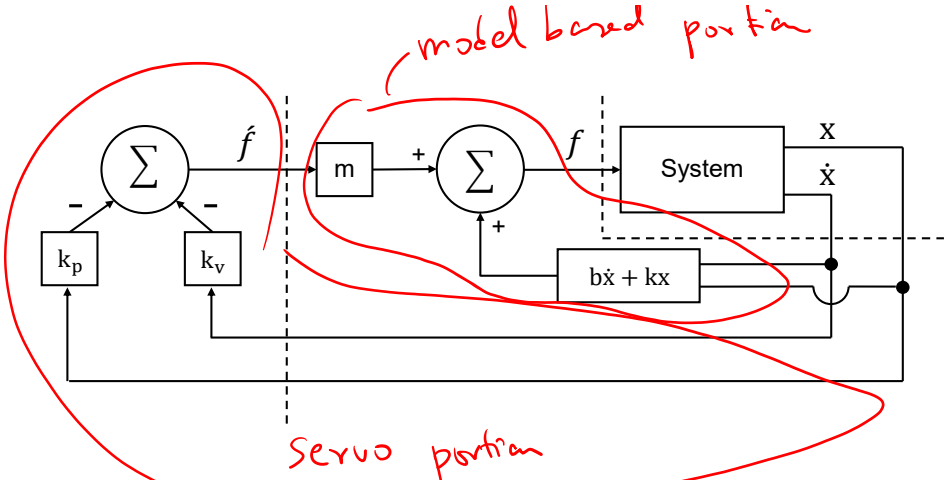
f	actual actuator force sent to the joint	N
f'	servo output: a desired *acceleration* (acute accent on the slide)	m/s^2
α	model-based gain (= mass)	kg
β	model-based force $b\dot{x} + kx$: friction and spring terms	N
k_p, k_v	servo gains, now in s^{-2} and s^{-1}	–

Use: regulation to $x = 0$. The result is independent of m, b, k (if the model is right). Poles p_1, p_2: $k_v = -(p_1 + p_2)$, $k_p = p_1 p_2$ [added].



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] partitioned controller for the single mass (slide p. 37, redrawn). Servo part: k_p, k_v acting on the measured state. Model part: $\alpha = m$, $\beta = b\dot{x} + kx$.



[W08 lec p. 37] — the slide's version, annotated by the lecturer: "servo portion" (k_p, k_v, f') and "model-based portion" ($m, b\dot{x} + kx$)

SLIDE CHECK p.36 critical damping

Slide p. 36 prints the critical-damping condition as $k_p^2 = 4k_v$ (subscripts swapped). The characteristic equation $s^2 + k_v s + k_p = 0$ (written beside it) has equal roots iff $k_v^2 = 4k_p$, which agrees with p. 32 ($b'^2 = 4mk'$). With the slide's rule, $k_p = 16 \Rightarrow k_v = 64$ gives roots -63.75, -0.25 (heavily over-damped); correct: $k_v = 8$, double root -4. [checked]

STEPS Partitioned force at one instant

1. Plant $m\ddot{x} + b\dot{x} + kx = f$. Set $\alpha = m$, $\beta = b\dot{x} + kx$ (evaluated at the *measured* $x, \dot{x}$).
2. Servo: $f' = -k_v\dot{x} - k_p x$ (regulation) or $f' = \ddot{x}_d + k_v\dot{E} + k_p E$ (tracking).
3. Critical damping: $k_v = 2\sqrt{k_p}$; for a chosen ω_n: $k_p = \omega_n^2$.
4. Force $f = mf' + b\dot{x} + kx$. Label the model-based and servo parts on a block diagram.
5. Check: $(f - b\dot{x} - kx)/m$ must equal f'.

WORKED force at a state

$m, b, k = 2, 3, 8$; $k_p = 25$, $k_v = 10$ (critical, double pole -5); $x = 0.5$ m, $\dot{x} = 1$ m/s. $f' = -10(1) - 25(0.5) = -22.5$; $f = 2(-22.5) + 3(1) + 8(0.5) = -38$ N. Check: $\ddot{x} = (-38 - 3 - 4)/2 = -22.5 = f'$. [checked]

ALTERNATIVE APPROACHES

A. PD on the raw plant	B. Partitioned (model + servo)
$f = -k_p x - k_v\dot{x}$.	$f = mf' + b\dot{x} + kx$, $f' = -k_v\dot{x} - k_p x$.
Closed loop $m\ddot{x} + (b + k_v)\dot{x} + (k + k_p)x = 0$.	Closed loop $\ddot{x} + k_v\dot{x} + k_p x = 0$.
Gains depend on m, b, k:	Gains independent of m, b, k:
$k_p = m\omega_n^2 - k$, $k_v = 2m\omega_n - b$.	$k_p = \omega_n^2$, $k_v = 2\omega_n$.

WORKED scenario: critical damping at a given stiffness

Plant $m = 0.5$ kg, $b = 1$ N s/m, $k = 18$ N/m (a stiff spring). Want a critical response with $\omega_n = 12$ rad/s.

A (raw PD): $k' = m\omega_n^2 = 72$, $b' = 2m\omega_n = 12$, so $k_p = 72 - 18 = 54$ N/m, $k_v = 12 - 1 = 11$ N s/m. Double pole -12.

B (partitioned): $k_p = \omega_n^2 = 144$ s^{-2}, $k_v = 2\omega_n = 24$ s^{-1}, with $f = 0.5f' + \dot{x} + 18x$. Double pole -12.

Mixing them (B's gains on the raw plant) gives $s^2 + 50s + 324$: poles -42.35, -7.65, not critical. [checked]

Wrong model: what changes [added]

Controller uses $\hat{m}, \hat{b}, \hat{k}$; plant has m, b, k. With $\hat{k} = k$, $\hat{m} = m$: $\ddot{x} + \left(k_v + \frac{b - \hat{b}}{m}\right)\dot{x} + k_p x = 0$. With $\hat{b} = b$, $\hat{k} = k$ and $\hat{m} \neq m$: $\ddot{x} + \frac{\hat{m}}{m}k_v\dot{x} + \frac{\hat{m}}{m}k_p x = 0$ (both gains scaled by $\hat{m}/m$).

Example $m = 2, b = 3, k = 8, k_p = 25, k_v = 10$: $\hat{b} = 1 \Rightarrow k_v^{eff} = 11$, $11^2 = 121 > 100$: slightly over-damped. $\hat{m} = 1.5 \Rightarrow s^2 + 7.5s + 18.75$, $56.25 - 75 < 0$: under-damped. [checked]

Choosing the gains from the poles [added]

Gains ↔ poles (partitioned servo, unit mass)

$$s^2 + k_v s + k_p = 0 \Rightarrow \omega_n = \sqrt{k_p}, \quad \zeta = \frac{k_v}{2\sqrt{k_p}}, \quad k_p = p_1 p_2, \; k_v = -(p_1 + p_2)$$

$$\text{overshoot } (0 < \zeta < 1) = \exp\left(-\frac{\pi\zeta}{\sqrt{1 - \zeta^2}}\right)$$

$$t_{s,2\%} \approx \frac{4}{\zeta\omega_n} \; (\zeta < 1), \qquad t_{s,2\%} \approx \frac{4}{|s_{slow}|} \; (\zeta > 1), \qquad t_{s,2\%} = \frac{5.83}{\omega_n} \; (\zeta = 1)$$

Use: design: critical with speed ω_n: $k_p = \omega_n^2$, $k_v = 2\omega_n$. Given ζ: $k_v = 2\zeta\omega_n$. Settling-time rules are approximations (conservative); the critical value 5.83 = root of $(1 + u)e^{-u} = 0.02$. $t_{s,2\%}$ = time after which $|x|$ stays within 2% of the initial offset.

Poles	Behaviour
both real, negative, unequal	no overshoot; decay rate set by the *slower* pole (time constant $1/\lvert s_{slow}\rvert$)
real, equal	critical: fastest without overshoot
complex, Re < 0	overshoot, oscillating decay: envelope $e^{-\zeta\omega_n t}$, frequency $\omega_n\sqrt{1 - \zeta^2}$
purely imaginary $\pm i\omega$	sustained oscillation, never settles ($k_v = 0$)
any pole with Re > 0	unstable: grows

STEPS Settling behaviour from pole positions

1. Poles → $\omega_n = \sqrt{p_1 p_2}$ and $\zeta = -(p_1 + p_2)/(2\omega_n)$.
2. Classify (real/complex/equal); state overshoot (complex only) and the slowest decay.
3. Settling time from the rule for that case.
4. Gains if asked: $k_p = p_1 p_2$, $k_v = -(p_1 + p_2)$.

WORKED scenario: read the response off the poles

(a) $-3 \pm 4i$: $\omega_n = 5$, $\zeta = 0.6$, $k_p = 25$, $k_v = 6$; overshoot 9.48%; $t_s \approx 4/3 = 1.33$ s (exact 1.19 s).
(b) -2, -8: $k_p = 16$, $k_v = 10$, $\zeta = 1.25$; from $x(0) = 1$: $x = (8e^{-2t} - 2e^{-8t})/6$; slow pole -2, time constant 0.5 s; $t_s \approx 4/2 = 2.0$ s (exact 2.10 s).
(c) -5, -5: $k_p = 25$, $k_v = 10$; $t_s = 5.83/5 = 1.17$ s.
(d) $k_v = 0$, $k_p = 9$: poles $\pm 3i$; sustained oscillation, period $2\pi/3 = 2.09$ s. [checked]

Trajectory following with the servo error [p. 38–40]

PREMISE tracking **Given:** a smooth, twice-differentiable $x_d(t)$ with $\dot{x}_d, \ddot{x}_d$; the model m, b, k; measured $x, \dot{x}$

Find: the force $f(t)$ and the error equation that governs $E = x_d - x$

Servo law with feed-forward [p. 39]

$$f' = \ddot{x}_d + k_v\dot{E} + k_pE \implies \ddot{E} + k_v\dot{E} + k_pE = 0, \qquad f = mf' + b\dot{x} + kx$$

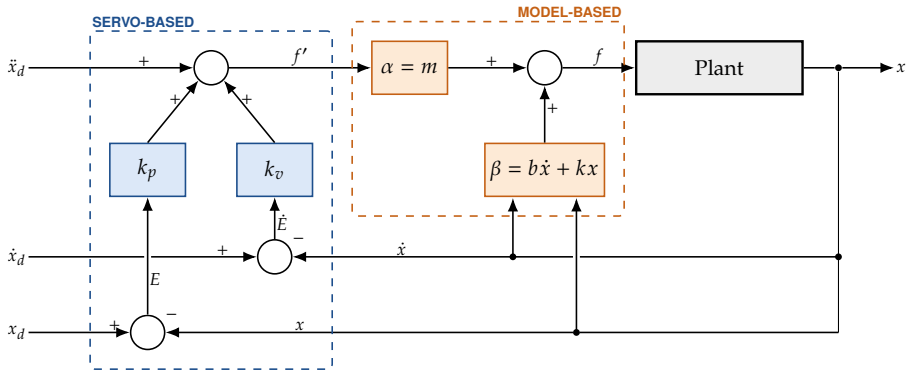
SYMBOLS

$x_d, \dot{x}_d, \ddot{x}_d$	desired position, velocity, acceleration	m, m/s, m/s^2
$E, \dot{E}$	$E = x_d - x$, $\dot{E} = \dot{x}_d - \dot{x}$	m, m/s

Use: needs $\ddot{x}_d$ ("second order in error space"). Without the feed-forward $\ddot{x}_d$, a constant desired acceleration a leaves $E_{ss} = a/k_p$.

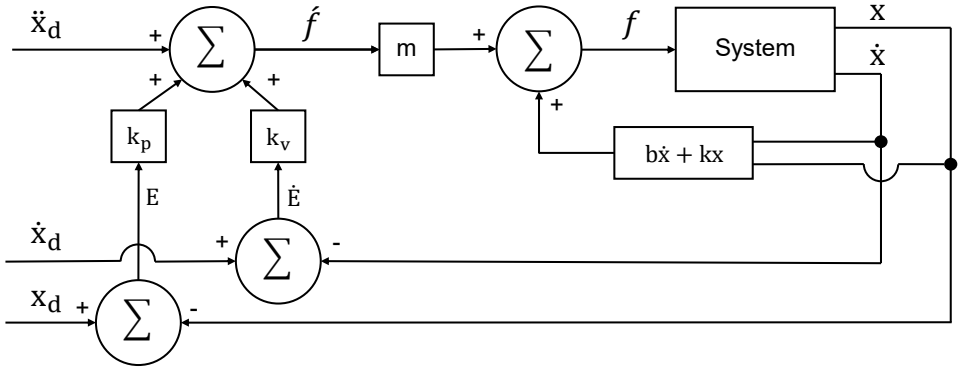
DERIVATION error equation [p. 39]

Partitioning gives $\ddot{x} = f'$. Insert the servo law: $\ddot{x} = \ddot{x}_d + k_v\dot{E} + k_pE$. Move $\ddot{x}_d$: $\ddot{x}_d - \ddot{x} = -k_v\dot{E} - k_pE$; but $\ddot{x}_d - \ddot{x} = \ddot{E}$, so $\ddot{E} + k_v\dot{E} + k_pE = 0$. The error obeys the same homogeneous equation as the regulation problem, for *any* trajectory.



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] partitioned controller with servo error (slide p. 40, redrawn). Feedback of $x, \dot{x}$ enters the two error junctions with − and the model part β.



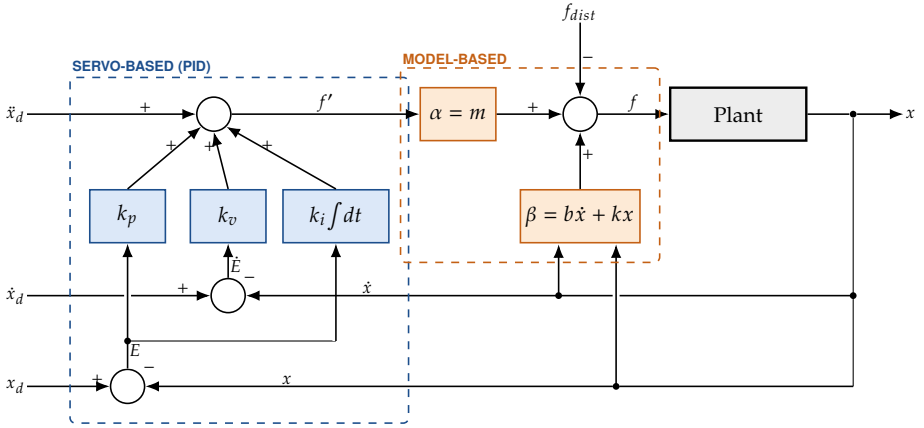
[W08 lec p. 40] — the slide's diagram: $\ddot{x}_d$ goes straight to the servo sum; E and $\dot{E}$ are formed by subtracting the feedback

WORKED tracking numbers [checked]

$k_p = 100$, $k_v = 20$ ($k_v^2 = 400 = 4k_p$: critical, $\omega_n = 10$). $E(0) = 0.5$ m, $\dot{E}(0) = 0$: $E = E_0(1 + 10t)e^{-10t}$, so $E(0.5) = 0.5(6)e^{-5} = 0.0202$ m. If $E(0) = 0$ and $\dot{E}(0) = 0$ the error stays 0 for any x_d.
Without feed-forward on $x_d = t^2$ ($\ddot{x}_d = 2$): $\ddot{E} + k_v\dot{E} + k_pE = 2 \Rightarrow E_{ss} = 2/100 = 0.02$ m.

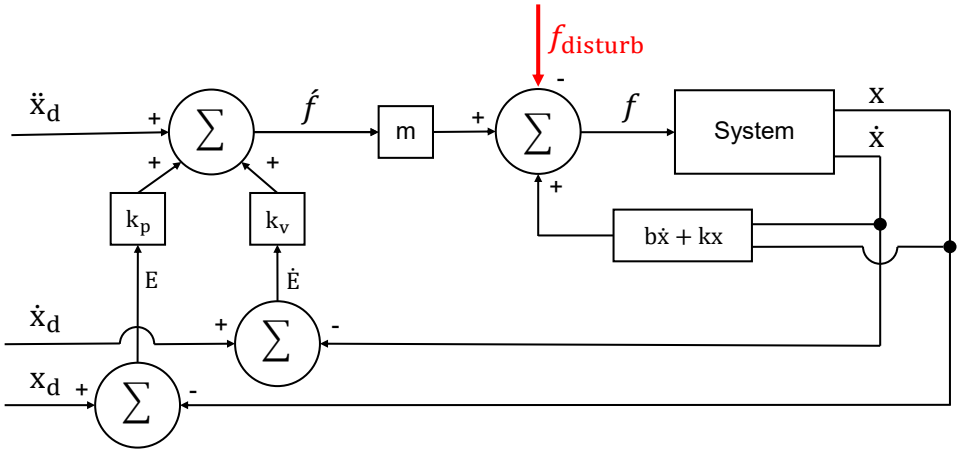
Disturbance rejection and steady-state error [p. 41–43]

PREMISE disturbance **Given:** the partitioned tracking controller, plus an unmodelled force f_{dist} acting on the joint
Find: the error equation, a bound on E (BIBO) and the steady-state error E_{ss} for constant f_{dist}



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] PID with disturbance input (slides p. 41 and p. 44 combined). Without the $k_i \int dt$ block this is the PD loop of p. 41; f_{dist} enters the force junction with −.



[W08 lec p. 41] — the slide's disturbance diagram: $f_{disturb}$ (red) enters the force junction with a minus sign

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DERIVATION error equation with a disturbance [p. 41–42]

The plant now feels $f - f_{dist}$: $m\ddot{x}+b\dot{x}+kx = f - f_{dist}$. Put in $f = mf' + b\dot{x} + kx$: $m\ddot{x} = mf' - f_{dist}$, so $\ddot{x} = f' - f_{dist}/m$. With $f' = \ddot{x}_d + k_v\dot{E} + k_pE$ and $\ddot{E} = \ddot{x}_d - \ddot{x}$:

$$\ddot{E} + k_v\dot{E} + k_pE = \frac{f_{dist}}{m}$$

(the minus at the junction becomes a plus because $E = x_d - x$ flips the sign).

Bounded disturbance and steady-state error [p. 42–43]

$$\text{bounded } f_{dist} \Rightarrow \text{bounded } E \text{ (BIBO)}$$

$$f_{dist} = \text{const},\ \ddot{E} = \dot{E} = 0: \quad k_pE = \frac{f_{dist}}{m} \Rightarrow E_{ss} = \frac{f_{dist}}{k_p\, m}$$

SYMBOLS

f_{dist}	constant disturbance force (a joint: torque)	N
E_{ss}	steady-state error	m

Use: k_v, b, k do not appear: only k_p and m. Tolerance E_{max}: $k_p \geq f_{dist}/(mE_{max})$, then $k_v = 2\sqrt{k_p}$. Heavier mass or bigger k_p means smaller error.

STEPS Steady-state error and its cure

1. Write the error equation with f_{dist}/m on the right.
2. Set $\ddot{E} = \dot{E} = 0$ and solve: $E_{ss} = f_{dist}/(k_pm)$.
3. State the sign: the joint sags in the direction of the disturbance; PD cannot reach 0 unless $k_p \to \infty$.
4. Cure: integral action (next section); stable only if $k_vk_p > k_i$.

WORKED scenario: steady-state error and tolerance [checked]

$m = 1.5$ kg, $f_{dist} = 6$ N, $k_p = 64\ \text{s}^{-2}$, $k_v = 16$ (critical): $E_{ss} = 6/(64 \times 1.5) = 0.0625$ m, and the same for $k_v = 16$ or 40 (simulated).
To get $E_{ss} \leq 5$ mm: $k_p \geq 6/(1.5 \times 0.005) = 800\ \text{s}^{-2}$, $k_v = 2\sqrt{800} = 56.57\ \text{s}^{-1}$ (large gains: noise and actuator limits, so integral action is better).

TRAP

"E_{ss} does not depend on the mass", "$E_{ss} = f/k_v$", "raising k_v cuts the steady error" (it only changes the speed), "PD removes it".

The integral term: PID [p. 44–45]

PID law and third-order error equation [p. 44, 45]

$$f' = \ddot{x}_d + k_v\dot{E} + k_pE + k_i\int E\,dt \Rightarrow \dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_iE = \frac{\dot{f}_{dist}}{m}$$

$$\ddot{E} + k_v\ddot{E} + k_p\dot{E} + k_iE = \frac{\dot{f}_{dist}}{m} \qquad \text{(differentiate once)}$$

SYMBOLS

k_i	integral gain	s^{-3}
$\int E\,dt$	accumulated error	m s

Use: for constant f_{dist}: $\dot{f}_{dist} = 0$ and at steady state $k_iE = 0$, so $E = 0$. The slide's condition "$E = 0$ for $t < 0$" is only a loose justification; the result is what is examined.

Stability of the PID loop [added]

$$s^3 + k_vs^2 + k_ps + k_i = 0, \qquad \text{stable} \iff k_v > 0,\ k_p > 0,\ 0 < k_i < k_vk_p$$

Use: Routh array: rows s^3: 1, k_p; s^2: k_v, k_i; s^1: $(k_vk_p - k_i)/k_v$; s^0: k_i. All first-column entries > 0 gives $k_i < k_vk_p$. At $k_i = k_vk_p$: $(s + k_v)(s^2 + k_p)$, a sustained oscillation at $\sqrt{k_p}$.

STEPS Check PID stability / limit

1. Write $s^3 + k_vs^2 + k_ps + k_i$ (unit-mass servo).
2. Compare k_i with k_vk_p: below stable, equal marginal, above unstable.
3. If asked for roots, solve the cubic (calculator polynomial solver).
4. Steady-state error for constant f_{dist}: 0; $\int E \to f_{dist}/(mk_i)$.

WORKED scenario: is it stable? [checked]

$k_p = 64$, $k_v = 16$ (critical PD), limit $k_vk_p = 1024$. $k_i = 900$: roots -15.60, $-0.20 \pm 7.59i$ (stable, lightly damped); $k_i = 1024$: $(s + 16)(s^2 + 64)$, roots -16, $\pm 8i$ (marginal); $k_i = 1100$: -16.23, $+0.12 \pm 8.23i$ (unstable).
$k_p = 30$, $k_v = 5$, $k_i = 200$: $k_vk_p = 150 < 200$, unstable (roots -5.79, $+0.39 \pm 5.87i$); $k_i = 100$ stable. With $f_{dist}/m = 4$, $k_p = 64$, $k_v = 16$, $k_i = 900$ the simulated error goes to 0 and $\int E \to 4/900$.

TRAP

"integral never hurts": a large k_i destabilises. "PI is enough": on a unit mass it is unstable (no s^2 term).

DC motor, friction and gearing [p. 46–50]

PREMISE single joint **Given:** a DC motor (armature current i_a, torque constant k_4, rotor inertia I_m, motor friction c_m), gear ratio η, link-side inertia I and friction c
Find: the axle torque Q, the motor torque Q_m and the armature current i_a for a planned $\ddot{q}, \dot{q}$ at the axle

Motor model [p. 49]

$$Q_m(t) = k_1Bi_a = k_4i_a, \qquad Q_{net} = Q_m - f_m, \qquad I_m\ddot{q}_m = k_4i_a - c_m\dot{q}_m$$

SYMBOLS

i_a	armature current	A
k_4	torque constant (= k_1B)	N m/A
Q_m	motor torque	N m
f_m	friction torque lost in the motor	N m
I_m	rotor inertia	kg m^2
c_m	motor viscous friction coefficient	N m s/rad
$\dot{q}_m, \ddot{q}_m$	motor shaft speed, acceleration	rad/s, rad/s^2

Use: the electrical side (armature R_a, L_a, back-emf V_b, field circuit) is only drawn on p. 48; no calculation uses it.

Gearing [p. 50]

$$Q = \eta Q_m, \qquad \dot{q}_{axle} = \frac{1}{\eta}\dot{q}_m, \qquad Q = (I + \eta^2 I_m)\,\ddot{q}_{axle} + (c + \eta^2 c_m)\,\dot{q}_{axle}$$

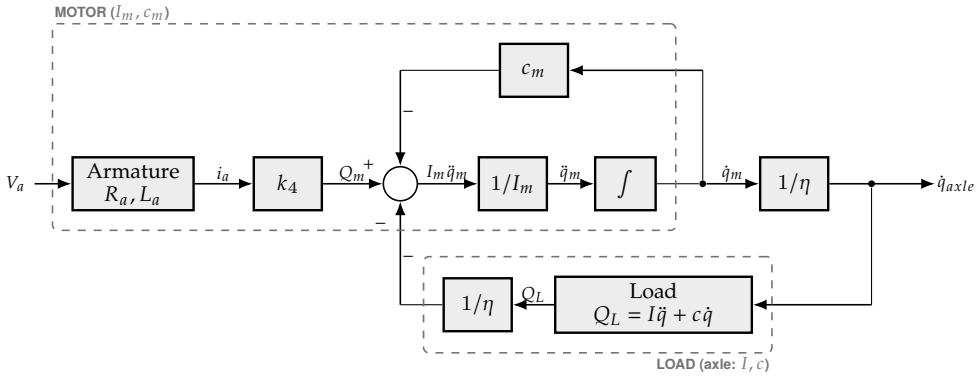
SYMBOLS

η	gear ratio ($\eta > 1$: motor faster than the axle)	–
Q	torque at the axle	N m
I, c	link-side inertia and friction	kg m^2, N m s/rad
I_{eff}, c_{eff}	$I + \eta^2 I_m$, $c + \eta^2 c_m$: effective values seen at the axle	–

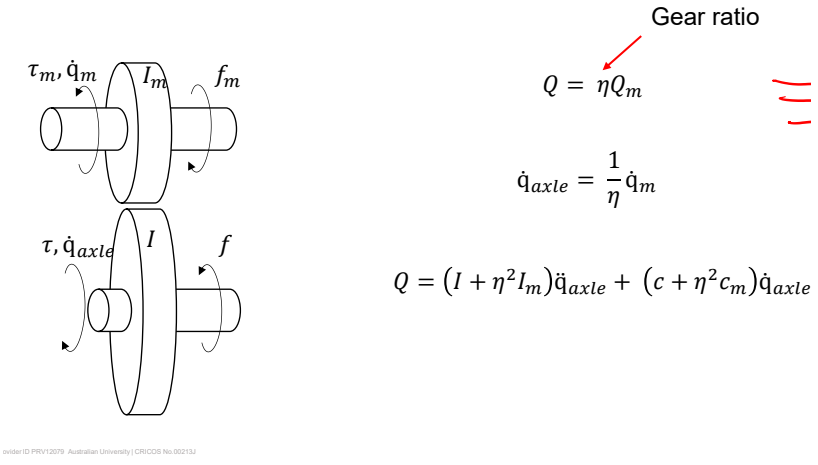
Use: the motor's inertia and friction count η^2 times: not η, not ηI_m. Then $Q_m = Q/\eta$ and $i_a = Q_m/k_4$.

DERIVATION the η^2 [added]

Motor shaft: $Q_m = I_m\ddot{q}_m + c_m\dot{q}_m + Q_L/\eta$, where the load torque $Q_L = I\ddot{q} + c\dot{q}$ appears divided by η at the motor. With $q_m = \eta q$: $Q_m = I_m\eta\ddot{q} + c_m\eta\dot{q} + (I\ddot{q} + c\dot{q})/\eta$. Multiply by η: $\eta Q_m = (I + \eta^2 I_m)\ddot{q} + (c + \eta^2 c_m)\dot{q}$. The motor's own inertia torque $I_m\eta\ddot{q}$ is multiplied by η once to reach the axle, and it already contains one η from $\ddot{q}_m = \eta\ddot{q}$.



[drawn for this booklet] DC motor and gear joint: torque balance at the motor shaft, $Q_m = I_m\ddot{q}_m + c_m\dot{q}_m + Q_L/\eta$ with $q_m = \eta q_{axle}$. Not a controller, so no servo/model colouring. The slides (p. 48) draw only the circuit; the block form is added.



[W08 lec p. 50] — the slide's gearing figure and equations: motor side (τ_m, $\dot{q}_m$, I_m, f_m) above the axle side (τ, $\dot{q}_{axle}$, I, f)

WORKED gear numbers [checked]

$\eta = 50$, $I_m = 2\times10^{-5}$, $I = 0.04$ kg m^2, $c = 0.1$, $c_m = 10^{-4}$, $k_4 = 0.04$; $\ddot{q} = 10$, $\dot{q} = 2$ at the axle. $I_{eff} = 0.04 + 2500(2\times10^{-5}) = 0.09$; $c_{eff} = 0.1 + 2500(10^{-4}) = 0.35$; $Q = 0.09(10) + 0.35(2) = 1.6$ N m; $Q_m = 1.6/50 = 0.032$ N m; $i_a = 0.032/0.04 = 0.8$ A; motor speed 100 rad/s. Wrong: 1.6 (no $\div\eta$), 0.012 (no η^2).

SLIDE CHECK p.50, p.51

The slide gives $Q = \eta Q_m$ without saying that Q is the torque the motor delivers *at the axle*; the η^2 result above fixes this. Slide p. 51 ("Bringing it all together") is blank: there is no combined PID + motor formula, so do not look for one.

How this is examined: worked solutions

W13 has no control question; expect the sentences below. Practical sheet W8 (title "Dynamics", three questions): solved here in full; the theory is in the W7 part.

EXAM-STYLE QUESTION Practical sheet Q1: single link [marks not published]

A 1-DoF manipulator: $m = 1$ kg, ${}^1\mathbf{P}_C = [3,0,0]^\top$ m, ${}^C\mathbf{I} = \mathrm{diag}(1,2,3)$ kg m^2. The joint starts at rest and moves as $q(t) = bt + ct^2$ about z_1. Give the angular acceleration of the link and the linear acceleration of the centre of mass in frame $\{1\}$.

Solution

$\dot{q} = b + 2ct$, $\ddot{q} = 2c$ (at rest at $t = 0$ means $b = 0$; keep b for generality). Frame $\{1\}$, joint axis $\hat{z}_1$: $\boldsymbol{\omega} = {}^1[0,0,\dot{q}]^\top$, $\dot{\boldsymbol{\omega}} = {}^1[0,0,2c]^\top$ rad/s^2 [angular acceleration]

Linear acceleration of the COM (fixed to the link, radius $\mathbf{P}_C$): $\dot{\mathbf{v}}_C = \dot{\boldsymbol{\omega}}\times\mathbf{P}_C + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{P}_C)$.

Tangential: $[0,0,2c]\times[3,0,0] = [0,\ 6c,\ 0]$. Centripetal: $\boldsymbol{\omega}\times\mathbf{P}_C = [0, 3\dot{q}, 0]$, then $\boldsymbol{\omega}\times[0,3\dot{q},0] = [-3\dot{q}^2, 0, 0]$.

$$ {}^1\dot{\mathbf{v}}_C = \left[-3(b+2ct)^2,\ 6c,\ 0\right]^\top \text{ m/s}^2 $$

Extra (joint torque, no gravity): $\mathbf{F} = m\dot{\mathbf{v}}_C$, $\mathbf{N} = {}^C\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}\times{}^C\mathbf{I}\boldsymbol{\omega} = [0,0,6c]$ ($\boldsymbol{\omega}$ is along a principal axis); $\tau = N_z + (\mathbf{P}_C\times\mathbf{F})_z = 6c + 3(6c) = 24c$. Check: $(I_{zz} + m\,3^2)\ddot{q} = 12(2c) = 24c$.

Check at $b = 0$, $c = 0.5$, $t = 2$ s: $\dot{\boldsymbol{\omega}} = [0,0,1]$, $\dot{\mathbf{v}}_C = [-12, 3, 0]$, $\tau = 12$ N m. [checked]

EXAM-STYLE QUESTION Practical sheet Q2: 3D RRR equations of motion [marks not published]

The arm has a waist q_1 about z, a shoulder q_2 and an elbow q_3 about horizontal axes, wrist point (x_W, y_W, z_W). Derive the equations of motion and give the joint torques for a prescribed trajectory. (Assume, as in the lecturer's W13 Q4 picture: link L_3 shoulder→elbow, L_4 elbow→wrist, point mass m_2 at the elbow, m_3 at the wrist, base rotor inertia I_1 about z.)

Solution

$c_2 = \cos q_2$, $c_{23} = \cos(q_2 + q_3)$, etc. Horizontal distance from the z axis and height of each mass: $\rho_2 = L_3c_2$, $h_2 = L_3s_2$; $\rho_3 = L_3c_2 + L_4c_{23}$, $h_3 = L_3s_2 + L_4s_{23}$ (plus the constant shoulder height).

Speed^2 of a point at (ρ, h) on the plane that spins at $\dot{q}_1$: $v^2 = \dot{\rho}^2 + \dot{h}^2 + \rho^2\dot{q}_1^2$.

$$ K = \tfrac{1}{2}I_1\dot{q}_1^2 + \tfrac{1}{2}m_2\left(L_3^2\dot{q}_2^2 + \rho_2^2\dot{q}_1^2\right) + \tfrac{1}{2}m_3\left(L_3^2\dot{q}_2^2 + L_4^2(\dot{q}_2+\dot{q}_3)^2 + 2L_3L_4c_3\dot{q}_2(\dot{q}_2+\dot{q}_3) + \rho_3^2\dot{q}_1^2\right) $$

$$ P = g\left[m_2L_3s_2 + m_3(L_3s_2 + L_4s_{23})\right], \qquad \mathcal{L} = K - P, \qquad Q_i = \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}_i} - \frac{\partial\mathcal{L}}{\partial q_i} $$

Result $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{V}(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q})$ with

$$ \mathbf{M} = \begin{bmatrix} I_1 + m_2\rho_2^2 + m_3\rho_3^2 & 0 & 0 \\ 0 & (m_2+m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3 & m_3(L_4^2 + L_3L_4c_3) \\ 0 & m_3(L_4^2 + L_3L_4c_3) & m_3L_4^2 \end{bmatrix} $$

$$ \mathbf{G} = g\begin{bmatrix} 0 \\ (m_2+m_3)L_3c_2 + m_3L_4c_{23} \\ m_3L_4c_{23} \end{bmatrix} $$

EXAM-STYLE QUESTION Practical sheet Q2 (cont.): velocity terms and torques

$$V_1 = -2\dot{q}_1\left[m_2\rho_2 L_3 s_2\dot{q}_2 + m_3\rho_3\left(L_3 s_2\dot{q}_2 + L_4 s_{23}(\dot{q}_2+\dot{q}_3)\right)\right]$$
$$V_2 = \dot{q}_1^2\left[m_2\rho_2 L_3 s_2 + m_3\rho_3(L_3 s_2 + L_4 s_{23})\right] - m_3 L_3 L_4 s_3\,(2\dot{q}_2\dot{q}_3 + \dot{q}_3^2)$$
$$V_3 = \dot{q}_1^2\, m_3\rho_3 L_4 s_{23} + m_3 L_3 L_4 s_3\,\dot{q}_2^2$$

$\dot{q}_1\dot{q}_2$ in V_1: Coriolis (raising the arm while spinning changes the inertia about z); $\dot{q}_1^2$ terms in V_2, V_3: centrifugal (spin flings the arm out); gravity loads joints 2, 3 only.
Prescribed trajectory: take $\mathbf{q}(t), \dot{\mathbf{q}}(t), \ddot{\mathbf{q}}(t)$ (cubic from W5), substitute at each t in $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{V} + \mathbf{G}$. This is inverse dynamics: the open-loop block of p. 11.
Check ($L_3 = 0.4$, $L_4 = 0.3$ m, $m_2 = 2$, $m_3 = 1$ kg, $I_1 = 0.5$, $g = 9.81$, $\mathbf{q} = (0°, 30°, 40°)$): $\mathbf{G} = (0,\ 11.201,\ 1.007)$ N m; $\mathbf{M} = \begin{bmatrix} 0.9416 & 0 & 0 \\ 0 & 0.7539 & 0.1819 \\ 0 & 0.1819 & 0.09 \end{bmatrix}$; at rest with $\ddot{\mathbf{q}} = (1, 0.5, -0.5)$: $\mathbf{Q} = (0.942,\ 11.487,\ 1.053)$ N m. $\mathbf{M}$ symmetric positive definite. [checked]

EXAM-STYLE QUESTION Practical sheet Q3: planar 3-link arm along a cubic [marks not published]

$L = (3, 2, 1)$ m. Start at rest at tool pose $(5, 1, 90°)$, stop at $(-3, 3, 180°)$. Describe the kinematics, choose a cubic trajectory (take $T = 2$ s), and compute the joint torques at $t = 0, 1, 2$ s with point masses 3, 2, 1 kg at the distal ends of the links (assume the arm moves in a vertical plane, gravity along $-y$).

Solution
Kinematics: tool pose in $\{0\}$: $x = L_1 c_1 + L_2 c_{12} + L_3 c_{123}$, $y = L_1 s_1 + L_2 s_{12} + L_3 s_{123}$, $\varphi = q_1 + q_2 + q_3$.
IK: wrist $= (x - L_3\cos\varphi,\ y - L_3\sin\varphi)$; $\cos q_2 = (w^2 - L_1^2 - L_2^2)/(2L_1L_2)$. Start: wrist $(5, 0)$, $\cos q_2 = 1$: stretched, $q_0 = (0°, 0°, 90°)$ (one solution). Goal: wrist $(-2, 3)$, $\cos q_2 = 0$; take $q_2 = +90°$: $q_1 = \text{atan2}(3, -2) - \text{atan2}(2, 3) = 123.69° - 33.69° = 90°$, $q_3 = 180° - 180° = 0$: $q_f = (90°, 90°, 0°)$. (Other elbow: $(157.38°, -90°, 112.62°)$.)
Cubic (rest to rest): $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$, $\Delta = (90°, 90°, -90°)$: $a_2 = (67.5, 67.5, -67.5)°/\text{s}^2$, $a_3 = (-22.5, -22.5, 22.5)°/\text{s}^3$. $\ddot{q}(0) = 6\Delta/T^2 = \pm 2.356$ rad/s^2, peak speed $1.5\Delta/T = 1.178$ rad/s at $t = 1$ s.

EXAM-STYLE QUESTION Practical sheet Q3 (cont.): joint torques

Dynamics: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{V} + \mathbf{G}$, built from $K = \sum \frac{1}{2} m_i |\dot{\mathbf{p}}_i|^2$ and $P = \sum m_i g y_i$ with $\mathbf{p}_i$ the distal end of link i (as for the 2R, with three tips). At $t = 0$ the arm is at rest: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{G}$, and $G_1 = 9.81(3\cdot 3 + 2\cdot 5 + 1\cdot 5) = 235.44$ N m.

t (s)	$\mathbf{q}$ (°)	Q_1	Q_2	Q_3 (N m)
0	(0, 0, 90)	548.814	160.176	2.356
1	(45, 45, 45)	70.817	20.781	5.078
2	(90, 90, 0)	−268.947	−141.712	−21.591

[checked] Sanity: at $t = 0$, $Q_1 = 548.8$ N m is the gravity term $G_1 = 235.44$ plus the inertia term $\mathbf{M}\ddot{\mathbf{q}}$ for the largest acceleration of the move. In closed loop (this week) these torques are the model-based part $\mathbf{M}\ddot{\mathbf{q}}_d + \mathbf{C} + \mathbf{g}$ of the controller; the servo part adds $\mathbf{M}(k_v\dot{E} + k_p E)$ on top.

EXAM-STYLE QUESTION Mock 1 Q7: partitioned control of one joint [marks assumed 7]

$m\ddot{x} + b\dot{x} + kx = f$, $m = 2$ kg, $b = 0.5$ N s/m, $k = 4$ N/m. (a) Write the partitioned law for trajectory following and the closed-loop error equation. (b) k_p, k_v for critical damping with $\omega_n = 8$ rad/s. (c) Steady-state error under a constant 5 N disturbance and how to remove it. (d) f at $x = 0.10$ m, $\dot{x} = -0.20$ m/s, $x_d = 0.15$ m, $\dot{x}_d = \ddot{x}_d = 0$.

Solution
(a) $f = \alpha f' + \beta$, $\alpha = m$, $\beta = b\dot{x} + kx$; $f' = \ddot{x}_d + k_v\dot{E} + k_p E$, $E = x_d - x$; then $\ddot{x} = f'$ and $\ddot{E} + k_v\dot{E} + k_p E = 0$ [3]
(b) $k_p = \omega_n^2 = 64$ s^{-2}, $k_v = 2\omega_n = 16$ s^{-1} [1]
(c) $E_{ss} = f_{dist}/(k_p m) = 5/(64\cdot 2) = 0.0391$ m, in the direction of the push. Add integral action; keep $k_i < k_v k_p = 1024$ [2]
(d) $E = 0.05$, $\dot{E} = 0.20$: $f' = 16(0.2) + 64(0.05) = 6.4$ m/s^2; $f = 2(6.4) + 0.5(-0.2) + 4(0.10) =$ $\boxed{13.1\text{ N}}$ [1]. Check: $(13.1 + 0.1 - 0.4)/2 = 6.4 = f'$. [checked]

EXAM-STYLE QUESTION Mock 2 Q7: PID gains and stability [marks assumed 7]

A partitioned joint uses $f' = \ddot{x}_d + k_v\dot{E} + k_p E + k_i\int E\,dt$. (a) With $k_p = 49$ s^{-2} choose k_v for critical damping of the PD part. (b) Derive the third-order error equation and the characteristic polynomial. (c) For which k_i is the loop stable? Is $k_i = 100$ stable? (d) Steady-state error under a constant disturbance?

Solution
(a) $k_v = 2\sqrt{49} = 14$ s^{-1} [1]
(b) $\dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_i E = \dot{f}_{dist}/m$; differentiate: $\dddot{E} + 14\ddot{E} + 49\dot{E} + k_i E = \dot{f}_{dist}/m$; polynomial $s^3 + 14s^2 + 49s + k_i$ [3]
(c) Routh: stable for $0 < k_i < k_v k_p = 14(49) = 686$. $k_i = 100 < 686$: stable; roots -10.140, $-1.930 \pm 2.477i$ [2]
(d) 0: at steady state $k_i E = 0$ [1]. [checked]

EXAM-STYLE QUESTION Practice Q8.4: motor, gearbox, armature current [marks assumed 6]

Gear ratio $\eta = 40$; $I_m = 3\times 10^{-5}$ kg m^2, $c_m = 2\times 10^{-4}$ N m s/rad; $I = 0.05$ kg m^2, $c = 0.2$ N m s/rad; $k_4 = 0.05$ N m/A. At the axle $\ddot{q} = 8$ rad/s^2, $\dot{q} = 3$ rad/s. Find Q and i_a.

Solution
$\eta^2 = 1600$. $I_{eff} = 0.05 + 1600(3\times 10^{-5}) = 0.098$ kg m^2; $c_{eff} = 0.2 + 1600(2\times 10^{-4}) = 0.52$ N m s/rad.
$Q = 0.098(8) + 0.52(3) = 0.784 + 1.56 = \boxed{2.344\text{ N m}}$; $Q_m = Q/\eta = 0.0586$ N m; $i_a = Q_m/k_4 = \boxed{1.172\text{ A}}$; motor speed 120 rad/s.
Check: $Q_m = I_m\eta\ddot{q} + c_m\eta\dot{q} + (I\ddot{q} + c\dot{q})/\eta = 0.0096 + 0.0240 + 0.0250 = 0.0586$ N m. Using η instead of η^2 gives 0.517 A (wrong). [checked]

Which law for which problem? [added]

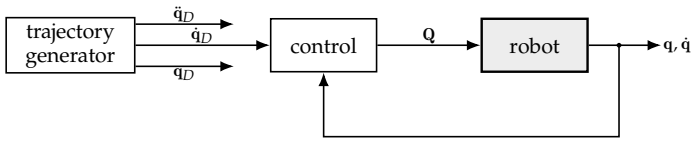
Law	needs (model)	needs (sensors, $\mathbf{J}$)	what it gives / when to use
joint PD / PID (W8)	none, or gravity only	$\mathbf{q}, \dot{\mathbf{q}}$	stable; fine for slow motion or highly geared joints; steady error without an integral
computed torque (partitioned) [p. 18–21]	$\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}$ exactly	$\mathbf{q}, \dot{\mathbf{q}}$	$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}$, decoupled; fast motion, good model, continuous law
computed torque, inexact [p. 27–28]	estimates $\hat{\mathbf{M}}, \hat{\mathbf{C}}, \hat{\mathbf{g}}, \hat{\mathbf{f}}$	$\mathbf{q}, \dot{\mathbf{q}}$	error eq. with a non-zero right side: small, not zero; the real case
Cartesian [p. 32–33]	$\mathbf{M}_x, \mathbf{c}_x, \mathbf{g}_x, \mathcal{K}$	$\mathbf{q}, \dot{\mathbf{q}} \to \mathbf{x}, \dot{\mathbf{x}}$; $\mathbf{J}$	same form in $\mathbf{x}$; no IK or $\mathbf{J}^{-1}$ along the path; fails near a singularity
force [p. 36–43]	m (and k_e)	force sensor f_e; $\dot{x}$	Option 2: $E_{ss} = f_{dist}/(1 + mk_e^{-1}k_p)$; pushing, polishing, assembly
hybrid [p. 47]	as Cartesian, plus force law	$\mathbf{x}$, $\mathcal{F}_E$, the constraints	each axis obeys exactly one law; surface or hole tasks

If the question says…

"write the control law / partitioned law" ..... recipe, p. 111
"closed-loop error equation" ..... §Computed torque (exact) or §Imprecise model
"the mass is wrong by 10 %" ..... p. 114
"choose K_p, K_v" / critical damping ..... p. 113
"Cartesian controller, joint torques from a tool force" ..... p. 115
"force control against a wall / steady-state force error" ..... p. 116
"which axes are force / position controlled" ..... p. 117

Recap and roadmap [p. 5–6]

Week 8 built the pipeline drawn below: a *trajectory generator* turns the task into $\mathbf{q}_D(t), \dot{\mathbf{q}}_D(t), \ddot{\mathbf{q}}_D(t)$; a *control* block turns these and the measured $\mathbf{q}, \dot{\mathbf{q}}$ into joint torques $\mathbf{Q}$; the *robot* moves. Today's questions [p. 6]: (1) non-linear systems; (2) one combined law for an n-joint manipulator; (3) force control; (4) force *and* position control.



[drawn for this booklet] slide p. 5; every line is an $n \times 1$ vector for an n-joint robot [p. 17]

Simple non-linear systems: the cubic spring [p. 7–11]

PREMISE partitioned control, scalar **Given:** the equation of motion of one joint or mass, e.g. $m\ddot{x} + b\dot{x} + kx^3 = f$; a reference $x_d(t), \dot{x}_d, \ddot{x}_d$; measured $x, \dot{x}$
Find: the actuator force $f(t)$ so that $x \to x_d$ with error dynamics you choose

Linear case [p. 8]**.** $m\ddot{x} + b\dot{x} + kx = f$ (mass m, damper b, spring k) is a linear second-order ODE; W8 gave control laws f that make it behave (e.g. critically damped). **Non-linear spring** [p. 9]**.** Replace the spring force by $f_{spring} = kx^3$: $m\ddot{x} + b\dot{x} + kx^3 = f$ is a second-order *non-linear* ODE with constant coefficients; the linear W8 gains no longer give a clean closed loop.

Partitioned (computed-torque) law, scalar [p. 10]

$$f = \alpha\acute{f} + \beta, \qquad \alpha = m, \quad \beta = b\dot{x} + kx^3, \qquad \acute{f} = \ddot{x}_d + k_v\dot{E} + k_pE, \quad E = x_d - x$$

SYMBOLS

f	actuator force on the mass	N
α	scaling: the coefficient of $\ddot{x}$ in the plant (here the mass)	kg
β	everything else in the plant: damper and non-linear spring, evaluated with the *measured* $x, \dot{x}$	N
$\acute{f}$	commanded acceleration ("f with an acute accent"): output of the servo law	$\mathrm{m/s^2}$
$x_d, \dot{x}_d, \ddot{x}_d$	desired position, velocity, acceleration (from the trajectory generator)	m, m/s, $\mathrm{m/s^2}$
$E, \dot{E}$	position and velocity error, $E = x_d - x$, $\dot{E} = \dot{x}_d - \dot{x}$	m, m/s
k_p, k_v	position and velocity gains of the servo law	$\mathrm{s^{-2}}$, $\mathrm{s^{-1}}$

Use: any plant you can write as (coefficient)$\ddot{x}$ + (rest) = f. α is the coefficient of $\ddot{x}$, β the rest, with its sign.

DERIVATION closed loop of the cubic spring, every line

Plant: $m\ddot{x} + b\dot{x} + kx^3 = f$. Insert the law $f = m\acute{f} + b\dot{x} + kx^3$:

$$m\ddot{x} + b\dot{x} + kx^3 = m\acute{f} + b\dot{x} + kx^3 \;\Rightarrow\; m\ddot{x} = m\acute{f} \;\Rightarrow\; \ddot{x} = \acute{f} = \ddot{x}_d + k_v\dot{E} + k_pE.$$

The damper and the cubic spring cancel exactly. With $\ddot{E} = \ddot{x}_d - \ddot{x}$:

$$\boxed{\ddot{E} + k_v\dot{E} + k_pE = 0}$$

a *linear* second-order equation, whatever the non-linearity. Roots $s^2 + k_vs + k_p = 0$: both in the left half-plane for $k_p, k_v > 0$.

STEPS Write a partitioned law [p. 15]

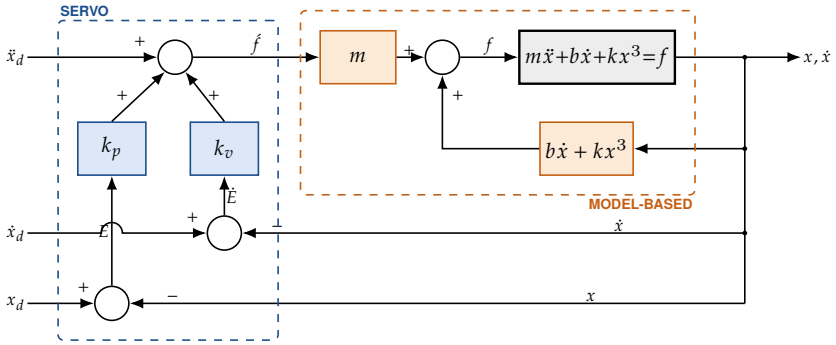
1. Write the equation of motion in the form (coefficient) $\ddot{q}$ + (rest) = input (Lagrange for a link, §p. 13).
2. α = coefficient of the acceleration; β = the rest, with signs, written with the measured state.
3. Servo law $\acute{f} = \ddot{q}_d + k_v\dot{E} + k_pE$, with $E = q_d - q$.
4. Law: input = $\alpha\acute{f} + \beta$. Closed loop: $\ddot{E} + k_v\dot{E} + k_pE = 0$.
5. Check: units of $\alpha\acute{f}$ equal units of the input; at rest on the path the law must reduce to β (e.g. $mgL\cos q$).

WORKED cubic spring, slide structure; $m = 2, b = 4, k = 50, k_p = 100, k_v = 20$

State $x = 0.10$ m, $\dot{x} = 0.30$ m/s; reference $x_d = 0.12$, $\dot{x}_d = 0.40$, $\ddot{x}_d = 0.50$. $E = 0.02$, $\dot{E} = 0.10$. $\acute{f} = 0.50 + 20(0.10) + 100(0.02) = 4.5\ \mathrm{m/s^2}$; $\beta = 4(0.30) + 50(0.10)^3 = 1.25$ N; $f = 2(4.5) + 1.25 = 10.25$ N. Check: plant $\ddot{x} = (10.25 - 1.25)/2 = 4.5 = \acute{f}$ ✓. [checked]

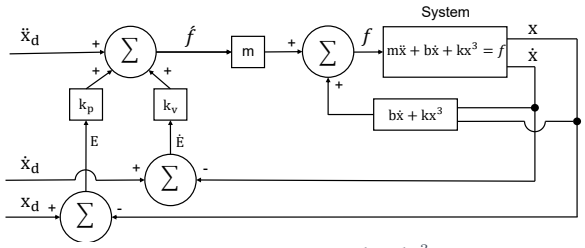
TRAP

$\acute{f}$ is an *acceleration*; writing $f = \acute{f} + \beta$ (no α) mixes $\mathrm{m/s^2}$ and N. $\ddot{x}_d$ belongs in $\acute{f}$, never in β. $E = x_d - x$, not $x - x_d$.



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

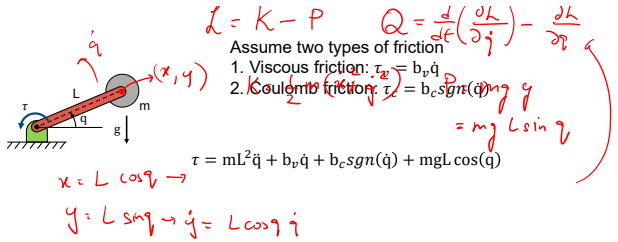
[drawn for this booklet] non-linear SISO partitioned controller (slide p. 11, redrawn). $\ddot{x}_d$ is a feed-forward; $\alpha = m$ scales f; β is fed back from the measured $x, \dot{x}$ and *added* to give f. The plant block is the slide's "System".



[W09 lec p. 11] — the slide's own block diagram (the slide's "m" and "$b\dot{x} + kx^3$" blocks are the orange ones above)

Single link with viscous and Coulomb friction [p. 12–14]

PREMISE single-link computed torque **Given:** link of length L, point mass m at the tip, joint angle q from the horizontal, gravity g acting down, friction b_v, b_c; reference $q_d, \dot{q}_d, \ddot{q}_d$
Find: joint torque τ



[W09 lec p. 13] — slide p. 13: link of length L, mass m at (x, y), angle q from the horizontal, gravity down; the handwriting is the Lagrange derivation below

DERIVATION equation of motion, as handwritten on p. 13

Frictions [p. 12]: viscous $\tau_v = b_v\dot{q}$, Coulomb $\tau_c = b_c\,\mathrm{sgn}(\dot{q})$. Position of the mass: $x = L\cos q$, $y = L\sin q$, so $\dot{x} = -L\sin q\,\dot{q}$, $\dot{y} = L\cos q\,\dot{q}$.

$$K = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \tfrac{1}{2}mL^2\dot{q}^2, \qquad P = mgy = mgL\sin q, \qquad \mathcal{L} = K - P.$$

Lagrange (slide): $Q = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q}$. Here $\frac{\partial \mathcal{L}}{\partial \dot{q}} = mL^2\dot{q}$, so $\frac{d}{dt}(\cdot) = mL^2\ddot{q}$, and $\frac{\partial \mathcal{L}}{\partial q} = -mgL\cos q$.
Adding the two friction torques:

$$\boxed{\tau = mL^2\ddot{q} + b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q} \quad [p. 13]$$

Control law of the single link [p. 14]

$$\tau = \alpha\acute{\tau} + \beta, \quad \alpha = mL^2, \quad \beta = b_v\dot{q} + b_c\,\mathrm{sgn}(\dot{q}) + mgL\cos q, \quad \acute{\tau} = \ddot{q}_d + k_v\dot{E} + k_pE$$

SYMBOLS

τ	joint torque	N m
m, L	tip mass, link length	kg, m
$\alpha = mL^2$	moment of inertia about the joint	kg m^2
b_v, b_c	viscous coefficient; Coulomb friction torque	N m s/rad; N m
$q, \dot{q}$	joint angle (from the horizontal), rate	rad, rad/s
$\acute{\tau}$	commanded joint acceleration	rad/s^2

Use: closed loop $\ddot{E} + k_v\dot{E} + k_pE = 0$. At rest on the path: $\tau = mgL\cos q$ (largest horizontal, zero when vertical).

WORKED $m = 2, L = 0.5, b_v = 0.3, b_c = 0.2, k_p = 64, k_v = 16$; radians

State $q = 30°$, $\dot{q} = 1$ rad/s, $\ddot{q}_d = 2$, $E = 0.05$, $\dot{E} = -0.1$. $\alpha = 2(0.25) = 0.5$. $\acute{\tau} = 2 + 16(-0.1) + 64(0.05) = 3.6$. $\beta = 0.3(1) + 0.2(+1) + 2(9.81)(0.5)\cos 30° = 0.5 + 8.4957 = 8.9957$. $\tau = 0.5(3.6) + 8.9957 = 10.7957$ N m. Plant check: $\ddot{q} = (\tau - \beta)/\alpha = 1.8/0.5 = 3.6 = \acute{\tau}$ ✓. [checked]

TRAP

Degrees in $\cos q$ on a radian calculator; sgn takes the sign of $\dot{q}$ (not q, not E); $\alpha = mL$ instead of mL^2; $\cos q$ dropped from β. If the link has its own inertia, mL^2 becomes the total inertia about the pivot. $\mathrm{sgn}(\dot{q})$ jumps at $\dot{q} = 0$: real controllers smooth it [added].

General procedure and its limits [p. 15]

The slide's three steps: (1) determine the equation of motion; (2) compute a non-linear *model-based* part that cancels the non-linearities; (3) the system is then linear and a linear *servo* law controls it. **Limit** [p. 15]**:** you must know the *parameters* (k, b_c, …) and the *structure* (cubic spring, Coulomb friction, …). "In practice, these are not always known": see §Imprecise model.

MIMO partitioning: n joints [p. 16–21]

PREMISE computed torque, n joints **Given:** equation of motion $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}(\dot{\mathbf{q}})$ (all $n \times 1$ except $\mathbf{M}$, $n \times n$); reference $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$; measured $\mathbf{q}, \dot{\mathbf{q}}$
Find: joint torque vector $\mathbf{Q}$ ($n \times 1$) that makes every joint error decay as you choose

W8 split the problem into n independent single-joint loops [p. 17]. Now each line of the diagram is an $n \times 1$ vector: positions, velocities, accelerations in; torques out [p. 18]. The model-based part is $\mathbf{Q} = \boldsymbol{\alpha}\acute{\mathbf{Q}} + \boldsymbol{\beta}$ with shapes $n{\times}1 = n{\times}n \cdot n{\times}1 + n{\times}1$.

Computed-torque law of a manipulator [p. 18–19]

$$\mathbf{Q} = \boldsymbol{\alpha}\dot{\mathbf{Q}} + \boldsymbol{\beta}, \quad \boldsymbol{\alpha} = \mathbf{M}(\mathbf{q}), \quad \boldsymbol{\beta} = \mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}(\dot{\mathbf{q}}), \quad \dot{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}, \quad \mathbf{E} = \mathbf{q}_d - \mathbf{q}$$

SYMBOLS

$\mathbf{Q}$	joint torques (N m for a revolute joint, N for a prismatic one), $n \times 1$	N m, N
$\mathbf{M}(\mathbf{q})$	inertia matrix, $n \times n$, symmetric positive definite	kg m^2
$\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$	Coriolis and centripetal torques, $n \times 1$	N m
$\mathbf{g}(\mathbf{q})$	gravity torques, $n \times 1$	N m
$\mathbf{f}(\dot{\mathbf{q}})$	friction torques (the slides write $\mathbf{f}(\dot{\mathbf{q}})$ on p. 19 and $\hat{\mathbf{f}}(\mathbf{q},\dot{\mathbf{q}})$ on p. 27)	N m
$\dot{\mathbf{Q}}$	commanded joint accelerations, $n \times 1$	rad/s^2
$\mathbf{K}_p, \mathbf{K}_v$	position, velocity gain matrices, *diagonal*	s^{-2}, s^{-1}
$\mathbf{E}, \dot{\mathbf{E}}$	joint position and velocity error	rad, rad/s

*Use: $\boldsymbol{\alpha}$ is chosen so that it decouples the n equations; it is **not** diagonal. $\mathbf{K}_v, \mathbf{K}_p$ are usually diagonal [p. 18].*

DERIVATION closed loop, every line [p. 21]

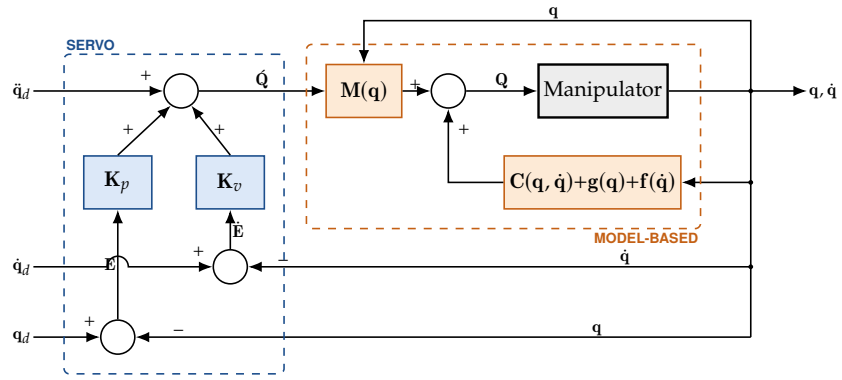
Plant: $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} = \mathbf{Q}$. Insert $\mathbf{Q} = \mathbf{M}\dot{\mathbf{Q}} + \mathbf{C} + \mathbf{g} + \mathbf{f}$:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} = \mathbf{M}\dot{\mathbf{Q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} \;\Rightarrow\; \mathbf{M}\ddot{\mathbf{q}} = \mathbf{M}\dot{\mathbf{Q}} \;\Rightarrow\; \ddot{\mathbf{q}} = \dot{\mathbf{Q}}$$

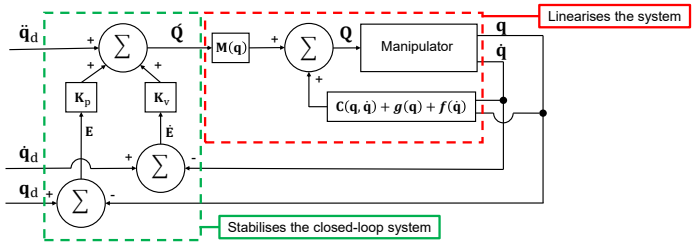
($\mathbf{M}$ is invertible, being positive definite). With $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d - \ddot{\mathbf{E}}$ and the servo law:

$$\boxed{\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}} \quad \Rightarrow \quad \ddot{e}_i + k_{vi}\dot{e}_i + k_{pi}e_i = 0 \;\; (i = 1, \ldots, n).$$

Because $\mathbf{K}_p, \mathbf{K}_v$ are diagonal this is n separate, linear, *decoupled* equations [p. 21]. The off-diagonal terms of $\mathbf{M}$ in $\boldsymbol{\alpha}$ are what cancel the inertial coupling $M_{ij}\ddot{q}_j$ of the plant.



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)
[drawn for this booklet] MIMO computed-torque controller (slide p. 20, redrawn). The slide's **red** dashed box "linearises the system" = the **orange** model-based group here; the slide's **green** dashed box "stabilises the closed-loop system" = the **blue** servo group. The arrow from $\mathbf{q}$ into $\mathbf{M}(\mathbf{q})$ is added (the slide omits it).



[W09 lec p. 20] — slide p. 20: red box "Linearises the system", green box "Stabilises the closed-loop system"

Reading the diagram. Inside the orange group $\mathbf{Q} = \mathbf{M}\dot{\mathbf{Q}} + \boldsymbol{\beta}$ and the robot is a copy of the same equation, so the combination from $\dot{\mathbf{Q}}$ to $\mathbf{q}$ is n double integrators ($\ddot{\mathbf{q}} = \dot{\mathbf{Q}}$): the non-linear coupled arm has been *linearised*. The blue group closes a PD loop around those integrators: it *stabilises*.

ALTERNATIVE APPROACHES

A. Independent joint PD (W8, p. 17)	B. Computed torque (this week)
$\tau_i = k_{pi}e_i + k_{vi}\dot{e}_i$ (plus gravity). Each joint is its own loop; other joints act as unmodelled disturbances. No model needed. Tracking error grows with speed and coupling.	$\mathbf{Q} = \mathbf{M}\dot{\mathbf{Q}} + \mathbf{C} + \mathbf{g} + \mathbf{f}$ with the full matrix $\mathbf{M}$. Needs the model. With an exact model $\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \mathbf{0}$ along the whole path, fast or slow.

WORKED $\boldsymbol{\alpha}$ is not diagonal: practical 3-link arm at $\mathbf{q}_0 = (0, 0, 90°)$

$\mathbf{M}(\mathbf{q}_0) = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ kg m^2 (derived in the practical, p. 118). To command $\dot{\mathbf{Q}} = (1, 0, 0)$ rad/s^2 the law needs $\boldsymbol{\alpha}\dot{\mathbf{Q}} = (103, 31, 1)$ N m: the 31 and 1 N m keep joints 2 and 3 *still* while joint 1 accelerates. A diagonal $\boldsymbol{\alpha}$ would apply only 103 and let joints 2, 3 swing. [checked]

STEPS Choose $\mathbf{K}_p, \mathbf{K}_v$ for critical damping, joint by joint [added]

1. Each joint: $\ddot{e}_i + k_{vi}\dot{e}_i + k_{pi}e_i = 0$, characteristic polynomial $s^2 + k_v s + k_p$.
2. Critical damping: double root $s = -\omega_n$: $k_p = \omega_n^2$, $k_v = 2\omega_n$ (equal roots when $k_v^2 = 4k_p$).
3. Pick ω_n from a settling time: $e = e_0(1 + \omega_n t)e^{-\omega_n t}$ reaches 2 % at $\omega_n t = 5.83$ and 5 % at 4.74. [checked]
4. Write $\mathbf{K}_p = \mathrm{diag}(\omega_i^2)$, $\mathbf{K}_v = \mathrm{diag}(2\omega_i)$. Units: s^{-2} and s^{-1} (they act on accelerations, so they do not depend on the inertia).
5. Check: roots of $s^2 + k_v s + k_p$ equal and negative; no overshoot.

WORKED gains from a settling time

Joint 1 must settle to 2 % in 0.5 s: $\omega_n \geq 5.83/0.5 = 11.67$; take $\omega_n = 12$: $k_p = 144$ s^{-2}, $k_v = 24$ s^{-1} ($24^2 = 4 \cdot 144$ ✓). [checked] For three joints with $\omega = (10, 15, 20)$ rad/s: $\mathbf{K}_p = \mathrm{diag}(100, 225, 400)$, $\mathbf{K}_v = \mathrm{diag}(20, 30, 40)$; 2 % settling 0.58, 0.39, 0.29 s. [checked] With $k_p = 100, k_v = 20$ and $e(0) = 0.1$ rad: $e(0.1\,\mathrm{s}) = 0.1(1 + 1)e^{-1} = 0.0736$ rad. [checked]

TRAP

"$\mathbf{K}_p, \mathbf{K}_v$ diagonal $\Rightarrow$ $\boldsymbol{\alpha}$ diagonal": false, $\boldsymbol{\alpha} = \mathbf{M}$ is full. "Guarantees $\ddot{\mathbf{E}} + \cdots = \mathbf{0}$ for any robot": only for an exact model and a continuous law. Gains of the *outer-loop* scheme (p. 25) are torque gains, not these.

Practical issue 1: an imprecise model [p. 22, 26–28]

Two practical issues [p. 22]: (1) the model $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}$ may be inaccurate; (2) the law runs on a computer in discrete time. Why the model is wrong [p. 26]: friction is hard to model; the robot wears

down; grasped objects change the weight distribution.

Law and closed loop with hats [p. 27–28]

$$\mathbf{Q} = \boldsymbol{\alpha}\hat{\mathbf{Q}} + \boldsymbol{\beta}, \quad \boldsymbol{\alpha} = \hat{\mathbf{M}}(\mathbf{q}), \quad \boldsymbol{\beta} = \hat{\mathbf{C}}(\mathbf{q},\dot{\mathbf{q}}) + \hat{\mathbf{g}}(\mathbf{q}) + \hat{\mathbf{f}}(\mathbf{q},\dot{\mathbf{q}}), \quad \hat{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$$

$$\ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \hat{\mathbf{M}}^{-1}(\mathbf{q})\Big[(\mathbf{M}(\mathbf{q}) - \hat{\mathbf{M}}(\mathbf{q}))\ddot{\mathbf{q}} + (\mathbf{C} - \hat{\mathbf{C}}) + (\mathbf{g} - \hat{\mathbf{g}}) + (\mathbf{f} - \hat{\mathbf{f}})\Big]$$

SYMBOLS

hat ($\hat{\cdot}$)	the controller's estimate of a model quantity; no hat = the true robot	–
$\mathbf{M} - \hat{\mathbf{M}}$	inertia error, multiplies the *actual* acceleration $\ddot{\mathbf{q}}$	$\mathrm{kg\,m^2}$
$\mathbf{C} - \hat{\mathbf{C}}$, $\mathbf{g} - \hat{\mathbf{g}}$, $\mathbf{f} - \hat{\mathbf{f}}$	Coriolis, gravity, friction errors	N m

Use: the right-hand side is a forcing term: if it is non-zero the error does not decay to zero. It vanishes only when every hatted quantity equals the true one.

DERIVATION of the error equation [added]: the slide prints only the result

True plant: $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} + \mathbf{f} = \mathbf{Q}$. Law: $\mathbf{Q} = \hat{\mathbf{M}}\hat{\mathbf{Q}} + \hat{\mathbf{C}} + \hat{\mathbf{g}} + \hat{\mathbf{f}}$. Equate and collect:

$$\hat{\mathbf{M}}\hat{\mathbf{Q}} = \mathbf{M}\ddot{\mathbf{q}} + (\mathbf{C} - \hat{\mathbf{C}}) + (\mathbf{g} - \hat{\mathbf{g}}) + (\mathbf{f} - \hat{\mathbf{f}}) \;\Rightarrow\; \hat{\mathbf{Q}} = \hat{\mathbf{M}}^{-1}\big[\mathbf{M}\ddot{\mathbf{q}} + \boldsymbol{\Delta}\big], \quad \boldsymbol{\Delta} = (\mathbf{C} - \hat{\mathbf{C}}) + (\mathbf{g} - \hat{\mathbf{g}}) + (\mathbf{f} - \hat{\mathbf{f}}).$$

Subtract $\ddot{\mathbf{q}}$ from both sides, using $\hat{\mathbf{Q}} - \ddot{\mathbf{q}} = (\ddot{\mathbf{q}}_d - \ddot{\mathbf{q}}) + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E} = \ddot{\mathbf{E}} + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$ and $\hat{\mathbf{M}}^{-1}\mathbf{M}\ddot{\mathbf{q}} - \ddot{\mathbf{q}} = \hat{\mathbf{M}}^{-1}(\mathbf{M} - \hat{\mathbf{M}})\ddot{\mathbf{q}}$: the boxed equation. [checked]

Single link, only the mass wrong [added]

With $\hat{m}$ in both $\hat{\alpha} = \hat{m}L^2$ and $\hat{\beta} = \hat{m}gL\cos q$ (no friction):

$$\ddot{e} + k_v\dot{e} + k_p e = \frac{m - \hat{m}}{\hat{m}}\left[\ddot{q} + \frac{g}{L}\cos q\right], \qquad e_{ss} \approx \frac{m - \hat{m}}{\hat{m}}\,\frac{g}{L}\,\frac{\cos q}{k_p} \quad \text{(held at } q\text{)}$$

Use: $\hat{m} < m$: gravity is under-compensated, the link sags, $e = q_d - q > 0$. $e_{ss} = 0$ at $q = 90°$ (no lever arm). If every model quantity is scaled by λ: right side $= \frac{1-\lambda}{\lambda}\left[\ddot{\mathbf{q}} + \mathbf{M}^{-1}(\mathbf{C} + \mathbf{g} + \mathbf{f})\right]$. [checked]

WORKED scenario: the mass is 10 % off; $m = 2$ kg, $L = 0.5$ m, $k_p = 100$, $k_v = 20$, $g/L = 19.62$ $\mathrm{s^{-2}}$

Estimate 10 % low, $\hat{m} = 1.8$ kg: $(m - \hat{m})/\hat{m} = 0.2/1.8 = 1/9$. Held horizontal ($q_d = 0$): forcing $= \frac{1}{9}(19.62) = 2.18$ rad/s^2; $e_{ss} = 2.18/100 = 0.0218$ rad $= 1.25°$. At $q_d = 30°$: 0.0189 rad from the formula with $\cos q_d$ (simulation 0.0191 rad, 1.09°). At 90°: 0. **Estimate 10 % high**, $\hat{m} = 2.2$: ratio $-0.2/2.2 = -1/11$, $e_{ss} = -0.0178$ rad $= -1.02°$ (over-compensated, the link rises above the reference). [checked]

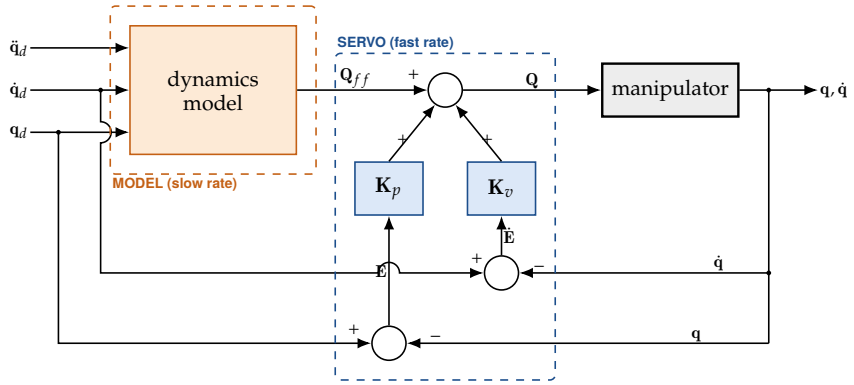
What remains: the error is *not* zero and does not decay; it is about (forcing)$/k_p$. Raising k_p to 1000 cuts it ten-fold (2.18 mrad) but is limited by the sampling rate (next section); integral action (W8) removes the constant part [added]. During motion the term $\frac{1}{9}\ddot{q}$ adds a transient. The error stays small because the *ratio* $(m - \hat{m})/\hat{m}$ is small.

SLIDE CHECK p. 19 vs p. 27

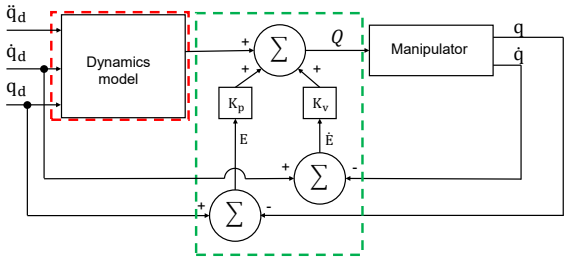
Friction is written $\mathbf{f}(\dot{\mathbf{q}})$ on p. 19 and $\hat{\mathbf{f}}(\mathbf{q},\dot{\mathbf{q}})$ in $\hat{\boldsymbol{\beta}}$ on pp. 27–28 and on the practical sheet. Treat it as $\mathbf{f}(\mathbf{q},\dot{\mathbf{q}})$ in general. Notation only; no change to the algebra.

Practical issue 2: discretisation, model in the outer loop [p. 22–25]

The controller runs on a computer [p. 23]: sensors are read at discrete times, actuator commands update at discrete times, and *computations take finite time*, which limits how often torques are refreshed. The model-based part (the dynamics $\mathbf{M}, \mathbf{C}, \mathbf{g}$, W7) is slow; the servo part (an error and a multiplication by gains) is fast [p. 24]. **Fix [p. 25]:** move the model-based part to the *outer loop*. It is evaluated from the *planned* trajectory $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$ at a slower rate (it can even be tabulated before the motion); the fast inner loop only adds the PD correction $\mathbf{K}_p\mathbf{E} + \mathbf{K}_v\dot{\mathbf{E}}$ from the measured $\mathbf{q}, \dot{\mathbf{q}}$.



[drawn for this booklet] model in the outer loop (slide p. 25, redrawn; the $\dot{\mathbf{q}}_d$ line hops over the $\mathbf{q}_d$ line with no connection). The slide labels the sum output Q and draws no label on the dynamics output; $\mathbf{Q}_{ff}$ is my name for it. The dynamics model reads *only* the plan.



[W09 lec p. 25] — slide p. 25: red = dynamics model (slow), green = servo (fast)

SLIDE CHECK p. 25: units of $\mathbf{K}_p, \mathbf{K}_v$

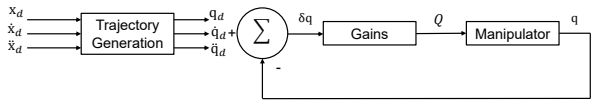
In this scheme the servo terms are added *directly to the torque* $\mathbf{Q}$, so $\mathbf{K}_p, \mathbf{K}_v$ are torque gains (N m/rad, N m s/rad), not the accelerations gains ($\mathrm{s^{-2}}$, $\mathrm{s^{-1}}$) of p. 20. They are no longer scaled by $\mathbf{M}$, so the error is only approximately decoupled. The slide does not say this; I add it. E.g. link 1 of the practical ($\alpha = 27$ kg m^2): equal behaviour needs $27 \cdot 100 = 2700$ N m/rad and $27 \cdot 20 = 540$ N m s/rad. [checked]

Numbers (site check script). Practical link 1 ($\alpha = 27$, $\beta = 88.29\cos q$, $k_p = 100$, $k_v = 20$), exact model, cubic $0 \to \pi/2$ in $T = 2$ s: continuous law $|e| < 10^{-10}$ rad; torque refreshed every 10 ms: peak error 0.26 mrad; every 50 ms: 1.13 mrad. The slower the refresh, the larger the error. [checked]

Cartesian-space control [p. 29–33]

PREMISE Cartesian controller **Given:** a desired tool path $\mathbf{x}_d(t), \dot{\mathbf{x}}_d, \ddot{\mathbf{x}}_d$ in the base frame $\{0\}$; the arm's FK $\mathcal{K}$, Jacobian $\mathbf{J}(\mathbf{q})$ and dynamics
Find: joint torques $\mathbf{Q}$ that make the *tool* follow the path

The joint-space route [p. 30]. A trajectory generator converts $\mathbf{x}_d, \dot{\mathbf{x}}_d, \ddot{\mathbf{x}}_d$ into $\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d$, then a joint-space servo ("Gains") uses $\delta\mathbf{q} = \mathbf{q}_d - \mathbf{q}$.



[W09 lec p. 30] — slide p. 30: Cartesian reference converted to joint space by a trajectory generator

Cost of that conversion [p. 31]

$$\mathbf{q} = \mathcal{K}^{-1}(\mathbf{x}), \qquad \dot{\mathbf{q}} = \mathbf{J}^{-1}(\mathbf{q})\,\dot{\mathbf{x}}, \qquad \ddot{\mathbf{q}} = \tfrac{d}{dt}\big(\mathbf{J}^{-1}(\mathbf{q})\big)\dot{\mathbf{x}} + \mathbf{J}^{-1}(\mathbf{q})\ddot{\mathbf{x}}$$

Use: "All these are computationally intensive" [p. 31]: inverse kinematics (possibly multi-valued or iterative), $\mathbf{J}^{-1}$ and its derivative at every point of the path, and $\mathbf{J}^{-1}$ does not exist at a singularity.

SLIDE CHECK p. 31

The slide prints $\dot{J}^{-1}(\mathbf{q})\,\dot{\mathbf{x}}$ (a dot over J). The correct derivative is $\frac{d}{dt}(\mathbf{J}^{-1}) = -\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}$, so $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}})$ (from $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$, $\ddot{\mathbf{x}} = \mathbf{J}\ddot{\mathbf{q}} + \dot{\mathbf{J}}\dot{\mathbf{q}}$). Checked numerically. [checked]

Alternative [p. 32]**:** write the equation of motion *in Cartesian space*, solve for the Cartesian control force $\mathcal{F}$ for the desired tool trajectory, and only then convert to torques.

Cartesian computed-torque law [p. 32–33]

$$\mathcal{F} = \mathbf{M}_x(\mathbf{q})\,\ddot{\mathbf{x}} + \mathbf{c}_x(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{g}_x(\mathbf{q}) \;\text{(plant)}, \qquad \mathbf{Q} = \mathbf{J}^{\top}(\mathbf{q})\,\mathcal{F}$$

$$\mathcal{F} = \mathbf{M}_x(\mathbf{q})\big(\ddot{\mathbf{x}}_d + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x\big) + \mathbf{c}_x + \mathbf{g}_x \;\text{(control law)}$$

SYMBOLS

$\mathcal{F}$	Cartesian force (and moment) the tool must exert, in $\{0\}$	N, N m
$\mathbf{x}, \dot{\mathbf{x}}$	tool position (and orientation), velocity: $\mathbf{x} = \mathcal{K}(\mathbf{q})$, $\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$	m, m/s
$\mathbf{M}_x(\mathbf{q})$	inertia seen at the tool	kg
$\mathbf{c}_x, \mathbf{g}_x$	velocity-dependent and gravity terms at the tool	N
$\mathbf{E}_x, \dot{\mathbf{E}}_x$	$\mathbf{x}_d - \mathbf{x}$, $\dot{\mathbf{x}}_d - \dot{\mathbf{x}}$	m, m/s
$\mathbf{K}_p, \mathbf{K}_v$	Cartesian gains	s^{-2}, s^{-1}
$\mathbf{J}^{\top}(\mathbf{q})$	transpose of the Jacobian: maps $\mathcal{F}$ to joint torques	m (N m per N)

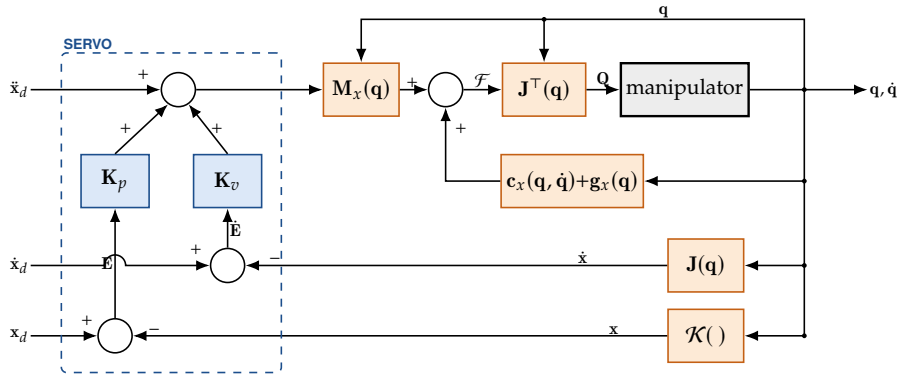
Use: the task is given as a path of the tool. The loop closes on $\mathbf{x}$ (from $\mathcal{K}$) and $\dot{\mathbf{x}}$ (from $\mathbf{J}\dot{\mathbf{q}}$): no IK, no $\mathbf{J}^{-1}$ along the path, and the closed loop is $\ddot{\mathbf{E}}_x + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x = \mathbf{0}$. $\mathbf{M}_x$ blows up near a singularity.

DERIVATION where $\mathbf{M}_x, \mathbf{c}_x, \mathbf{g}_x$ and $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ come from [added]

Statics (virtual work). A small joint motion $\delta\mathbf{q}$ moves the tool by $\delta\mathbf{x} = \mathbf{J}\delta\mathbf{q}$. Work of the torques = work of the tool force: $\mathbf{Q}^{\top}\delta\mathbf{q} = \mathcal{F}^{\top}\mathbf{J}\delta\mathbf{q}$ for every $\delta\mathbf{q}$, so $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$.
Dynamics. Put $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ into $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C} + \mathbf{g} = \mathbf{Q}$ and use $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}})$ (no friction):

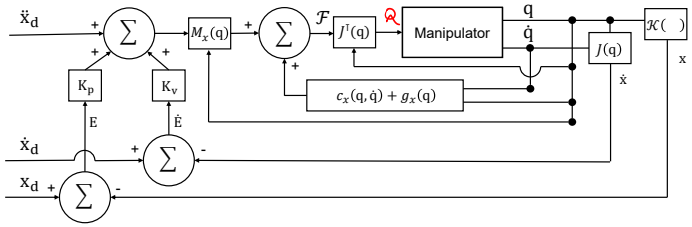
$$\mathcal{F} = \underbrace{\mathbf{J}^{-\top}\mathbf{M}\mathbf{J}^{-1}}_{\mathbf{M}_x}\ddot{\mathbf{x}} + \underbrace{\mathbf{J}^{-\top}\big(\mathbf{C} - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\dot{\mathbf{q}}\big)}_{\mathbf{c}_x} + \underbrace{\mathbf{J}^{-\top}\mathbf{g}}_{\mathbf{g}_x}.$$

Needs $\mathbf{J}$ square and invertible. [checked]



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] Cartesian-based controller (slide p. 33, redrawn). Orange here also includes the kinematic model ($\mathcal{K}, \mathbf{J}, \mathbf{J}^{\top}$) besides $\mathbf{M}_x, \mathbf{c}_x, \mathbf{g}_x$.
Feedback: $\mathbf{q} \to \mathcal{K} \to \mathbf{x}$ and $\dot{\mathbf{q}} \to \mathbf{J} \to \dot{\mathbf{x}}$. The slide's red "$Q$" labels the torque after $\mathbf{J}^{\top}$. $\mathbf{M}_x, \mathbf{J}^{\top}$ and $\mathbf{J}$ also read $\mathbf{q}$ (line added).



[W09 lec p. 33] — slide p. 33: the Cartesian controller

WORKED $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$: planar 2R, $l_1 = l_2 = 1$ m, $\mathbf{q} = (30°, 60°)$

$\mathbf{J} = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} = \begin{bmatrix} -1.5 & -1 \\ 0.8660 & 0 \end{bmatrix}$ ($s_{12} = 1$, $c_{12} = 0$). $\mathcal{F} = (3, -4)$ N: $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F} = (-1.5(3) + 0.8660(-4),\ -1(3) + 0) = (-7.9641, -3.0)$ N m. Check by moments: tool at $(0.8660, 1.5)$: $xF_y - yF_x = 0.8660(-4) - 1.5(3) = -7.9641$; about joint 2 the tool is at $(0, 1)$ from it: $0(-4) - 1(3) = -3$. [checked]

WORKED Cartesian law gives the same torques as joint space; point masses $m_1 = m_2 = 1$ kg at the link ends, same pose, at rest

$\mathbf{M} = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 1 \end{bmatrix}$ kg m^2, $\mathbf{g} = g(2c_1 + c_{12},\ c_{12}) = (16.991,\ 0)$ N m; $\det\mathbf{J} = l_1 l_2 s_2 = 0.866 \neq 0$. $\mathbf{M}_x = \mathbf{J}^{-\top}\mathbf{M}\mathbf{J}^{-1} = \mathrm{diag}(1,\ 2.333)$ kg, $\mathbf{g}_x = \mathbf{J}^{-\top}\mathbf{g} = (0,\ 19.62)$ N (the total weight, $2g$). Error $\mathbf{E}_x = (0.02, -0.01)$ m, $\dot{\mathbf{E}}_x = \ddot{\mathbf{x}}_d = \mathbf{0}$, $\mathbf{K}_p = 400$: $\mathcal{F} = \mathbf{M}_x(8, -4)^{\top} + \mathbf{g}_x = (8,\ 10.287)$ N, $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F} = (-3.0915, -8.0)$ N m. Joint-space route: $\ddot{\mathbf{q}} = \mathbf{J}^{-1}(8, -4)^{\top}$, $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g} = (-3.0915, -8.0)$: identical. [checked]

STEPS Cartesian controller on paper

1. State the frame: $\mathbf{x}, \mathbf{J}, \mathcal{F}$ all in $\{0\}$ (base).
2. Feedback: $\mathbf{x} = \mathcal{K}(\mathbf{q})$, $\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$; errors $\mathbf{E}_x = \mathbf{x}_d - \mathbf{x}$, $\dot{\mathbf{E}}_x = \dot{\mathbf{x}}_d - \dot{\mathbf{x}}$.
3. $\mathcal{F} = \mathbf{M}_x(\ddot{\mathbf{x}}_d + \mathbf{K}_v\dot{\mathbf{E}}_x + \mathbf{K}_p\mathbf{E}_x) + \mathbf{c}_x + \mathbf{g}_x$.
4. $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ (transpose, not inverse; $\mathbf{J}$ of the right pose).
5. Check: static pose $\Rightarrow \mathbf{Q} = \mathbf{g}$; torque about each joint = $\mathbf{r} \times \mathcal{F}$ (moment check); note singular poses.

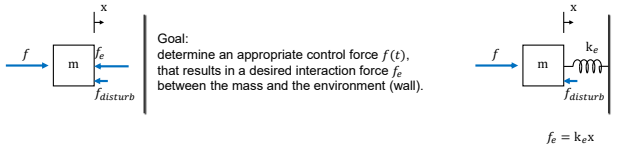
TRAP

$\mathbf{Q} = \mathbf{J}\mathcal{F}$ or $\mathbf{J}^{-1}\mathcal{F}$ (wrong map, wrong shape of units); $\mathbf{J}$ evaluated at the wrong $\mathbf{q}$; forgetting that the slide's Cartesian $\mathcal{F}$ may include moments for a 6-DoF arm (then $\mathbf{J}$ is $6 \times n$ and $\mathbf{J}^{\top}\mathcal{F}$ is still $n \times 1$).

Force control [p. 34–44]

PREMISE force control **Given:** a mass m in contact with a wall (stiffness k_e, an unknown disturbance f_{dist} acts); desired contact force $f_d(t)$; measured f_e and $\dot{x}$

Find: the actuator force $f(t)$ so that the interaction force $f_e \to f_d$



[W09 lec p. 36] — slide p. 36: mass m, wall, contact force f_e, disturbance f_{dist}; right: the wall as a spring k_e, $f_e = k_e x$; handwritten $f - f_e - f_{dist} = m\ddot{x}$

DERIVATION plant and partitioned law, every line [p. 36–37]

Wall as a spring: $f_e = k_e x \Leftrightarrow x = k_e^{-1} f_e$, so $\ddot{x} = k_e^{-1}\ddot{f}_e$. Forces on the mass [p. 36]: $f - f_e - f_{dist} = m\ddot{x}$, i.e.

$$f = m\ddot{x} + f_{dist} + f_e = mk_e^{-1}\ddot{f}_e + f_{dist} + f_e.$$

Now the *force* f_e is the controlled variable: $\alpha = mk_e^{-1}$, $\beta = f_{dist} + f_e$, with f_d the desired force and $E = f_d - f_e$:

$$f = mk_e^{-1}\left[\ddot{f}_d + k_v\dot{E} + k_pE\right] + f_{dist} + f_e \;\Rightarrow\; \ddot{E} + k_v\dot{E} + k_pE = 0 \text{ [p. 38]}.$$

Problem: the law contains f_{dist}, which we do not know: the law is infeasible [p. 38].

The two feasible options [p. 39–40]

Option 1 (drop f_{dist}): $f = mk_e^{-1}[\ddot{f}_d + k_v\dot{E} + k_pE] + f_e$

$$\ddot{E} + k_v\dot{E} + k_pE = \frac{f_{dist}}{mk_e^{-1}}, \qquad E_{ss} = \frac{f_{dist}}{mk_e^{-1}k_p}$$

Option 2 (use f_d for $f_{dist}+f_e$): $f = mk_e^{-1}[\ddot{f}_d + k_v\dot{E} + k_pE] + f_d$

$$\ddot{E} + k_v\dot{E} + \left(\frac{1}{mk_e^{-1}} + k_p\right)E = \frac{f_{dist}}{mk_e^{-1}}, \qquad E_{ss} = \frac{f_{dist}}{1 + mk_e^{-1}k_p}$$

SYMBOLS

$f_e = k_e x$	contact force the environment exerts, measured	N
f_d	desired interaction force	N
$E = f_d - f_e$	force error	N
f_{dist}	unknown disturbance force on the mass	N
k_e	environment (wall) stiffness	N/m
mk_e^{-1}	α: mass divided by stiffness	s^2
$X = mk_e^{-1}k_p$	dimensionless gain product	–

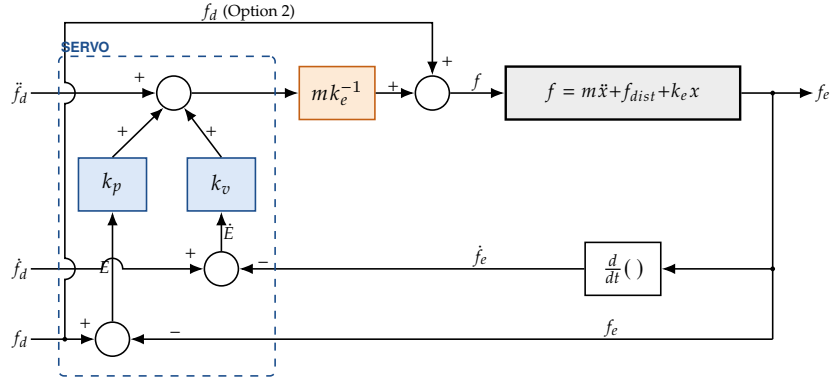
Use: $E_2/E_1 = X/(1+X) < 1$: *Option 2 is always better, most on a stiff wall [p. 40]. Both are non-zero unless* $f_{dist} = 0$. *Option 2 adds* $1/(mk_e^{-1}) = k_e/m$ *to the stiffness of the error equation (a stiffer loop).*

DERIVATION the two closed loops, every line [added]

Plant $mk_e^{-1}\ddot{f}_e + f_{dist} + f_e = f$ with $E = f_d - f_e$, $\ddot{E} = \ddot{f}_d - \ddot{f}_e$, $\alpha = mk_e^{-1}$.

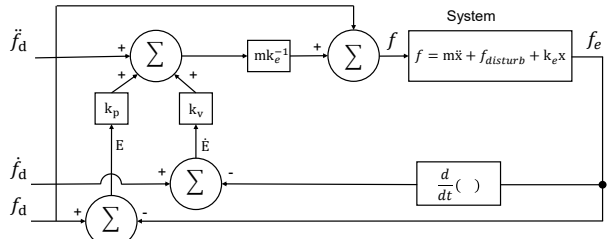
Option 1: $\alpha\ddot{f}_e + f_{dist} + f_e = \alpha[\ddot{f}_d + k_v\dot{E} + k_pE] + f_e \Rightarrow \alpha(\ddot{f}_d - \ddot{f}_e) + \alpha k_v\dot{E} + \alpha k_pE = f_{dist} \Rightarrow$ $\ddot{E} + k_v\dot{E} + k_pE = f_{dist}/\alpha$. Static: $k_pE = f_{dist}/\alpha$.

Option 2: $\alpha\ddot{f}_e + f_{dist} + f_e = \alpha[\dots] + f_d \Rightarrow \alpha\ddot{E} + \alpha k_v\dot{E} + \alpha k_pE = f_{dist} + f_e - f_d = f_{dist} - E \Rightarrow$ $\ddot{E} + k_v\dot{E} + (k_p + 1/\alpha)E = f_{dist}/\alpha$. Static: $E = \dfrac{f_{dist}/\alpha}{k_p + 1/\alpha} = \dfrac{f_{dist}}{1 + \alpha k_p}$.



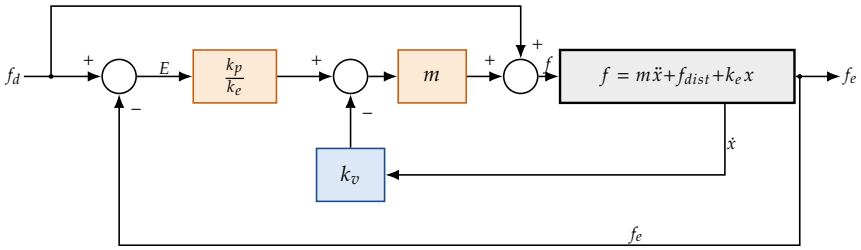
servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] force controller (slide p. 41, redrawn). The top path adds f_d at the final sum: this is Option 2; for Option 1 that path carries the measured f_e instead. The $\frac{d}{dt}$ block is the one the slide flags as "not a good idea" on p. 42.



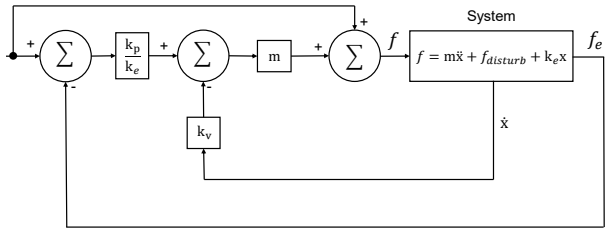
[W09 lec p. 41] — slide p. 41: force-control block diagram

On p. 42 the same diagram has $\ddot{f}_d$ and $\dot{f}_d$ set to 0 (constant f_d) and the $\frac{d}{dt}$ block flagged "Not a good idea": differentiating a noisy force measurement amplifies the noise. **Why this is the same law.** For constant f_d, $\dot{E} = -\dot{f}_e = -k_e\dot{x}$, so Option 2's $mk_e^{-1}k_v\dot{E} = -mk_v\dot{x}$ and $mk_e^{-1}k_pE = m[k_e^{-1}k_pE]$: the practical law *is* Option 2 with $\dot{f}_e$ replaced by $k_e\dot{x}$. [added] **Stiffness unknown [p. 44]:** k_e is usually chosen as a high value, and the controller is designed to be robust to small changes in k_e.



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] the practical force controller (slide p. 43, redrawn): $f = m\left[k_e^{-1} k_p E - k_v \dot{x}\right] + f_d$. Velocity $\dot{x}$ is measured directly, so f_e is never differentiated. Here the gains k_p/k_e and m are model-based (orange), k_v is servo (blue).



Control law is $f = m\left[k_e^{-1} k_p E - k_v \dot{x}\right] + f_d$

[W09 lec p. 43] — slide p. 43: the practical block diagram

WORKED numbers: $m = 2$ kg, $f_{dist} = 5$ N, $k_p = 100$ s^{-2}

$k_e = 1000$ N/m: $mk_e^{-1} = 0.002$ s^2, $X = 0.2$; $E_1 = 5/0.2 = 25$ N, $E_2 = 5/1.2 = 4.17$ N. Soft wall $k_e = 20$: $X = 10$, $E_1 = 0.5$, $E_2 = 0.455$ N. For $E_2 = 1\,\%$ of f_{dist} on the stiff wall: $1 + X = 100$, $k_p = 49\,500$ s^{-2}. Practical law at $f_d = 10$ N, $f_e = 7$ N, $\dot{x} = 0.003$ m/s ($k_v = 20$, $k_e = 1000$): $E = 3$; $f = 2[0.001(100)(3) - 20(0.003)] + 10 = 2(0.3 - 0.06) + 10 = 10.48$ N; Option 2 gives $0.002[20(-3) + 100(3)] + 10 = 10.48$ N: identical. Option 2 error loop: $\omega_n^2 = k_p + k_e/m = 600$, critical $k_v = 2\sqrt{600} = 49.0$ s^{-1}. [checked]

TRAP

E_{ss} of Option 1 quoted for Option 2 (missing "1+"); "Option 2 improves on a stiffer wall" (both errors grow as k_e rises; Option 2 saturates at f_{dist} while Option 1 grows without limit); differentiating the measured force; $E = f_e - f_d$; using the Option-1 critical $k_v = 2\sqrt{k_p}$ for Option 2's loop.

Hybrid force/position control [p. 45–47]

PREMISE hybrid control **Given:** a task with some directions blocked by a surface or hole (force control) and the others free (position control); desired motion $\ddot{\mathbf{x}}_d, \dot{\mathbf{x}}_d, \mathbf{x}_d$ and desired force $\mathcal{F}_d$; the constraint description

Find: the Cartesian force $\mathcal{F}$ to command, then $\mathbf{Q} = \mathbf{J}^\top \mathcal{F}$

[p. 46]: a humanoid wiping or pressing a wall needs both: slide along the wall under position control, push *into* it under force control.

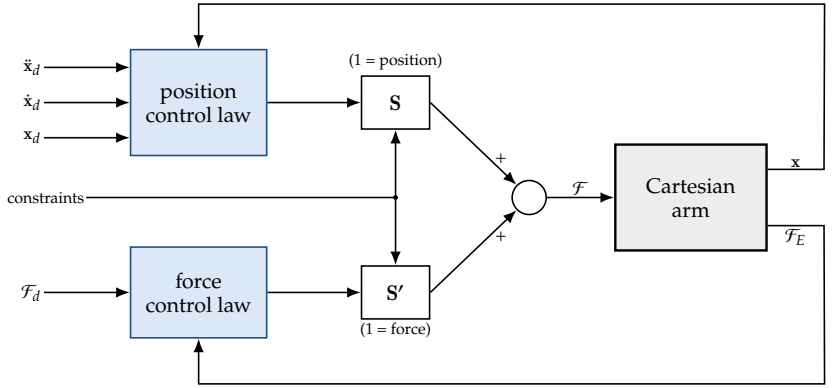
Selection matrices [p. 47]

$$\mathcal{F} = \mathbf{S}\mathcal{F}_{pos} + \mathbf{S}'\mathcal{F}_{force}, \qquad \mathbf{S}' = \mathbf{I} - \mathbf{S} \text{ (complementary)}, \qquad \mathbf{Q} = \mathbf{J}^\top \mathcal{F}$$

SYMBOLS

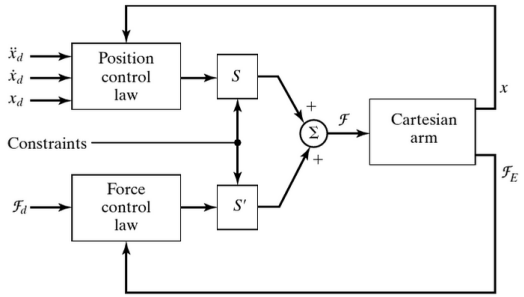
$\mathcal{F}_{pos}$	output of the position control law (inputs $\ddot{\mathbf{x}}_d, \dot{\mathbf{x}}_d, \mathbf{x}_d$, feedback $\mathbf{x}$)	N
$\mathcal{F}_{force}$	output of the force control law (input $\mathcal{F}_d$, feedback $\mathcal{F}_E$, the measured force)	N
$\mathbf{S}$	diagonal, entries 0 or 1: **1 = this axis is position-controlled**	–
$\mathbf{S}'$	diagonal, entries 0 or 1: **1 = this axis is force-controlled**; $\mathbf{S} + \mathbf{S}' = \mathbf{I}$	–
"constraints"	tell the controller which axes are blocked; they set $\mathbf{S}, \mathbf{S}'$	–

Use: the slide says S and S′ are complementary and "indicate whether a joint is in force control mode or position control mode"; $\mathbf{S}' = \mathbf{I} - \mathbf{S}$ is my reading of "complementary" and the figure is in Cartesian space. Both diagonal matrices are written in the constraint frame: if the surface is tilted, rotate $\mathcal{F}$ into the surface frame first [added].



servo-based (feedback on the error; gains k_p, k_v, k_i) model-based (uses the robot model: $\mathbf{M}, \mathbf{C}, \mathbf{g}, \mathbf{f}, \boldsymbol{\alpha}, \boldsymbol{\beta}$) plant (the robot)

[drawn for this booklet] hybrid position/force controller (slide p. 47, after Craig, redrawn). Both control laws are servo laws (blue); S and S′ pick, axis by axis, which law reaches the arm. The signals $\mathbf{x}$ and $\mathcal{F}_E$ are measured on the arm.



[W09 lec p. 47] — slide p. 47: hybrid force and position controller

Which entries of S are 1? [added]

Rule: **a direction the environment blocks → force control** ($\mathbf{S}$ entry 0, $\mathbf{S}'$ entry 1); a direction in which the tool is free to move → position control ($\mathbf{S}$ entry 1, $\mathbf{S}'$ entry 0). Translations (x, y, z) unless stated.

task	blocked / free	$\mathbf{S}$ (position)
wipe a table, z normal (slide p. 46 idea)	z blocked; x, y free	$\mathrm{diag}(1,1,0)$
push on a wall with normal x	x blocked; y, z free	$\mathrm{diag}(0,1,1)$
whiteboard eraser, normal = tool z	z blocked; x, y free	$\mathrm{diag}(1,1,0)$
pull a drawer open along x [added]	rails block y, z	$\mathrm{diag}(1,0,0)$
insert a peg in a vertical hole, 6 axes ($x, y, z, \theta_x, \theta_y, \theta_z$) [added]	walls block x, y, θ_x, θ_y (force/torque 0); z and θ_z free	$\mathrm{diag}(0,0,1,0,0,1)$

WORKED wipe a table, z normal; $\mathcal{F}_{pos} = (12,-3,9)$ N, $\mathcal{F}_{force} = (1,2,10)$ N

$\mathbf{S} = \mathrm{diag}(1,1,0)$, $\mathbf{S}' = \mathrm{diag}(0,0,1)$: $\mathcal{F} = (12,-3,0) + (0,0,10) = (12,-3,10)$ N. The position law's 9 N along z and the force law's 1, 2 N along x, y are discarded. Wall normal x: $\mathbf{S} = \mathrm{diag}(0,1,1)$ gives $\mathcal{F} = (1,-3,9)$ N. Peg (6 axes): position law $(1,2,-15,0.3,-0.2,0.8)$, force law $(-3,4,2,0.5,-0.6,0.1)$, $\mathbf{S} = \mathrm{diag}(0,0,1,0,0,1)$: $\mathcal{F} = (-3,4,-15,0.5,-0.6,0.8)$ (N, N m). [checked]

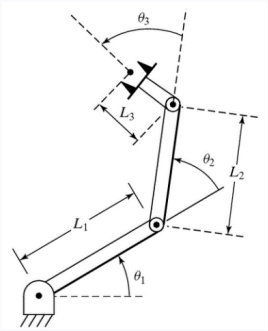
TRAP

$\mathbf{S}$ and $\mathbf{S}'$ swapped (the eraser slides freely and ignores the press); both entries 1 on one axis (two laws fighting); using the *world* axes when the surface is tilted; forgetting $\mathbf{Q} = \mathbf{J}^\top \mathcal{F}$ at the end.

How this is examined: worked solutions

The W9 practical sheet is the only lecturer-written question on this week (no W13 sample question covers it, but Q4 "derive the equations of motion" is the stepping stone: derive the EoM, then partition it). Below: the practical sheet as an exam question (times and numbers fixed where the sheet gives none), then five exam-style questions. Marks are not published; indicative marks are "about one per step".

EXAM-STYLE QUESTION Practical sheet (a)–(c): trajectory and the model [marks assumed]



[W09 prac p. 1] — the practical's 3-link planar arm: $L_1 = 3, L_2 = 2, L_3 = 1$ m, joint angles $\theta_1, \theta_2, \theta_3$ measured as relative angles

A planar 3-link arm ($L_1 = 3$, $L_2 = 2$, $L_3 = 1$ m; point masses $m_1 = 3$, $m_2 = 2$, $m_3 = 1$ kg at the *distal* ends of the links) starts at rest at the end-effector pose $(x, y, \varphi) = (5, 1, 90°)$ and must stop at $(-3, 3, 180°)$. (a) Find the joint angles at both poses and the cubic joint trajectories ($T = 2$ s assumed: the sheet gives no time). (b) Use Lagrange to get $\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q})$. (c) Evaluate $\mathbf{M}, \mathbf{C}, \mathbf{g}$ in the initial configuration.

Solution

(a) Joint angles (frame $\{0\}$ at the base, θ_i relative angles, tool orientation $\varphi = \theta_1 + \theta_2 + \theta_3$). Remove link 3: wrist $\mathbf{w} = (x - L_3\cos\varphi,\ y - L_3\sin\varphi)$; then a planar 2R.
Start: $\mathbf{w} = (5-0,\ 1-1) = (5,0)$; $\cos\theta_2 = (25 - 9 - 4)/12 = 1 \Rightarrow \theta_2 = 0$; $\theta_1 = \mathrm{atan2}(0,5) = 0$; $\theta_3 = 90° - 0 - 0$. $\mathbf{q}_0 = (0, 0, 90°)$ (arm stretched along x, last link up). Check FK: $(3+2,\ 0+1) = (5,1)$ ✓.
Goal: $\mathbf{w} = (-3 - \cos 180°,\ 3 - \sin 180°) = (-2, 3)$, $|\mathbf{w}|^2 = 13$; $\cos\theta_2 = (13 - 9 - 4)/12 = 0 \Rightarrow \theta_2 = 90°$ (elbow branch +); $\theta_1 = \mathrm{atan2}(3,-2) - \mathrm{atan2}(2,3) = 123.69° - 33.69° = 90°$; $\theta_3 = 180° - 90° - 90° = 0$. $\mathbf{q}_f = (90°, 90°, 0)$. (Other branch $\theta_2 = -90°$: $(157.38°, -90°, 112.62°)$; the sheet does not say; the integers favour the first.) [checked]

EXAM-STYLE QUESTION Practical sheet (a), continued: the cubics

Cubics (rest to rest, $q = q_0 + a_2 t^2 + a_3 t^3$, $a_2 = 3\Delta/T^2$, $a_3 = -2\Delta/T^3$), $\Delta = (\pi/2, \pi/2, -\pi/2)$, $T = 2$:

joints 1, 2: $a_2 = 1.1781$, $a_3 = -0.3927$; joint 3: $a_2 = -1.1781$, $a_3 = +0.3927$

(units rad/s^2 and rad/s^3). Peak speed $1.5\Delta/T = 1.1781$ rad/s at $t = 1$ s (not $\Delta/T = 0.785$); peak acceleration $6\Delta/T^2 = 2.3562$ rad/s^2 at $t = 0$. Plot: $q(t)$ an S-curve $0 \to 90°$; $\dot{q}$ a parabola with peak 1.18; $\ddot{q}$ a straight line from +2.36 to −2.36. [checked] Note: a joint-space cubic does *not* move the tool in a line: at $t = 1$ the joints are $(45°, 45°, 45°)$, tool $(1.414, 4.828)$. [checked]

EXAM-STYLE QUESTION Practical sheet (b): the model by Lagrange

(b) Model by Lagrange. Mass positions ($c_{12} = \cos(\theta_1 + \theta_2)$ etc.): $\mathbf{p}_1 = L_1(c_1, s_1)$, $\mathbf{p}_2 = \mathbf{p}_1 + L_2(c_{12}, s_{12})$, $\mathbf{p}_3 = \mathbf{p}_2 + L_3(c_{123}, s_{123})$. $K = \sum \tfrac{1}{2} m_i |\dot{\mathbf{p}}_i|^2$, $P = g \sum m_i y_i$, so $M_{ij} = \partial^2 K / \partial\dot{q}_i \partial\dot{q}_j$ and $g_i = \partial P / \partial q_i$. With $A = (m_2 + m_3)L_1 L_2 = 18$, $B = m_3 L_1 L_3 = 3$, $D = m_3 L_2 L_3 = 2$ (kg m^2):

EXAM-STYLE QUESTION Practical sheet (b), continued, and (c)

$$M_{11} = 67 + 36c_2 + 6c_{23} + 4c_3, \quad M_{12} = 13 + 18c_2 + 3c_{23} + 4c_3, \quad M_{13} = 1 + 3c_{23} + 2c_3,$$
$$M_{22} = 13 + 4c_3, \quad M_{23} = 1 + 2c_3, \quad M_{33} = 1 \quad (\mathbf{M} = \mathbf{M}^\top)$$

$$C_1 = -As_2(2\dot q_1\dot q_2 + \dot q_2^2) - Bs_{23}\big(2\dot q_1(\dot q_2{+}\dot q_3) + (\dot q_2{+}\dot q_3)^2\big) - Ds_3\big(2(\dot q_1{+}\dot q_2)\dot q_3 + \dot q_3^2\big)$$
$$C_2 = (As_2 + Bs_{23})\dot q_1^2 - Ds_3\big(2(\dot q_1{+}\dot q_2)\dot q_3 + \dot q_3^2\big), \qquad C_3 = Bs_{23}\dot q_1^2 + Ds_3(\dot q_1 + \dot q_2)^2$$
$$g_1 = g\,[18c_1 + 6c_{12} + c_{123}], \qquad g_2 = g\,[6c_{12} + c_{123}], \qquad g_3 = g\,c_{123}$$

($g = 9.81$ m/s^2, arm in a vertical plane.) (The last-link weight g_3 has lever arm L_3c_{123} about joint 3.) These were checked against a symbolic Lagrangian at random poses. [checked]

(c) At the start $\mathbf{q}_0 = (0, 0, 90°)$, $\dot{\mathbf{q}} = \mathbf{0}$: $c_2 = 1, c_3 = 0, c_{23} = 0, c_1 = 1, c_{12} = 1, c_{123} = 0$:

$$\mathbf{M} = \begin{bmatrix} 103 & 31 & 1 \\ 31 & 13 & 1 \\ 1 & 1 & 1 \end{bmatrix} \text{kg m}^2, \quad \mathbf{C} = \mathbf{0}, \quad \mathbf{g} = 9.81 \begin{bmatrix} 18+6+0 \\ 6+0 \\ 0 \end{bmatrix} = \begin{bmatrix} 235.44 \\ 58.86 \\ 0 \end{bmatrix} \text{N m}.$$

Sanity: g_1 is the moment of all three weights about joint 1, lever arms $3, 5, 5$ m: $9.81(3\cdot 3 + 2\cdot 5 + 1\cdot 5) = 9.81\cdot 24$ ✓; $g_3 = 0$ as link 3 points straight up. Inverse dynamics at $t = 0$ on the cubic: $\mathbf{Q} = \mathbf{M}\ddot{\mathbf{q}}_d + \mathbf{g} = (548.8, 160.2, 2.4)$ N m. At the goal: $\mathbf{M} = \begin{bmatrix} 71 & 17 & 3 \\ 17 & 17 & 3 \\ 3 & 3 & 1 \end{bmatrix}$, $\mathbf{g} = (-68.67, -68.67, -9.81)$. [checked]

EXAM-STYLE QUESTION Practical sheet (d)–(i): controller, simulator, imprecise mass [marks assumed]

(d)–(e) For the first link at the first instant, ignoring friction, write a partitioned law. (f) Describe a forward-dynamics simulator. (g) What does the position error look like along the trajectory? (h) The controller uses $\hat m = 2.8$ kg. (i) Extend to all joints.

Solution

(d) The reference supplies $q_d, \dot q_d, \ddot q_d$ at every servo step (q_d is the cubic sampled at the *next* time step); the servo uses $E = q_d - q$, $\dot E = \dot q_d - \dot q$, and $\ddot q_d$ as feed-forward.

(e) Link 1 alone ($m_1 = 3$ kg at the tip, $L_1 = 3$ m): $\tau = m_1L_1^2\ddot q + m_1gL_1\cos q$. $\alpha = 27$ kg m^2, $\beta = 88.29\cos q$ N m (radians). With $k_p = 100$, $k_v = 20$ ($\omega_n = 10$, critical):

$$\tau = 27\big(\ddot q_d + 20\dot E + 100E\big) + 88.29\cos q.$$

At $t = 0$: $q = 0, E = \dot E = 0, \ddot q_d = 2.3562$: $\tau(0) = 27(2.3562) + 88.29 = 63.62 + 88.29 = 151.91$ N m. (Reading B: joint 1 of the full arm with joints 2, 3 held: $\alpha = M_{11} = 103$, $\beta = g_1 = 235.44$ N m. Same structure.) [checked]

(f) Forward dynamics for the *plant* (it maps torque to motion), inverse dynamics inside the *controller* (it asks which torque gives an acceleration). Loop: $\mathbf{Q} = \text{controller}(t_k, \mathbf{q}_k, \dot{\mathbf{q}}_k)$; $\ddot{\mathbf{q}} = \mathbf{M}_{true}^{-1}(\mathbf{Q} - \mathbf{C} - \mathbf{g} - \mathbf{f})$ (solve, do not invert); integrate (RK4) to $\mathbf{q}_{k+1}, \dot{\mathbf{q}}_{k+1}$; log $E = q_d - q$. Keep the controller's model and the simulator's "true" model separate. At $t = 0$: $\ddot q = (151.91 - 88.29)/27 = 2.3562 = \ddot q_d$. [checked]

(g) Exact model: $\ddot E + 20\dot E + 100E = 0$, roots $-10, -10$. Starting on the path, $E \equiv 0$ (a flat line: the plot is not broken). Starting 0.1 rad behind: $E = 0.1(1 + 10t)e^{-10t}$, $E(0.1) = 0.0736$ rad, no overshoot, settled by about 0.5 s. If the torque is refreshed only every 10 ms or 50 ms the cancellation is imperfect: peak error 0.26 mrad or 1.13 mrad. [checked]

EXAM-STYLE QUESTION Practical sheet (h), (i) [marks assumed]

(h) Scalar link 1 with $m = 3$, $\hat m = 2.8$, $L = 3$, $k_p = 100$, $k_v = 20$: $\ddot E + 20\dot E + 100E = \dfrac{m - \hat m}{\hat m}\Big[\ddot q + \dfrac{g}{L}\cos q\Big] = \dfrac{1}{14}\big[\ddot q + 3.27\cos q\big]$. Held horizontal ($q_d = 0$): $E_{ss} = \frac{1}{14}(3.27)/100 = 2.34$ mrad (0.134°); held vertical: $\cos q = 0$, no static error. Sign: $\hat m < m$, gravity under-compensated, link sags, $E > 0$. On the cubic: forcing at $t = 0$ is $\frac{1}{14}(2.3562 + 3.27) = 0.402$ rad/s^2; simulated peak error 3.35 mrad (0.19°) at $t \approx 0.47$ s, −1.21 mrad at $t = 2$ s (deceleration term), then dies out. Full-arm reading ($m_1 = 2.8$ in joint 1 of the 3-link model): $\hat M_{11} = 103 - 0.2(9) = 101.2$, $\hat g_1 = 235.44 - 0.2(9.81)(3) = 229.554$ N m. [checked]

(i) All joints. $\mathbf{Q} = \mathbf{M}(\mathbf{q})(\ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q})$ with $\mathbf{K}_p = 100\mathbf{I}$, $\mathbf{K}_v = 20\mathbf{I}$ and $\mathbf{M}, \mathbf{C}, \mathbf{g}$ from (b) re-evaluated every step; closed loop $\ddot{\mathbf{E}} + 20\dot{\mathbf{E}} + 100\mathbf{E} = \mathbf{0}$, three decoupled critically damped loops. New compared with one joint: $\boldsymbol{\alpha}$ is a full matrix; $\mathbf{C}, \mathbf{g}$ depend on the whole pose; the model must be recomputed each step (hence the outer-loop idea). If a student used $\boldsymbol{\alpha} = \text{diag}(\mathbf{M})$: at $t = 0$ row 1 gives forcing $\big(31(2.3562) + 1(-2.3562)\big)/103 = 0.6863$ rad/s^2, 29 % of the commanded 2.36, an error of order 6.9 mrad. [checked]

EXAM-STYLE QUESTION EQ1: cubic spring (derive, calculate) [marks assumed: 8]

A mass obeys $m\ddot x + b\dot x + kx^3 = f$ with $m = 2$ kg, $b = 0.5$ N s/m, $k = 4$ N/m^3. (a) Write a partitioned control law for trajectory following and show the closed-loop error equation [3]. (b) Choose k_p, k_v for critical damping with $\omega_n = 8$ rad/s [1]. (c) Find f at $x = 0.10$ m, $\dot x = -0.20$ m/s when $x_d = 0.15$ m, $\dot x_d = \ddot x_d = 0$ [2]. (d) An unmodelled constant force of 5 N pushes the mass in the $+x$ direction. Find the steady-state error and say how to remove it [2].

EXAM-STYLE QUESTION EQ1: solution

Solution

(a) $\alpha = m$, $\beta = b\dot x + kx^3$, $\acute f = \ddot x_d + k_v\dot E + k_pE$, $E = x_d - x$; $f = m\acute f + b\dot x + kx^3$. Insert in the plant: $m\ddot x = m\acute f \Rightarrow \ddot x = \acute f \Rightarrow \boxed{\ddot E + k_v\dot E + k_pE = 0}$ [3].

(b) $k_p = \omega_n^2 = 64$ s^{-2}, $k_v = 2\omega_n = 16$ s^{-1} [1].

(c) $E = 0.15 - 0.10 = 0.05$, $\dot E = 0 - (-0.20) = 0.20$. $\acute f = 0 + 16(0.20) + 64(0.05) = 6.4$ m/s^2. $\beta = 0.5(-0.20) + 4(0.10)^3 = -0.1 + 0.004 = -0.096$ N. $f = 2(6.4) - 0.096 = \boxed{12.704\text{ N}}$ [2]. Check: plant $\ddot x = (12.704 + 0.096)/2 = 6.4 = \acute f$ ✓.

(d) Plant becomes $m\ddot x + \cdots = f + 5$, so $\ddot x = \acute f + 5/m$ and $\ddot E + k_v\dot E + k_pE = -5/m$. Steady state: $E_{ss} = -5/(mk_p) = -5/(2\cdot 64) = \boxed{-0.0391\text{ m}}$: the mass sits 3.9 cm past the target (in the direction of the push). Remove it with integral action (W8: $\acute f$ gets $+k_i\int E\,dt$, stable for $k_i < k_vk_p$) or by measuring and cancelling the force [2]. [checked]

EXAM-STYLE QUESTION EQ2: imprecise gravity term (derive, calculate) [marks assumed: 7]

A single link (point mass m at the tip, length L, angle q from the horizontal) obeys $mL^2\ddot q + mgL\cos q = \tau$. The controller uses $\tau = \alpha\acute\tau + \beta$ with the correct $\alpha = mL^2$ but $\beta = \hat mgL\cos q$, and $\acute\tau = \ddot q_d + k_v\dot E + k_pE$. (a) Derive the error equation [3]. (b) For $m = 3$ kg, $\hat m = 2.5$ kg, $L = 0.6$ m, $k_p = 81$ s^{-2}, find E_{ss} with the link held at $q = 20°$ [2]. (c) Say what changes if $q = 90°$ and why raising k_p never gives zero error [2].

EXAM-STYLE QUESTION EQ2: solution

Solution

(a) Equate: $mL^2\ddot{q} + mgL\cos q = mL^2\ddot{\tau} + \hat{m}gL\cos q$. Divide by mL^2: $\ddot{q} = \ddot{\tau} - \dfrac{(m-\hat{m})g\cos q}{mL}$. With $\ddot{E} = \ddot{q}_d - \ddot{q}$:

$$\boxed{\ddot{E} + k_v\dot{E} + k_pE = \frac{(m-\hat{m})\,g\cos q}{mL}} \quad [3].$$

(b) Static: $E_{ss} = \dfrac{(m-\hat{m})g\cos q}{mLk_p} = \dfrac{0.5(9.81)(0.9397)}{3(0.6)(81)} = \dfrac{4.609}{145.8} = \boxed{0.0316\text{ rad}} = 1.81°$; positive: $\hat{m} < m$ so the link sags [2]. Wrong answer to avoid: omitting $\cos q$ gives 0.0336.

(c) At $q = 90°$, $\cos q = 0$: the gravity mismatch has no lever arm, $E_{ss} = 0$. Raising k_p shrinks the error ($\propto 1/k_p$) but never removes it: the right side is a constant forcing, and k_p is limited by the sampling rate [2]. [checked]

EXAM-STYLE QUESTION EQ3: computed torque for a 2R arm (calculate, explain) [marks assumed: 8]

A planar 2R arm ($l_1 = l_2 = 1$ m) has point masses $m_1 = m_2 = 1$ kg at the link ends; its dynamics are $\mathbf{M} = \begin{bmatrix} 3+2c_2 & 1+c_2 \\ 1+c_2 & 1 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} -s_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ s_2\dot{q}_1^2 \end{bmatrix}$, $\mathbf{g} = g\begin{bmatrix} 2c_1 + c_{12} \\ c_{12} \end{bmatrix}$. At $\mathbf{q} = (30°, 60°)$, $\dot{\mathbf{q}} = (1, -0.5)$ rad/s the reference has $\ddot{\mathbf{q}}_d = (0.5, 0)$, $\mathbf{E} = (0.02, -0.01)$ rad, $\dot{\mathbf{E}} = \mathbf{0}$; gains $\mathbf{K}_p = 100\mathbf{I}$, $\mathbf{K}_v = 20\mathbf{I}$. (a) Write the law [2]. (b) Evaluate $\mathbf{M}, \mathbf{C}, \mathbf{g}$ [3]. (c) Find $\mathbf{Q}$ [2]. (d) If only joint 1 is to accelerate ($\acute{\mathbf{Q}} = (0.5, 0)$), what torque does joint 2 still receive, and why [1].

EXAM-STYLE QUESTION EQ3: solution

Solution

(a) $\mathbf{Q} = \mathbf{M}\acute{\mathbf{Q}} + \mathbf{C} + \mathbf{g}$, $\acute{\mathbf{Q}} = \ddot{\mathbf{q}}_d + \mathbf{K}_v\dot{\mathbf{E}} + \mathbf{K}_p\mathbf{E}$ [2].

(b) $c_2 = 0.5$, $s_2 = 0.866$, $c_1 = 0.866$, $c_{12} = \cos 90° = 0$. $\mathbf{M} = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 1 \end{bmatrix}$. $C_1 = -0.866\big(2(1)(-0.5) + 0.25\big) = 0.6495$, $C_2 = 0.866(1) = 0.866$. $\mathbf{g} = 9.81(1.732,\ 0) = (16.991,\ 0)$ N m [3].

(c) $\acute{\mathbf{Q}} = (0.5 + 100(0.02),\ 0 + 100(-0.01)) = (2.5,\ -1.0)$. $\mathbf{M}\acute{\mathbf{Q}} = (4(2.5) + 1.5(-1),\ 1.5(2.5) + (-1)) = (8.5,\ 2.75)$. $\mathbf{Q} = (8.5 + 0.6495 + 16.991,\ 2.75 + 0.866 + 0) = \boxed{(26.14,\ 3.62)\text{ N m}}$ [2].

(d) $\mathbf{M}(0.5, 0)^\top = (2.0,\ 0.75)$: joint 2 still gets 0.75 N m, because $\boldsymbol{\alpha} = \mathbf{M}$ is not diagonal: this torque cancels the inertial coupling $M_{21}\ddot{q}_1$ so that joint 2 stays on its path [1]. [checked]

EXAM-STYLE QUESTION EQ6: explain and spot the errors [marks assumed: 6]

(a) In 2–3 sentences: why does the slide move the model-based part to the outer loop, and what is the cost? [2] (b) A student writes: "$\boldsymbol{\alpha} = \mathrm{diag}(\mathbf{M})$ is enough because $\mathbf{K}_p, \mathbf{K}_v$ are diagonal; with a 10 % mass error the error still goes to zero; for the Cartesian law $\mathbf{Q} = \mathbf{J}\mathcal{F}$." Find three errors [3]. (c) Why can the Cartesian controller not start at the 3-link practical's start pose $(5, 1, 90°)$? [1]

EXAM-STYLE QUESTION EQ6: solution

Solution

(a) The dynamics (computing $\mathbf{M}, \mathbf{C}, \mathbf{g}$) takes time and limits how often torques can be updated; the servo part is cheap. Evaluating the model along the plan at a slow rate and keeping only the PD correction fast allows a faster servo loop. Cost: the model no longer reacts to the measured state, so decoupling is only approximate and the gains become torque gains [2].

(b) (i) $\boldsymbol{\alpha} = \mathbf{M}(\mathbf{q})$ is a full matrix; its off-diagonals cancel the inertial coupling; a diagonal $\boldsymbol{\alpha}$ leaves forcing $\hat{\mathbf{M}}^{-1}(\mathbf{M} - \hat{\mathbf{M}})\ddot{\mathbf{q}} \neq 0$ (e.g. 0.686 rad/s^2 on joint 1 of the practical). (ii) With a mass error the right-hand side of the error equation is non-zero, so $E_{ss} \approx$ (forcing)$/k_p \neq 0$. (iii) $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$ (transpose) [3].

(c) The start is fully stretched ($\theta_2 = 0$), a singularity: $\det\mathbf{J} \propto \sin\theta_2 = 0$, so $\mathbf{J}^{-1}$ and $\mathbf{M}_x$ do not exist there [1]. [checked]

EXAM-STYLE QUESTION EQ4: force control against a wall (calculate, explain) [marks assumed: 7]

A mass $m = 1.5$ kg presses on a wall modelled as a spring $k_e = 300$ N/m; an unknown disturbance $f_{dist} = 8$ N acts; $k_p = 64$ s^{-2}. (a) Why is the law with f_{dist} infeasible? [1] (b) Find the steady-state force error for Option 1 and Option 2 [3]. (c) Which would you build, and what is the practical form of the law? [2] (d) Give k_v for critical damping of the Option 2 loop [1].

EXAM-STYLE QUESTION EQ4: solution

Solution

(a) The ideal law contains f_{dist}, which is not measured [1].

(b) $mk_e^{-1} = 1.5/300 = 0.005$ s^2; $X = 0.005(64) = 0.32$. $E_1 = f_{dist}/X = 8/0.32 = \boxed{25\text{ N}}$; $E_2 = f_{dist}/(1+X) = 8/1.32 = \boxed{6.06\text{ N}}$ [3]; ratio $X/(1+X) = 0.24$. Option 1 leaves an error three times the disturbance.

(c) Option 2 (it uses the desired force and always has the smaller error). Practical form: $f = m\left[k_e^{-1}k_pE - k_v\dot{x}\right] + f_d$, which avoids differentiating the noisy force [2].

(d) Option 2 loop: $\ddot{E} + k_v\dot{E} + (k_p + k_e/m)E = \ldots$, $\omega_n^2 = 64 + 200 = 264$, $k_v = 2\sqrt{264} = \boxed{32.5\text{ s}^{-1}}$ [1]. [checked]

EXAM-STYLE QUESTION EQ5: Cartesian torques and hybrid control (calculate, sketch, explain) [marks assumed: 8]

(a) Show with virtual work that $\mathbf{Q} = \mathbf{J}^\top\mathcal{F}$ [2]. (b) A planar 2R arm ($L_1 = 0.5$, $L_2 = 0.4$ m) is at $\mathbf{q} = (60°, -30°)$; the Cartesian controller asks for the tool force $\mathcal{F} = (-2, 6)$ N. Find $\mathbf{Q}$ [3]. (c) A robot erases a whiteboard (normal = tool z), following a circle and pressing with 10 N. Give $\mathbf{S}, \mathbf{S}'$; the position law outputs $(2, -1, 6)$ N and the force law $(0.5, 0.3, 12)$ N: find $\mathcal{F}$ [3]. Sketch the block diagram.

EXAM-STYLE QUESTION EQ5: solution

Solution

(a) $\delta\mathbf{x} = \mathbf{J}\delta\mathbf{q}$; work of torques $\mathbf{Q}^{\top}\delta\mathbf{q}$ equals work of the tool force $\mathcal{F}^{\top}\mathbf{J}\delta\mathbf{q}$ for all $\delta\mathbf{q}$, so $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F}$ [2].

(b) $s_1 = 0.866$, $c_1 = 0.5$, $s_{12} = \sin 30^\circ = 0.5$, $c_{12} = 0.866$. $\mathbf{J} = \begin{bmatrix} -0.5(0.866) - 0.4(0.5) & -0.4(0.5) \\ 0.5(0.5) + 0.4(0.866) & 0.4(0.866) \end{bmatrix} = \begin{bmatrix} -0.633 & -0.2 \\ 0.5964 & 0.3464 \end{bmatrix}$. $\mathbf{Q} = \mathbf{J}^{\top}\mathcal{F} = (-0.633(-2) + 0.5964(6),\ -0.2(-2) + 0.3464(6)) = \boxed{(4.845,\ 2.478)\ \text{N m}}$ [3]. Moments: tool $(0.5964, 0.633)$: $0.5964(6) - 0.633(-2) = 4.845$ ✓.

(c) The board blocks z: $\mathbf{S}' = \text{diag}(0, 0, 1)$ (force), $\mathbf{S} = \text{diag}(1, 1, 0)$ (position). $\mathcal{F} = (2, -1, 0) + (0, 0, 12) = \boxed{(2, -1, 12)\ \text{N}}$ [3]; the 6 N and the 0.5, 0.3 N are discarded. Swapped matrices would give $(0.5, 0.3, 6)$: the eraser would ignore the press. Diagram: the hybrid controller above with $\mathcal{F}_d = (0, 0, 10)$ N. [checked]

Week 10 at a glance [added]

The formulas to know (each is derived and explained on the page given)

frame change	$\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$, $\boldsymbol{\zeta} = \mathbf{J}^{\top}\boldsymbol{\nu}$ $(u, v$ along $\hat{x}_C, \hat{y}_C)$
any wheel, $\gamma = 0$	$r\dot{\varphi}_i = \cos\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i})$
standard wheel: no sliding	$0 = -\sin\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \cos\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i})$
differential drive	$u = \frac{r}{2}(\dot{\varphi}_1 + \dot{\varphi}_2)$, $\dot{\theta} = \frac{r}{2a}(\dot{\varphi}_2 - \dot{\varphi}_1)$, $v = 0$, $y_{ICR} = u/\dot{\theta}$
omni triangle	$r\dot{\varphi}_1 = u - d\dot{\theta}$, $r\dot{\varphi}_{2,3} = -\frac{1}{2}u \mp \frac{\sqrt{3}}{2}v - d\dot{\theta}$
bicycle	$\dot{\theta} = u\tan\beta/L$, $R = L/\tan\beta$, $v = 0$
holonomic?	a standard wheel gives a no-sliding row $\Rightarrow$ non-holonomic

SYMBOLS

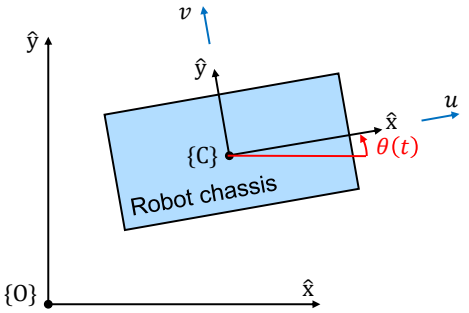
r, a, d, L	wheel radius; half track of a differential drive; centre-to-wheel distance of the omni triangle; wheelbase	m
β_i, β	drive direction of wheel i from $\hat{x}_C$; steering angle of the bicycle front wheel	rad
${}^{C}x_{W_i}$, ${}^{C}y_{W_i}$	position of wheel i in $\{C\}$	m
$\dot{\varphi}_i$	speed of wheel i	rad/s

Use: start here in the exam: identify the robot, write the row of each wheel, then stack and invert.

The kinematics problem: frames $\{O\}$ and $\{C\}$ [p. 11–13]

PREMISE wheeled robot, planar **Given:** pose (x, y, θ) of the robot in $\{O\}$; wheel layout and radii; wheel speeds $\dot{\boldsymbol{\varphi}}$ (forward problem) *or* a wanted motion (inverse problem)

Find: chassis velocity $\boldsymbol{\zeta} = (u, v, \dot{\theta})$ in $\{C\}$ and $\boldsymbol{\nu} = (\dot{x}, \dot{y}, \dot{\theta})$ in $\{O\}$; or the wheel speeds $\dot{\boldsymbol{\varphi}}$ that produce it



[W10 lec p. 13] — inertial frame $\{O\}$, chassis frame $\{C\}$, chassis velocity (u, v) along the chassis axes, heading $\theta(t)$

Frames and the two velocity vectors [p. 13]

$${}^{O}\mathbf{v}_C = \boldsymbol{\nu} = \begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix} \text{ (components along } \hat{x}_O, \hat{y}_O), \qquad {}^{C}\mathbf{v}_C = \boldsymbol{\zeta} = \begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix} \text{ (components along } \hat{x}_C, \hat{y}_C)$$

SYMBOLS

$\{O\}$	inertial (world) frame, fixed to the ground, z up	–
$\{C\}$	chassis frame, fixed to the robot at its reference point C; $\hat{x}_C$ points forwards, $\hat{y}_C$ to the left	–
$\theta(t)$	heading: angle from $\hat{x}_O$ to $\hat{x}_C$, anticlockwise positive	rad
$\dot{x}, \dot{y}$	velocity of C along the world axes	m/s
u, v	velocity of C along $\hat{x}_C$ (forwards) and $\hat{y}_C$ (sideways, left)	m/s
$\dot{\theta}$	heading rate (the same number in both frames, both z axes are parallel)	rad/s

Use: every velocity in this week is a planar twist (two velocity components, $\dot{\theta}$). Always say which frame the numbers are in.

Rotation between $\{C\}$ and $\{O\}$ [p. 14–16]

Forward and inverse differential kinematics [p. 14–16]

$$\underbrace{\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix}}_{\boldsymbol{\nu}} = \underbrace{\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}}_{\mathbf{J}(\theta)} \underbrace{\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix}}_{\boldsymbol{\zeta}}, \qquad \boldsymbol{\zeta} = \mathbf{J}^{-1}(\theta)\boldsymbol{\nu} = \mathbf{J}^{\top}(\theta)\boldsymbol{\nu} = \begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix}$$

SYMBOLS

$\mathbf{J}(\theta)$	rotation of the chassis frame into the world frame, 3×3; $\mathbf{J}^{-1} = \mathbf{J}^{\top}$, $\det\mathbf{J} = 1$	–
$\boldsymbol{\nu}$, $\boldsymbol{\zeta}$	chassis velocity in $\{O\}$, in $\{C\}$ (above)	m/s, rad/s

Use: forward $\boldsymbol{\nu} = \mathbf{J}\boldsymbol{\zeta}$ ([p. 15]): the robot is commanded $(u, v, \dot{\theta})$, where does it go in the world? Used for simulation. Inverse $\boldsymbol{\zeta} = \mathbf{J}^{-1}\boldsymbol{\nu}$ ([p. 16]): the world velocity you want, what to command? Used for control. The slides' generalised model ([p. 36], [p. 68]) writes $\mathbf{J}^{-1}$ as $\mathbf{R}^{-1}(\theta)$ or $\mathbf{R}^{-1}(\theta)$ in front of $\boldsymbol{\nu}$: same matrix, different name.

DERIVATION of $\mathbf{J}(\theta)$ [added]

Write the unit vectors of $\{C\}$ in $\{O\}$: $\hat{x}_C = (\cos\theta, \sin\theta)$, $\hat{y}_C = (-\sin\theta, \cos\theta)$. The velocity of C is $u\,\hat{x}_C + v\,\hat{y}_C$, so in $\{O\}$

$$\dot{x} = u\cos\theta - v\sin\theta, \qquad \dot{y} = u\sin\theta + v\cos\theta, \qquad \dot{\theta} = \dot{\theta}.$$

That is $\boldsymbol{\nu} = \mathbf{J}\boldsymbol{\zeta}$. $\mathbf{J}$ is a rotation (orthonormal columns, $\det = +1$), so $\mathbf{J}^{-1} = \mathbf{J}^{\top}$: $u = \dot{x}\cos\theta + \dot{y}\sin\theta$ (the part of the world velocity along $\hat{x}_C$) and $v = -\dot{x}\sin\theta + \dot{y}\cos\theta$ (the part along $\hat{y}_C$).

WORKED world ↔ chassis, $\theta = 30°$ [added]

$\boldsymbol{\nu} = (0.5, 0.2, 0.3)$ m/s, rad/s, $\cos 30° = 0.8660$, $\sin 30° = 0.5$: $u = 0.8660(0.5) + 0.5(0.2) = \mathbf{0.5330}$ m/s, $v = -0.5(0.5) + 0.8660(0.2) = \mathbf{-0.0768}$ m/s, $\dot{\theta} = 0.3$ rad/s. Back: $\dot{x} = 0.8660(0.5330) - 0.5(-0.0768) = 0.5$, $\dot{y} = 0.5(0.5330) + 0.8660(-0.0768) = 0.2$ ✓. Speed is preserved: $\sqrt{0.5^2 + 0.2^2} = \sqrt{0.5330^2 + 0.0768^2} = 0.5385$ m/s. A robot heading 90° with $\boldsymbol{\zeta} = (0.4, 0, 0)$ moves $\boldsymbol{\nu} = (0, 0.4, 0)$: straight up the world y axis. [checked]

TRAP
Using $\mathbf{J}$ where $\mathbf{J}^{-1} = \mathbf{J}^{\top}$ is needed (or the reverse): $\theta \to -\theta$. Test: at $\theta = 90°$ a robot driving forward ($u > 0$) must move along $+y_O$.

Pose update (simulation). Euler: $x_{k+1} = x_k + \dot{x}_k \Delta t$, $y_{k+1} = y_k + \dot{y}_k \Delta t$, $\theta_{k+1} = \theta_k + \dot{\theta}_k \Delta t$ with $(\dot{x}, \dot{y})$ recomputed from $\mathbf{J}(\theta_k)$ each step. An exact version for constant $\boldsymbol{\zeta}$ is in the ICR section.

Wheel speeds to chassis velocity [p. 17]

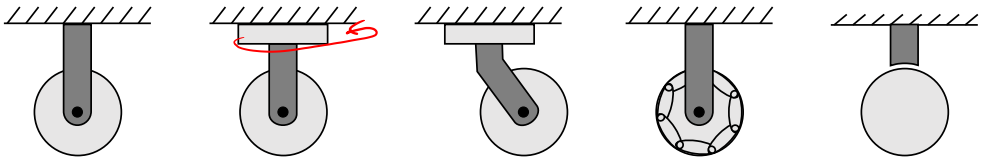
Only the wheel speeds $\dot{\boldsymbol{\varphi}} = (\dot{\varphi}_1, \ldots, \dot{\varphi}_n)$ can be commanded, not $u, v, \dot{\theta}$ ([p. 17]). The robot's kinematics is the map between them. The slides write $\boldsymbol{\zeta} = \mathbf{W}\dot{\boldsymbol{\varphi}}$ (wheels → chassis); the model derived next gives the other direction, $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$ (one row per wheel), and $\mathbf{W}$ is its (pseudo)inverse where it exists.

The two maps [p. 17, 47]

$$\boldsymbol{\zeta} = \mathbf{W}\dot{\boldsymbol{\varphi}} \quad (3 \times n \text{ matrix } \mathbf{W}), \qquad \dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta} \quad (n \times 3 \text{ matrix } \mathbf{M},\ \text{row } i \text{ from the wheel model})$$

Use: write $\mathbf{M}$ first (one row per wheel, always possible). Then $\mathbf{W} = \mathbf{M}^{-1}$ if $n = 3$ and $\det \mathbf{M} \neq 0$; for $n = 2$ (differential drive) use the no-slide constraint $v = 0$ to close the system.

Types of wheels and the constraints they impose [p. 18–20]



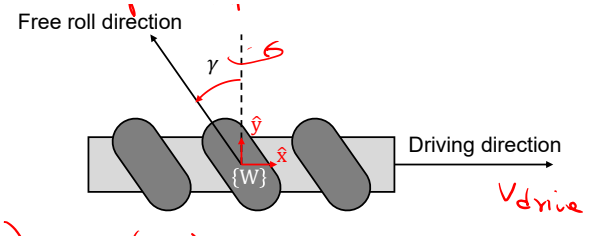
[W10 lec p. 18] — the five wheel types

Wheel (slide [p. 18])	What it is	Constrains chassis?	Rows
Fixed standard	axle fixed to chassis; β_i constant, $\gamma_i = 0$; rolls forwards/back, *cannot slide sideways*	yes	rolling + no-slide
Steerable standard	as above, axle turns about a vertical axis; $\beta_i(t)$, $\gamma_i = 0$	yes	rolling + no-slide
Castor	offset steering axis, passive, follows the motion	no	none
Swedish (omni)	rollers on the rim: free roll along γ_i from the wheel-plane normal	no	rolling only
Spherical	a ball, rolls any way	no	not modelled

Constraints ([p. 19]): Swedish wheels, spherical balls and castors impose *no* constraint on the chassis motion. Only fixed and steerable standard wheels impose constraints: lateral motion is not possible. **Assumption ([p. 20]):** pure rolling, no slipping at the contact point, so the contact point moves at $v_{drive} = r\,\dot{\varphi}$ along the drive direction.

Generalised wheel model: start with an omni wheel [p. 21–25]

PREMISE one wheel, own frame $\{W\}$ **Given:** contact-point velocity (v_x, v_y) in the wheel frame; radius r; roller angle γ
Find: wheel angular speed $\dot{\varphi}$



[W10 lec p. 22] — omni wheel in its frame $\{W\}$: $\hat{x}_W$ is the driving direction, $\hat{y}_W$ the wheel-plane normal; the rollers let the contact point roll freely in the direction at angle γ from $\hat{y}_W$

Omni wheel in its own frame [p. 23–25]

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = v_{drive} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + v_{free\text{-}roll} \begin{bmatrix} -\sin\gamma \\ \cos\gamma \end{bmatrix} \Rightarrow v_{drive} = v_x + v_y \tan\gamma$$

$$\dot{\varphi} = \frac{v_{drive}}{r} = \begin{bmatrix} \frac{1}{r} & \frac{\tan\gamma}{r} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

SYMBOLS

v_x, v_y	contact-point velocity along $\hat{x}_W$ (drive), $\hat{y}_W$ (wheel-plane normal), in $\{W\}$	m/s
v_{drive}	component the wheel's rotation produces, along $\hat{x}_W$	m/s
$v_{free\text{-}roll}$	component the rollers absorb, along $(-\sin\gamma, \cos\gamma)$	m/s
γ	angle between the free-roll direction and the wheel-plane normal $\hat{y}_W$	rad
r, $\dot{\varphi}$	wheel radius; wheel angular speed ($v_{drive} = r\dot{\varphi}$, [p. 24])	m, rad/s

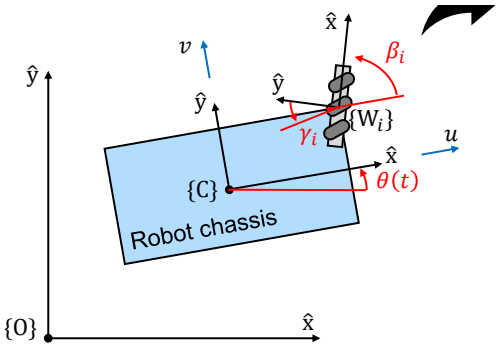
Use: $\gamma = 0$ (rollers $\perp$ drive direction): $\dot{\varphi} = v_x/r$, the sideways part v_y is ignored. A standard wheel is the limit where the free direction does not exist: slide [p. 39] takes $\gamma = 0$ and adds the no-slide rule.

DERIVATION solving for v_{drive} [p. 23–25]
Component equations: $v_x = v_{drive} - \sin\gamma\, v_{free}$ and $v_y = \cos\gamma\, v_{free}$. From the second, $v_{free} = v_y/\cos\gamma$. Put it in the first: $v_x = v_{drive} - \sin\gamma\, v_y/\cos\gamma$, so $v_{drive} = v_x + v_y \tan\gamma$. With $v_{drive} = r\dot{\varphi}$: $\dot{\varphi} = (v_x + v_y \tan\gamma)/r$.

WORKED [added]
$r = 0.05$ m, contact velocity $(v_x, v_y) = (0.3, 0.1)$ m/s. $\gamma = 0$: $\dot{\varphi} = 0.3/0.05 = 6$ rad/s. $\gamma = 45°$ ($\tan\gamma = 1$): $\dot{\varphi} = (0.3 + 0.1)/0.05 = 8$ rad/s. [checked]

Generalised model: wheel in a robot, step by step [p. 26–36]

PREMISE any wheel i on the chassis **Given:** wheel position $({}^{C}x_{W_i}, {}^{C}y_{W_i})$ in $\{C\}$; drive direction β_i; roller angle γ_i; radius r_i; chassis velocity $\boldsymbol{\zeta}$ or $\boldsymbol{\nu}$
Find: wheel speed $\dot{\varphi}_i$



[W10 lec p. 26] — wheel $\{W_i\}$ fixed to the chassis: β_i is the angle from $\hat{x}_C$ to the wheel's drive axis $\hat{x}_{W_i}$

DERIVATION Step 1 and Step 2, then combine [p. 27–34]

Step 1 ([p. 28]). The contact-point velocity in the wheel frame is ${}^{W_i}\mathbf{v} = (v_x, v_y)^\top$. Express it in $\{C\}$ by rotating through β_i:

$$ {}^{C}\mathbf{v}_{W_i} = \begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}. $$

Step 2 ([p. 29–30]). The wheel is a point of the rigid chassis at $({}^{C}x_{W_i}, {}^{C}y_{W_i})$, and $\{C\}$ turns at $\dot\theta$. Velocity of a body point = velocity of C + $\dot\theta\hat{z}\times$(position), and $\dot\theta\hat{z}\times(x,y) = \dot\theta(-y,x)$:

$$ {}^{C}\mathbf{v}_{W_i} = \begin{bmatrix} u - \dot\theta\,{}^{C}y_{W_i} \\ v + \dot\theta\,{}^{C}x_{W_i} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot\theta \end{bmatrix}. $$

Combine ([p. 32]). Both are ${}^{C}\mathbf{v}_{W_i}$, so $\begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \boldsymbol{\zeta}$. **Solve ([p. 34]).** Multiply by the inverse rotation (its transpose):

$$ \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \boldsymbol{\zeta}. $$

Wheel speed. $\dot\varphi_i = \frac{1}{r_i}[1 \;\; \tan\gamma_i](v_x, v_y)^\top$ (omni result [p. 25]). **World frame ([p. 35]).** $\boldsymbol{\zeta} = \mathbf{J}^{-1}(\theta)\boldsymbol{\nu}$, giving the model below.

Generalised model of a wheeled robot [p. 36]

$$ \dot\varphi_i = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \underbrace{\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{R}^{-1}(\theta)=\mathbf{J}^{-1}(\theta)} \begin{bmatrix} \dot x \\ \dot y \\ \dot\theta \end{bmatrix} $$

SYMBOLS

$\dot\varphi_i$	angular velocity of wheel i (positive when it drives the robot along its drive direction)	rad/s
r_i	radius of wheel i	m
γ_i	angle between $v_{free\text{-}roll}$ and the normal to the plane of the wheel	rad
β_i	angle between v_{drive} and the robot chassis ($\hat{x}_C$)	rad
${}^{C}x_{W_i}$, ${}^{C}y_{W_i}$	position of wheel i with respect to the chassis, in $\{C\}$	m
θ	orientation of the chassis with respect to the inertial frame	rad
$\dot x, \dot y, \dot\theta$	velocity of the chassis in the inertial frame	m/s, rad/s

Use: read it right to left: world → chassis velocity; twist → velocity of the wheel's contact point; chassis axes → wheel axes; keep the driven part; divide by the radius. If the question gives $\boldsymbol{\zeta} = (u, v, \dot\theta)$ directly, drop the $\mathbf{R}^{-1}(\theta)$ matrix.

Same model, multiplied out, and stacked ([added])

$$ \dot\varphi_i = \frac{1}{r_i \cos\gamma_i} \Big[\cos(\beta_i+\gamma_i) \;\; \sin(\beta_i+\gamma_i) \;\; {}^{C}x_{W_i}\sin(\beta_i+\gamma_i) - {}^{C}y_{W_i}\cos(\beta_i+\gamma_i) \Big] \boldsymbol{\zeta} $$

For $\gamma_i = 0$:

$$ r_i\dot\varphi_i = \cos\beta_i\,(u - \dot\theta\,{}^{C}y_{W_i}) + \sin\beta_i\,(v + \dot\theta\,{}^{C}x_{W_i}) $$

Stacked ($n = 3$ wheels): $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$ with

$$ \begin{bmatrix} \dot\varphi_1 \\ \dot\varphi_2 \\ \dot\varphi_3 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot\theta \end{bmatrix}, \quad m_{i1} = \frac{\cos\beta_i - \tan\gamma_i\sin\beta_i}{r_i}, \quad m_{i2} = \frac{\sin\beta_i + \tan\gamma_i\cos\beta_i}{r_i}, \quad m_{i3} = -{}^{C}y_{W_i}\,m_{i1} + {}^{C}x_{W_i}\,m_{i2}. $$

Use: the first line says: the wheel only measures the component of the contact-point velocity along the direction at angle $\beta_i + \gamma_i$ (perpendicular to its free-roll direction), divided by $\cos\gamma_i$. For n wheels stack n rows. For world-frame data use $\dot{\boldsymbol{\varphi}} = \mathbf{M}\,\mathbf{R}^{-1}(\theta)\,\boldsymbol{\nu}$. [checked]

STEPS wheel speeds of any wheeled robot (W13 Q10)

1. Draw $\{C\}$ ($\hat{x}_C$ forward) and every wheel. For each wheel write $({}^{C}x_{W_i}, {}^{C}y_{W_i})$, r_i, β_i (direction of positive rolling, from $\hat{x}_C$, anticlockwise), γ_i.
2. Write the generalised model, or for $\gamma_i = 0$ the row $r_i\dot\varphi_i = \cos\beta_i(u - \dot\theta\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot\theta\,{}^{C}x_{W_i})$. Substitute *each wheel's numbers on its own line.*
3. Standard wheels (fixed/steerable): add the no-slide row $0 = -\sin\beta_i(u - \dot\theta\,{}^{C}y_{W_i}) + \cos\beta_i(v + \dot\theta\,{}^{C}x_{W_i})$.
4. Stack the rows into $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$; $\det\mathbf{M} \neq 0$ (3 wheels) means every $(u, v, \dot\theta)$ is reachable; invert for $\boldsymbol{\zeta} = \mathbf{W}\dot{\boldsymbol{\varphi}}$.
5. If the world velocity is needed: $\boldsymbol{\nu} = \mathbf{J}(\theta)\boldsymbol{\zeta}$; if the world velocity is given, $\boldsymbol{\zeta} = \mathbf{J}^\top\boldsymbol{\nu}$ first.
6. **Check:** straight ahead ($u \neq 0$ only) and spin on the spot ($\dot\theta \neq 0$ only) must give sensible signs.

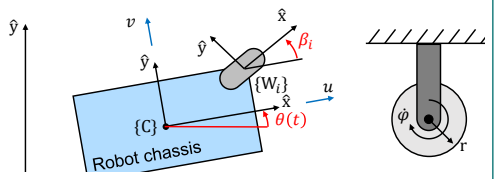
TRAP

(1) Sign of the ${}^{C}y_{W_i}$ term: it is $-\dot\theta\,{}^{C}y_{W_i}$ in the first component. (2) $\mathbf{J}$ (or $\mathbf{R}$) where $\mathbf{R}^{-1}(\theta) = \mathbf{J}^\top$ is needed. (3) $\gamma \neq 0$ for a standard wheel: it is 0. (4) Forgetting the drive direction β_i of each wheel (state your convention: the drive arrow you chose).

Fixed and steerable standard wheels: rolling and no-sliding [p. 37–42, 68]

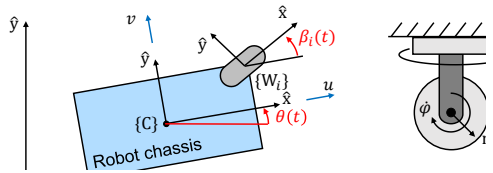
ALTERNATIVE APPROACHES

Fixed standard wheel [p. 39]



[W10 lec p. 39] — β_i constant

Steerable standard wheel [p. 42]



[W10 lec p. 42] — $\beta_i(t)$ a function of time

Standard wheel ($\gamma_i = 0$: "this wheel cannot slide sideways", [p. 39, 42, 68])

Slide form ([p. 39]):

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & 0 \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \mathbf{R}^{-1}(\theta) \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

The bottom component of the product $\begin{bmatrix} v_x \\ v_y \end{bmatrix}$ (the sideways velocity v_y in the wheel frame) is not used by $\dot{\varphi}_i$ because $\tan\gamma_i = 0$; instead it must be *zero* ([p. 68]).

Standard wheel, multiplied out: the rolling and the no-sliding constraint [p. 68]

$$\begin{aligned} \text{rolling:} \quad r_i\dot{\varphi}_i &= \cos\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i}) \\ &= \begin{bmatrix} \cos\beta_i & \sin\beta_i & -{}^{C}y_{W_i}\cos\beta_i + {}^{C}x_{W_i}\sin\beta_i \end{bmatrix} \mathbf{R}^{-1}(\theta)\,\boldsymbol{\nu} \\ \text{no-sliding:} \quad 0 &= -\sin\beta_i(u - \dot{\theta}\,{}^{C}y_{W_i}) + \cos\beta_i(v + \dot{\theta}\,{}^{C}x_{W_i}) \\ &= \begin{bmatrix} -\sin\beta_i & \cos\beta_i & {}^{C}y_{W_i}\sin\beta_i + {}^{C}x_{W_i}\cos\beta_i \end{bmatrix} \mathbf{R}^{-1}(\theta)\,\boldsymbol{\nu} \end{aligned}$$

SYMBOLS

β_i	fixed wheel: constant; steerable wheel: $\beta_i(t)$, an extra input $\dot{\beta}_i$	rad
${}^{C}x_{W_i}, {}^{C}y_{W_i}$	wheel position in $\{C\}$	m

Use: rolling row = component of the contact velocity along the drive direction (gives the wheel speed); no-sliding row = component along the axle (must vanish). The no-sliding rows are constraints on $\boldsymbol{\zeta}$ and are the source of non-holonomy ([p. 68]: "these are non-holonomic constraints").

All standard wheels at once ([added])

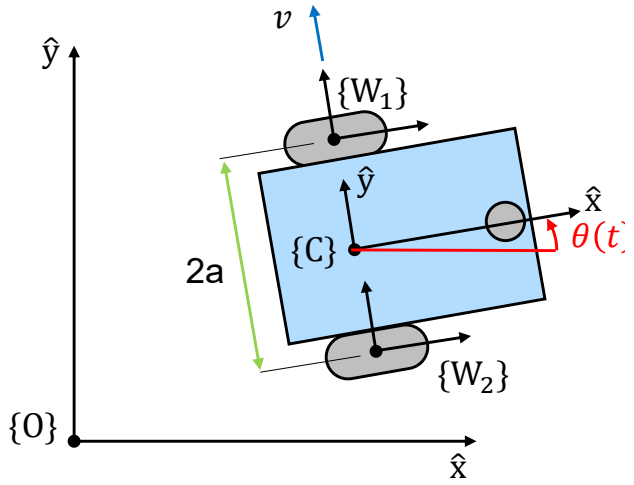
$$\underbrace{\begin{bmatrix} -\sin\beta_1 & \cos\beta_1 & {}^{C}y_{W_1}\sin\beta_1 + {}^{C}x_{W_1}\cos\beta_1 \\ \vdots & \vdots & \vdots \\ -\sin\beta_k & \cos\beta_k & {}^{C}y_{W_k}\sin\beta_k + {}^{C}x_{W_k}\cos\beta_k \end{bmatrix}}_{\mathbf{C}_1(\beta)\ (k \text{ rows, one per standard wheel})} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix} = \mathbf{0} \quad (\text{in } \{O\}:\ \mathbf{C}_1\mathbf{R}^{-1}(\theta)\boldsymbol{\nu} = \mathbf{0})$$

Use: $\operatorname{rank}\mathbf{C}_1$ is how many chassis velocity directions are forbidden; $3 - \operatorname{rank}\mathbf{C}_1$ is the degree of mobility (next sections).

Differential-drive robot [p. 43–47]

PREMISE forward **Given:** wheel speeds $\dot{\varphi}_1$ (left), $\dot{\varphi}_2$ (right); radius r; half track a
Find: u, v, $\dot{\theta}$ (and the ICR)

PREMISE inverse **Given:** wanted u and $\dot{\theta}$
Find: $\dot{\varphi}_1, \dot{\varphi}_2$



[W10 lec p. 45] — differential drive: two fixed standard wheels at $\{W_1\} = (0, a)$ and $\{W_2\} = (0, -a)$, no rollers, each independently driven

wheel ([p. 45])	r_i	γ_i	β_i	${}^{C}x_{W_i}$	${}^{C}y_{W_i}$
W_1 (left)	r	0	0	0	a
W_2 (right)	r	0	0	0	$-a$

DERIVATION each wheel's row, in full [p. 45–47]

Wheel 1 (${}^{C}y_{W_1} = a$): $\dot{\varphi}_1 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix} = \frac{1}{r}(u - a\dot{\theta})$.

Wheel 2 (${}^{C}y_{W_2} = -a$): $\dot{\varphi}_2 = \begin{bmatrix} \frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix} = \frac{1}{r}(u + a\dot{\theta})$.

Stacked (2×3, [p. 47]), then invert. Add and subtract: $\dot{\varphi}_1 + \dot{\varphi}_2 = 2u/r$ and $\dot{\varphi}_2 - \dot{\varphi}_1 = 2a\dot{\theta}/r$. The no-sliding row of either fixed wheel ($\beta = 0$, ${}^{C}x_{W_i} = 0$) is $0 = v$, which fills the middle row of $\mathbf{W}$.

Differential drive: $\dot{\varphi} = \mathbf{M}\zeta$ and $\zeta = \mathbf{W}\dot{\varphi}$ [p. 47]

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\end{bmatrix} = \begin{bmatrix}\frac{1}{r} & 0 & -\frac{a}{r}\\ \frac{1}{r} & 0 & \frac{a}{r}\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \begin{bmatrix}\frac{r}{2} & \frac{r}{2}\\ 0 & 0\\ -\frac{r}{2a} & \frac{r}{2a}\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\end{bmatrix}$$

SYMBOLS

$\dot\varphi_1, \dot\varphi_2$	left ($y_C = +a$) and right ($y_C = -a$) wheel speeds	rad/s
r, a	wheel radius; half the distance between the wheels ($2a$ on [p. 45])	m
$u = \frac{r}{2}(\dot\varphi_1 + \dot\varphi_2)$	forward speed of C (mean rim speed)	m/s
$\dot\theta = \frac{r}{2a}(\dot\varphi_2 - \dot\varphi_1)$	turn rate, anticlockwise positive	rad/s
$v = 0$	no sideways velocity (fixed standard wheels)	m/s

Use: left wheel at $+a$ has contact velocity $u - a\dot\theta$: turning left ($\dot\theta > 0$) slows the left wheel.

WORKED slide-style numbers [checked]

$r = 0.05$ m, $a = 0.15$ m, $\dot{\boldsymbol{\varphi}} = (4, 6)$ rad/s: $u = 0.05(10)/2 = \mathbf{0.25}$ m/s, $\dot\theta = 0.05(6-4)/(2\cdot 0.15) = \mathbf{0.3333}$ rad/s (anticlockwise), $v = 0$. Inverse check: $\dot\varphi_1 = (0.25 - 0.15\cdot 0.3333)/0.05 = 4$ ✓, $\dot\varphi_2 = (0.25 + 0.05)/0.05 = 6$ ✓.

Instantaneous centre of rotation (ICR) [p. 61; practical Q3]

At every instant a planar rigid body rotates about one point, the ICR; every point moves perpendicular to its line to the ICR at speed $\dot\theta\times$ distance. A wheel that rolls without slipping has its axle line through the ICR (as in the Ackermann figure, [p. 61]).

ICR in the chassis frame; ICR of a differential drive (practical Q3)

$$(x_{ICR}, y_{ICR}) = \left(-\frac{v}{\dot\theta}, \frac{u}{\dot\theta}\right), \qquad \text{differential drive } (v=0): \quad y_{ICR} = \frac{u}{\dot\theta} = a\,\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1}, \quad x_{ICR} = 0$$

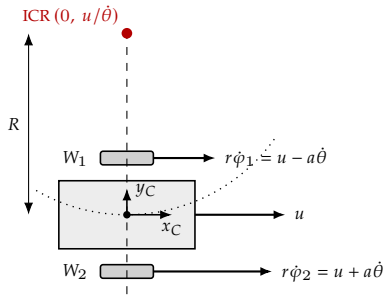
SYMBOLS

x_{ICR}, y_{ICR}	ICR coordinates in $\{C\}$; $y_{ICR} > 0$ is on the *left* of the robot	m
$R = \lvert y_{ICR}\rvert$	radius of the circle traced by C about the ICR	m

Use: $\dot\varphi_1 = \dot\varphi_2$: ICR at infinity (straight line). $\dot\varphi_2 = -\dot\varphi_1$: ICR at C (spin on the spot). $\dot\varphi_1 = 0$: $y_{ICR} = a$, the robot pivots about its left wheel. The slower wheel is on the ICR side. r cancels.

DERIVATION of the ICR [added]

A chassis point (x, y) has velocity $(u - \dot\theta y,\ v + \dot\theta x)$ (Step 2 above). It is zero when $y = u/\dot\theta$ and $x = -v/\dot\theta$. For $v = 0$ substitute $u = \frac{r}{2}(\dot\varphi_1 + \dot\varphi_2)$ and $\dot\theta = \frac{r}{2a}(\dot\varphi_2 - \dot\varphi_1)$: $y_{ICR} = a(\dot\varphi_1 + \dot\varphi_2)/(\dot\varphi_2 - \dot\varphi_1)$. Each wheel's rim speed is $\dot\theta\times$ its distance to the ICR: $r\dot\varphi_1 = \dot\theta(y_{ICR} - a)$, $r\dot\varphi_2 = \dot\theta(y_{ICR} + a)$.



[drawn for this booklet] differential drive turning left: arrow lengths are proportional to $\dot\theta\times$ distance to the ICR; the right wheel is faster

Differential drive: simple scenarios [added]

STEPS circle of radius R at speed v_0 (anticlockwise)

1. $u = v_0$, $\dot\theta = v_0/R$ (use − for clockwise), $v = 0$.
2. $\dot\varphi_1 = \frac{v_0}{r}\left(1 - \frac{a}{R}\right)$ (left), $\dot\varphi_2 = \frac{v_0}{r}\left(1 + \frac{a}{R}\right)$ (right).
3. ICR: $y_{ICR} = R$ on the left. Lap time $T = 2\pi R/v_0$.
4. Check: $a(\dot\varphi_1 + \dot\varphi_2)/(\dot\varphi_2 - \dot\varphi_1) = R$; $R = a$ gives $\dot\varphi_1 = 0$; $R < a$ makes the left wheel *reverse*.

WORKED circle: $r = 0.05$ m, $a = 0.15$ m, $R = 1$ m, $v_0 = 0.4$ m/s [checked]

$\dot\theta = 0.4/1 = 0.4$ rad/s. $\dot\varphi_1 = \frac{0.4}{0.05}(1 - 0.15) = \mathbf{6.8}$ rad/s, $\dot\varphi_2 = 8(1.15) = \mathbf{9.2}$ rad/s. Rim speeds 0.34 and 0.46 m/s $= 0.4 \mp 0.06$. ICR: $0.15\cdot 16/2.4 = 1.0$ m ✓. Lap time $2\pi/0.4 = \mathbf{15.71}$ s. The left wheel follows a circle of radius 0.85 m (17 revolutions per lap), the right wheel radius 1.15 m (23 revolutions). Clockwise instead: swap to (9.2, 6.8). $R = a = 0.15$: (0, 16) rad/s, pivot on the left wheel.

WORKED turning on the spot, and a rectangle path [checked]

Spin at $\dot\theta = \omega$, $u = 0$: $\dot\varphi_1 = -a\omega/r$, $\dot\varphi_2 = +a\omega/r$. With $\omega = 0.5$ rad/s: (−1.5, +1.5) rad/s; a 90° turn takes $(\pi/2)/0.5 = 3.142$ s; each wheel rolls $a\pi/2 = 0.2356$ m (4.712 rad).
Rectangle 2.0 × 1.0 m, straight at $u = 0.25$ m/s, corners as 90° spins at $\omega = 0.5$ rad/s, same r, a:

segment	$(\dot\varphi_1, \dot\varphi_2)$ rad/s	time
long side (2.0 m)	(5, 5)	$2.0/0.25 = 8$ s
short side (1.0 m)	(5, 5)	$1.0/0.25 = 4$ s
each corner (90°)	(−1.5, 1.5)	3.142 s

Straight: $\dot\varphi = u/r = 5$. Total $= 2(8 + 4) + 4(3.142) = \mathbf{36.57}$ s. Clockwise corners: swap the signs of the corner row.

WORKED wheel speeds to motion, ICR outside the axle [checked]

$r = 0.1$ m, $a = 0.25$ m, $\dot{\boldsymbol{\varphi}} = (10, 2)$ rad/s: $u = 0.1(12)/2 = \mathbf{0.6}$ m/s, $\dot\theta = 0.1(2 - 10)/0.5 = \mathbf{-1.6}$ rad/s (clockwise), $y_{ICR} = 0.25(12)/(2 - 10) = \mathbf{-0.375}$ m: 0.375 m to the *right*, beyond the slower right wheel (0.125 m past it), so both wheels turn the same way. World constraint for any pose: $\dot x = u\cos\theta$, $\dot y = u\sin\theta$, so $-\dot x\sin\theta + \dot y\cos\theta = 0$. Example $\theta = 30°$, $\boldsymbol{\zeta} = (0.4, 0, 0.25)$: $\boldsymbol{\nu} = (0.3464, 0.2, 0.25)$ and $-0.3464(0.5) + 0.2(0.8660) = 0$ ✓.

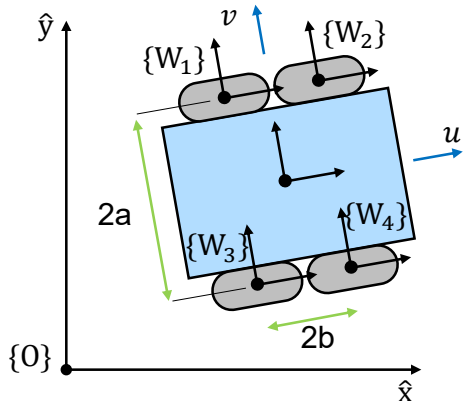
Exact pose after time T for constant $(u, \dot\theta)$, $v = 0$ ([added]; Practice Q10.2)

$$\theta_1 = \theta_0 + \dot\theta T, \quad x_1 = x_0 + \frac{u}{\dot\theta}\left(\sin\theta_1 - \sin\theta_0\right), \quad y_1 = y_0 - \frac{u}{\dot\theta}\left(\cos\theta_1 - \cos\theta_0\right)$$

Use: integrates $\dot x = u\cos\theta(t)$, $\dot y = u\sin\theta(t)$ exactly: a circle of radius $u/\dot\theta$. Radians. Example: $u = 0.4$, $\dot\theta = 0.8$, start $(0, 0, 30°)$, $T = 1.5$ s: $(0.2442,\ 0.5091)$ m. [checked]

Skid-steer robot [p. 48–52]

PREMISE **Given:** wheel speeds of 4 standard wheels (left pair, right pair); radius r; wheels $W_1(-b, a)$, $W_2(b, a)$, $W_3(-b, -a)$, $W_4(b, -a)$
Find: $u, \dot\theta$ (and the wheel rows)



[W10 lec p. 52] — skid-steer: four standard wheels at $\{W_1\}(-b,a)$, $\{W_2\}(b,a)$, $\{W_3\}(-b,-a)$, $\{W_4\}(b,-a)$

Skid-steer rows and inverse [p. 50, 52]

All $r_i = r$, $\gamma_i = \beta_i = 0$. Each row is $\frac{1}{r}[1,\ 0,\ -{}^{C}y_{W_i}]$, because $\tan\gamma = 0$ removes ${}^{C}x_{W_i}$:

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\\ \dot\varphi_4\end{bmatrix} = \begin{bmatrix}1/r & 0 & -a/r\\ 1/r & 0 & -a/r\\ 1/r & 0 & a/r\\ 1/r & 0 & a/r\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \begin{bmatrix}\frac{r}{4} & \frac{r}{4} & \frac{r}{4} & \frac{r}{4}\\ 0 & 0 & 0 & 0\\ -\frac{r}{4a} & -\frac{r}{4a} & \frac{r}{4a} & \frac{r}{4a}\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\\ \dot\varphi_4\end{bmatrix}$$

Use: the 3×4 matrix is the least-squares (pseudo-)inverse: average the wheel speeds. Wheels 1,2 are the left pair at $y_C = +a$, 3,4 the right pair. Example $r = 0.1$, $a = 0.2$, $\dot{\boldsymbol\varphi} = (4,4,6,6)$: $u = 0.5$ m/s, $\dot\theta = 0.5$ rad/s. [checked]

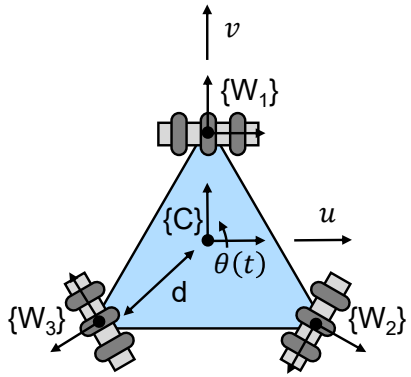
TRAP

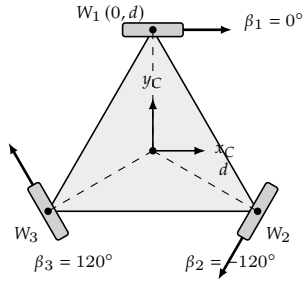
Four standard wheels give four no-sliding rows $[0,\ 1,\ {}^{C}x_{W_i}]$ with ${}^{C}x_{W_i} = \mp b$ (rank 2: $v = 0$ and $\dot\theta = 0$), so a real skid-steer *cannot* turn without slipping. The slide's model keeps only $v = 0$ and assumes pure rolling for turning anyway [added].

Robots with Swedish wheels: the triangular robot [p. 53–54, W13 Q10]

PREMISE W13 Q10 / practical Q1 **Given:** three omni wheels at the vertices of an equilateral triangle, distance d from C; rollers $\perp$ drive direction; radius r

Find: $\dot{\boldsymbol\varphi} = \mathbf{M}\boldsymbol\zeta$ and $\boldsymbol\zeta = \mathbf{W}\dot{\boldsymbol\varphi}$ (and are $u, v, \dot\theta$ all free?)





[W10 lec p. 54] — the robot of the slide and of W13 Q10 ([W13 lec p. 20])

[drawn for this booklet] wheel positions and drive-direction arrows (clockwise tangential, read from the slide figure); free roll is along the dashed radius

DERIVATION Step 1: geometry of each wheel [added]

Wheel i sits at distance d from C at position angle ϕ_i (W_1: 90°, W_2: −30°, W_3: 210°). The drive arrows in the figure are tangential and clockwise, so $\beta_i = \phi_i - 90°$. Rollers $\perp$ drive $\Rightarrow \gamma_i = 0$. State this convention in your answer: if you drew the drive arrows anticlockwise, every $-d$ below becomes $+d$.

i	${}^{C}x_{W_i}$	${}^{C}y_{W_i}$	β_i	$\cos\beta_i$	$\sin\beta_i$
1	0	d	0°	1	0
2	$\frac{\sqrt3}{2}d$	$-\frac{d}{2}$	−120°	$-\frac12$	$-\frac{\sqrt3}{2}$
3	$-\frac{\sqrt3}{2}d$	$-\frac{d}{2}$	120°	$-\frac12$	$\frac{\sqrt3}{2}$

DERIVATION Step 2: each wheel's row from $r\dot\varphi_i = \cos\beta_i(u - \dot\theta\,{}^{C}y_{W_i}) + \sin\beta_i(v + \dot\theta\,{}^{C}x_{W_i})$ [added]

Wheel 1: $r\dot\varphi_1 = 1\cdot(u - \dot\theta d) + 0\cdot(v+0) = u - d\dot\theta$.

Wheel 2: $r\dot\varphi_2 = -\frac12\left(u + \frac{d}{2}\dot\theta\right) - \frac{\sqrt3}{2}\left(v + \frac{\sqrt3}{2}d\dot\theta\right) = -\frac12 u - \frac{\sqrt3}{2}v - \left(\frac14 + \frac34\right)d\dot\theta = -\frac12 u - \frac{\sqrt3}{2}v - d\dot\theta$.

Wheel 3: $r\dot\varphi_3 = -\frac12\left(u + \frac{d}{2}\dot\theta\right) + \frac{\sqrt3}{2}\left(v - \frac{\sqrt3}{2}d\dot\theta\right) = -\frac12 u + \frac{\sqrt3}{2}v - \left(\frac14 + \frac34\right)d\dot\theta = -\frac12 u + \frac{\sqrt3}{2}v - d\dot\theta$.

Same for wheel 2 with the matrix chain of the slide ([p. 36]): $\left[\frac{1}{r}\ 0\right]\begin{bmatrix}-\frac12 & -\frac{\sqrt3}{2}\\ \frac{\sqrt3}{2} & -\frac12\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{d}{2}\\ 0 & 1 & \frac{\sqrt3}{2}d\end{bmatrix} =$

$\frac{1}{r}\left[-\frac12\ \ -\frac{\sqrt3}{2}\ \ -\frac{d}{4} - \frac{3d}{4}\right] = \frac{1}{r}\left[-\frac12\ \ -\frac{\sqrt3}{2}\ \ -d\right]$.

Omni triangle: the full 3×3 matrix, its inverse (Sample Q10) [p. 54]

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\end{bmatrix} = \frac{1}{r}\begin{bmatrix}1 & 0 & -d\\ -\frac12 & -\frac{\sqrt3}{2} & -d\\ -\frac12 & \frac{\sqrt3}{2} & -d\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}, \qquad \begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix} = \frac{r}{3}\begin{bmatrix}2 & -1 & -1\\ 0 & -\sqrt3 & \sqrt3\\ -\frac1d & -\frac1d & -\frac1d\end{bmatrix}\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\end{bmatrix}$$

SYMBOLS

d	distance from C to each wheel	m
r	wheel radius	m
det (unscaled 3×3)	$= \frac{3\sqrt3}{2}\,d \neq 0$	m

Use: $\det \neq 0$ *and no no-slide rows:* $(u, v, \dot\theta)$ *can all be set freely (holonomic,* $\delta_m = 3$*). Check: all wheels equal* $\dot\varphi_i = -d\dot\theta/r$ *for a pure spin;* $\dot\varphi_1 + \dot\varphi_2 + \dot\varphi_3 = 0$ *for a pure translation.* [checked]

DERIVATION Step 3: the inverse by elimination [added]

Let $a_i = r\dot{\varphi}_i$. Then $a_1 = u - d\dot{\theta}$, $a_2 = -\frac{1}{2}u - \frac{\sqrt{3}}{2}v - d\dot{\theta}$, $a_3 = -\frac{1}{2}u + \frac{\sqrt{3}}{2}v - d\dot{\theta}$. Add all three: $a_1 + a_2 + a_3 = -3d\dot{\theta} \Rightarrow \dot{\theta} = -\dfrac{a_1 + a_2 + a_3}{3d}$. Subtract: $a_3 - a_2 = \sqrt{3}v \Rightarrow v = \dfrac{a_3 - a_2}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}(a_3 - a_2)$. From row 1: $u = a_1 + d\dot{\theta} = a_1 - \dfrac{a_1 + a_2 + a_3}{3} = \dfrac{2a_1 - a_2 - a_3}{3}$. These are the rows of the inverse above. Determinant (expand along row 1): $1\left[(-\frac{\sqrt{3}}{2})(-d) - (-d)(\frac{\sqrt{3}}{2})\right] + (-d)\left[(-\frac{1}{2})(\frac{\sqrt{3}}{2}) - (-\frac{\sqrt{3}}{2})(-\frac{1}{2})\right] = \sqrt{3}d + \frac{\sqrt{3}}{2}d = \frac{3\sqrt{3}}{2}d$.

WORKED numbers: $d = 0.10$ **m,** $r = 0.05$ **m** [checked]

Forward at $u = 0.2$ **m/s** ($v = \dot{\theta} = 0$): $\dot{\varphi} = \frac{1}{0.05}(0.2, -0.1, -0.1) = (\mathbf{4}, \mathbf{-2}, \mathbf{-2})$ rad/s (W13 Q10 numbers). Wheel 1 carries all of u; wheels 2,3 see $-\frac{1}{2}u$ along their drive directions and their rollers absorb the rest.

Sideways $v = 0.2$**:** $(0, -3.464, +3.464)$ rad/s. **Spin** $\dot{\theta} = 0.5$**:** $(-1, -1, -1)$ rad/s.

Practical Q2 (P10.2): $\dot{\varphi} = (4, -1, -1)$: $u = \frac{0.05}{3}(2\cdot 4 + 1 + 1) = 0.1667$ m/s, $v = \frac{0.05}{\sqrt{3}}(\dot{\varphi}_3 - \dot{\varphi}_2) = 0$, $\dot{\theta} = -\frac{0.05}{0.3}(4 - 1 - 1) = -0.3333$ rad/s. ICR in $\{C\}$: $(-v/\dot{\theta}, u/\dot{\theta}) = (0, -0.5)$: a clockwise circle of radius 0.5 m below the robot. From the origin with $\theta = 0$, after 2 s: $\theta = -0.6667$, $x = 0.5\sin 0.6667 = 0.3092$ m, $y = -0.5(1 - \cos 0.6667) = -0.1071$ m.

WORKED world velocity to wheel speeds: heading matters [added] [checked]

$d = 0.10$, $r = 0.05$, heading $\theta = 30°$, wanted $\boldsymbol{\nu} = (0.3, 0.1, 0)$ m/s. Chassis: $u = 0.8660(0.3) + 0.5(0.1) = 0.3098$, $v = -0.5(0.3) + 0.8660(0.1) = -0.0634$ m/s. Then $\dot{\varphi}_1 = u/r = 6.196$, $\dot{\varphi}_2 = (-\frac{1}{2}(0.3098) - \frac{\sqrt{3}}{2}(-0.0634))/0.05 = -2.000$, $\dot{\varphi}_3 = -4.196$ rad/s. (Wrongly ignoring the heading, $\theta = 0$, would give $(6, -4.732, -1.268)$.) The full chain of [p. 36] with $\mathbf{R}^{-1}(30°)$ gives the same three numbers.

STEPS omni robot following a path without rotating ($\dot{\theta} = 0$, $\theta = 0$)

1. The world velocity is the chassis velocity: $\boldsymbol{\zeta} = \boldsymbol{\nu}$.
2. Wheel speeds: $\dot{\varphi}_1 = u/r$, $\dot{\varphi}_2 = -(\frac{1}{2}u + \frac{\sqrt{3}}{2}v)/r$, $\dot{\varphi}_3 = (-\frac{1}{2}u + \frac{\sqrt{3}}{2}v)/r$.
3. Time = path length / speed. **Check:** $\dot{\varphi}_1 + \dot{\varphi}_2 + \dot{\varphi}_3 = 0$ at every instant.

WORKED square 1×1 **m at** 0.2 **m/s, then a circle keeping the heading** [checked]

side	(u, v) m/s	$\dot{\varphi}_1$	$\dot{\varphi}_2$	$\dot{\varphi}_3$
east	$(0.2, 0)$	4	−2	−2
north	$(0, 0.2)$	0	−3.464	3.464
west	$(-0.2, 0)$	−4	2	2
south	$(0, -0.2)$	0	3.464	−3.464

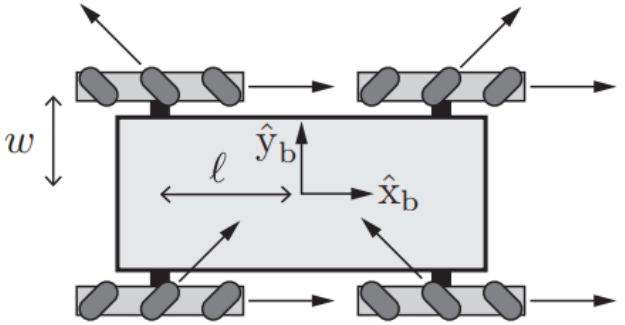
(rad/s.) $1/0.2 = 5$ s per side, 20 s in all; the robot never rotates. **Circle** of radius $R = 0.5$ m at $s = 0.2$ m/s, heading fixed: $\omega = s/R = 0.4$ rad/s, period $2\pi/0.4 = 15.71$ s, centre velocity $(u, v) = s(-\sin\omega t, \cos\omega t)$. Then $\dot{\varphi}_1 = -\frac{s}{r}\sin\omega t$, $\dot{\varphi}_2 = \frac{s}{r}\sin(\omega t - 60°)$, $\dot{\varphi}_3 = \frac{s}{r}\sin(\omega t + 60°)$, with $s/r = 4$ rad/s: three sinusoids 120° apart. At $t = 0$: $(0, -3.464, 3.464)$ (moving north); at $t = 3.93$ s: $(-4, 2, 2)$ (moving west). **Circle with the heading following the path** ($u = 0.2$, $\dot{\theta} = 0.4$, $v = 0$): $\dot{\varphi}_1 = (0.2 - 0.04)/0.05 = 3.2$, $\dot{\varphi}_2 = \dot{\varphi}_3 = (-0.1 - 0.04)/0.05 = -2.8$ rad/s; ICR $(0, 0.5)$.

A robot with 4 Swedish wheels [p. 55]

PREMISE radius r

Given: wheel positions $(\pm\ell, \pm w)$, drive direction $\beta = 0$ for all, roller angles $\gamma_i = \pm 45°$,

Find: $\dot{\boldsymbol{\varphi}} = \mathbf{M}\boldsymbol{\zeta}$ (a 4×3 matrix) and the best $\boldsymbol{\zeta}$ for given $\dot{\boldsymbol{\varphi}}$



[W10 lec p. 55] — four Swedish wheels; ℓ and w are the lengths shown (the slide gives no equations)

4-Swedish-wheel rows ([added]), $\beta_i = 0$

$$\dot{\varphi}_i = \frac{1}{r}\begin{bmatrix}1 & \tan\gamma_i & \tan\gamma_i\,{}^{C}x_{W_i} - {}^{C}y_{W_i}\end{bmatrix}\boldsymbol{\zeta}$$

With the pattern given in the mock exam (FL $\gamma = -45°$, FR $+45°$, RL $+45°$, RR $-45°$; wheels at $(\pm\ell, \pm w)$):

$$\begin{bmatrix}\dot{\varphi}_{FL}\\ \dot{\varphi}_{FR}\\ \dot{\varphi}_{RL}\\ \dot{\varphi}_{RR}\end{bmatrix} = \frac{1}{r}\begin{bmatrix}1 & -1 & -(\ell + w)\\ 1 & 1 & \ell + w\\ 1 & 1 & -(\ell + w)\\ 1 & -1 & \ell + w\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot{\theta}\end{bmatrix}, \qquad \boldsymbol{\zeta} = \frac{r}{4}\begin{bmatrix}1 & 1 & 1 & 1\\ -1 & 1 & 1 & -1\\ -\frac{1}{\ell+w} & \frac{1}{\ell+w} & -\frac{1}{\ell+w} & \frac{1}{\ell+w}\end{bmatrix}\dot{\boldsymbol{\varphi}}$$

Use: rank 3: every $(u, v, \dot{\theta})$ is reachable, so it is holonomic; 4 wheels for 3 unknowns, so the 3×4 inverse is the least-squares one. Always take the signs of γ_i from the question: the slide's figure shows only the free-roll arrows. Example $r = 0.05$, $\ell = 0.20$, $w = 0.15$: sideways $v = 0.2$ gives $(-4, 4, 4, -4)$ rad/s; forward $u = 0.3$ gives $(6, 6, 6, 6)$ rad/s.

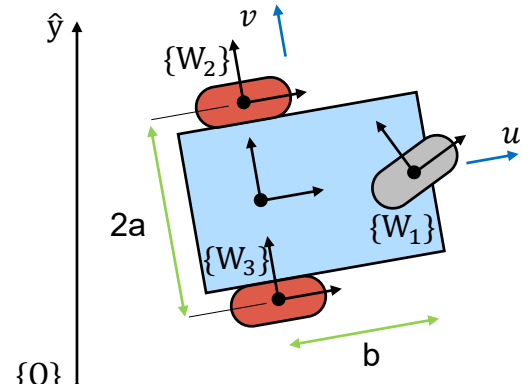
[checked]

Tricycle robot [p. 56–58]

PREMISE radius r

Given: wanted u and $\dot{\theta}$ at C (the mid-point of the rear axle); front wheel at $(b, 0)$;

Find: front wheel speed $\dot{\varphi}_1$ and steering angle β_1



[W10 lec p. 58] — tricycle: front wheel $\{W_1\}$ driven and steered at $(b, 0)$; rear wheels $\{W_2\}, \{W_3\}$ at $(0, \pm a)$ not driven

DERIVATION [p. 58], filled in

Front wheel: $r_1 = r$, $\gamma_1 = 0$, $\beta_1 = \beta_1(t)$, ${}^C x_{W_1} = b$, ${}^C y_{W_1} = 0$. The rear wheels cannot slide laterally (no-slide row with $\beta = 0$, ${}^C x_{W_i} = 0$: $v = 0$). Step 2 with $v = 0$: ${}^C\mathbf{v}_{W_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & b \end{bmatrix} (u, 0, \dot\theta)^\top = (u,\ b\dot\theta)$. Step 1: ${}^C\mathbf{v}_{W_1} = \begin{bmatrix} \cos\beta_1 & -\sin\beta_1 \\ \sin\beta_1 & \cos\beta_1 \end{bmatrix} (r\dot\varphi_1,\ 0)^\top = r\dot\varphi_1(\cos\beta_1, \sin\beta_1)$. Equate: $r\dot\varphi_1 \cos\beta_1 = u$ and $r\dot\varphi_1 \sin\beta_1 = b\dot\theta$. Square and add; divide.

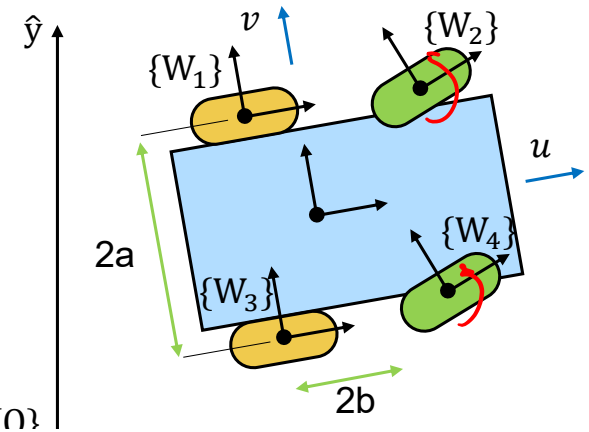
Tricycle: front-wheel command [p. 58]

$$\dot\varphi_1 = \frac{1}{r}\sqrt{u^2 + b^2\dot\theta^2}, \qquad \beta_1 = \operatorname{atan2}(b\dot\theta,\ u), \qquad v = 0$$

Use: the slide assumes the rear axle passes through C (so $v = 0$). Equivalent: $\tan\beta_1 = b\dot\theta/u = b/R$ *with* $R = u/\dot\theta$. *Rear wheels roll at* $(u \mp a\dot\theta)/r$ *[added]. Example (P10.5):* $b = 0.6$, $r = 0.1$, $u = 0.4$, $\dot\theta = 0.5$*: contact velocity* $(0.4, 0.3)$, 0.5 *m/s,* $\dot\varphi_1 =$ **5** *rad/s,* $\beta_1 = \operatorname{atan2}(0.3, 0.4) =$ **36.87°**; $R = 0.8$ *m and* $\arctan(0.6/0.8) = 36.87°$ ✓. *[checked]*

Homework: car-type robot [p. 59–60]

PREMISE derived [added] **Given:** rear wheels W_1, W_3 driven, fixed, at $(-b, \pm a)$; front wheels W_2, W_4 steered, not driven, at $(b, \pm a)$ (C at the body centre, as on [p. 60]); speed u, turn rate $\dot\theta$
Find: rear wheel speeds, front steering angles β_2, β_4, sideways v



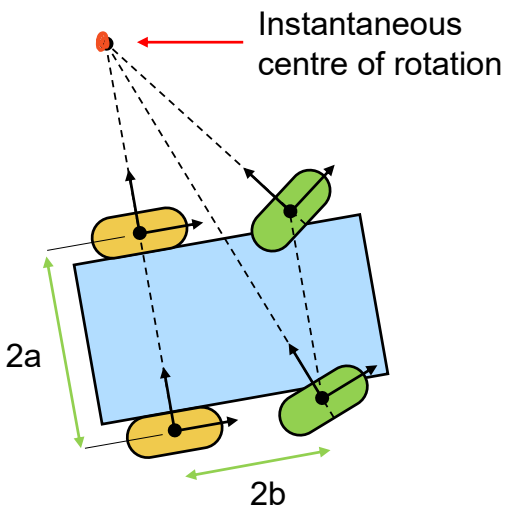
[W10 lec p. 60] — car-type robot (homework slide): rear wheels yellow (driven, fixed), front wheels green (steered)

Car-type robot, from the generalised model ([added])

$$\text{rear no-slide } (\beta = 0,\ x = -b): \quad 0 = v - b\dot\theta \implies v = b\dot\theta,$$
$$\text{rear rolling:} \quad r\dot\varphi_1 = u - a\dot\theta, \quad r\dot\varphi_3 = u + a\dot\theta,$$
$$\text{front no-slide (wheel } (b, \pm a),\ \beta_k): \quad \tan\beta_2 = \frac{v + b\dot\theta}{u - a\dot\theta} = \frac{2b\dot\theta}{u - a\dot\theta}, \quad \tan\beta_4 = \frac{2b\dot\theta}{u + a\dot\theta}.$$

Use: ICR $= (-v/\dot\theta,\ u/\dot\theta) = (-b,\ R)$ *lies on the rear-axle line; with* $R = u/\dot\theta$: $\tan\beta_2 = 2b/(R - a)$, $\tan\beta_4 = 2b/(R + a)$, *and* $\cot\beta_4 - \cot\beta_2 = 2a/(2b)$. *Example* $a = 0.75$, $b = 1.25$, $r = 0.3$, $u = 2$, $R = 6$ *m:* $\dot\theta = 0.3333$, $v = 0.4167$ *m/s, rear* $(5.833, 7.5)$ *rad/s,* $\beta_2 = 25.46°$ *(inner),* $\beta_4 = 20.32°$ *(outer). [checked]*

Ackermann steering [p. 61]



[W10 lec p. 61] — Ackermann: the two front wheels are steered at different angles so that all four axle lines meet at one ICR

Ackermann condition

$$\tan\beta_{in} = \frac{L}{R - t/2}, \qquad \tan\beta_{out} = \frac{L}{R + t/2}, \qquad \cot\beta_{out} - \cot\beta_{in} = \frac{t}{L}$$

SYMBOLS

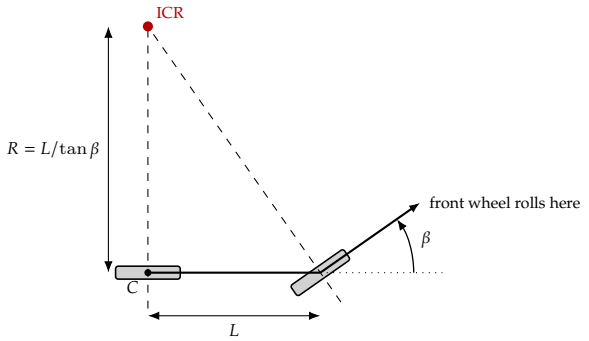
L	wheelbase (rear axle to front axle; $2b$ on the slide)	m
t	track (between the two wheels of an axle; $2a$ on the slide)	m
R	distance from the middle of the rear axle to the ICR, along the rear-axle line	m

Use: purpose ([p. 61]): all wheels rotate without slipping. Example $L = 2.5$, $t = 1.5$, $R = 6$: *inner* $\arctan(2.5/5.25) = \mathbf{25.46}°$, *outer* $\arctan(2.5/6.75) = \mathbf{20.32}°$ *(same numbers as the car above). One common angle would make the tyres scrub.* [checked]

Homework: bicycle, and jammed steering [p. 69]

PREMISE practical Q4 **Given:** rear wheel fixed, driven, at C; front wheel at $(L, 0)$, steered by β, not driven

Find: kinematics $(\dot{x}, \dot{y}, \dot{\theta})$ in terms of u and β; is it holonomic?



[drawn for this booklet] bicycle: both axle lines (the rear one is the y_C axis, the front one is perpendicular to the steered wheel) meet at the ICR

DERIVATION [added]

Rear wheel $(0,0)$, $\beta = 0$: rolling $r\dot{\varphi}_r = u$; no-slide $0 = v$. Front wheel $(L, 0)$, drive direction at angle β: no-slide $0 = -\sin\beta\,(u - 0) + \cos\beta\,(v + \dot{\theta}L)$. With $v = 0$: $L\dot{\theta}\cos\beta = u\sin\beta$.

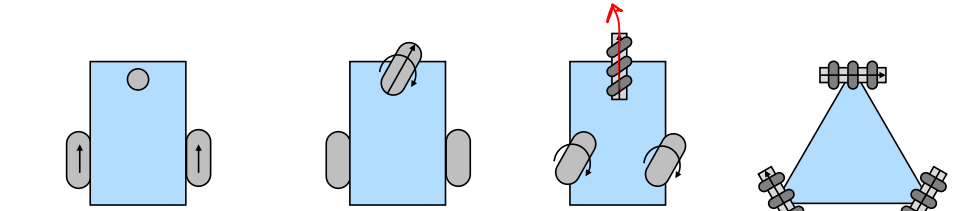
Bicycle kinematics

$$\dot{\theta} = \frac{u\tan\beta}{L}, \quad R = \frac{L}{\tan\beta}, \quad \dot{x} = u\cos\theta,\ \dot{y} = u\sin\theta, \quad v = 0; \qquad \text{front wheel: } r_f\dot{\varphi}_f = \frac{u}{\cos\beta}$$

Use: the front wheel's rim speed follows from the rolling row: $\sqrt{u^2 + (L\dot{\theta})^2} = u/\cos\beta$. *Example (P10.4):* $u = 2$, $\beta = 20°$, $L = 1.2$: $\dot{\theta} = \mathbf{0.6066}$ *rad/s,* $R = \mathbf{3.297}$ *m. Given* $R = 3$ *m instead:* $\beta = \arctan(1.2/3) = 21.80°$, $\dot{\theta} = 0.6667$ *rad/s, one lap* $2\pi(3)/2 = 9.425$ *s, front rim speed* 2.154 *m/s.* [checked]

Jammed steering ($\beta = 0$). The front wheel is then a fixed wheel with no-slide row $0 = v + L\dot{\theta}$; with the rear row $v = 0$ this forces $\dot{\theta} = 0$ too: $\operatorname{rank}\mathbf{C}_1 = 2$, only forward/backward motion remains ($\delta_m = 1$, $\delta_s = 0$). [added]

Manoeuvrability: $\delta_M = \delta_m + \delta_s$ [p. 62–64]



[W10 lec p. 64] — the four examples of the slide, left to right

Degree of manoeuvrability [p. 63]

$$\delta_M = \delta_m + \delta_s$$

SYMBOLS

δ_m	degree of mobility: slide's aid, "the number of driven wheels"; precisely ([added]) $\delta_m = 3 - \operatorname{rank} \mathbf{C}_1$, the number of chassis velocity directions the standard wheels leave free	–
δ_s	degree of steerability: "the number of steerable wheels"; precisely ([added]) the rank of the no-slide rows of the steered wheels	–
δ_M	overall degrees of freedom the robot can manipulate	–

Use: $\delta_M = 3$ is the most a planar robot can have.

Robot (slide [p. 64], left to right)	δ_m	δ_s	δ_M
2 fixed standard on one axle + castor (diff. drive)	2	0	2
2 fixed + 1 steerable standard (tricycle, car)	1	1	2
2 steerable standard + 1 Swedish	1	2	3
3 Swedish (omni triangle)	3	0	3

Why ([added]). Diff. drive: rows $[0, 1, 0]$ twice, rank 1, so $\delta_m = 3-1 = 2$. Tricycle: rear $[0, 1, 0]$, front $[-\sin\beta, \cos\beta, b\cos\beta]$, rank 2, so $\delta_m = 1$, $\delta_s = 1$. Omni: no standard wheel, $\mathbf{C}_1$ empty, $\delta_m = 3$. [checked]

Holonomic vs non-holonomic [p. 65–68]

Definitions [p. 66–68]

Holonomic constraint: can be written as a function of position only, $f(\beta, \gamma, \varphi, \theta, x, y) = 0$; no derivatives. **Non-holonomic constraint:** contains derivatives of position terms and *cannot be integrated* to a position-only form (non-integrable). The no-sliding rows of a standard wheel are non-holonomic ([p. 68]); for a diff. drive in the world frame:

$$-\dot{x}\sin\theta + \dot{y}\cos\theta = 0$$

Use: a robot with such a constraint is non-holonomic; with none (omni, mecanum) it is holonomic and $\delta_M = \delta_m = 3$.

Reason (explain-type). The constraint forbids sideways *velocity* at every instant, but not any final pose: parallel parking reaches every (x, y, θ), just not directly. If it were holonomic, $f(x, y, \theta) = 0$ would confine the robot to a surface in the 3-D pose space and some poses would be unreachable. Test (optional, [added]): for $\omega = -\sin\theta\, dx + \cos\theta\, dy$, $\boldsymbol{\omega}\cdot\nabla\times\boldsymbol{\omega} = -1 \neq 0$, so it is not integrable. [checked]

STEPS "Is robot X holonomic?"

1. List its wheels. Any standard (fixed or steerable) wheel? Write its no-sliding row $-\sin\beta(u - \dot{\theta}y) + \cos\beta(v + \dot{\theta}x) = 0$.
2. Say: it involves *velocities*, and cannot be integrated to a position-only equation.
3. Consequence: it can still reach any pose by manoeuvring, but not along any path. Quote $\delta_m < 3$.
4. Omni/Swedish only: no no-slide row, $\det \mathbf{M} \neq 0$, $\delta_m = 3$: holonomic.

ALTERNATIVE APPROACHES

Holonomic (omni, mecanum)	Non-holonomic (diff. drive, bicycle, car)
Constraints on velocity: none. $(u, v, \dot{\theta})$ all free; can translate sideways; path of any shape directly. $\delta_M = 3$.	No-slide rows $\Rightarrow$ $v = 0$ (diff. drive) or $\dot{\theta} = u\tan\beta/L$ (bicycle). $\delta_m < 3$. Reaches any pose, but only by manoeuvres (arcs, reversing).

Which wheel model? Decision table [added]

If the question gives …	use …
wheel positions and drive directions, any layout	generalised row $r\dot{\varphi}_i = \cos\beta(u - \dot{\theta}y) + \sin\beta(v + \dot{\theta}x)$
two fixed wheels at $\pm a$	$\dot{\varphi}_{1,2} = (u \mp a\dot{\theta})/r$, $v = 0$
wheel speeds, "where is the ICR?"	$y_{ICR} = u/\dot{\theta} = a\frac{\dot{\varphi}_1+\dot{\varphi}_2}{\dot{\varphi}_2-\dot{\varphi}_1}$
circle of radius R at speed v_0	$\dot{\theta} = v_0/R$, then wheel rows; lap $2\pi R/v_0$
steering angle, wheelbase	bicycle $\dot{\theta} = u\tan\beta/L$; Ackermann $\tan\beta = L/(R \mp t/2)$
three Swedish wheels, d	3×3 matrix with rows $[\cos\beta, \sin\beta, -d]$ …inverse by add/subtract
world velocity given	$\boldsymbol{\zeta} = \mathbf{J}^\top(\theta)\boldsymbol{\nu}$ first; then wheel rows
"holonomic?"	does a standard wheel (no-slide row) exist?

SLIDE CHECK W10: where the slides leave a choice (no numerical error found)

Every matrix on [p. 36, 39, 42, 45, 47, 50, 52, 58, 68] was re-derived and agrees with this booklet. Points to state in an answer: (1) [p. 36] prints the world → chassis matrix $\begin{bmatrix}\cos\theta & \sin\theta & 0\\ -\sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$ without a name and [p. 68] calls it $\mathbf{R}^{-1}(\theta)$; [p. 14] calls the other direction $\mathbf{J}(\theta)$. They are inverses of each other. (2) The drive direction β_i of the omni triangle ([p. 54], W13 Q10) is *not* written on the slides: the figure gives tangential arrows, so $\beta = (0°, -120°, 120°)$ clockwise; anticlockwise gives the same physics with the sign of d (and of the wheel speeds) flipped. (3) [p. 58]: $\dot{\varphi}_1 = +\frac{1}{r}\sqrt{\cdot}$ and $\beta_1 = \operatorname{atan2}(b\dot{\theta}, u)$ is one of two equivalent answers (the other has the wheel running backwards, $\beta_1 + 180°$). (4) [p. 63] describes δ_m and δ_s only informally ("number of driven / steerable wheels"); the rank form used here reproduces the four examples on [p. 64]. (5) [p. 55] and [p. 60] give figures and no equations.

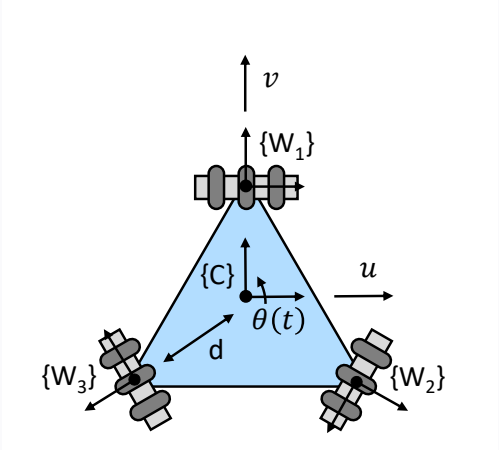
TRAP typical lost marks, W10 [added]

(1) Forgetting to say *which frame* a velocity is in. (2) Degrees in a calculator set to radians (and vice versa) when evaluating $\tan\beta$ or $\sin\theta$. (3) a is the *half* track ($2a$ is the track on the slides). (4) Left wheel is at $+a$ and *slows* when the robot turns left. (5) ICR on the wrong side: $y_{ICR} > 0$ is left of the robot for $\dot{\theta} > 0$. (6) Writing the no-slide row but not using it ($v = 0$). (7) Saying "holonomic because it reaches every pose". (8) Giving $\dot{\varphi}$ without the drive-direction convention.

How this is examined: worked solutions

EXAM-STYLE QUESTION (W13 Q10) [marks not published; indicative 1 per step]

Consider a robot with three omnidirectional wheels, each independently driven, rollers perpendicular to each wheel's driving direction. The robot is an equilateral triangle; the distance from the centre to each vertex is d. Derive a kinematic expression that relates the angular velocities of the wheels $(\dot\varphi_1, \dot\varphi_2, \dot\varphi_3)$ to the velocity of the robot chassis frame $(u, v, \dot\theta)$. Is the robot holonomic? Find the wheel speeds for $u = 0.2$ m/s with $d = 0.10$ m, $r = 0.05$ m.



[W13 lec p. 20] — W13 p. 20: the question's figure

Solution

Setup. $\{C\}$ at the centre, $\hat x_C$ to the right, $\hat y_C$ up. Wheels at $W_1(0, d)$, $W_2(\frac{\sqrt3}{2}d, -\frac{d}{2})$, $W_3(-\frac{\sqrt3}{2}d, -\frac{d}{2})$; drive directions tangential (clockwise, as drawn): $\beta_1 = 0°$, $\beta_2 = -120°$, $\beta_3 = 120°$; $\gamma_i = 0$ (rollers $\perp$ drive). [2]

General formula (slide [p. 36]). $\dot\varphi_i = \begin{bmatrix}\frac1r & \frac{\tan\gamma_i}{r}\end{bmatrix}\mathbf{R}(\beta_i)^\top \begin{bmatrix}1 & 0 & -{}^C y_{W_i} \\ 0 & 1 & {}^C x_{W_i}\end{bmatrix}\boldsymbol\zeta$; with $\gamma_i = 0$: $r\dot\varphi_i = \cos\beta_i(u - \dot\theta\, {}^C y_{W_i}) + \sin\beta_i(v + \dot\theta\, {}^C x_{W_i})$. [1]

Rows. $r\dot\varphi_1 = u - d\dot\theta$; $r\dot\varphi_2 = -\frac12(u + \frac{d}{2}\dot\theta) - \frac{\sqrt3}{2}(v + \frac{\sqrt3}{2}d\dot\theta) = -\frac12 u - \frac{\sqrt3}{2}v - d\dot\theta$; $r\dot\varphi_3 = -\frac12 u + \frac{\sqrt3}{2}v - d\dot\theta$. [3]

$$\begin{bmatrix}\dot\varphi_1\\ \dot\varphi_2\\ \dot\varphi_3\end{bmatrix} = \frac1r\begin{bmatrix}1 & 0 & -d\\ -\frac12 & -\frac{\sqrt3}{2} & -d\\ -\frac12 & \frac{\sqrt3}{2} & -d\end{bmatrix}\begin{bmatrix}u\\ v\\ \dot\theta\end{bmatrix}$$

Inverse and holonomic. $\det = \frac{3\sqrt3}{2}d \neq 0$; $\boldsymbol\zeta = \frac r3\begin{bmatrix}2 & -1 & -1\\ 0 & -\sqrt3 & \sqrt3\\ -\frac1d & -\frac1d & -\frac1d\end{bmatrix}\dot{\boldsymbol\varphi}$. Swedish wheels give no no-slide row, so $u, v, \dot\theta$ are free: holonomic, $\delta_m = 3$. World frame: $\boldsymbol\nu = \mathbf{J}(\theta)\boldsymbol\zeta$. [2]

Numbers. $u = 0.2$: $\dot\varphi = \frac{1}{0.05}(0.2, -0.1, -0.1) = \boxed{(4, -2, -2)\ \text{rad/s}}$. [1]

Check. Wheel 1 drives along $\hat x_C$ so it carries all of u; wheels 2 and 3 drive at $\pm120°$ and see $-\frac12 u$; a pure spin gives $-d\dot\theta/r$ on every wheel; sum of the three speeds is 0 for pure translation. [checked]

EXAM-STYLE QUESTION (practical Q2) [marks not published]

The robot above has $d = 0.10$ m, $r = 0.05$ m. Constant wheel speeds $\dot{\boldsymbol\varphi} = (4, -1, -1)$ rad/s, start at the origin with $\theta = 0$. Find $(u, v, \dot\theta)$, describe the path, and give the pose after 2 s.

Solution

Inverse: $u = \frac r3(2\dot\varphi_1 - \dot\varphi_2 - \dot\varphi_3) = \frac{0.05}{3}(8 + 1 + 1) = 0.1667$ m/s; $v = \frac{r}{\sqrt3}(\dot\varphi_3 - \dot\varphi_2) = 0$; $\dot\theta = -\frac{r}{3d}(\dot\varphi_1 + \dot\varphi_2 + \dot\varphi_3) = -\frac{0.05}{0.3}(2) = -0.3333$ rad/s. [2]

Constant $\boldsymbol\zeta$: a circle about the ICR $(-v/\dot\theta,\ u/\dot\theta) = (0, -0.5)$ in $\{C\}$, radius 0.5 m, clockwise ($\dot\theta < 0$). [1]

Pose, $\theta_0 = 0$, $T = 2$: $\theta = -0.6667$ rad; $x = \frac{u}{\dot\theta}(\sin\theta - \sin 0) = \frac{0.1667}{-0.3333}\sin(-0.6667) = 0.5\sin 0.6667 = \boxed{0.3092\ \text{m}}$; $y = -\frac{u}{\dot\theta}(\cos\theta - 1) = 0.5(\cos 0.6667 - 1) = \boxed{-0.1071\ \text{m}}$. Euler with $\Delta t = 0.01$ s is within 0.55 mm. [2]

Check: $\sqrt{0.3092^2 + (-0.1071 + 0.5)^2} = 0.5$ m from the ICR $(0, -0.5)$ ✓. [checked]

EXAM-STYLE QUESTION (practical Q3 and Q4) [marks not published]

(a) Derive the ICR of a differential drive for wheel speeds $\dot\varphi_1$ (left, at $+a$), $\dot\varphi_2$ (right, at $-a$); evaluate for $a = 0.3$ m, $\dot{\boldsymbol\varphi} = (2, 5)$ rad/s. (b) Derive the kinematics of a bicycle robot (rear wheel fixed and driven, front wheel steered, not driven). Is it holonomic? Evaluate $\dot\theta$ for $u = 2$ m/s, $\beta = 20°$, $L = 1.2$ m.

Solution

(a) Rows $r\dot\varphi_1 = u - a\dot\theta$, $r\dot\varphi_2 = u + a\dot\theta$, $v = 0 \Rightarrow u = \frac r2(\dot\varphi_1 + \dot\varphi_2)$, $\dot\theta = \frac{r}{2a}(\dot\varphi_2 - \dot\varphi_1)$. ICR: the chassis point of zero velocity, $u - \dot\theta y = 0$, $v + \dot\theta x = 0$, so $x = 0$, $y = u/\dot\theta$:

$$\boxed{y_{ICR} = a\,\frac{\dot\varphi_1 + \dot\varphi_2}{\dot\varphi_2 - \dot\varphi_1}} = 0.3 \cdot \frac73 = \boxed{0.7\ \text{m left of } C}$$

Check: equal speeds $\to \infty$ (straight), opposite $\to 0$ (spin). [3]

(b) Rear (at C, $\beta = 0$): $r\dot\varphi_r = u$, $v = 0$. Front (at $(L, 0)$, angle β) no-slide: $0 = -u\sin\beta + L\dot\theta\cos\beta \Rightarrow \dot\theta = u\tan\beta/L$. World: $\dot x = u\cos\theta$, $\dot y = u\sin\theta$. Numbers: $\dot\theta = 2\tan 20°/1.2 = \boxed{0.6066\ \text{rad/s}}$ ($R = 3.297$ m). [3]

Non-holonomic: $-\dot x\sin\theta + \dot y\cos\theta = 0$ restricts velocity and cannot be integrated; the bicycle can still reach any pose by manoeuvring. [1] [checked]

EXAM-STYLE QUESTION (new, scenario) [marks assumed]

A differential-drive robot ($r = 0.05$ m, $a = 0.15$ m) must (i) drive an anticlockwise circle of radius 1 m at 0.4 m/s, then (ii) trace a 2×1 m rectangle at 0.25 m/s, turning each corner on the spot at 0.5 rad/s. Give the wheel speeds and times, and the ICR in (i).

Solution

(i) $u = 0.4$, $\dot\theta = u/R = 0.4$ rad/s. $\dot\varphi_1 = (u - a\dot\theta)/r = (0.4 - 0.06)/0.05 = \boxed{6.8}$ rad/s (left), $\dot\varphi_2 = (0.4 + 0.06)/0.05 = \boxed{9.2}$ rad/s (right). ICR $y = a\frac{6.8 + 9.2}{9.2 - 6.8} = 1.0$ m left ✓. Lap $2\pi R/u = 15.71$ s. [3]

(ii) Straights: $\dot\varphi_1 = \dot\varphi_2 = u/r = \boxed{5}$ rad/s for 8 s (long sides) and 4 s (short sides). Corners: $u = 0$, $\dot\varphi_{1,2} = \mp a\omega/r = \boxed{\mp1.5}$ rad/s for $(\pi/2)/0.5 = 3.142$ s. Total $2(8 + 4) + 4(3.142) = \boxed{36.57\ \text{s}}$. [3]

Check: during a corner each wheel rolls $1.5 \times 3.142 = 4.712$ rad $= a\pi/2/r$ ✓. [checked]

EXAM-STYLE QUESTION (Practice Q10.1) [marks assumed]

Three omni wheels at $d = 0.12$ m, at position angles $\phi_i = 0°, 120°, 240°$, each driving *anticlockwise* tangentially ($\beta_i = \phi_i + 90°$), $\gamma_i = 0$, $r = 0.04$ m. Derive $\dot{\varphi}_i$ and find the wheel speeds for $(u, v, \dot{\theta}) = (0.3, 0.1, 0.5)$.

Solution

Position $(x_i, y_i) = d(\cos\phi_i, \sin\phi_i)$; $\cos\beta_i = -\sin\phi_i$, $\sin\beta_i = \cos\phi_i$. Then $r\dot{\varphi}_i = -\sin\phi_i(u - \dot{\theta}d\sin\phi_i) + \cos\phi_i(v + \dot{\theta}d\cos\phi_i) = -u\sin\phi_i + v\cos\phi_i + d\dot{\theta}$. Rows $\frac{1}{r}[-\sin\phi_i,\ \cos\phi_i,\ d]$; $\det = \frac{3\sqrt{3}}{2}d \neq 0$. [3]

$\dot{\varphi}_1 = (0 + 0.1 + 0.06)/0.04 = \boxed{4.000}$; $\dot{\varphi}_2 = (-0.8660(0.3) - 0.5(0.1) + 0.06)/0.04 = \boxed{-6.245}$; $\dot{\varphi}_3 = (+0.8660(0.3) - 0.5(0.1) + 0.06)/0.04 = \boxed{6.745}$ rad/s. [2]

Check: pure spin gives $+d\dot{\theta}/r$ on all three. The slides' clockwise layout (W13 Q10 matrix) gives $-d$ and different signs: state which drive direction you chose. [checked]

EXAM-STYLE QUESTION (mock Q8-style) [marks assumed]

Four Swedish wheels ($r = 0.05$ m) at $(\pm 0.20, \pm 0.15)$ m, all driving along $\hat{x}_C$; roller angles FL $-45°$, FR $+45°$, RL $+45°$, RR $-45°$. (a) Write the four wheel rows. (b) Wheel speeds for pure sideways motion $v = 0.2$ m/s. (c) Holonomic?

Solution

(a) $r\dot{\varphi}_i = (u - \dot{\theta}y_i) + \tan\gamma_i(v + \dot{\theta}x_i)$. FL $(0.2, 0.15, -45°)$: $[20, -20, -7]$; FR $(0.2, -0.15, +45°)$: $[20, 20, 7]$; RL $(-0.2, 0.15, +45°)$: $[20, 20, -7]$; RR $(-0.2, -0.15, -45°)$: $[20, -20, 7]$ (rows are $\frac{1}{r}[\cdot]$ with $\frac{1}{r} = 20$). [3]

(b) $v = 0.2$: $\boxed{\dot{\varphi} = (-4,\ 4,\ 4,\ -4)\ \text{rad/s}}$: diagonal pairs turn the same way. [2]

(c) Rank 3: any $(u, v, \dot{\theta})$ reachable, no no-slide rows: holonomic, $\delta_M = 3$. [1] [checked]

EXAM-STYLE QUESTION (explain / spot the errors) [marks assumed]

A student writes: "For a differential drive with $r = 0.05$ m and $\dot{\varphi} = (8, 12)$ rad/s, $u = r(\dot{\varphi}_1 + \dot{\varphi}_2) = 1.0$ m/s. The bicycle is holonomic because it can park anywhere. For a standard wheel $\gamma = \pi/2$." Find the errors; also explain why a diff. drive cannot move sideways.

Solution

(1) u is the *mean* rim speed: $u = r(\dot{\varphi}_1 + \dot{\varphi}_2)/2 = 0.05(20)/2 = \boxed{0.5}$ m/s. (2) The bicycle is non-holonomic: the rear no-slide gives $v = 0$ and the front gives $\dot{\theta} = u\tan\beta/L$; both restrict velocities and cannot be integrated; reaching every pose by manoeuvring does not make it holonomic. (3) A standard wheel has no rollers: $\gamma = 0$ and it cannot slide sideways.

Why no sideways motion: the no-sliding row of a fixed wheel at $x = 0, \beta = 0$ is $0 = v$: the contact point may only move along the wheel's rolling direction; that is a constraint on velocities. [checked]

EXAM-STYLE QUESTION (mock Q8, differential drive) [6 marks assumed]

Wheels of radius $r = 0.08$ m sit at $(0, \pm a)$, $a = 0.20$ m (left wheel 1 at $+a$). (a) From the generalised wheel model derive the chassis velocity from the wheel speeds. (b) For $\dot{\boldsymbol{\varphi}} = (5, 7.5)$ rad/s find u, $\dot{\theta}$ and the ICR. (c) Wheel speeds for $u = 0.3$ m/s, $\dot{\theta} = -0.5$ rad/s. (d) Write the non-holonomic constraint in the inertial frame.

Solution

(a) Rows $r\dot{\varphi}_1 = u - a\dot{\theta}$, $r\dot{\varphi}_2 = u + a\dot{\theta}$; no-sliding of the fixed wheels gives $v = 0$. So $u = \frac{r}{2}(\dot{\varphi}_1 + \dot{\varphi}_2)$, $\dot{\theta} = \frac{r}{2a}(\dot{\varphi}_2 - \dot{\varphi}_1)$. [2]

(b) $u = 0.08(12.5)/2 = \boxed{0.5}$ m/s; $\dot{\theta} = 0.08(2.5)/0.4 = \boxed{0.5}$ rad/s; ICR $y = u/\dot{\theta} = \boxed{1.0}$ m to the left on the axle (check $0.2 \cdot 12.5/2.5 = 1.0$). [2]

(c) $\dot{\varphi}_1 = (0.3 - 0.2(-0.5))/0.08 = \boxed{5.0}$, $\dot{\varphi}_2 = (0.3 + 0.2(-0.5))/0.08 = \boxed{2.5}$ rad/s (clockwise: left wheel faster). [1]

(d) $-\dot{x}\sin\theta + \dot{y}\cos\theta = 0$: no sideways velocity; it contains velocities and cannot be integrated to a position-only relation. [1] [checked]

EXAM-STYLE QUESTION (Practice Q10.4 and a tricycle) [marks assumed]

(a) A car-like robot has wheelbase $L = 2.7$ m and track $t = 1.5$ m and drives a circle of radius $R = 8$ m about an ICR on the rear-axle line (measured to the middle of the rear axle). Find the steering angles of the inner and outer front wheels. (b) A tricycle with the front wheel at $b = 0.6$ m ($r = 0.1$ m) must move at $u = 0.4$ m/s with $\dot{\theta} = 0.5$ rad/s: find the front wheel speed and steering angle.

Solution

(a) Every axle line passes through the ICR. The inner wheel is $R - t/2 = 7.25$ m from the ICR along the rear-axle line, the outer one $R + t/2 = 8.75$ m: $\beta_{in} = \arctan\frac{2.7}{7.25} = \boxed{20.43°}$, $\beta_{out} = \arctan\frac{2.7}{8.75} = \boxed{17.15°}$. Check: $\cot\beta_{out} - \cot\beta_{in} = 0.5556 = t/L$ ✓. A single angle $\arctan(2.7/8) = 18.65°$ would make the tyres scrub. [3]

(b) Front contact velocity $(u,\ b\dot{\theta}) = (0.4,\ 0.3)$ m/s, so $r\dot{\varphi}_1 = 0.5$: $\dot{\varphi}_1 = \boxed{5\ \text{rad/s}}$, $\beta_1 = \text{atan2}(0.3, 0.4) = \boxed{36.87°}$. Check $R = u/\dot{\theta} = 0.8$ m and $\arctan(b/R) = 36.87°$ ✓. [3] [checked]

Week 11 at a glance [added]

The formulas to know (each is derived on the page given)

balance at G	${}^C\mathbf{F}_G = m\,{}^C\ddot{\mathbf{P}}_G, \quad {}^C N_z = {}^G I\,\ddot{\theta}, \quad {}^C N_z = {}^C n_z + ({}^C y_G\,{}^C f_x - {}^C x_G\,{}^C f_y)$
acceleration of G	${}^C\ddot{x}_G = \dot{u} - {}^C y_G\ddot{\theta} - v\dot{\theta} - {}^C x_G\dot{\theta}^2, \quad {}^C\ddot{y}_G = \dot{v} + {}^C x_G\ddot{\theta} + u\dot{\theta} - {}^C y_G\dot{\theta}^2$
body-frame EoM	$\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^C\mathbf{Q}, \quad \boldsymbol{\zeta} = (u, v, \dot{\theta})$
$C = G$	$\mathbf{M} = \mathrm{diag}(m, m, {}^G I), \quad c = (-mv\dot{\theta},\ mu\dot{\theta},\ 0)$
frames	$\dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\zeta}, \quad {}^C\mathbf{Q} = \mathbf{J}^{\top\,O}\mathbf{Q}$ (power invariance)
inertial-frame EoM	${}^O\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^O c = {}^O\mathbf{Q}, \quad {}^O\mathbf{M} = (\mathbf{J}^{\top})^{-1}\mathbf{M}\mathbf{J}^{-1}$
differential drive	${}^C\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}, \quad F_i = \tau_i / r$

Use: given forces/torques find accelerations ($\dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}({}^C\mathbf{Q} - c)$), or the inverse. Frame names: $\{O\}$ world, $\{C\}$ chassis, $\{G\}$ centre of mass.

TRAP

There is *no control* in W11 (no feed-forward, no PID, no Lyapunov law). The deck is: Newton–Euler for one planar rigid chassis, the $\{C\} \leftrightarrow \{O\}$ conversion, an integration recipe, a differential-drive and a Swedish-wheel example. Control of such robots belongs to W8–W9.

Recap of week 10, and what is new [p. 4–7]

W10 (kinematics) gave the wheel model $\dot{\varphi}_i = f_i(\dot{x}, \dot{y}, \dot{\theta})$ ([p. 6]) and $\dot{\boldsymbol{\xi}} = \mathbf{J}(\theta)\boldsymbol{\zeta}$ with $\boldsymbol{\zeta} = (u, v, \dot{\theta})$ the chassis velocity in $\{C\}$. W11 asks the *cause* of motion: which forces and torques produce which accelerations.

STEPS The W11 chain, in the order the deck builds it

1. Frames $\{O\}, \{C\}, \{G\}$ and the free-body picture ([p. 12–17]).
2. Velocity then acceleration of G in $\{C\}$ ([p. 18–24]).
3. Newton and Euler: $m\mathbf{a}_G$, ${}^G I\ddot{\theta}$, move the moment from G to $C \Rightarrow \mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^C\mathbf{Q}$ ([p. 25–26]).
4. Re-express in $\{O\}$ with $\mathbf{J}$ ([p. 28–34]).
5. Simulate ([p. 35]); wheel forces to ${}^C\mathbf{Q}$ ([p. 36–38]).

The dynamics problem: forward and inverse [p. 9–11]

Definition [p. 9]

$$\mathbf{Q} = f(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}}), \qquad \boldsymbol{\xi} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}, \qquad \mathbf{Q} = \begin{bmatrix} f_x \\ f_y \\ \tau_z \end{bmatrix}$$

SYMBOLS

$\boldsymbol{\xi}$	pose of the robot in the inertial frame $\{O\}$	m, m, rad
$\mathbf{Q}$	net force and torque on the robot, expressed in an inertial frame	N, N, N m

Use: one equation, read two ways: forward (simulation) or inverse (control).

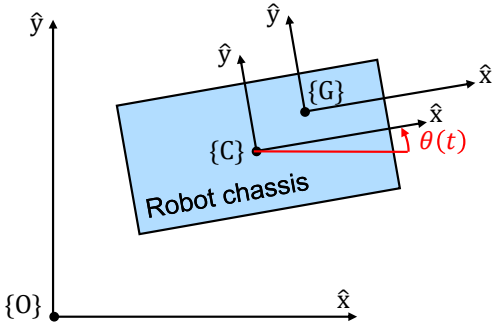
PREMISE forward dynamics [p. 10] **Given:** input force–torque vector $\mathbf{Q}$, the robot's m, ${}^G I$, geometry, the starting state
Find: the trajectory $(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$: used to *simulate or analyse* the robot

PREMISE inverse dynamics [p. 11] **Given:** a desired trajectory $(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$
Find: the force and torque $\mathbf{Q}$ that the robot needs: used for *control*

TRAP

Mixing up $\boldsymbol{\xi}$ (pose, always in $\{O\}$) and $\boldsymbol{\zeta}$ (chassis velocity, in $\{C\}$). $\mathbf{Q}$ also has two forms, ${}^O\mathbf{Q}$ and ${}^C\mathbf{Q}$ ([p. 26, 31]): always write the frame.

Setup: the frames $\{O\}$, $\{C\}$, $\{G\}$ [p. 12]



[W11 lec p. 12] — $\{O\}$ inertial, $\{C\}$ on the chassis, $\{G\}$ at the centre of mass with the same orientation as $\{C\}$

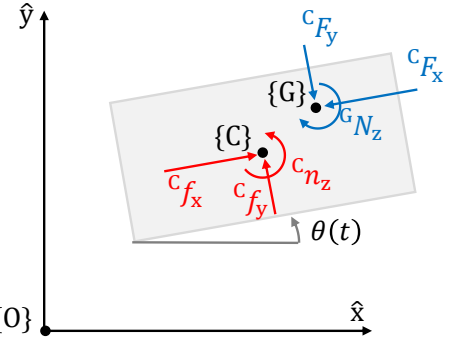
Quantities of the problem [p. 12]

SYMBOLS

$\{O\}$	global inertial frame, fixed to the ground	–
$\{C\}$	frame attached to the chassis (origin C, $\hat{x}$ forwards)	–
$\{G\}$	frame at the centre of mass, *same orientation as* $\{C\}$	–
m	mass of the robot	kg
${}^G I$	${}^G I_{zz}$: moment of inertia about the vertical axis through G	kg m^2
${}^C\mathbf{P}_G = ({}^C x_G, {}^C y_G, 0)$	position of G in $\{C\}$ (the slides also write ${}^C\mathbf{P}_{CG}$: same vector, $C \to G$)	m
$\theta(t)$	heading: angle of $\hat{x}_C$ from $\hat{x}_O$, anticlockwise positive	rad
$\mathbf{J} = {}^O\mathbf{R}_C$	rotation of $\{C\}$ in $\{O\}$ (rotation about z by θ, as W10)	–

Use: C is the reference point where wheel loads are applied; G is where the mass acts. If $C = G$ everything simplifies. Both frames rotate with the robot, so ${}^C\mathbf{P}_G$ is constant.

Newton–Euler: force and moment balance [p. 14–17]



[W11 lec p. 15] — red: applied force and torque at C; blue: inertial force ${}^C\mathbf{F}_G$ and torque ${}^G N_z$ at G

Balance, all terms in $\{C\}$ [p. 15]

$$ {}^{C}\mathbf{F}_G - {}^{C}\mathbf{f}_C = \mathbf{0} \;\Rightarrow\; {}^{C}F_x = {}^{C}f_x, \quad {}^{C}F_y = {}^{C}f_y $$

$$ {}^{C}N_G - {}^{C}n_C - (-{}^{C}\mathbf{P}_{CG}) \times {}^{C}\mathbf{f}_C = 0 \;\Rightarrow\; {}^{C}N_z = {}^{C}n_z + ({}^{C}y_G\,{}^{C}f_x - {}^{C}x_G\,{}^{C}f_y) $$

SYMBOLS

${}^{C}\mathbf{f}_C = ({}^{C}f_x, {}^{C}f_y)$	applied force from wheels and ground, acting at C (red)	N
${}^{C}n_C = {}^{C}n_z$	applied torque (about z) acting at C	N m
${}^{C}\mathbf{F}_G = ({}^{C}F_x, {}^{C}F_y)$	inertial force at G (what the body "needs", blue)	N
${}^{C}N_G = {}^{C}N_z$	inertial torque about G	N m
${}^{C}x_G, {}^{C}y_G$	coordinates of G in $\{C\}$	m

Use: the applied force acts at C but the inertial force acts at G. Moving the applied force's moment from C to G gives the bracket $(y_G f_x - x_G f_y)$.

DERIVATION of the moment transfer, every line [added]

The lever arm from G to C is $-{}^{C}\mathbf{P}_{CG} = (-x_G, -y_G, 0)$. The moment of the applied force about G is the cross product

$$ (-{}^{C}\mathbf{P}_{CG}) \times {}^{C}\mathbf{f}_C = \begin{bmatrix} -x_G \\ -y_G \\ 0 \end{bmatrix} \times \begin{bmatrix} f_x \\ f_y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ (-x_G)f_y - (-y_G)f_x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ y_G f_x - x_G f_y \end{bmatrix}. $$

Torque balance about G: inertial torque = applied torque + moment of the applied force about G, so ${}^{C}N_z = {}^{C}n_z + (y_G f_x - x_G f_y)$. Rearranged for the *applied* torque: ${}^{C}n_z = {}^{C}N_z - (y_G f_x - x_G f_y)$.

Inertial force and torque [p. 17]

$$ {}^{C}\mathbf{F}_G = m\,{}^{C}\dot{\mathbf{v}}_G \quad \left({}^{C}F_x = m\,{}^{C}\ddot{x}_G,\ {}^{C}F_y = m\,{}^{C}\ddot{y}_G\right), \qquad {}^{C}N_G = {}^{G}I\dot{\omega} + \omega \times {}^{G}I\omega \;\Rightarrow\; {}^{C}N_z = {}^{G}I\,\ddot{\theta} $$

Use: m and ${}^{G}I$ are known; $\ddot{x}_G, \ddot{y}_G, \ddot{\theta}$ are the unknowns (the slide's own note). In planar motion $\boldsymbol{\omega} = (0, 0, \dot{\theta})$ is parallel to ${}^{G}I\boldsymbol{\omega}$, so $\boldsymbol{\omega} \times {}^{G}I\boldsymbol{\omega} = \mathbf{0}$. Missing piece: the acceleration of G.

WORKED moment transfer, numbers ours

G is 0.05 m ahead of C ($x_G = 0.05$, $y_G = 0$), ${}^{C}\mathbf{f} = (10, 4)$ N, ${}^{C}n_z = 0.4$ N m: ${}^{C}N_z = 0.4 + (0 \cdot 10 - 0.05 \cdot 4) = \mathbf{0.2}$ N m. A sideways push at C is a clockwise lever about G when G is ahead. [checked]

Velocity of a point on a rotating body [p. 18–20]

DERIVATION velocity of G, as the slides do it [p. 19–20]

Position of G in $\{O\}$: ${}^{O}\mathbf{P}_G = {}^{O}\mathbf{P}_C + {}^{O}\mathbf{R}_C\,{}^{C}\mathbf{P}_G$. Differentiate:

$$ {}^{O}\dot{\mathbf{P}}_G = {}^{O}\dot{\mathbf{P}}_C + {}^{O}\dot{\mathbf{R}}_C\,{}^{C}\mathbf{P}_G + {}^{O}\mathbf{R}_C\,{}^{C}\dot{\mathbf{P}}_G, \qquad {}^{C}\dot{\mathbf{P}}_G = \mathbf{0} $$

because G is fixed in the chassis (the slide strikes this term out). For a rotating frame ${}^{O}\dot{\mathbf{R}}_C\,\mathbf{p} = {}^{O}\boldsymbol{\omega}_C \times ({}^{O}\mathbf{R}_C\,\mathbf{p})$, so

$$ {}^{O}\dot{\mathbf{P}}_G = {}^{O}\dot{\mathbf{P}}_C + {}^{O}\boldsymbol{\omega}_C \times {}^{O}\mathbf{R}_C\,{}^{C}\mathbf{P}_G. $$

Write the velocity of C and the rotation rate through $\mathbf{J}(\theta) = {}^{O}\mathbf{R}_C$: ${}^{O}\dot{\mathbf{P}}_C = \mathbf{J}\,{}^{C}\dot{\mathbf{P}}_C$ and ${}^{O}\boldsymbol{\omega}_C = {}^{O}\mathbf{R}_C\,{}^{C}\boldsymbol{\omega}_C$; a rotation preserves cross products, $\mathbf{R}(\mathbf{a} \times \mathbf{b}) = \mathbf{R}\mathbf{a} \times \mathbf{R}\mathbf{b}$. Factor ${}^{O}\mathbf{R}_C$:

Velocity of G [p. 19–20]

$$ {}^{O}\dot{\mathbf{P}}_G = {}^{O}\mathbf{R}_C\left[{}^{C}\dot{\mathbf{P}}_C + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G\right], \qquad {}^{C}\dot{\mathbf{P}}_C = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix},\ {}^{C}\boldsymbol{\omega}_C = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix},\ {}^{C}\mathbf{P}_G = \begin{bmatrix} {}^{C}x_G \\ {}^{C}y_G \\ 0 \end{bmatrix} $$

$$ {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G = \begin{bmatrix} 0 \cdot 0 - \dot{\theta}\,{}^{C}y_G \\ \dot{\theta}\,{}^{C}x_G - 0 \cdot 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot{\theta}\,{}^{C}y_G \\ \dot{\theta}\,{}^{C}x_G \\ 0 \end{bmatrix} \;\Rightarrow\; \text{velocity of } G \text{ in the chassis axes} = \begin{bmatrix} u - \dot{\theta}\,{}^{C}y_G \\ v + \dot{\theta}\,{}^{C}x_G \end{bmatrix} $$

SYMBOLS

${}^{C}\dot{\mathbf{P}}_C$	velocity of the origin C, written in the chassis axes: $(u, v, 0)$ (the position of C seen from $\{C\}$ is zero, so this is the velocity *expressed* in $\{C\}$)	m/s
u, v	components of that velocity along $\hat{x}_C$ (forwards) and $\hat{y}_C$ (left)	m/s
${}^{C}\boldsymbol{\omega}_C$	angular velocity of the chassis, along z	rad/s

Use: every point of a rigid body has velocity (velocity of C) + ($\boldsymbol{\omega}\times$ position). The cross product $(0, 0, k) \times (a, b, 0) = (-kb, ka, 0)$ is used in every line of this week.

WORKED velocity of G, numbers ours

$G = (0.3, 0.2)$ m, $u = 1.0$, $v = 0.5$ m/s, $\dot{\theta} = 2$ rad/s: in $\{C\}$: $(1.0 - 2(0.2),\ 0.5 + 2(0.3)) = (\mathbf{0.6}, \mathbf{1.1})$ m/s. At $\theta = 30°$ in $\{O\}$: $\mathbf{J}(0.6, 1.1) = (0.6(0.8660) - 1.1(0.5),\ 0.6(0.5) + 1.1(0.8660)) = (\mathbf{-0.0304},\ \mathbf{1.2526})$ m/s. Speed 1.253 in both frames. [checked]

Acceleration of G [p. 21–24]

DERIVATION differentiate the velocity again [p. 22]

Start from ${}^{O}\dot{\mathbf{P}}_G = {}^{O}\mathbf{R}_C\left[{}^{C}\dot{\mathbf{P}}_C + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G\right]$ and apply the product rule:

$$ {}^{O}\ddot{\mathbf{P}}_G = {}^{O}\dot{\mathbf{R}}_C\left[{}^{C}\dot{\mathbf{P}}_C + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G\right] + {}^{O}\mathbf{R}_C\left[{}^{C}\ddot{\mathbf{P}}_C + {}^{C}\dot{\boldsymbol{\omega}}_C \times {}^{C}\mathbf{P}_G + \underbrace{{}^{C}\boldsymbol{\omega}_C \times {}^{C}\dot{\mathbf{P}}_G}_{=0}\right] $$

The last term is the slide's red 0 (G does not move in $\{C\}$). Replace ${}^{O}\dot{\mathbf{R}}_C\,\mathbf{p} = {}^{O}\mathbf{R}_C\,({}^{C}\boldsymbol{\omega}_C \times \mathbf{p})$ and take ${}^{O}\mathbf{R}_C$ out as a common factor:

$$ {}^{O}\ddot{\mathbf{P}}_G = {}^{O}\mathbf{R}_C\left[{}^{C}\boldsymbol{\omega}_C \times ({}^{C}\dot{\mathbf{P}}_C + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G) + {}^{C}\ddot{\mathbf{P}}_C + {}^{C}\dot{\boldsymbol{\omega}}_C \times {}^{C}\mathbf{P}_G\right]. $$

Multiply on the left by ${}^{O}\mathbf{R}_C^{-1}$ to read the acceleration *in the chassis axes*, and expand the bracket:

Acceleration of G in $\{C\}$: four terms [p. 22–23]

$$ \boxed{{}^{C}\ddot{\mathbf{P}}_G = {}^{C}\ddot{\mathbf{P}}_C + {}^{C}\dot{\boldsymbol{\omega}}_C \times {}^{C}\mathbf{P}_G + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\dot{\mathbf{P}}_C + {}^{C}\boldsymbol{\omega}_C \times {}^{C}\boldsymbol{\omega}_C \times {}^{C}\mathbf{P}_G} $$

SYMBOLS

${}^{C}\ddot{\mathbf{P}}_C = (\dot{u}, \dot{v}, 0)$	rate of change of the velocity components of C	m/s^2
${}^{C}\dot{\boldsymbol{\omega}}_C = (0, 0, \ddot{\theta})$	angular acceleration	rad/s^2
$\dot{\boldsymbol{\omega}} \times \mathbf{P}_G$	*tangential* term, from angular acceleration; $\perp$ to the arm $C \to G$	m/s^2
$\boldsymbol{\omega} \times \dot{\mathbf{P}}_C$	*Coriolis-like* term: velocity of C turns with the body	m/s^2
$\boldsymbol{\omega} \times \boldsymbol{\omega} \times \mathbf{P}_G$	*centripetal* term, points from G towards C	m/s^2

Use: the slide writes ${}^{C}\mathbf{P}_{CG}$ in the second term and ${}^{C}\mathbf{P}_G$ in the last; both are the same vector.

DERIVATION every cross product expanded: skew-symmetric matrix [added]

In the plane $[\boldsymbol{\omega}\times] = \begin{bmatrix}0 & -\dot\theta & 0\\ \dot\theta & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$ (the cross product $\boldsymbol{\omega}\times\mathbf{a}$ equals this matrix times $\mathbf{a}$) and $[\dot{\boldsymbol{\omega}}\times]$ is the same with $\ddot\theta$. Then

$$\dot{\boldsymbol{\omega}}\times\mathbf{P}_G = \begin{bmatrix}0 & -\ddot\theta & 0\\ \ddot\theta & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_G\\ y_G\\ 0\end{bmatrix} = \begin{bmatrix}-y_G\ddot\theta\\ x_G\ddot\theta\\ 0\end{bmatrix}, \qquad \boldsymbol{\omega}\times\dot{\mathbf{P}}_C = \begin{bmatrix}0 & -\dot\theta & 0\\ \dot\theta & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}u\\ v\\ 0\end{bmatrix} = \begin{bmatrix}-v\dot\theta\\ u\dot\theta\\ 0\end{bmatrix}$$

$$\boldsymbol{\omega}\times\boldsymbol{\omega}\times\mathbf{P}_G = \boldsymbol{\omega}\times\begin{bmatrix}-y_G\dot\theta\\ x_G\dot\theta\\ 0\end{bmatrix} = \begin{bmatrix}-\dot\theta(x_G\dot\theta)\\ \dot\theta(-y_G\dot\theta)\\ 0\end{bmatrix} = \begin{bmatrix}-x_G\dot\theta^2\\ -y_G\dot\theta^2\\ 0\end{bmatrix}$$

Add the four columns (the slide's matrix equation, [p. 24]):

$$\begin{bmatrix}{}^C\ddot x_G\\ {}^C\ddot y_G\\ 0\end{bmatrix} = \begin{bmatrix}\dot u\\ \dot v\\ 0\end{bmatrix} + \begin{bmatrix}-y_G\ddot\theta\\ x_G\ddot\theta\\ 0\end{bmatrix} + \begin{bmatrix}-v\dot\theta\\ u\dot\theta\\ 0\end{bmatrix} + \begin{bmatrix}-x_G\dot\theta^2\\ -y_G\dot\theta^2\\ 0\end{bmatrix}$$

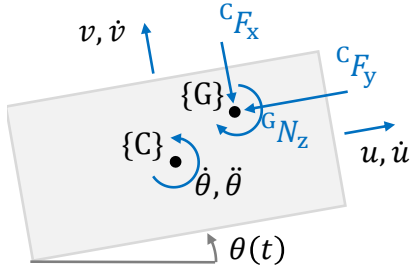
Components [p. 24]

$$ {}^C\ddot x_G = \dot u - {}^C y_G\,\ddot\theta - v\,\dot\theta - {}^C x_G\,\dot\theta^2$$
$$ {}^C\ddot y_G = \dot v + {}^C x_G\,\ddot\theta + \boxed{u}\,\dot\theta - {}^C y_G\,\dot\theta^2$$

Use: pattern: the Coriolis-like pair is $\dot\theta\,(-v,\ u)$: x gets $-v\dot\theta$, y gets $+u\dot\theta$. With $C = G$ and no accelerations this is $\dot\theta(-v, u)$, perpendicular to the velocity: the centripetal acceleration of any turn.

SLIDE CHECK pp. 24–25: $v\dot\theta$ should be $u\dot\theta$ in y

The slides print ${}^C\ddot y_G = \dot v + x_G\ddot\theta + \mathbf{v}\dot{\boldsymbol{\theta}} - y_G\dot\theta^2$ (and the same $+v\dot\theta$ in ${}^C F_y$ and ${}^C f_y$, [p. 24, 25]). The correct term is $+u\dot\theta$: (1) the slide's own matrix equation gives $(0,0,\dot\theta)\times(u,v,0) = (-v\dot\theta,\ u\dot\theta, 0)$; (2) the matrix form on p. 26 already uses u in rows 2 and 3; (3) at constant u on a circle ($v = 0$) the printed form gives $a_y = 0$, no centripetal acceleration. The x equation is right. In the exam write the cross-product line so $u\dot\theta$ is justified. [checked]



[W11 lec p. 24] — velocity components u, v, rates $\dot\theta, \ddot\theta$ at C; ${}^C F_x, {}^C F_y, {}^G N_z$ at G

WORKED all four terms, numbers ours

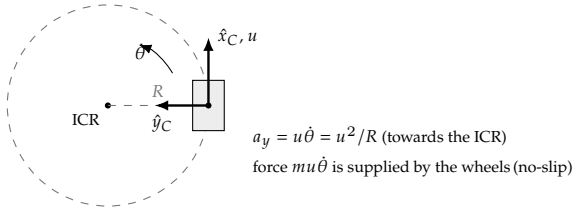
$u = 1.5$, $v = 0.2$ m/s, $\dot\theta = 0.8$ rad/s, $\dot u = 0.3$, $\dot v = -0.1$ m/s^2, $\ddot\theta = 0.5$ rad/s^2, $G = (0.2, 0.1)$ m in $\{C\}$:

term	x	y
$\ddot{\mathbf{P}}_C$	0.3	−0.1
tangential $\ddot\theta(-y_G, x_G)$	−0.05	0.10
Coriolis-like $\dot\theta(-v, u)$	−0.16	1.20
centripetal $-\dot\theta^2(x_G, y_G)$	−0.128	−0.064
sum (m/s^2)	**−0.038**	**1.136**

(The printed $+v\dot\theta$ would give $a_y = 0.096$: wrong.) A finite-difference derivative of the world-frame motion, rotated back into $\{C\}$, gives the same $(-0.038,\ 1.136)$. [checked]

TRAP

Treating $\dot u, \dot v$ as the whole acceleration. They are only the rate of change of the *components*; the axes themselves turn, which adds $\dot\theta(-v, u)$. A robot driving a circle at constant speed still accelerates.



[drawn for this booklet] top view: $C = G$, constant $u, v = 0$, $\dot\theta = u/R$ (circle at constant speed)

Equations of motion in the chassis frame [p. 25–26]

Newton–Euler equations of motion [p. 25]

$$ {}^C f_x = {}^C F_x = m\left(\dot u - {}^C y_G\,\ddot\theta - v\dot\theta - {}^C x_G\,\dot\theta^2\right)$$
$$ {}^C f_y = {}^C F_y = m\left(\dot v + {}^C x_G\,\ddot\theta + u\dot\theta - {}^C y_G\,\dot\theta^2\right) \quad \text{(slide prints } v\dot\theta\text{; see the slide check)}$$
$$ {}^C n_z = {}^C N_z - \left({}^C y_G\,{}^C f_x - {}^C x_G\,{}^C f_y\right) = {}^G I\,\ddot\theta - \left({}^C y_G\,{}^C f_x - {}^C x_G\,{}^C f_y\right)$$

Use: this is the whole model: the two force equations come from $m\mathbf{a}_G$, the third from the torque balance. Three equations, three unknowns $(\dot u, \dot v, \ddot\theta)$.

DERIVATION from the three equations to the matrix, every line [added]

Insert ${}^C f_x, {}^C f_y$ into ${}^C n_z = {}^G I\ddot\theta - y_G f_x + x_G f_y$ (x_G, y_G short for ${}^C x_G, {}^C y_G$):

$$\begin{aligned} {}^C n_z &= {}^G I\ddot\theta - y_G\,m\left(\dot u - y_G\ddot\theta - v\dot\theta - x_G\dot\theta^2\right) + x_G\,m\left(\dot v + x_G\ddot\theta + u\dot\theta - y_G\dot\theta^2\right)\\ &= -m y_G\,\dot u + m x_G\,\dot v + \left({}^G I + m(x_G^2 + y_G^2)\right)\ddot\theta + m\dot\theta\left(y_G v + y_G x_G\dot\theta + x_G u - x_G y_G\dot\theta\right)\\ &= -m y_G\,\dot u + m x_G\,\dot v + \left({}^G I + m(x_G^2 + y_G^2)\right)\ddot\theta + m\dot\theta\,(x_G u + y_G v).\end{aligned}$$

The $x_G y_G\dot\theta^2$ terms cancel. Collect $\dot u, \dot v, \ddot\theta$ on the left and velocity terms on the right.

Matrix form $\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^{C}\mathbf{Q}$ [p. 26]

$$\underbrace{\begin{bmatrix} m & 0 & -m\,{}^{C}y_G \\ 0 & m & m\,{}^{C}x_G \\ -m\,{}^{C}y_G & m\,{}^{C}x_G & {}^{G}I + m\left({}^{C}x_G^2 + {}^{C}y_G^2\right) \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{\theta} \end{bmatrix} + \underbrace{\begin{bmatrix} -m\dot{\theta}\,(v + {}^{C}x_G\,\dot{\theta}) \\ m\dot{\theta}\,(u - {}^{C}y_G\,\dot{\theta}) \\ m\dot{\theta}\,({}^{C}x_G\,u + {}^{C}y_G\,v) \end{bmatrix}}_{c(\boldsymbol{\zeta})} = \underbrace{\begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix}}_{{}^{C}\mathbf{Q}}$$

SYMBOLS

$\mathbf{M}$	depends only on geometry and mass distribution: constant, symmetric; $M_{33} = {}^{G}I + m(x_G^2 + y_G^2)$ is the inertia about C (parallel-axis theorem)	kg, kg m^2
$c(\boldsymbol{\zeta})$	function of the instantaneous velocity (centripetal, Coriolis-like); zero at rest	N, N m
${}^{C}\mathbf{Q}$	forces and moment on the chassis, in $\{C\}$, applied at C	N, N m
$\boldsymbol{\zeta}$	$(u, v, \dot{\theta})^\top$, chassis velocity in $\{C\}$	m/s, rad/s

Use: simulation: $\dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}\left({}^{C}\mathbf{Q} - c(\boldsymbol{\zeta})\right)$. *Inverse dynamics: given* $\dot{\boldsymbol{\zeta}}, \boldsymbol{\zeta}$ *read* ${}^{C}\mathbf{Q}$ *directly. Special case* $C = G$ $(x_G = y_G = 0)$: $\mathbf{M} = \mathrm{diag}(m, m, {}^{G}I)$, $c = (-mv\dot{\theta},\ mu\dot{\theta},\ 0)^\top$, *i.e.* $m(\dot{u} - v\dot{\theta}) = f_x$, $m(\dot{v} + u\dot{\theta}) = f_y$, ${}^{G}I\ddot{\theta} = n_z$.

STEPS Derive the equations of motion (W13 Q4 style)

1. Draw $\{O\}, \{C\}, \{G\}$; applied ${}^{C}f_x, {}^{C}f_y, {}^{C}n_z$ at C; inertial ${}^{C}F_x, {}^{C}F_y, {}^{C}N_z$ at G; mark $({}^{C}x_G, {}^{C}y_G)$.
2. Velocity of G: ${}^{C}\dot{\mathbf{P}}_G = (u - \dot{\theta}y_G,\ v + \dot{\theta}x_G)$.
3. Acceleration of G by the four terms (expand each cross product).
4. Newton: ${}^{C}f_x = m\ddot{x}_G$, ${}^{C}f_y = m\ddot{y}_G$. Euler about G: ${}^{C}N_z = {}^{G}I\ddot{\theta}$; transfer to C: ${}^{C}n_z = {}^{C}N_z - (y_G f_x - x_G f_y)$.
5. Substitute, collect $\rightarrow \mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^{C}\mathbf{Q}$; state which of $\mathbf{M}, c$ are zero or diagonal when $C = G$.
6. **Check:** $\mathbf{M}$ symmetric; $c = 0$ at rest; $C = G$ gives $\mathrm{diag}(m, m, {}^{G}I)$; units N and N m.

ALTERNATIVE APPROACHES

A. Newton–Euler (the slides)

Body frame $\{C\}$. Unknowns $\dot{u}, \dot{v}, \ddot{\theta}$. Steps: $m\mathbf{a}_G$, ${}^{G}I\ddot{\theta}$, move the moment $G \rightarrow C$. Gives $\mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^{C}\mathbf{Q}$ directly, in the axes where wheel forces are natural. Needs the acceleration of G (four terms) and care with $u\dot{\theta}$.

B. Lagrange ([p. 27] homework; [added])

World coordinates $\boldsymbol{\xi} = (x, y, \theta)$ of C. Kinetic energy $T = \frac{1}{2}m|\dot{\mathbf{P}}_G|^2 + \frac{1}{2}{}^{G}I\dot{\theta}^2 = \frac{1}{2}\boldsymbol{\zeta}^\top \mathbf{M}\boldsymbol{\zeta} = \frac{1}{2}\dot{\boldsymbol{\xi}}^\top {}^{O}\mathbf{M}\dot{\boldsymbol{\xi}}$. No potential (flat floor). $\frac{d}{dt}\frac{\partial T}{\partial \dot{\boldsymbol{\xi}}} - \frac{\partial T}{\partial \boldsymbol{\xi}} = {}^{O}\mathbf{Q}$. Gives ${}^{O}\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^{O}c = {}^{O}\mathbf{Q}$ with no cross products. For $C = G$: $m\ddot{x} = f_x$, $m\ddot{y} = f_y$, ${}^{G}I\ddot{\theta} = \tau_z$.

DERIVATION the kinetic energy is $\frac{1}{2}\boldsymbol{\zeta}^\top \mathbf{M}\boldsymbol{\zeta}$; why the two methods agree [added]

With ${}^{C}\dot{\mathbf{P}}_G = (u - \dot{\theta}y_G,\ v + \dot{\theta}x_G)$ from the velocity page:

$$|\dot{\mathbf{P}}_G|^2 = u^2 + v^2 - 2\dot{\theta}y_G u + 2\dot{\theta}x_G v + \dot{\theta}^2(x_G^2 + y_G^2),$$

so $T = \frac{1}{2}m(u^2 + v^2) - m y_G\,u\dot{\theta} + m x_G\,v\dot{\theta} + \frac{1}{2}\left({}^{G}I + m(x_G^2 + y_G^2)\right)\dot{\theta}^2$, which is $\frac{1}{2}\boldsymbol{\zeta}^\top \mathbf{M}\boldsymbol{\zeta}$ with exactly the matrix above (the cross terms $-my_G$, mx_G appear in rows 1 and 2 of column 3). Lagrange and Newton–Euler describe the same motion: for an offset G I checked numerically that the Euler–Lagrange load equals $\mathbf{J}\,(\mathbf{M}\dot{\boldsymbol{\zeta}} + c)$ (the Newton–Euler load turned into world axes). [checked]

WORKED offset centre of mass, solved at rest

$m = 8$ kg, ${}^{G}I = 0.3$ kg m^2, $G = (0.05,\ 0.02)$ m, applied at C: ${}^{C}\mathbf{f} = (6, 0)$ N, ${}^{C}n_z = 0.5$ N m, robot at rest ($c = \mathbf{0}$). $M_{33} = 0.3 + 8(0.0025 + 0.0004) = 0.3232$:

$$\begin{bmatrix} 8 & 0 & -0.16 \\ 0 & 8 & 0.4 \\ -0.16 & 0.4 & 0.3232 \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0.5 \end{bmatrix} \Rightarrow \dot{v} = -0.05\ddot{\theta},\ \dot{u} = 0.75 + 0.02\ddot{\theta},$$

row 3: $-0.12 + 0.3\ddot{\theta} = 0.5$, so $\ddot{\theta} = \mathbf{2.0667}$ rad/s^2, $\dot{v} = \mathbf{-0.1033}$, $\dot{u} = \mathbf{0.7913}$ m/s^2. (Using $M_{33} = {}^{G}I$ alone gives 2.24: wrong.) [checked]

From $\{C\}$ to $\{O\}$: $\zeta, \dot{\zeta}$ in terms of $\xi, \dot{\xi}, \ddot{\xi}$ [p. 28–30]

Kinematic link and its derivative [p. 30]

$$\dot{\boldsymbol{\xi}} = {}^{O}\mathbf{v}_C = \mathbf{J}(\boldsymbol{\xi})\,\boldsymbol{\zeta}, \qquad \ddot{\boldsymbol{\xi}} = \dot{\mathbf{J}}(\boldsymbol{\xi})\boldsymbol{\zeta} + \mathbf{J}(\boldsymbol{\xi})\dot{\boldsymbol{\zeta}}$$

$$\dot{\boldsymbol{\zeta}} = \mathbf{J}^{-1}(\boldsymbol{\xi})\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}(\boldsymbol{\xi})\boldsymbol{\zeta}\right] = \mathbf{J}^{-1}(\boldsymbol{\xi})\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}(\boldsymbol{\xi})\mathbf{J}^{-1}(\boldsymbol{\xi})\dot{\boldsymbol{\xi}}\right]$$

$$\mathbf{J} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \dot{\mathbf{J}} = \dot{\theta}\begin{bmatrix} -\sin\theta & -\cos\theta & 0 \\ \cos\theta & -\sin\theta & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{J}^{-1}\dot{\mathbf{J}} = \dot{\theta}\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Use: aim of the slide ([p. 28]): write ${}^{O}\mathbf{Q} = f(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}})$. *Two ingredients: (1)* $\boldsymbol{\zeta}, \dot{\boldsymbol{\zeta}}$ *from* $\boldsymbol{\xi}$ *and its derivatives (this slide), (2) a relation between* ${}^{C}\mathbf{Q}$ *and* ${}^{O}\mathbf{Q}$ *(next). The product* $\mathbf{J}^{-1}\dot{\mathbf{J}}$ *is the constant skew matrix* $[\boldsymbol{\omega}\times]$, *so* $\ddot{\boldsymbol{\zeta}} = \mathbf{J}^{-1}\ddot{\boldsymbol{\xi}} - [\boldsymbol{\omega}\times]\boldsymbol{\zeta}$.

SLIDE CHECK p. 30: "$\mathbf{J}(\eta)$"

The line "Taking the derivative of $\dot{\boldsymbol{\xi}} = \mathbf{J}(\eta)\boldsymbol{\zeta}$" should read $\mathbf{J}(\boldsymbol{\xi})$ (the slide's η is a typo; $\mathbf{J}$ depends on the heading θ only). The three displayed equations are right.

Loads between frames: power invariance [p. 31–32]

${}^C\mathbf{Q} = \mathbf{J}^\top {}^O\mathbf{Q}$ [p. 32]

$$ {}^O\mathbf{Q}\cdot\dot{\boldsymbol{\xi}} = {}^C\mathbf{Q}\cdot\boldsymbol{\zeta} \;\Rightarrow\; {}^O\mathbf{Q}^\top\dot{\boldsymbol{\xi}} = {}^C\mathbf{Q}^\top\boldsymbol{\zeta} \;\Rightarrow\; {}^O\mathbf{Q}^\top\mathbf{J}(\boldsymbol{\xi})\boldsymbol{\zeta} = {}^C\mathbf{Q}^\top\boldsymbol{\zeta} $$
$$ \Rightarrow\; {}^O\mathbf{Q}^\top\mathbf{J}(\boldsymbol{\xi}) = {}^C\mathbf{Q}^\top \;\Rightarrow\; \boxed{\mathbf{J}(\boldsymbol{\xi})^\top {}^O\mathbf{Q} = {}^C\mathbf{Q}} $$

SYMBOLS

${}^O\mathbf{Q}\cdot\dot{\boldsymbol{\xi}}$	power of the load in world axes ($f_x\dot{x} + f_y\dot{y} + \tau_z\dot{\theta}$)	W
${}^C\mathbf{Q}\cdot\boldsymbol{\zeta}$	power of the same load in chassis axes ($f_x u + f_y v + n_z\dot{\theta}$)	W

Use: the slide's argument: "conservation of energy (power, to be precise)": the power consumed by the robot is the same in any frame. The step ${}^O\mathbf{Q}^\top\mathbf{J}\boldsymbol{\zeta} = {}^C\mathbf{Q}^\top\boldsymbol{\zeta}$ holds for every $\boldsymbol{\zeta}$, so the row vectors are equal. Velocities map with $\mathbf{J}$, loads with $\mathbf{J}^\top$ (same idea as $\boldsymbol{\tau} = \mathbf{J}^\top\mathbf{F}$ for an arm). Since $\mathbf{J}$ is a rotation, $\mathbf{J}^\top = \mathbf{J}^{-1}$ and ${}^O\mathbf{Q} = \mathbf{J}\,{}^C\mathbf{Q}$.

WORKED turn a load between frames

(a) ${}^C\mathbf{Q} = (10, 0, 0.4)$ (N, N, N m), $\theta = 30°$: ${}^O\mathbf{Q} = \mathbf{J}{}^C\mathbf{Q} = (10\cos 30°,\ 10\sin 30°,\ 0.4) = (\mathbf{8.660},\ \mathbf{5.000},\ 0.4)$. **(b)** ${}^O\mathbf{Q} = (4, -2, 0.7)$, $\theta = 120°$: ${}^C\mathbf{Q} = \mathbf{J}^\top{}^O\mathbf{Q}$: ${}^Cf_x = 4(-0.5) + (-2)(0.8660) = -3.732$, ${}^Cf_y = -4(0.8660) + (-2)(-0.5) = -2.464$, ${}^Cn_z = 0.7$. $|\mathbf{f}| = 4.472$ in both frames. Using $\mathbf{J}$ instead gives $(-0.268,\ 4.464)$: wrong. [checked]

Equation of motion in the inertial frame [p. 33–34]

Plug into $\mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^C\mathbf{Q}$ [p. 34]

$$ \mathbf{M}\mathbf{J}^{-1}\left[\ddot{\boldsymbol{\xi}} - \dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] + c(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) = \mathbf{J}^\top {}^O\mathbf{Q} $$
$$ (\mathbf{J}^\top)^{-1}\mathbf{M}\mathbf{J}^{-1}\ddot{\boldsymbol{\xi}} + (\mathbf{J}^\top)^{-1}\left[c(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] = {}^O\mathbf{Q} $$
$$ \boxed{{}^O\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^Oc(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) = {}^O\mathbf{Q}}, \qquad {}^O\mathbf{M} = (\mathbf{J}^\top)^{-1}\mathbf{M}\mathbf{J}^{-1}, \quad {}^Oc = (\mathbf{J}^\top)^{-1}\left[c - \mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}\right] $$

Use: the result is "entirely in terms of $\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}, \ddot{\boldsymbol{\xi}}$". First line: multiply $\mathbf{M}\dot{\boldsymbol{\zeta}}$ with the formula for $\dot{\boldsymbol{\zeta}}$ above; second: multiply the line by $(\mathbf{J}^\top)^{-1}$. Since $\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}} = [\boldsymbol{\omega}\times]\boldsymbol{\zeta}$, ${}^Oc = \mathbf{J}\left[c - \mathbf{M}[\boldsymbol{\omega}\times]\boldsymbol{\zeta}\right]$ (a rotation has $(\mathbf{J}^\top)^{-1} = \mathbf{J}$).

SLIDE CHECK p. 34, third line: a missing $\mathbf{J}^{-1}$

The slide prints ${}^Oc = (\mathbf{J}^\top)^{-1}[c - \mathbf{M}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}]$, without the $\mathbf{J}^{-1}$ in front of $\dot{\mathbf{J}}$ (expand line 2 of the same slide: $\mathbf{M}\mathbf{J}^{-1}\dot{\mathbf{J}}\mathbf{J}^{-1}\dot{\boldsymbol{\xi}}$). The two agree only at $\theta = 0$. Test: $C = G$, $m = 10$, ${}^GI = 0.25$, $\theta = 90°$, $u = 1$, $v = 0$, $\dot{\theta} = 2$: correct ${}^Oc = (0, 0, 0)$ (as it must be, a free body obeys $m\ddot{x} = f_x$), the printed form gives $(-20,\ 20,\ 0)$. [checked]

WORKED $C = G$: the velocity terms vanish in world axes

$c = (-mv\dot{\theta},\ mu\dot{\theta},\ 0) = m\dot{\theta}(-v, u, 0) = \mathbf{M}[\boldsymbol{\omega}\times]\boldsymbol{\zeta}$, so $c - \mathbf{M}[\boldsymbol{\omega}\times]\boldsymbol{\zeta} = \mathbf{0}$ and ${}^O\mathbf{M} = \mathrm{diag}(m, m, {}^GI)$, ${}^Oc = \mathbf{0}$: $m\ddot{x} = f_x$, $m\ddot{y} = f_y$, ${}^GI\ddot{\theta} = \tau_z$ in world axes. The c in $\{C\}$ exists only because the axes rotate. [checked]

	Chassis frame $\mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^C\mathbf{Q}$	Inertial frame ${}^O\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^Oc = {}^O\mathbf{Q}$
unknown	$\dot{u}, \dot{v}, \ddot{\theta}$	$\ddot{x}, \ddot{y}, \ddot{\theta}$
$\mathbf{M}$	constant	depends on θ (full for offset G)
loads	wheel forces natural here	need ${}^O\mathbf{Q} = \mathbf{J}{}^C\mathbf{Q}$
use when	simulating with wheel forces ([p. 35])	you want ${}^O\mathbf{Q}$ along world axes

[drawn for this booklet] choice of frame, added

Using the model: forward simulation [p. 35]

Slide recipe [p. 35]

$$ \dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}\left[{}^C\mathbf{Q} - c(\boldsymbol{\zeta})\right], \qquad \boldsymbol{\zeta} \leftarrow \int \dot{\boldsymbol{\zeta}}\,dt, \qquad \dot{\boldsymbol{\xi}} = \mathbf{J}(\boldsymbol{\xi})\,\boldsymbol{\zeta}, \qquad \boldsymbol{\xi} \leftarrow \int \dot{\boldsymbol{\xi}}\,dt $$

Use: the chassis frame is used because the forces on the robot come from wheel–ground interaction, naturally written in $\{C\}$. Start from a known configuration with known wheel forces. Order: acceleration, velocity, pose.

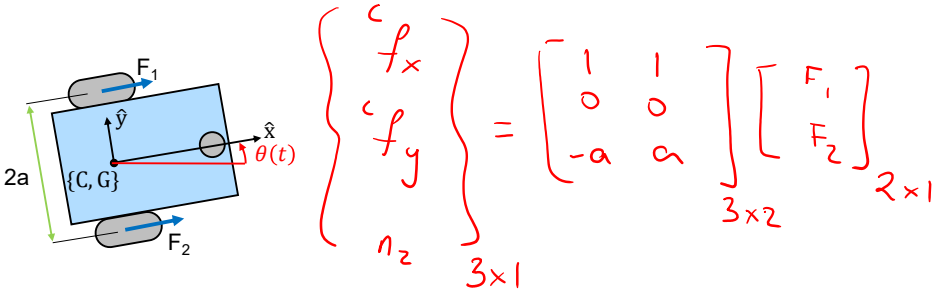
STEPS One Euler step

1. From $\boldsymbol{\zeta}_k$ compute $c(\boldsymbol{\zeta}_k) = (-m\dot{\theta}v,\ m\dot{\theta}u,\ 0)$ if $C = G$ (zero at rest).
2. Wheel forces → ${}^C\mathbf{Q}$ (differential drive: next section); add the lateral no-slip reaction if the question says the wheels do not slip.
3. $\dot{\boldsymbol{\zeta}} = \mathbf{M}^{-1}({}^C\mathbf{Q} - c)$.
4. $\boldsymbol{\zeta}_{k+1} = \boldsymbol{\zeta}_k + \Delta t\,\dot{\boldsymbol{\zeta}}$; then $\dot{\boldsymbol{\xi}} = \mathbf{J}(\theta_k)\boldsymbol{\zeta}_{k+1}$; $\boldsymbol{\xi}_{k+1} = \boldsymbol{\xi}_k + \Delta t\,\dot{\boldsymbol{\xi}}$.
5. **Check:** θ in radians; v stays 0 for a no-slip diff drive; with $\boldsymbol{\zeta} = \mathbf{0}$ the pose does not move at all in the first step.

WORKED one step with a moving robot, numbers ours

$C = G$, $m = 10$, ${}^GI = 0.25$, $r = 0.1$, $a = 0.2$; state $\boldsymbol{\xi} = (1.0,\ 2.0,\ \pi/6)$, $\boldsymbol{\zeta} = (1.0,\ 0,\ 1.6)$; torques $\tau_L = 0.4$, $\tau_R = 0.6$ N m; $\Delta t = 0.1$ s. $F = \tau/r = (4, 6)$ N; ${}^C\mathbf{Q} = (10,\ 0,\ 0.2(6-4)) = (10, 0, 0.4)$. $c = (-0,\ 10(1.0)(1.6),\ 0) = (0,\ 16,\ 0)$. Free body: $\dot{\boldsymbol{\zeta}} = (10/10,\ (0-16)/10,\ 0.4/0.25) = (1,\ -1.6,\ 1.6)$. With no slip the ground adds ${}^Cf_y = 16$ N, so $\dot{\boldsymbol{\zeta}} = (\mathbf{1},\ \mathbf{0},\ \mathbf{1.6})$. $\boldsymbol{\zeta}_1 = (1.0 + 0.1,\ 0,\ 1.6 + 0.16) = (1.1,\ 0,\ 1.76)$. $\dot{\boldsymbol{\xi}} = \mathbf{J}(30°)\boldsymbol{\zeta}_1 = (1.1\cos 30°,\ 1.1\sin 30°,\ 1.76) = (0.9526,\ 0.55,\ 1.76)$. $\boldsymbol{\xi}_1 = (1.0953,\ 2.055,\ 0.6996)$ ($\theta = 40.1°$). Using the old $\boldsymbol{\zeta}_k$ in $\dot{\boldsymbol{\xi}}$ instead gives $(1.0866,\ 2.05,\ 0.6836)$: the two orders differ at $O(\Delta t)$ here; state which you use. [checked]

Where does the chassis load come from? Differential drive [p. 36–37]



[W11 lec p. 37] — two driven wheels $2a$ apart, $\{C, G\}$ at the axle midpoint, forces F_1 (left, $+\hat{y}$) and F_2 (right, $-\hat{y}$), handwritten ${}^C\mathbf{Q} = \mathbf{B}\mathbf{F}$

PREMISE wheel forces → chassis loads **Given:** wheel forces F_1, F_2 (or torques τ_i and radius r), half-track a, m, ${}^GI$

Find: loads ${}^C\mathbf{Q} = ({}^Cf_x, {}^Cf_y, {}^Cn_z)$, then accelerations $\dot{u}, \ddot{\theta}$

Differential drive [p. 37]

$$\begin{bmatrix} {}^{C}f_x \\ {}^{C}f_y \\ {}^{C}n_z \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix}}_{3\times 2} \underbrace{\begin{bmatrix} F_1 \\ F_2 \end{bmatrix}}_{2\times 1}, \qquad F_i = \frac{\tau_i}{r}$$

$$\dot{u} = \frac{F_1 + F_2}{m}, \qquad \ddot{\theta} = \frac{a\,(F_2 - F_1)}{{}^{G}I} \qquad (C = G, \text{ from rest})$$

SYMBOLS

F_1, F_2	forward ground force of the left wheel (at $+a$) and right wheel (at $-a$)	N
τ_i, r	wheel torque, wheel radius (wheel inertia neglected)	N m, m
a	half the wheel separation $2a$ (lever arm of each wheel about C)	m

Use: two inputs, three loads: the second row is zero, so wheel torques cannot push sideways. The sideways force that turns the path (the $mu\dot{\theta}$ of c) is a no-slip reaction, which is the non-holonomic constraint of W10 again.

DERIVATION of the matrix: each wheel is one column [added]

A force F along unit vector $\mathbf{d} = (d_x, d_y)$ applied at position (x_i, y_i) contributes the load $(Fd_x,\ Fd_y,\ F(x_i d_y - y_i d_x))$ (moment = $xf_y - yf_x$). Both wheels push along $\hat{x}$, $\mathbf{d} = (1, 0)$: left wheel at $(0, +a)$: $(F_1,\ 0,\ -aF_1)$; right wheel at $(0, -a)$: $(F_2,\ 0,\ +aF_2)$. Add the two columns to get the matrix. In terms of W10: wheel speeds are $\dot{\boldsymbol{\varphi}} = \mathbf{J}_w\boldsymbol{\zeta}$ with $\mathbf{J}_w = \frac{1}{r}\begin{bmatrix} 1 & 0 & -a \\ 1 & 0 & a \end{bmatrix}$ and ${}^{C}\mathbf{Q} = \mathbf{J}_w^{\top}\boldsymbol{\tau} = \mathbf{BF}$: the same matrix transposed, because power $\boldsymbol{\tau}\cdot\dot{\boldsymbol{\varphi}} = {}^{C}\mathbf{Q}\cdot\boldsymbol{\zeta}$ is the same.

STEPS Wheel torques → $\dot{u}, \ddot{\theta}$

1. Top view: $C = G$, wheels at $(0, \pm a)$, F_1 left, F_2 right, both along $+\hat{x}$.
2. $F_i = \tau_i / r$.
3. $f_x = F_1 + F_2$, $f_y = 0$, $n_z = a(F_2 - F_1)$ (lever arm a, not $2a$).
4. $C = G$, no slip: $\dot{u} = f_x/m$, $\ddot{\theta} = n_z/{}^{G}I$ (from rest $c = 0$), $v = 0$.
5. **Check sign:** the right wheel pushing harder turns the robot left ($\ddot{\theta} > 0$). Check power: $\boldsymbol{\tau}\cdot\dot{\boldsymbol{\varphi}} = f_x u + n_z\dot{\theta}$. Path radius $R = u/\dot{\theta} = {}^{G}I f_x/(m\, n_z)$.

WORKED wheel torques, numbers ours (the slide has none)

$m = 10\,\text{kg}$, ${}^{G}I = 0.25\,\text{kg m}^2$, $r = 0.1\,\text{m}$, $a = 0.2\,\text{m}$, $\tau_L = 0.4$, $\tau_R = 0.6\,\text{N m}$, from rest. $F = (4, 6)\,\text{N}$; $f_x = 10\,\text{N}$, $n_z = 0.2(6 - 4) = 0.4\,\text{N m}$; $\dot{u} = \mathbf{1.0}\,\text{m/s}^2$, $\ddot{\theta} = \mathbf{1.6}\,\text{rad/s}^2$; $R = {}^{G}I f_x/(m n_z) = 0.625\,\text{m}$. At $t = 1\,\text{s}$: $u = 1$, $\dot{\theta} = 1.6$; $\dot{\varphi}_L = (1 - 0.2\cdot 1.6)/0.1 = 6.8$, $\dot{\varphi}_R = 13.2\,\text{rad/s}$; power $0.4(6.8) + 0.6(13.2) = 10.64\,\text{W} = 10(1) + 0.4(1.6)$. Holding $v = 0$ needs a sideways ground force $mu\dot{\theta} = 16\,\text{N}$ (if the wheels were free to slide: $\dot{v} = -u\dot{\theta} = -1.6$). At $t = 2\,\text{s}$: $(x, y) = (-0.036,\ 1.249)\,\text{m}$, $\theta = 3.2\,\text{rad}$. [checked]

SLIDE CHECK gap, not an error: no-slip is not in the deck

The deck never states the no-slip condition inside the dynamics. The p. 37 matrix lists the forward forces only; the sideways force that keeps $v = 0$ is a constraint reaction. If wheel forces are given and $\dot{v}$ is asked: say "no slip ⇒ $v = 0$, $\dot{v} = 0$"; for a body with no lateral grip use $\mathbf{M}\dot{\boldsymbol{\zeta}} + c = {}^{C}\mathbf{Q}$ with ${}^{C}f_y = 0$.

TRAP

$2a$ as the lever arm (doubles $\ddot{\theta}$); swapping the signs of the n_z row (right push turns right); forgetting $F = \tau/r$; using $\mathbf{J}$ instead of $\mathbf{J}^{\top}$ to bring loads to $\{C\}$.

Scenario: force to drive a circle of radius R at speed v [added]

PREMISE circle at constant speed **Given:** mass m, speed u (call it v in the question), radius R; $C = G$, no slip

Find: the loads ${}^{C}\mathbf{Q}$ needed, which of them the wheels supply, wheel speeds for half-track a and wheel radius r

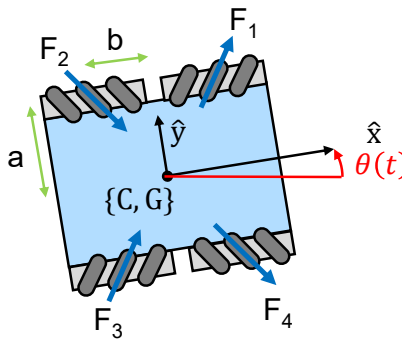
STEPS Circle of radius R at speed u

1. Anticlockwise circle (centre on the $+\hat{y}$ side): $\dot{\theta} = u/R$, $v = 0$, $\dot{u} = \ddot{\theta} = 0$.
2. Steady state: ${}^{C}\mathbf{Q} = c = (-mv\dot{\theta},\ mu\dot{\theta},\ 0) = (0,\ mu^2/R,\ 0)$ (for $C = G$).
3. Meaning: the force is mu^2/R *towards the centre*, sideways in $\{C\}$. Drive forces $f_x = n_z = 0$ (apart from friction losses): both wheels give equal forces $F_1 = F_2$ that only balance friction. The sideways force is the wheel–ground no-slip reaction.
4. Wheel speeds: $\dot{\varphi}_{1,2} = (u \mp a\dot{\theta})/r$ (left, right; W10).
5. **Check:** $R \to \infty$ gives 0 N; double the speed means 4 times the force.

WORKED numbers ours

$m = 12\,\text{kg}$, $u = 0.6\,\text{m/s}$, $R = 1.5\,\text{m}$: $\dot{\theta} = 0.4\,\text{rad/s}$, $f_y = mu\dot{\theta} = 12(0.6)(0.4) = \mathbf{2.88}\,\text{N}$ $(= mu^2/R)$. With $a = 0.2$, $r = 0.1$: $\dot{\varphi}_L = (0.6 - 0.08)/0.1 = \mathbf{5.2}$, $\dot{\varphi}_R = \mathbf{6.8}\,\text{rad/s}$. (Slide-style: $m = 10$, $u = 2$, $\dot{\theta} = 0.5$: $R = 4\,\text{m}$, $f_y = 10\,\text{N}$; the printed $v\dot{\theta}$ would give 0.) [checked]

Swedish-wheel robot [p. 38]



[W11 lec p. 38] — four Swedish wheels, forces $F_1 \ldots F_4$ along the roller directions, offsets a and b, $\{C, G\}$ at the centre

The slide is a figure only (the answer was done in class). The method is the same as for the differential drive.

DERIVATION load matrix for the four wheels [added]

Read from the figure: wheel i at (x_i, y_i) in $\{C\}$ with roller (force) direction $\mathbf{d}_i$ at $\pm 45°$, $s = 1/\sqrt{2}$. I took F_1 at (b, a), $\mathbf{d}_1 = s(1, 1)$; F_2 at $(-b, a)$, $\mathbf{d}_2 = s(1, -1)$; F_3 at $(-b, -a)$, $\mathbf{d}_3 = s(1, 1)$; F_4 at $(b, -a)$, $\mathbf{d}_4 = s(1, -1)$ (the exact label positions of a, b on the slide are my reading). Each column is $(d_x,\ d_y,\ x_i d_y - y_i d_x)$:

$$\begin{bmatrix} {}^C f_x \\ {}^C f_y \\ {}^C n_z \end{bmatrix} = s \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ b-a & b-a & a-b & a-b \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

3×4, rank 3: any load is reachable, and the null space $\mathbf{F} \propto (1, -1, -1, 1)$ gives internal forces with zero net load (wheels fighting each other).

WORKED numbers ours

$a = 0.2$, $b = 0.25$ m. $\mathbf{F} = (5, -5, 5, -5)$ N: ${}^C\mathbf{f} = (0,\ 14.14)$ N, $n_z = 0$: pure sideways push. $\mathbf{F} = (5, 5, 5, 5)$: $\mathbf{f} = (14.14,\ 0)$, $n_z = 0$: forwards. $\mathbf{F} = (5, 5, -5, -5)$: $\mathbf{f} = \mathbf{0}$, $n_z = 0.0354 \cdot 20 = 0.707$ N m: pure spin. With $m = 10$, $\mathbf{F} = (5, -5, 5, -5)$ from rest: $\dot{v} = 1.414$ m/s^2. A holonomic robot can accelerate sideways without turning, unlike the differential drive. [checked]

Summary [p. 39]

Two equations, one link

body frame: $\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^C\mathbf{Q}$ inertial frame: ${}^O\mathbf{M}\ddot{\boldsymbol{\xi}} + {}^O c(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) = {}^O\mathbf{Q}$ ${}^C\mathbf{Q} = \mathbf{J}^\top {}^O\mathbf{Q}$

Use: the slide lists only these two ("dynamics equation in the body frame", "in the inertial frame"). Wheel forces enter through ${}^C\mathbf{Q} = \mathbf{B}\mathbf{F}$.

TRAP

Spot-the-error list: $+v\dot{\theta}$ in the y row (should be u); $M_{33} = {}^G I$ for an offset G (needs $+m(x_G^2 + y_G^2)$); ${}^C\mathbf{Q} = \mathbf{J}\,{}^O\mathbf{Q}$ (should be $\mathbf{J}^\top$); $2a$ as lever arm; calling a simulation from given forces "inverse dynamics" (it is forward).

How this is examined: worked solutions

No W11 question is among the W13 samples (W13 pp. 4–20). The nearest are Q4 (derive the equations of motion) and Q10 (wheel model). Marks are not published; indicative, about 1 mark per step.

EXAM-STYLE QUESTION 1: derive and solve, offset centre of mass [indicative: 6 steps]

A planar chassis has mass $m = 8$ kg and moment of inertia ${}^G I = 0.3$ kg m^2 about its centre of mass G, which lies at $({}^C x_G, {}^C y_G) = (0.05,\ 0.02)$ m from the reference point C. Wheels apply at C the force ${}^C\mathbf{f} = (6, 0)$ N and torque ${}^C n_z = 0.5$ N m; the robot is at rest. (a) Derive $\mathbf{M}$ in $\mathbf{M}\dot{\boldsymbol{\zeta}} + c(\boldsymbol{\zeta}) = {}^C\mathbf{Q}$. (b) Find $\ddot{\theta}$ and $\dot{v}$.

Solution

(a) Frames $\{O\}, \{C\}, \{G\}$ drawn, applied loads at C, inertial loads at G. At rest $u = v = \dot{\theta} = 0$, so the acceleration of G is $({}^C\ddot{x}_G, {}^C\ddot{y}_G) = (\dot{u} - y_G\ddot{\theta},\ \dot{v} + x_G\ddot{\theta})$ and $c = \mathbf{0}$. Newton: ${}^C f_x = m(\dot{u} - y_G\ddot{\theta})$, ${}^C f_y = m(\dot{v} + x_G\ddot{\theta})$. Euler about G and transfer to C: ${}^C n_z = {}^G I\ddot{\theta} - y_G\,{}^C f_x + x_G\,{}^C f_y$. Substitute: ${}^C n_z = -m y_G\dot{u} + m x_G\dot{v} + ({}^G I + m(x_G^2 + y_G^2))\ddot{\theta}$. So

$$\mathbf{M} = \begin{bmatrix} m & 0 & -m y_G \\ 0 & m & m x_G \\ -m y_G & m x_G & {}^G I + m(x_G^2 + y_G^2) \end{bmatrix} = \begin{bmatrix} 8 & 0 & -0.16 \\ 0 & 8 & 0.4 \\ -0.16 & 0.4 & 0.3232 \end{bmatrix}$$

(symmetric, constant; $M_{33} = 0.3 + 8(0.0025 + 0.0004) = 0.3232$). (b) Row 2: $8\dot{v} + 0.4\ddot{\theta} = 0$, $\dot{v} = -0.05\ddot{\theta}$. Row 1: $8\dot{u} - 0.16\ddot{\theta} = 6$, $\dot{u} = 0.75 + 0.02\ddot{\theta}$. Row 3: $-0.16(0.75 + 0.02\ddot{\theta}) + 0.4(-0.05\ddot{\theta}) + 0.3232\ddot{\theta} = -0.12 + 0.3\ddot{\theta} = 0.5$.

$$\boxed{\ddot{\theta} = 2.067 \text{ rad/s}^2, \quad \dot{v} = -0.103 \text{ m/s}^2, \quad \dot{u} = 0.791 \text{ m/s}^2}$$

Check: the coefficient of $\ddot{\theta}$ is exactly ${}^G I = 0.3$ after elimination (torque about G sets the spin); $M_{33} = {}^G I$ would give 2.24, wrong. [checked]

EXAM-STYLE QUESTION 2: differential drive from wheel torques [indicative: 5 steps]

A differential-drive robot ($C = G$ at the axle midpoint) has $m = 12$ kg, ${}^G I = 0.4$ kg m^2, wheel radius $r = 0.08$ m and half-track $a = 0.25$ m. From rest the left and right wheel torques are $\tau_L = 0.5$ and $\tau_R = 0.9$ N m (wheel inertia neglected, no slip). Find $\dot{u}$ and $\ddot{\theta}$, and the radius of the path.

Solution

Frames: $\hat{x}_C$ forwards, left wheel at $(0, +a)$. Forces $F_1 = \tau_L/r = 6.25$ N, $F_2 = \tau_R/r = 11.25$ N, both along $+\hat{x}_C$.

$${}^C\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -a & a \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} = \begin{bmatrix} 17.5 \\ 0 \\ 0.25(11.25 - 6.25) \end{bmatrix} = \begin{bmatrix} 17.5 \text{ N} \\ 0 \\ 1.25 \text{ N m} \end{bmatrix}$$

From rest $c = \mathbf{0}$ and $C = G$: $\mathbf{M} = \mathrm{diag}(12, 12, 0.4)$.

$$\boxed{\dot{u} = 17.5/12 = 1.458 \text{ m/s}^2, \qquad \ddot{\theta} = 1.25/0.4 = 3.125 \text{ rad/s}^2}$$

Both are constant, so $R = u/\dot{\theta} = {}^G I f_x/(m n_z) = 0.4(17.5)/(12 \cdot 1.25) = 0.467$ m: a circle. Check: the stronger right wheel turns the robot left ($\ddot{\theta} > 0$); using $2a$ would give 6.25, wrong. [checked]

EXAM-STYLE QUESTION 3: the force needed to drive a circle [indicative: 5 steps]

A robot of mass 12 kg ($C = G$) drives a circle of radius 1.5 m at a constant 0.6 m/s without sliding. (a) What force, in the chassis frame, does the ground exert? (b) Which part of ${}^{C}\mathbf{Q}$ do the drive wheels supply? (c) Wheel speeds for $a = 0.2$ m, $r = 0.1$ m. (d) Explain why constant speed still needs a force.

Solution

(a) Steady circle: $\dot\theta = u/R = 0.6/1.5 = 0.4$ rad/s, $v = 0$, $\dot u = \ddot\theta = 0$. Equations of motion with $C = G$: ${}^{C}f_x = m(\dot u - v\dot\theta) = 0$, ${}^{C}f_y = m(\dot v + u\dot\theta) = 12(0.6)(0.4)$, ${}^{C}n_z = {}^{G}I\ddot\theta = 0$.

$$\boxed{{}^{C}f_y = mu\dot\theta = mu^2/R = 2.88\ \text{N},\ \text{towards the centre } (+\hat y_C)}$$

(b) The drive wheels supply no net load in steady state ($f_x = n_z = 0$, only friction); the sideways 2.88 N is the no-slip reaction (row 2 of the diff-drive matrix is zero). (c) $\dot\varphi_L = (0.6 - 0.2(0.4))/0.1 = 5.2$, $\dot\varphi_R = (0.6 + 0.08)/0.1 = 6.8$ rad/s. (d) The velocity vector turns with the body: the acceleration is $u\dot\theta$ even though $|\mathbf{v}|$ is constant; Newton needs a force. Check: $R \to \infty$ gives zero. [checked]

EXAM-STYLE QUESTION 4: acceleration of the centre of mass [indicative: 4 steps]

The chassis velocity is $u = 1.5$, $v = 0.2$ m/s, $\dot\theta = 0.8$ rad/s, and $\dot u = 0.3$, $\dot v = -0.1$ m/s^2. (a) With $C = G$ and $\ddot\theta = 0$, find the acceleration of C in the chassis axes. (b) Now G is at $(0.2, 0.1)$ m and $\ddot\theta = 0.5$ rad/s^2: find the acceleration of G and the force $m\mathbf{a}_G$ for $m = 12$ kg.

Solution

(a) $\mathbf{v} = u\hat x + v\hat y$; the unit vectors turn: $\dot{\hat x} = \dot\theta\hat y$, $\dot{\hat y} = -\dot\theta\hat x$. So $\mathbf{a} = (\dot u - v\dot\theta)\hat x + (\dot v + u\dot\theta)\hat y$:

$$\boxed{a_x = 0.3 - 0.2(0.8) = 0.14\ \text{m/s}^2, \qquad a_y = -0.1 + 1.5(0.8) = 1.10\ \text{m/s}^2}$$

(b) Four terms (cross products of $\boldsymbol\omega = (0, 0, 0.8)$, $\dot{\boldsymbol\omega} = (0, 0, 0.5)$): $\ddot{\mathbf{P}}_C = (0.3, -0.1)$; $\dot{\boldsymbol\omega} \times \mathbf{P}_G = (-0.5(0.1),\ 0.5(0.2)) = (-0.05, 0.10)$; $\boldsymbol\omega \times \dot{\mathbf{P}}_C = 0.8(-0.2,\ 1.5) = (-0.16, 1.20)$; $\boldsymbol\omega \times \boldsymbol\omega \times \mathbf{P}_G = -0.64(0.2, 0.1) = (-0.128, -0.064)$.

$$\boxed{\mathbf{a}_G = (-0.038,\ 1.136)\ \text{m/s}^2, \qquad m\mathbf{a}_G = (-0.456,\ 13.63)\ \text{N}}$$

Check: the y entry is dominated by $u\dot\theta = 1.2$ (centripetal); the printed $+v\dot\theta$ would give 0.06 in (a), 18 times too small. [checked]

EXAM-STYLE QUESTION 5: one integration step [indicative: 5 steps]

The robot of Worked example "wheel torques" ($m = 10$, ${}^{G}I = 0.25$, $r = 0.1$, $a = 0.2$, $C = G$, no slip) is at $\boldsymbol\xi = (1.0,\ 2.0,\ \pi/6)$ with chassis velocity $\boldsymbol\zeta = (1.0,\ 0,\ 1.6)$. The wheel torques are $\tau_L = 0.4$, $\tau_R = 0.6$ N m. Using the slide recipe with $\Delta t = 0.1$ s find $\dot{\boldsymbol\zeta}$, $\boldsymbol\zeta$ and $\boldsymbol\xi$ after one step.

Solution

$\{C\}$ axes. Wheel forces $F = (4, 6)$ N, so ${}^{C}\mathbf{Q} = (10,\ 0,\ 0.2(6 - 4)) = (10, 0, 0.4)$. With $C = G$: $\mathbf{M} = \text{diag}(10, 10, 0.25)$ and $c = (-mv\dot\theta,\ mu\dot\theta,\ 0) = (0,\ 16,\ 0)$. No slip: the ground supplies ${}^{C}f_y = 16$ N (so that $\dot v = 0$), giving $\dot{\boldsymbol\zeta} = \mathbf{M}^{-1}({}^{C}\mathbf{Q} + (0, 16, 0) - c) = (1.0,\ 0,\ 1.6)$. Integrate: $\boldsymbol\zeta_1 = (1.0 + 0.1(1.0),\ 0,\ 1.6 + 0.1(1.6)) = (1.1,\ 0,\ 1.76)$. Kinematics: $\dot{\boldsymbol\xi} = \mathbf{J}(\pi/6)\boldsymbol\zeta_1 = \begin{bmatrix}\cos 30^\circ & -\sin 30^\circ & 0\\ \sin 30^\circ & \cos 30^\circ & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1.1\\ 0\\ 1.76\end{bmatrix} = (0.9526,\ 0.55,\ 1.76)$. Integrate: $\boldsymbol\xi_1 = (1.0 + 0.0953,\ 2.0 + 0.055,\ 0.5236 + 0.176)$:

$$\boxed{\dot{\boldsymbol\zeta} = (1,\ 0,\ 1.6), \quad \boldsymbol\zeta_1 = (1.1,\ 0,\ 1.76), \quad \boldsymbol\xi_1 = (1.095\ \text{m},\ 2.055\ \text{m},\ 0.700\ \text{rad})}$$

Check: radians throughout; speed $|\mathbf{v}|$ goes from 1.0 to 1.1 m/s. If the lateral reaction were omitted the free body would give $\dot v = -1.6$ (it would slide). [checked]

EXAM-STYLE QUESTION 6: loads between frames; spot the errors [indicative: 4 steps]

(a) Show that ${}^{C}\mathbf{Q} = \mathbf{J}^{\top\,O}\mathbf{Q}$. (b) At $\theta = 120^\circ$ the world-frame load is ${}^{O}\mathbf{Q} = (4, -2, 0.7)$ (N, N, N m): find ${}^{C}f_x, {}^{C}f_y$. (c) A student writes "for $a = 0.2$ m, $F_1 = 3$ N, $F_2 = 7$ N: $n_z = 2a(F_2 - F_1) = 1.6$ N m; the simulation from these forces is inverse dynamics; ${}^{C}\mathbf{Q} = \mathbf{J}^{O}\mathbf{Q}$". Find the three errors.

Solution

(a) Power is the same in any frame: ${}^{O}\mathbf{Q}^{\top}\dot{\boldsymbol\xi} = {}^{C}\mathbf{Q}^{\top}\boldsymbol\zeta$. With $\dot{\boldsymbol\xi} = \mathbf{J}\boldsymbol\zeta$: ${}^{O}\mathbf{Q}^{\top}\mathbf{J}\boldsymbol\zeta = {}^{C}\mathbf{Q}^{\top}\boldsymbol\zeta$ for every $\boldsymbol\zeta$, so ${}^{O}\mathbf{Q}^{\top}\mathbf{J} = {}^{C}\mathbf{Q}^{\top}$, i.e. ${}^{C}\mathbf{Q} = \mathbf{J}^{\top\,O}\mathbf{Q}$ ($= \mathbf{J}^{-1\,O}\mathbf{Q}$ for a rotation). (b) $\cos 120^\circ = -0.5$, $\sin 120^\circ = 0.8660$. ${}^{C}f_x = 4(-0.5) + (-2)(0.8660) = -3.732$, ${}^{C}f_y = -4(0.8660) + (-2)(-0.5) = -2.464$:

$$\boxed{{}^{C}f_x = -3.73\ \text{N}, \quad {}^{C}f_y = -2.46\ \text{N}, \quad {}^{C}n_z = 0.7\ \text{N m}}$$

Check: $\sqrt{3.732^2 + 2.464^2} = 4.472 = \sqrt{4^2 + 2^2}$ (rotation keeps magnitude); torque about z unchanged. (c) Error 1: the lever arm is a not $2a$: $n_z = 0.2(7 - 3) = 0.8$ N m. Error 2: forces in, motion out is *forward* dynamics. Error 3: ${}^{C}\mathbf{Q} = \mathbf{J}^{\top\,O}\mathbf{Q}$, not $\mathbf{J}^{O}\mathbf{Q}$ (the latter would rotate the load the wrong way, by $+\theta$). [checked]

Which planner? Look-up for the exam [added]

If the question says…

"find a path / shortest path on this graph" ..... BFS, Dijkstra or A* (p. 144–145)
"show the queue / stack / table at every step" ..... solved samples 1–4 (p. 144–145)
"is the heuristic admissible / consistent?" ..... checks on p. 146
"what changes if all edge costs are equal?" ..... p. 147
"robot has radius r, can it pass?" ..... inflate obstacles (p. 143)
"2-link arm, where are the obstacles in joint space?" ..... p. 142
"6-DoF arm / many joints, which planner?" ..... sampling-based (p. 150)
"car / diff-drive that cannot slide sideways" ..... RRT with sampled controls (p. 149)
"one PRM or RRT step with numbers" ..... p. 148–149
"the W12 practical sheet" ..... p. 150

Tie-break used throughout this part: nodes with equal priority are taken in **alphabetical** order; on the grid, neighbours are visited up, right, down, left. Write your own rule at the top of an exam answer: marks go to a stated, consistent rule.

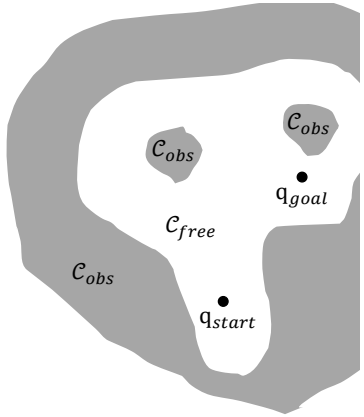
The planning problem: premise [p. 5–9]

PREMISE motion / path planning **Given:** map (geometry of the robot and of the obstacles), robot kinematics (degrees of freedom, constraints), start and goal configurations $\mathbf{q}_{start}$, $\mathbf{q}_{goal}$ [p. 7]
Find: a continuous sequence of collision-free configurations from $\mathbf{q}_{start}$ to $\mathbf{q}_{goal}$, or the report "no path exists" [p. 7]

Slide assumptions [p. 8]: the representation of robot and world is expressive enough; we know where we are and where we want to go; we have a motion model; the problem is usually posed as an optimisation (minimise time or distance within the constraints). Trends [p. 9]: classical (mid-70s, exact models, no sensing), reactive (mid-80s, no models, heavy sensing), hybrid (90s, models at high level, reactive at low level), probabilistic (since mid-90s, inaccurate models *and* sensors). **Roadmap of this part (slides [p. 6]):** (1) transform the problem to configuration space, (2) represent it as a graph, (3) search the graph, (4) grid-based versus sampling-based planning.

Task space and configuration space [p. 12–17]

Task space [p. 12]: the frame in which the robot performs its task (a frame on the end effector, or on a mobile robot). **Configuration space** C [p. 14]: the set of generalised coordinates that completely describes the robot: the joint parameters of a manipulator, the position and orientation of a mobile robot. A *configuration* $\mathbf{q}$ is one point of C; the robot is a point there.



[W12 lec p. 15] — C_{obs} (grey), C_{free} (white), and the path-planning problem from $\mathbf{q}_{start}$ to $\mathbf{q}_{goal}$

Free space [p. 15]

$$C_{free} = \{\mathbf{q} \in C \mid \mathbf{q} \cap C_{obs} = \varnothing\}$$

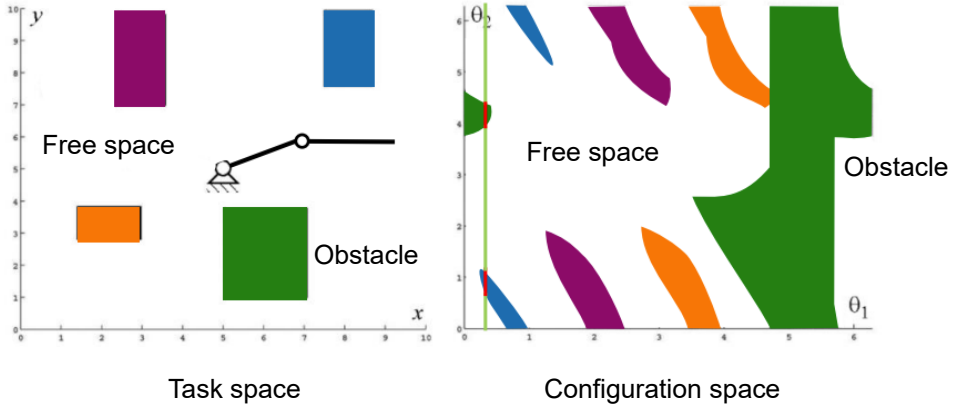
SYMBOLS

C	configuration space (all configurations of the robot)	–
$\mathcal{W}$	workspace of the robot (the physical space, task space)	–
C_{obs}	obstacle region: configurations in which the robot touches an obstacle	–
C_{free}	free space: configurations with no contact	–
$\mathbf{q}$	one configuration, a point of C (joint angles; or x, y, θ)	rad, m
$\varnothing$	empty set	–

Use: planning means finding a curve inside C_{free} from $\mathbf{q}_{start}$ to $\mathbf{q}_{goal}$. (The slide writes $C_{obs} \in C$; the intended symbol is $\subset$.)

Why C-space [p. 17]: configurations that are near each other are near for the robot; collision checks and joining nearby poses become easy; it is an abstraction so one solver handles many robots; it helps with wrap-around of rotary joints ($q \equiv q + 2\pi$, so the C-space of an RR arm is a torus).

C-space example: the 2-link arm [p. 18–19]



[W12 lec p. 19] — task space (x, y) with four obstacles (left); the same obstacles in configuration space (θ_1, θ_2) (right). Each obstacle becomes a curved blob;

the green box becomes a large region

Planar 2R arm [p. 18]: lengths L_1, L_2, θ_1 from the horizontal, θ_2 relative to link 1, tip (x, y) in the base frame $\{0\}$. A workspace point can be reached by *two* configurations (elbow up / down), so one point obstacle becomes **two** C-space points, and a solid obstacle becomes blobs with curved edges. The vertical line on the slide is one slice $\theta_1 = \text{const}$: the arm can have any θ_2 in the white gaps of that slice and none inside a blob.

C-space image of a point obstacle [added]

$$\cos\theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}, \qquad \theta_2 = \pm\arccos(\cdot)$$

$$\theta_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2 \sin\theta_2,\ L_1 + L_2\cos\theta_2)$$

SYMBOLS

(x, y)	obstacle point, in the base frame $\{0\}$	m
L_1, L_2	link lengths	m
θ_1, θ_2	joint angles; the two signs of θ_2 give the two C-space points	rad

Use: map one workspace point to C-space (this is the planar RR inverse kinematics of W3). If $|\cos\theta_2| > 1$ the point is out of reach: no C-space point.

WORKED $L_1 = L_2 = 1$ m, obstacle point $P = (1, 1)$

$\cos\theta_2 = (1 + 1 - 1 - 1)/2 = 0 \Rightarrow \theta_2 = \pm 90°$. For $+90°$: $\theta_1 = 45° - \operatorname{atan2}(1, 1) = 45° - 45° = 0°$. For $-90°$: $\theta_1 = 45° - \operatorname{atan2}(-1, 1) = 45° + 45° = 90°$. So P is the two C-space points $(\theta_1, \theta_2) = (0°, 90°)$ and $(90°, -90°)$; FK of each returns $(1, 1)$. [checked] A point at $(1.5, 1.5)$ ($r = 2.121 > 2$) has no C-space point.

Inflating the obstacles [p. 20–24]



[W12 lec p. 22] — the robot (blue disc) is shrunk to a point and the obstacle grows by the robot size (blue margin)

For a **disc** robot the C-space obstacle is the obstacle grown by r (Minkowski sum). A rectangular robot gives a polygonal margin [p. 23]; the point is then free to move anywhere outside the grown shape [p. 24].

Area of an obstacle grown by a disc of radius r [added]

$$A_{grown} = A + P\,r + \pi r^2$$

SYMBOLS

A	area of the convex polygon obstacle	m^2
P	perimeter of the polygon	m
r	robot (disc) radius	m

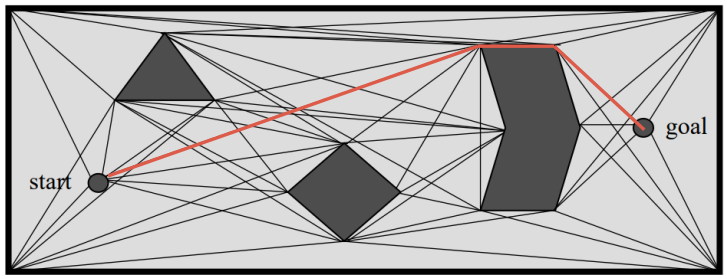
Use: strips of width r along each edge (Pr) plus corner sectors that add to one disc (πr^2). Corridor of width w: free width for the point robot is $w - 2r$; negative means the robot cannot pass.

WORKED triangle $(5, 3), (6, 3), (5, 1)$, $r = 0.5$ m

$A = 1$, $P = 1 + \sqrt{5} + 2 = 5.236$, $Pr = 2.618$, $\pi r^2 = 0.785$: $A_{grown} = 4.403\ \text{m}^2$. A 2×2 m square with $r = 0.5$: $4 + 4 + 0.785 = 8.785$ (the square margin $3 \times 3 = 9$ over-blocks). Corridor $w = 1.0$: $r = 0.4$ leaves 0.2 m, $r = 0.55$ gives $1.0 - 1.1 = -0.1 < 0$: blocked. [checked]

Framework, visibility graph, Voronoi, cells [p. 25–46]

Slide [p. 25]: from a continuous C-space, *discretise*, then either **combinatorial** planning (describe C_{free} explicitly as a graph, then search) or **sampling-based** planning (use a collision detector to probe C incrementally). Why discretise [p. 26]: computers store things digitally, and a graph plus an existing search algorithm is the common route.



[W12 lec p. 34] — visibility graph: lines between start, goal and polygon vertices; the shortest path (red) touches corners

	Visibility graph [p. 28–36]	Voronoi diagram [p. 37–40]
Idea	join start, goal and every vertex pair whose segment is not blocked; search for the shortest path	paths equidistant from the obstacles (straight and parabolic pieces)
Gives	shortest path	maximum clearance ("maximise sensor readings")
Cons	hugs obstacles; many edges	longer; fails with short-range sensors

Reduced visibility graph [p. 36]: keep only lines that go around an object or pass through its extremes. **Cell decomposition** [p. 41–46]: *exact* (vertical trapezoids and their adjacency graph), *approximate* (uniform grid, a cell touching an obstacle is blocked) and *adaptive* (big cells in open space, small near obstacles); search the cells containing start and goal.

WORKED practical map, visibility optimum [added]

$S = (1.5, 1.5)$, $G = (9, 9)$ (disc of radius 1), blocked on the diagonal by the rectangle $(6, 6)$–$(8, 8)$. Via the corner $(8, 6)$ (touching only): $\sqrt{6.5^2 + 4.5^2} = 7.906$ and $\sqrt{1^2 + 3^2} = 3.162$, total 11.068 to the centre; to the goal-disc edge 10.068. [checked]

Graph search methods: overview [p. 47–49]

PREMISE graph search **Given:** a graph $G = (V, E)$ (nodes V, edges E, optional edge costs $d(n, m) \geq 0$), a start node and a goal node
Find: a path (list of nodes) from start to goal, its cost, or "no path"

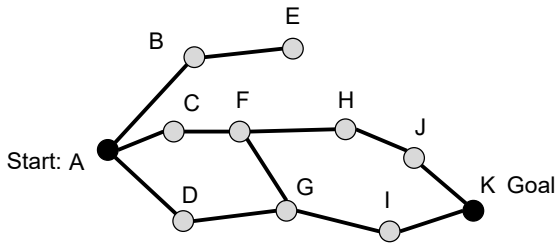
Slide [p. 48]: a least-cost problem on a (directed) graph. Limits: the state space is discretised, so completeness is at stake; feasibility of paths is seldom encoded.

Method [p. 49]	Kind	Frontier (open set) is a…	Uses
BFS	uninformed	queue, FIFO	number of edges
DFS	uninformed	stack, LIFO	nothing (dives)
Dijkstra	uninformed	priority queue on $g(n)$	edge costs
A*	informed	priority queue on $f = g + h$	costs and a goal heuristic
PRM, RRT	randomised	graph / tree built from samples	collision checker

Vocabulary. *Frontier* (open set): discovered nodes waiting to be expanded. *Expanded* (closed, visited): nodes already taken from the frontier. *Discovered ≠ expanded*. *Parent table*: for each node, the node it was reached from; the path is read backwards from the goal.

Grid as a graph [p. 64]: each free cell is a node; edges join neighbours. *4-connected*: up, right, down, left (2 per dimension). *8-connected*: also the four diagonals (diagonal step costs $\sqrt{2}$ when a straight step costs 1). Connectivity is set by the grid shape; the resolution trades missing narrow gaps (too coarse) against run time (too fine).

Breadth-first search (BFS) [p. 50–51]



[W12 lec p. 50] — the slide graph (unweighted): start A, goal K

STEPS BFS by hand

1. Write the tie-break (alphabetical). Queue = [start], visited = {start}.
2. Remove the *front* node n. If n is the goal: stop.
3. For each neighbour m of n (alphabetical) that is not visited: set parent(m) = n, add m to the *back*, mark it visited.
4. Repeat from 2. Read the path by following parents from the goal back to the start.
5. **Check:** the path has the fewest edges; edge costs were ignored, so add them up separately.

Properties. Expands level by level (all nodes k edges away before k+1), so it finds the path with the *fewest edges*; it is cost-optimal only if all edges cost the same. Time $O(|V| + |E|)$ [p. 65]. Complete on a finite graph.

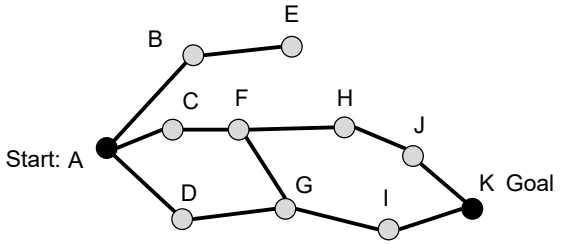
WORKED SOLVED SAMPLE 1: BFS on the slide graph, A to K, alphabetical

Edges (weights are ignored by BFS): A–B, B–E, A–C, C–F, F–H, H–J, J–K, A–D, D–G, F–G, G–I, I–K. Queue: front is on the left. A node is marked visited when it is *added* to the queue.

step	pop	new (parent)	queue after	visited
init	–	–	A	A
1	A	B(A), C(A), D(A)	B, C, D	A, B, C, D
2	B	E(B)	C, D, E	A, B, C, D, E
3	C	F(C)	D, E, F	A, B, C, D, E, F
4	D	G(D)	E, F, G	A, B, C, D, E, F, G
5	E	–	F, G	A, B, C, D, E, F, G
6	F	H(F)	G, H	A, B, C, D, E, F, G, H
7	G	I(G)	H, I	A, B, C, D, E, F, G, H, I
8	H	J(H)	I, J	A, B, C, D, E, F, G, H, I, J
9	I	K(I)	J, K	A, B, C, D, E, F, G, H, I, J, K
10	J	–	K	A, B, C, D, E, F, G, H, I, J, K
11	K	–	–	A, B, C, D, E, F, G, H, I, J, K

Parents: B←A, C←A, D←A, E←B, F←C, G←D, H←F, I←G, J←H, K←I. Path (K back to A): K←I←G←D←A, i.e. **A–D–G–I–K**, 4 edges, cost 8 + 9 + 7 + 5 = **29**. J would also reach K, but K was already visited via I. Levels: A (0); B, C, D (1); E, F, G (2); H, I (3); J, K (4). Fewest edges but **not** cheapest (best is 25, see Dijkstra). [checked]

Depth-first search (DFS) [p. 52–54]



[W12 lec p. 50] — same graph, searched depth first (slide 52 uses this unweighted graph)

STEPS DFS by hand (explicit stack)

1. State the rule: *mark visited when popped*; push neighbours in *reverse* alphabetical order so the alphabetically first is on top and is popped first. Stack = [start].
2. Pop the *top* node n. If already visited: skip. Else mark visited and record its parent (the node that pushed that copy).
3. If n is the goal: stop. Else push every unvisited neighbour of n.
4. Repeat. Read the path from the parent table.
5. **Check:** DFS gives *a* path, not the shortest or cheapest; another neighbour order gives another path. A depth limit [p. 54] stops it diving for ever.

WORKED SOLVED SAMPLE 2: DFS on the slide graph, A to K, alphabetical

Top of the stack is on the *right*. Entries are nodes; a node can be on the stack twice.

step	pop	action	stack after (top at right)
init	–	push A	A
1	A	push D, C, B	DCB
2	B	push E	DCE
3	E	nothing new	DC
4	C	push F	DF
5	F	push H, G	DHG
6	G	push I, D	DHID
7	D	nothing new	DHI
8	I	push K	DHK
9	K	GOAL: stop	DH

Visit order A, B, E, C, F, G, D, I, K. Parents: B←A, E←B, C←A, F←C, G←F, D←G, I←G, K←I. Path **A–C–F–G–I–K**, 5 edges, cost 6 + 5 + 3 + 7 + 5 = **26**. Not optimal (25) and not fewest edges (BFS: 4). The dive A–B–E ended in a dead end and the stack backed up to C. [checked]

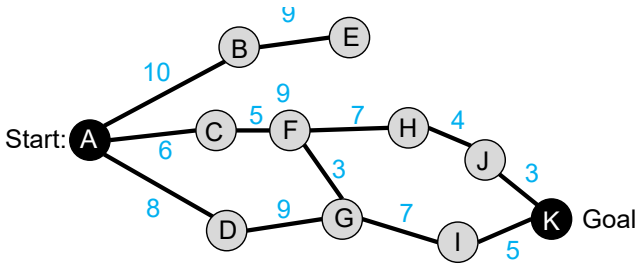
TRAP

Using a queue instead of a stack (that is BFS). Marking visited when *pushed* in DFS changes the order. Saying DFS is complete: on an infinite or very deep graph it can dive for ever.

Weighted graphs [p. 55]

Each edge has a numerical value, the weight (a cost, a distance in miles, a time). The **path cost** is the sum of the edge weights along the path, not the number of edges. BFS and DFS ignore weights; Dijkstra and A* use them. All weights must be ≥ 0 for Dijkstra and A*.

Dijkstra's algorithm [p. 56–58]



[W12 lec p. 57] — weighted slide graph (start A, goal K); the slide's table (rows Past cost, Parent node; columns A–K) is blank and is filled in below. The blue 9 above F has no edge: a stray label

Relaxation (Dijkstra's one idea) [p. 56]

$$\text{if } g(n) + d(n,m) < g(m) \text{ then } g(m) \leftarrow g(n) + d(n,m), \quad \text{parent}(m) \leftarrow n$$

SYMBOLS

$g(n)$	"past cost": cheapest cost found so far from the start to n	cost units
$d(n,m)$	weight of the edge between neighbours n and m	cost units
n	node being expanded (smallest g among unexpanded nodes)	–
m	a neighbour of n not yet expanded	–

Use: search outwards from the start in order of cost. Unreached nodes have $g = \infty$. Dijkstra does not know where the goal is until it reaches it [p. 58].

STEPS Dijkstra by hand

1. Table with one column per node and rows "past cost" and "parent". g(start) = 0, all others ∞.
2. Pick the unexpanded node n with the smallest g (tie: alphabetical); mark it expanded.
3. Relax every non-expanded neighbour m with the box above (strict <). Never change an expanded node.
4. **Stop when the goal is expanded**, not when it is first discovered. Read the path from the parents.
5. **Check:** g(goal) equals the sum of the edges on the path; expansions come out in non-decreasing g.

WORKED SOLVED SAMPLE 3: Dijkstra on the slide graph, A to K, alphabetical

Each row is the table *after* expanding the node in the second column (its g in brackets). Cell = g/parent; **bold** = expanded (final); underlined = changed in this step; – = not reached yet (∞).

#	expand	A	B	C	D	E	F	G	H	I	J	K
1	A (0)	**0**	10/A	6/A	8/A	–	–	–	–	–	–	–
2	C (6)	**0**	10/A	**6/A**	8/A	–	11/C	–	–	–	–	–
3	D (8)	**0**	10/A	**6/A**	**8/A**	–	11/C	17/D	–	–	–	–
4	B (10)	**0**	**10/A**	**6/A**	**8/A**	19/B	11/C	17/D	–	–	–	–
5	F (11)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	14/F	18/F	–	–	–
6	G (14)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	**14/F**	18/F	21/G	–	–
7	H (18)	**0**	**10/A**	**6/A**	**8/A**	19/B	**11/C**	**14/F**	**18/F**	21/G	22/H	–
8	E (19)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	21/G	22/H	–
9	I (21)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	22/H	26/I
10	J (22)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	**22/H**	25/J
11	K (25)	**0**	**10/A**	**6/A**	**8/A**	**19/B**	**11/C**	**14/F**	**18/F**	**21/G**	**22/H**	**25/J**

The slide's table, filled in (final values):

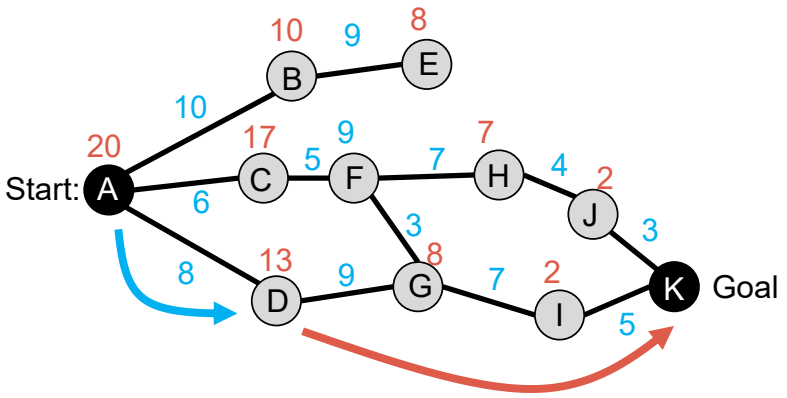
	A	B	C	D	E	F	G	H	I	J	K
Past cost g	0	10	6	8	19	11	14	18	21	22	25
Parent node	–	A	A	A	B	C	F	F	G	H	J

Expansion order A (0), C (6), D (8), B (10), F (11), G (14), H (18), E (19), I (21), J (22), K (25): 10 nodes before K, 11 in all. G first gets 17 (via D) then 14 (via F). K is first *discovered* at 26 (via I, step 9) and improves to 25 (via J, step 10), so stopping at discovery would be wrong. Path A–C–F–H–J–K, cost 6 + 5 + 7 + 4 + 3 = **25**. [checked]

TRAP

Stopping when the goal is first discovered (gives 26); relaxing an already expanded node; negative edge weights; confusing the *expanded* order with the *path*.

A* algorithm [p. 59–61]



[W12 lec p. 59] — A* slide: blue = cost to arrive g (edge weights), red = optimistic cost to goal h. There is no red label at F or K

A* priority [p. 59]

$$f(n) = g(n) + h(n)$$

SYMBOLS

$g(n)$	cost to arrive at node n (cheapest found so far, start to n)	cost units
$h(n)$	optimistic cost from n to the goal (heuristic; the slide uses straight-line style estimates)	cost units
$f(n)$	estimated total cost of the cheapest path through n	cost units

Use: Dijkstra with the priority changed from g to $g + h$: the search is pulled towards the goal. Stop when the goal is expanded.

Admissible and consistent heuristics [p. 61]

admissible: $h(n) \le d(n, \text{goal})\ \ \forall n,$ consistent: $h(n) \le d(n, m) + h(m)\ \ \forall$ adjacent n, m

SYMBOLS

$d(n, \text{goal})$	true cheapest cost from n to the goal	cost units
$d(n, m)$	cost of the edge between adjacent nodes n, m	cost units

Use: admissible = never overestimates (non-strict $\le$, and $h(goal) = 0$): keeps A optimal. Consistent = triangle inequality along every edge, in both directions: also makes the popped f non-decreasing. Consistent $\Rightarrow$ admissible, not the reverse.*

STEPS A* by hand

1. Write h beside every node. Check admissible (compare with the true remaining cost) and consistent (edge by edge, both directions).
2. Pop the open node with the smallest $f = g + h$ (tie: smaller h, then alphabetical).
3. For each neighbour m: if $g(n)+d(n,m) < g(m)$ set $g(m)$ and parent; then $f(m) = g(m)+h(m)$ uses the *child's* h.
4. Stop when the goal is popped. Path from the parents.
5. **Check:** $f(\text{goal}) = g(\text{goal})$ equals the path cost; nodes never expanded are A*'s saving over Dijkstra.

SLIDE CHECK pp. 56–60

(1) The A* slide prints red h for A 20, B 10, C 17, D 13, E 8, G 8, H 7, I 2, J 2 only: **F and K have none.** A blue 9 sits above F (also on the Dijkstra slides, where there is no h), so it is probably a leftover. We use $h(\text{K}) = 0$ (the only valid value) and $h(\textbf{F}) = 9$ (the number printed beside F); any integer h(F) from 0 to 13 gives the same expansion order and cost, checked. (2) The tables on pp. 57 and 60 are blank (the p. 60 table stops at column F); the fills here are ours. (3) The slide asks for h admissible *and* consistent, but its own labels are not consistent (below).

WORKED SOLVED SAMPLE 4a: A* on the slide graph, A to K, h from p. 59

Tie-break: lowest f, then lowest h, then alphabetical (no ties actually occur). Open list written as node g+h=f (parent), sorted by f.

#	expand	g	h	f	open list after expanding
1	A	0	20	20	B 10+10=20 (A); D 8+13=21 (A); C 6+17=23 (A)
2	B	10	10	20	D 8+13=21 (A); C 6+17=23 (A); E 19+8=27 (B)
3	D	8	13	21	C 6+17=23 (A); G 17+8=25 (D); E 19+8=27 (B)
4	C	6	17	23	F 11+9=20 (C); G 17+8=25 (D); E 19+8=27 (B)
5	F	11	9	20	G 14+8=22 (F); H 18+7=25 (F); E 19+8=27 (B)
6	G	14	8	22	I 21+2=23 (G); H 18+7=25 (F); E 19+8=27 (B)
7	I	21	2	23	H 18+7=25 (F); K 26+0=26 (I); E 19+8=27 (B)
8	H	18	7	25	J 22+2=24 (H); K 26+0=26 (I); E 19+8=27 (B)
9	J	22	2	24	K 25+0=25 (J); E 19+8=27 (B)
10	K	25	0	25	E 19+8=27 (B)

The slide's table (rows Past cost, Optimistic cost to goal, Estimated total cost, Parent node), all nodes:

	A	B	C	D	E	F	G	H	I	J	K
$g(n)$	0	10	6	8	19	11	14	18	21	22	25
$h(n)$	20	10	17	13	8	9	8	7	2	2	0
$f = g + h$	20	20	23	21	27	20	22	25	23	24	25
parent	–	A	A	A	B	C	F	F	G	H	J

Order A B D C F G I H J K (**10** expansions; Dijkstra 11). E was discovered ($f = 27$) but never expanded. Path A–C–F–H–J–K, cost **25** = f(K). Popped f: 20, 20, 21, 23, 20, 22, 23, 25, 24, 25: it *drops* twice, the signature of an inconsistent h. [checked]

WORKED SOLVED SAMPLE 4b: admissibility and consistency checks, h from p. 59 ($h(\mathrm{F}) = 9$, $h(\mathrm{K}) = 0$)

True cost-to-go $d(n, \mathrm{K})$ from a backwards Dijkstra from K.

node	$h(n)$	$d(n, \mathrm{K})$	$d - h$	$h \le d$?
A	20	25	5	yes
B	10	35	25	yes
C	17	19	2	yes
D	13	21	8	yes
E	8	44	36	yes
F	9	14	5	yes
G	8	12	4	yes
H	7	7	0	yes
I	2	5	3	yes
J	2	3	1	yes
K	0	0	0	yes

Every $h \le d$: **admissible**, tight at H (7) and K (0). Consistency, each edge both ways, $h(n) \le w + h(m)$:

edge (w)	first direction	second direction
A–B (10)	A→B: 20≤10+10=20	B→A: 10≤10+20=30
A–C (6)	A→C: 20≤6+17=23	C→A: 17≤6+20=26
A–D (8)	A→D: 20≤8+13=21	D→A: 13≤8+20=28
B–E (9)	B→E: 10≤9+8=17	E→B: 8≤9+10=19
C–F (5)	C→F: **17>5+9=14 FAIL**	F→C: 9≤5+17=22
D–G (9)	D→G: 13≤9+8=17	G→D: 8≤9+13=22
F–G (3)	F→G: 9≤3+8=11	G→F: 8≤3+9=12
F–H (7)	F→H: 9≤7+7=14	H→F: 7≤7+9=16
G–I (7)	G→I: 8≤7+2=9	I→G: 2≤7+8=15
H–J (4)	H→J: **7>4+2=6 FAIL**	J→H: 2≤4+7=11
I–K (5)	I→K: 2≤5+0=5	K→I: 0≤5+2=7
J–K (3)	J→K: 2≤3+0=3	K→J: 0≤3+2=5

Two failures: C→F ($17 > 5 + 9$) and H→J ($7 > 4 + 2$). So h is admissible but **not consistent**. (No value of h(F) repairs it: C needs $h(\mathrm{F}) \ge 12$, G needs $h(\mathrm{F}) \le 11$.) A* still found the optimum because h is admissible. [checked]

TRAP

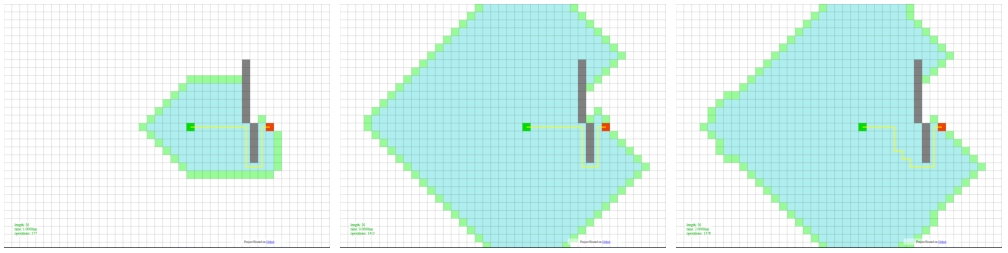
f built with the *parent's* h; reporting 11 expansions for A*; calling h consistent after testing one direction only; saying every heuristic speeds A* up ($h = 0$ is Dijkstra); "admissible means $h < d$" (it is $\le$).

Heuristics and the comparison slide [p. 61–62]

A heuristic must be [p. 61] fast to compute, reasonably close to the real cost, admissible and consistent. Defined from knowledge of the workspace: *Euclidean* distance; *Manhattan* distance $|\Delta x| + |\Delta y|$; *zero* ($h = 0$ gives Dijkstra).

Grid	Admissible and consistent heuristic
4-connected, unit cost	Manhattan (exact when no walls); Euclidean also OK (smaller)
8-connected, diagonal $\sqrt{2}$	Euclidean. **Not** Manhattan: it says 2 for one diagonal step that costs 1.414
Any graph	zero (safe but uninformed)

Check, 8-connected (practice Q12.4): offset $(7, 5)$: 5 diagonal + 2 straight = $5\sqrt{2} + 2 = 9.071$; Euclidean $8.602 \le 9.071$ fine; Manhattan $12 > 9.071$ **overestimates**. [checked]



A* (277 operations) BFS (1413 operations) Dijkstra's (1378 operations)

[W12 lec p. 62] — same grid with a wall: A* 277 operations, BFS 1413, Dijkstra 1378 (qiao.github.io/PathFinding.js). BFS and Dijkstra expand large diamonds; A* a thin region towards the goal. Not a general ratio

Overestimating breaks A*. On the slide graph, $h(\mathrm{F}) = 30$ makes A* skip F: order A B D C G I E K, path A–D–G–I–K, cost 29 instead of 25. A perfect $h = d$ expands only A, C, F, H, J, K. [checked]

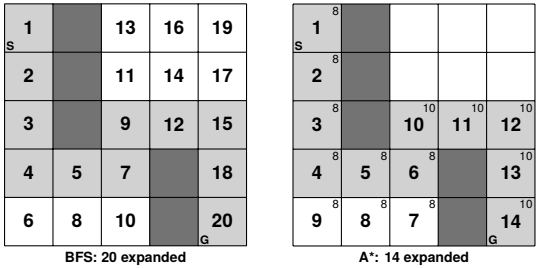
WORKED "What changes if all edge costs are equal?" [added]

Every edge cost 1 on the slide graph: Dijkstra's g is the edge count, so it expands in BFS order A, B, C, D, E, F, G, H, I, J, K and returns A–D–G–I–K, cost 4 (the optimum). So **BFS becomes cost-optimal and equals Dijkstra**; A* still helps only through h. Unequal costs: BFS cost 29, optimum 25. [checked]

Solved sample 5: BFS and A* on a 5×5 grid [added]

PREMISE grid search **Given:** 5×5 grid, black cells = obstacles, 4-connected, every move costs 1, start S = (0, 0) (top-left), goal G = (4, 4); cells are (row, col)

Find: expansion order and shortest path for BFS and for A* with h = Manhattan distance to G



[drawn for this booklet] numbers = expansion order; small number in A* = f; shaded = the 10-move path (dark cells = obstacles). Neighbour order up, right, down, left; A* ties: lowest f, lowest h, first discovered

WORKED both searches, step by step

The wall in column 1 (rows 0–2) forces a detour; the second wall (column 3, rows 3–4) blocks the bottom route, so the path must go up to row 2. $h(\text{S}) = |4-0| + |4-0| = 8$, true cost 10.

BFS (20 of the 20 free cells expanded: it reaches G last). Pops: $\text{S}(0,0) \to$ queue $[(1,0)]$; $(1,0) \to [(2,0)]$; $(2,0) \to [(3,0)]$; $(3,0) \to [(3,1),(4,0)]$; and so on level by level. Order: (0,0), (1,0), (2,0), (3,0), (3,1), (4,0), (3,2), (4,1), (2,2), (4,2), (1,2), (2,3), (0,2), (1,3), (2,4), (0,3), (1,4), (3,4), (0,4), (4,4).

A* (14 expansions). Table of g, h, f at each expansion:

#	cell	g	h	f
1	(0,0)	0	8	8
2	(1,0)	1	7	8
3	(2,0)	2	6	8
4	(3,0)	3	5	8
5	(3,1)	4	4	8
6	(3,2)	5	3	8
7	(4,2)	6	2	8
8	(4,1)	5	3	8
9	(4,0)	4	4	8
10	(2,2)	6	4	10
11	(2,3)	7	3	10
12	(2,4)	8	2	10
13	(3,4)	9	1	10
14	(4,4)	10	0	10

f stays 8 while A* walks down column 0 and along row 3, and it *rises* to 10 once the dead end at the bottom is exhausted (cells $(4,0),(4,1),(4,2)$ are explored first because they look closer to G); after that A* runs straight to G. Cells discovered but never expanded: (1,2), (1,3), (1,4). Path (both): (0,0) → (1,0) → (2,0) → (3,0) → (3,1) → (3,2) → (2,2) → (2,3) → (2,4) → (3,4) → (4,4), **10 moves**. Both are optimal; A* expanded 14 cells against 20. Popped f never decreases because Manhattan is consistent here. [checked]

Dimensionality: why grids fail [p. 63–65]

Size of a grid in d dimensions [p. 65]

$$|V| = N^d, \qquad \text{neighbours} = 2d \text{ (axis moves) or } 3^d - 1 \text{ (with diagonals)}, \qquad t_{BFS} = O(|V| + |E|)$$

SYMBOLS

N	cells per dimension (resolution)	–
d	number of dimensions of the C-space (joints for an arm)	–
$\lvert V\rvert, \lvert E\rvert$	number of nodes, edges of the grid graph	–
$m \times n$	4-connected grid: $\lvert E\rvert = m(n-1) + n(m-1)$	–

Use: show that a grid is hopeless for a many-joint arm: the node count grows exponentially with d.

Grid [p. 65]	Nodes	Neighbours	Note
2D, 100^2	10^4	4 or 8	
3D, 100^3	10^6	6 to 26	
6D, 100^6	10^{12}	12 or 728	slide leaves "Neighbours?" open; $2d = 12$, $3^6 - 1 = 728$ [added]

Example: 4-connected 100×100: $|E| = 2 \cdot 100 \cdot 99 = 19\,800$, so $|V| + |E| = 29\,800$. A 40×25 grid has $40 \cdot 24 + 25 \cdot 39 = 1935$ edges. A 5D grid with 50 cells per dimension has $312\,500\,000$ nodes. [checked]

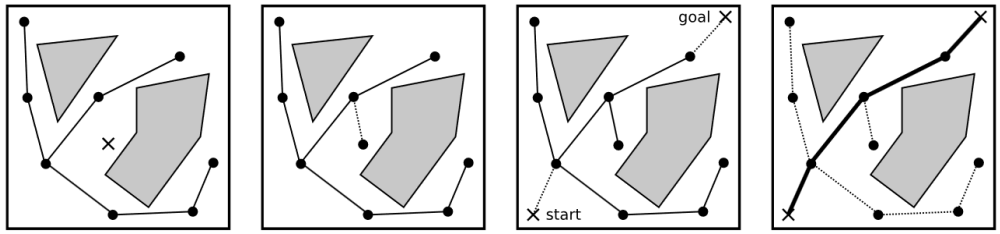
Slide [p. 64]: resolution is a trade-off (fine enough to keep narrow solutions, coarse enough to be fast); connectivity is limited by the grid shape; and grids suffer the curse of dimensionality. So we *sample*.

Sampling-based planning: framework [p. 66–69]

Idea [p. 66]: pick random points, connect nearby ones; there is no explicit split of C into C_{free} and C_{obs}; a *collision detector* only answers "is this point in C_{free}?". The two most influential methods: PRM (1996) and RRT (1998) [p. 68]. The framework [p. 69], with what each step needs:

1. $G = (V, E)$ starts empty.
2. Sample a point and add it to V (needs a **sampler**).
3. Choose nodes of V that are "near" the new vertex v and try to connect them with a **local planner** (needs a **distance function**).
4. If the local path is collision-free, add it to E (needs a **collision detector**).
5. Repeat from 2 until satisfied.
6. **Query**: find a connection from start to goal.

Probabilistic roadmap (PRM) [p. 70–72]



(a) The *learning* phase: a random sample, denoted by ×, is generated

(b) A local planner is used to connect the new sample to nearby roadmap vertices.

(c) The *query* phase: the start and goal configurations are added to the roadmap.

(d) A graph search algorithm is used to connect the start and goal through the roadmap.

[W12 lec p. 70] — PRM: (a) learning phase, a random sample ×; (b) a local planner connects it to nearby roadmap vertices; (c) query phase: start and goal are added; (d) graph search (e.g. A*) through the roadmap (thick)

STEPS PRM

1. **Learning phase.** Repeat: sample **q** uniformly in C; discard it if it is in C_{obs} (point-in-polygon test); add it to V.
2. For each new vertex, take the roadmap vertices within distance r (or the k nearest); add an edge if the straight segment is collision-free (segment-polygon test).
3. **Query phase.** Add start and goal (connect them the same way); run A* or Dijkstra on the roadmap; read the path.
4. **Check:** the path is valid but rarely the shortest; if start and goal are in different components, sample more.

Strengths: simple, works in high dimensions (snake robots). Limits [p. 72]: assumes holonomic motion; can fail in narrow passages; unsuited to dynamic environments (the roadmap is made first, then searched).

Probability of hitting the goal region [added]

$$p = \frac{A_{goal}}{A_C}, \qquad E[N] = \frac{1}{p}, \qquad P(\text{at least one hit in } N) = 1 - (1-p)^N$$

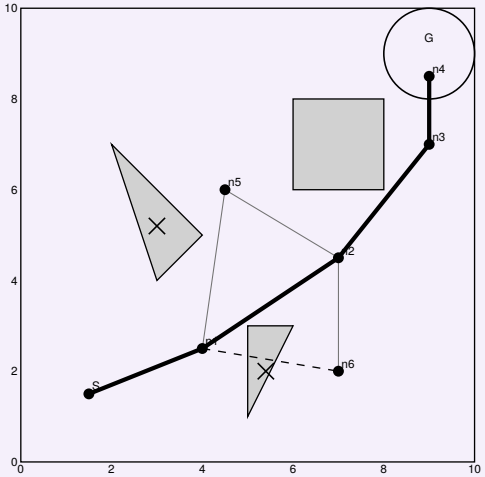
SYMBOLS

A_{goal}	area of the goal region (practical map: disc of radius 1, π)	m^2
A_C	area of the C-space sampled uniformly ($10 \times 10 = 100$)	m^2
N	number of independent uniform samples	–

Use: "how many samples before the goal is hit?" Practical map: $p = \pi/100 = 0.0314$, $E[N] = 31.8$ samples. As $N \to \infty$ the probability $\to 1$: this is what "probabilistically complete" means.

WORKED SOLVED SAMPLE 6a: one PRM build and query on the practical map, $r = 4$

Map $C = [0, 10]^2$ in the frame $\{0\}$ at the lower-left corner. Obstacles: T1 $(2, 7), (4, 5), (3, 4)$; rectangle R $(6, 6), (6, 8), (8, 8), (8, 6)$; T2 $(5, 3), (6, 3), (5, 1)$. $S = (1.5, 1.5)$, goal disc centre $(9, 9)$, radius 1.



[drawn for this booklet] map of the practical sheet; roadmap edges grey, query path thick, rejected edge dashed; × = rejected samples

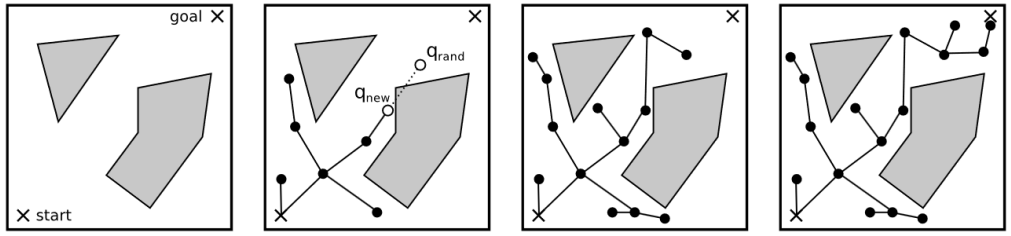
Samples. Hand-picked free nodes S, n1 $(4, 2.5)$, n2 $(7, 4.5)$, n3 $(9, 7)$, n4 $(9, 8.5)$, n5 $(4.5, 6)$, n6 $(7, 2)$; n4 is inside the goal disc (distance 0.5). Rejected samples: $(3, 5.2)$ lies inside T1 (ray to $+x$ crosses 1 edge = odd = inside); $(5.4, 2)$ inside T2; $(7, 7)$ inside R.

Local planner. Pairs within $r = 4$ whose segment misses every obstacle edge:

edge	length	segment test
S–n1	2.693	free
n1–n2	3.606	free
n1–n5	3.536	free
n2–n3	3.202	free
n2–n5	2.915	free
n2–n6	2.500	free
n3–n4	1.500	free
n1–n6	3.041	**rejected**: crosses T2

S to n2 (6.265) and n3 (9.301) are too far; n5–n3 (4.610) too far, and would cross R anyway; the direct S→G segment crosses R. **Query** (Dijkstra on the roadmap, order and g): S (0.000), n1 (2.693), n5 (6.228), n2 (6.298), n6 (8.798), n3 (9.500), n4 (11.000). Path **S–n1–n2–n3–n4**, legs $2.693 + 3.606 + 3.202 + 1.500 = \mathbf{11.00}$. The straight S–n2 would be shorter than S–n1–n2 but is farther than r, so it was never tried: that is why PRM paths are rarely shortest (11.00 vs the visibility optimum 10.068, 9.3 % longer). [checked]

Rapidly-exploring random tree (RRT) [p. 73–76]



(a) A tree is grown from the start configuration towards the goal.

(b) The planner generates a configuration q_{rand}, and grows from the nearest node towards it to create q_{new}.

(c) The tree rapidly explores the free space.

(d) The planner terminates when a node is close to the goal node. Common implementations will connect directly to the goal.

[W12 lec p. 74] — RRT: (a) tree grown from the start towards the goal; (b) a random $\mathbf{q}_{rand}$ is generated and the tree grows from the nearest node towards it, creating $\mathbf{q}_{new}$; (c) the tree rapidly explores free space; (d) stops when a node is near the goal; implementations often then connect directly to the goal

RRT extension step [p. 74]

$$\mathbf{q}_{new} = \mathbf{q}_{near} + \delta \frac{\mathbf{q}_{rand} - \mathbf{q}_{near}}{\|\mathbf{q}_{rand} - \mathbf{q}_{near}\|} \quad (\text{if } \|\mathbf{q}_{rand} - \mathbf{q}_{near}\| \le \delta : \ \mathbf{q}_{new} = \mathbf{q}_{rand})$$

SYMBOLS

$\mathbf{q}_{rand}$	random sample in C (with probability p_{goal} use the goal itself: *goal bias*, [added])	m
$\mathbf{q}_{near}$	node of the tree closest (Euclidean) to $\mathbf{q}_{rand}$	m
δ	step size (extension length)	m
$\mathbf{q}_{new}$	new node, added with parent $\mathbf{q}_{near}$ if the segment $\mathbf{q}_{near}\mathbf{q}_{new}$ is collision-free	m

Use: one RRT iteration. All points in the map frame $\{0\}$. Terminate when a node falls in the goal region.

STEPS RRT

1. Tree = $\{\mathbf{q}_{start}\}$.
2. Sample $\mathbf{q}_{rand}$ (goal bias: with probability p_{goal}, e.g. 5–10 %, take $\mathbf{q}_{goal}$).
3. Find $\mathbf{q}_{near}$ by Euclidean distance; compute $\mathbf{q}_{new}$ with the box above.
4. Keep $\mathbf{q}_{new}$ only if it and the segment are collision-free; else discard.
5. Stop when a node is in the goal region; read the path from the parents. **Check:** every edge $\le \delta$; no edge crosses an obstacle.

WORKED SOLVED SAMPLE 6b: RRT steps on the practical map, $\delta = 1$

Step 1. Tree $\{S\}$, $S = (1.5, 1.5)$, $\mathbf{q}_{rand} = (6, 2)$. Nearest node: S (only one). $\|\cdot\| = \sqrt{4.5^2 + 0.5^2} = 4.528$; unit vector $(0.9939, 0.1104)$; $\mathbf{q}_{new} = (1.5 + 0.994,\ 1.5 + 0.110) = (\mathbf{2.494}, \mathbf{1.610})$; free. [checked]

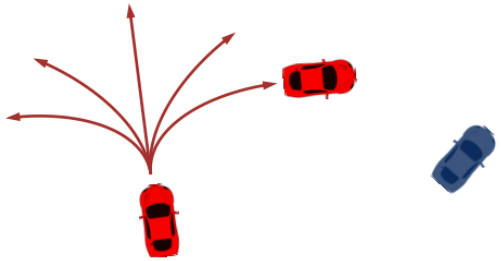
Step 2. Tree $\{S, \mathbf{n}_1 = (2.494, 1.610)\}$, $\mathbf{q}_{rand} = (4, 5)$. Distances: to S $\sqrt{2.5^2 + 3.5^2} = 4.301$; to $\mathbf{n}_1$ $\sqrt{1.506^2 + 3.390^2} = 3.709$. Nearest is $\mathbf{n}_1$; unit vector $(0.4061, 0.9138)$; $\mathbf{q}_{new} = (\mathbf{2.900}, \mathbf{2.524})$; segment and point free. [checked]

A rejected step. From $(4.5, 2.5)$ towards $(6.5, 2)$: $\mathbf{q}_{new} = (5.470, 2.257)$ is inside T2, so it is discarded. **Goal bias:** $\mathbf{q}_{rand} = G = (9, 9)$ from the tree $\{S, \mathbf{n}_1, \mathbf{n}_2\}$: nearest is $(2.900, 2.524)$, distance 8.896, $\mathbf{q}_{new} = (3.586, 3.252)$. [checked]

TRAP

"RRT finds the shortest path" (no); "PRM stops when it reaches the goal" (that is RRT; PRM builds the roadmap first); forgetting to test the *segment*, not just the endpoint.

RRT for non-holonomic robots [p. 77–79]



[W12 lec p. 78] — sample controls to generate feasible local paths and keep the end configuration closest to the target

Slide [p. 77]: drawing straight lines in C assumes the point robot can move in any direction (holonomic), but a car or diff-drive cannot slide sideways. **Fix** [p. 78]: instead of joining to a specific sample, *sample controls* (speed, steering), simulate the robot's kinematic model for a short time to get feasible local paths, and add the end configuration that gets *closest to the target*.

WORKED one non-holonomic extension [added]

Unicycle (diff-drive, W10) from pose $(x, y, \theta) = (0, 0, 0)$ in $\{0\}$, speed $v = 1$ m/s, duration $T = 1$ s, steering rates $\omega \in \{-1, 0, +1\}$ rad/s. Constant v, ω: $x = \frac{v}{\omega}\sin\omega T$, $y = \frac{v}{\omega}(1-\cos\omega T)$, $\theta = \omega T$ (straight line for $\omega = 0$). Target $\mathbf{q}_{rand} = (2, 1)$:

ω (rad/s)	x (m)	y (m)	θ	distance to $(2,1)$ (m)
-1	0.841	-0.460	-57.3°	1.864
+0	1.000	+0.000	+0.0°	1.414
+1	0.841	+0.460	+57.3°	1.278

Keep the $\omega = +1$ end pose $(0.841, 0.460, 57.3°)$, the closest (1.278 m). Every candidate obeys $\dot{y}\cos\theta - \dot{x}\sin\theta = 0$ (no sideways velocity), which straight-line RRT cannot guarantee. [checked]

Completeness; choosing a planner [p. 80–83]

Complete [p. 80]	finds a collision-free path whenever one exists, otherwise reports failure (visibility graph, Voronoi, exact cells)
Resolution complete	finds a path whenever one exists *at the chosen resolution* (BFS on a grid, approximate cells); a gap narrower than a cell is invisible
Probabilistically complete	P(path found) $\to 1$ as samples $\to \infty$; may never terminate if no path exists (PRM, RRT) [p. 81]

DFS is not complete on infinite trees. Slide [p. 81] says "probability of samples approaches 1": it means the probability of *finding a path*.

Planner	Complete?	Optimal?	Data structure	Needs / use when
BFS	resolution complete (finite graph)	fewest edges; cost-optimal only if equal costs	queue	graph or grid, equal moves
DFS	not on infinite/deep graphs	no	stack	any path will do; low memory
Dijkstra	resolution complete	cheapest, costs ≥ 0	priority queue on g	edge costs, no goal estimate
A*	resolution complete	cheapest if h admissible	priority queue on f	costs plus a good distance-to-goal h; 2D grids
Visibility graph	complete	shortest	graph of corners	2D polygon map (inflate first)
Voronoi	complete	max clearance, not shortest	curves	range-sensor following
PRM	probabilistic	no	roadmap graph	static map, many queries, high DoF
RRT	probabilistic	no	tree	single query, differential constraints

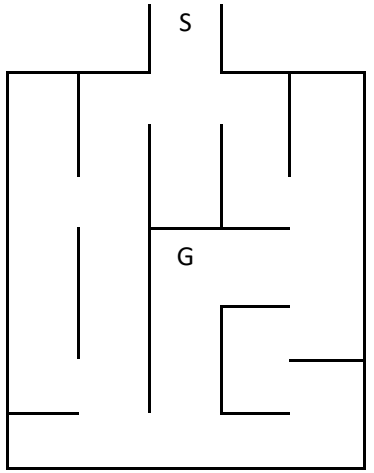
STEPS "Which planner would you use?" scenario [added]

1. Count the C-space dimension d. Small ($d \leq 3$) and known map: graph search on a grid or visibility graph. Large: grids need N^d nodes, so sample.
2. Does the robot have differential constraints (car, diff-drive)? Then RRT with sampled controls.
3. Static map and many queries: PRM. One query or a changing map: RRT.
4. Costs and a good distance guess: A*; costs only: Dijkstra; equal costs: BFS.
5. Name the method *and* the property that justifies it (complete / probabilistically complete / optimal given admissible h).

6-DoF arm: 100 cells per joint is $100^6 = 10^{12}$ nodes (slide [p. 65]), and C_{obs} is curved and hard to build: use PRM (repeated queries, static workspace) or RRT, only a collision checker is needed; guarantee: probabilistically complete, path not shortest. **2D grid robot:** N^2 nodes is small, so A* with Manhattan (4-connected) or Euclidean (8-connected) h: complete at the resolution and optimal because h is admissible.

The W12 practical sheet, solved

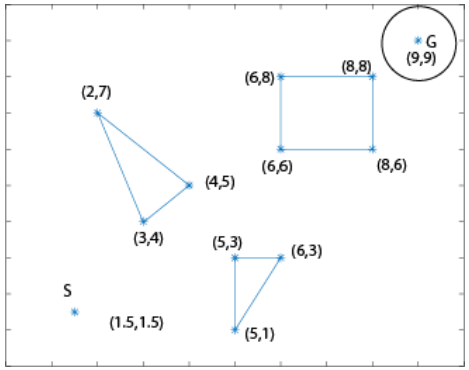
Sheet title says "mobile robot kinematics"; the content is planning. **Q1** map (no scale given; our grid reading is 5 columns wide with an entrance cell for S):



[W12 prac p. 1] — practical Q1: maze with start S (top) and goal G

Q1a. *Represent the map as a graph.* Each open cell is a node, each opening between two neighbouring cells is an undirected edge (weight 1 per move). On our reading: 36 open cells, 38 edges; BFS 8 moves (26 cells popped); DFS 16 moves (19 popped); A* 8 moves (16 popped) (the numbers depend on how the figure is gridded). **Q1b.** BFS finds the shortest route (fewest moves); DFS (order up, right, down, left) finds a route but a longer one. **Q1c. Observe:** BFS explores outward in rings and guarantees the fewest moves; DFS can wander a long way (here twice as long) and its answer depends on the neighbour order; the goal is only 4 cells away in Manhattan terms but a wall forces an 8-move detour. **Q1d. Upgrade to A*:** keep the edge weights (cost per move, or Euclidean distance between cell centres) as g; add a heuristic $h(n)$ = Manhattan (or Euclidean) distance from n to G, admissible because walls only lengthen paths; replace the queue by a priority queue on $f = g + h$. A* then expands fewer cells for the same 8-move path.

Q2. $C = [0, 10]^2$, $S = (1.5, 1.5)$, goal disc radius 1 at $(9, 9)$; obstacles on p. 148.



[W12 prac p. 1] — practical Q2 map (obstacle vertices as printed)

(a) **Graph**: vertex list (name → coordinates) and a dictionary of weighted edges (length of the segment); add S first.

(b) **Sampler**: uniform $x, y \in [0, 10]$. Hit chance in the goal $\pi/100 = 0.0314$: about 32 samples on average.

(c) **Collision of a point**: ray to $+x$, count edge crossings; odd means inside. $(3, 5.2)$ in T1: one crossing, inside.

(d) **Nearest vertex**: Euclidean. For $(6, 4)$: n2 $(7, 4.5)$ at 1.118, n1 at 2.500.

(e) **Edge test**: segment ab meets edge cd if c, d lie on opposite sides of line ab and a, b on opposite sides of line cd (orientation signs); test all edges of all polygons. S→G meets R; n1→n6 meets T2.

(f) **Goal check**: $\|\mathbf{q} - (9, 9)\| \le 1$ after every added vertex (n4: 0.5).

(g) **Path**: by parents, S–n1–n2–n3–n4, length 11.00 (p. 148).

Q3 (bonus) RRT on the same map: use the RRT recipe; the first steps are in solved sample 6b (step $\delta = 1$; step 1 gives $(2.494, 1.610)$).

How this is examined: worked solutions

EXAM-STYLE QUESTION (W12) Graph search [marks not published; indicative 8]

Undirected edges with costs: S–A 2, S–B 5, A–C 4, A–D 7, B–D 3, C–G 6, D–G 2. Heuristic h: S 7, A 6, B 4, C 5, D 2, G 0. (a) Run A* from S to G with the table (expand, g, h, f, open list) and give the path and cost. (b) How many nodes does Dijkstra expand? (c) Is h admissible? Consistent?

Solution

Tie-break: lowest f, then alphabetical.

#	expand	g	h	f	open after
1	S	0	7	7	A 2+6=8 (S); B 5+4=9 (S)
2	A	2	6	8	B 5+4=9 (S); D 9+2=11 (A); C 6+5=11 (A)
3	B	5	4	9	D 8+2=10 (B); C 6+5=11 (A)
4	D	8	2	10	G 10+0=10 (D); C 6+5=11 (A)
5	G	10	0	10	C 6+5=11 (A)

(a) Order S A B D G; D improves from $g = 9$ (via A) to 8 (via B). Path **S–B–D–G**, cost $5 + 3 + 2 = \boxed{10}$, $f(G) = g(G) = 10$. [4]

(b) Dijkstra expands S (0), A (2), B (5), C (6), D (8), G (10): 6 nodes (A* 5); it also expands C. [1]

(c) True costs to G: S 10, A 9, B 5, C 6, D 2, G 0. Every $h \le d$: admissible. Every edge satisfies $h(n) \le d(n, m) + h(m)$ in both directions: consistent. [3] [checked]

EXAM-STYLE QUESTION (W12) Spot the errors [indicative 4]

Graph S–A 2, S–B 3, A–G 9, B–C 2, C–G 4. A student runs Dijkstra from S to G: "Expanding S discovers A (2) and B (3). Expanding A discovers G at 11, so I stop: the shortest path is S–A–G, cost 11. BFS would also be optimal, and A* is always optimal." Find the errors and the true cost.

Solution

Error 1: stopping at discovery. After A, the open list is B 3, G 11; B is expanded next and discovers C (5); C improves G to $5 + 4 = 9$. G is expanded last with $g = 9$: path S–B–C–G, cost $\boxed{9}$ (order S, A, B, C, G). **Error 2**: BFS returns the fewest edges, S–A–G (2 edges, cost 11), not the cheapest. **Error 3**: A* is optimal only if h is admissible (and consistent for a simple closed set). [checked]

EXAM-STYLE QUESTION (W12) C-space and corridor [indicative 5]

(a) A disc robot of radius 0.5 m must pass a corridor 0.9 m wide. Show in configuration space that it cannot, and give the largest radius that fits. (b) A 2R arm, $L_1 = L_2 = 1$ m, must avoid a point obstacle at $(1, 1)$. Give the C-space image of the obstacle. (c) The obstacle grows by r to a triangle with $A = 1$ m^2, $P = 5.236$ m, $r = 0.5$ m: find the grown area.

Solution

(a) Shrink the robot to a point and grow each wall by r: free width $0.9 - 2(0.5) = -0.1$ m < 0, no free configuration across. Need $0.9 - 2r \ge 0$: $r \le 0.45$ m. [2]

(b) $\cos\theta_2 = (1 + 1 - 2)/2 = 0$, so $\theta_2 = \pm 90°$; $\theta_1 = 45° - \text{atan2}(\pm 1, 1)$ gives $0°$ and $90°$: the points $(0°, 90°)$ and $(90°, -90°)$ (two points, elbow up and down). [2]

(c) $A_{grown} = A + Pr + \pi r^2 = 1 + 2.618 + 0.785 = \boxed{4.403 \text{ m}^2}$. [1] [checked]

EXAM-STYLE QUESTION (W12) Grid and heuristic [indicative 5]

(a) A 5-dimensional C-space is gridded with 50 cells per dimension: how many nodes, and how many neighbours (axis only; with diagonals)? (b) How many edges has a 4-connected 40×25 grid? (c) On an 8-connected grid (straight 1, diagonal $\sqrt{2}$) the goal is 7 rows and 5 columns away on an empty grid. Is Manhattan distance admissible? Give a better h.

Solution

(a) $50^5 = 312\,500\,000$ nodes; $2d = 10$ or $3^5 - 1 = 242$ neighbours. [1]

(b) $m(n-1) + n(m-1) = 40(24) + 25(39) = 960 + 975 = \boxed{1935}$. [1]

(c) True cost $= 5\sqrt{2} + 2 = 9.071$. Manhattan $= 12 > 9.071$: overestimates, **not** admissible, so A* may return a non-optimal path. Euclidean $\sqrt{49+25} = 8.602 \leq 9.071$ is admissible (and consistent). [3] [checked]

EXAM-STYLE QUESTION (W12) Sampling-based planners [indicative 6]

(a) A planner draws uniform samples in a 10×10 m map; the goal is a disc of radius 0.8 m. Find p for one sample and the expected number of samples to the first hit. (b) From $\mathbf{q}_{near} = (1.5, 1.5)$ with $\mathbf{q}_{rand} = (6, 2)$ and $\delta = 1$, find $\mathbf{q}_{new}$. (c) Which planner for a 6-DoF arm in a static workshop, and what does its guarantee mean?

Solution

(a) $p = \pi(0.8)^2/100 = 0.0201$; $E[N] = 1/p = 49.7$ samples. [2]

(b) $\|\mathbf{q}_{rand} - \mathbf{q}_{near}\| = \sqrt{4.5^2 + 0.5^2} = 4.528$; $\mathbf{q}_{new} = (1.5, 1.5) + 1 \cdot (4.5, 0.5)/4.528 = \boxed{(2.494,\ 1.610)}$ (then test the segment against the obstacles). [2]

(c) PRM (static map, repeated queries) or RRT (single query): grids would need N^6 nodes (10^{12} at 100 cells). Probabilistically complete: if a path exists the probability of finding it $\to 1$ as the samples grow; there is no termination guarantee if none exists, and the path is not shortest. [2]

[checked]

EXAM-STYLE QUESTION (W12) Explain: BFS or Dijkstra? [indicative 3]

On the slide graph (A to K), BFS returns A–D–G–I–K and Dijkstra A–C–F–H–J–K. Explain the difference, and say when they agree.

Solution

BFS orders the frontier by *number of edges*, so it returns the path with the fewest edges (4): cost $8 + 9 + 7 + 5 = 29$. Dijkstra orders by *accumulated cost* g, so it returns the cheapest path: 5 edges but cost $6 + 5 + 7 + 4 + 3 = 25$. They agree when all edge costs are equal (then g is proportional to the number of edges); with unit costs on the same graph both return A–D–G–I–K, cost 4. [3]

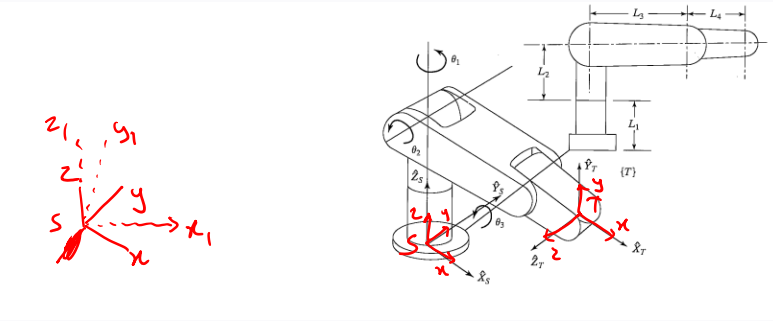
[checked]

The ten questions of the W13 revision lecture, in order, each with the model answer from its week. Marks are not published; indicative only. The week page has the method, traps and variations.

Sample Q1 (W2, p. 33)

EXAM-STYLE QUESTION W13 Q1: FK of an RRR arm by ETS [p. 4–5 of W13] [marks not published; indicative]

Write down the transformation matrix relating the tool frame $\{T\}$ to the station frame $\{S\}$ for the RRR manipulator below. Indicate intermediate frames. (Link data: L_1, L_2 vertical, L_3 upper arm, L_4 forearm; $L = (0.6, 0.5, 0.4, 0.3)$ m for the numbers.)



[W13 lec p. 4] — W13 Q1 arm with the lecturer's frames and ETS

Solution

Frames. $\{S\}$ on the base, z_S up along joint 1. $\{1\}$: $\{S\}$ turned θ_1 about z_S. $\{2\}$: shoulder, height $L_1 + L_2$ above $\{S\}$, with y_2 along the axis of joint 2. $\{3\}$: elbow, L_3 along the upper arm. $\{T\}$: tip, L_4 further; at $\theta = \mathbf{0}$ x_T points along the forearm and y_T up.
General rule: ${}^S\mathbf{T}_T = {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2\,{}^2\mathbf{T}_3\,{}^3\mathbf{T}_T$ (the lecturer's ETS, W13 p.4). The arrows of joints 2 and 3 lift the arm, which is *negative* about $+y$, hence the minus signs:

$$ {}^S\mathbf{T}_T = \mathbf{R}_z(\theta_1)\; t_z(L_1{+}L_2)\; \mathbf{R}_y(-\theta_2)\; t_x(L_3)\; \mathbf{R}_y(-\theta_3)\; t_x(L_4)\; \mathbf{R}_x(90^\circ) $$

Intermediate frames ($c_i = \cos\theta_i$, $s_i = \sin\theta_i$):

$$ {}^S\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^1\mathbf{T}_2 = \begin{bmatrix} c_2 & 0 & -s_2 & 0 \\ 0 & 1 & 0 & 0 \\ s_2 & 0 & c_2 & L_1{+}L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^2\mathbf{T}_3 = \begin{bmatrix} c_3 & 0 & -s_3 & L_3 \\ 0 & 1 & 0 & 0 \\ s_3 & 0 & c_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^3\mathbf{T}_T = \begin{bmatrix} 1 & 0 & 0 & L_4 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

EXAM-STYLE QUESTION W13 Q1 (cont.): multiplication, result, check

Multiply step by step ($c_{23} = \cos(\theta_2 + \theta_3)$, $s_{23} = \sin(\theta_2 + \theta_3)$, using $c_2c_3 - s_2s_3 = c_{23}$, $s_2c_3 + c_2s_3 = s_{23}$):

$$ {}^S\mathbf{T}_2 = {}^S\mathbf{T}_1\,{}^1\mathbf{T}_2 = \begin{bmatrix} c_1c_2 & -s_1 & -c_1s_2 & 0 \\ s_1c_2 & c_1 & -s_1s_2 & 0 \\ s_2 & 0 & c_2 & L_1{+}L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^S\mathbf{T}_3 = {}^S\mathbf{T}_2\,{}^2\mathbf{T}_3 = \begin{bmatrix} c_1c_{23} & -s_1 & -c_1s_{23} & c_1c_2L_3 \\ s_1c_{23} & c_1 & -s_1s_{23} & s_1c_2L_3 \\ s_{23} & 0 & c_{23} & L_1{+}L_2{+}s_2L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Last factor: $\mathbf{R}_x(90^\circ)$ sends column 2 of the rotation to +column 3 and column 3 to −column 2, and $t_x(L_4)$ adds $L_4\times$ column 1 to the position. With $r = L_3c_2 + L_4c_{23}$:

$$ {}^S\mathbf{T}_T = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1\,r \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1\,r \\ s_{23} & c_{23} & 0 & L_1{+}L_2{+}L_3s_2{+}L_4s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Check at $\theta = \mathbf{0}$: tip $(L_3{+}L_4,\ 0,\ L_1{+}L_2)$ and $x_T = x_S$, $y_T = z_S$, $z_T = -y_S$, as drawn. At $\theta_2 = 90^\circ$, $\theta_3 = 0$: tip $(0, 0, L_1{+}L_2{+}L_3{+}L_4)$, arm straight up (as $\mathbf{R}_y(-\theta_2)$ predicts).
Numbers $\theta = (30^\circ, 20^\circ, 40^\circ)$: frame $\{2\}$ at $(0, 0, 1.1)$, frame $\{3\}$ at $(0.3255,\ 0.1879,\ 1.2368)$ m,

$$ {}^S\mathbf{T}_T = \begin{bmatrix} 0.4330 & -0.7500 & 0.5000 & 0.4554 \\ 0.2500 & -0.4330 & -0.8660 & 0.2629 \\ 0.8660 & 0.5000 & 0 & 1.4966 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$ [checked]

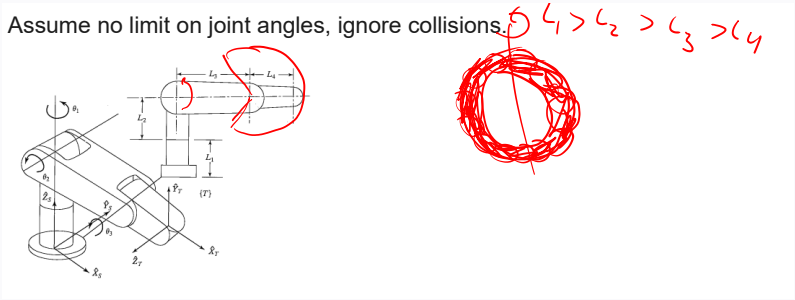
Marks lost by: $\mathbf{R}_y(+\theta_2)$ (arm dips when θ_2 grows), dropping the final $\mathbf{R}_x(90^\circ)$ (position right, orientation wrong), giving only the product with no intermediate frames when the question asks for them.

Sample Q2 (W3, p. 45)

SAMPLE

EXAM-STYLE QUESTION W13 Q2 [marks not published; ~1 mark per step]

Sketch the workspace of the manipulator (RRR arm of Q1: rotation θ_1 about z_S, then two joints about horizontal axes, links L_3, L_4 above a shoulder at height L_1+L_2). Assume no limit on joint angles, ignore collisions.



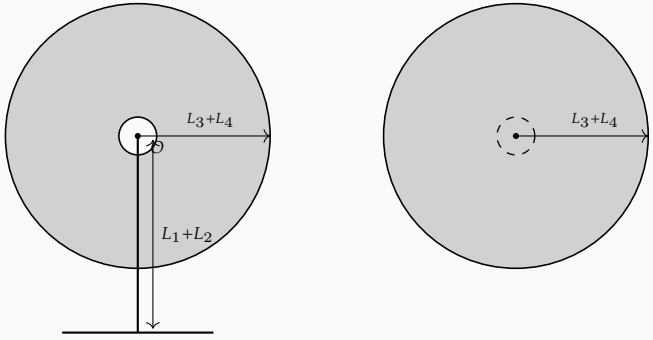
[W13 lec p. 6] — the question and the lecturer's "donut": a ring in the side view

Solution

Frames. $\{S\}$ at the base, z_S up. Shoulder point $O = (0, 0, L_1 + L_2)$.

Reasoning. Joints 2 and 3 are a planar RR arm (links L_3, L_4) in a vertical plane through z_S, pivoting about O. By the cosine rule the tip is at distance $\rho = \sqrt{L_3^2 + L_4^2 + 2L_3L_4\cos\theta_3}$ from O; with θ_3 free, ρ takes every value from $|L_3 - L_4|$ (folded) to $L_3 + L_4$ (stretched). With θ_2 free, the tip sweeps a full annulus in that plane. Joint 1 spins the plane about z_S, which passes through O.

Result. The workspace is a **hollow sphere (spherical shell) centred on** O, outer radius $L_3 + L_4$, inner radius $|L_3 - L_4|$; L_1, L_2 only lift the centre. For $L_3 = 0.4$, $L_4 = 0.3$ m: $0.1 \le \rho \le 0.7$ m about O. If $L_3 = L_4$ the hole closes: a solid ball. The dexterous workspace is empty (a position fixes the tool direction up to the branch).



side view (x_Sz_S plane) top view (disc)

[drawn for this booklet] side view: ring about O (hole radius $|L_3 - L_4|$, to scale; height of O not to scale). Top view: full disc of radius $L_3 + L_4$ (hole not visible from above: those points are reached at other heights).

Check at a simple pose: $\theta_2 = \theta_3 = 0$, arm horizontal, tip at $(L_3 + L_4, 0, L_1 + L_2)$, on the outer sphere ✓; folded $\theta_3 = \pi$: $\rho = |L_3 - L_4|$ ✓ (500 random poses give $\rho \in [0.1, 0.7]$). [checked]

Sample Q3 (W3, p. 45)

EXAM-STYLE QUESTION W13 Q3 [marks not published]

Derive the inverse kinematics of the RRR arm of Q1. You are given the position of the end effector (x_T, y_T, z_T).

Solution

FK (from Q1). ${}^S\mathbf{p}_T = (c_1 r,\ s_1 r,\ L_1 + L_2 + L_3 s_2 + L_4 s_{23})$, $r = L_3c_2 + L_4c_{23}$; $c_{23} = \cos(\theta_2 + \theta_3)$.

Top-down view gives θ_1: $\theta_1 = \operatorname{atan2}(y_T, x_T)$, $r = +\sqrt{x_T^2 + y_T^2}$ (arm faces the point). Branch 2: $\theta_1 + \pi$ with $r = -\sqrt{x_T^2 + y_T^2}$ (arm reaches back over the base axis). atan2, not the $\tan^{-1}(y/x)$ of the slide.

Side-on view: planar RR in the plane (r, z'), $z' = z_T - L_1 - L_2$:

$$r = L_3c_2 + L_4c_{23}, \qquad z' = L_3s_2 + L_4s_{23}$$

Square and add (the cross terms give $2L_3L_4(c_2c_{23} + s_2s_{23}) = 2L_3L_4\cos\theta_3$):

$$r^2 + z'^2 = L_3^2 + L_4^2 + 2L_3L_4\cos\theta_3 \ \Rightarrow\ \cos\theta_3 = \frac{r^2 + z'^2 - L_3^2 - L_4^2}{2L_3L_4},$$

$$\theta_3 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2\theta_3}, \cos\theta_3\right)$$

Expand c_{23}, s_{23} and write $r = (L_3 + L_4c_3)c_2 - (L_4s_3)s_2$, $z' = (L_4s_3)c_2 + (L_3 + L_4c_3)s_2$; the point (r, z') is $(L_3 + L_4c_3,\ L_4s_3)$ rotated by θ_2:

$$\theta_2 = \operatorname{atan2}(z', r) - \operatorname{atan2}(L_4\sin\theta_3,\ L_3 + L_4\cos\theta_3)$$

Number and existence. 2 (shoulder) $\times$ 2 (elbow) = 4 for an interior point. No solution if $|\cos\theta_3| > 1$, i.e. the point lies outside the shell of Q2 (outside $|L_3 - L_4| \le \rho \le L_3 + L_4$ about the shoulder). On the shell surface the two elbow branches merge. If $x_T = y_T = 0$, θ_1 is arbitrary.

Numbers. $L = (0.6, 0.5, 0.4, 0.3)$ m, $\mathbf{p}_T = (0.4554, 0.2629, 1.4966)$ m. $\theta_1 = \operatorname{atan2}(0.2629, 0.4554) = 30°$, $r = 0.5259$, $z' = 1.4966 - 1.1 = 0.3966$; $\cos\theta_3 = \dfrac{0.2766 + 0.1573 - 0.16 - 0.09}{0.24} = 0.7660$, $\theta_3 = \pm 40°$; $\sigma = \operatorname{atan2}(0.3966, 0.5259) = 37.024°$, $\mu = \pm 17.024°$. Solutions $(\theta_1, \theta_2, \theta_3)$ in degrees:

$$(30,\ 20,\ 40),\quad (30,\ 54.047,\ -40),\quad (-150,\ 125.953,\ 40),\quad (-150,\ 160,\ -40)$$

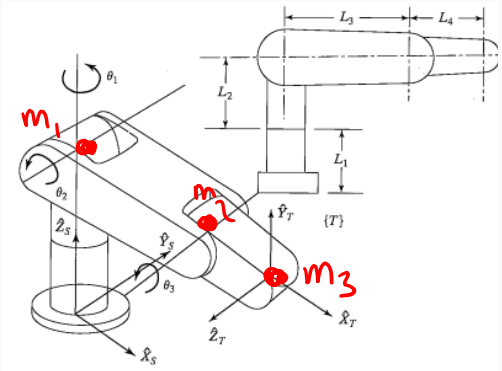
Each reproduces $\mathbf{p}_T$ by FK; the first is the Q1 pose, so the check closes. Reach test: a target $(0.5, 0.5, 1.8)$ is 0.995 m from the shoulder > 0.7: no solution. [checked]

Sanity: $\theta_2 = \theta_3 = 0$ gives $r = L_3 + L_4$, $z = L_1 + L_2$ (stretched, on the outer sphere) ✓.

Sample Q4 (W7, p. 91)

SAMPLE

EXAM-STYLE QUESTION W13 Q4 (a): derive the equations of motion, 3D RRR arm [marks not published; indicative, 1 per step]



[W13 lec p. 8] — W13 p. 8: RRR arm; the lecturer's red dots mark m_1 (shoulder), m_2 (elbow), m_3 (tool). θ_1 about z_S, θ_2, θ_3 in the vertical arm plane

"Mass, moments of inertia and positions of centre of mass are given. Derive the equations of motion."

Solution

Set-up (state it). Frame $\{S\}$: z_S up, gravity along $-z_S$. θ_1 turns the whole arm about z_S. The arm plane is vertical, at azimuth θ_1; shoulder at height $H = L_1 + L_2$ on the axis. θ_2: elevation of the upper arm (length L_3) from the horizontal; θ_3: forearm (length L_4) relative to the upper arm, so the forearm's absolute elevation is $\theta_2 + \theta_3$. (If the exam measures angles from the vertical, swap $c \leftrightarrow s$ in heights.) Point masses first (the red dots): m_2 at the elbow, m_3 at the tool; m_1 sits on the axis, so it only adds a constant to P and nothing to the equations. Use $\rho =$ horizontal distance from z_S and $h =$ height.

Positions. $\rho_2 = L_3 c_2$, $h_2 = H + L_3 s_2$; $\rho_3 = L_3 c_2 + L_4 c_{23}$, $h_3 = H + L_3 s_2 + L_4 s_{23}$.

Velocities. A point at (ρ, h) on a plane spinning at $\dot\theta_1$: $v^2 = \dot\rho^2 + \dot h^2 + \rho^2\dot\theta_1^2$. $\dot\rho_2^2 + \dot h_2^2 = L_3^2\dot\theta_2^2$. For mass 3: $(\dot\rho_3, \dot h_3) = L_3\dot\theta_2(-s_2, c_2) + L_4(\dot\theta_2 + \dot\theta_3)(-s_{23}, c_{23})$, so $\dot\rho_3^2 + \dot h_3^2 = L_3^2\dot\theta_2^2 + L_4^2(\dot\theta_2 + \dot\theta_3)^2 + 2L_3L_4c_3\,\dot\theta_2(\dot\theta_2 + \dot\theta_3)$ (using $s_2s_{23} + c_2c_{23} = c_3$).

Energies.

$$K = \tfrac{1}{2}\big(m_2\rho_2^2 + m_3\rho_3^2\big)\dot\theta_1^2 + \tfrac{1}{2}\big[(m_2+m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3\big]\dot\theta_2^2 + m_3\big(L_4^2 + L_3L_4c_3\big)\dot\theta_2\dot\theta_3 + \tfrac{1}{2}m_3L_4^2\dot\theta_3^2$$

$$P = m_2g(H + L_3s_2) + m_3g(H + L_3s_2 + L_4s_{23}) + \text{const}, \qquad \mathcal{L} = K - P$$

Reading $\mathbf{M}$ off $K = \tfrac{1}{2}\dot{\boldsymbol\theta}^\top\mathbf{M}\dot{\boldsymbol\theta}$: $M_{11} = m_2\rho_2^2 + m_3\rho_3^2$; $M_{22} = (m_2+m_3)L_3^2 + m_3L_4^2 + 2m_3L_3L_4c_3$; $M_{23} = m_3(L_4^2 + L_3L_4c_3)$; $M_{33} = m_3L_4^2$; $M_{12} = M_{13} = 0$.

EXAM-STYLE QUESTION W13 Q4 (b): Euler–Lagrange lines [cont.]

Joint 1. $\partial\mathcal{L}/\partial\dot\theta_1 = M_{11}\dot\theta_1$, $\frac{d}{dt} = M_{11}\ddot\theta_1 + \dot M_{11}\dot\theta_1$; $\partial\mathcal{L}/\partial\theta_1 = 0$ (nothing depends on θ_1). With $\dot\rho_2 = -L_3s_2\dot\theta_2$, $\dot\rho_3 = -L_3s_2\dot\theta_2 - L_4s_{23}(\dot\theta_2+\dot\theta_3)$, $\dot M_{11} = 2m_2\rho_2\dot\rho_2 + 2m_3\rho_3\dot\rho_3$:

$$Q_1 = M_{11}\ddot\theta_1 - 2\dot\theta_1\Big[m_2\rho_2L_3s_2\dot\theta_2 + m_3\rho_3\big(L_3s_2\dot\theta_2 + L_4s_{23}(\dot\theta_2+\dot\theta_3)\big)\Big]$$

Joint 2. $\partial\mathcal{L}/\partial\dot\theta_2 = M_{22}\dot\theta_2 + M_{23}\dot\theta_3$; $\frac{d}{dt} = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 - m_3L_3L_4s_3\dot\theta_3(2\dot\theta_2+\dot\theta_3)$ (using $\dot M_{22} = -2m_3L_3L_4s_3\dot\theta_3$, $\dot M_{23} = -m_3L_3L_4s_3\dot\theta_3$). $\partial\mathcal{L}/\partial\theta_2 = \tfrac{1}{2}\dot\theta_1^2\,\partial M_{11}/\partial\theta_2 - g\big[(m_2+m_3)L_3c_2 + m_3L_4c_{23}\big]$ with $\tfrac{1}{2}\partial M_{11}/\partial\theta_2 = -\big[m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2 + L_4s_{23})\big]$.

$$Q_2 = M_{22}\ddot\theta_2 + M_{23}\ddot\theta_3 + \dot\theta_1^2\big[m_2\rho_2L_3s_2 + m_3\rho_3(L_3s_2 + L_4s_{23})\big] - m_3L_3L_4s_3(2\dot\theta_2\dot\theta_3 + \dot\theta_3^2) + g\big[(m_2+m_3)L_3c_2 + m_3L_4c_{23}\big]$$

Joint 3. $\partial\mathcal{L}/\partial\dot\theta_3 = M_{23}\dot\theta_2 + M_{33}\dot\theta_3$; $\frac{d}{dt} = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 - m_3L_3L_4s_3\dot\theta_3\dot\theta_2$. $\partial K/\partial\theta_3 = -m_3\rho_3L_4s_{23}\dot\theta_1^2 - m_3L_3L_4s_3(\dot\theta_2^2 + \dot\theta_2\dot\theta_3)$; $\partial P/\partial\theta_3 = m_3gL_4c_{23}$. The $\dot\theta_2\dot\theta_3$ terms cancel:

$$Q_3 = M_{23}\ddot\theta_2 + M_{33}\ddot\theta_3 + m_3\rho_3L_4s_{23}\dot\theta_1^2 + m_3L_3L_4s_3\dot\theta_2^2 + g\,m_3L_4c_{23}$$

SAMPLE

EXAM-STYLE QUESTION W13 Q4 (c): M, C, g, general links, numbers [cont.]

$$\mathbf{M} = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33} \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ g[(m_2+m_3)L_3c_2 + m_3L_4c_{23}] \\ g\,m_3L_4c_{23} \end{bmatrix}$$

$$\mathbf{C} = (V_1, V_2, V_3)^\top = \text{the velocity terms of } Q_1, Q_2, Q_3$$

Centrifugal: the $\dot\theta_1^2$ terms in V_2, V_3 and $m_3L_3L_4s_3\dot\theta_2^2$ (squares). Coriolis: $\dot\theta_1\dot\theta_2$, $\dot\theta_1\dot\theta_3$ in V_1; $\dot\theta_2\dot\theta_3$ in V_2 (products). Physical reading: spinning about z_S flings the arm outwards; raising the arm while spinning changes the inertia about z_S, so joint 1 must push; gravity loads joints 2 and 3 only, most when the arm is horizontal.

With link inertias (slender links, COM at l_2 from the shoulder and l_3 from the elbow, I_2, I_3 about axes $\perp$ the link through the COM, ≈ 0 about the link axis, I_1 the column about z_S): replace $L_3 \to l_2$ in link 2's terms and $L_4 \to l_3$ in link 3's (keep L_3 where the elbow position enters ρ_3), $\rho_2 = l_2c_2$, $\rho_3 = L_3c_2 + l_3c_{23}$, and

$$M_{11} = I_1 + m_2\rho_2^2 + m_3\rho_3^2 + I_2c_2^2 + I_3c_{23}^2, \quad M_{22} = m_2l_2^2 + I_2 + m_3(L_3^2 + l_3^2 + 2L_3l_3c_3) + I_3$$

$$M_{23} = m_3(l_3^2 + L_3l_3c_3) + I_3, \quad M_{33} = m_3l_3^2 + I_3, \quad \mathbf{g} = \begin{bmatrix} 0 \\ g[(m_2l_2 + m_3L_3)c_2 + m_3l_3c_{23}] \\ g\,m_3l_3c_{23} \end{bmatrix}$$

$$V_1 = -2\dot\theta_1\left[m_2\rho_2l_2s_2\dot\theta_2 + m_3\rho_3(L_3s_2\dot\theta_2 + l_3s_{23}(\dot\theta_2+\dot\theta_3)) + I_2c_2s_2\dot\theta_2 + I_3c_{23}s_{23}(\dot\theta_2+\dot\theta_3)\right]$$

$$V_2 = \dot\theta_1^2\left[m_2\rho_2l_2s_2 + m_3\rho_3(L_3s_2 + l_3s_{23}) + I_2c_2s_2 + I_3c_{23}s_{23}\right] - m_3L_3l_3s_3(2\dot\theta_2\dot\theta_3 + \dot\theta_3^2)$$

$$V_3 = \dot\theta_1^2\left[m_3\rho_3l_3s_{23} + I_3c_{23}s_{23}\right] + m_3L_3l_3s_3\dot\theta_2^2$$

(The red-dot case is $l_2 = L_3$, $l_3 = L_4$, $I_2 = I_3 = 0$.) Rotational energy added: $\frac12 I_1\dot\theta_1^2 + \frac12 I_2(\dot\theta_2^2 + c_2^2\dot\theta_1^2) + \frac12 I_3((\dot\theta_2+\dot\theta_3)^2 + c_{23}^2\dot\theta_1^2)$.

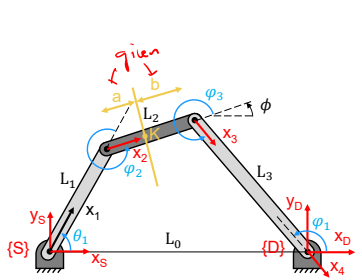
Numbers (point masses): $m_2 = 2$, $m_3 = 1$ kg, $L_3 = 0.6$, $L_4 = 0.4$ m, $\boldsymbol\theta = (20°, 30°, -40°)$, $\dot{\boldsymbol\theta} = (1, 0.5, -0.5)$ rad/s, $\ddot{\boldsymbol\theta} = (0.2, 1, 0.5)$ rad/s^2. $\rho_2 = 0.5196$, $\rho_3 = 0.9135$ m, $c_3 = 0.7660$, $s_3 = -0.6428$, $c_{23} = 0.9848$, $s_{23} = -0.1736$.

$$\mathbf{M} = \begin{bmatrix} 1.3746 & 0 & 0 \\ 0 & 1.6077 & 0.3439 \\ 0 & 0.3439 & 0.1600 \end{bmatrix}, \ \mathbf{C} = \begin{bmatrix} -0.5858 \\ 0.4838 \\ -0.1020 \end{bmatrix}, \ \mathbf{g} = \begin{bmatrix} 0 \\ 19.157 \\ 3.864 \end{bmatrix}$$

$Q_1 = 1.3746(0.2) - 0.5858 = \boxed{-0.311}$; $Q_2 = 1.6077(1) + 0.3439(0.5) + 0.4838 + 19.157 = \boxed{21.420}$; $Q_3 = 0.3439(1) + 0.16(0.5) - 0.1020 + 3.864 = \boxed{4.186}$ N m. Hold still: $\mathbf{Q} = \mathbf{g} = (0, 19.157, 3.864)$; arm stretched out horizontally: $(0, 21.582, 3.924)$ N m (largest). [checked]

Checks. $\mathbf{M}$ symmetric, positive definite; $\mathbf{g}$ contains no $\dot\theta$; $\mathbf{C} = \mathbf{0}$ at rest; power $\dot{\boldsymbol\theta}^\top\mathbf{Q} = 8.306$ W $= \dot K + \dot P$; free swing conserves $K + P$ to 10^{-9} J over 3 s. [checked]

Sample Q5 (W4, p. 56)



(x, y) in {S}

- Given:
  - The position of a point K on the output link
  - The orientation ϕ of the output link (alternatively, ${}^S\mathbf{R}_2$ of link 2 is given)
- Known:
  - L_0, L_1, L_2, L_3
  - L_2 = a + b
  - a and b are known
- Determine:
  - θ_1, φ_1, φ_2 and φ_3

[W13 lec p. 9] — W13 p. 9: the question (K and ϕ given; find $\theta_1, \varphi_1, \varphi_2, \varphi_3$)

EXAM-STYLE QUESTION W13 Q5: four-bar IK [marks not published]

A planar four-bar has $L_0 = 1.0$, $L_1 = 0.5$, $L_3 = 0.6$ m; the output link (coupler) has $a = 0.2$, $b = 0.25$ m. Point K on the output link is at $(0.3710, 0.4678)$ m in S, and the link is oriented at $\phi = -0.600°$. Determine $\theta_1, \varphi_1, \varphi_2, \varphi_3$.

Solution

Frames: S at the left pivot, D at $(L_0, 0)$, angles counter-clockwise from $+x$. Write each chain's K equations: $x = L_1\cos\theta_1 + a\cos(\theta_1+\varphi_2)$, $y = L_1\sin\theta_1 + a\sin(\theta_1+\varphi_2)$, $\theta_1 + \varphi_2 = \phi$ (mod 2π). [setup]

Substitute: $L_1\cos\theta_1 = x - a\cos\phi$, $L_1\sin\theta_1 = y - a\sin\phi$: $\theta_1 = \operatorname{atan2}(0.4699, 0.1710) = \boxed{70.0°}$;

$\varphi_2 = \phi - \theta_1 = \boxed{289.4°}$ ($\equiv -70.6°$). [left chain]

Right chain: $x = L_0 + L_3\cos\varphi_1 + b\cos(\varphi_1+\varphi_3)$ with $\varphi_1 + \varphi_3 = \phi + \pi$: $L_3\cos\varphi_1 = x - L_0 + b\cos\phi = -0.3790$, $L_3\sin\varphi_1 = y + b\sin\phi = 0.4652$: $\varphi_1 = \operatorname{atan2}(0.4652, -0.3790) = \boxed{129.2°}$;

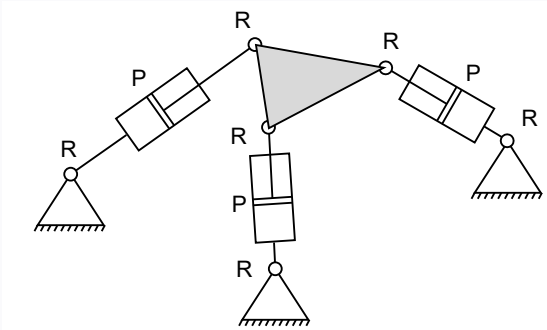
$\varphi_3 = \phi + \pi - \varphi_1 = \boxed{50.2°}$. [right chain]

Check: $\|(-0.379, 0.4652)\| = 0.600 = L_3$; $\theta_1 + \varphi_2 = 359.4° = \varphi_1 + \varphi_3 + 180°$. Say: IK is easy because each leg is solved alone with atan2; one answer per leg. Using the slide's printed $\varphi_3 = \varphi_1 - \phi + \pi$ gives 309.8° and K misses by 0.4 m. [check]

Sample Q6 (W1, p. 26)

SAMPLE

EXAM-STYLE QUESTION W13 Q6: mobility of a planar parallel manipulator [marks not published]



[W13 lec p. 13] — W13 pp.12–13: the same figure; on p.12 the lecturer numbers the links 1 ground, 2 platform, 3–4 left leg, 5–6 middle leg, 7–8 right leg

Find the mobility of the mechanism drawn (planar, three R–P–R legs).

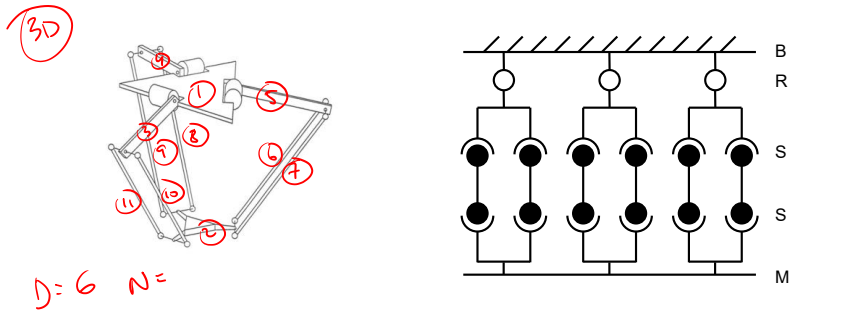
Solution

$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$ (G–K). Planar, so $D = 3$.
Links: 1 ground, 2 platform, and each prismatic joint joins a cylinder to a rod (two links per leg: 3,4; 5,6; 7,8): $N = 8$.
Joints: per leg R, P, R: $J = 9$, each $F_i = 1$, $\sum F_i = 9$.

$$M = 3(8 - 9 - 1) + 9 = -6 + 9 = \boxed{3} \qquad \text{(form 1: } 3(8-1) - 2 \cdot 9 = 3\text{)}$$

Three DoF: platform x, y and rotation, driven by the three prismatic actuators. Check: $N = 7$ (ground forgotten) would give 0. [checked]

Sample Q7 (W4, p. 56)



Note: the mechanism is still 3D despite this depiction

[W13 lec p. 14] — W13 p. 14: the delta mechanism and its joint graph (B base, R, S, S, M moving platform)

EXAM-STYLE QUESTION W13 Q7: delta mobility, plain vs extended [marks not published]

A delta robot has a base, three R joints, each followed by a parallelogram of two parallel bars with spherical joints at both ends of every bar, and a moving platform. (a) Find the mobility with plain G–K. (b) The platform has only 3 translational DoF. Reconcile with the extended equation.

Solution

(a) 3D, $D = 6$. $N = 1 + 3 + 6 + 1 = 11$. $J = 3 + 3 \times 4 = 15$. $\sum F_i = 3(1 + 4 \times 3) = 39$.

$$M = 6(11 - 15 - 1) + 39 = \boxed{9}$$

(b) Each of the 6 bars has an S joint at both ends, so it can spin about its own axis with nothing else moving: $R_M = 6$. No constraint is repeated: $R_C = 0$.

$$M = 6(11 - 15 - 1) + 39 + 0 - 6 = \boxed{3}$$

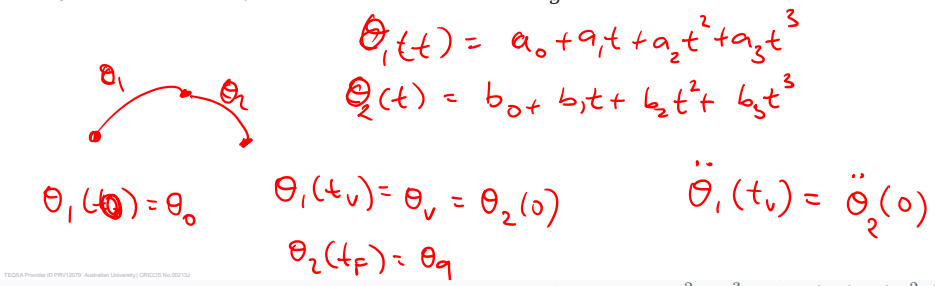
Explain: plain G–K counts every DoF as moving the output; the 6 bar spins are internal, so they are subtracted. The result 3 equals the 3 motors and the 3 platform translations.

Sample Q8 (W5, p. 67)

SAMPLE

EXAM-STYLE QUESTION W13 Q8, symbolic via-point cubics [marks not published]

Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is θ_0 the via point θ_v is and the goal point is θ_g.



[W13 lec p. 18] — W13 p. 18: question and the lecturer's set-up (the slide's cubics are $\theta_1 = a_0+a_1t+a_2t^2+a_3t^3$ and $\theta_2 = b_0+b_1t+b_2t^2+b_3t^3$)

Solution

Assume (state it): start and end at rest, each segment on its own clock, equal durations t_f; frames: joint angle θ in rad (or deg).

Conditions (8 for 8 unknowns): $\theta_1(0) = \theta_0$, $\dot\theta_1(0) = 0$, $\theta_1(t_f) = \theta_v$, $\theta_2(0) = \theta_v$, $\theta_2(t_f) = \theta_g$, $\dot\theta_2(t_f) = 0$, $\dot\theta_1(t_f) = \dot\theta_2(0)$, $\ddot\theta_1(t_f) = \ddot\theta_2(0)$.

Step 1. $a_0 = \theta_0$, $a_1 = 0$, $b_0 = \theta_v$. Let $w = \dot\theta_2(0) = \dot\theta_1(t_f)$.

Step 2. Segment 1 has end speeds $(0, w)$ and $\Delta = \theta_v - \theta_0$: $a_2 = \dfrac{3\Delta - wt_f}{t_f^2}$, $a_3 = \dfrac{-2\Delta + wt_f}{t_f^3}$.

Segment 2 has $(w, 0)$ and $\Delta' = \theta_g - \theta_v$: $b_1 = w$, $b_2 = \dfrac{3\Delta' - 2wt_f}{t_f^2}$, $b_3 = \dfrac{-2\Delta' + wt_f}{t_f^3}$.

Step 3. $\ddot\theta_1(t_f) = 2a_2 + 6a_3t_f = \dfrac{-6\Delta + 4wt_f}{t_f^2}$, $\ddot\theta_2(0) = 2b_2 = \dfrac{6\Delta' - 4wt_f}{t_f^2}$. Equal: $8wt_f = 6(\Delta + \Delta') \Rightarrow \boxed{w = \dfrac{3(\theta_g - \theta_0)}{4t_f}}$.

Step 4. Substitute w:

$$\boxed{\begin{aligned} a_2 &= \tfrac{12\theta_v - 3\theta_g - 9\theta_0}{4t_f^2}, \quad a_3 = \tfrac{-8\theta_v + 3\theta_g + 5\theta_0}{4t_f^3}, \\ b_1 &= \tfrac{3(\theta_g - \theta_0)}{4t_f}, \quad b_2 = \tfrac{-12\theta_v + 6\theta_g + 6\theta_0}{4t_f^2}, \quad b_3 = \tfrac{8\theta_v - 5\theta_g - 3\theta_0}{4t_f^3} \end{aligned}}$$

Check with $\theta = (10, 40, 20)°$, $t_f = 1$ s: $w = 7.5°/\text{s}$, $\theta_1 = 10 + 82.5t^2 - 52.5t^3$, $\theta_2 = 40 + 7.5t - 75t^2 + 47.5t^3$; $\theta_1(1) = 40$, $\theta_2(1) = 20$, $\ddot\theta_1(1) = -150 = \ddot\theta_2(0)$ ✓. Variant $(0, 40, 30)°$, $t_f = 1$: $w = 22.5$; segment 1 $97.5t^2 - 57.5t^3$, segment 2 $40 + 22.5t - 75t^2 + 42.5t^3$; peak speed in segment 1 $55.11°/\text{s}$ at $t = 0.565$ s. [checked]

Common slips: one cubic for both segments; forgetting $a_1 = 0$; using total time instead of t_f per segment; dropping the acceleration condition (then w is undetermined).

Sample Q9 (W5, p. 67)

EXAM-STYLE QUESTION W13 Q9, cubic for a prismatic joint [marks not published]

A prismatic joint, initially at rest, is programmed to move from an initial position d=0.02m to a resting goal position for d=0.15m in 0.8 seconds. Calculate the coefficients of a cubic spline that accomplishes this motion.

[W13 lec p. 19] — W13 p. 19 (question text)

Solution

Prismatic joint, so $q = d$ in m. Conditions: $d(0) = 0.02$, $d(0.8) = 0.15$, $\dot d(0) = 0$, $\dot d(0.8) = 0$. $\Delta d = 0.13$ m, $t_F = 0.8$ s.

$a_0 = 0.02$ m, $a_1 = 0$, $a_2 = \dfrac{3(0.13)}{0.8^2} = 0.609375\,\text{m/s}^2$, $a_3 = \dfrac{-2(0.13)}{0.8^3} = -0.5078125\,\text{m/s}^3$.

$$\boxed{d(t) = 0.02 + 0.609375\,t^2 - 0.5078125\,t^3 \text{ m}} \qquad \dot d = 1.21875t - 1.5234375t^2 \text{ m/s}$$

Check: $d(0.8) = 0.02 + 0.39 - 0.26 = 0.15$ ✓; $\dot d(0.8) = 0.975 - 0.975 = 0$ ✓. Peak speed $1.5\Delta d/t_F = 0.24375$ m/s at $t = 0.4$ s ($d = 0.085$ m); end accelerations $\pm 1.21875\,\text{m/s}^2$. [checked]

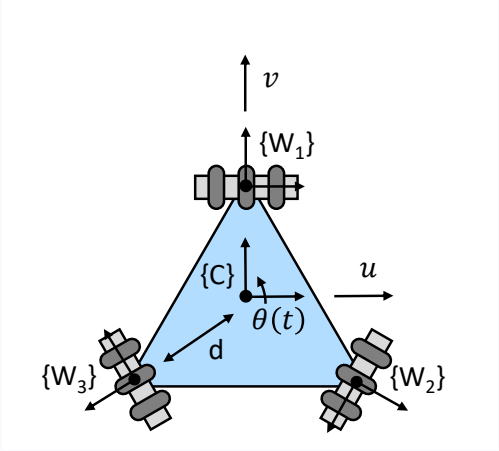
Common slips: a_3 divided by t_F^2; cm/m mixed; forgetting $a_0 = 0.02$.

Sample Q10 (W10, p. 122)

SAMPLE

EXAM-STYLE QUESTION (W13 Q10) [marks not published; indicative 1 per step]

Consider a robot with three omnidirectional wheels, each independently driven, rollers perpendicular to each wheel's driving direction. The robot is an equilateral triangle; the distance from the centre to each vertex is d. Derive a kinematic expression that relates the angular velocities of the wheels $(\dot\varphi_1, \dot\varphi_2, \dot\varphi_3)$ to the velocity of the robot chassis frame $(u, v, \dot\theta)$. Is the robot holonomic? Find the wheel speeds for $u = 0.2$ m/s with $d = 0.10$ m, $r = 0.05$ m.



[W13 lec p. 20] — W13 p. 20: the question's figure

Solution

Setup. $\{C\}$ at the centre, $\hat x_C$ to the right, $\hat y_C$ up. Wheels at $W_1(0, d)$, $W_2(\frac{\sqrt3}{2}d, -\frac{d}{2})$, $W_3(-\frac{\sqrt3}{2}d, -\frac{d}{2})$; drive directions tangential (clockwise, as drawn): $\beta_1 = 0°$, $\beta_2 = -120°$, $\beta_3 = 120°$; $\gamma_i = 0$ (rollers $\perp$ drive). [2]

General formula (slide [p. 36]). $\dot\varphi_i = \begin{bmatrix}\frac{1}{r} & \frac{\tan\gamma_i}{r}\end{bmatrix} \mathbf{R}(\beta_i)^\top \begin{bmatrix}1 & 0 & -{}^C y_{W_i} \\ 0 & 1 & {}^C x_{W_i}\end{bmatrix} \boldsymbol\zeta$; with $\gamma_i = 0$: $r\dot\varphi_i = \cos\beta_i(u - \dot\theta\, {}^C y_{W_i}) + \sin\beta_i(v + \dot\theta\, {}^C x_{W_i})$. [1]

Rows. $r\dot\varphi_1 = u - d\dot\theta$; $r\dot\varphi_2 = -\frac12(u + \frac{d}{2}\dot\theta) - \frac{\sqrt3}{2}(v + \frac{\sqrt3}{2}d\dot\theta) = -\frac12 u - \frac{\sqrt3}{2}v - d\dot\theta$; $r\dot\varphi_3 = -\frac12 u + \frac{\sqrt3}{2}v - d\dot\theta$. [3]

$$\begin{bmatrix}\dot\varphi_1 \\ \dot\varphi_2 \\ \dot\varphi_3\end{bmatrix} = \frac{1}{r}\begin{bmatrix}1 & 0 & -d \\ -\frac12 & -\frac{\sqrt3}{2} & -d \\ -\frac12 & \frac{\sqrt3}{2} & -d\end{bmatrix}\begin{bmatrix}u \\ v \\ \dot\theta\end{bmatrix}$$

Inverse and holonomic. $\det = \frac{3\sqrt3}{2}d \neq 0$; $\boldsymbol\zeta = \frac{r}{3}\begin{bmatrix}2 & -1 & -1 \\ 0 & -\sqrt3 & \sqrt3 \\ -\frac{1}{d} & -\frac{1}{d} & -\frac{1}{d}\end{bmatrix}\dot{\boldsymbol\varphi}$. Swedish wheels give no no-slide row, so $u, v, \dot\theta$ are free: holonomic, $\delta_m = 3$. World frame: $\boldsymbol\nu = \mathbf{J}(\theta)\boldsymbol\zeta$. [2]

Numbers. $u = 0.2$: $\dot\varphi = \frac{1}{0.05}(0.2, -0.1, -0.1) = \boxed{(4, -2, -2)\ \text{rad/s}}$. [1]

Check. Wheel 1 drives along $\hat x_C$ so it carries all of u; wheels 2 and 3 drive at $\pm120°$ and see $-\frac12 u$; a pure spin gives $-d\dot\theta/r$ on every wheel; sum of the three speeds is 0 for pure translation. [checked]

SAMPLE